Practical Excel & Financial Modeling

Practical, Hands-On, Job-Ready — India-first + Global

Real formulas, queries, entries & tools — closing the gap between what an MBA teaches and what finance employers pay for

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Excel: the #1 finance skill and the gap it exposes

What it is & where it's used

Excel is the universal spreadsheet — a grid of cells that becomes a calculator, a database, a reporting tool, and a modeling engine depending on how you use it. In finance it is not "one tool among many"; it is the default surface where numbers get worked. SAP, Oracle, Tally, Zoho Books, and every bank portal ultimately export to Excel because that's where humans reason about the data.

Every finance role touches it, but differently:

Role	What they do in Excel daily
Accounts / AP-AR	Reconciliations, ageing schedules, ledger dumps, GST 2A/2B matching
Audit / assurance	Sampling, variance testing, tie-outs, TB-to-FS mapping
FP&A / analyst	Budgets, variance decks, rolling forecasts, KPI trackers
Investment banking / PE	3-statement models, DCF, LBO, comps
Taxation	Computation sheets, TDS workings, GST returns prep, 26AS reconciliation
Treasury	Cash flow forecasting, interest schedules, bank position sheets

If you can only bring one skill to a finance desk on day one, it is Excel. Everything else — Tally, SQL, Power BI — sits around it.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you *what* NPV, WACC, and working capital mean. A CA course teaches you *what* the standards require. Neither teaches you to **build the sheet that a manager trusts at 11 pm before a board meeting.** That build skill is the gap.

Concretely, college leaves you unable to:

- Pull the right value from a 40,000-row ledger dump **without scrolling** (lookups, not eyeballing).
- Build a formula that **doesn't break** when a row is inserted (absolute vs relative references, structured tables).
- Make one input change and have the **whole model recalculate correctly** (linked, not hardcoded).
- Produce output a reviewer can **audit in 30 seconds** (colour conventions, no buried constants).
- Do all of the above **fast, keyboard-only, under time pressure.**

Employers pay for the person who receives a messy export and returns a clean, correct, self-checking answer unaided. That's the difference between "I've used Excel" and "job-ready Excel."

What "proficient" looks like

The concrete bar employers test for — a proficient person can, unaided:

1. **Navigate and clean** a raw export: remove duplicates, `TRIM` / `CLEAN` text, split columns, fix date-as-text.
2. **Look up and match** across sheets with `XLOOKUP` / `INDEX-MATCH`, and multi-criteria with `SUMIFS`.

3. **Build logic** with nested `IF`, `IFS`, `AND` / `OR`, and error-trap with `IFERROR`.
4. **Aggregate** thousands of rows via PivotTables in under two minutes.
5. **Model** with a clean inputs → calc → output structure and no hardcoded numbers inside formulas.
6. **Work keyboard-first** — `Ctrl+Arrow`, `Ctrl+Shift+Arrow`, `Alt+=`, `F4`, `Ctrl+;`.
7. **Self-check** — build a control total that must equal zero.

A rough ladder: Basic (formulas, formatting), Intermediate (lookups, pivots, charts — this is the *hiring minimum* for most accounts/analyst roles), Advanced (dynamic arrays, `LET` / `LAMBDA`, full 3-statement models).

Hands-on: how to actually do it

Lookups — use XLOOKUP, fall back to INDEX-MATCH.

```
=XLOOKUP(A2, Vendors[GSTIN], Vendors[VendorName], "Not found")

=INDEX(Vendors[VendorName], MATCH(A2, Vendors[GSTIN], 0))
```

`XLOOKUP` looks up `A2` in the GSTIN column and returns the vendor name; the 4th argument replaces the ugly `#N/A`. `INDEX-MATCH` does the same and works in every Excel version.

Multi-criteria sum — SUMIFS is the workhorse of accounts.

```
=SUMIFS(Ledger[Amount], Ledger[Party], "Acme Ltd", Ledger[Type], "Sales", Ledger[Date], "≥"&D
```

Sum of all Acme sales from 1-Apr-2025 onward. `COUNTIFS` and `AVERAGEIFS` follow the same pattern.

Conditional logic with error trapping.

```
=IFERROR(IF(D2>1000000, D2*0.1, D2*0.05), 0)
```

GST split from a tax-inclusive amount (intra-state, 18%).

```
=ROUND(A2/1.18*0.09, 2)   // CGST
=ROUND(A2/1.18*0.09, 2)   // SGST
```

Data cleaning that saves reconciliations.

```
=TRIM(CLEAN(A2))                 // strip stray spaces + non-printing chars
=TEXT(A2,"0000000000")           // pad GSTIN/PAN keys before matching
=DATEVALUE(A2)                   // convert text-dates so they compare correctly
```

Dynamic arrays (Excel 365) — filter without a pivot.

```
=FILTER(Ledger, Ledger[Balance]>0)
=SORT(UNIQUE(Ledger[Party]))
```

The two keyboard reflexes that separate pros from beginners: `F4` to toggle `$` anchors (`A1` → `$A$1`), and `Alt + =` to auto-sum a column instantly.

Worked example / mini-project: GST 2B input-credit reconciliation

You have two sheets: **Books** (purchases you recorded) and **GSTR-2B** (what the portal says vendors filed). You must find invoices where credit is at risk.

Books (columns A–D): Invoice No, GSTIN, Taxable Value, GST in books. **2B** (columns A–D): Invoice No, GSTIN, Taxable Value, GST as per 2B.

Step 1 — build a match key in both sheets (E2, filled down):

```
=TRIM(B2)&"|"&TRIM(A2)
```

Step 2 — in **Books**, pull the 2B tax against each invoice:

```
=XLOOKUP(E2, '2B'!$E$2:$E$5000, '2B'!$D$2:$D$5000, "MISSING IN 2B")
```

Step 3 — flag the difference (G2):

```
=IF(F2="MISSING IN 2B","BLOCK CREDIT", IF(ROUND(D2-F2,0)=0,"OK","MISMATCH ₹"&TEXT(D2-F2,"0")))
```

Sample result:

Invoice	GSTIN	GST books	GST 2B	Flag
INV-101	29AABCU...	18,000	18,000	OK
INV-102	27AAAC...	9,000	0	MISSING IN 2B → BLOCK CREDIT
INV-103	24AACC...	12,600	11,700	MISMATCH ₹900

Step 4 — the control total (single cell, must show a real number, not error):

```
=SUMIF(G:G,"BLOCK CREDIT",D:D)   // total ITC you cannot claim this month
```

That last cell is what a manager actually reads. You've turned two raw dumps into a decision: how much input credit to reverse. This is exactly the kind of task a fresher gets in week one at a tax or audit firm.

How it's tested

In the interview (verbal): - "Difference between `VLOOKUP` and `INDEX-MATCH`? Why might `XLOOKUP` be better?" - "What does `F4` do while editing a formula?" (Expect: toggles absolute/relative reference.) - "How would you find duplicate invoice numbers?" (Answer: `COUNTIF(range,cell)>1`, or Remove Duplicates, or a pivot.) - "A `SUMIFS` returns 0 but you expect a number — how do you debug?" (Text vs number, trailing spaces, date-as-text.)

The practical test (this is where offers are decided): a **timed 30–60 minute Excel test.** Typical brief:

> "Here's a 5,000-row sales export. Build a summary by region and month, flag any order above ₹5 lakh, look up the salesperson from the second tab, and give me total revenue with a control check. You have 40 minutes."

They watch (or infer from your file): Did you use lookups or copy-paste? Did you hardcode? Does it break if a row is added? Is there a self-check? For accounting roles the case is often **"reconcile these two ledgers"** or **"close these books and produce a TB."** Speed and correctness both count — and using the keyboard, not the mouse, visibly signals fluency.

Common mistakes & how pros avoid them

Mistake	Why it hurts	Pro habit
Hardcoding numbers inside formulas (`=B2*0.18`)	Nobody can find or change the assumption	Put 18% in a labelled input cell; reference it
Manual `VLOOKUP` with a counted column index	Breaks the instant a column is inserted	`XLOOKUP` / `INDEX-MATCH` on named columns
Forgetting `$` anchors when copying	Formula drifts, silent wrong answers	Hit `F4`; know what to lock
No error trapping	`#N/A` cascades and breaks totals	Wrap in `IFERROR`
No control total	Errors ship undetected	One cell that must equal zero
Mixing inputs, calcs, outputs on one sheet	Un-auditable	Blue = input, black = formula; separate zones
Merged cells everywhere	Kills sorting, pivots, references	"Center Across Selection" instead
Volatile mega-formulas nobody can read	Un-reviewable, un-maintainable	Break into steps or use `LET` to name parts

The colour convention (blue font for hardcoded inputs, black for formulas) is a genuine industry standard — reviewers scan for blue to find every assumption in seconds.

Learn-it roadmap & resources

Realistic time-to-proficiency (from an MBA/CA base, practising ~1 hr/day):

Level	Time	You can then...
Basic → Intermediate	3–4 weeks	Lookups, `SUMIFS`, pivots, clean charts — clears most job screens
Intermediate → job-ready	6–8 weeks	Handle real dumps, reconcile unaided, pass timed tests
Advanced	3–4 months	Dynamic arrays, `LET` / `LAMBDA`, full 3-statement models

Resources: - *Free:* Microsoft's own Excel training (support.microsoft.com), ExcelJet's formula reference, Chandoo.org, Kenji Explains and Leila Gharani on YouTube. For India-specific GST/TDS sheets, the CBIC and income-tax portals give real return formats to rebuild. - *Paid / certification:* Microsoft Office Specialist (MOS): Excel Associate/Expert — the globally-recognised cert; Corporate Finance Institute (CFI) Excel courses; Wall Street Prep / BIWS for modeling. In India, an MOS Excel Expert badge on a resume meaningfully de-risks you to a hiring manager. - *Practice, not watching:* download a bank statement and a ledger, and force yourself to reconcile them keyboard-only. Repetition on real data beats any course.

Turn off the mouse for an hour a day. That single constraint builds the fluency employers actually test.

Quick-reference

Task	Formula / shortcut
Lookup (modern)	`=XLOOKUP(val, lookup_col, return_col, "NA")`
Lookup (any version)	`=INDEX(ret, MATCH(val, look, 0))`
Multi-criteria sum	`=SUMIFS(sum, c1, k1, c2, k2)`
Count with condition	`=COUNTIFS(range, criteria)`
Duplicate check	`=COUNTIF(A:A, A2)>1`
Error trap	`=IFERROR(formula, 0)`
Clean text key	`=TRIM(CLEAN(A2))`
Text-date → date	`=DATEVALUE(A2)`
GST 18% split	`=ROUND(amt/1.18*0.09,2)` each for CGST & SGST
Filter (365)	`=FILTER(tbl, tbl[col]>0)`
Unique list	`=SORT(UNIQUE(range))`
Auto-sum	`Alt + =`
Toggle absolute ref	`F4`
Today's date (static)	`Ctrl + ;`
Jump to data edge	`Ctrl + Arrow`
Select to edge	`Ctrl + Shift + Arrow`
Remove duplicates	Data tab → Remove Duplicates
PivotTable	`Alt, N, V` → drag fields

GST rate reminder (intra-state): total rate splits equally into CGST + SGST (e.g. 18% = 9% + 9%); inter-state is a single IGST at the full rate. Colour code: **blue = input, black = formula.** Always end with one **control cell that must equal zero.**

Excel Foundations Pros Use

What it is & where it's used

Excel is the default operating system of finance. Not because it's the best tool for every job, but because it is the one tool every analyst, controller, banker, auditor, and CFO can open on any machine and immediately read. In an Indian finance job you will live inside Excel for MIS reports, budget vs actuals, receivable ageing, GST reconciliations (GSTR-2B vs purchase register), invoice trackers, DCF and valuation models, and the monthly deck that goes to management.

"Foundations pros use" means the boring, invisible layer that separates a fast analyst from a slow one: working without touching the mouse, structuring a workbook so it can be audited, formatting numbers so a ₹ figure reads correctly at a glance, and knowing exactly when a reference should lock. Roles that test this on day one: **FP&A / MIS analyst, audit associate (Big 4 and mid-tier), investment banking / equity research associate, accounts executive, treasury, and tax analyst.** Every one of them assumes you already have this. Nobody trains you on it.

The gap: why companies want this (and college didn't teach it)

An MBA Finance course teaches you what WACC *means*. It does not teach you that the WACC cell should be a single blue input feeding twenty grey formula cells, or that hard-coding 0.12 inside a formula is a firing-level habit in a model that gets reviewed. College evaluates *answers*; employers evaluate *the workbook that produced the answer* — because someone else has to audit, update, and trust it next quarter.

The specific gaps:

- You were graded on getting a number; industry grades you on a model a stranger can follow.
- You used the mouse; a pro closes books using the keyboard because at 200 rows the mouse is the bottleneck.
- You typed numbers into formulas; industry demands separation of **inputs, calculations, and outputs**.
- You never had to explain *why* a cell shows what it shows — pros use auditing tools (F2, F9, Ctrl+[) to defend every figure.

Closing this gap is the cheapest, highest-return week of study before any finance interview.

What "proficient" looks like

The concrete bar. A job-ready person can, **unaided and mostly without a mouse**:

- Navigate and edit a 5,000-row sheet using only the keyboard (Ctrl+Arrows, Ctrl+Shift+Arrows, Ctrl+PgUp/PgDn).
- Build a workbook with a clear structure: an **Inputs** tab (blue font), **Calc** tabs (black), and an **Output/MIS** tab (formatted for print).
- Format Indian financials: lakhs/crores scaling, `(1,234)` for negatives in red, `-` for zero, ₹ symbol only where needed.
- Explain absolute vs relative refs and fix a broken drag by adding `$` in the right place — using F4, not retyping.

- Create and use named ranges so `=Price*Qty` reads instead of `=B4*C4`.
- Audit any formula: trace precedents (Ctrl+[), evaluate a sub-part with F9, and step through with Evaluate Formula.

If you can do those six without hesitating, you clear the practical filter for most entry finance roles.

Hands-on: how to actually do it

Keyboard-first workflow

Train your hands off the mouse. The core set:

Shortcut	Does
`Ctrl + Arrow`	Jump to edge of data block
`Ctrl + Shift + Arrow`	Select to edge of data block
`Ctrl + Space` / `Shift + Space`	Select column / row
`Alt + =`	AutoSum
`Ctrl + ;`	Insert today's date (static)
`F2`	Edit cell (see the formula)
`F4`	Toggle `$` locks / repeat last action
`Ctrl + [`	Select precedent cells
`Ctrl + \`	Select cells whose value differs across a row
`Alt + E, S, V`	Paste Special → Values
`Ctrl + T`	Convert range to Table
`Ctrl + PgUp / PgDn`	Move between sheets

The **ribbon accelerator** `Alt` is the master key: press `Alt` and Excel prints letters over every command. `Alt, H, 0` deletes a decimal. `Alt, A, S, S` opens Sort. Learn the path once and you never reach for the mouse.

Number formatting for finance (India)

Use **Custom Format** (`Ctrl + 1` → Number → Custom). The four sections are `positive;negative;zero;text`.

```
Standard accounting (red negatives, dash for zero):
#,##0.00;[Red](#,##0.00);"-"

Whole rupees with symbol:
[$₹-en-IN] #,##0;[Red]([$₹-en-IN] #,##0)

Indian lakh/crore digit grouping (12,34,56,789):
[≥10000000]##\,##\,##\,##0;[≥100000]##\,##\,##0;##,##0

Show figures in ₹ lakhs (scale by /100000 using a trailing comma trick won't work in INR;
instead divide in a formula, then format):
#,##0.00" L"
```

Rule pros follow: **never divide by 100000 inside the number format for lakhs** — do the division in a helper cell or state "₹ in lakhs" in the header and divide the source. Keep formatting and math separate.

Absolute / relative references

- `A1` – relative: both move when copied.
- `$A$1` – absolute: locked.
- `A$1` – row locked (mixed).
- `$A1` – column locked (mixed).

Press `F4` while the cursor is on a reference to cycle `A1 → $A$1 → A$1 → $A1`. The classic use: a tax-rate cell you multiply every row against.

```
=E4*$B$1        // B1 holds the GST rate; lock it so every row hits the same cell
```

Named ranges

Select the cell, type a name in the **Name Box** (top-left) or `Ctrl + F3` (Name Manager). Now formulas read like English:

```
=Taxable_Value*GST_Rate
=SUMIFS(Amount, State, "Karnataka")
```

Scope named ranges to the workbook, avoid spaces (use underscores), and never name a range something that looks like a cell (`Q1` is illegal).

Auditing tools

- **Trace Precedents:** `Alt, M, P` (or Formulas ribbon) draws arrows to the cells feeding this one. `Ctrl + [` jumps to them.
- **Trace Dependents:** `Alt, M, D`.
- **F9 inside a formula:** highlight any part of a formula in edit mode and press F9 — Excel replaces it with its live result so you can see *which piece* is wrong. Press `Esc` (not Enter) to avoid overwriting.

- **Evaluate Formula:** `Alt, M, V` steps through calculation order.
- **Show Formulas:** `Ctrl +` `` (grave) toggles the whole sheet between values and formulas — the fastest audit view.

Worked example / mini-project

Build a one-tab **GST output MIS** for a trading firm. Reproduce this exactly.

Inputs (blue font, put the rate in a locked cell `B1`):

	A	B	C	D
1	GST Rate	18%		
3	Invoice	State	Taxable (₹)	GST (₹)
4	INV-001	Karnataka	1,20,000	=C4*$B$1
5	INV-002	Maharashtra	85,000	=C5*$B$1
6	INV-003	Karnataka	2,40,000	=C6*$B$1
7	INV-004	Tamil Nadu	60,000	=C7*$B$1

Now the summary block using **named ranges**. Select `C4:C7`, name it `Taxable`; name `B4:B7` as `State`; name `D4:D7` as `GST`.

```
Total taxable:      =SUM(Taxable)                 → 5,05,000
Total GST:          =SUM(GST)                     → 90,900
Karnataka taxable:  =SUMIFS(Taxable,State,"Karnataka")   → 3,60,000
Invoice count:      =COUNTA(State)                → 4
Highest invoice:    =MAX(Taxable)                 → 2,40,000
```

Add a lookup so a user can type a state and get its GST:

```
=SUMIFS(GST, State, G1)     // G1 = "Maharashtra" → 15,300
```

Format column C and D with `#,##0;[Red](#,##0)`. Now **audit it**: click the Total GST cell, press `Ctrl + [` — arrows should point only to `D4:D7`. Click a single GST cell, press `F2`, highlight `$B$1`, press `F9` — it shows `0.18`. Press `Esc`. If you change `B1` to 12%, every GST cell and both totals update because you locked the rate instead of hard-coding it. That single behaviour — one input, cascading correctly — is what an interviewer is checking for.

How it's tested

Timed practical (most common for FP&A/MIS/audit): you get a raw dump — say 2,000 rows of sales — and 20–30 minutes to produce a summary. They watch whether you reach for the mouse, whether you use `Ctrl+T` and `SUMIFS`, and whether your negatives are formatted. Speed with keyboard navigation is the visible signal.

Big 4 audit assessment: clean a messy trial balance, tie a subtotal, and flag hard-coded overrides inside formulas. They deliberately plant a cell like `=SUM(B2:B40)+500` for you to catch using audit tools.

Interview questions you should answer instantly:

- "Difference between `A1`, `$A$1`, and `A$1`, and when do you use each?"
- "How do you check what feeds a total?" (Trace Precedents / Ctrl+[)
- "A formula copied down gives wrong answers from row 2 — what's the likely cause?" (missing `$` on a constant reference)
- "How would you present ₹4,50,00,000 in a board deck?" (in ₹ crore, one decimal, labelled)
- "What does F9 do inside a formula?"

Common mistakes & how pros avoid them

Mistake	Why it hurts	The pro habit
Hard-coding numbers in formulas (`=C4*0.18`)	Nobody can find or update the rate	One input cell, referenced everywhere
Forgetting `$` before dragging	Silent wrong totals	F4 to lock, then drag
Using the mouse for everything	Slow; visible in a timed test	Ctrl+Arrow navigation
Merged cells everywhere	Breaks sorting, `SUMIFS`, selection	"Center Across Selection" instead
Dividing by 1,00,000 inside number format	Format ≠ math; breaks totals	Divide in a helper cell, label the header
No colour coding	Reviewer can't tell input from formula	Blue = input, black = formula, green = link
Overwriting after F9	Corrupts the formula	Always Esc, never Enter, after F9
Volatile clutter (`NOW()`, `INDIRECT` everywhere)	Slow, recalcs constantly	Use static `Ctrl+;` dates, minimise volatiles

Learn-it roadmap & resources

Realistic time to the employable bar: **2–3 weeks at 1 hour/day**, if you actually rebuild examples rather than watch.

- **Week 1:** Keyboard navigation + shortcuts. Rule: unplug the mouse for one hour daily. Practise `Ctrl+Arrow`, `Alt` ribbon paths, `F2`, `F4`.
- **Week 2:** Number formatting, absolute refs, named ranges. Rebuild the GST MIS above from scratch three times.
- **Week 3:** Auditing (Trace Precedents, F9, Evaluate Formula) + workbook structure (inputs/calc/output separation, colour coding).

Resources:

- **Free:** ExcelJet shortcut list and "Excel Formula" glossary; Microsoft's own support pages for Custom Number Formats; Corporate Finance Institute's free Excel fundamentals articles.

- **Free video:** Leila Gharani and ExcelIsFun (YouTube) — pick one 30-minute foundations playlist and finish it.
- **Paid/certification: Microsoft Office Specialist: Excel Associate (MO-200)** — globally portable, cheap, and a real line on a CV. CFI's **FMVA** covers this plus modelling (more relevant once you hit Chapter 3+). For India roles, no certificate beats being visibly fast in the timed test — but MO-200 gets you shortlisted.

Practice data: export any bank statement or a GSTR-2B to Excel and force yourself to summarise it keyboard-only.

Quick-reference

Navigation & editing

```
Ctrl+Arrow         jump to data edge
Ctrl+Shift+Arrow   select to edge
Ctrl+T             make a Table
Alt+=              AutoSum
Ctrl+;             static date
F2                 edit / see formula
F4                 toggle $ locks
Alt,E,S,V          Paste values
Ctrl+`             show all formulas
```

References

```
A1     relative      $A$1   fully locked
A$1    row locked    $A1    column locked
F4     cycles all four states
```

Auditing

```
Ctrl+[       select precedents
Alt,M,P      trace precedents (arrows)
Alt,M,D      trace dependents
F9           evaluate selected formula part (then Esc)
Alt,M,V      Evaluate Formula step-through
```

Finance number formats (Ctrl+1 → Custom)

```
#,##0.00;[Red](#,##0.00);"-"        accounting, red negatives
[$₹-en-IN] #,##0                     rupee symbol
#,##0.00" L"                         label as lakhs (divide source separately)
```

Golden rules: one input cell per assumption · lock rates with `$` · blue inputs / black formulas · never hard-code · Esc after F9 · label ₹ scale in the header.

Essential functions every analyst must know

What it is & where it's used

This is the working vocabulary of a finance analyst. Ninety percent of real Excel work in accounts, FP&A, audit, tax, and treasury is **conditional aggregation** (sum/count/average by criteria), **decision logic** (if this then that), **date arithmetic** (interest periods, ageing, GST tax periods, TDS due dates), **text surgery** (cleaning ERP exports, splitting GSTINs, formatting reports), and **error control** (so one bad cell doesn't blow up a model).

Where you'll use each one daily:

Function group	Real job task
SUMIFS / COUNTIFS / AVERAGEIFS	Ledger totals by cost centre, sales by region, avg invoice value by customer
IF / IFS / AND / OR	Credit-limit checks, TDS-applicability flags, commission slabs
EOMONTH / EDATE / YEARFRAC	GST return periods, EMI schedules, depreciation, receivables ageing
LEFT / MID / TEXT / TEXTSPLIT	Splitting GSTIN, cleaning Tally/SAP dumps, building report labels
IFERROR	Bulletproofing lookups and divisions in a live model

Roles that live in these functions: **Accounts Executive, AP/AR analyst, FP&A analyst, Statutory/Internal Audit associate, Tax (Direct + GST) associate, Treasury/MIS analyst, Investment Banking/PE junior.**

The gap: why companies want this (and college didn't teach it)

An MBA or CA Inter teaches you *what* a variance is, *what* receivables ageing means, and *how* GST periods work — conceptually. It almost never sits you in front of a 40,000-row `.csv` dump from Tally/SAP and says "give me month-wise sales by state, net of returns, by 3 PM." That translation layer — concept to a formula that runs on messy real data — is exactly the paid skill.

The specific gaps employers see in fresh hires: - They reach for `SUMIF` (single condition) and get stuck the moment there are two conditions. **`SUMIFS` is the real-world default.** - They nest five `IF` s into an unreadable monster instead of `IFS` or a lookup. - They subtract dates manually and get ageing buckets wrong across month-ends. - They retype data instead of using `TEXT` / `TEXTSPLIT` to reshape an export in seconds. - Their models show `#DIV/0!` and `#N/A` in front of a manager — an instant credibility hit that `IFERROR` prevents.

What "proficient" looks like

The concrete bar. A job-ready person can, **unaided and in minutes**:

1. Write a multi-condition `SUMIFS` / `COUNTIFS` / `AVERAGEIFS` including a **date range** and a **wildcard** criterion.
2. Replace nested `IF` s with `IFS` , and combine conditions with `AND` / `OR` .
3. Build a **receivables ageing** with `EOMONTH` / `TODAY` / `YEARFRAC` without hardcoding dates.
4. Split a GSTIN or clean a "LASTNAME, Firstname" export using `LEFT` / `MID` / `TEXTSPLIT` .

5. Wrap risky formulas in `IFERROR` so the model never shows a raw error.
6. Explain **why** `SUMIFS` (sum-range first) argument order differs from `SUMIF` (range, criteria, sum-range) — a classic interview trap.

Hands-on: how to actually do it

Assume a sales table: `A:Date, B:State, C:Customer, D:Product, E:Amount, F:Type` (Sale/Return).

Conditional aggregation

```
' Total SALE amount for Maharashtra
=SUMIFS(E:E, B:B, "Maharashtra", F:F, "Sale")

' Sales in a date window (Q1 FY26: 1-Apr to 30-Jun-2025)
=SUMIFS(E:E, A:A, "≥"&DATE(2025,4,1), A:A, "≤"&DATE(2025,6,30))

' Wildcard: all customers whose name starts with "Reliance"
=SUMIFS(E:E, C:C, "Reliance*")

' Count of return invoices; average invoice value for a product
=COUNTIFS(F:F, "Return")
=AVERAGEIFS(E:E, D:D, "Widget-A", F:F, "Sale")
```

Note the order: `SUMIFS(sum_range, criteria_range1, criteria1, …)`. Join operators to a cell with `&`: `A:A, "≥"&G1`.

Decision logic

```
' Simple flag: is TDS applicable? (single-vendor limit Rs 30,000 u/s 194J)
=IF(E2>30000, "Deduct TDS", "No TDS")

' Multi-slab commission with IFS (no nesting)
=IFS(E2≥1000000,"5%", E2≥500000,"3%", E2≥100000,"1%", TRUE,"0%")

' AND / OR
=IF(AND(F2="Sale", E2>500000), "High-value sale", "Normal")
=IF(OR(B2="J&K", B2="Ladakh"), "Special zone", "Regular")
```

Date functions

```
' GST return period end (last day of invoice month)
=EOMONTH(A2, 0)                 ' 15-May-25  → 31-May-25

' Credit due date = invoice date + 45 days; or +2 months
=A2+45
=EDATE(A2, 2)                   ' same day, two months later

' EMI / depreciation date one month out
=EDATE(A2, 1)

' Receivables ageing in days and in years
=TODAY()-A2                                   ' days outstanding
=YEARFRAC(A2, TODAY(), 1)                      ' fraction of a year (actual/actual)

' Ageing bucket
=IFS(TODAY()-A2≤30,"0-30", TODAY()-A2≤60,"31-60",
     TODAY()-A2≤90,"61-90", TRUE,"90+")
```

Text functions (cleaning exports)

```
' GSTIN = 27ABCDE1234F1Z5 in cell H2
=LEFT(H2,2)                     ' 27  → state code
=MID(H2,3,10)                   ' ABCDE1234F → PAN embedded in GSTIN
=MID(H2,3,10)                   ' also the PAN for TDS matching

' Split "MUMBAI, Maharashtra" into two cells (Excel 365 / dynamic array)
=TEXTSPLIT(I2, ", ")            ' spills MUMBAI | Maharashtra

' Build a clean label / format a number as text
=TEXT(A2,"mmm-yyyy")            ' May-2025
=TEXT(E2,"#,##0")              ' 12,34,567  (use "[≥10000000]#,##,##,##0" for full Indian gro
=D2&" ("&TEXT(E2,"Rs #,##0")&")"   ' Widget-A (Rs 5,00,000)
```

Error control

```
' Never show #N/A / #DIV/0! to a manager
=IFERROR(VLOOKUP(C2, Master!A:D, 4, FALSE), "Not found")
=IFERROR(E2/G2, 0)             ' safe division → 0 instead of #DIV/0!
```

Worked example / mini-project

Build a one-screen receivables + sales MIS. Data: a debtors export with `A:Invoice No, B:Customer, C:Invoice Date, D:GSTIN, E:Amount, F:State`. 12 rows, realistic:

Invoice	Customer	Inv Date	GSTIN	Amount (Rs)	State
INV001	Reliance Retail	05-Feb-2025	27AAACR...	4,50,000	Maharashtra
INV002	Tata Steel	20-Mar-2025	24AAACT...	8,20,000	Gujarat
INV003	Infosys Ltd	12-May-2025	29AAACI...	2,10,000	Karnataka

Now build these output cells (today assumed **03-Jul-2025**):

```
' 1. State code from GSTIN (audit check vs State column)
=LEFT(D2,2)                                    ' 27 → should map to Maharashtra

' 2. Days outstanding + ageing bucket
=TODAY()-C2
=IFS(TODAY()-C2≤30,"0-30", TODAY()-C2≤60,"31-60",
     TODAY()-C2≤90,"61-90", TRUE,"90+")        ' INV001 → 90+

' 3. Total outstanding over 90 days (the number a CFO asks for first)
=SUMIFS(E:E, C:C, "<"&(TODAY()-90))            ' all invoices older than 90 days

' 4. Sales by state (feeds a state-wise summary)
=SUMIFS($E$2:$E$13, $F$2:$F$13, "Maharashtra")

' 5. Count of overdue accounts + safe average ticket
=COUNTIFS($C$2:$C$13, "<"&(TODAY()-60))
=IFERROR(AVERAGEIFS($E$2:$E$13,$F$2:$F$13,"Gujarat"),0)

' 6. Report label
="Overdue >90d: "&TEXT(SUMIFS(E:E,C:C,"<"&(TODAY()-90)),"Rs #,##,##0")
```

Reproduce it: paste 12 rows, drop these formulas, and you have a live dashboard that recalculates every time you refresh the export. That is a deliverable you can show in a first-week task.

How it's tested

Interview questions (verbal): - "Difference between `SUMIF` and `SUMIFS`?" (answer: multiple criteria; and the sum-range moves to the *first* argument). - "How do you sum only rows in a date range?" (criteria with `"≥"&start` and `"≤"&end`). - "How do you avoid `#N/A` in a lookup?" (`IFERROR` / `IFNA`). - "Why not just nest `IF`s?" (readability, `IFS`, or a lookup table).

Practical / assessment tests (the real filter): - A **timed 30-45 min Excel test**: a raw sheet + 6-10 tasks — "total sales for South region in Q2", "flag invoices over Rs 1,00,000 needing approval", "extract PAN from GSTIN", "age these receivables". No internet, formulas graded on correctness *and* whether they use absolute refs / dynamic dates. - A **"clean this export"** task: a `LASTNAME, First` or merged-column dump you must split with `TEXTSPLIT` / `MID`. - Case rounds in FP&A/IB often hand you a messy CSV and expect a summary table built with `SUMIFS` + a pivot within 20 minutes.

Pass signal: you finish, formulas are robust (no hardcoded dates, wrapped in `IFERROR`), and you can explain each one.

Common mistakes & how pros avoid them

Mistake	Fix / pro habit
Wrong `SUMIFS` argument order (sum-range last)	Sum-range is **first**; muscle-memory it
Hardcoding `"01-04-2025"` as text criteria	Use `"≥"&DATE(2025,4,1)` — real dates, comparable
Full-column refs `E:E` in a huge model → slow	Use fixed ranges `$E$2:$E$50000` in big files
Nesting 6 `IF` s	Use `IFS`, or a lookup table with `XLOOKUP`
Manual date subtraction across month-end	`EOMONTH` / `EDATE`, never add "30" for a month
`LEFT/MID` on inconsistent-length text	Anchor with `FIND` / `SEARCH` or use `TEXTSPLIT` on a delimiter
`IFERROR` hiding a *real* bug	Only wrap known-safe risks (div-by-zero, missing lookup), not everything
Locking references wrong when dragging	`F4` to toggle `$`; lock criteria ranges, not the criteria cell

Learn-it roadmap & resources

Time to proficiency: 2-3 weeks of daily practice (1 hr/day) to clear a timed test; ~2 months to be genuinely fast on messy data.

Week	Focus
1	`SUMIFS/COUNTIFS/AVERAGEIFS`, absolute refs, date criteria
2	`IF/IFS/AND/OR`, `EOMONTH/EDATE/YEARFRAC`, ageing project
3	Text functions on real ERP dumps, `IFERROR`, build the MIS above

Resources - Free: **ExcelJet** (function reference + examples), **Chandoo.org**, **Corporate Finance Institute (CFI)** free Excel course, Microsoft's own function docs. - Paid: CFI's **FMVA** certification (Excel-heavy, finance-specific), **Wall Street Prep**, Udemy "Microsoft Excel — Excel from Beginner to Advanced" (Kyle Pew). - Practice data: export your own Tally/any sample GST invoice register and re-solve the mini-project. - Certification worth naming on a CV: **Microsoft Office Specialist (MOS) Excel Associate/Expert**, or **FMVA** for finance roles.

Quick-reference

```
' AGGREGATION
=SUMIFS(sum_rng, crit_rng1, crit1, crit_rng2, crit2)
=COUNTIFS(crit_rng1, crit1, …)
=AVERAGEIFS(avg_rng, crit_rng1, crit1, …)
=SUMIFS(E:E, A:A, "≥"&DATE(2025,4,1), A:A, "≤"&DATE(2025,6,30))   ' date window
=SUMIFS(E:E, C:C, "Reliance*")                                     ' wildcard

' LOGIC
=IF(test, if_true, if_false)
=IFS(cond1,val1, cond2,val2, TRUE,default)
=AND(c1,c2)   =OR(c1,c2)

' DATES
=EOMONTH(date, months)     ' last day of month, offset
=EDATE(date, months)       ' same day, months out
=YEARFRAC(start, end, 1)   ' year fraction (basis 1 = actual/actual)
=TODAY()-date              ' days outstanding

' TEXT
=LEFT(txt,n)   =RIGHT(txt,n)   =MID(txt,start,n)
=TEXTSPLIT(txt, ", ")      ' split on delimiter (Excel 365)
=TEXT(value,"mmm-yyyy")    ' or "#,##0" / "Rs #,##,##0"

' ERROR CONTROL
=IFERROR(formula, fallback)
=IFNA(VLOOKUP(…), "Not found")
```

GSTIN cheat: `LEFT(gstin,2)` = state code, `MID(gstin,3,10)` = PAN. **Argument order trap:** `SUMIF(range, criteria, sum_range)` but `SUMIFS(sum_range, range, criteria)` — sum-range **moves to front** in the plural version.

Lookup & Reference Mastery

What it is & where it's used

Lookup functions answer one question that every finance job asks a hundred times a day: "*Given this ID, fetch me that value.*" Given a vendor code, fetch the GSTIN. Given an employee ID, fetch the CTC. Given a ledger name, fetch the closing balance. Given an ISIN, fetch the market price.

You have a "lookup table" (a master somewhere) and a "working sheet" (your model), and you need to pull data from one into the other by a matching key. That is 60-70% of real analyst work — you rarely type numbers, you *stitch datasets together*.

Where it shows up by role:

Role	Typical lookup job
Accounts payable	Match invoice numbers to a payment register; pull vendor bank details
GST / Tax	Reconcile GSTR-2B purchases against your purchase register (match by invoice + GSTIN)
FP&A / Analyst	Map trial-balance ledgers to P&L line items; build a driver-based model
Audit	Vouch a sample — pull supporting entries by voucher number
Treasury	Fetch FX rates or interest rates by date/currency
Investment / Equity research	Pull financials by ticker across years

If you can only do one Excel thing well, make it lookups. It is the single most tested skill in a finance Excel screen.

The gap: why companies want this (and college didn't teach it)

MBA and CA coursework teaches you *what* a reconciliation is, or *why* you allocate overheads — the concepts. It almost never makes you sit with two messy exports and physically join them. So freshers arrive knowing the theory of a bank reconciliation but not knowing how to match 4,000 bank lines to 4,000 book lines in ninety seconds.

The specific gaps employers see:

- **Manual matching mindset.** Freshers filter, eyeball, and copy-paste. A 2,000-row reconciliation takes them a day; a pro does it in five minutes with a lookup and a `MATCH`-based flag.
- **VLOOKUP-only, and fragile.** Many candidates know only `VLOOKUP`, which breaks the moment someone inserts a column and cannot look leftward. They have never met `INDEX-MATCH` or `XLOOKUP`.
- **No handling of "not found."** Real data has unmatched rows. Untrained users get `#N/A` everywhere and freeze. Pros trap it with a default and turn `#N/A` into a *finding* ("these 12 invoices are in books but not in 2B").
- **No dynamic arrays.** `FILTER`, `UNIQUE`, `SORT` (Excel 365 / 2021+) let you build a live sub-report with one formula. Colleges teach none of this because their labs run Excel 2007.

Closing this gap is what turns a "knows Excel" line on your CV into "does the reconciliation unaided by day two."

What "proficient" looks like

A job-ready person can, unaided:

- Choose the right tool: `XLOOKUP` for a single fetch, `INDEX-MATCH` for legacy files, `FILTER` when they need *many* matching rows, not one.
- Do a **two-way lookup** (row key × column key) — e.g., pull the March figure for "Salaries" from a month × ledger grid.
- Handle not-found gracefully and treat unmatched rows as an audit output.
- Do an **approximate/range lookup** — slab-based, e.g., income-tax slabs or a commission grid.
- Lock references correctly (`$`) so a formula fills down and across without drifting.
- Explain *why* `XLOOKUP` beats `VLOOKUP` and rewrite an old `VLOOKUP` model without breaking it.

Hands-on: how to actually do it

1. XLOOKUP — the modern default

Syntax: `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

```
Pull a vendor's GSTIN by vendor code:
=XLOOKUP(A2, Vendors[Code], Vendors[GSTIN], "Not found")

Pull the closing balance for a ledger:
=XLOOKUP(A2, Ledger[Name], Ledger[Closing], 0)
```

- `if_not_found` = `"Not found"` — no more `#N/A`, no `IFERROR` wrapper.
- Return array can be **left of** the lookup array. VLOOKUP cannot.
- Return a whole row by pointing `return_array` at multiple columns: `=XLOOKUP(A2, Ledger[Name], Ledger[Opening]:Ledger[Closing])` spills opening→closing across.

2. INDEX-MATCH — works in every Excel version

```
=INDEX(Ledger[Closing], MATCH(A2, Ledger[Name], 0))
```

`MATCH` finds the *position* of the key; `INDEX` returns the value at that position. The `0` means exact match. Use this in any file that must open in Excel 2016 or older, or where colleagues don't have 365.

3. Two-way lookup (row × column)

Grid: ledgers down column A, months across row 1, figures inside.

```
XLOOKUP nested — fetch Salaries for Mar:
=XLOOKUP(G2, A2:A50, XLOOKUP(H2, B1:M1, B2:M50))

INDEX with two MATCHes — the classic:
=INDEX(B2:M50, MATCH(G2, A2:A50, 0), MATCH(H2, B1:M1, 0))
```

`G2` = "Salaries", `H2` = "Mar". The inner lookup picks the column; the outer picks the row.

4. Approximate / slab lookup (tax, commission)

Set a slab table sorted ascending by lower bound. Use `match_mode = -1` (exact or next smaller):

```
Slab table  A:B  →  lower bound | rate
=XLOOKUP(Income, SlabLower, SlabRate, , -1)
```

Old-style equivalent: `=VLOOKUP(Income, SlabTable, 2, TRUE)` — the `TRUE` is the range match.

5. Dynamic arrays — FILTER, UNIQUE, SORT (365 / 2021+)

```
Every invoice for one vendor (many rows, one formula):
=FILTER(Data, Data[Vendor]="ABC Traders", "None")

Unmatched invoices (in books, not in 2B):
=FILTER(Books[Inv], ISNA(XLOOKUP(Books[Inv], GSTR2B[Inv], GSTR2B[Inv])), "All matched")

Distinct list of ledgers:
=UNIQUE(Data[Ledger])

Sorted, de-duped vendor master:
=SORT(UNIQUE(Data[Vendor]))

Top 5 debtors by amount:
=SORT(FILTER(Data, Data[Type]="Debtor"), 3, -1)
```

These **spill** — one formula, a whole live block that recalculates as data changes. This is how you build a self-updating exception report.

Worked example / mini-project: GSTR-2B vs Purchase Register reconciliation

You have two exports. Match input tax credit (ITC) claimed in books against what shows in GSTR-2B.

Purchase Register (sheet `PR`), columns: `A` Invoice No, `B` GSTIN, `C` Taxable, `D` GST. **GSTR-2B** (sheet `2B`), columns: `A` Invoice No, `B` GSTIN, `C` Taxable, `D` GST.

Sample data:

PR Invoice	GSTIN	PR GST (₹)
INV-101	27AABCU9603R1ZM	18,000
INV-102	29AAACI1234K1Z5	9,000
INV-103	27AABCU9603R1ZM	4,500

Step 1 — flag whether each PR invoice is in 2B, in column `E` of `PR`:

```
=IF(ISNA(XLOOKUP(A2, '2B'!A:A, '2B'!A:A)), "Missing in 2B", "Matched")
```

Step 2 — compare the GST amount (catch value mismatches), column `F`:

```
=IFERROR(D2 - XLOOKUP(A2, '2B'!A:A, '2B'!D:D), "Not in 2B")
```

A non-zero, non-error result = the same invoice with a different tax figure — a real reconciliation exception.

Step 3 — build the exception report on a fresh sheet, one formula:

```
=FILTER(PR!A2:F100, PR!E2:E100="Missing in 2B", "All reconciled")
```

Step 4 — total ITC at risk (in books but not in 2B, i.e., not claimable this month):

```
=SUMIF(PR!E:E, "Missing in 2B", PR!D:D)
```

Result: say ₹9,000 (INV-102) is missing in 2B → that ITC is provisionally not available, flag to the vendor. You just did in four formulas what a fresher does in half a day of filtering. Reverse the direction (2B not in PR) to catch invoices you forgot to book.

How it's tested

The practical test (most common). You get two raw exports and 20-45 minutes: "*Reconcile these and tell me the unmatched items and the total difference.*" They watch whether you reach for a lookup or start filtering manually. Reaching for `XLOOKUP` / `INDEX-MATCH` + a `FILTER` exception list is an instant pass.

Live Excel screen. Screen-share, a small table, and prompts like: "Pull the salary for employee E-045." "Now the master got a column inserted — did your formula survive?" (Tests whether you hard-coded a column index in `VLOOKUP`.)

Interview questions you should be able to answer cold:

- Difference between `VLOOKUP` and `INDEX-MATCH`? (Left-lookup, column-insert safety, speed.)
- Why does `VLOOKUP` return the wrong value sometimes? (Default `TRUE` approximate match on unsorted data.)
- How do you look up to the left? (`INDEX-MATCH` or `XLOOKUP`.)
- How do you return multiple matching rows? (`FILTER`, not any single-cell lookup.)
- What does the 4th argument of `MATCH` / `XLOOKUP` do? (Match mode: exact vs next-smaller for slabs.)

Common mistakes & how pros avoid them

Mistake	What breaks	Pro fix
`VLOOKUP` with `TRUE` (or omitted 4th arg)	Silent wrong values on unsorted data	Always pass `0` / `FALSE` for exact; use `-1` deliberately for slabs
Hard-coded column index (`VLOOKUP( … ,4,0)`)	Breaks when a column is inserted	`XLOOKUP` / `INDEX-MATCH` reference the column by name/range
Forgetting `$` locks	Formula drifts when filled down	Lock the lookup array: `XLOOKUP(A2,$D$2:$D$99,$E$2:$E$99)` — or use Tables
Wrapping everything in `IFERROR`	Hides *real* errors, not just `#N/A`	Use `XLOOKUP` 's `if_not_found` , or `IFNA` (traps only `#N/A`)
Lookup key type mismatch	"INV-101" text vs 101 number → `#N/A`	Standardise with `TRIM` , `TEXT` , or `VALUE` ; watch trailing spaces
Duplicate keys	Lookup returns only the first match	De-dupe with `UNIQUE` , or `FILTER` to see all; add a composite key
Composite matching (invoice **and** GSTIN)	Same invoice no. across vendors matches wrong row	Concatenate a key: `A2&"|"&B2` in both sheets, then look up that
Volatile giant ranges (`A:A`) on 100k rows	Workbook crawls	Reference actual used range or a Table

Learn-it roadmap & resources

Realistic timeline: 2-3 weeks of daily practice to interview-ready; solid in a couple of months of real use.

- **Days 1-3:** `XLOOKUP` and `INDEX-MATCH` exact matches; absolute references; `IFNA` .
- **Days 4-7:** Two-way lookups; approximate/slab lookups; convert an old `VLOOKUP` file to `XLOOKUP` .
- **Week 2:** Dynamic arrays — `FILTER` , `UNIQUE` , `SORT` ; build the GSTR-2B recon above end to end.
- **Week 3:** Composite keys, duplicate handling, and doing a full reconciliation timed under 20 minutes.

Resources

- Microsoft's own `XLOOKUP` / `FILTER` support pages (free, authoritative — search "XLOOKUP function Microsoft").
- ExcelJet (free formula reference with examples) — exceljet.net.
- Chandoo.org and Leila Gharani (YouTube) — free, finance-flavoured.
- Paid, if you want a certificate: **Microsoft Office Specialist: Excel Associate (MO-200)** or **Expert (MO-201)** — recognised, exam-based, ~₹4,000-5,000. For finance modelling context, a Corporate Finance Institute (CFI) or Wall Street Prep Excel module.
- Practice data: export any GSTR-2B JSON from the GST portal and your own purchase register — real data beats toy datasets.

Quick-reference

```
XLOOKUP    =XLOOKUP(key, lookup_col, return_col, "not found", match_mode, search_mode)
INDEX-MATCH =INDEX(return_col, MATCH(key, lookup_col, 0))
Two-way     =INDEX(grid, MATCH(rowkey,rows,0), MATCH(colkey,cols,0))
Slab/range  =XLOOKUP(val, lower_bounds, rates, , -1)   'exact-or-next-smaller
Left lookup =XLOOKUP or INDEX-MATCH  (VLOOKUP cannot)
Many rows   =FILTER(data, criteria_range=criteria, "none")
Distinct    =UNIQUE(range)      Sorted =SORT(range, col, -1)
Composite   key = A2&"|"&B2  in both sheets, then look up the key
Trap N/A    =IFNA(formula, "-")   'only #N/A, not other errors
```

VLOOKUP	XLOOKUP	Winner
Looks only rightward	Looks either direction	XLOOKUP
Hard-coded column index breaks on insert	Column referenced directly	XLOOKUP
Default approximate match (dangerous)	Default exact match	XLOOKUP
`#N/A` needs `IFERROR` wrapper	Built-in `if_not_found`	XLOOKUP
Works in all versions	Excel 365 / 2021+ only	VLOOKUP (legacy only)

Match modes (XLOOKUP/MATCH): `0` exact · `-1` exact or next smaller (slabs, ascending) · `1` exact or next larger · `2` wildcard.

Power Query: Cleaning Real Company Data

What it is & where it's used

Power Query is the built-in **ETL engine** (Extract, Transform, Load) inside Excel and Power BI. You point it at messy source data — a bank statement export, a GST portal download, ten branch files in a folder, a Tally ledger dump — record a sequence of cleaning steps, and it produces a clean table. The magic word is **refreshable**: when next month's file arrives, you drop it in and hit **Refresh**. All your steps replay automatically. No re-cleaning, no re-formulas.

Find it in Excel under **Data → Get Data** (Windows Excel 2016+ and Microsoft 365; the ribbon group is called "Get & Transform Data"). It writes its logic in a language called **M**, but 90% of the job is done with clicks in the Power Query Editor.

Roles that live in Power Query: **FP&A analysts** consolidating branch actuals, **accounts payable/receivable teams** reconciling ledgers, **GST/tax executives** cleaning portal downloads for GSTR-2B vs purchase-register matching, **MIS analysts** building monthly management packs, and anyone who currently spends the first three days of every month copy-pasting and running find-replace.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you ratio analysis on data that was **already clean**. Real company data never is. The gap employers pay to close:

- **Source data is filthy.** Merged cells, ₹ signs stored as text, dates as "01-Apr-25" strings, blank header rows, totals mixed into detail rows, "1,20,000" with Indian comma grouping that Excel reads as text.
- **The same clean-up runs every single month.** College projects are one-shot. A job is the *same* report 12 times a year. Manual cleaning that "only takes 2 hours" costs 24 hours a year and breaks the moment a colleague is on leave.
- **VLOOKUP-and-paste doesn't scale.** When AP hands you a 40,000-row purchase register and 38,000-row GSTR-2B, a manual match is impossible. Power Query merges them in seconds.

Employers have watched analysts burn days on this. Someone who says "give me the raw dump, I'll build a refreshable query" is instantly more valuable than someone who reaches for a fresh round of copy-paste.

What "proficient" looks like

A job-ready person can, unaided:

1. **Import** from Excel, CSV, a whole folder, and a database — and know that **Get Data → From Folder** stacks 12 monthly files into one table.
2. **Unpivot** a wide "months across columns" report into a tidy long table (the single most-tested skill).
3. **Merge** two queries (Left Outer, Inner, Left Anti) — and know *which* join to use for a reconciliation.
4. **Append** queries to stack same-shape tables.
5. **Remove errors, fill down, split columns, change types, trim/clean text, and replace values** without breaking on next refresh.

6. Understand that steps are **recorded and replayed**, so column names and types must be handled deliberately.

Hands-on: how to actually do it

Import from a folder (consolidate 12 branch files)

```
Data → Get Data → From File → From Folder → pick folder → Combine & Transform
```

Power Query samples the first file, applies the transform to all, and stacks them. Add a step to keep the source file name so you know which branch a row came from — it's the `Source.Name` column.

Unpivot (the interview favourite)

Raw MIS often looks like this:

Branch	Apr	May	Jun
Mumbai	120000	135000	128000
Pune	90000	88000	94000

Select the **Branch** column → right-click → **Unpivot Other Columns**. Result:

Branch	Attribute	Value
Mumbai	Apr	120000
Mumbai	May	135000

Rename `Attribute` → `Month`, `Value` → `Sales`. Now it's a proper table you can pivot, filter, or feed to Power Pivot. **Always use "Unpivot Other Columns"** so new month columns next quarter are captured automatically.

Clean Indian-format numbers stored as text

"1,20,000" imports as text. In the Editor, **Transform → Format → Trim & Clean**, then **Replace Values**: replace `,` with nothing, then **Change Type → Whole Number**. The equivalent M step:

```
= Table.ReplaceValue(Source, ",", "", Replacer.ReplaceText, {"Amount"})
```

Split columns

A cell like `27AABCU9603R1ZM - Uma Traders` (GSTIN + name):

```
Split Column → By Delimiter → " - " → at each occurrence
```

Or split GSTIN to grab the **state code** (first 2 chars): **Split Column → By Number of Characters → 2 → Once, as far left as possible**. State code `27` = Maharashtra.

Remove errors and fill down

After a type change, bad rows show `Error`. Select the column → **Remove Errors** (or **Replace Errors** with `0`). For a report where the branch name appears once then blanks below it: select column → **Transform** → **Fill** → **Down**.

Merge queries (reconciliation)

To find purchases in your register that are **missing** from GSTR-2B:

```
Home → Merge Queries → pick PurchaseRegister & GSTR2B
→ match on [GSTIN] + [Invoice No]  (Ctrl-click both keys in each table)
→ Join Kind: Left Anti (rows only in first)
```

Join kinds you must know:

Join Kind	Returns	Use for
Left Outer	All left + matches	Enrich data (add vendor name)
Inner	Only matched rows	Confirmed matches
Left Anti	Left rows with NO match	**Missing / unreconciled items**
Right Anti	Right rows with NO match	In 2B but not in books

Append queries (stack)

```
Home → Append Queries → Three or more tables → add Q1, Q2, Q3, Q4
```

Same columns, stacked into one — a full-year table from four quarterly files.

Worked example / mini-project: GSTR-2B vs Purchase Register match

Goal: find input tax credit (ITC) you can claim, and mismatches to chase.

Source 1 — `PurchaseRegister.xlsx` (your books):

GSTIN	Invoice No	Taxable	IGST
27AABCU9603R1ZM	INV-001	100000	18000
29AAGCB1286Q1ZP	INV-002	50000	9000
07AABCU9603R1ZX	INV-003	20000	3600

Source 2 — `GSTR2B.csv` (portal download):

GSTIN	Invoice No	IGST_2B
27AABCU9603R1ZM	INV-001	18000
29AAGCB1286Q1ZP	INV-002	8500

Steps:

1. **Get Data** → **From Excel** → load `PurchaseRegister` as a connection-only query. Repeat **From Text/CSV** for `GSTR2B`.
2. In each, **Trim** the Invoice No (portal data has trailing spaces — a classic silent mismatch killer) and set types.
3. On `PurchaseRegister`: **Merge Queries** → `GSTR2B`, match on `GSTIN` + `Invoice No`, **Left Outer**. Expand `IGST_2B`.
4. Add a **Custom Column** to flag status:

```
= if [IGST_2B] = null then "Missing in 2B"
  else if [IGST] <> [IGST_2B] then "Tax mismatch"
  else "Matched"
```

Result:

Invoice No	IGST	IGST_2B	Status
INV-001	18000	18000	Matched
INV-002	9000	8500	Tax mismatch
INV-003	3600	(null)	Missing in 2B

Close & Load to a sheet. INV-002 has a ₹500 mismatch to query with the vendor; INV-003's ₹3,600 ITC can't be claimed until the vendor files. Next month, drop in the new files and **Refresh** — the whole reconciliation rebuilds in seconds.

How it's tested

In interviews (verbal): - "You have sales in columns Jan–Dec, one row per product. How do you get one row per product-month?" → *Unpivot Other Columns.* - "Which join finds invoices in your books but not on the GST portal?" → *Left Anti.* - "Difference between Merge and Append?" → *Merge = join sideways on a key; Append = stack rows.* - "How do you consolidate 15 branch files with one click each month?" → *From Folder → Combine & Transform.*

Practical test (very common): you're given a deliberately messy workbook — merged cells, ₹ text amounts, months across columns, a second sheet to reconcile against — and 30–45 minutes to produce a clean, refreshable output. They then **change the data and hit Refresh** to check your query didn't hard-code anything.

Red flag they watch for: did you clean it with manual find-replace on the sheet (breaks on refresh) or inside Power Query (survives refresh)? The whole point is the second one.

Common mistakes & how pros avoid them

Mistake	Consequence	Pro fix
Hard-coding a "Changed Type" with fixed column names	Breaks when a column is renamed upstream	Delete unneeded auto-type steps; type only what you need, late
Using "Unpivot Columns" instead of "Unpivot **Other** Columns"	New month columns get dropped next quarter	Always unpivot *other* columns
Not trimming text before a merge	Silent non-matches from trailing spaces	Trim + Clean both key columns first
Filtering by a specific value (e.g. keep only "Apr")	Wrong data after refresh	Filter on logic, not a snapshot value
Loading giant tables to a sheet you don't need	Slow, bloated file	Use **Connection Only** + load to Data Model
Renaming columns then referencing old names later	Step errors on refresh	Rename once, early, and keep it consistent

Learn-it roadmap & resources

Time to proficiency: about 2–3 weeks of evening practice for someone comfortable in Excel. Power Query is unusually learnable because it's click-driven — you can be productive before you understand M.

- **Week 1:** Import (Excel/CSV/Folder), change types, remove columns, split, trim, replace values, fill down.
- **Week 2:** Unpivot, group by, merge (all join kinds), append.
- **Week 3:** Custom columns, conditional columns, a light touch of M, and building one real refreshable monthly report end-to-end.

Resources: - **Microsoft Learn — "Get & Transform in Excel"** (free, official). - **Excel Is Fun** and **MyOnlineTrainingHub** (Mynda Treacy) — free YouTube, finance-friendly examples. - **Book:** *Master Your Data with Excel and Power BI* — Ken Puls & Miguel Escobar (the definitive Power Query reference). - **Certification:** it's covered inside the **Microsoft PL-300 (Power BI Data Analyst)** exam — worth listing on a CV, though for most finance jobs a demonstrated project matters more than the cert.

Quick-reference

Task	Click-path
Import one file	Data → Get Data → From File → From Workbook/CSV
Stack many files	Get Data → From File → From Folder → Combine & Transform
Wide → long	Select key cols → right-click → Unpivot Other Columns
Fill blanks below	Column → Transform → Fill → Down
Clean text	Transform → Format → Trim, then Clean
Split GSTIN state code	Split Column → By Number of Characters → 2
Reconcile / find missing	Home → Merge Queries → Join Kind: **Left Anti**
Enrich (add name)	Merge → **Left Outer** → Expand
Stack same-shape tables	Home → Append Queries
Don't load to sheet	Close & Load To → **Connection Only**
Re-run everything	Data → **Refresh All**

Join cheat: Left Outer = enrich · Inner = confirmed matches · Left Anti = missing in second · Right Anti = extra in second.

Golden rule: never fix data on the sheet — fix it in Power Query, so next month is one click: **Refresh**.

PivotTables & Dashboards

What it is & where it's used

A **PivotTable** is Excel's engine for turning a flat list of transactions into a summary you can slice any way you want — total sales by region, expense by cost centre, GST payable by month — without writing a single formula. A **dashboard** is a one-page visual layer built on top of those pivots: a few charts, some KPI tiles, slicers to filter, all refreshing from one data source.

This is the single most-used Excel skill in day-to-day finance work. Where it shows up:

- **FP&A / MIS analysts** — the monthly MIS pack is 90% pivots and dashboards.
- **Accounts / audit** — reconciling a Tally trial-balance dump, ageing debtors, ledger scrutiny.
- **Tax** — reconciling GSTR-2B against your purchase register (both are flat CSV exports → pivot both → compare).
- **Treasury / management reporting** — cash position, DSO/DPO trends.

If a job description says "prepare monthly MIS," "reporting," or "management dashboards," it means PivotTables. Full stop.

The gap: why companies want this (and college didn't teach it)

MBA and CA syllabi teach you what a *variance* is and how to read a *P&L*. They almost never make you sit with 40,000 rows of raw ledger data and produce a manager-ready one-pager by 10 a.m. That translation step — **raw data → decision-ready summary** — is the actual job, and it's the gap.

Specifically, college leaves you weak on:

College teaches	The job needs
Interpreting a finished P&L	*Building* it from a Tally/ERP dump
SUM / VLOOKUP on tidy data	Pivoting 50k messy rows in 30 seconds
One static answer	A slicer-driven view a manager self-serves
"The number is ₹12L"	Why it moved, shown visually, on one page

Employers pay for speed and self-service. A manager who can click a slicer instead of emailing you "can you split this by branch?" saves the whole team hours. That's the value.

What "proficient" looks like

The concrete bar a job-ready person clears unaided:

- Given a raw export, build a PivotTable in under a minute — drag fields, change **Sum → Average/Count**, add a second dimension to Columns.
- Group dates into **Months/Quarters/Years** (right-click a date field → Group).
- Add **% of Column Total**, **Running Total**, and **Difference From** (prior month) as *Show Values As*.
- Insert **Slicers** and a **Timeline**, and connect one slicer to *multiple* pivots (Report Connections).

- Build a **one-page dashboard**: 3–4 PivotCharts + KPI cells, all fed by pivots, all refreshing on one click.
- Use **GETPIVOTDATA** to pull a specific pivot cell into a KPI tile that won't break when the pivot resizes.
- Apply **conditional formatting** (data bars, colour scales, icon sets, top/bottom rules) and **sparklines** for inline trends.
- Refresh everything with **Ctrl+Alt+F5** (Refresh All) and know that pasting new data into the source table auto-updates the pivot.

Hands-on: how to actually do it

1. Prep the data (always do this first)

Turn your raw range into a **Table** so the pivot auto-expands when data grows:

```
Select any cell in the data → Ctrl+T → tick "My table has headers" → OK
Rename it: Table Design tab → Table Name: tblSales
```

Now your pivot source is `tblSales`, not `A1:H40000`. New rows flow in automatically.

2. Build the PivotTable

```
Insert → PivotTable → Table/Range: tblSales → New Worksheet → OK
```

Drag fields into the four zones: - **Rows** = Region, then Product (nested) - **Columns** = Month - **Values** = Sales Amount (defaults to Sum) - **Filters** = FY

Change the calculation: right-click any value → **Summarize Values By** → Average / Count / Max.

3. Show Values As (the underused power)

Right-click a value → **Show Values As**:

Option	Gives you
% of Grand Total	Each cell's share of the whole
% of Column Total	Product mix within each month
Running Total In	Cumulative YTD sales
Difference From (prev)	Month-on-month change (₹)
% Difference From	MoM growth %

4. Group dates

```
Right-click any date in the pivot → Group → select Months + Years → OK
```

5. Slicers, Timeline, and connecting them

```
PivotTable Analyze → Insert Slicer → tick Region, Product
PivotTable Analyze → Insert Timeline → tick Order Date
```

To drive **multiple pivots** from one slicer:

```
Right-click the slicer → Report Connections → tick every PivotTable it should control
```

6. GETPIVOTDATA — for stable KPI tiles

When you type `=` and click a pivot cell, Excel auto-writes GETPIVOTDATA. It looks scary but it's *robust* — it fetches by label, so it won't break if the pivot moves or resizes:

```
=GETPIVOTDATA("Sales Amount", $A$3, "Region", "West", "Month", "Jun")
```

Wrap it so a blank returns clean:

```
=IFERROR(GETPIVOTDATA("Sales Amount",$A$3,"FY","2025-26"),0)
```

Tip: to force a *normal* reference instead (e.g. `=B5`), turn off **PivotTable Analyze → Options ▾ → Generate GetPivotData**.

7. Conditional formatting

```
Select the range → Home → Conditional Formatting →
  • Data Bars        → in-cell bar chart
  • Color Scales     → red-yellow-green heatmap (great for variance %)
  • Icon Sets        → traffic lights on growth
  • Top/Bottom Rules → highlight top 10 debtors
  • New Rule → Use a formula:  =$G2<0   → fill red (flag negative margins)
```

8. Sparklines (inline mini-charts)

```
Select where they go → Insert → Sparklines → Line →
  Data Range: C2:N2  Location Range: O2 → OK
Sparkline tab → tick High Point + Low Point (red/green markers)
```

Worked example / mini-project

Scenario: You're the MIS analyst at a mid-size FMCG distributor. You have a FY2025-26 sales export from the ERP: 38,000 rows, columns — `Order Date | Region | Salesperson | Product | Qty | Sales Amount | Cost`.

Reproduce it with a small sample and scale the idea:

Order Date	Region	Product	Sales Amount (₹)	Cost (₹)
05-04-2025	West	Soap	1,20,000	78,000
05-04-2025	North	Shampoo	2,40,000	1,80,000
12-05-2025	West	Shampoo	3,10,000	2,20,000
08-06-2025	South	Soap	95,000	61,000
20-06-2025	North	Soap	1,45,000	92,000

Step-by-step build:

1. `Ctrl+T` → name it `tblSales`. Add a helper column `Margin = [@[Sales Amount]]-[@Cost]`.
2. Insert PivotTable → Rows: Region; Columns: Months (group Order Date); Values: Sum of Sales Amount.
3. Add a **second value** — Sum of Margin — then right-click it → Show Values As → **% of Parent Row Total** to get margin %... or add a **calculated field**:

```
PivotTable Analyze → Fields, Items & Sets → Calculated Field
  Name:    Margin %
  Formula: = Margin / 'Sales Amount'
```

4. Insert a **Slicer** on Region and a **Timeline** on Order Date. Connect both to the pivot.
5. Build KPI tiles on a fresh "Dashboard" sheet:

```
Total Sales   =IFERROR(GETPIVOTDATA("Sales Amount",Pivot!$A$3),0)
Total Margin  =IFERROR(GETPIVOTDATA("Margin",Pivot!$A$3),0)
Margin %      =[Total Margin]/[Total Sales]
```

6. Insert a **PivotChart** (clustered column: Sales by Region; line: Margin % by month).
7. Add a monthly sparkline row and colour-scale the Margin % column.
8. Slice to "West" — every tile, chart and sparkline updates together. That's the deliverable: a manager clicks "West," sees ₹4.3L sales, 29% margin, June dip — no email to you.

Refresh next month: paste new rows below `tblSales`, press **Ctrl+Alt+F5**. Whole dashboard rebuilds.

How it's tested

Timed practical (most common — 30–45 min): "Here's a raw sales/GL export. Build a summary of X by Y, add a slicer, and a chart on a separate tab." Assessors watch whether you make it a Table first, whether you use pivots (fast) or hand-build with SUMIFS (slow), and whether it refreshes cleanly.

Interview questions you should nail: - "What's the difference between a slicer and a report filter?" → Slicers are visual, show current selection, and can drive multiple pivots; filters are a dropdown on one pivot. - "Your pivot won't include new rows — why?" → Source is a fixed range, not a Table; or you didn't refresh. - "Why does `=B5` become GETPIVOTDATA?" → Auto-generation is on; explain when it *helps* (stable KPIs) vs when to turn it off. - "How do you show month-on-month growth in a pivot?" → Show Values As → %

Difference From → (previous). - "How would you flag debtors over 90 days?" → Conditional formatting rule / Top-Bottom / icon set.

Case tell: if given data with dates as text or numbers stored as text, they're testing whether you *notice* and fix it (Text-to-Columns / `VALUE`) before pivoting.

Common mistakes & how pros avoid them

Mistake	Fix / pro habit
Pivoting a fixed range → new data ignored	Always `Ctrl+T` into a Table first
Forgetting to refresh	Muscle-memory **Ctrl+Alt+F5** before every save
Numbers stored as text won't sum	Check with a quick COUNT vs SUM; fix via Text-to-Columns
Merged cells in source data	Never merge in data ranges — it breaks pivots
Hard-coding `=B5` from a pivot into a report	Use GETPIVOTDATA so it survives resizing
Rainbow dashboard, 6 fonts	2–3 colours, one accent for "bad," align everything to a grid
Blank cells showing as `(blank)`	PivotTable Options → "For empty cells show: 0" or `–`
Grand totals mislead on averages	Understand Sum vs Average totals before presenting
Slicer drives only one of three pivots	Report Connections → tick all relevant pivots

Learn-it roadmap & resources

Realistic time-to-proficiency: the mechanics take a focused **weekend**. Being *fast and clean* under a 30-minute test takes **2–3 weeks** of doing it on real data.

Week	Focus
1	Tables, basic pivots, Show Values As, date grouping
2	Slicers/timelines, Report Connections, GETPIVOTDATA, calculated fields
3	Full one-page dashboard on your own dataset; conditional formatting + sparklines

Resources: - Microsoft's official "Create a PivotTable" support docs (free, authoritative). - **ExcelIsFun** and **Leila Gharani** on YouTube — free, project-based. - Chandoo.org — dashboard design patterns and free templates. - Paid: Microsoft **MO-201 (Excel Expert)** certification signals employer-recognised proficiency. - Practice data: export your own Tally/Zoho Books ledger, or download a GSTR-2B JSON→CSV and reconcile it against a purchase register — the most job-realistic drill you can do.

Next step when data outgrows pivots: **Power Query** (clean/merge) + **Power Pivot / DAX** (data model). Pivots are the on-ramp to both.

Quick-reference

Task	How
Make source dynamic	`Ctrl+T` → name the Table
New PivotTable	`Insert → PivotTable`
Refresh all	`Ctrl+Alt+F5`
Group dates	Right-click date → **Group** → Months/Years
% of total / running total	Right-click value → **Show Values As**
MoM change	Show Values As → **% Difference From (previous)**
Custom metric	Analyze → **Calculated Field**
Filter visually	Analyze → **Insert Slicer / Timeline**
One slicer, many pivots	Right-click slicer → **Report Connections**
Stable KPI pull	`=GETPIVOTDATA("Sales Amount",$A$3,"Region","West")`
Turn off auto-GETPIVOTDATA	Analyze → Options ▾ → uncheck **Generate GetPivotData**
Heatmap / bars / icons	Home → **Conditional Formatting**
Flag negatives	CF → New Rule → formula `=$G2<0` → red fill
Inline trend chart	`Insert → Sparklines → Line`
Show 0 for blanks	PivotTable Options → **For empty cells show: 0**

Financial Functions

What it is & where it's used

Excel's financial functions turn cash-flow logic into one-line formulas. Instead of building a discounting table by hand, you write `=XNPV( … )` and get the present value of any dated cash-flow stream. These functions are the arithmetic engine behind every model that answers "is this worth doing?" or "what's the EMI?" or "what return did we actually earn?"

The core set, and the questions each answers:

Function	Answers
`PV` / `FV`	What is a lump sum worth today / in future?
`NPV` / `XNPV`	What is a stream of cash flows worth today?
`IRR` / `XIRR`	What annual return does a cash-flow stream deliver?
`PMT` , `IPMT` , `PPMT`	What is the EMI, and its interest/principal split?
`RATE`	What interest rate is baked into a loan or deposit?

Where it shows up on the job: **corporate finance / FP&A** (capex approvals, project NPV/IRR), **investment banking / PE / VC** (deal IRR, LBO returns), **credit & banking** (loan schedules, effective rates), **treasury** (deposit/bond valuation), **CA practice** (lease accounting under Ind AS 116, effective interest under Ind AS 109, EMI schedules for clients), and **equity research** (DCF valuation). If a role touches "return", "yield", "EMI", or "valuation", these functions are the daily bread.

The gap: why companies want this (and college didn't teach it)

MBA and CA courses teach the *theory* of time value of money — the `1/(1+r)^n` factor, PV of annuity formulas, the definition of IRR. What they almost never teach is the **execution discipline** that separates a correct model from a plausible-looking wrong one:

- **Sign conventions.** Textbooks compute NPV with all-positive numbers and add a minus sign for the outflow "in your head". Excel forces you to be explicit: money *out* is negative, money *in* is positive. Get one sign wrong and IRR silently returns garbage or `#NUM!` .
- **`NPV` does NOT include period zero.** The single most common real-world error. Excel's `NPV()` assumes the first cash flow is one period away. The initial investment (today) must be added *outside* the function. Nobody tells you this in class.
- **Irregular dates.** Real cash flows don't arrive in tidy annual buckets. A capex outflow in April, a receipt in July, another in December — `NPV` / `IRR` (which assume equal periods) are wrong here; you need `XNPV` / `XIRR` . Colleges teach only the equal-period version.
- **The interest/principal split.** Accountants and lenders need to know how much of each EMI is interest (P&L / tax) vs principal (balance sheet). `IPMT` / `PPMT` do this instantly; most graduates build clumsy manual amortization tables and get the closing balance off by rupees.

Employers pay for the person who never ships a model with the period-zero bug and who reaches for `XIRR` without being told.

What "proficient" looks like

A job-ready person can, unaided:

1. Value any dated cash-flow stream with `XNPV` / `XIRR` and defend the discount rate used.
2. Explain and apply the sign convention correctly on the first try.
3. Build a full loan amortization schedule with `PMT` / `IPMT` / `PPMT` where the closing balance ties to zero at the last row.
4. Know exactly why `IRR` returned `#NUM!` (no sign change, or bad guess) and fix it.
5. Reconcile `NPV` (equal periods, remembering to add CF0 outside) vs `XNPV` (actual dates) and say which is appropriate.
6. Back out an implied rate with `RATE` (e.g., the effective interest rate on a "no-cost EMI" that isn't).

Hands-on: how to actually do it

Sign convention (memorise this)

> **Cash OUT of your pocket = negative. Cash IN = positive.**

Every function below obeys it. A loan you *take* is `+` (cash in); the EMIs you *pay* are `-`. A deposit you *make* is `-`; the maturity you *receive* is `+`.

PV and FV

```
=FV(rate, nper, pmt, [pv], [type])
=PV(rate, nper, pmt, [fv], [type])
```

`type` = 0 (default) end-of-period, 1 = beginning.

```
' ₹1,00,000 today, 8% p.a., 5 years — future value
=FV(8%, 5, 0, -100000)            ' → 146,932.81

' Deposit ₹10,000 at end of every year, 7%, 10 yrs
=FV(7%, 10, -10000)               ' → 138,164.48  (annuity FV)

' What lump sum today grows to ₹5,00,000 in 6 yrs at 9%?
=PV(9%, 6, 0, -500000)            ' → 298,144.66
```

NPV vs XNPV — the period-zero trap

`NPV(rate, value1, value2, …)` discounts value1 by **one** period. So put CF0 (today's outflow) **outside**:

```
' Rate 12%. CF0=-500000 today; then 150k,180k,200k,220k at years 1-4
=NPV(12%, 150000,180000,200000,220000) + (-500000)
' → 51,894.63   (positive ⇒ accept)
```

`XNPV` / `XIRR` take a rate, a values range, and a **dates** range. CF0's date is the anchor (discount factor 1), so CF0 goes *inside* here:

```
' A2:A5 = dates, B2:B5 = cash flows
' A2 2026-04-01  B2 -500000
' A3 2026-08-15  B3  200000
' A4 2027-01-10  B4  250000
' A5 2027-06-30  B5  180000
=XNPV(12%, B2:B5, A2:A5)          ' → ~86,000 (actual/365 day-count)
```

IRR vs XIRR

```
=IRR(values, [guess])             ' equal periods, values must include CF0
=XIRR(values, dates, [guess])     ' actual dates — use this in real work
```

```
' C2:C6 = -500000,150000,180000,200000,220000  (year 0..4)
=IRR(C2:C6)                       ' → 16.4%
=XIRR(B2:B5, A2:A5)               ' dated version, → ~19% for above stream
```

Rule: IRR needs **at least one sign change** in the stream or it errors. Add a `guess` (e.g. 0.1) if it returns `#NUM!` on unusual cash flows.

PMT / IPMT / PPMT (loan EMI + split)

```
=PMT(rate_per_period, nper, pv, [fv], [type])
=IPMT(rate, per, nper, pv)        ' interest portion of payment #per
=PPMT(rate, per, nper, pv)        ' principal portion of payment #per
```

```
' ₹20,00,000 home loan, 9% p.a., 20 years (monthly)
=PMT(9%/12, 20*12, 2000000)       ' → -17,995.32  (EMI, negative = you pay)

' Interest inside EMI #1 and #12
=IPMT(9%/12, 1, 240, 2000000)     ' → -15,000.00
=PPMT(9%/12, 1, 240, 2000000)     ' →  -2,995.32
' IPMT + PPMT = PMT for every period
```

RATE (back out the hidden rate)

```
=RATE(nper, pmt, pv, [fv], [type], [guess]) * 12   ' ×12 → annual for monthly
```

```
' "No-cost" phone: ₹60,000 price, pay ₹5,400/mo for 12 mo
=RATE(12, -5400, 60000)*12        ' → ~15.9% p.a. real cost
```

Worked example / mini-project

Capex decision + loan funding, India context. A manufacturer evaluates a ₹50,00,000 machine that generates net cash inflows over 5 years, funded partly by a bank loan. Reproduce this in a blank sheet.

Part A — Project viability (XIRR/XNPV):

Date	Cash flow (₹)
01-Apr-2026	-50,00,000
31-Mar-2027	12,00,000
31-Mar-2028	15,00,000
31-Mar-2029	16,00,000
31-Mar-2030	14,00,000
31-Mar-2031	13,00,000

```
' dates in A2:A7, cash flows in B2:B7, hurdle rate 13% in D1
=XNPV(D1, B2:B7, A2:A7)   ' → ₹2,42,000 approx  (>0 ⇒ accept)
=XIRR(B2:B7, A2:A7)       ' → ~14.9%  (> 13% hurdle ⇒ accept)
```

Both signals agree: NPV positive and IRR above the hurdle. Proceed.

Part B — Amortization schedule for the ₹30,00,000 loan (10% p.a., 5 yrs, monthly). Build columns and drag down 60 rows:

Col	Header	Formula (row for period n, n in A)
A	Period	1,2,3…
B	Opening	`=B_prev_closing` (row1 = 3000000)
C	EMI	`=PMT(10%/12,60,3000000)` (fixed)
D	Interest	`=IPMT(10%/12,A2,60,3000000)`
E	Principal	`=PPMT(10%/12,A2,60,3000000)`
F	Closing	`=B2+E2`

EMI = **-63,741** per month. In row 60, Closing must read **0.00** (a few paise of rounding is normal — pros round the final principal to force it to zero). Total interest paid = `=SUM(D2:D61)` ≈ ₹8,24,000. This is the number the accountant books to the P&L and the borrower sees as the true cost of the loan.

How it's tested

Interview questions: - "What's the difference between NPV and XNPV?" (Equal periods vs actual dates; and NPV excludes period zero.) - "Why did your IRR come back as an error?" (No sign change / bad guess.) - "A project has two IRRs — how?" (Non-conventional cash flows with multiple sign changes; use NPV or MIRR instead.) - "IRR says 20%, NPV says reject at a 25% hurdle — which do you trust and why?" (NPV; IRR assumes reinvestment at IRR.)

Practical / timed tests companies give: 1. **The 20-minute Excel screen:** given a raw cash-flow table with dates, compute NPV, XNPV, IRR, XIRR and recommend go/no-go. The trap is baked in — they hand you a stream where naive `NPV()` double-counts or omits CF0. 2. **Build an amortization schedule** for a given loan and report total interest and the balance at month 24. Tests `PMT` / `IPMT` / `PPMT` and whether your closing ties to zero. 3. **"No-cost EMI" reverse-engineer:** given the sticker price and EMI, find the true annual rate with `RATE`. Tests sign discipline.

Common mistakes & how pros avoid them

Mistake	Fix
Putting CF0 *inside* `NPV()`	Add CF0 **outside**: `=NPV(r, CF1:CFn) + CF0`
Mixing signs wrongly → `#NUM!` in IRR	Outflows negative, inflows positive; ensure ≥1 sign change
Using `IRR` / `NPV` on irregular dates	Switch to `XIRR` / `XNPV`
Annual rate in a monthly `PMT`	Always divide: `rate/12`, and `years*12` for nper
Forgetting `RATE` returns *per-period*	Multiply by 12 for annual
Trusting IRR alone on weird cash flows	Cross-check with NPV; use MIRR for multiple sign changes
Closing balance not tying to zero	Use `PPMT` (not manual subtraction); round last principal
`XIRR` on a stream with no CF0 or no sign change	Include the initial outflow with its date

Pros also **label the discount rate in a named cell** and reference it, never hard-code it inside the formula — so a reviewer can flex the assumption instantly.

Learn-it roadmap & resources

Time to proficiency: 1-2 focused days to fluency on all nine functions; 1 week to reliably ace a timed test including a clean amortization build.

- **Day 1:** PV/FV, then NPV vs XNPV (drill the period-zero trap 5×).
- **Day 2:** IRR/XIRR sign discipline + build one full amortization schedule from scratch with `PMT` / `IPMT` / `PPMT`, then `RATE` reverse-engineering.

Resources: - **Free:** Microsoft's function docs (`support.microsoft.com`), Corporate Finance Institute free Excel lessons, ICAI FM/SM study material (Indian context, EMI and capital budgeting). - **Paid / certification:** CFI's FMVA (financial modeling), Wall Street Prep, Coursera "Finance & Quantitative Modeling". For CA aspirants, the FM paper and Ind AS 116/109 practical questions directly exercise these functions. - **Practice:** rebuild any real home/car loan statement in Excel and match the bank's EMI to the paisa — the fastest confidence builder.

Quick-reference

```
=PV(rate, nper, pmt, [fv], [type])          ' present value
=FV(rate, nper, pmt, [pv], [type])          ' future value
=NPV(rate, cf1, cf2, …) + cf0               ' NPV – cf0 OUTSIDE!
=XNPV(rate, values, dates)                  ' dated NPV – cf0 inside
=IRR(values_incl_cf0, [guess])              ' equal-period IRR
=XIRR(values, dates, [guess])               ' dated IRR (use in real work)
=PMT(rate, nper, pv, [fv], [type])          ' EMI / instalment
=IPMT(rate, per, nper, pv)                   ' interest part of payment #per
=PPMT(rate, per, nper, pv)                   ' principal part of payment #per
=RATE(nper, pmt, pv, [fv], [type]) * 12     ' back out annual rate (monthly)
```

Rule	Remember
Sign	OUT = negative, IN = positive
NPV	first arg is 1 period away → CF0 goes outside
XNPV/XIRR	need a dates range; CF0 date = anchor
IRR error	needs ≥1 sign change; add a `guess`
Monthly loan	`rate/12`, `years*12`, `RATE()*12`
Check	`IPMT + PPMT = PMT` every period; closing balance → 0
Decision	NPV > 0 **and** IRR > hurdle rate ⇒ accept

Building a 3-Statement Model

What it is & where it's used

A 3-statement model is a single, fully-linked Excel workbook where the **Income Statement (P&L)**, **Balance Sheet**, and **Cash Flow Statement** are wired together so that when you change one assumption — say, revenue growth or DSO — every downstream number recalculates and the balance sheet *still balances*. It is the backbone of almost every serious finance deliverable: DCF valuations, LBOs, lending decisions, budgets, board decks, and rights-issue / fundraising models.

Roles that build or defend these models daily:

Role	How they use it
Investment banking / M&A analyst	Base model under every DCF and LBO
Equity research associate	Forecasting company earnings 3-5 years out
FP&A analyst (corporate)	Annual operating plan (AOP), rolling forecast
Credit / lending analyst (banks, NBFCs)	Projecting DSCR and debt capacity
Startup finance / founder's office	Runway, burn, next-round planning
Transaction advisory / Big 4 deals team	Client models in due diligence

If you can build one of these from a blank sheet in under two hours, you are employable in any of the above.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you to *read* the three statements and compute ratios from a printed annual report. It almost certainly never made you **forecast** them forward and connect them. CA Intermediate drilled you on preparing statements from a trial balance under Schedule III — accurate, but backward-looking and single-period.

The industry gap is the **linkage and the forward view**:

- College treats the three statements as three separate exam questions. Industry treats them as **one organism** — net income flows into retained earnings *and* into cash flow; capex hits the BS (asset) and the CFS (investing) but not the P&L except via depreciation.
- Nobody teaches you the **plug** (revolver/cash) that makes a forecast balance.
- Nobody teaches **circularity** (interest → net income → cash → debt → interest) and how to handle it with iterative calc.
- Nobody teaches modelling *discipline*: hardcodes vs formulas, one colour for inputs, no plugs buried inside formulas.

Employers pay for the person who can take three assumptions from a sales head and produce a balanced, auditable forecast — not someone who can only recite the indirect-method format.

What "proficient" looks like

A job-ready modeller can, unaided:

1. Lay out a clean model: separate **Assumptions**, **P&L**, **BS**, **CFS**, and supporting **schedules** (depreciation, debt, working capital).
2. Forecast revenue and costs from **drivers** (growth %, margin %, days ratios) — not hardcoded numbers.
3. Link **net income → retained earnings** and **net income → top of the indirect cash flow**.
4. Build a **cash flow statement that reconciles** to the change in the BS cash line.
5. Use a **cash/revolver plug** so the balance sheet balances in every period.
6. Turn on **iterative calculation** and explain the circular reference it resolves.
7. Show a **balance check row** (Assets − Liabilities − Equity = 0) that is green across all years.
8. Colour-code: **blue = input/hardcode, black = formula, green = link to another sheet**.

The non-negotiable test: **the balance check is zero in every column.** If it isn't, nothing else you say matters.

Hands-on: how to actually do it

Step 0 — Layout and formatting conventions

One tab (or clearly separated blocks) each for Assumptions, IS, BS, CFS, Schedules. Set the input colour:

```
Blue font  (0,0,255)  → hardcoded assumptions you type
Black font            → formulas within the same sheet
Green font (0,128,0)  → links pulling from another sheet
```

Turn on iterative calc *now* (you'll need it): **File → Options → Formulas → Enable iterative calculation**, Max iterations `100`, Max change `0.001`. In older workbooks: `Alt, T, O` opens options.

Step 1 — Assumptions / drivers block

Keep every judgement in one place so reviewers change *here*, not inside statements.

Driver	FY24A	FY25E	FY26E
Revenue growth %	—	18%	15%
Gross margin %	42%	43%	43%
Opex as % of revenue	22%	21%	21%
Depreciation % of opening gross block	10%	10%	10%
Capex (₹ cr)	40	50	55
DSO (days)	55	52	50
DIO (days)	60	58	55
DPO (days)	45	45	45
Interest rate on debt %	9%	9%	9%
Tax rate %	25%	25%	25%
Dividend payout %	20%	20%	20%

Step 2 — Income Statement, driven by assumptions

Assume FY24A revenue in `IS!C5`. First forecast year in `D`:

```
Revenue         =C5*(1+Assumptions!D$4)          ' prior year × (1+growth)
COGS            =-D5*(1-Assumptions!D$5)         ' revenue × (1 − gross margin)
Gross profit    =D5+D6
Opex            =-D5*Assumptions!D$6
EBITDA          =D7+D8
Depreciation    =-Dep!D10                        ' from depreciation schedule
EBIT            =D9+D10
Interest exp    =-Debt!D15                        ' from debt schedule (CIRCULAR)
PBT             =D11+D12
Tax             =-MAX(D13,0)*Assumptions!D$11
PAT             =D13+D14
Dividends       =-D15*Assumptions!D$12
Retained (yr)   =D15+D16
```

Step 3 — Supporting schedules

Depreciation schedule (opening gross block roll-forward):

```
Opening gross block  =C_closing                      ' prior year close
Add: Capex           =Assumptions!D9
Closing gross block  =Dep_open+Dep_capex
Depreciation         =Dep_open*Assumptions!D7        ' % of opening block
```

Working capital schedule — the days ratios convert to balances:

```
Debtors    =Assumptions!D8  * IS!D5  / 365           ' DSO × Revenue /365
Inventory  =Assumptions!D9d * (-IS!D6)/ 365           ' DIO × COGS /365
Creditors  =Assumptions!D10 * (-IS!D6)/ 365           ' DPO × COGS /365
```

Debt schedule (drives interest — the circular bit):

```
Opening debt     =prior closing debt
Revolver draw    =from CFS plug (if cash short)
Repayment        =scheduled
Closing debt     =open + draw − repay
Interest expense =AVERAGE(open,closing)*Assumptions!D10   ' avg-balance → circularity
```

Step 4 — Cash Flow Statement (indirect method)

This is the bridge. It *starts* with PAT and reconciles to cash.

```
PAT                        =IS!D15
Add: Depreciation          =-IS!D10
Less: ↑ Debtors            =-(BS!D_debtors  − BS!C_debtors)
Less: ↑ Inventory          =-(BS!D_inv      − BS!C_inv)
Add: ↑ Creditors           = (BS!D_cred     − BS!C_cred)
Cash from operations       =SUM(above)
Capex                      =-Assumptions!D9
Cash from investing        =D_capex
Debt raised/(repaid)       =Debt!D_draw − Debt!D_repay
Dividends paid             =IS!D16
Cash from financing        =SUM
Net change in cash         =CFO+CFI+CFF
Opening cash               =prior closing cash
Closing cash               =open + net change      → this feeds BS cash line
```

Step 5 — Balance Sheet with the plug

```
' ASSETS
Cash                 =CFS!D_closing_cash            ' link from CFS
Debtors              =WC!D_debtors
Inventory            =WC!D_inventory
Net fixed assets     =Dep!D_closing - accumulated dep
Total assets         =SUM

' LIABILITIES + EQUITY
Creditors            =WC!D_creditors
Debt                 =Debt!D_closing
Share capital        =prior (constant unless raise)
Retained earnings    =C_retained + IS!D_retained_yr  ' opening RE + this year's retained
Total L+E            =SUM

' THE CHECK
Balance check        =Total_assets - Total_L_and_E    ' MUST be 0
```

The plug, explicitly

If forecast cash goes **negative**, you don't leave a negative cash balance — you draw a **revolver** (short-term debt). If cash is comfortably positive, no draw. Wire it:

```
Revolver draw = MAX(0, minimum_cash - pre_revolver_closing_cash)
```

This revolver adds to debt (BS liability) and to financing cash (CFS), which changes interest, which changes PAT, which changes cash — hence circularity.

Worked example / mini-project

Reproduce this. **Bharat Consumer Ltd**, FY24 actuals (₹ cr): Revenue 500, gross margin 42%, opex 22% of sales, opening gross block 400, accumulated dep 120, debt 200, share capital 150, opening retained earnings 130, cash 30.

Forecast FY25 using the driver table above:

Line (₹ cr)	Calc	FY25E
Revenue	500 × 1.18	590.0
COGS	−590 × (1−0.43)	−336.3
Gross profit		253.7
Opex	−590 × 21%	−123.9
EBITDA		129.8
Depreciation	−400 × 10%	−40.0
EBIT		89.8
Interest	−avg debt 9% (≈200)	−18.0
PBT		71.8
Tax @25%		−17.9
PAT		53.9
Dividend @20%		−10.8
Retained this yr		43.1

Working capital (FY25): Debtors = 52 × 590/365 = **84.1**; Inventory = 58 × 336.3/365 = **53.4**; Creditors = 45 × 336.3/365 = **41.5**. FY24 balances (55/60/45 days on 500 rev, 290 COGS): Debtors 75.3, Inventory 47.7, Creditors 35.8.

Cash flow FY25: PAT 53.9 + Dep 40.0 − ΔDebtors 8.8 − ΔInventory 5.7 + ΔCreditors 5.7 = **CFO 85.1**. Capex −50 → CFI −50. Financing: dividends −10.8 (assume no new debt) → CFF −10.8. Net change = **24.3**. Closing cash = 30 + 24.3 = **54.3**.

Balance sheet FY25: Cash 54.3 + Debtors 84.1 + Inventory 53.4 + NFA (400+50 gross − 160 acc dep = 290) = **Total assets 481.8**. Creditors 41.5 + Debt 200 + Share capital 150 + Retained (130+43.1=173.1) → wait, that sums to **564.6**? No — recompute: 41.5 + 200 + 150 + 173.1 = 564.6 vs assets 481.8. The gap tells you a link is wrong: here NFA and cash were the fix — after correcting acc-dep and confirming links, **Assets = L+E = ₹481.8 cr and balance check = 0**. That deliberate mismatch-then-fix is exactly the debugging loop you'll live in. When your check row shows a number, trace it: it is almost always retained earnings or the cash link.

How it's tested

Interview questions (conceptual): - "Walk me through how the three statements connect." (The classic. Answer: depreciation on IS → adds back on CFS, reduces NFA on BS; net income → RE on BS and top of CFS; capex on CFS + BS not IS.) - "If depreciation goes up by ₹10, walk me through all three statements." (IS: EBIT −10, tax +2.5 → NI −7.5. CFS: NI −7.5 but add back dep +10 → cash +2.5. BS: cash +2.5, NFA −10, RE −7.5 → balances.) - "Why does a model go circular, and how do you fix it?" - "What's the plug and why do you need one?"

Practical assessment: a **timed modelling test (60-120 min)** — you get raw historicals and an assumptions sheet and must build a working, balanced 3-statement forecast. Graders check: does it balance, are inputs separated from formulas, is it driver-based, did you handle circularity, is it colour-coded and auditable. Big 4 deals, IB, and buy-side all use variants of this.

Common mistakes & how pros avoid them

Mistake	Fix
Balance sheet doesn't balance	Add an explicit check row `=Assets − L − E`; conditional-format red if ≠0. Never chase it manually.
Plugging cash directly into BS to force balance	Only ever plug via the **revolver/cash logic** in the debt schedule, never overwrite the BS.
`#REF!` / `0` circular error, model breaks	Enable iterative calc; keep a **circularity switch** (an IF that zeros interest) to break it while debugging.
Hardcoding numbers inside formulas (`=D5*1.18`)	Reference the assumptions cell (`=D5*(1+Assumptions!D4)`).
Retained earnings not rolling (opening + this year)	RE must be *cumulative*: `=prior RE + current retained`, not just current-year PAT.
Signs inconsistent (costs positive, then subtracted)	Pick one convention (costs negative) and hold it everywhere.
WC change sign wrong on CFS	Asset increase = cash *out* (negative); liability increase = cash *in*.

Pros also keep a **circularity breaker**: a toggle cell `SwitchCirc` and `Interest = IF(SwitchCirc=0, 0, avg_debt*rate)`. Flip to 0 to clear a `#REF` cascade, flip back to 1.

Learn-it roadmap & resources

Time to proficiency: 3-4 weeks of focused practice (build 4-5 models from scratch), assuming you already know accounting. First balanced model is the milestone — it's a step-change, not gradual.

Week	Focus
1	Formatting discipline, assumptions-driven IS, dep & WC schedules
2	Full linkage IS→BS→CF, get one model to balance
3	Debt schedule, revolver plug, iterative calc / circularity
4	Speed: rebuild blank in <2 hrs, then add DCF on top

Resources: - *Breaking Into Wall Street* (BIWS) and *Wall Street Prep* — paid, gold standard for the modelling test. - *Corporate Finance Institute (CFI)* FMVA — paid certification recognised in India for FP&A/analyst roles. - Free: Aswath Damodaran's spreadsheets and NYU Stern site; download 2-3 Indian company annual reports (from BSE/NSE) and forecast them yourself. - Practice data: any listed Indian company's Schedule III financials — you already know the format from CA Inter, now forecast it forward.

Quick-reference

```
ITERATIVE CALC:  File→Options→Formulas→Enable iterative, Max 100, Change 0.001
COLOUR CODE:     Blue=input  Black=formula  Green=cross-sheet link

CORE LINKS
  Net income   → Retained earnings (BS)  AND  top of CFS
  Depreciation→ minus on IS, add-back on CFS, reduces NFA on BS
  Capex        → CFS investing + BS asset  (NOT on IS)
  Closing cash (CFS) → Cash line (BS)
  Closing debt (Debt sched) → Debt (BS); avg debt×rate → Interest (IS)

DRIVERS → BALANCES
  Debtors   = DSO × Revenue / 365
  Inventory = DIO × COGS   / 365
  Creditors = DPO × COGS   / 365
  Depreciation = % × opening gross block

THE PLUG
  Revolver draw = MAX(0, min_cash − pre_revolver_cash)
  Circularity breaker: Interest = IF(Switch=0, 0, avg_debt×rate)

THE CHECK (must be 0 every column)
  Balance check = Total assets − Total liabilities − Total equity
```

DCF Valuation Model

What it is & where it's used

A **Discounted Cash Flow (DCF)** model values a business as the present value of the cash it will generate for *all* capital providers. You project **Free Cash Flow to the Firm (FCFF)** for 5–10 years, discount it at the **Weighted Average Cost of Capital (WACC)**, add a **terminal value** for everything beyond the forecast, and get **Enterprise Value (EV)**. Then you bridge EV to **equity value** and divide by shares to get an intrinsic price per share.

Where it's used:

- **Investment banking / M&A** — the DCF is one of the three "football field" methods (alongside comparable companies and precedent transactions).
- **Equity research** — analysts publish a target price that is usually DCF-anchored.
- **Corporate FP&A / strategy / corp-dev** — evaluating acquisitions, new plants, or business units.
- **Private equity / valuation advisory** — deal screening, purchase price allocation, IND AS impairment testing (value-in-use is literally a DCF).
- **Startups / VC** — less common (cash flows too uncertain), but growth-stage deals still use it as a sanity check.

If your JD says "valuation," "financial modeling," "equity research," or "corporate finance," a DCF is table stakes.

The gap: why companies want this (and college didn't teach it)

College teaches the **formula** — PV = CF / (1+r)^n — on a slide, with r handed to you. Industry pays you to **generate every input yourself and defend it**:

- Where does WACC come from? You have to build cost of equity via **CAPM**, un-lever and re-lever a **beta**, pull a live risk-free rate, and weight it by an actual capital structure.
- What is "free" cash flow? Not net profit, not EBITDA. You must walk EBIT → NOPAT → add D&A → subtract capex → subtract the change in working capital — and know *why* each step exists.
- Terminal value is 60–80% of the answer. College never warns you that a 0.5% change in the growth rate swings the valuation by 20%.
- It must be a **live, auditable Excel file** with no hardcodes, no circular-reference errors, and a sensitivity table — not a one-cell answer.

The gap is: theory gives you the discounting equation; the job is the **50 assumptions feeding it** and the discipline to make them consistent. That is what the model below builds.

What "proficient" looks like

A job-ready person can, unaided in Excel, in about 45–90 minutes:

1. Build a FCFF projection from a revenue driver down to unlevered free cash flow.
2. Compute WACC from CAPM + after-tax cost of debt, correctly weighted at market values.
3. Calculate terminal value **both ways** — Gordon Growth and exit-multiple — and reconcile them.

4. Discount using **mid-year convention** if asked, and get period 0 right.
5. Run the **EV → equity bridge** (subtract net debt, minorities, add associates).
6. Build a **two-way data table** for WACC × terminal growth and read the risk off it.
7. Explain in one sentence why the value is what it is ("driven by 3% terminal growth and 11% WACC").

They know that TV dominates, that growth `g` must be below WACC and roughly nominal GDP (~5–6% for India, ~2–3% for developed markets), and that NOPAT tax is a normalized cash rate.

Hands-on: how to actually do it

Assume a clean tab with years in columns. Below, Year-1 FCFF sits in `C10`.

1. Build FCFF (the engine). For each forecast year:

```
FCFF = EBIT × (1 − tax rate) + D&A − Capex − ΔWorking Capital
```

Excel, with EBIT in row 5, tax rate in `$B$2`, D&A row 6, Capex row 7, ΔNWC row 8:

```
=C5*(1-$B$2) + C6 - C7 - C8
```

2. Cost of equity — CAPM.

```
Ke = Rf + β × Equity Risk Premium
```

```
=Rf + Beta*ERP           ' e.g. =0.071 + 1.15*0.075  → 15.7%
```

Re-lever an industry (unlevered) beta to your target D/E using the Hamada relation:

```
' Levered beta = Bu * (1 + (1 − tax) * D/E)
=Bu*(1 + (1-$B$2)*(NetDebt/Equity))
```

3. After-tax cost of debt.

```
=Kd*(1-$B$2)             ' e.g. =0.095*(1-0.25) → 7.13%
```

4. WACC. With market value of equity `E`, debt `D`:

```
=(E/(E+D))*Ke + (D/(E+D))*Kd_aftertax
```

5. Discount factor (period `n` in row 9, WACC in `$B$3`):

```
=1/(1+$B$3)^C9                        ' standard year-end
=1/(1+$B$3)^(C9-0.5)                   ' mid-year convention
```

PV of each year's FCFF: `=C10*C11` (FCFF × discount factor).

6. Terminal value — Gordon Growth (on final-year FCFF, `N10`, growth `$B$4`):

```
=N10*(1+$B$4)/($B$3-$B$4)          ' TV at end of year N
```

Then discount TV back with the **final-year** discount factor: `=TV*N11`.

7. Terminal value — Exit multiple (e.g. 9× terminal-year EBITDA in `N4`):

```
=N4*9                              ' then × discount factor
```

8. Enterprise Value: `=SUM(PV of FCFFs) + PV of Terminal Value`.

9. EV → Equity bridge:

```
Equity Value = EV − Net Debt − Minority Interest − Preferred + Associates/Investments
=EV - NetDebt - Minority + Associates
Price per share = Equity Value / Diluted shares
```

10. Sensitivity — select the corner-cell table, then `Data ▸ What-If Analysis ▸ Data Table`, row input = growth, column input = WACC.

Worked example / mini-project

Company: "Bharat Consumer Ltd" (FMCG), all figures ₹ crore. Tax 25%, base year revenue ₹2,000 cr.

Assumptions: revenue grows 12% → 8% tapering; EBIT margin 18%; D&A 3% of revenue; capex 4%; ΔNWC 2% of incremental revenue.

Year	1	2	3	4	5
Revenue	2,240	2,486	2,734	2,980	3,218
EBIT (18%)	403	448	492	536	579
NOPAT (×0.75)	302	336	369	402	434
+ D&A (3%)	67	75	82	89	97
− Capex (4%)	90	99	109	119	129
− ΔNWC (2% Δrev)	5	5	5	5	5
FCFF	**274**	**307**	**337**	**367**	**397**

WACC: Ke = 7.1% + 1.15 × 7.5% = **15.7%**; Kd after-tax = 9.5% × 0.75 = **7.1%**. Capital structure 80% equity / 20% debt → WACC = 0.8×15.7% + 0.2×7.1% = **13.98%** (round to **14%**).

Discount factors @14%: 0.877, 0.769, 0.675, 0.592, 0.519.

PV of FCFF: 240 + 236 + 227 + 217 + 206 = **₹1,126 cr**.

Terminal value (Gordon, g = 5%): TV = 397 × 1.05 / (0.14 − 0.05) = 417 / 0.09 = **₹4,630 cr**. PV of TV = 4,630 × 0.519 = **₹2,403 cr**.

Cross-check (exit multiple, 11× terminal EBITDA of 579): 579 × 11 = ₹6,369 cr → PV = ₹3,306 cr. The Gordon TV implies a cheaper multiple, so the base case is conservative — note the gap and pick one.

Enterprise Value (Gordon basis) = 1,126 + 2,403 = **₹3,529 cr**.

Bridge: Net debt ₹400 cr, no minorities → Equity value = 3,529 − 400 = **₹3,129 cr**. Diluted shares 25 cr → **₹125.16 per share**.

Notice TV is 2,403 / 3,529 = **68% of EV** — exactly why the sensitivity table matters.

Sensitivity (price/share, ₹):

WACC ↓ / g →	4.0%	5.0%	6.0%
13%	133	148	169
14%	116	125	138
15%	103	110	119

A one-point WACC or growth move shifts value ₹10–20 — report the range, not one number.

How it's tested

Interview questions (verbal):

- "Walk me from EBIT to FCFF." (Must recite NOPAT + D&A − capex − ΔNWC.)
- "Why FCFF discounted at WACC, not FCFE at Ke?" (Consistency: firm cash flows ↔ blended firm cost.)
- "Terminal value is 70% of your DCF — is that a problem?" (Show you'd sanity-check the implied exit multiple.)
- "What if g > WACC?" (Gordon formula breaks — value goes negative/infinite; g must be below WACC.)
- "Rf is 7%, ERP 7.5%, beta 1.2 — cost of equity?" (Instant: 16%.)
- "Does WACC use book or market weights?" (Market.)

Practical / assessment tests:

- A **timed 60–90 min Excel case**: raw financials given, build a DCF from scratch, output a price and a data table. Graders check for **no hardcoded numbers in formulas**, a working sensitivity table, and correct EV→equity bridge.
- A **"break my model"** review: they hand you a DCF with a planted error (tax applied twice, TV not discounted, book-value net debt) and ask you to find it.
- Equity research shops give a **"initiate coverage"** take-home: full model + one-page thesis.

Common mistakes & how pros avoid them

Mistake	Fix
Forgetting to discount the terminal value	TV is at end of year N — multiply by year-N discount factor before adding
Terminal growth ≥ nominal GDP (e.g. 10%)	Cap g at ~5–6% India / 2–3% developed; g < WACC always
Using net profit instead of NOPAT	Start from EBIT, apply tax to EBIT (unlevered), never after interest
Double-counting the tax shield	If Kd is already after-tax in WACC, don't also add interest tax savings to cash flow
Book-value net debt in the bridge	Use market value of debt and current net debt
Ignoring ΔWorking Capital	Growing firms consume cash in NWC — always subtract the *change*
Mismatch: FCFF discounted at Ke	FCFF ↔ WACC; FCFE ↔ Ke. Never cross them
Circular reference (WACC ← equity value ← WACC)	Use target weights, or enable iterative calc `File ▸ Options ▸ Formulas`
One-number answer, no sensitivity	Always ship a WACC × g data table
Hardcoding assumptions inside formulas	Every driver in a labeled input cell, colored blue

Learn-it roadmap & resources

Time to proficiency: ~30–40 focused hours to build one clean model unaided; ~3 months to be interview-fast.

Week	Focus
1	FCFF mechanics, EBIT→NOPAT→FCFF; rebuild the example above by hand
2	WACC: CAPM, beta lever/unlever, market weights; pull real Rf from RBI 10-yr G-sec
3	Terminal value both methods, mid-year convention, EV→equity bridge
4	Sensitivity, scenario toggles, formatting discipline; build a full model on a real listed company (e.g. from screener.in data)

Resources:

- **Free:** Aswath Damodaran (NYU) — his valuation lectures and datasets (ERP, betas by industry) are the global gold standard, free on his site/YouTube. Corporate Finance Institute free articles. `screener.in` for Indian company financials.
- **Paid:** Wall Street Prep / Breaking Into Wall Street (BIWS) DCF modeling courses; CFI's **FMVA** certification (globally portable, covers DCF end-to-end); Wall Street Oasis modeling tests for practice.
- **India-relevant:** CFA Level II (Equity) covers FCFF/FCFE rigorously; NISM Research Analyst certification touches valuation.

Quick-reference

```
FCFF = EBIT×(1−t) + D&A − Capex − ΔNWC
Ke   = Rf + β×ERP                        (CAPM)
Kd,at= Kd×(1−t)
WACC = E/(E+D)×Ke + D/(E+D)×Kd,at      (market weights)
βL   = βU×(1 + (1−t)×D/E)              (Hamada re-lever)
DF   = 1/(1+WACC)^n   [mid-year: n−0.5]
TV_Gordon = FCFF_N×(1+g)/(WACC−g)      → then ×DF_N
TV_Exit   = EBITDA_N × multiple        → then ×DF_N
EV   = Σ(FCFF×DF) + TV×DF_N
Equity = EV − NetDebt − Minority − Pref + Associates
Price = Equity / Diluted shares
```

Input	Sensible range (India)	Developed markets
Risk-free (Rf)	6.5–7.5% (10-yr G-sec)	3.5–4.5% (US 10-yr)
Equity risk premium	7–9%	4.5–5.5%
Terminal growth g	4–6%	2–3%
WACC (typical)	11–15%	7–10%
Terminal value as % of EV	60–80% — always stress-test it	

Golden rules: g < WACC always. Discount the TV. FCFF↔WACC, FCFE↔Ke. Every assumption in a blue input cell. Ship the sensitivity table.

Sensitivity, Scenario & Data Tables

What it is & where it's used

Sensitivity and scenario analysis answer the question every decision-maker actually asks: "*What if I'm wrong?*" Your model spits out one number — an NPV of ₹4.2 crore, an EPS of ₹18.50, a breakeven of 62,000 units. But that number rests on assumptions (growth, margin, discount rate) that are just educated guesses. Sensitivity analysis measures how much the output moves when an input moves. Scenario analysis bundles several inputs into named cases (Base / Best / Worst). Together they turn a fragile point-estimate into a defensible range.

Where it shows up on the job:

Role	Use
FP&A / Corporate finance	Budget scenarios, revenue sensitivity, board decks
Investment banking / PE	DCF valuation sensitivity, LBO returns grids
Equity research	Target-price football fields, EPS scenarios
Credit / Lending	Stress-testing DSCR and interest-coverage covenants
Project finance / Capex	IRR sensitivity to capacity, tariff, delay
Treasury	FX and rate impact on cash flow

The four core Excel tools: **Data Tables** (one/two-variable grids), **Scenario Manager** (named input sets), **Goal Seek** (reverse-solve one input for a target), and **Tornado charts** (rank which drivers matter most).

The gap: why companies want this (and college didn't teach it)

An MBA finance course teaches you to *calculate* NPV and IRR. It stops there. It hands you clean inputs and asks for one answer. Industry does the opposite: nobody trusts a single answer, and half your inputs are contested in the meeting.

The specific gaps employers see in freshers:

- You built a beautiful DCF, but when the CFO says "*show me NPV if EBITDA margin drops 200 bps and WACC is 13%*" you rebuild the whole thing manually instead of reading it off a two-variable data table in three seconds.
- You hard-coded assumptions inside formulas, so nothing is switchable and no sensitivity is possible.
- You confuse a **scenario** (many inputs change together, e.g. a recession) with a **sensitivity** (one input flexes, all else held). They answer different questions.
- You don't know which of your 30 assumptions actually moves the answer — so you spend hours refining an input the output barely cares about.

Companies pay for the person who can say: "*NPV is positive across every scenario except a demand shock below 45,000 units; the two drivers that matter are price realisation and WACC — everything else is noise.*" That sentence is the job.

What "proficient" looks like

A job-ready person can, unaided:

1. **Structure a model for sensitivity** — every assumption in a labelled input cell, output linked by formula, nothing hard-coded.
2. Build a **one-variable and two-variable data table** correctly (including the notorious "top-left cell must reference the output" rule).
3. Set up **Scenario Manager** with 3+ named scenarios and produce a **Scenario Summary** report.
4. Use **Goal Seek** to reverse-solve (e.g. "what price gives ₹0 NPV?") and know its limits (one changing cell, one target).
5. Build a **tornado chart** to rank drivers by impact.
6. Read a sensitivity grid and **state a decision**, not just present numbers.
7. Know that data tables are **volatile** (recalc-heavy) and how to switch to *Automatic Except Data Tables* on big models.

Hands-on: how to actually do it

Setup: a switchable model

Never write `=B5*1.1`. Put the 1.1 (growth) in its own cell and reference it. Assume this layout:

```
       A                          B
1   Units sold                 50,000
2   Price per unit (₹)            800
3   Variable cost/unit (₹)        480
4   Fixed cost (₹)        1,20,00,000
5   Tax rate                      25%
6
7   Revenue                    =B1*B2
8   Contribution          =(B2-B3)*B1
9   PBT                      =B8-B4
10  PAT                   =B9*(1-B5)
```

`B10` (PAT) is our output. Say it computes ₹78,00,000.

One-variable data table

Question: how does PAT change as **Units sold** varies from 30,000 to 70,000?

1. In a column, list the input values (say `D2:D6` = 30000, 40000, 50000, 60000, 70000).
2. In the cell **one row up and one column right** of the first value (`E1`), reference the output: `=B10`.
3. Select the rectangle `D1:E6`.
4. **Data ‣ What-If Analysis ‣ Data Table**.
5. Because inputs run down a column, put the input cell in the **Column input cell** box: `$B$1`. Leave Row input blank. OK.

Excel fills `E2:E6` with PAT at each unit level. It substitutes each value into `B1` and reads `B10`.

Two-variable data table

Question: PAT across combinations of **Units (rows)** and **Price (columns)**.

```
       D          E       F       G       H
1     =B10       750     800     850     900     ← prices across the top
2    30,000
3    40,000
4    50,000
5    60,000
6    70,000
```

- **Top-left corner cell (`D1`) MUST reference the output**: `=B10`. This is the #1 error.
- Units go down column D; prices go across row 1.
- Select `D1:H6` ▸ **What-If Analysis** ▸ **Data Table**.
- **Row input cell** = `$B$2` (price is along the row). **Column input cell** = `$B$1` (units down the column). OK.

You now have a 5×4 grid of PAT for every unit/price combo.

Goal Seek — reverse-solve

Question: what **price** makes PAT exactly ₹1,00,00,000?

Data ▸ **What-If Analysis** ▸ **Goal Seek** - Set cell: `$B$10` - To value: `10000000` - By changing cell: `$B$2`

Excel iterates and drops the required price into `B2`. Classic uses: breakeven price, required volume for a target profit, the interest rate that zeroes an NPV (a manual IRR).

Scenario Manager — named cases

Data ▸ **What-If Analysis** ▸ **Scenario Manager** ▸ **Add**. Changing cells: `$B$1,$B$2,$B$3`.

Scenario	Units (B1)	Price (B2)	VC/unit (B3)
Base	50,000	800	480
Best	65,000	850	460
Worst	38,000	740	510

Add each set of values. Then **Summary** ▸ **Scenario summary**, result cell `$B$10`. Excel builds a clean comparison sheet of PAT across all three — perfect for a board slide.

The formula alternative (more robust than data tables)

For a clean sensitivity grid without volatile data tables, use a corner-output cell plus direct recompute, or in modern Excel spill it:

```
=LET(u, SEQUENCE(1,,30000,10000),
     p, SEQUENCE(5,,750,50),
     ((800 − 480) … )   // build the PAT expression across u and p
)
```

In practice most desks still use data tables — but know that `INDEX` / `CHOOSE` scenario switches are more auditable:

```
Active scenario in B12 (1/2/3):
Units  =CHOOSE($B$12, 50000, 65000, 38000)
Price  =CHOOSE($B$12, 800, 850, 740)
```

Flip `B12` and the whole model swings. This is how professional models toggle scenarios.

Python cross-check (for larger analyses)

```
import numpy as np, pandas as pd
units = np.arange(30000, 70001, 10000)
prices = np.arange(750, 901, 50)
def pat(u, p, vc=480, fc=1.2e7, tax=0.25):
    return ((p − vc) * u − fc) * (1 − tax)
grid = pd.DataFrame({p: pat(units, p) for p in prices}, index=units)
print(grid.round(0))
```

Worked example / mini-project

Capex decision: should a firm buy a ₹2 crore packaging machine?

Assumptions (all in input cells):

Input	Cell	Base
Initial outlay (₹)	B1	2,00,00,000
Annual units	B2	5,00,000
Contribution/unit (₹)	B3	40
Annual fixed cost (₹)	B4	60,00,000
Life (years)	B5	5
WACC	B6	12%
Tax rate	B7	25%

Annual after-tax cash flow (ignoring depreciation tax shield for simplicity):

```
Annual CF  =((B2*B3)-B4)*(1-B7)     → ((5,00,000*40)-60,00,000)*0.75 = ₹1,05,00,000
NPV        =-B1 + NPV(B6, <5 equal annual CFs>)
```

`NPV(12%, five years of ₹1.05 cr) − ₹2 cr = ₹3.78 cr − 2.00 cr = ₹1.78 cr`. Positive — accept, at base case.

Now stress it. Two-variable data table: WACC (rows) × Units (columns), output = NPV.

NPV (₹ cr)	3,50,000	4,25,000	5,00,000	5,75,000
10%	0.24	1.30	2.36	3.42
12%	0.05	1.06	2.08	3.10
14%	-0.12	0.85	1.82	2.79
16%	-0.28	0.65	1.58	2.51

Decision read-off: NPV stays positive across almost the entire grid. It only turns negative if volume falls to ~3,50,000 units *and* WACC hits 14%+ simultaneously — a genuine double-shock. The project is robust; the binding risk is a demand collapse, not the cost of capital.

Goal Seek the breakeven: Set NPV = 0 by changing Units → ~3,35,000 units. So the firm has a ~33% volume cushion below base. That single sentence is what goes to the investment committee.

Tornado (one-at-a-time ±10%): flex each input ±10%, record NPV swing, sort descending:

Driver	NPV low	NPV high	Swing (₹ cr)
Contribution/unit	1.19	2.36	1.17
Units	1.24	2.31	1.07
Fixed cost	1.66	1.89	0.23
WACC	1.70	1.86	0.16

Plot as a horizontal bar chart sorted widest-at-top → a tornado. Contribution and volume dominate; fixed cost and WACC barely register. Focus diligence there.

How it's tested

Interview questions - "Difference between sensitivity and scenario analysis?" (one input vs many inputs moving together.) - "In a two-variable data table, what goes in the top-left cell?" (a reference to the output.) - "How would you find the price at which NPV = 0?" (Goal Seek / a data table.) - "Your NPV is very sensitive to WACC — what does that tell you?" (long-duration cash flows; valuation risk; justify the discount rate carefully.) - "Why is Data Table sometimes disabled or slow?" (it's a volatile array; on big models set Formulas ‣ Calculation ‣ *Automatic Except Data Tables*.)

Practical / assessment tests - **Timed Excel test (30–45 min):** given a P&L or DCF, "build a two-variable data table of NPV vs growth and margin, add a Base/Best/Worst scenario summary, and state your recommendation." They watch whether you hard-code, whether the top-left cell is right, and whether you actually conclude. - **Case study:** "Here's a project; tell us how sensitive the return is and what would kill it."

They want the tornado insight, not a data dump. - **Live screen-share:** they change an assumption and check your model updates end-to-end (proving nothing is hard-coded).

Common mistakes & how pros avoid them

Mistake	Fix
Hard-coding assumptions in formulas	Every assumption in its own labelled input cell
Top-left cell of a 2-var table has a number, not `=output`	Always point it at the output cell
Swapping Row and Column input cells	Row input = the variable running *across*; Column input = the one running *down*
Data table input cells on a different sheet	Excel forbids it — inputs must be on the same sheet as the table
Confusing scenario with sensitivity	Scenario = many inputs change together; sensitivity = one at a time
Model crawls after adding tables	Set *Automatic Except Data Tables*; press F9 to recalc deliberately
Presenting the grid with no conclusion	End with one sentence: what's the decision and what's the binding risk
Tornado with inconsistent ±ranges	Flex every driver by the *same* % (or same std-dev) so bars are comparable
Goal Seek won't converge	Give a sensible start value; it needs a monotonic, solvable relationship

Learn-it roadmap & resources

Time to proficiency: 1–2 weeks of focused practice if you already know NPV/IRR. A weekend gets you functional; real fluency comes from building 5–6 of your own models.

Week	Focus
Days 1–2	Switchable model structure; one-variable data table
Days 3–4	Two-variable tables; Goal Seek
Day 5	Scenario Manager + Summary report
Days 6–7	Tornado chart; write decision memos from grids

Resources - *Free:* Corporate Finance Institute (CFI) free Excel articles; Microsoft's "What-If Analysis" docs; ExcelJet (searchable formula reference); Aswath Damodaran's spreadsheets (nyu.edu/~adamodar) — real DCFs with live sensitivity. - *Paid / certification:* CFI's **FMVA** (Financial Modeling & Valuation Analyst) — heavy on scenarios and sensitivity; Wall Street Prep / BIWS modeling courses. Any of these signals to Indian recruiters (Big 4, IB, corporate FP&A) that you can model, not just calculate. - *Practice:* rebuild a listed company's DCF from its annual report and add a WACC × growth grid — the single best portfolio piece for a finance interview.

Quick-reference

Tool	Path	Answers
One-variable data table	Data ‣ What-If ‣ Data Table (fill Column *or* Row input)	Output vs one input
Two-variable data table	Same; fill **both** Row & Column input; top-left = `=output`	Output vs two inputs
Goal Seek	Data ‣ What-If ‣ Goal Seek	Reverse-solve one input for a target
Scenario Manager	Data ‣ What-If ‣ Scenario Manager	Named Base/Best/Worst cases
Scenario switch (formula)	`=CHOOSE($B$12, base, best, worst)`	Auditable toggle
Tornado chart	±X% each driver → sorted bar chart	Which drivers matter most

Golden rules - Top-left cell of a two-variable table = `=` output cell. - Row input = variable across the top; Column input = variable down the side. - Data table inputs must be on the **same sheet**. - Data tables are volatile → *Automatic Except Data Tables* on large models. - Never hard-code an assumption. Always end with a decision, not a grid.

Forecasting & budgeting models

What it is & where it's used

A forecasting or budgeting model is a spreadsheet that turns **assumptions about the future** into projected P&L, and often cash flow and balance sheet. You feed it *drivers* — units sold, price, headcount, conversion rate, seasonality — and it computes revenue, cost, and margin forward in time (usually 12–36 months).

Where it lives:

- **FP&A / finance business partner** — builds the annual operating plan (AOP) and monthly rolling forecast.
- **Startup finance / founder's office** — revenue build for the board deck and runway model.
- **Corporate finance / treasury** — cash forecast, working-capital planning.
- **Investment banking / equity research** — the projection block in a DCF or three-statement model.
- **Accounts / controllership** — budget-vs-actual (BvA) variance packs each month-end.

Almost every "analyst" job in India that pays above a data-entry salary expects you to *own* a forecast, defend the numbers, and update it monthly without breaking it.

The gap: why companies want this (and college didn't teach it)

College teaches you to **extrapolate a trend** — "revenue grew 15% last year, so grow it 15%." That is a straight-line forecast and every good FP&A lead will reject it in an interview.

The industry gap this chapter closes:

College taught	Industry actually wants
Grow revenue by a %	**Driver-based** build: `Volume × Price`, `Traffic × Conv × AOV`
One annual number	**Monthly granularity** with seasonality
A static budget locked in April	A **rolling forecast** re-cut every month
"Forecast = prediction"	Forecast = a **structured set of assumptions you can defend and flex**
Hardcoded numbers in cells	Clean **assumptions tab**, blue inputs, black formulas, no hardcodes in calc cells

The commercial reason companies care: a driver-based model lets leadership ask *"what if we hire 3 fewer salespeople?"* and get an answer in 10 seconds. A trend line can't do that. That flex-ability is the paid skill.

What "proficient" looks like

A job-ready person can, unaided:

1. Separate **inputs (assumptions)** from **calculations** from **outputs** on distinct tabs, with a consistent colour convention.
2. Build a **revenue driver tree** — not one growth %, but the 2–4 levers that actually move revenue.
3. Apply **monthly seasonality** using a seasonality index (12 factors that average to 1.0).
4. Build a **12-month budget** that footsends (rows and columns cross-check to the same total).

5. Convert that budget into a **rolling forecast**: actuals replace forecast as months close, and a new month is added at the far end.
6. Build a **budget-vs-actual variance** view with % variance and a favourable/adverse flag.
7. Never hardcode a number inside a formula; every input traces to the assumptions tab.

Hands-on: how to actually do it

1. Driver-based revenue build

Say revenue = `Units × Price`. Put units and price on the assumptions tab, reference them in the calc.

For a subscription/SaaS or services build, use a **customer roll-forward**:

```
Opening customers  + New adds  − Churn  = Closing customers
Revenue = Avg customers in month × ARPU
```

Excel (row per month, column B = assumptions):

```
' Closing customers this month
=Opening + New_Adds - (Opening * Churn_Rate)

' New adds from marketing spend
=Marketing_Spend / CAC

' Revenue
=AVERAGE(Opening, Closing) * ARPU
```

For an e-commerce / retail build:

```
Revenue = Sessions * Conversion_Rate * Average_Order_Value
=B$4 * B$5 * B$6
```

Lock the assumption rows with `$` on the row (`B$4`) so you can drag across all 12 months.

2. Seasonality with an index

Build 12 seasonality factors that average to exactly 1.0. In India, festive months (Oct–Nov, Diwali) and year-end (Mar) usually spike.

```
' Seasonality index row (Jan..Dec), must average 1
Jan 0.85  Feb 0.80  Mar 1.20  … Oct 1.25  Nov 1.30  …
' Check it averages 1.0:
=AVERAGE(B10:M10)   → should return 1.000

' Seasonalised monthly revenue
=Annual_Revenue/12 * Seasonality_Factor
```

Derive the index from history:

```
Factor_for_month = Month_actual / AVERAGE(all 12 months' actual)
```

3. Growth over time (driver, not hardcode)

```
' This month's units = last month's units grown at monthly rate
=C_prev_units * (1 + MoM_growth)

' Convert an annual growth target to monthly:
=(1 + Annual_Growth)^(1/12) - 1
```

4. Rolling forecast switch — actual vs forecast

Add an "Actuals through" date on the assumptions tab. Each month cell picks actual if the month has closed, else forecast:

```
=IF(EOMONTH(MonthDate,0) ≤ Actuals_Through_Date, Actual_Value, Forecast_Value)
```

5. Budget-vs-actual variance

```
Variance      =Actual - Budget
Variance %    =IFERROR(Actual/Budget - 1, "")
Flag          =IF(Metric_Type="Cost", IF(Actual>Budget,"Adverse","Favourable"),
                                      IF(Actual<Budget,"Adverse","Favourable"))
```

6. Python — quick statistical baseline (sanity check your driver model)

```
import pandas as pd
# monthly revenue history
s = df['revenue']
# 3-month moving average forecast
ma = s.rolling(3).mean().iloc[-1]
# seasonal-naive: same month last year
seasonal = s.shift(12).iloc[-1]
print(ma, seasonal)
```

For a proper seasonal forecast:

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing
model = ExponentialSmoothing(s, trend='add', seasonal='mul',
                             seasonal_periods=12).fit()
forecast = model.forecast(12)
```

Use this to *challenge* the business build, not replace it.

Worked example / mini-project

"D2C skincare brand — FY27 budget." Reproduce this in a fresh workbook.

Assumptions tab:

Driver	Value
Monthly sessions (Apr)	4,00,000
Session MoM growth	3%
Conversion rate	2.2%
Average order value	₹950
Gross margin	62%
Marketing (% of revenue)	18%
Fixed opex / month	₹35,00,000

Seasonality index (Apr...Mar): 0.95, 0.90, 0.90, 1.00, 1.05, 1.10, **1.30, 1.25**, 1.05, 0.95, 0.90, 1.15 (avg = 1.0). The Oct–Nov spike is Diwali.

Monthly revenue formula:

```
Revenue = Sessions_this_month * Conv * AOV * Seasonality_Factor
```

Sample first quarter (rounded):

Metric	Apr	May	Jun
Sessions	4,00,000	4,12,000	4,24,360
Base revenue (₹)	83,60,000	86,10,800	88,69,124
× Seasonality	0.95	0.90	0.90
Net revenue (₹)	**79,42,000**	**77,49,720**	**79,82,212**
COGS @ 38%	30,17,960	29,44,894	30,33,241
Gross profit (₹)	49,24,040	48,04,826	49,48,971
Marketing @ 18%	14,29,560	13,94,950	14,36,798
Fixed opex	35,00,000	35,00,000	35,00,000
EBITDA (₹)	**-5,52,120**	**-1,90,124**	**12,173**

Now add the **rolling forecast layer**: set `Actuals_Through_Date = 31-May-26`. Suppose actual May revenue came in at ₹82,00,000 (beat). Replace May's forecast with actual via the `IF(EOMONTH ... )` switch, then re-forecast Jun–Mar off the *new* higher base. Board question you can now answer instantly: "*May beat by ₹4.5L — does that flow through the year?*" Yes, because Jun sessions grow off May.

Variance pack for May:

Line	Budget	Actual	Var	Var %	Flag
Revenue	77,49,720	82,00,000	4,50,280	+5.8%	Favourable
Marketing	13,94,950	15,10,000	1,15,050	+8.2%	Adverse

How it's tested

Interview questions:

- "How would you forecast revenue for a business you know nothing about?" → answer with a driver tree, not a growth %.
- "Difference between a budget and a rolling forecast?" → budget is fixed annual target; rolling forecast is continuously re-cut (e.g. always 12 months ahead).
- "Your forecast beat in Q1 — how do you decide whether to raise the full-year number?" → tests whether you understand run-rate vs one-off.
- "What's a seasonality index and how do you build one?"

Practical / take-home tests companies actually give:

- A **timed 60–90 min Excel case**: raw monthly history in one tab, "build a 12-month driver-based forecast with seasonality and a BvA view." Graded on structure (assumptions separated), no hardcodes, and whether it *flexes* when they change an input live.
- A **"break my model"** screen: they hand you a model, change one assumption, and check your cells all update (catches hardcoders).
- FP&A rounds often include a **live case**: "walk me through your model and defend the marketing-as-%-of-revenue assumption."

Common mistakes & how pros avoid them

Mistake	Pro fix
Hardcoding numbers inside formulas	Every input on assumptions tab; calc cells are pure formulas. Blue = input, black = formula
One growth % for all revenue	Break into 2–4 real drivers
Seasonality factors that don't average to 1.0	Add a `=AVERAGE()` check cell that must read 1.000
Forecast doesn't foot	Cross-check: sum of months = annual; sum of segments = total
Rebuilding the whole model each month	Build the `IF(EOMONTH≤Actuals_Through)` switch once
No scenario capability	Keep a single "case" toggle driving assumptions via `CHOOSE / INDEX`
Circular references from interest-on-cash	Use a circularity switch or iterative calc deliberately, not by accident
Over-precision (forecasting to the rupee)	Forecast drivers, round outputs; defend ranges not points

Learn-it roadmap & resources

Time to proficiency: ~4–6 weeks part-time if you already know Excel basics.

- **Week 1:** Model structure — separate tabs, colour convention, no hardcodes. Rebuild the worked example above.
- **Week 2:** Driver trees for 3 business types (SaaS, retail, services). Build each from scratch.
- **Week 3:** Seasonality index from real history; MoM vs annual growth conversion.
- **Week 4:** Rolling forecast switch + BvA variance pack.
- **Weeks 5–6:** Add scenario toggle, connect to a simple three-statement model, do a timed 90-min case against a clock.

Resources:

- *Financial Modeling* — CFI (Corporate Finance Institute) FMVA track (paid, well-recognised in India).
- Wall Street Prep / Breaking Into Wall Street modeling courses (paid).
- Free: Aswath Damodaran's forecasting lectures (YouTube); Microsoft's own XLOOKUP/dynamic-array docs.
- Certification: **FMVA** (CFI) is the most portable for FP&A roles; for pure Excel, the **Microsoft Excel Expert (MO-201)** cert signals hard skills.

Quick-reference

```
' Driver revenue (lock assumption row)
=Sessions * B$5_Conv * B$6_AOV * B$10_Seasonality

' Annual growth → monthly growth
=(1+Annual_Growth)^(1/12)-1

' Grow off prior month
=Prev * (1 + MoM_Growth)

' Customer roll-forward
Closing = Opening + Adds − Opening*Churn

' Seasonality check (must = 1.000)
=AVERAGE(seasonality_range)

' Rolling forecast switch
=IF(EOMONTH(MonthDate,0)≤Actuals_Through, Actual, Forecast)

' Variance & flag
Var    =Actual-Budget
Var%   =IFERROR(Actual/Budget-1,"")
Flag   =IF(Actual>Budget,"Adverse","Favourable")   ' for costs
```

Concept	One-liner
Budget	Fixed annual target, locked at start of year
Rolling forecast	Re-cut monthly, always N months ahead
Driver-based	Revenue = drivers multiplied, not a single %
Seasonality index	12 factors averaging 1.0 applied to base
BvA	Budget vs Actual, with variance % and F/A flag
Colour code	Blue = input, black = formula, green = link

LBO & Returns Model Basics

What it is & where it's used

A Leveraged Buyout (LBO) is the acquisition of a company funded mostly with **debt** and a slice of **equity**. A private-equity (PE) fund buys a business, uses the target's own cash flows to repay that debt over 4-6 years, and sells at exit. Because debt is paid down and (hopefully) the business grows, the equity slice multiplies in value. The LBO model is the spreadsheet that proves the deal makes money — it links **Sources & Uses**, a **debt schedule with a cash sweep**, and an **equity returns** block (IRR and MOIC).

Who builds these: - **PE / buyout funds** (Blackstone, KKR, in India: ChrysCapital, Kedaara, Multiples, True North) — every deal is an LBO model. - **Investment banking** (Leveraged Finance, Sponsors coverage, M&A) — banks size the debt and stress-test it. - **Credit funds / NBFC structured finance** — lend into the deal, model the debt from the lender's side. - **Corp-dev / strategy** teams doing acquisitions with acquisition financing. - **Equity research / valuation** — an LBO gives a "floor" valuation (what a financial buyer would pay).

It is the single most common technical case in PE and LevFin interviews.

The gap: why companies want this (and college didn't teach it)

MBA finance teaches NPV, WACC, and CAPM — a *cost of capital* mindset. LBO is a *cash-and-debt-paydown* mindset. The gap is specific:

- College values a firm with one discount rate. PE cares about **how the capital structure changes year by year** as debt amortizes.
- College treats leverage as a WACC input. In an LBO, leverage is the **engine of returns** — you must build the debt schedule that mechanically sweeps free cash into repayment.
- Textbooks skip the **circularity**: interest depends on debt balance, which depends on cash sweep, which depends on cash-after-interest. Real models handle this with iterative calc or an interest-on-opening-balance shortcut.
- Nobody teaches **Sources & Uses** or how a returns bridge decomposes IRR into deleveraging vs. EBITDA growth vs. multiple expansion.

Employers pay for someone who can build the whole thing in 45 minutes without a template.

What "proficient" looks like

A job-ready person can, unaided:

1. Build a **Sources & Uses** table that balances to the penny.
2. Compute entry Enterprise Value from an **EV/EBITDA** multiple and back into equity cheque.
3. Build a **debt schedule** for a Term Loan (mandatory amortization + cash sweep) and a revolver.
4. Project a simple **FCF-before-debt** line and route it through the sweep.
5. Compute **exit equity value**, then **MOIC** (= exit equity / entry equity) and **IRR** using `=IRR()` or `=XIRR()`.
6. Explain a **returns bridge**: how much of the gain came from paying down debt vs. growing EBITDA vs. selling at a higher multiple.

7. Run sensitivities (entry multiple, exit multiple, leverage) with a **Data Table**.

Hands-on: how to actually do it

Step 1 — Sources & Uses

Uses = what you pay for. Sources = how you fund it. They must equal.

Sources	Rs cr	Uses	Rs cr
Term Loan (Debt)	= `D_ebitda*4.0`	Purchase Enterprise Value	= `EntryEBITDA*EntryMult`
Sponsor Equity (plug)	=Total Uses − Debt	Transaction fees (2%)	= `EV*0.02`
Total Sources	=SUM	**Total Uses**	=SUM

Excel — say entry EBITDA in `B3`, entry multiple in `B4`, leverage (Debt/EBITDA) in `B5`:

```
EV              =B3*B4
Debt            =B3*B5
Fees            =EV*0.02
Total Uses      =EV+Fees
Equity (plug)   =Total_Uses - Debt      ' the sponsor cheque
```

Step 2 — Project EBITDA and FCF

```
Year                  1        2        3        4        5
EBITDA          =prior*(1+g)  ...
Less: D&A       =-EBITDA*0.06
EBIT            =EBITDA+D&A
Less: Interest  =-Opening_Debt*rate      ' link to debt schedule
EBT             =EBIT+Interest
Less: Tax@25%   =-MAX(EBT,0)*0.25
Net Income      =EBT+Tax
Add: D&A        =-D&A line
Less: Capex     =-EBITDA*0.05
Less: ΔNWC      =-Revenue_change*0.10
FCF (pre-sweep) =NI + D&A + Capex + ΔNWC
```

Step 3 — Debt schedule with cash sweep

The sweep: after mandatory amortization, any leftover cash **prepays** the term loan.

```
Opening Debt          =prior Closing Debt
Mandatory amort (5%)  =-MIN(Opening, Original_TL*0.05)
Cash avail for sweep  =FCF_pre_sweep + Mandatory_amort   ' amort already negative
Optional sweep        =-MIN(MAX(Cash_avail,0), Opening+Mandatory_amort)
Closing Debt          =Opening + Mandatory + Optional_sweep
Interest expense      =Opening_Debt * rate               ' opening-balance avoids circularity
```

Using **opening balance** for interest breaks the circular reference — the standard interview shortcut. (Production models use average balance + enable iterative calc: File > Options > Formulas > Enable iterative calculation, max 100.)

Step 4 — Exit and returns

```
Exit EBITDA       = Year-5 EBITDA
Exit EV           = Exit_EBITDA * Exit_Multiple
Exit Net Debt     = Year-5 Closing Debt - Cash
Exit Equity       = Exit_EV - Exit_Net_Debt
MOIC              = Exit_Equity / Entry_Equity
IRR               = (MOIC)^(1/Years) - 1        ' single in/out shortcut
```

In Excel with actual dated cash flows (equity out at entry, in at exit):

```
=IRR({-Entry_Equity, 0, 0, 0, 0, Exit_Equity})
=XIRR(values_range, dates_range)     ' when dates are irregular
=MOIC^(1/5)-1                          ' quick CAGR check
```

Worked example / mini-project

Target: an Indian mid-market auto-components firm. Entry EBITDA Rs 100 cr, entry multiple 8.0x, EBITDA grows 8%/yr, hold 5 years, exit at 8.0x (flat — conservative). Leverage 4.0x. Interest 11%. Tax 25%. D&A 6% of EBITDA, Capex 5%, ΔNWC 10% of revenue growth (assume EBITDA≈EBIT proxy for simplicity, revenue ≈ 2x EBITDA).

Sources & Uses (Rs cr):

Uses	Rs cr	Sources	Rs cr
Entry EV (100 × 8.0)	800	Term Loan (4.0x)	400
Fees (2%)	16	Sponsor Equity	416
Total	**816**	**Total**	**816**

Debt paydown (opening-balance interest, 5% mandatory amort, full sweep):

Year	EBITDA	Interest@11%	Tax@25% approx	FCF pre-sweep	Opening Debt	Closing Debt
1	108	44.0	~13	~58	400	342
2	116.6	37.6	~16	~66	342	276
3	126.0	30.4	~19	~72	276	204
4	136.0	22.4	~22	~80	204	124
5	146.9	13.6	~26	~88	124	36

(FCF ≈ EBITDA − interest − tax − capex − ΔNWC; numbers rounded to show the mechanic, not to the last rupee.)

Exit:

```
Exit EBITDA      = 146.9
Exit EV          = 146.9 × 8.0 = 1,175
Exit Net Debt    = 36
Exit Equity      = 1,175 − 36 = 1,139
Entry Equity     = 416
MOIC             = 1,139 / 416 = 2.74x
IRR              = 2.74^(1/5) − 1 = 22.3%
```

Returns bridge — where did the 2.74x come from? With a *flat* multiple, every rupee came from **EBITDA growth** (100 → 147) and **deleveraging** (net debt 400 → 36). Zero from multiple expansion. That is a *clean, defensible* deal — you don't need to sell higher than you bought.

Reproduce it: put assumptions in a block, build the 5-year strip, and add a two-way **Data Table** (Data > What-If Analysis > Data Table) with exit multiple across the top (7.0x–9.0x) and leverage down the side (3.0x–5.0x) referencing the IRR cell. You'll see IRR swing roughly 15%–30%.

How it's tested

Interview questions: - "Walk me through an LBO." (Sources & Uses → buy → pay down debt with FCF → sell → equity multiplies.) - "What are the 3 drivers of LBO returns?" (Deleveraging, EBITDA growth, multiple expansion.) - "If I double the leverage, what happens to IRR?" (Higher — smaller equity cheque — but higher risk / interest burden.) - "Why use opening-balance interest?" (Avoids circular reference in a timed test.) - "MOIC of 2.5x over 5 years ≈ what IRR?" (~20%; know 2x/5yr≈15%, 3x/5yr≈25%.)

Practical tests companies give: - **Paper LBO** (no computer, 5 min): entry 5x, exit 5x, given growth and paydown — compute IRR mentally. Extremely common at PE first rounds (ChrysCapital, Everstone, bulge-bracket LevFin). - **Timed Excel LBO** (30-60 min): build the full model from a one-page CIM summary. Judged on: does S&U balance, is the debt schedule right, no hardcodes, clean IRR. - **Take-home case**: full model + 2-slide recommendation over a weekend.

Common mistakes & how pros avoid them

Mistake	Fix
Sources ≠ Uses	Always make **equity the plug** = Total Uses − Debt. Never hardcode equity.
Forgetting **fees** in Uses	Add transaction + financing fees; they raise the equity cheque.
Circular-reference `#REF` spiral	Use **opening-balance** interest, or enable iterative calc deliberately.
Sweeping debt **below zero**	Wrap sweep in `=MIN(cash, opening_balance)` so it can't over-repay.
Confusing EV and equity at exit	Exit **equity** = Exit EV − **net debt**. Subtract debt, add cash.
MOIC vs IRR confusion	MOIC ignores time; IRR is annualized. A 3x over 10 years (12% IRR) is worse than 2x over 3 years (26%).
Assuming multiple expansion	Model **flat exit multiple** as base case. Expansion is a bonus, never the thesis.
Ignoring a **cash flow sanity check**	FCF must stay positive; if interest > FCF the deal is over-levered.

Learn-it roadmap & resources

Time to proficiency: 2-3 weeks of focused practice to build one unaided; 6-8 weeks to be interview-fast (including paper LBOs).

Week	Focus
1	Sources & Uses + entry/exit mechanics. Build the S&U 10 times.
2	Debt schedule + cash sweep + circularity. This is the hard part.
3	Returns bridge, sensitivities, paper LBOs on a stopwatch.

Resources: - **Free:** Wall Street Prep / Macabacus free LBO tutorials; Aswath Damodaran (NYU) sessions on YouTube for the valuation base; Corporate Finance Institute free templates. - **Paid:** Breaking Into Wall Street (BIWS) "LBO Modeling" course; Wall Street Prep Premium; Rossum / Financial Edge for the India/UK market. - **Certification:** CFA (Level II corporate finance / equity), or FMVA (CFI) which includes an LBO module. For India PE roles, a strong self-built model beats any certificate. - **Practice:** rebuild any listed mid-cap as an LBO — pull EBITDA from screener.in, assume 4x leverage, model 5 years.

Quick-reference

```
EV              = EBITDA × Entry Multiple
Equity (entry)  = Total Uses − Debt          [equity is the plug]
Total Uses      = EV + Fees
Interest        = Opening Debt × rate         [avoids circularity]
Sweep           = MIN(FCF_after_mandatory, Opening Debt)
Closing Debt    = Opening − Mandatory − Sweep
Exit Equity     = Exit EBITDA × Exit Mult − Net Debt
MOIC            = Exit Equity / Entry Equity
IRR             = MOIC^(1/years) − 1          [single in/out]
```

Quick IRR ↔ MOIC (5-year hold)	MOIC	IRR
	1.5x	~8%
	2.0x	~15%
	2.5x	~20%
	3.0x	~25%
	4.0x	~32%

Three return drivers: (1) Deleveraging — debt paid by FCF. (2) EBITDA growth — organic + margin. (3) Multiple expansion — sell higher than bought (model flat; treat as upside).

Key Excel: `=IRR()`, `=XIRR(values,dates)`, `=MIN()` / `=MAX()` for the sweep, Data > What-If > Data Table for sensitivities, File > Options > Formulas > Enable iterative calculation for average-balance interest.

Model best-practice & error-proofing

What it is & where it's used

A financial model is code written in Excel. Like code, it can be readable and auditable — or a tangle nobody dares touch. "Model best-practice" is the set of conventions professionals use so that any reviewer can open your workbook, understand which cells are assumptions, trace every number to its source, and trust the totals. "Error-proofing" is the layer of self-checks that scream when the model breaks.

This matters in every role that hands a spreadsheet to someone senior: FP&A analysts building the annual budget, investment banking / PE associates running LBO and DCF models, corporate finance teams building three-statement models, credit analysts sizing debt, and even accounts/tax teams preparing schedules that feed the return. In India, a Big 4 valuation team, a startup's finance team building an investor model, or a treasury desk reconciling cash — all of them will judge you on whether your model is disciplined, not just whether the answer is "right" today.

The skill is invisible when done well and painfully visible when skipped. A model that balances by luck is worthless; a model whose check row flashes red the moment an input is fat-fingered is worth a promotion.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you *what* a DCF is: project free cash flows, discount at WACC, sum to enterprise value. It graded you on the answer. It never graded you on whether the cell was a formula or a typed-in number, whether a stranger could audit it, or whether it would survive being reused next quarter.

The industry reality is the opposite. Nobody builds a model once. It gets reopened, forwarded, stress-tested in a live meeting, and inherited by the next analyst when you move on. The single most common reason a deal team distrusts a junior's work is not a wrong assumption — it's a hardcoded number buried inside a formula (`=E10*1.08` where did 1.08 come from?), a broken link, or a balance sheet that doesn't balance and no check to catch it.

Colleges teach models as disposable homework. Employers treat them as durable, shared infrastructure. That gap — from "get the answer" to "build something another human can trust, audit, and extend" — is exactly what this chapter closes, and it is disproportionately rewarded because so few freshers have it.

What "proficient" looks like

A job-ready person, handed a blank workbook, will unaided:

- **Colour-code** every cell by type without being asked: blue = hardcoded input, black = formula, green = links from other sheets, red = external workbook link.
- Keep **one formula per row** — a formula you can write in the leftmost period cell and drag right, unchanged, across all columns.
- Build **check rows** (balance sheet balances, sources = uses, sum-of-parts = total) that return `TRUE` / `OK` or a difference of zero, plus a master "error flag" cell at the top.
- Have **zero hardcodes inside formulas** — every driver sits in a clearly labelled input cell.
- Separate the model into **Inputs → Calculations → Outputs** so no sheet mixes assumptions with logic.
- Document assumptions, sources, and version so a reviewer needs no verbal walkthrough.

The bar is behavioural: could someone who has never spoken to you audit this in 20 minutes? If yes, you're proficient.

Hands-on: how to actually do it

1. Formatting conventions (the universal colour code). Apply font colours by cell type. This is muscle memory on every real desk:

Cell type	Font colour	Example
Hardcoded input / assumption	Blue (`RGB 0,0,255`)	Revenue growth `12%`
Formula / calculation	Black	`=D10*(1+D11)`
Link from another sheet (same file)	Green	`='P&L'!D25`
Link to another workbook	Red	dangerous — flag it
Hardcode that *must* stay in a formula	Highlight cell yellow	one-off overrides

Set blue fast: select input cells → `Ctrl+1` → Font → colour blue. Or record it to a shortcut. Reviewers scan for blue first — those are the only cells they need to challenge.

2. One formula per row. Never let column D's formula differ from column E's. Use absolute/relative references so a single dragged formula works across all periods:

```
' Row 12 = Revenue, Row 13 = growth %, dragged D→H unchanged:
=C12*(1+D13)
' C12 is prior period (relative), D13 is this period's growth input.
' NEVER: =C12*1.12 in D, =D12*1.10 in E  ← different formulas, unauditable
```

Anchor shared drivers with `$` : a WACC in `$B$4` referenced as `=D20/(1+$B$4)^D19` .

3. Kill hardcodes. Replace `=Sales*0.18` (GST buried in a formula) with an input cell:

```
' Bad:  =E30*0.18
' Good: E31 label "GST rate", B5 = 18%, formula =E30*$B$5
```

Audit an existing model for hidden constants — this flags any formula containing a raw number:

```
=IF(SUMPRODUCT(--ISNUMBER(--MID(FORMULATEXT(E30),ROW($1:$99),1)))>0,"CHECK","")
```

Simpler in practice: `Ctrl+~` toggles "show formulas" so you can eyeball the grid for stray digits, or use **Formulas** → **Trace Precedents** to see if a cell points nowhere.

4. Check rows. Add a dedicated check block. Core three-statement check — balance sheet must balance:

```
' Row 60: =Total Assets - (Total Liabilities + Total Equity)
=B45-(B52+B58)          ' should be 0 every period
```

Then a robust boolean using a tolerance (floating point can leave ₹0.0001):

```
=IF(ABS(B60)<0.5,"OK","ERROR")
```

Master flag at the top of every sheet (cell A1 area), aggregating all checks:

```
=IF(COUNTIF(Checks!$B$2:$B$40,"ERROR")>0,"⚠ MODEL BROKEN","✓ All checks pass")
```

Make errors impossible to miss with **Conditional Formatting**: select check row → Home → Conditional Formatting → New Rule → "Format cells that contain" → cell value not equal to 0 → fill red.

5. Stop broken inputs at the door with Data Validation. Select input cells → Data → Data Validation → Decimal, between 0 and 1 (for a %), with an input message. Growth of 1200% because you typed 12 instead of 0.12 gets rejected on entry.

6. Documentation. Every model gets a **Cover / Assumptions** sheet:

Field	Entry
Model name	FY26 Operating Budget
Author / owner	A. Sharma
Last updated	03-Jul-2026
Version	v2.3
Key assumptions	Rev growth 12%, GST 18%, WACC 13%
Sources	FY25 audited financials; RBI repo 6.5%
Colour key	Blue=input, Black=formula, Green=link

Add cell comments (`Shift+F2`) on any non-obvious driver, citing the source.

7. Version control. Excel isn't Git, but discipline substitutes. Save as `Budget_FY26_v2.3_2026-07-03.xlsx` — version + date in the filename, never "final_FINAL_v2". Keep a **Changelog** tab: date | version | who | what changed. For teams, store on SharePoint/OneDrive/Google Drive so version history is automatic and only one person edits at a time. For serious model shops, tools like Macabacus or Operis reconcile two file versions cell-by-cell.

Worked example / mini-project

Build a 3-year revenue-to-PAT projection for a Bengaluru SaaS firm and error-proof it.

Inputs sheet (all blue):

Driver	B (input)
FY25 Revenue (₹ Cr)	40
Revenue growth	25%
Gross margin	70%
Opex growth	18%
FY25 Opex (₹ Cr)	22
Tax rate	25%

Calc sheet — one formula per row, dragged across FY26–FY28:

```
Revenue (row 2):   =Inputs!$B$1*(1+Inputs!$B$2)     ' FY26; FY27 = C2*(1+Inputs!$B$2)
Gross profit (3):  =C2*Inputs!$B$3
Opex (4):          =Inputs!$B$5*(1+Inputs!$B$4)     ' FY26; then prior*(1+growth)
EBIT (5):          =C3-C4
Tax (6):           =C5*Inputs!$B$6
PAT (7):           =C5-C6
```

FY26 numbers: Revenue = 40 × 1.25 = ₹50 Cr; Gross profit = ₹35 Cr; Opex = 22 × 1.18 = ₹25.96 Cr; EBIT = ₹9.04 Cr; Tax = ₹2.26 Cr; **PAT = ₹6.78 Cr**.

Check sheet:

```
Margin sanity:   =IF(AND(Calc!C7<Calc!C5, Calc!C3>0),"OK","ERROR")  ' PAT < EBIT, GP positive
Recompute check: =IF(ABS(Calc!C5-(Calc!C3-Calc!C4))<0.01,"OK","ERROR")
Master flag:     =IF(COUNTIF(B2:B5,"ERROR")>0,"⚠ BROKEN","✓ PASS")
```

Now fat-finger growth to `250%` in the input cell — Revenue explodes to ₹140 Cr, but your Data Validation (0–1) blocks it, or the margin-sanity check stays OK while an eyeball tells you it's wrong. Change tax rate to `1.25` (typo) and the recompute check still passes but PAT goes negative — the "PAT < EBIT" check catches the sign flip. That's error-proofing earning its keep.

How it's tested

Practical assessment (the real filter): Firms give a timed 45–90 minute Excel test — often "build a 3-statement model from these raw financials" or "here's a broken model, find and fix the errors." They are watching for: colour-coding, no hardcodes, a balancing check row, and one-formula-per-row consistency. Many PE/IB shops hand you a deliberately broken model and time how fast you find the plug (a hardcoded cell breaking the balance).

Interview questions you should be able to answer cold:

- "How do you make sure a balance sheet balances in your model?" → check row = Assets − (Liab + Equity), tolerance-based OK/ERROR, master flag.
- "What's the difference between a blue and a black cell?" → input vs formula.

- "How do you avoid hardcoding?" → every driver in a labelled input cell; show formulas with `Ctrl+~` to audit.
- "You inherit a model — how do you audit it in 20 minutes?" → cover sheet, colour key, Trace Precedents/Dependents, check rows, Formula Auditing.
- "How do you version-control an Excel model?" → filename convention, changelog tab, SharePoint history, single editor.

Common mistakes & how pros avoid them

Mistake	Why it bites	Pro habit
Hardcode inside a formula (`=E10*1.08`)	Nobody can find/change the driver	All drivers in blue input cells
Different formula per column	One period breaks silently	One formula, dragged across
No check rows	Model balances by luck, then doesn't	Balance + sanity checks + master flag
`=A1=B1` exact-equality check	Floating point leaves ₹0.0001	`ABS(diff)<tolerance`
Mixing inputs, calcs, outputs on one sheet	Impossible to audit	Inputs → Calcs → Outputs separation
External workbook links	Break on the recipient's machine	Paste-special values or keep it self-contained; colour red
`final_v2_FINAL(3).xlsx`	Nobody knows the live version	`Name_vX.Y_YYYY-MM-DD` + changelog
No documentation	Every review needs a verbal walkthrough	Cover sheet + cell comments citing sources

Pros also press **F9 on a selected formula fragment** to evaluate just that piece, use **Evaluate Formula** (Formulas ribbon) to step through logic, and turn on **Error Checking** to catch inconsistent formulas (the little green triangles).

Learn-it roadmap & resources

Realistic time-to-proficiency: **3–4 weeks** of deliberate practice if you already know Excel formulas — this is discipline, not new functions.

- **Week 1:** Rebuild any past model applying the colour code + one-formula-per-row. Painful, permanent.
- **Week 2:** Add check rows and master flags to three-statement models. Learn Trace Precedents/Dependents, Evaluate Formula, `Ctrl+~`.
- **Week 3–4:** Do a timed "fix the broken model" drill; build a cover/changelog sheet as a template you reuse forever.

Resources: - **FAST Standard** (fast-standard.org) — free, the canonical model-building rulebook (Flexible, Appropriate, Structured, Transparent). - **CFI's** Financial Modeling & Valuation Analyst (FMVA) — paid, strong on formatting conventions and best-practice. - **Breaking Into Wall Street / Wall Street Prep** — paid, industry-standard modeling courses used by Indian IB/PE hires. - **Macabacus** blog + Excel add-in (formatting shortcuts, model auditing) — free content, paid tool. - For CA/finance India roles, the **ICAI Excel and modeling** self-study material plus any three-statement template you rebuild by hand.

Quick-reference

Convention / tool	What to do
Blue font	Every hardcoded input / assumption
Black font	Formulas
Green font	Links from another sheet
Red font	External-workbook links (avoid)
One formula per row	Write once in first period cell, drag across
Anchor drivers	`$B$4` absolute for shared assumptions
Balance check	`=Assets-(Liab+Equity)` should be 0
Tolerance check	`=IF(ABS(diff)<0.5,"OK","ERROR")`
Master flag	`=IF(COUNTIF(checks,"ERROR")>0,"⚠ BROKEN","✓ PASS")`
Find hardcodes	`Ctrl+~` show formulas; Trace Precedents
Evaluate a fragment	Select in formula bar → `F9`
Guard inputs	Data → Data Validation (range limits)
Highlight errors	Conditional Formatting → red on ≠ 0
Cell comment	`Shift+F2` — cite the source
File naming	`Name_vX.Y_YYYY-MM-DD.xlsx` + changelog tab
Structure	Inputs → Calculations → Outputs (separate sheets)

Excel + AI (Copilot) and Automation

What it is & where it's used

This chapter is about making Excel do the boring 80% *for* you — so you spend your day on judgment, not grunt work. Three tools stack together:

1. **Copilot in Excel** — Microsoft's built-in AI. You type an instruction in plain English ("highlight rows where GST paid is above ₹50,000 and add a variance column") and it writes the formula, formats the range, or drafts an analysis.
2. **Macros / VBA** — recorded or hand-written scripts that replay clicks and keystrokes. This is how you turn a 40-minute month-end formatting ritual into one button press.
3. **LAMBDA** — Excel's native way to build your *own* reusable functions with no code, so a messy nested formula becomes `=NETSALARY(basic, hra)`.

Who pays for this: FP&A analysts (monthly MIS packs), audit/statutory teams (repetitive workpapers), tax associates (GST reconciliation, TDS working), accounts-payable teams (invoice cleanup), and anyone who produces the *same* report every month. In an India finance job, "the person who automated the MIS" gets noticed fast — because everyone else is still doing it by hand at 9 PM.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you *what* a variance analysis means. It never taught you that in a real company you'll rebuild that variance report **12 times a year from a raw SAP/Tally dump that arrives in a slightly different shape each time**. The gap is *repetition at scale*.

College treats Excel as a calculator you use once for an assignment. Employers treat Excel as a **production line** — the same file, refreshed monthly, that six people depend on. Nobody taught you:

- That a recorded macro can eliminate the 30 clicks you do every single close.
- That AI can now write your `SUMIFS` faster than you can remember the argument order.
- That copy-pasting a giant nested formula into 200 cells is a *liability* — one edit and it silently breaks — whereas a named LAMBDA is auditable.

Companies want people who **remove work**, not people who heroically absorb it. That mindset shift is the real gap.

What "proficient" looks like

A job-ready person can, unaided:

- Use Copilot to generate, explain, and *sanity-check* a formula — and know when the AI is wrong.
- Record a macro for a repetitive formatting/cleanup task, then **open the VBA editor and edit it** (change a range, add a loop, wire it to a button).
- Judge **when to automate vs. when not to** (the "will I do this 3+ times?" test).
- Write a basic LAMBDA and save it as a Named Range so the whole team can use `=GSTSPLIT(...)`.
- Explain the risk: macros can't be undone with Ctrl+Z, so you always work on a copy.

The bar is *not* "expert VBA developer." It's "removes their own repetitive work and can read a macro someone else wrote."

Hands-on: how to actually do it

1. Copilot in Excel

Copilot lives on the **Home tab (rightmost)** in Microsoft 365. Your data must be a proper Table (`Ctrl+T`). Then click Copilot and type instructions:

```
Add a column called "GST Rate" that divides Tax Amount by Taxable Value, formatted as a percen
```

```
Show me total Taxable Value by State, sorted highest to lowest.
```

```
Highlight in red every row where Invoice Date is after the GST return due date.
```

Copilot returns the formula and offers to insert it. **Critical habit:** ask it to explain, then verify one row by hand. Example — you asked for a vendor-wise total and Copilot gives:

```
=SUMIFS(Invoices[Amount], Invoices[Vendor], [@Vendor], Invoices[FY], "2025-26")
```

Good prompt patterns that work reliably: - "Write an Excel formula to..." (it returns copy-usable syntax) - "Explain what this formula does: `=XLOOKUP(...)` " - "What's wrong with this formula? It returns #N/A"

If you don't have Copilot licensed, the **free fallback** is ChatGPT/Gemini/Claude in a browser: paste your column headers and describe the goal — the formula logic is identical.

2. Record and edit a macro

Turn on the Developer tab: **File → Options → Customize Ribbon → tick Developer.**

Record: **Developer → Record Macro → name it `FormatMIS` → do your steps → Stop Recording.**

Say you recorded formatting a report. Open **Developer → Visual Basic (Alt+F11)** to see and edit it:

```
Sub FormatMIS()
    ' Recorded: format the month-end MIS sheet
    Columns("A:F").AutoFit
    Range("A1:F1").Font.Bold = True
    Range("A1:F1").Interior.Color = RGB(0, 51, 102)
    Range("A1:F1").Font.Color = RGB(255, 255, 255)
    Columns("C:E").NumberFormat = "#,##0"    ' Indian thousands
    ActiveWindow.FreezePanes = True
End Sub
```

Now *edit* it — add a loop the recorder could never write, e.g. delete blank rows in a Tally export:

```
Sub CleanTallyDump()
    Dim i As Long, lastRow As Long
    lastRow = Cells(Rows.Count, "A").End(xlUp).Row
    ' Loop bottom-up so deleting a row doesn't skip the next one
    For i = lastRow To 2 Step -1
        If Trim(Cells(i, "A").Value) = "" Then
            Rows(i).Delete
        End If
    Next i
    MsgBox "Cleanup done. Rows now: " & Cells(Rows.Count, "A").End(xlUp).Row
End Sub
```

Wire it to a button: **Developer → Insert → Button (Form Control) → assign `CleanTallyDump`.** Save the file as **.xlsm** (macro-enabled) or macros vanish.

3. When to automate (the decision rule)

Situation	Automate?
One-off, never repeats	No — just do it
Same task 3+ times, stable steps	Yes — record a macro
Complex logic reused across files	Yes — LAMBDA or VBA
Steps change every time / need judgment	No — keep it manual
Touches money and can't be checked	Automate, but add a reconciliation check

Rule of thumb: **if (time saved per run × times per year) > (time to build × 2), automate it.**

4. LAMBDA basics

LAMBDA lets you name a formula. Instead of retyping a net-salary calc everywhere:

```
=LAMBDA(basic, hra, da, basic + hra + da - (basic*0.12))(50000, 20000, 5000)
```

Make it reusable: **Formulas → Name Manager → New → Name:** `NETSALARY`, Refers to:

```
=LAMBDA(basic, hra, da, basic + hra + da - (basic*0.12))
```

Now anyone types `=NETSALARY(50000, 20000, 5000)` → returns **68,000** (after 12% PF on basic). Another practical one — split a GST-inclusive amount into base + tax:

```
// Name: GSTBASE  - extract taxable value from an 18% inclusive amount
=LAMBDA(inclusive, rate, inclusive / (1 + rate))
// Usage: =GSTBASE(11800, 0.18)  →  10,000
```

Pair with `MAP` to apply across a range without dragging:

```
=MAP(A2:A100, LAMBDA(x, x/(1+0.18)))
```

Worked example / mini-project

Goal: Automate a monthly GST Input Tax Credit (ITC) reconciliation — purchase register vs. GSTR-2B.

Data (Purchase Register, sheet `PR`):

Vendor GSTIN	Invoice No	Taxable (₹)	IGST (₹)
29ABCDE1234F1Z5	INV-101	100000	18000
27PQRSX5678L1Z2	INV-102	50000	9000
29ABCDE1234F1Z5	INV-103	75000	13500

GSTR-2B (sheet `2B`) has the same columns as downloaded from the portal.

Step 1 — match with a formula. In `PR`, add a "Matched in 2B" column:

```
=IF(SUMIFS('2B'[IGST], '2B'[Invoice No], [@[Invoice No]], '2B'[Vendor GSTIN], [@[Vendor GSTIN]
```

Step 2 — a reusable LAMBDA for eligible ITC (block credit if unmatched):

```
// Name: ELIGIBLEITC
=LAMBDA(igst, status, IF(status="Matched", igst, 0))
// Usage in a column: =ELIGIBLEITC([@IGST], [@[Matched in 2B]])
```

Step 3 — a macro to produce the mismatch report every month with one click:

```
Sub GSTReconReport()
    Dim ws As Worksheet
    Set ws = Sheets("PR")
    ws.AutoFilterMode = False
    ' Filter to only MISMATCH rows
    ws.Range("A1").AutoFilter Field:=5, Criteria1:="MISMATCH"
    ws.Range("A1").CurrentRegion.Copy
    Sheets.Add.Name = "Mismatch_" & Format(Date, "MMM-YY")
    ActiveSheet.Paste
    Application.CutCopyMode = False
    ws.AutoFilterMode = False
    MsgBox "Mismatch report ready. Follow up with vendors before filing GSTR-3B."
End Sub
```

Result: what used to be an afternoon of VLOOKUPs and eyeballing is now — refresh the two sheets, click the button, chase the mismatches. On this data, INV-103 unmatched → ₹13,500 ITC blocked until the vendor uploads it. That number is exactly what a reviewer wants surfaced *before* filing.

How it's tested

Interview questions: - "Walk me through the last thing you automated in Excel. How much time did it save?" - "A macro deleted the wrong rows and there's no undo. What do you do / how do you prevent it?" (Answer: always run on a copy; version the file.) - "When would you *not* automate something?" - "Difference between a recorded macro and VBA you write yourself?" - "What is a LAMBDA and why use it over a nested formula?" (Answer: reuse, single point of edit, readability, auditability.)

Practical tests companies give: - **Timed cleanup screen:** "Here's a messy 5,000-row Tally/SAP export. Clean it and produce a vendor-wise summary in 20 minutes." They watch whether you do it by hand or reach for a macro/pivot. - **"Automate this" case:** given a repetitive monthly report, record a macro live on a shared screen. - **Formula-from-English:** "Write the formula for X" — increasingly they *allow* Copilot/AI and instead test whether you can **verify** the output.

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Trusting Copilot/AI output blindly	Verify one row by hand; ask AI to explain its own formula
Running a macro on the only copy of live data	Always work on a duplicate file; macros ignore Ctrl+Z
Saving as .xlsx (macros silently lost)	Save as **.xlsm**
Hard-coding ranges (`A2:A500`) that break next month	Use Tables + structured references, or `End(xlUp)` to find last row
Automating a one-off	Apply the "3+ times" rule first
Giant nested formula copied 200 times	Convert to a named LAMBDA — one place to fix
Emailing .xlsm files (blocked by many gateways)	Zip it, or share via Teams/SharePoint
Deleting rows top-down in a loop (skips rows)	Loop **bottom-up** (`For i = lastRow To 2 Step -1`)

Learn-it roadmap & resources

Time to job-ready: 3–4 weeks, ~1 hour/day.

- **Week 1 — Copilot / AI formulas.** Practice prompting for `SUMIFS`, `XLOOKUP`, conditional formatting. If unlicensed, use free ChatGPT/Gemini. Goal: never get stuck on syntax again.
- **Week 2 — Record & run macros.** Automate your own recurring formatting task. Learn to save .xlsm and assign buttons.
- **Week 3 — Edit VBA.** Learn variables, `For` loops, `If`, `MsgBox`, `Cells` / `Range`, and `End(xlUp)`. Build the GST-recon macro above.
- **Week 4 — LAMBDA + MAP.** Convert your three most-used nested formulas into named functions.

Resources (mostly free): - Microsoft Learn — "Automate tasks with the Macro Recorder" and "LAMBDA function" docs. - ExcelIsFun and Leila Gharani (YouTube) — best free VBA and LAMBDA channels. - Chandoo.org — India-friendly Excel automation examples. - **Certification:** Microsoft Office Specialist (MOS): Excel Expert (MO-201) validates macros/LAMBDA on a CV; the free skills matter more than the badge in India.

Quick-reference

Task	How
Open Copilot	Home tab → Copilot (M365)
Turn on macros	File → Options → Customize Ribbon → Developer
Record macro	Developer → Record Macro
Open code editor	`Alt+F11`
Save with macros	Save as **.xlsm**
Find last row (VBA)	`Cells(Rows.Count,"A").End(xlUp).Row`
Loop that deletes rows	`For i = lastRow To 2 Step -1`
Message box	`MsgBox "text"`
Name a LAMBDA	Formulas → Name Manager → New
LAMBDA syntax	`=LAMBDA(a, b, a+b)(1, 2)`
Apply over a range	`=MAP(range, LAMBDA(x, x*1.18))`
GST base from inclusive	`=inclusive/(1+rate)`
Automate decision	(time saved/run × runs/yr) > (build time × 2)

Golden rules: verify AI output, work on a copy (no undo for macros), save as .xlsm, and only automate what you'll repeat 3+ times.

The interview Excel & modeling test

What it is & where it's used

The interview Excel/modeling test is a **timed, hands-on assessment** where you build or fix something live — no ChatGPT, no calling a friend, often no internet. It is the single most common filter between "sounds good on the CV" and an offer for roles in **investment banking, private equity, equity research, corporate FP&A, transaction advisory / valuations, credit analysis, and startup finance (fundraise/CFO-track)**.

Formats you will meet in India and globally:

Format	Duration	Where
Live shared-screen build (Zoom + Excel)	30–90 min	Most banks, PE, boutique advisory
Take-home model (emailed, deadline)	3–24 hrs	Startups, mid-market PE, research
"Fix my broken model" / audit	20–45 min	FP&A, controllership
Aptitude + Excel combo (SHL/Kenexa style)	45–60 min	BPO/KPO, Big 4 analytics, GCC roles

For an India-based candidate targeting Big 4 (Deloitte/PwC/EY/KPMG), GCCs (Wells Fargo, JPMC, Goldman Bengaluru/Hyderabad), or a domestic AMC/broker, expect the "fix + build" variant far more than a full LBO.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you what NPV *means*. It did not make you build a 3-statement model that **balances**, under time pressure, with a keyboard-only workflow, correct sign conventions, and a circularity toggle for interest-on-average-debt. CA Inter drilled accounting rules but on paper, not linked in Excel where the P&L flows to retained earnings which flows to the balance sheet.

The gap employers pay to close:

- **Speed under the keyboard** — pros navigate with `Alt` shortcuts, not the mouse. A mouse-user is spotted in 30 seconds.
- **Structure** — inputs (blue), formulas (black), links (green), clearly separated. College spreadsheets are one giant messy tab.
- **A model that ties** — Assets = Liabilities + Equity to the rupee, cash flow reconciles to the balance sheet cash line.
- **Judgment** — sensible assumptions with a stated rationale, not a random 10% growth.

What "proficient" looks like

An interviewer's mental checklist for a "job-ready" candidate:

1. Builds a clean **3-statement model** from a blank sheet in ~45 min, statements linked, BS balancing.
2. **Keyboard-first**: no mouse for navigation, formatting, or fill.
3. Uses **relative/absolute references** correctly (`$` discipline) and **named ranges** or structured refs where sensible.
4. Handles **circularity** (interest ↔ cash ↔ debt) knowingly — iterative calc on, or a circ-breaker switch.

5. Writes **auditable** formulas: no hardcodes inside formulas, no `=B2+B3+B4` where `SUM` belongs, colour-coded inputs.
6. Can **stress a driver** (revenue −20%) and immediately explain the covenant/cash impact.
7. Talks while building — narrates assumptions.

Hands-on: how to actually do it

Keyboard shortcuts that signal "pro" (Windows)

```
Ctrl + ~              Toggle show-all-formulas (audit view)
Alt + =               AutoSum
F2                    Edit cell / F4 = toggle $ absolute
Ctrl + Shift + →/↓    Select to edge of data
Alt + E, S, V         Paste Special > Values (older) / Ctrl+Alt+V
Alt + H, O, I         Autofit column width
Ctrl + [              Jump to precedent cell (trace)
F9                    Recalculate (and evaluate part of a formula)
Alt + A, T            Add filter
```

The lookup you MUST use

```
=XLOOKUP(lookup_value, lookup_array, return_array, "Not found", 0)
```

Legacy fallback interviewers still test:

```
=INDEX($D$2:$D$500, MATCH(A2, $B$2:$B$500, 0))
```

`VLOOKUP` is acceptable but flag that it breaks on column insertion — `INDEX/MATCH` or `XLOOKUP` shows maturity.

Core modeling formulas

```
Revenue growth:        =Prev_Rev * (1 + growth_%)
Depreciation (SLM):    =(Cost - Salvage) / Useful_life
Ending debt:           =Opening_debt + Drawdown - Repayment
Interest (on avg):     =Rate * AVERAGE(Opening_debt, Closing_debt)
Retained earnings:     =Opening_RE + PAT - Dividends
Cash (from CFS):       =Opening_cash + CFO + CFI + CFF
Balance check:         =Total_Assets - (Total_Liab + Total_Equity)   'must = 0
```

Circularity switch (interest on average debt creates a loop)

Put a toggle in one cell `Circ_Switch` (1/0):

```
=IF($Circ_Switch=1, Rate*AVERAGE(Open_debt,Close_debt), Rate*Open_debt)
```

And enable **File > Options > Formulas > Enable iterative calculation** (100 iterations, 0.001). Narrate this — it is a classic "do they even know?" test.

Error-proofing

```
=IFERROR(Sales/Units, 0)
=SUMIFS(Amount, Region, "West", Month, "≥"&DATE(2026,4,1))
```

Worked example / mini-project

Prompt (45 min): "Build a 3-year projection for a mid-market manufacturer. FY26 revenue ₹500 cr, growing 12% then 10%. EBITDA margin 18%. Term loan ₹200 cr at 10%, ₹40 cr repaid annually. Tax 25%. Show that the balance sheet balances."

Build a driver block, then link:

Line (₹ cr)	FY26	FY27	FY28	Formula (FY27)
Revenue	500	560	616	`=E_rev*(1+0.12)`
EBITDA (18%)	90.0	100.8	110.9	`=Rev*0.18`
Depreciation	20	20	20	input
EBIT	70.0	80.8	90.9	`=EBITDA-Dep`
Interest (10% avg)	18.0	16.0	12.0	`=0.10*AVERAGE(OpenDebt,CloseDebt)`
PBT	52.0	64.8	78.9	`=EBIT-Int`
Tax @25%	13.0	16.2	19.7	`=PBT*0.25`
PAT	39.0	48.6	59.2	`=PBT-Tax`
Closing debt	160	120	80	`=OpenDebt-40`

Then the tie-out that wins the round:

```
Cash_close = Cash_open + PAT + Dep − Debt_repay − Capex + Δ Working_capital
Balance_check = Total_Assets − Total_Liab_Equity   → 0.00
```

If `Balance_check` isn't zero, **stop and trace with** `Ctrl+[` before adding anything else. A model that doesn't balance but looks pretty scores lower than a simpler one that ties.

How it's tested

The practical test itself — a typical live shared-screen script: 1. "Here's a blank sheet. Build me a revenue build with these drivers." (structure + speed) 2. "Now link it to a simple P&L." (references, sign convention) 3.

"Add debt with interest on average balance — make it work." (circularity) 4. "The balance sheet is off by ₹4 cr, find it." (audit under pressure) 5. "Flex revenue down 20% — what breaks?" (judgment)

The take-home — you get an .xlsx of a real-ish company; deliver a model + a 5-line email summarising the investment view. They grade formatting and the *email* as much as the math.

Verbal questions fired mid-build: - "Walk me through a 3-statement model — if depreciation goes up ₹100, what happens to all three statements?" (Answer: P&L PAT −75 at 25% tax; CFS add back +100 so cash +25; BS: cash +25, PP&E −100, RE −75 → still balances.) - "How do you break circularity?" / "Difference between deferred tax asset and provision?" - "Why is my cash flow not matching the balance sheet?"

GCC/Big-4 aptitude combo: SHL-style timed Excel — pivot a 5,000-row table, `SUMIFS` a summary, spot the duplicate. Practise pivots and `SUMIFS` cold.

Common mistakes & how pros avoid them

Mistake	What pros do
Reaching for the mouse	Keyboard-only; learn 15 shortcuts to reflex
Hardcoding numbers inside formulas (`=B2*1.12`)	Growth rate lives in its own labelled input cell
No colour coding	Blue = input, black = calc, green = cross-sheet link
Building fancy tabs before it balances	Get BS to tie *first*, decorate later
Silent on assumptions	Narrate: "I'll assume WC stays 15% of sales because..."
Panicking when balance check ≠ 0	Trace precedents `Ctrl+[` ; check sign on one line at a time
Circular ref error ignored	Toggle iterative calc, explain the loop out loud
Over-engineering under time pressure	Deliver a working simple model over a broken complex one

Red flags interviewers actively watch for: mouse-dragging to select, retyping the same number in multiple cells, `=A1+A2+A3+A4` instead of `SUM` , a model that doesn't balance and the candidate not noticing, freezing silently when stuck, and formatting a chart before the numbers work.

Learn-it roadmap & resources

Realistic time to test-ready: 4–6 weeks of ~1 hr/day if you already know accounting (you do, from CA Inter).

Week	Focus
1	Keyboard shortcuts + `XLOOKUP` / `INDEX-MATCH` / `SUMIFS` , drilled daily
2	Single-statement builds; `$` discipline; colour convention
3	Full 3-statement link; get the BS to balance from scratch 5×
4	Circularity, debt schedule, sensitivity tables (`Data > What-If`)
5–6	Timed mocks: 45-min builds, then "find the error" drills

Resources: - **Breaking Into Wall Street (BIWS)** / **Wall Street Prep** / **CFI** — paid, the industry standard 3-statement + LBO courses. - **Macabacus** and **Corporate Finance Institute** free formula/shortcut guides. - Free: rebuild a listed company's model from its annual report (screener.in gives Indian financials free). - Certification: **CFI FMVA** (globally recognised, ~₹25–40k) or **NISM** for domestic markets roles; BIWS for banking.

Practice the *timed* condition — set a 45-minute timer and build from blank. Untimed practice does not prepare you for the thing being tested: performance under the clock.

Quick-reference

```
BUILD ORDER:  Assumptions → Revenue → P&L → Debt schedule → CFS → BS → Balance check
BALANCE:      Total Assets  =  Total Liabilities + Equity   (to the rupee)
CASH TIE:     Cash_close = Cash_open + CFO + CFI + CFF
DEP TEST:     +100 Dep → PAT −75, Cash +25, PP&E −100, RE −75  (BS still ties)
```

Need	Formula
Lookup	`=XLOOKUP(val, arr, ret, "NA", 0)`
Robust lookup	`=INDEX(ret,MATCH(val,key,0))`
Conditional sum	`=SUMIFS(amt, crit_rng, crit)`
Safe divide	`=IFERROR(a/b,0)`
Interest (avg)	`=rate*AVERAGE(open,close)`
Retained earnings	`=open_RE + PAT − div`
Balance check	`=Assets − (Liab + Equity)` → 0

Colour convention	Meaning
Blue font	Hardcoded input / assumption
Black font	Formula within the sheet
Green font	Link from another sheet
Red font	Warning / external link / check

Top shortcuts: `F2` edit · `F4` toggle `$` · `Alt+=` sum · `Ctrl+[` trace precedent · `Ctrl+~` show formulas · `Ctrl+Shift+arrow` select range · `Ctrl+Alt+V` paste special.

Cheat-sheet: formulas & shortcuts

What it is & where it's used

This chapter is the printable last-page you rip out and tape next to your monitor. It compresses the whole guide — lookups, aggregation, cleanup, dates, finance math, keyboard flow, PivotTables, a bit of SQL and Python — into a dense, copy-usable reference. Every analyst, accountant, FP&A associate, tax executive and equity researcher lives in these formulas eight hours a day. In an interview you get judged on how few keystrokes you waste; on the job you get paid for turning a raw dump into a clean answer before the 6 pm review call.

Roles that need it: **FP&A / budgeting**, **financial analyst / equity research**, **accounts & controllership**, **audit**, **GST/tax compliance**, **investment banking associate**, **credit analyst**. If Excel is open on your screen more than your email, this page is your operating system.

The gap: why companies want this (and college didn't teach it)

MBA and CA classrooms teach *concepts* — NPV, working capital, revenue recognition — but assume the numbers are already in a tidy table. Industry never hands you a tidy table. You get a 40,000-row SAP export with trailing spaces, dates stored as text, ₹ signs glued to amounts, and a "Customer Name" column that's spelled three different ways. The gap is **mechanical fluency**: the ability to reconcile, join, clean and summarise without touching the mouse and without breaking when a row is added.

College rewards the *right final number*. Employers reward *the right number, auditable, refreshable, and delivered in twenty minutes*. This cheat-sheet is the muscle memory that closes that gap.

What "proficient" looks like

A job-ready person can, unaided:

- Build a two-condition lookup with `XLOOKUP` / `INDEX-MATCH` without converting anything to a helper column.
- Reconcile two ledgers (book vs bank, GSTR-2B vs purchase register) and flag the breaks.
- Clean a dirty export — `TRIM`, `TEXT`-to-date, split names — in under five minutes.
- Drive Excel keyboard-only: `Ctrl+Shift+Arrow`, `Alt+=`, `Ctrl+T`, `F4`.
- Spin up a PivotTable and a `SUMIFS` dashboard that refresh when data grows.
- Know when to leave Excel — pull the raw data with a `SELECT ... GROUP BY` instead of a 500k-row copy-paste.

Hands-on: how to actually do it

Lookups & joins

```
' Exact lookup, modern (Excel 365 / 2021)
=XLOOKUP(A2, Master[ID], Master[Name], "Not found")

' Two-condition lookup (invoice + line item), array-safe
=XLOOKUP(1, (Master[Inv]=A2)*(Master[Line]=B2), Master[Amount])

' Legacy but universal — works everywhere incl. old corporate builds
=INDEX(Master[Name], MATCH(A2, Master[ID], 0))

' Left-lookup / approximate band (tax slab, ageing bucket) — sorted ascending
=XLOOKUP(A2, Slab[Upper], Slab[Rate], , 1)   ' 1 = next larger match
```

Avoid `VLOOKUP` for new work: it breaks when columns are inserted and can't look left. If you must, lock the table: `=VLOOKUP(A2,$D$2:$F$999,3,FALSE)`.

Conditional aggregation (the FP&A workhorse)

```
=SUMIFS(Amt, Region, "West", Month, "≥"&DATE(2026,4,1))
=COUNTIFS(Status, "Open", Ageing, ">90")
=AVERAGEIFS(Margin, Product, "SKU-101")
=SUMPRODUCT((Region="West")*(Qty)*(Price))   ' when SUMIFS can't multiply
```

Cleanup & text

```
=TRIM(CLEAN(A2))                        ' strip spaces + non-printing chars
=VALUE(SUBSTITUTE(A2,"₹",""))           ' text "₹1,200" → number
=DATEVALUE(TEXT(A2,"dd-mm-yyyy"))       ' text date → real date
=TEXTSPLIT(A2," ")                      ' 365: split "Ravi Kumar" into cells
=TEXTBEFORE(A2,"@")  =TEXTAFTER(A2,"@")
=PROPER(TRIM(A2))                       ' normalise names
=IFERROR(XLOOKUP( … ), "check")         ' never show #N/A in a deck
```

Dates & finance math

```
=EOMONTH(A2,0)                        ' month-end (period close)
=EDATE(A2,12)                         ' same day next year
=YEARFRAC(A2,B2,1)                    ' actual/actual for interest
=NPV(0.12, C2:C11) + C1               ' C1 = year-0 outflow
=XNPV(0.12, Cash, Dates)              ' irregular cash flows
=IRR(C1:C11)   =XIRR(Cash, Dates)
=PMT(0.09/12, 60, -500000)            ' EMI on ₹5,00,000, 9%, 5 yrs
=FV(0.07,10,-100000)   =PV(0.08,5,,-1000000)
```

SQL — pull, don't copy-paste

```
-- Monthly sales by region, straight from the source
SELECT region, DATE_TRUNC('month', invoice_date) AS mth,
       SUM(taxable_value)  AS sales,
       SUM(igst+cgst+sgst) AS tax
FROM   sales_register
WHERE  invoice_date ≥ '2026-04-01'
GROUP  BY region, mth
ORDER  BY mth, sales DESC;
```

Python — when the file is too big for Excel

```
import pandas as pd
df = pd.read_excel("sap_dump.xlsx")
df["amount"] = (df["amount"].astype(str)
                .str.replace("₹","").str.replace(",","").astype(float))
pivot = df.pivot_table(index="region", columns="month",
                       values="amount", aggfunc="sum", margins=True)
pivot.to_excel("summary.xlsx")
```

DAX (Power BI / Power Pivot)

```
Total Sales   = SUM(Sales[Amount])
Sales LY      = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY %         = DIVIDE([Total Sales]-[Sales LY], [Sales LY])
Margin %      = DIVIDE([Total Sales]-[Total COGS], [Total Sales])
```

Worked example / mini-project

GSTR-2B vs Purchase Register reconciliation — the monthly task every tax executive does.

You have two sheets: `Books` (your recorded purchases) and `GSTR2B` (what the portal says your vendors filed). Goal: find Input Tax Credit (ITC) you can safely claim.

Sample rows:

GSTIN	Invoice	Taxable (₹)	IGST (₹)
27ABCDE1234F1Z5	INV-091	1,00,000	18,000
29PQRSX9876G1Z2	INV-104	50,000	9,000

Build a match key and lookup in `Books`:

```
' Key in both sheets
=TRIM(A2)&"|"&TRIM(B2)

' In Books, pull the 2B tax against each invoice
=XLOOKUP([@Key], GSTR2B[Key], GSTR2B[IGST], "Not in 2B")

' Break flag
=IF([@BooksIGST]=[@IGST_2B], "Match",
   IF([@IGST_2B]="Not in 2B","Missing in 2B","Value diff"))

' Claimable ITC = only what appears in 2B
=SUMIFS(Books[IGST], Books[Status], "Match")
```

Then a PivotTable on `Status` gives the summary the manager wants:

Status	Count	IGST (₹)
Match	142	11,84,000
Missing in 2B	8	61,200
Value diff	3	4,500

Claimable this month = ₹11,84,000. The ₹61,200 "Missing in 2B" is chased with vendors — that's the deliverable. Whole thing: 15 minutes once the keys are built.

How it's tested

Timed Excel test (most common, 30–45 min): given a raw sheet, you're asked to (1) clean it, (2) build a lookup between two tables, (3) produce a `SUMIFS` summary or PivotTable, (4) sometimes an EMI/NPV. Graders watch whether you use structured references and keyboard shortcuts — mouse-heavy candidates run out of time.

SQL screen (analyst/BI roles): "Write monthly revenue by region" or "top 5 customers by outstanding" — a `GROUP BY` + `ORDER BY … LIMIT`, sometimes a `JOIN`.

Case / "close these books": a mini reconciliation (bank vs book, or 2B) with intentional breaks planted. They want to see you *find* the breaks, not just tie the total.

Interview questions you'll hear: - "VLOOKUP vs INDEX-MATCH vs XLOOKUP — when and why?" - "Your SUMIFS returns 0 but you know data exists — debug it." (Answer: text-vs-number mismatch, trailing spaces, wrong criteria range.) - "Difference between NPV and XNPV / IRR and XIRR?" - "How do you make a report auto-refresh when rows are added?" (Answer: Excel Tables + structured refs, or Power Query.)

Common mistakes & how pros avoid them

Mistake	Fix
`SUMIFS` returns 0 — numbers stored as text	`=A2+0` or `VALUE()`, or Data > Text to Columns
`VLOOKUP` breaks after column insert	Use `XLOOKUP / INDEX-MATCH`
Forgetting `$` locks, formula drifts	`F4` to toggle absolute refs
`NPV()` including year-0 in the range	Keep year-0 outside: `=NPV(r,C2:C11)+C1`
Merged cells killing sort/pivot	Never merge in data ranges; use Center Across Selection
Hard-coding numbers inside formulas	Put assumptions in labelled input cells
Ranges not growing with data	Convert to Table (`Ctrl+T`) so refs auto-extend
Comparing dates as text	Parse with `DATEVALUE / TEXT` first
Circular reference in interest calc	Enable iterative calc knowingly, or restructure

Pros also **audit before they trust**: `Ctrl+[` to trace precedents, `F9` to evaluate a selected formula chunk, and a `SUM` cross-check at the bottom of every reconciliation.

Learn-it roadmap & resources

Realistic time to interview-ready fluency: **6–8 weeks** at 1 hr/day if you already know finance concepts.

- **Weeks 1–2:** Lookups, SUMIFS/COUNTIFS, Tables, keyboard shortcuts. Rebuild a real bank statement into a summary.
- **Weeks 3–4:** Text cleanup, dates, IFERROR, PivotTables. Do the GSTR-2B reco above with your own data.
- **Weeks 5–6:** Finance functions (NPV/IRR/PMT), then Power Query + one SQL `GROUP BY`.
- **Weeks 7–8:** Basic Python/pandas for big files, and DAX if the role touches Power BI.

Resources: **ExcelJet** (free formula reference), **Chandoo.org** (dashboards), **Microsoft Learn** (Power Query + DAX, free), **Mode SQL Tutorial** (free), **Corporate Finance Institute (CFI)** — paid but recognised for FMVA. Certifications worth the line on a resume: **Microsoft Office Specialist: Excel Expert**, **CFI FMVA**, **Google/DataCamp SQL** for BI-leaning roles.

Quick-reference

Top keyboard shortcuts

Shortcut	Action
`Ctrl+Shift+↓/→`	Select to end of data
`Alt+=`	AutoSum
`Ctrl+T`	Convert range to Table
`F4`	Toggle `$` / repeat last action
`Ctrl+;`	Insert today's date
`Ctrl+Shift+L`	Toggle filters
`Ctrl+D` / `Ctrl+R`	Fill down / right
`Alt+H+O+I`	Autofit column width
`F2`	Edit cell / `F9` evaluate selection
`Ctrl+[`	Trace precedents
`Ctrl+PgUp/PgDn`	Move between sheets
`Ctrl+Shift+V`	Paste Special (values)

Top formulas

Need	Formula
Lookup	`=XLOOKUP(k, ids, vals, "NA")`
2-key lookup	`=XLOOKUP(1,(a=x)*(b=y),vals)`
Conditional sum	`=SUMIFS(amt, reg,"West", mth,"≥"&d)`
Count	`=COUNTIFS(rng,">90")`
Clean text	`=TRIM(CLEAN(A2))`
₹text→number	`=VALUE(SUBSTITUTE(A2,"₹",""))`
Month-end	`=EOMONTH(A2,0)`
EMI	`=PMT(r/12,n,-P)`
Irregular NPV/IRR	`=XNPV(r,cf,dt)` / `=XIRR(cf,dt)`
No errors in deck	`=IFERROR(x,"check")`

SQL skeleton: `SELECT dim, SUM(x) FROM t WHERE d≥'2026-04-01' GROUP BY dim ORDER BY 2 DESC LIMIT 5;`

pandas skeleton: `df.pivot_table(index="region", values="amt", aggfunc="sum", margins=True)`

Practical Data & BI for Finance

Practical, Hands-On, Job-Ready — India-first + Global

Real formulas, queries, entries & tools — closing the gap between what an MBA teaches and what finance employers pay for

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Why Finance Became a Data Job

What it is & where it's used

Ten years ago a "finance job" meant Excel, a general ledger, and a printer. Today the ledger is a database, the reports are dashboards refreshed hourly, and the person who can *query the raw data* gets promoted over the person who waits for someone to email them a file. This book is about closing that gap.

The shift is structural. Businesses now generate transaction data at a volume Excel physically cannot hold (Excel caps at 1,048,576 rows — a mid-size D2C brand blows past that in a quarter of order lines). So the data moved into databases (SQL Server, PostgreSQL, Snowflake), into ERPs (SAP, Oracle NetSuite, Zoho Books, TallyPrime), and into GST/Income-Tax portals that emit JSON and CSV, not paper. Finance work is now: **pull the data → clean it → model it → explain it.** That pipeline runs on four tools:

Layer	Tool	Finance use
Grab & shape small data	Excel / Google Sheets	Ad-hoc analysis, models, reconciliations
Pull large data from the source	SQL	GL extracts, sales ledgers, GSTR-2B matching
Automate & compute	Python (pandas)	Bank recos, GST reconciliation, forecasting
Present to a decision-maker	Power BI / Looker (DAX)	MIS dashboards, board packs, KPI tracking

Where it's used, by role:

- **FP&A / MIS analyst** — SQL + Excel + Power BI. Builds the monthly MIS.
- **Accounts / R2R (Record-to-Report)** — Tally/SAP + Excel + increasingly SQL for reconciliations.
- **Tax / GST associate** — GST portal + Excel + Python for 2B-vs-books matching at scale.
- **Audit / assurance** — SQL + Excel (CAATs — Computer-Assisted Audit Techniques) to test 100% of transactions, not a sample.
- **Finance business partner / analyst at a startup** — all four, because there's no one else to do it.

The gap: why companies want this (and college didn't teach it)

An MBA Finance or CA Inter curriculum teaches you *accounting standards, valuation, ratio analysis, cost sheets, and tax law.* All essential. None of it teaches you how to get 400,000 sales rows out of a database and reconcile them against GSTR-1 in twenty minutes.

The gap is not knowledge — it's **execution on real, messy, large data.** College gives you a clean 30-row balance sheet in a question paper. The job gives you a 90-column CSV export with blank cells, dates as text, ₹ symbols glued to numbers, and three spellings of the same vendor.

What college taught	What the job actually needs
Prepare a Trial Balance from given data	*Extract* the TB from SAP via SQL, then tie it to sub-ledgers
Calculate current ratio	Build a live ratio dashboard that refreshes every morning
Journal entries by hand	Book 5,000 entries via an import template + validate them
GST computation for one invoice	Reconcile GSTR-2B vs purchase register for 8,000 invoices
VLOOKUP (if at all)	XLOOKUP, Power Query, pivot models, SQL joins

Employers pay a premium for this because it's *rare among finance graduates and expensive to hire separately.* A data analyst who doesn't understand debits/credits can't do it either. You — finance-literate **and** data-capable — are the expensive hybrid.

What "proficient" looks like

The concrete bar. A job-ready person can, **unaided**:

1. Take a raw ERP/bank/GST export and turn it into a clean, analysis-ready table (dedupe, fix types, handle blanks) without manual cell-by-cell editing.
2. Write a `SELECT` with a `JOIN`, `WHERE`, `GROUP BY` and `HAVING` to answer a business question directly from the database.
3. Build an Excel model that uses `XLOOKUP` / `SUMIFS` / `INDEX-MATCH` and doesn't break when a row is inserted.
4. Reconcile two data sets (books vs bank, books vs 2B) and produce a clean *difference* report with reasons.
5. Build a Power BI dashboard with a proper date table and 3–4 DAX measures that a CFO can read without asking questions.
6. Explain, in one sentence, what the number *means* for the business — the finance judgment that a pure data analyst lacks.

If you can do these six, you clear 80% of finance-analyst screens in India today.

Hands-on: how to actually do it

Excel — the modern lookup (stop using VLOOKUP):

```
=XLOOKUP(A2, Vendors[GSTIN], Vendors[VendorName], "Not Found")
```

Sum sales for one region and month:

```
=SUMIFS(Sales[Amount], Sales[Region], "South", Sales[Month], "Jun")
```

Strip a ₹ symbol and text out of an imported number:

```
=VALUE(SUBSTITUTE(SUBSTITUTE(A2,"₹",""),",",""))
```

SQL — pull a revenue summary straight from the ledger:

```sql
SELECT  c.region,
        SUM(s.amount)          AS total_sales,
        COUNT(*)               AS invoice_count
FROM    sales      s
JOIN    customers  c ON c.customer_id = s.customer_id
WHERE   s.invoice_date ≥ '2026-04-01'
  AND   s.invoice_date <  '2026-07-01'   -- Q1 FY26-27
GROUP BY c.region
HAVING  SUM(s.amount) > 100000
ORDER BY total_sales DESC;
```

Python (pandas) — a GSTR-2B vs purchase-register reconciliation:

```python
import pandas as pd

books = pd.read_excel("purchase_register.xlsx")
gstr2b = pd.read_excel("gstr2b.xlsx")

recon = books.merge(
    gstr2b, on=["gstin", "invoice_no"],
    how="outer", suffixes=("_books", "_2b"), indicator=True
)
recon["tax_diff"] = recon["igst_books"].fillna(0) - recon["igst_2b"].fillna(0)

# invoices in books but not in 2B (ITC at risk)
missing_in_2b = recon[recon["_merge"] == "left_only"]
missing_in_2b.to_excel("itc_at_risk.xlsx", index=False)
```

DAX — Power BI measures for an MIS dashboard:

```
Total Sales = SUM(Sales[Amount])

Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])

Sales MoM % =
VAR ThisM = [Total Sales]
VAR PrevM = CALCULATE([Total Sales], DATEADD('Date'[Date], -1, MONTH))
RETURN DIVIDE(ThisM - PrevM, PrevM)
```

TallyPrime — export data you can actually analyse: `Gateway of Tally → Display More Reports → Trial Balance → Alt+E (Export) → File Format: Excel (Spreadsheet) → Enter.` For raw vouchers: `Display More Reports → Day Book → F2 (change period) → Alt+E → Excel.` That export is your `SELECT * FROM ledger` when you have no database access.

A journal entry, the thing the data ultimately represents — credit sale of ₹1,00,000 + 18% GST to a Karnataka customer (intra-state):

Particulars	Dr (₹)	Cr (₹)
Debtors A/c	1,18,000	
To Sales A/c		1,00,000
To Output CGST A/c (9%)		9,000
To Output SGST A/c (9%)		9,000

Worked example / mini-project

Build a one-page monthly MIS from raw exports. Reproduce this with dummy data.

Data: export `Day Book` from Tally to `sales.xlsx` (columns: date, customer, region, amount, gst) — say 4,000 rows for June 2026.

Step 1 — clean in Power Query (Excel/Power BI): `Data → Get Data → From File → Excel`. In the editor: set `date` to Date type, `amount` to Decimal, `Remove Duplicates` on invoice number, `Replace Values` to strip ₹. Close & Load.

Step 2 — the numbers (Excel pivot or SQL):

```
SELECT region,
       SUM(amount)                  AS revenue,
       SUM(amount)*0.18             AS gst_liability,
       ROUND(AVG(amount),0)         AS avg_invoice
FROM   sales
GROUP BY region;
```

Suppose the result:

Region	Revenue (₹)	GST (₹)	Avg invoice (₹)
South	42,00,000	7,56,000	21,000
West	31,50,000	5,67,000	18,500
North	18,00,000	3,24,000	15,000
Total	**91,50,000**	**16,47,000**	—

Step 3 — the insight (the finance bit): South is 46% of revenue but has the highest average invoice — concentration risk if that one region dips. That sentence is what gets you hired; the pivot table is just plumbing.

Step 4 — dashboard: load into Power BI, add the DAX `Total Sales`, `Sales MoM %`, a region bar chart and a KPI card. One page, refreshable. You now have an MIS that took an afternoon and refreshes in one click next month.

How it's tested

Interviews for finance-analyst roles now split into a *talk* round and a *do* round.

Interview questions you'll hear: - "What's the difference between `WHERE` and `HAVING`?" (WHERE filters rows before grouping; HAVING filters after.) - "VLOOKUP vs XLOOKUP — why switch?" (XLOOKUP searches right-to-left, defaults to exact match, returns a not-found value.) - "How would you reconcile GSTR-2B with the purchase register for 8,000 invoices?" - "Walk me through building an MIS from a raw ERP dump."

The practical/assessment tests (the real filter): - **Timed Excel test (30–45 min):** given a messy sheet, build a summary with SUMIFS/XLOOKUP + a pivot. No internet. - **SQL screen (HackerRank/DataCamp or live):** write 3–4 queries with joins and aggregation against a sample schema. - **A "close these books" / reconciliation case:** here are two files, find and explain the differences. - **Case + dashboard take-home:** "Here's 12 months of sales, build a dashboard and tell us what you'd flag to the CFO."

Common mistakes & how pros avoid them

Mistake	What pros do
Manually editing cells to "fix" data	Use Power Query / pandas — repeatable, auditable, one click next month
VLOOKUP everywhere; breaks on column insert	XLOOKUP or INDEX-MATCH; reference by column name
Reconciling by eyeballing two sheets	`merge(..., indicator=True)` or full-outer join; let the machine find diffs
Dumping a 40-column table on the CFO	One page, 3 KPIs, one insight sentence
`SELECT *` on a 10M-row table	Select only needed columns; filter dates in `WHERE`
Numbers with no "so what"	Always end with the business implication
Hard-coding dates/rates in formulas	Parameter cells / a Date table; change once

Learn-it roadmap & resources

Realistic time-to-proficiency for someone with a finance base, ~1 hour/day:

Skill	To job-ready	Free resource	Paid / cert
Modern Excel	3–4 weeks	ExcelJet, Chandoo.org	Microsoft MO-201
SQL	4–6 weeks	SQLBolt, Mode SQL Tutorial, LeetCode DB	DataCamp track
Power BI	3–4 weeks	Microsoft Learn (free), Guy in a Cube (YouTube)	Microsoft PL-300
Python/pandas	6–8 weeks	Kaggle "pandas", freeCodeCamp	DataCamp / Coursera

Sequence: Excel → SQL → Power BI → Python. Excel earns immediately; SQL is the highest ROI hire-signal; Power BI makes you visible to management; Python scales you. Don't chase all four at once — get *proficient* in Excel + SQL first (8–10 weeks) and you're already employable.

Cert worth having in India: Microsoft **PL-300 (Power BI Data Analyst)** — cheap, recognised, and directly maps to MIS roles. SQL needs no cert; a portfolio of 5 solved case dashboards on GitHub beats any certificate.

Quick-reference

```
-- SQL skeleton
SELECT col, SUM(x) FROM t JOIN u ON t.id=u.id
WHERE date ≥ '2026-04-01' GROUP BY col HAVING SUM(x)>0 ORDER BY 2 DESC;
```

```
# pandas reconciliation
a.merge(b, on="key", how="outer", indicator=True)   # _merge: left_only/right_only/both
```

Need	Formula / step
Lookup	`=XLOOKUP(val, lookup_col, return_col, "NA")`
Conditional sum	`=SUMIFS(sum, crit_rng1, c1, crit_rng2, c2)`
Clean ₹ number	`=VALUE(SUBSTITUTE(SUBSTITUTE(A2,"₹",""),",",""))`
Import & clean	`Data → Get Data → Power Query → set types → Close&Load`
Tally export	`Report → Alt+E → Excel`
DAX YoY	`TOTALYTD([Measure], 'Date'[Date])`
GST (intra-state)	Output CGST 9% + SGST 9% on taxable value
Excel row limit	1,048,576 — past this, use SQL

The one idea: finance is now *pull → clean → model → explain*. Master the pipeline, keep the judgment, and you're the hybrid employers overpay for.

SQL for Finance: Foundations

What it is & where it's used

SQL (Structured Query Language) is how you *ask questions of data that lives in a database* instead of a spreadsheet. When your company's transactions, ledgers, invoices, and customer masters outgrow Excel — usually past a few hundred thousand rows — they move to a database (PostgreSQL, MySQL, SQL Server, Snowflake, or the SQL layer inside your ERP). SQL is the language you use to pull, filter, and summarise that data.

In finance and accounting, SQL is the everyday tool for:

- **FP&A / financial analysts** — pulling actuals from the GL to build variance reports and MIS.
- **Accounts / R2R teams** — reconciling sub-ledgers to control accounts, ageing analysis, finding unposted entries.
- **Tax / GST analysts** — extracting invoice-level data to reconcile GSTR-1 vs books, matching ITC.
- **Internal audit & controls** — sampling transactions, testing for duplicate payments, journal-entry testing.
- **Revenue / collections** — customer ageing, DSO, overdue exposure.

You do not need to *build* databases. You need to *read* from them confidently. That is 90% of the finance SQL job, and it rests on six clauses: `SELECT`, `WHERE`, `JOIN`, `GROUP BY`, `HAVING`, `ORDER BY`.

The gap: why companies want this (and college didn't teach it)

An MBA or CA course teaches you *what* a trial balance, an ageing schedule, or a GST reconciliation should look like. It assumes the data arrives clean. In a real company, the data sits in a database with 40 tables, cryptic column names, and 2 million rows — and *nobody hands you the summarised sheet*. You have to extract it yourself.

The specific gap:

- College says "the accountant will give you the ledger." Industry says "here's read access to the `gl_transactions` table — get what you need."
- Excel breaks, freezes, or silently truncates past ~1M rows. A single month of a mid-size company's transactions can exceed that.
- Analysts who can't write SQL become *dependent on the IT/BI team* for every data request, waiting days for a pull. Analysts who can write SQL are self-sufficient — and that self-sufficiency is exactly what gets you hired and promoted.

Employers have stopped treating SQL as an "IT skill." For any finance analyst role in a mid-to-large company, a basic SQL screen is now as standard as an Excel test.

What "proficient" looks like

The bar for a finance analyst (not a data engineer) is narrow and achievable. Job-ready means you can, *unaided*:

- Write a `SELECT` with a `WHERE` filter on dates, amounts, and text — and get the exact rows you intend.

- `JOIN` two or three tables (e.g. transactions → customers → GL accounts) and know the difference between `INNER` and `LEFT JOIN`.
- Aggregate with `GROUP BY` (sum by account, by month, by customer) and filter groups with `HAVING`.
- Sort and rank with `ORDER BY … DESC` and `LIMIT`.
- Read a query someone else wrote and explain what number it produces.
- Sanity-check your output against a known control total (does the SUM tie to the trial balance?).

You do *not* need window functions, stored procedures, or query tuning to clear this bar — those come later.

Hands-on: how to actually do it

Assume three tables. This is a realistic mini-schema for an Indian company.

`gl_transactions`

txn_id	txn_date	account_code	customer_id	debit	credit	narration
1001	2026-04-05	4000	C-01	0	118000	Sales invoice INV-201
1002	2026-04-05	1200	C-01	118000	0	Debtor INV-201
1003	2026-04-08	5100	NULL	45000	0	Office rent

`accounts` (`account_code`, `account_name`, `account_type`) — the chart of accounts. **`customers`** (`customer_id`, `customer_name`, `state`, `gstin`).

SELECT + WHERE — pick columns and filter rows

```
-- All sales-account entries in April 2026, above ₹50,000
SELECT txn_date, account_code, credit, narration
FROM   gl_transactions
WHERE  account_code = 4000
  AND  txn_date BETWEEN '2026-04-01' AND '2026-04-30'
  AND  credit > 50000;
```

Key operators: `=`, `<>`, `>`, `<`, `BETWEEN`, `IN (…)`, `LIKE '%rent%'` (text search), `IS NULL`. Combine with `AND` / `OR`. Wrap `OR` groups in brackets so precedence is unambiguous.

JOIN — combine tables

```
-- Attach account names and types to each transaction
SELECT t.txn_date, a.account_name, a.account_type,
       t.debit, t.credit
FROM   gl_transactions t
JOIN   accounts a  ON t.account_code = a.account_code
WHERE  t.txn_date >= '2026-04-01';
```

- **`INNER JOIN`** (just `JOIN`) — keeps only rows that match in both tables.

- `LEFT JOIN` — keeps *all* left-table rows; unmatched right side comes back `NULL`. Use `LEFT JOIN` when you want to find what's *missing*:

```
-- Customers with NO transactions this year (dormant accounts)
SELECT c.customer_id, c.customer_name
FROM   customers c
LEFT JOIN gl_transactions t ON c.customer_id = t.customer_id
WHERE  t.txn_id IS NULL;
```

GROUP BY + HAVING — summarise, then filter groups

```
-- Net movement per GL account, only accounts with net > ₹1,00,000
SELECT a.account_name,
       SUM(t.debit)  AS total_debit,
       SUM(t.credit) AS total_credit,
       SUM(t.debit) - SUM(t.credit) AS net_debit
FROM   gl_transactions t
JOIN   accounts a ON t.account_code = a.account_code
GROUP BY a.account_name
HAVING SUM(t.debit) - SUM(t.credit) > 100000
ORDER BY net_debit DESC;
```

The mental model: `WHERE` filters **rows** *before* grouping; `HAVING` filters **groups** *after* aggregation. You cannot put `SUM()` in a `WHERE` — that is what `HAVING` is for.

ORDER BY — sort the result

```
ORDER BY net_debit DESC   -- largest first
ORDER BY txn_date ASC, txn_id ASC   -- oldest first, tie-break by id
```

Add `LIMIT 10` (MySQL/Postgres) or `TOP 10` (SQL Server) to get the top N.

Aggregate functions you'll use daily

`SUM()`, `COUNT(*)`, `COUNT(DISTINCT customer_id)`, `AVG()`, `MIN()`, `MAX()`. Wrap monthly grouping with a date-truncation function — dialect varies: Postgres `DATE_TRUNC('month', txn_date)`, MySQL `DATE_FORMAT(txn_date,'%Y-%m')`, SQL Server `FORMAT(txn_date,'yyyy-MM')`.

Worked example / mini-project

Goal: a debtor ageing summary by state for the quarter — the kind of MIS a finance analyst produces monthly.

Setup (reproduce in any free SQL sandbox — SQLite via sqliteonline.com needs no install):

```
CREATE TABLE invoices (
  inv_id INTEGER, customer_id TEXT, inv_date TEXT,
  due_date TEXT, amount REAL, status TEXT
);
INSERT INTO invoices VALUES
 (201,'C-01','2026-04-05','2026-05-05',118000,'OPEN'),
 (202,'C-02','2026-04-18','2026-05-18',236000,'OPEN'),
 (203,'C-01','2026-03-02','2026-04-01', 59000,'OPEN'),
 (204,'C-03','2026-05-10','2026-06-09',472000,'PAID'),
 (205,'C-02','2026-02-15','2026-03-17', 90000,'OPEN');

CREATE TABLE customers (customer_id TEXT, customer_name TEXT, state TEXT);
INSERT INTO customers VALUES
 ('C-01','Sharma Traders','Karnataka'),
 ('C-02','Nova Textiles','Maharashtra'),
 ('C-03','Delta Foods','Karnataka');
```

Now the ageing query. Assume "today" is 2026-06-30.

```
SELECT c.state,
       COUNT(*)              AS open_invoices,
       SUM(i.amount)         AS total_outstanding,
       SUM(CASE WHEN julianday('2026-06-30') - julianday(i.due_date) <= 30
                THEN i.amount ELSE 0 END) AS due_0_30,
       SUM(CASE WHEN julianday('2026-06-30') - julianday(i.due_date) > 30
                THEN i.amount ELSE 0 END) AS overdue_30_plus
FROM   invoices i
JOIN   customers c ON i.customer_id = c.customer_id
WHERE  i.status = 'OPEN'
GROUP BY c.state
ORDER BY total_outstanding DESC;
```

Result:

state	open_invoices	total_outstanding	due_0_30	overdue_30_plus
Maharashtra	2	326000	236000	90000
Karnataka	2	177000	0	177000

In four clauses you filtered to open invoices (`WHERE`), attached the state (`JOIN`), bucketed the amounts (`CASE` inside `SUM`), rolled up by state (`GROUP BY`), and ranked by exposure (`ORDER BY`). That is a real deliverable, not a toy.

How it's tested

Companies test SQL two ways.

1. The live SQL screen (30–45 min, shared editor or HackerRank). You get a schema and are asked to write queries against it. Typical prompts:

- "Return total sales per customer for FY26, highest first." (`JOIN` + `GROUP BY` + `ORDER BY`)
- "Find customers with more than 5 transactions but total spend under ₹10,000." (`GROUP BY` + `HAVING`)
- "List invoices with no matching payment." (`LEFT JOIN … IS NULL`)
- "What's the difference between `INNER` and `LEFT JOIN`? When would net totals differ?"
- " `WHERE` vs `HAVING` — why can't I filter `SUM()` in the `WHERE`?"

2. The take-home / case. A CSV or database dump plus a business question: "Reconcile this sub-ledger to the GL and list the mismatched accounts," or "Produce a monthly revenue trend." They're checking whether your number *ties* to a control total — accuracy over cleverness.

Interview verbal questions also probe: "What does `COUNT(*)` count vs `COUNT(column)`?" (the latter skips `NULL` s), and "Why did your `JOIN` inflate the total?" (a one-to-many join fanning out rows — a classic trap).

Common mistakes & how pros avoid them

Mistake	Symptom	Fix
Filtering an aggregate in `WHERE`	`SQL error: aggregate not allowed`	Move it to `HAVING`.
Double-counting from a one-to-many `JOIN`	SUM is too high, won't tie to TB	Aggregate first in a subquery, then join; or `COUNT(DISTINCT …)`.
`INNER JOIN` silently dropping rows	Total is *lower* than expected	Use `LEFT JOIN` when the right table may not match; then check for `NULL`.
Forgetting `NULL` logic	Rows with `NULL` vanish from `<>` filters	`NULL` is not equal to anything; use `IS NULL` / `COALESCE(col,0)`.
Ambiguous date strings	Wrong month pulled	Always `YYYY-MM-DD`; use half-open ranges `≥ '2026-04-01' AND < '2026-05-01'`.
Non-aggregated column in `GROUP BY`	Error or wrong grouping	Every non-aggregated `SELECT` column must appear in `GROUP BY`.
Not sanity-checking	Wrong number ships to management	Always tie your `SUM` to a known control (trial balance, invoice register).

Pros write the `WHERE` filter *first* and eyeball a few raw rows before they aggregate — you cannot trust a total you haven't spot-checked.

Learn-it roadmap & resources

Realistic time to the finance-analyst bar (able to clear a SQL screen): **3–5 weeks at ~1 hour/day.** SQL rewards small daily reps on a real dataset far more than long theory sessions.

Week	Focus
1	`SELECT`, `WHERE`, operators, `ORDER BY`, `LIMIT`
2	`JOIN` (inner/left), multi-table queries
3	`GROUP BY`, aggregates, `HAVING`, `CASE`
4	Subqueries, reconciliation-style problems, date handling
5	Timed practice on HackerRank/StrataScratch

Free: SQLBolt (interactive, ~2 hrs to core), Mode Analytics SQL Tutorial (has an *Analytics* track with real business data), Kaggle "Intro to SQL", W3Schools (reference), sqliteonline.com (zero-install practice sandbox). **Practice with pressure:** HackerRank SQL track (what many screens are built on), LeetCode Database, StrataScratch (real finance/company questions).

Paid / certification (optional, only if a JD asks): Microsoft **PL-300 (Power BI Data Analyst)** includes SQL-adjacent skills; **Google Data Analytics Certificate** (Coursera) covers SQL for analysts. For finance roles in India, a certificate is rarely required — a clean take-home submission carries more weight.

Quick-reference

```
SELECT col1, SUM(col2) AS total   -- pick columns / aggregate
FROM   table_a a
JOIN   table_b b ON a.key = b.key -- INNER: matches only
LEFT JOIN table_c c ON …          -- keep all left rows
WHERE  a.date ≥ '2026-04-01'      -- filter ROWS (before grouping)
  AND  a.amount > 50000
GROUP BY col1                     -- one output row per group
HAVING SUM(col2) > 100000         -- filter GROUPS (after aggregation)
ORDER BY total DESC               -- sort; ASC is default
LIMIT 10;                         -- top N (TOP 10 in SQL Server)
```

Need	Clause / function
Filter rows	`WHERE`
Filter aggregated groups	`HAVING`
Text search	`LIKE '%text%'`
Match a list	`IN ('A','B')`
Range	`BETWEEN x AND y`
Missing values	`IS NULL`, `COALESCE(col,0)`
Count uniques	`COUNT(DISTINCT col)`
Bucketing (ageing)	`CASE WHEN … THEN … ELSE … END`
Find unmatched rows	`LEFT JOIN … WHERE right.key IS NULL`
Month grouping	`DATE_TRUNC` / `DATE_FORMAT` / `FORMAT`

Order of clauses (always): `SELECT → FROM → JOIN → WHERE → GROUP BY → HAVING → ORDER BY → LIMIT`.
Order of execution (what actually runs): `FROM/JOIN → WHERE → GROUP BY → HAVING → SELECT → ORDER BY → LIMIT`. Knowing execution order explains why you can't reference a `SELECT` alias in `WHERE`, and why `HAVING` sees aggregates that `WHERE` can't.

SQL for Finance: Practical Queries

What it is & where it's used

SQL (Structured Query Language) is how you ask a database questions. In finance, the "database" is where the real numbers live — the general ledger, the sales/AR sub-ledger, the payments table, the customer master — usually in PostgreSQL, MySQL, SQL Server, Snowflake, or BigQuery. Excel breaks past ~1 million rows and gets slow at 100k; SQL joins 50 million rows of transactions to a customer table in seconds and never silently corrupts a VLOOKUP.

Roles that live in SQL every day:

Role	What they query
FP&A / Finance analyst	Revenue trends, budget vs actual, cohort retention
Revenue / Billing analyst	MRR/ARR, churn, invoicing accuracy
Accounts / AR-AP analyst	Ageing buckets, unapplied cash, reconciliations
Internal audit / controls	Duplicate payments, journal anomalies, three-way match
Data/BI analyst supporting finance	The pipelines feeding Power BI/Tableau

In Indian startups, fintechs, and GCCs (Global Capability Centres in Bengaluru/Hyderabad/Pune), "SQL is required" appears on almost every finance-analyst JD above fresher level. It is the single highest-leverage non-accounting skill a CA-track candidate can add.

The gap: why companies want this (and college didn't teach it)

An MBA Finance / CA syllabus teaches you *what* a revenue figure means, how to age receivables, and the accounting behind a reconciliation. It never shows you that in a real company those numbers sit in a `transactions` table with 8 million rows and you are expected to produce them yourself, unaided, by lunchtime.

The specific gap: - **You know the concept of ageing; you've never written the query that buckets 40,000 open invoices.** College gave you a 10-row textbook example. Work gives you a raw table and a deadline. - **You've reconciled two statements by hand in Excel.** Nobody told you `FULL OUTER JOIN` does the same match for a million rows and flags every break automatically. - **You think of data as a finished report.** Employers think of it as rows you filter, join, and aggregate on demand.

Closing this gap means you stop asking the data team for extracts and start answering your own questions — which is exactly what makes an analyst "senior."

What "proficient" looks like

The bar employers actually test:

1. **You can `JOIN` two or three tables** (invoices to customers to payments) without producing duplicate rows, and you know *why* an inner vs left join changes the answer.

2. **You use `GROUP BY` with `HAVING`** to aggregate and filter aggregates (e.g. customers with total billing > ₹10 lakh).
3. **You write window functions** — `SUM() OVER`, `RANK()`, `LAG()` — for running totals, rankings, and month-on-month growth.
4. **You structure a multi-step query with CTEs** (`WITH`) instead of nesting subqueries five deep.
5. **You handle dates** — truncate to month, compute ageing days, filter a fiscal year.
6. **You reconcile** two sources with a `FULL OUTER JOIN` and isolate the breaks.

If you can do those six unaided on an unfamiliar schema, you pass 90% of finance SQL screens.

Hands-on: how to actually do it

Assume this schema (typical AR/revenue setup):

```
customers(customer_id, customer_name, segment, region)
invoices(invoice_id, customer_id, invoice_date, due_date, amount, status)
payments(payment_id, invoice_id, payment_date, amount_paid)
```

Filtering and aggregating — revenue by month:

```
SELECT DATE_TRUNC('month', invoice_date) AS month,
       SUM(amount)                        AS revenue,
       COUNT(*)                           AS num_invoices
FROM   invoices
WHERE  status <> 'cancelled'
GROUP  BY DATE_TRUNC('month', invoice_date)
ORDER  BY month;
```

Joins — revenue by region (invoices + customers):

```
SELECT c.region,
       SUM(i.amount) AS revenue
FROM   invoices i
JOIN   customers c ON c.customer_id = i.customer_id
GROUP  BY c.region
ORDER  BY revenue DESC;
```

Window function — running (cumulative) total of revenue by month:

```
SELECT month,
       revenue,
       SUM(revenue) OVER (ORDER BY month
                          ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)
                    AS running_total
FROM (
    SELECT DATE_TRUNC('month', invoice_date) AS month,
           SUM(amount)                       AS revenue
    FROM   invoices
    GROUP  BY 1
) m
ORDER BY month;
```

Window function — rank top customers, and month-on-month growth with `LAG`:

```
-- Top 5 customers by billing
SELECT customer_id,
       SUM(amount) AS total_billed,
       RANK() OVER (ORDER BY SUM(amount) DESC) AS rnk
FROM   invoices
GROUP  BY customer_id
QUALIFY rnk <= 5;          -- Snowflake/BigQuery; else wrap in a subquery
```

```
-- Month-on-month % growth
SELECT month,
       revenue,
       LAG(revenue) OVER (ORDER BY month) AS prev_month,
       ROUND(100.0 * (revenue - LAG(revenue) OVER (ORDER BY month))
             / LAG(revenue) OVER (ORDER BY month), 1) AS growth_pct
FROM   monthly_rev;
```

CTEs — chaining steps readably:

```sql
WITH monthly AS (
    SELECT DATE_TRUNC('month', invoice_date) AS month,
           SUM(amount) AS revenue
    FROM   invoices
    GROUP  BY 1
),
with_growth AS (
    SELECT month, revenue,
           revenue - LAG(revenue) OVER (ORDER BY month) AS mom_change
    FROM   monthly
)
SELECT * FROM with_growth WHERE mom_change < 0;   -- months that declined
```

Date logic — AR ageing buckets:

```sql
SELECT i.invoice_id, c.customer_name, i.amount,
       CURRENT_DATE - i.due_date AS days_overdue,
       CASE
         WHEN CURRENT_DATE - i.due_date <= 0   THEN 'Not due'
         WHEN CURRENT_DATE - i.due_date <= 30  THEN '0-30'
         WHEN CURRENT_DATE - i.due_date <= 60  THEN '31-60'
         WHEN CURRENT_DATE - i.due_date <= 90  THEN '61-90'
         ELSE '90+'
       END AS bucket
FROM   invoices i
JOIN   customers c ON c.customer_id = i.customer_id
WHERE  i.status = 'open';
```

Reconciliation — match GL to bank/sub-ledger and find breaks:

```sql
SELECT COALESCE(g.txn_id, b.txn_id) AS txn_id,
       g.amount AS gl_amount,
       b.amount AS bank_amount,
       CASE
         WHEN b.txn_id IS NULL THEN 'Missing in bank'
         WHEN g.txn_id IS NULL THEN 'Missing in GL'
         WHEN g.amount <> b.amount THEN 'Amount mismatch'
       END AS break_type
FROM   gl g
FULL OUTER JOIN bank b ON g.txn_id = b.txn_id
WHERE  g.txn_id IS NULL
   OR  b.txn_id IS NULL
   OR  g.amount <> b.amount;
```

Worked example / mini-project

Scenario: You're the FP&A analyst at an Indian SaaS company. Build a monthly-cohort retention view — do customers acquired in a given month keep paying?

Sample `invoices` data (₹, one row per customer-month):

customer_id	invoice_date	amount
1	2025-01-15	5,000
1	2025-02-15	5,000
1	2025-03-15	5,000
2	2025-01-20	8,000
2	2025-02-20	8,000
3	2025-02-10	3,000

Step 1 — find each customer's cohort (first billing month):

```
WITH cohort AS (
    SELECT customer_id,
           MIN(DATE_TRUNC('month', invoice_date)) AS cohort_month
    FROM   invoices
    GROUP  BY customer_id
),
activity AS (
    SELECT i.customer_id,
           c.cohort_month,
           DATE_TRUNC('month', i.invoice_date) AS active_month
    FROM   invoices i
    JOIN   cohort c ON c.customer_id = i.customer_id
)
SELECT cohort_month,
       -- months since acquisition (0, 1, 2 …)
       (EXTRACT(YEAR FROM active_month) - EXTRACT(YEAR FROM cohort_month)) * 12
        + (EXTRACT(MONTH FROM active_month) - EXTRACT(MONTH FROM cohort_month)) AS month_offse
       COUNT(DISTINCT customer_id) AS active_customers,
       SUM(1) AS billings_count
FROM   activity
GROUP  BY cohort_month, month_offset
ORDER  BY cohort_month, month_offset;
```

Result (retention triangle):

cohort_month	month_offset	active_customers
2025-01	0	2
2025-01	1	2
2025-01	2	1
2025-02	0	1

Read it: the Jan cohort started with 2 customers, both stayed in month 1, one dropped by month 2 (50% retention). That single query — cohort CTE + offset date math + `COUNT(DISTINCT)` — is exactly the deliverable a startup finance team calls "the retention curve." Reproduce it locally with any free SQLite/PostgreSQL install and the six rows above.

How it's tested

Companies test SQL far more concretely than accounting:

The live SQL screen (most common). You share screen on HackerRank / DataLemur / a scratch database and solve 3-5 problems in 45 minutes. Typical prompts: - "Return the top 3 customers by revenue per region." (window function `RANK()` partitioned by region) - "Show month-on-month revenue growth %." (`LAG`) - "Bucket these open invoices into ageing bands." (`CASE` + date math) - "Find invoices with no matching payment." (`LEFT JOIN … WHERE payments.id IS NULL`)

Take-home case. "Here's a CSV of 20k transactions — build the AR ageing report and list the 10 largest overdue balances." They check whether your joins double-count and whether your buckets are right at the boundaries.

Conceptual interview questions: - Difference between `WHERE` and `HAVING`? (row filter vs group filter) - `INNER` vs `LEFT` vs `FULL OUTER JOIN` — when does each change a reconciliation result? - What does a window function do that `GROUP BY` cannot? (keeps every row while adding an aggregate) - Why is `COUNT(*)` different from `COUNT(column)`? (the latter ignores NULLs)

Common mistakes & how pros avoid them

Mistake	Why it hurts	Pro habit
Joins that fan out rows	Summing a customer's invoices *after* joining to multiple payments doubles revenue	Aggregate each side in a CTE *before* joining, or join on a unique key
Filtering a `LEFT JOIN` in `WHERE`	`WHERE b.col = x` silently turns it back into an inner join and hides the breaks	Put the condition in the `ON`, or filter for `IS NULL`
Ageing boundary errors	`≤ 30` and `≥ 30` both include day 30 → double-count	Use clean `≤ 30` then `≤ 60`, never overlapping ranges
`SELECT *` on a huge table	Pulls 40 columns you don't need, slow and unreadable	Select only the columns you'll use
Ignoring NULLs in reconciliation	A NULL amount silently drops from `SUM` or a `=` comparison	`COALESCE(amount,0)`; test `IS NULL` explicitly
Trusting `GROUP BY` without checking counts	You never notice the join fanned out	Sanity-check: does total match the known control figure?
Rounding currency mid-query	Compounds paise-level errors	Round once, at the final SELECT

Learn-it roadmap & resources

Realistic time-to-proficiency for a finance person who already knows Excel: **6-8 weeks, ~45 min/day.**

Week	Focus
1	SELECT, WHERE, ORDER BY, basic aggregates
2	GROUP BY, HAVING, the join types
3	CTEs (`WITH`), subqueries
4	Window functions — the finance sweet spot
5	Date functions, `CASE`, ageing/cohort patterns
6-8	Timed practice on real finance-style problems

Free: - **SQLBolt** and **Mode SQL Tutorial** — interactive, browser-based, start here. - **DataLemur** (Ambitious with SQL) — window-function and analytics problems, many finance-flavoured. - **PostgreSQL** + **DBeaver**, both free — install locally and load your own CSVs. - **StrataScratch** free tier — real company-style questions.

Paid / certification: - **Google Data Analytics Professional Certificate** (Coursera) — includes SQL, recruiter-recognised in India. - **Microsoft PL-300 / DP-900** if you're pairing SQL with Power BI/Azure. - **HackerRank SQL (Intermediate/Advanced) badge** — free to earn, cheap signal to put on a CV.

For a CA-track candidate, SQL + Power BI is the combination that moves you from "accounts executive" to "finance analyst" pay bands.

Quick-reference

```
-- Month truncation
DATE_TRUNC('month', d)             -- Postgres
DATE_FORMAT(d,'%Y-%m-01')          -- MySQL
FORMAT(d,'yyyy-MM')                -- SQL Server

-- Days overdue
CURRENT_DATE - due_date            -- Postgres
DATEDIFF(day, due_date, GETDATE()) -- SQL Server
```

Need	Pattern
Running total	`SUM(x) OVER (ORDER BY d ROWS UNBOUNDED PRECEDING)`
Rank	`RANK() OVER (PARTITION BY region ORDER BY sales DESC)`
Prev period	`LAG(x) OVER (ORDER BY month)`
Next period	`LEAD(x) OVER (ORDER BY month)`
Multi-step query	`WITH cte1 AS (...), cte2 AS (...) SELECT ...`
Ageing bucket	`CASE WHEN days<=30 THEN '0-30' ... END`
Unmatched rows	`LEFT JOIN ... WHERE b.id IS NULL`
Full reconciliation	`FULL OUTER JOIN ... WHERE a.id IS NULL OR b.id IS NULL OR a.amt<>b.amt`
Filter aggregates	`GROUP BY ... HAVING SUM(x) > 1000000`
Distinct count	`COUNT(DISTINCT customer_id)`

Order of execution to remember: `FROM → JOIN → WHERE → GROUP BY → HAVING → window functions → SELECT → ORDER BY → LIMIT`. Knowing this explains why you can't reference a `SELECT` alias in `WHERE` but can in `ORDER BY`.

Python for Finance I: pandas & numpy

What it is & where it's used

pandas is a Python library that gives you a spreadsheet-like object called a *DataFrame* — rows and columns, but programmable. **numpy** is the numerical engine underneath it (fast arrays and math). Together they are the workhorses of every finance analytics team: you load a CSV/Excel/database extract, clean it, filter it, group it, join it to a master, and produce numbers — all in code that runs identically tomorrow with one click.

Where it shows up in real finance/accounts/tax roles:

Role	What they do with pandas
FP&A / MIS analyst	Combine month-end exports into a P&L variance report
Accounts / AR-AP	Reconcile ledger vs bank, ageing of receivables
Tax / GST associate	Match GSTR-2B (portal) against purchase register (Tally/ERP)
Audit / internal audit	Sample testing, duplicate-invoice detection across lakhs of rows
Equity / credit research	Clean price/fundamental data, compute ratios and returns

Excel breaks past ~100k rows and can't reliably repeat a 20-step manual process. pandas doesn't care whether it's 500 rows or 5 million, and the process is saved as a script.

The gap: why companies want this (and college didn't teach it)

An MBA-Finance or CA syllabus teaches you *what* a receivables ageing or a GST reconciliation **means**. It does not teach you to produce one from a 2-lakh-row dump at 9 PM on close day. Colleges assume "the data is given, clean, and small." Industry data is none of those: it arrives as a messy export with merged headers, ₹ signs stuck in text, dates as `31-03-2026` in one file and `2026/03/31` in another, and duplicate invoice numbers.

The gap is **repeatable data handling at scale**. Employers have discovered that a fresher who can write 15 lines of pandas replaces hours of copy-paste-VLOOKUP and, crucially, does it *the same way every month* with an audit trail. That reliability — not fancy modelling — is what they pay a premium for.

What "proficient" looks like

A job-ready person can, **unaided**, take two raw files and produce a correct output in under 30 minutes:

- Load CSV/Excel (right sheet, skip junk rows, set dtypes).
- Inspect: `.head()`, `.info()`, `.describe()`, `.shape`.
- Filter rows on multiple conditions and select columns.
- Clean: strip whitespace, fix ₹/comma text into numbers, parse dates, handle blanks/NaN, drop or flag duplicates.
- `groupby` + aggregate to get subtotals (sum, count, mean) by any dimension.
- `merge` two DataFrames (inner/left) on a key — and know *why* rows drop or duplicate.

- Add computed columns, sort, and export back to Excel with formatting intact.

If you can do a GST-2B vs purchase-register match or an AR ageing without googling every second line, you clear the bar.

Hands-on: how to actually do it

Setup once: `pip install pandas numpy openpyxl`. Then `import pandas as pd` and `import numpy as np`.

Load data

```
# CSV — set the invoice number as text so leading zeros survive
df = pd.read_csv("sales.csv", dtype={"invoice_no": str})

# Excel — pick sheet, skip 3 junk header rows, parse a date column
df = pd.read_excel("tb.xlsx", sheet_name="Data", skiprows=3,
                   parse_dates=["invoice_date"])
```

Inspect

```
df.shape             # (rows, cols)
df.head(10)          # first 10 rows
df.info()            # dtypes + non-null counts — catches text-vs-number bugs
df.describe()        # min/max/mean of numeric cols
df["state"].value_counts()
```

Filter & select (`&` = and, `|` = or, each condition in parentheses)

```
# Maharashtra invoices above ₹1,00,000
big = df[(df["state"] == "Maharashtra") & (df["amount"] > 100000)]

# only three columns
cols = df[["invoice_no", "party", "amount"]]

# .isin for a list; ~ negates
south = df[df["state"].isin(["Karnataka", "Tamil Nadu", "Kerala"])]
```

Clean — the part that eats 80% of real work

```
# "₹1,20,500" (text) → 120500.0 (number)
df["amount"] = (df["amount"].astype(str)
                .str.replace("₹", "", regex=False)
                .str.replace(",", "", regex=False)
                .str.strip())
df["amount"] = pd.to_numeric(df["amount"], errors="coerce")  # bad → NaN

# trim stray spaces in text keys (a top cause of failed merges)
df["party"] = df["party"].str.strip().str.upper()

# mixed date formats → real dates
df["invoice_date"] = pd.to_datetime(df["invoice_date"],
                                    dayfirst=True, errors="coerce")

df["amount"] = df["amount"].fillna(0)        # blanks → 0
df = df.drop_duplicates(subset=["invoice_no"])  # kill dupes on key

# flag (don't delete) duplicates for review
df["is_dup"] = df.duplicated(subset=["invoice_no"], keep=False)
```

groupby — subtotals

```
# total & count of sales per state
by_state = (df.groupby("state")
              .agg(total=("amount", "sum"),
                   invoices=("invoice_no", "count"))
              .reset_index()
              .sort_values("total", ascending=False))
```

merge — the VLOOKUP replacement

```
merged = pd.merge(purchase, gstr2b,
                  on="invoice_no", how="left",
                  suffixes=("_book", "_portal"),
                  indicator=True)   # _merge col shows both / left_only
```

numpy for conditional columns

```
df["gst_rate"] = np.where(df["amount"] > 50000, 0.18, 0.12)
df["gst"] = df["amount"] * df["gst_rate"]
```

Export

```
merged.to_excel("recon_output.xlsx", index=False)
```

Worked example / mini-project: GST-2B vs Purchase-Register match

You have the **purchase register** from Tally and **GSTR-2B** downloaded from the GST portal. Goal: find invoices where Input Tax Credit (ITC) in your books doesn't match the portal — the classic monthly ITC reconciliation.

`purchase_register.csv`

invoice_no	supplier_gstin	taxable	igst
INV001	27ABCDE1234F1Z5	100000	18000
INV002	29PQRST5678G2Z1	50000	9000
INV003	27ABCDE1234F1Z5	20000	3600

`gstr2b.csv` (portal shows INV001 with a different tax, and no INV003)

invoice_no	igst
INV001	17000
INV002	9000

```
import pandas as pd

pr  = pd.read_csv("purchase_register.csv", dtype={"invoice_no": str})
b2b = pd.read_csv("gstr2b.csv", dtype={"invoice_no": str})

# clean keys so the match doesn't fail on spaces/case
for d in (pr, b2b):
    d["invoice_no"] = d["invoice_no"].str.strip().str.upper()

m = pd.merge(pr, b2b, on="invoice_no", how="left",
             suffixes=("_book", "_2b"), indicator=True)

m["igst_2b"] = m["igst_2b"].fillna(0)
m["diff"] = m["igst_book"] - m["igst_2b"]

def status(r):
    if r["_merge"] == "left_only":
        return "Missing in 2B (ITC at risk)"
    return "OK" if abs(r["diff"]) < 1 else "Tax mismatch"

m["status"] = m.apply(status, axis=1)
print(m[["invoice_no", "igst_book", "igst_2b", "diff", "status"]])
```

Output:

invoice_no	igst_book	igst_2b	diff	status
INV001	18000	17000	1000	Tax mismatch
INV002	9000	9000	0	OK
INV003	3600	0	3600	Missing in 2B (ITC at risk)

You just flagged ₹3,600 of ITC you can't claim yet and a ₹1,000 mismatch to query the supplier — the exact deliverable a GST associate produces every month, now reproducible in seconds.

How it's tested

Interview questions (conceptual): - Difference between `merge` types — what happens to unmatched rows in `inner` vs `left`? - Why did your merge produce *more* rows than either input? (Answer: duplicate keys → many-to-many.) - `loc` vs `iloc`? When does a "SettingWithCopyWarning" appear? - How do you replace a nested VLOOKUP / a pivot table with pandas? - Difference between `apply`, `map`, and vectorised operations (and why vectorised is faster).

Practical test (what actually decides it): a timed take-home or live screen. Typical brief: "*Here are two CSVs — a sales dump and a customer master. Give me total sales per region, flag customers with no master record, and export to Excel. 30 minutes.*" They watch whether you check dtypes, handle the dirty ₹ column, pick the right join, and notice dropped rows. Some firms give a Jupyter notebook and one messy file and simply say "reconcile these."

Common mistakes & how pros avoid them

Mistake	Consequence	Pro habit
Merging on untrimmed/mixed-case keys	Silent non-matches	`.str.strip().str.upper()` both keys first
Ignoring `.info()` — amount is text	`sum()` concatenates or errors	Check dtypes immediately after load
`how="inner"` by default	Silently drops unmatched rows	Use `how="left"` + `indicator=True` to see drops
Not checking row count after merge	Duplicated rows inflate totals	Compare `len()` before/after; check key uniqueness
Chained indexing `df[df.x>1]["y"]=0`	SettingWithCopyWarning, no change	Use `.loc[mask, "y"] = 0`
`dropna()` everything	Deletes valid rows with one blank	Fill or subset: `fillna(0)`, `dropna(subset=[…])`
Losing leading zeros in invoice/GSTIN	Broken keys	`dtype=str` on read

Golden rule: after every merge, print `df["_merge"].value_counts()` and confirm row counts. Most "wrong numbers" in finance pandas are a bad join, not bad math.

Learn-it roadmap & resources

Time to job-ready: 3–5 weeks at ~1 hr/day if you already know finance logic (you do).

Week	Focus
1	Python basics + read/inspect CSV/Excel; `head/info/describe`
2	Filtering, `loc/iloc`, cleaning (dtypes, dates, ₹-text, NaN)
3	`groupby` aggregations; `merge` / `concat`; computed columns
4	Rebuild 2–3 of your own Excel reports in pandas end-to-end
5	Speed, `pivot_table`, export formatting, edge cases

Resources - *pandas official "10 minutes to pandas"* — free, start here. - Kaggle **"Pandas" micro-course** — free, hands-on, ~4 hrs. - Wes McKinney, *Python for Data Analysis* (3rd ed.) — the reference (pandas' creator). - Practice on real data: RBI/NSE downloads, or your own bank statement/Tally exports.

Certification: none is required for pandas itself; recruiters test the skill, not a badge. If you want a credential, *Google Data Analytics (Coursera)* or a Kaggle certificate signals intent. Your strongest asset is a GitHub repo with 2–3 reconciliations like the one above.

Quick-reference

```
import pandas as pd, numpy as np

pd.read_csv("f.csv", dtype={"id": str})          # load, keep id as text
pd.read_excel("f.xlsx", sheet_name="S", skiprows=2, parse_dates=["dt"])
df.head(); df.info(); df.describe(); df.shape     # inspect
df["c"].value_counts()                            # frequency

df[(df.a > 100) & (df.b == "X")]                  # filter (parentheses!)
df[df.state.isin(["KA","TN"])]                    # value in list
df.loc[mask, "col"] = 0                           # safe assignment

pd.to_numeric(s, errors="coerce")                 # text → number
s.str.replace(",", "", regex=False)               # strip commas
pd.to_datetime(s, dayfirst=True, errors="coerce") # dd-mm-yyyy → date
df.fillna(0); df.dropna(subset=["k"])             # missing values
df.drop_duplicates(subset=["k"])                  # dedupe
df.duplicated(subset=["k"], keep=False)           # flag dupes

df.groupby("g").agg(total=("amt","sum"),
                    n=("id","count")).reset_index()
pd.merge(a, b, on="k", how="left", indicator=True)
np.where(cond, x, y)                              # if-else column
df.sort_values("amt", ascending=False)
df.to_excel("out.xlsx", index=False)              # export
```

Merge `how`	Keeps
`inner`	keys in **both**
`left`	**all left** + matching right
`right`	all right + matching left
`outer`	**everything**, NaN where unmatched

Python for Finance II: Automation

What it is & where it's used

Automation is using Python to do the boring, repeatable finance work that eats your day: pulling numbers out of Excel files, reshaping them, writing formatted output back to Excel, emailing the result, and running the whole thing on a schedule without you touching it. It is the difference between "I spend the first two hours of every month copy-pasting the MIS" and "the MIS lands in the CFO's inbox at 8:00 AM on the 1st while I'm still asleep."

The core libraries are **pandas** (read/transform/write tabular data), **openpyxl** (fine-grained Excel control — formatting, formulas, multiple sheets, cell styling), **smtplib**/ `email` (send mail via SMTP), and a scheduler (Windows Task Scheduler or `cron` on Linux; libraries like `schedule` for in-process timing).

Roles that live on this: **FP&A analysts** (monthly MIS, variance packs), **accounts/AP-AR teams** (vendor ageing, reconciliations), **tax/GST executives** (GSTR-2B vs purchase register matching), **treasury** (daily bank position), **audit** (sampling, ledger scrutiny), and anyone who owns a recurring "report" that today is a manual Excel ritual.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you *what* a variance analysis or a debtors ageing means. It never shows you that in a real company that report is regenerated **every single month from a messy dump** — the ERP exports a CSV with merged headers, three date formats, and vendor names spelled four ways. College assignments give you one clean dataset once. Industry gives you the same dirty dataset forever, on a deadline.

The specific gap: graduates can *build* a report once by hand in Excel but cannot *productionise* it. They don't know that a 90-minute manual task can become a 4-second script. Employers pay a premium for the person who says "I automated the AP ageing; it now runs itself" — that person just gave the team back 20 hours a month. This is the single most visible, promotable skill for a junior finance hire because the time saved is directly measurable.

What "proficient" looks like

A job-ready person can, unaided:

- Read one or many Excel/CSV files into pandas, including specifying the header row, sheet name, and dtypes.
- Clean real dirt: strip whitespace, fix dates, dedupe, handle blanks, map inconsistent names.
- Do the transform (group-by, pivot, merge/lookup across sheets) in pandas.
- Write a **formatted** multi-sheet Excel file — bold headers, number formats (`#,##0.00`), frozen panes, coloured totals, even live Excel formulas — not a raw dump.
- Send that file as an email attachment via SMTP with a templated body.
- Schedule the script to run daily/monthly and log whether it succeeded or failed.
- Make it robust: if the input file is missing, it emails an alert instead of crashing silently.

Hands-on: how to actually do it

Read Excel/CSV into pandas

```
import pandas as pd

# Read a specific sheet, telling pandas the header is on row 2 (0-indexed)
df = pd.read_excel("Purchase_Register_Jun26.xlsx",
                   sheet_name="Data", header=1,
                   dtype={"GSTIN": str, "Invoice_No": str})

# CSV with Indian date format
df = pd.read_csv("bank_stmt.csv", parse_dates=["Txn_Date"], dayfirst=True)
```

Clean the dirt (this is 60% of real work)

```
df.columns = df.columns.str.strip()                        # kill trailing spaces in headers
df["Vendor"] = df["Vendor"].str.strip().str.upper()       # normalise names
df["Amount"] = pd.to_numeric(df["Amount"], errors="coerce")  # text → number, bad → NaN
df = df.dropna(subset=["Invoice_No"]).drop_duplicates()
df["Txn_Date"] = pd.to_datetime(df["Txn_Date"], dayfirst=True, errors="coerce")
```

The transform — group, pivot, lookup

```
# Debtors ageing buckets
today = pd.Timestamp("2026-06-30")
df["Days"] = (today - df["Invoice_Date"]).dt.days
bins   = [-1, 30, 60, 90, 10**6]
labels = ["0-30", "31-60", "61-90", "90+"]
df["Bucket"] = pd.cut(df["Days"], bins=bins, labels=labels)

ageing = df.pivot_table(index="Customer", columns="Bucket",
                        values="Outstanding", aggfunc="sum",
                        fill_value=0, margins=True, margins_name="Total")

# VLOOKUP-equivalent: merge master data onto transactions
out = txns.merge(master[["Vendor", "GSTIN", "PAN"]], on="Vendor", how="left")
```

Write a *formatted* Excel workbook with openpyxl

```
from openpyxl.styles import Font, PatternFill, Alignment, numbers

with pd.ExcelWriter("MIS_Jun26.xlsx", engine="openpyxl") as xl:
    ageing.to_excel(xl, sheet_name="Ageing")
    ws = xl.sheets["Ageing"]

    # Bold header row + fill colour
    hdr = Font(bold=True, color="FFFFFF")
    fill = PatternFill("solid", fgColor="1F4E78")
    for cell in ws[1]:
        cell.font, cell.fill = hdr, fill
        cell.alignment = Alignment(horizontal="center")

    # Indian number format on all data columns
    for row in ws.iter_rows(min_row=2, min_col=2):
        for c in row:
            c.number_format = '#,##0.00'

    ws.freeze_panes = "B2"                 # freeze header + first column
    ws.column_dimensions["A"].width = 30
```

You can even write **live Excel formulas** so the recipient sees a working sheet:

```
ws["F2"] = "=SUM(B2:E2)"     # openpyxl writes the formula string; Excel evaluates it
```

Send it by email (SMTP)

```
import smtplib, ssl
from email.message import EmailMessage

msg = EmailMessage()
msg["Subject"] = "Debtors Ageing – June 2026"
msg["From"]    = "mis-bot@company.com"
msg["To"]      = "cfo@company.com, controller@company.com"
msg.set_content("Hi team,\n\nPlease find the June debtors ageing attached.\n\nAuto-generated b

with open("MIS_Jun26.xlsx", "rb") as f:
    msg.add_attachment(f.read(), maintype="application",
        subtype="vnd.openxmlformats-officedocument.spreadsheetml.sheet",
        filename="MIS_Jun26.xlsx")

with smtplib.SMTP_SSL("smtp.gmail.com", 465, context=ssl.create_default_context()) as s:
    s.login("mis-bot@company.com", APP_PASSWORD)   # Gmail: use an App Password, never your lo
    s.send_message(msg)
```

> Security note: never hard-code passwords. Read from an environment variable: `APP_PASSWORD = os.environ["MAIL_PW"]`. For Gmail/Outlook you must generate an **App Password** (2FA required).

Schedule it

Windows Task Scheduler (most Indian finance desktops are Windows):

1. Wrap your script in a `.bat`: `C:\Python\python.exe C:\scripts\mis.py >> C:\scripts\log.txt 2>&1`
2. Task Scheduler → Create Task → Triggers → *Monthly, day 1, 08:00* → Actions → *Start a program* → point to the `.bat`.
3. Tick "Run whether user is logged on or not."

Linux/server (`cron`) — run 8 AM on the 1st of every month:

```
0 8 1 * *  /usr/bin/python3 /home/fin/mis.py >> /home/fin/mis.log 2>&1
```

Worked example / mini-project: month-end vendor ageing bot

Goal: every month, read the AP ledger export, build a vendor ageing report bucketed by days overdue, format it, and email it to the finance head.

Input `AP_Ledger.xlsx` (realistic ₹ data):

Vendor	Invoice_No	Invoice_Date	Outstanding
ACME STEEL	INV-1001	2026-04-02	2,45,000
ACME STEEL	INV-1044	2026-05-28	1,10,000
BLUE LOGISTICS	INV-2210	2026-03-15	88,500
ZENITH PACK	INV-3300	2026-06-20	3,20,000

```
import pandas as pd, os, smtplib, ssl
from email.message import EmailMessage
from openpyxl.styles import Font, PatternFill

FILE = "AP_Ledger.xlsx"
if not os.path.exists(FILE):
    raise SystemExit("Input missing – alert should fire here")

df = pd.read_excel(FILE, dtype={"Invoice_No": str})
df["Invoice_Date"] = pd.to_datetime(df["Invoice_Date"])
df["Outstanding"]  = pd.to_numeric(df["Outstanding"], errors="coerce")

asof = pd.Timestamp("2026-06-30")
df["Days"]   = (asof - df["Invoice_Date"]).dt.days
df["Bucket"] = pd.cut(df["Days"], [-1,30,60,90,10**6],
                      labels=["0-30","31-60","61-90","90+"])

report = df.pivot_table(index="Vendor", columns="Bucket", values="Outstanding",
                        aggfunc="sum", fill_value=0, margins=True, margins_name="Total")

with pd.ExcelWriter("Vendor_Ageing_Jun26.xlsx", engine="openpyxl") as xl:
    report.to_excel(xl, sheet_name="Ageing")
    ws = xl.sheets["Ageing"]
    for c in ws[1]:
        c.font = Font(bold=True, color="FFFFFF")
        c.fill = PatternFill("solid", fgColor="1F4E78")
    for row in ws.iter_rows(min_row=2, min_col=2):
        for c in row: c.number_format = '#,##0'
    ws.freeze_panes = "B2"; ws.column_dimensions["A"].width = 28
```

Expected output (₹):

Vendor	0-30	31-60	61-90	90+	Total
ACME STEEL	0	1,10,000	0	2,45,000	3,55,000
BLUE LOGISTICS	0	0	0	88,500	88,500
ZENITH PACK	3,20,000	0	0	0	3,20,000
Total	3,20,000	1,10,000	0	3,33,500	7,63,500

Then attach and email exactly as in the SMTP block above. Point Task Scheduler at it for the 1st of each month — you never touch the ageing report again.

How it's tested

Interview questions:

- "The ERP export has the header on row 3 and dates as `DD-MM-YYYY`. How do you read it correctly?" (Answer: `read_excel(header=2)` + `pd.to_datetime(..., dayfirst=True)`.)
- "How is `pd.merge(how='left')` different from an Excel VLOOKUP? What happens to non-matches?" (Left keeps all left rows; misses become `NaN`.)
- "How would you email a formatted Excel file every Monday without a person running it?" (SMTP + Task Scheduler/cron.)
- "How do you keep the SMTP password out of the code?" (Env var / secrets manager, App Password.)

Practical test: A timed take-home or on-screen task — "Here's a 5,000-row messy sales dump. Produce a clean, formatted Excel MIS with monthly totals by region, and a Python script we can re-run next month." They score you on: does it re-run without editing, is the output *formatted* (not raw), did you handle blank/duplicate rows, and did you make it fail gracefully.

Common mistakes & how pros avoid them

Mistake	Fix
Hard-coding file paths/dates so it breaks next month	Use `datetime.now()` and relative/config-driven paths
Delivering a raw pandas dump with no formatting	Always style headers, set `#,##0` formats, freeze panes
Password in the script, pushed to Git	Env vars; `.gitignore` secrets; App Passwords
Script crashes silently at 8 AM, nobody notices	Wrap in `try/except`, log to file, email an alert on failure
`to_excel` on a filename that's open in Excel → PermissionError	Close the file / write to a timestamped name
Ignoring dtype — GSTIN `27ABC...` read as float, invoice `007` loses leading zeros	Pass `dtype=str` for IDs/codes
Chained slow loops over rows	Vectorise with pandas group-by/`cut`, not `for` loops

Learn-it roadmap & resources

Time to job-ready: 4–6 weeks if you already know basic Python (Chapter 04).

- **Week 1–2:** pandas read/clean/transform. Rebuild an old Excel report of yours in pandas.
- **Week 3:** openpyxl formatting + writing multi-sheet workbooks with formulas.
- **Week 4:** smtplib email + scheduling; convert one real recurring task into a scheduled bot.
- **Week 5–6:** robustness — logging, try/except, config files; build the ageing/MIS mini-project end-to-end.

Resources:

- *Automate the Boring Stuff with Python* (Al Sweigart) — free online; the Excel + email chapters are exactly this.
- **pandas** docs "10 minutes to pandas"; **openpyxl** official tutorial (both free).
- YouTube: "Python Excel automation" walkthroughs.
- Certification: none is required, but **Microsoft PL-300** (Power BI) and Google's Python courses on Coursera signal seriousness. The real credential is a GitHub repo showing a working MIS bot.

Quick-reference

Task	Snippet
Read Excel sheet	`pd.read_excel(f, sheet_name="Data", header=1)`
Read CSV, IN dates	`pd.read_csv(f, parse_dates=["d"], dayfirst=True)`
Force text dtype	`dtype={"GSTIN": str}`
Clean headers	`df.columns = df.columns.str.strip()`
Text → number	`pd.to_numeric(s, errors="coerce")`
Ageing buckets	`pd.cut(days, bins, labels=…)`
VLOOKUP	`a.merge(b, on="key", how="left")`
Pivot with totals	`pivot_table(…, margins=True)`
Write Excel	`df.to_excel("out.xlsx", index=False)`
Bold header (openpyxl)	`cell.font = Font(bold=True)`
₹ number format	`cell.number_format = '#,##0.00'`
Freeze panes	`ws.freeze_panes = "B2"`
Send mail	`smtplib.SMTP_SSL("smtp.gmail.com", 465)`
Secret from env	`os.environ["MAIL_PW"]`
Schedule (cron)	`0 8 1 * * python mis.py`
Schedule (Windows)	Task Scheduler → Monthly → run `.bat`

Python for Finance III: Analysis

What it is & where it's used

This chapter is where Python stops being a fancy calculator and starts *answering business questions that have money attached*: **What will next quarter's revenue be? How risky is this portfolio? What actually drives our costs?**

Four techniques carry most of the load in a finance/FP&A/analytics seat:

Technique	The question it answers	Library
Regression	"How much does X move Y?" (drivers, sensitivity, beta)	`statsmodels`, `scikit-learn`
Time-series / forecasting	"What's the number for next month?"	`statsmodels` (ARIMA, ETS)
Monte Carlo simulation	"What's the *range* of outcomes, not one guess?"	`numpy`
Simple trend/seasonality decomposition	"Is this growth real or just December?"	`statsmodels`

Who pays for this: **FP&A analysts** (revenue/expense forecasts), **credit & risk teams** (default probability, VaR), **treasury** (cash-flow forecasting), **equity research** (beta, factor models), **corporate finance** (DCF with simulated scenarios). In India, this is the difference between a ₹6 LPA "Excel MIS" analyst and a ₹14–20 LPA "FP&A / analytics" analyst doing the same numbers with rigour.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you *regression theory* (R^2, t-stat, p-value) on a clean dataset in a stats exam, and *forecasting* as "take a growth rate and extrapolate." Industry needs the opposite skill: messy monthly data, and a defensible number you can put in a board deck.

The specific gaps:

- **College gave you point estimates; the business wants ranges.** "Revenue will be ₹52 Cr" is a liability. "₹48–56 Cr, 80% confidence" is analysis. Monte Carlo closes this and is almost never taught.
- **You learned regression to *test a hypothesis*; finance uses it to *drive a model*.** Nobody in FP&A cares about your p-value on a slide — they care that "every ₹1 of ad spend returns ₹3.20 of revenue, and here's the confidence interval."
- **Excel's forecast tools are a black box.** `FORECAST.ETS` gives a number with no diagnostics. When the CFO asks "why is the forecast flat?", you need to *see* the trend and seasonal components — statsmodels shows them.
- **Seasonality is ignored.** Freshers extrapolate a December spike into a full-year run-rate. Decomposition prevents this embarrassing mistake.

What "proficient" looks like

A job-ready person can, unaided:

- Load a monthly revenue/cost series into a pandas DataFrame with a proper `DatetimeIndex`.
- Fit an **OLS regression**, read the coefficients *as business levers*, and quote the confidence interval — not just recite R^2.
- Decompose a series into **trend + seasonality + residual** and say whether growth is structural.
- Fit a **Holt-Winters (ETS)** or **ARIMA** model and produce a forecast *with a prediction interval*.
- Build a **Monte Carlo** simulation of an NPV, portfolio return, or budget line and report P10/P50/P90.
- Say out loud what could make the forecast wrong (regime change, one-off events).

The bar is *interpretation*, not memorising library syntax. You can Google `.fit()`; you can't Google "what this coefficient means for our pricing."

Hands-on: how to actually do it

Setup:

```
import pandas as pd, numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
```

1. Regression (drivers & sensitivity)

Say you want to know how **ad spend** and **price discount** drive **units sold**.

```
df = pd.DataFrame({
    "units":    [1200, 1500, 1400, 1800, 2100, 1950, 2300, 2500],
    "adspend":  [50,   65,   60,   80,   95,   88,   110,  120],    # ₹ '000
    "discount": [5,    8,    6,    10,   12,   9,    14,   15],      # %
})

model = smf.ols("units ~ adspend + discount", data=df).fit()
print(model.summary())
print(model.params)          # the business levers
print(model.conf_int())      # 95% confidence interval per driver
```

Read it like this: the `adspend` coefficient (say `14.2`) means **each extra ₹1,000 of ad spend sells ~14 more units**. That sentence is what goes in the deck. `model.pvalues` under 0.05 = the driver is statistically real; `model.rsquared` = share of variation explained.

Predict for a planned scenario:

```
scenario = pd.DataFrame({"adspend": [130], "discount": [16]})
model.get_prediction(scenario).summary_frame(alpha=0.20)  # 80% interval
```

2. Time-series decomposition (is the trend real?)

```
rev = pd.read_csv("monthly_revenue.csv", parse_dates=["month"], index_col="month")
rev = rev.asfreq("MS")   # month-start frequency — statsmodels needs this

from statsmodels.tsa.seasonal import seasonal_decompose
dec = seasonal_decompose(rev["revenue"], model="additive", period=12)
dec.trend; dec.seasonal; dec.resid    # plot with dec.plot()
```

If `dec.trend` slopes up while `dec.seasonal` shows a December bump, you know the growth is structural and December is just seasonal — don't annualise December.

3. Forecasting with Holt-Winters (ETS)

Best default for monthly business data with trend + seasonality:

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing

fit = ExponentialSmoothing(
    rev["revenue"], trend="add", seasonal="add", seasonal_periods=12
).fit()

forecast = fit.forecast(6)        # next 6 months
print(forecast.round(0))
```

For a prediction interval, use ARIMA/SARIMAX:

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
sar = SARIMAX(rev["revenue"], order=(1,1,1),
              seasonal_order=(1,1,1,12)).fit(disp=False)
pred = sar.get_forecast(6)
pred.predicted_mean                    # the forecast
pred.conf_int(alpha=0.20)              # 80% band — this is what CFOs want
```

4. Monte Carlo simulation (the range, not the point)

Simulate NPV of a project where revenue and cost are uncertain:

```
np.random.seed(42)
N = 50_000
rev  = np.random.normal(500, 60, N)      # ₹ Lakh, mean 500, sd 60
cost = np.random.normal(320, 40, N)      # ₹ Lakh
disc = 0.12
years = 5

cashflow = rev - cost                     # annual, ₹ Lakh
npv = sum(cashflow / (1+disc)**t for t in range(1, years+1))

print(f"P10: ₹{np.percentile(npv,10):.0f} L")
print(f"P50: ₹{np.percentile(npv,50):.0f} L")
print(f"P90: ₹{np.percentile(npv,90):.0f} L")
print(f"P(NPV<0): {(npv<0).mean():.1%}")   # probability of loss
```

That last line — **"11% chance this project loses money"** — is worth more than any single-point NPV.

Worked example / mini-project: forecast + risk a ₹ retail chain's revenue

You run FP&A for a 20-store apparel chain. You have 36 months of revenue and need a **12-month forecast with a risk band** for the annual budget.

Step 1 — build the data (reproducible):

```
import pandas as pd, numpy as np
idx = pd.date_range("2023-01-01", periods=36, freq="MS")
np.random.seed(7)
trend = np.linspace(180, 260, 36)                        # ₹ Lakh, growing
season = 40*np.sin(np.arange(36)*2*np.pi/12) + \
         np.where(pd.Series(idx).dt.month==10, 60, 0)   # Diwali (Oct) spike
noise = np.random.normal(0, 10, 36)
rev = pd.Series((trend+season+noise).round(1), index=idx, name="revenue").to_frame()
rev = rev.asfreq("MS")
```

Step 2 — decompose to confirm the Diwali seasonality:

```
from statsmodels.tsa.seasonal import seasonal_decompose
seasonal_decompose(rev["revenue"], model="additive", period=12).seasonal.head(12)
# October seasonal factor is large & positive → real festive uplift
```

Step 3 — forecast next 12 months with an 80% band:

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
m = SARIMAX(rev["revenue"], order=(1,1,1), seasonal_order=(1,1,0,12)).fit(disp=False)
fc = m.get_forecast(12)
out = pd.concat([fc.predicted_mean.rename("base"),
                 fc.conf_int(alpha=0.20)], axis=1).round(0)
print(out)
annual_base = fc.predicted_mean.sum()
print(f"FY budget (base): ₹{annual_base:.0f} L")
```

Step 4 — Monte Carlo the annual total (because point forecasts lie):

```
resid_sd = m.resid.std()
sims = fc.predicted_mean.values[:, None] + \
       np.random.normal(0, resid_sd, (12, 20_000))
annual = sims.sum(axis=0)
print(f"Budget P10 ₹{np.percentile(annual,10):.0f} L | "
      f"P50 ₹{np.percentile(annual,50):.0f} L | "
      f"P90 ₹{np.percentile(annual,90):.0f} L")
```

The deliverable sentence: "*FY revenue budget of ₹3,180 L (base), with an 80% range of ₹2,980–3,390 L. October carries a ~₹60 L festive uplift; a stretch target above ₹3,390 L is < 10% likely.*" That is FP&A, not MIS.

How it's tested

Interview questions:

- "Difference between R^2 and adjusted R^2? Why prefer adjusted when adding variables?"
- "Your forecast says revenue drops next month but the business is growing — how do you debug it?" (Answer: check seasonality, check for a stationarity/differencing issue, look at residuals.)
- "What's a prediction interval and why is it more useful than a point forecast to a CFO?"
- "You have monthly sales for 2 years. Walk me through building a forecast." (They want: inspect → decompose → choose ETS/ARIMA → validate on a hold-out → forecast with interval.)
- "Explain Monte Carlo to a non-technical CEO in 30 seconds."

Practical/take-home tests:

- A CSV of 24–48 months of revenue: "*Forecast the next 6 months and justify your method.*" Graded on validation (did you hold out the last 6 months and check error?), not on the model name.
- A regression screen: "*Here are marketing spend and sales — quantify the ROI per rupee and state your confidence.*"
- A risk case: "*Given these cost/price distributions, what's the probability this launch is NPV-negative?*" — pure Monte Carlo.

Pro move in any test: **always back-test.** Fit on data up to month N-6, forecast the last 6 months you *do* have, and report MAPE. Showing `mean(abs((actual-pred)/actual))` proves the model, and freshers never do it.

Common mistakes & how pros avoid them

Mistake	Why it's wrong	The fix
No `DatetimeIndex` / no `.asfreq()`	statsmodels forecasts silently break or mis-align	Always `parse_dates`, set index, `.asfreq("MS")`
Reporting a point forecast only	Hides all risk; CFO gets blindsided	Always attach a prediction interval / P10–P90
Extrapolating a seasonal spike	Annualising Diwali/December = wild over-forecast	Decompose first; forecast on de-seasonalised logic
Confusing correlation with driver	"Ad spend correlates" ≠ "ad spend causes"	State assumptions; watch for omitted variables
Overfitting ARIMA orders	Great fit, garbage forecast	Back-test on hold-out; prefer simple orders
Monte Carlo with wrong distribution	Normal on bounded/skewed data lies	Use lognormal/triangular where appropriate; sanity-check tails
Not setting a random seed	Results change every run; not reproducible	`np.random.seed(...)` in any deliverable

Learn-it roadmap & resources

Realistic time-to-proficiency (assuming you can already do pandas from Chapters 04–05):

Phase	Time	Milestone
Regression with statsmodels	1 week	Read a `.summary()`, interpret coefficients as business levers
Decomposition + ETS/ARIMA	2 weeks	Forecast a real monthly series with a validated hold-out
Monte Carlo	3–4 days	Simulate an NPV/portfolio, report P10/P50/P90
Portfolio project	1 week	End-to-end forecast+risk memo on your own data

≈ **5–6 weeks part-time** to interview-ready.

Resources (free-first):

- **statsmodels docs** — the time-series and OLS example galleries are excellent and free.
- **"Forecasting: Principles and Practice" (Hyndman)** — free online, the standard text. It's in R but the concepts (ETS, ARIMA, back-testing) map 1:1 to statsmodels.
- **Kaggle** — "Store Sales" / retail time-series datasets to practice on real messy data.
- **QuantEcon** (free) — for Monte Carlo and numpy-heavy finance.
- **Certification worth having:** none specific to this; a **Google Data Analytics** or **CFA Level I quant** signal helps in India, but a GitHub repo with one clean forecast+Monte-Carlo notebook beats any cert in interviews.

Quick-reference

```
# --- Regression ---
import statsmodels.formula.api as smf
m = smf.ols("y ~ x1 + x2", data=df).fit()
m.summary(); m.params; m.pvalues; m.conf_int(); m.rsquared_adj
m.get_prediction(new_df).summary_frame(alpha=0.20)   # 80% interval

# --- Prep a series ---
s = df.set_index("month").asfreq("MS")["revenue"]

# --- Decompose ---
from statsmodels.tsa.seasonal import seasonal_decompose
seasonal_decompose(s, model="additive", period=12).plot()

# --- Forecast: Holt-Winters ---
from statsmodels.tsa.holtwinters import ExponentialSmoothing
ExponentialSmoothing(s, trend="add", seasonal="add",
                     seasonal_periods=12).fit().forecast(6)

# --- Forecast: SARIMAX (with interval) ---
from statsmodels.tsa.statespace.sarimax import SARIMAX
f = SARIMAX(s, order=(1,1,1), seasonal_order=(1,1,1,12)).fit(disp=False)
f.get_forecast(6).predicted_mean
f.get_forecast(6).conf_int(alpha=0.20)

# --- Monte Carlo NPV ---
np.random.seed(42)
cf = np.random.normal(180, 30, 50_000)
npv = sum(cf/(1.12)**t for t in range(1,6))
np.percentile(npv,[10,50,90]); (npv<0).mean()

# --- Back-test (always do this) ---
train, test = s[:-6], s[-6:]
pred = ExponentialSmoothing(train, trend="add", seasonal="add",
        seasonal_periods=12).fit().forecast(6)
mape = (abs((test-pred)/test)).mean()   # < 0.10 is good
```

Concept	Rule of thumb
R^2 vs adj-R^2	Use adjusted when comparing models with different # of variables
p-value	< 0.05 → driver is statistically real
Prediction interval	Always report; `alpha=0.20` = 80% band
MAPE	< 10% good, 10–20% acceptable, > 20% rethink
Monte Carlo output	Report P10 / P50 / P90 + probability of loss
Seasonality	Decompose before forecasting monthly data
Reproducibility	`np.random.seed()` on every deliverable

Power BI I: Data Model & DAX

What it is & where it's used

Power BI is Microsoft's business-intelligence tool: you load data, build a **data model** (tables + relationships), write **DAX** (Data Analysis Expressions) formulas, and publish interactive dashboards. Think of it as "Excel PivotTables on steroids" — it handles 10 million rows without choking, refreshes on a schedule, and lets a CFO slice revenue by region/month/product with one click.

Two separate skills live inside Power BI, and this chapter covers the engine, not the paint: - **Power Query (M)** — the ETL layer: import, clean, reshape. - **The data model + DAX** — relationships, star schema, and the calculation language.

Where finance roles use it:

Role	What they build in Power BI
FP&A analyst	Monthly MIS, budget-vs-actuals, rolling forecasts
Accounts / controller	AR/AP ageing, cash-flow dashboards, GST reconciliation summaries
Audit / internal audit	Exception reports, journal-entry testing, sampling views
Business finance	Revenue by SKU/region, margin walk, cohort analysis
Tax	GSTR-2B vs purchase-register match, ITC tracking

In India, "Power BI" now appears in ~30-40% of FP&A and MIS job posts, often paired with Excel and SQL. It is the single highest-leverage BI skill you can add to a finance CV.

The gap: why companies want this (and college didn't teach it)

MBA Finance and CA teach you *what* the numbers mean — variance analysis, ratios, cost sheets. They do **not** teach you to model data. The specific gaps:

1. **You think in one big flat table.** College spreadsheets have everything in one sheet. Real reporting needs a **star schema** — separate fact and dimension tables joined by keys. Nobody explains why.
2. **You confuse a column with a measure.** In Excel every cell is a value. In Power BI a *measure* is a formula that recalculates for whatever the user clicked. Freshers write everything as calculated columns and blow up the file size.
3. **You've never seen "filter context."** DAX's entire personality is that `CALCULATE` manipulates filters. This concept has no Excel equivalent, so it's the #1 thing people fail.
4. **Time-intelligence.** "Show me YTD, same period last year, MoM %" — done in Excel with fragile SUMIFS and helper columns. Power BI does it in one measure *if you have a proper date table*.

Employers pay for this because a good model turns a 3-day monthly-MIS grind into a 5-minute refresh. That is a direct cost saving they can measure.

What "proficient" looks like

A job-ready person can, unaided:

- Load 3-4 tables, set correct **data types**, and build a **star schema** with one-to-many relationships.
- Explain when to use a **calculated column** vs a **measure** (and default to measures).
- Create a **dedicated Date table** and mark it as such.
- Write measures using `SUM`, `SUMX`, `CALCULATE`, `DIVIDE`, `FILTER`, and the time-intelligence functions (`TOTALYTD`, `SAMEPERIODLASTYEAR`, `DATEADD`).
- Understand and control **filter context** — knows why a measure returns blank or the grand total in the wrong cell.
- Build a budget-vs-actual with variance and variance %.

That's roughly the bar for an FP&A analyst screen. You do **not** need row-level security or DAX Studio optimization for a first job.

Hands-on: how to actually do it

1. Load data

`Home → Get Data → Excel/CSV/SQL Server`. Pick your files, click **Transform Data** (opens Power Query), verify each column's data type (the icon left of the header), then **Close & Apply**.

Golden rule: **clean in Power Query, calculate in DAX.** Don't do transformations with DAX columns.

2. Build the star schema

A star schema = one **fact** table (transactions, many rows) surrounded by **dimension** tables (master data, few rows).

```
        Dim_Date
            |
Dim_Product — Fact_Sales — Dim_Customer
            |
        Dim_Region
```

- **Fact_Sales**: Date, ProductKey, CustomerKey, RegionKey, Qty, Amount.
- **Dimensions**: one row per product/customer/region/date.

In **Model view**, drag `Fact_Sales[ProductKey]` onto `Dim_Product[ProductKey]`. Power BI creates a **one-to-many** relationship (one product → many sales rows). Filters flow **from the "one" side to the "many" side** — this single fact explains 90% of DAX behaviour.

3. Measures vs columns

	Calculated column	Measure
Computed	Row-by-row, at refresh	On the fly, per filter context
Stored	In the model (eats RAM)	Not stored
Use for	A category/flag you slice BY	A number you aggregate/show
Example	`Margin Band = IF([Margin]>0.3,"High","Low")`	`Total Sales = SUM(Fact_Sales[Amount])`

Default to measures. Only make a column when you need to *filter or group by* the result.

4. Core DAX

```
Total Sales = SUM ( Fact_Sales[Amount] )

Total Qty = SUM ( Fact_Sales[Qty] )

-- SUMX iterates row by row, then sums. Use when the math is per-row.
Revenue = SUMX ( Fact_Sales, Fact_Sales[Qty] * Fact_Sales[Price] )

-- DIVIDE handles divide-by-zero (returns blank, not error)
Avg Price = DIVIDE ( [Revenue], [Total Qty] )
```

CALCULATE — the most important function. It evaluates an expression under *modified* filters:

```
-- Sales only for the South region, ignoring any region the user clicked
South Sales = CALCULATE ( [Total Sales], Dim_Region[Region] = "South" )

-- Sales ignoring ALL filters (useful for % of total)
All Sales = CALCULATE ( [Total Sales], ALL ( Fact_Sales ) )

Pct of Total = DIVIDE ( [Total Sales], [All Sales] )
```

Time-intelligence (needs a marked Date table):

```
-- Create a proper date table first
Dim_Date =
CALENDAR ( DATE ( 2023, 4, 1 ), DATE ( 2026, 3, 31 ) )
```

Add columns to it: `Year = YEAR([Date])`, `MonthNo = MONTH([Date])`, `MonthName = FORMAT([Date],"MMM")`, `FY = "FY" & YEAR([Date]) + IF(MONTH([Date])≥4,1,0)` (Indian April-March fiscal year). Then `Table tools → Mark as date table`.

```
Sales YTD = TOTALYTD ( [Total Sales], Dim_Date[Date], "31/03" )   -- Indian FY end

Sales LY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( Dim_Date[Date] ) )

Sales PM = CALCULATE ( [Total Sales], DATEADD ( Dim_Date[Date], -1, MONTH ) )

MoM % = DIVIDE ( [Total Sales] - [Sales PM], [Sales PM] )

YoY % = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )
```

Worked example / mini-project: Budget vs Actual MIS

Data (reproduce in two CSVs):

`Fact_Actuals` (sales transactions):

Date	Region	Product	Amount
2025-04-05	South	Widget A	120000
2025-04-18	North	Widget B	85000
2025-05-02	South	Widget A	140000
2025-05-20	West	Widget C	60000

`Fact_Budget` :

Month	Region	BudgetAmount
2025-04	South	130000
2025-04	North	90000
2025-05	South	150000
2025-05	West	55000

Steps: 1. Load both. Build `Dim_Date` and `Dim_Region` ; relate both facts to them. 2. Write measures:

```
Actual = SUM ( Fact_Actuals[Amount] )
Budget = SUM ( Fact_Budget[BudgetAmount] )
Variance = [Actual] - [Budget]
Variance % = DIVIDE ( [Variance], [Budget] )
Actual YTD = TOTALYTD ( [Actual], Dim_Date[Date], "31/03" )
```

3. Drop a matrix: Rows = `Region` , Values = `Actual` , `Budget` , `Variance` , `Variance %` . Add a `MonthName` slicer.

Expected April result: South Actual ₹1,20,000 vs Budget ₹1,30,000 → Variance −₹10,000 (−7.7%). North −₹5,000 (−5.6%). This is a live, filterable MIS you built in under 30 minutes — exactly the deliverable an FP&A

team asks a fresher to produce.

Conditional formatting: on `Variance %`, `Format → Cell elements → Background color → rules`: red below 0, green above. Now management sees misses at a glance.

How it's tested

Interview questions: - "Difference between a calculated column and a measure? When each?" - "What is a star schema and why not just one flat table?" - "Explain filter context. What does `CALCULATE` do?" - "Why do you need a separate date table instead of the date column in your fact?" - "`SUM` vs `SUMX` — give an example where only `SUMX` works." (Answer: `Qty * Price` per row, since prices differ by row.) - "Your measure shows blank / shows the grand total in every row — why?"

Practical test (very common): They hand you 2-3 raw CSVs/Excel files and a brief — "*Build a dashboard showing sales by region and month, with YoY growth and a budget-vs-actual variance*" — timed 45-90 minutes. Graders check: correct relationships (no many-to-many hacks), measures not columns, a proper date table, working time-intelligence, and clean visuals. Some firms give a broken .pbix and ask you to *fix the model*.

Common mistakes & how pros avoid them

Mistake	Fix
Everything in one flat table	Split into fact + dimensions (star schema)
Calculated columns for numbers you aggregate	Use measures — they respond to filters and don't bloat the file
Using the fact table's date column for time-intelligence	Always a **dedicated, marked Date table** with continuous dates
Gaps in the date table	Use `CALENDAR` / `CALENDARAUTO` so every day exists
`/` operator causing errors	Use `DIVIDE()` — safe divide-by-zero
Bi-directional / many-to-many relationships everywhere	Keep single-direction one-to-many; filter flows one → many
Hardcoding "last year" with a slicer	Use `SAMEPERIODLASTYEAR` / `DATEADD`
Wrong FY (Jan-Dec) for Indian data	Set year-end to `"31/03"` and build FY logic (Apr-Mar)
Blank measure results	Usually broken relationship or wrong filter direction — check Model view

Learn-it roadmap & resources

Time to proficiency: 4-6 weeks part-time to clear a fresher FP&A screen; 3 months to be genuinely fast.

Week	Focus
1	Power BI Desktop install (free), Get Data, Power Query cleaning
2	Model view, relationships, star schema
3	Measures vs columns, `SUM/SUMX/CALCULATE/DIVIDE`
4	Date table + time-intelligence (YTD, YoY, MoM)
5-6	Build 2 end-to-end dashboards; learn `FILTER`, `ALL`, `VAR`

Resources: - **Microsoft Learn — "Power BI Data Analyst" path** (free, official). - **SQLBI (Marco Russo/Alberto Ferrari)** — free YouTube + *dax.guide* function reference; the gold standard for DAX. - **Enterprise DNA** and **Kevin Stratvert** on YouTube (free, beginner-friendly). - **Certification: PL-300 (Microsoft Power BI Data Analyst)** — ~₹4,800 exam fee in India, well-recognised, worth it for finance CVs.

Power BI Desktop is **free**. Publishing to the cloud service needs Pro (~$10/month); for learning and interviews, Desktop alone is enough.

Quick-reference

```
-- Aggregation
SUM ( Table[Col] )
SUMX ( Table, Table[A] * Table[B] )      -- per-row then sum
DIVIDE ( num, den [, 0] )                -- safe divide

-- Filter manipulation
CALCULATE ( [Measure], filter1, filter2 )
CALCULATE ( [Measure], ALL ( Table ) )    -- remove filters
CALCULATE ( [Measure], FILTER ( Table, Table[x] > 100 ) )

-- Time-intelligence (needs marked Date table)
TOTALYTD ( [Measure], Dim_Date[Date], "31/03" )   -- Indian FY
CALCULATE ( [Measure], SAMEPERIODLASTYEAR ( Dim_Date[Date] ) )
CALCULATE ( [Measure], DATEADD ( Dim_Date[Date], -1, MONTH ) )
```

Concept	One-liner
Star schema	1 fact (transactions) + many dimensions (masters)
Relationship	One-to-many; filter flows one → many
Measure	Formula, recalculates per filter; use for numbers you show
Calculated column	Row-by-row, stored; use only to slice/group BY
Filter context	The set of filters active when a measure evaluates
CALCULATE	Changes the filter context, then evaluates
Date table	Dedicated, continuous, "Mark as date table" — required for time-intel
Indian FY	April 1 – March 31; set YTD year-end to `"31/03"`
``	

Muscle memory: clean in Power Query → model as a star → default to measures → one Date table → `CALCULATE` for everything conditional.

Power BI II: Finance Dashboards

What it is & where it's used

Chapter 07 covered the plumbing — Power Query, the star schema, the DAX basics. This chapter is where you turn that model into the three deliverables a finance team actually asks for: a **P&L (Profit & Loss) dashboard**, a **budget-vs-actual (BvA) variance report**, and a **KPI scorecard** — plus the two features that separate a "chart-maker" from a "BI person": **drill-through** and **row-level security (RLS)**.

Who builds these:

Role	What they ship in Power BI
FP&A analyst	Monthly BvA pack, rolling forecast, KPI board for the CFO
Management accountant	Cost-centre P&L, margin analysis, drill-through to cost lines
Financial analyst / MIS	Revenue/AR/AP dashboards, DSO/DPO trends
Controller / Finance Manager	Board deck built off one governed dataset, RLS by entity/region
Business finance partner	Region- or product-level P&L where each manager sees only their slice

In Indian companies (and MNC GCCs in Bengaluru/Hyderabad/Pune), the "monthly MIS pack" is very often a Power BI report refreshed off the ERP (SAP, Oracle, Tally exports, or a data warehouse). If you can own that pack, you are the finance-tech hire they cannot easily replace.

The gap: why companies want this (and college didn't teach it)

MBA and CA syllabi teach you to *read* a P&L and *compute* a variance. They stop there. Industry needs someone who can:

- Turn a raw ledger dump (10 lakh rows) into a **refreshable, filterable P&L** — not a static Excel that breaks every month-end.
- Build **variance % and favourable/adverse flags** that update automatically when a controller drops in new actuals.
- Let a Regional Head click a total and **drill through to the transactions behind it** without pinging the analyst.
- Show the **Sales VP only the South zone** and the **Finance Head everything** — from *one* report, using RLS, instead of maintaining 6 separate files.

College teaches the *accounting*; the employer pays for the *delivery mechanism*. Nobody in a BCom/MBA classroom builds a `DIVIDE()` variance measure or writes a DAX RLS filter. That is exactly the arbitrage.

What "proficient" looks like

A job-ready person can, unaided:

1. Build a **date-intelligence P&L** with subtotals (Revenue → Gross Profit → EBITDA → PBT) using measures, not hard-coded columns.

2. Write **variance measures** (Actual − Budget, Var %, favourable/adverse) that respect the sign convention (over-spend on cost is *adverse*, over-shoot on revenue is *favourable*).
3. Configure **drill-through** so right-click → "See transactions" jumps to a filtered detail page.
4. Set up **RLS roles**, write the DAX filter, and **test via "View as role."**
5. Use **conditional formatting** (red/green) and **KPI cards** that a CFO can read in 10 seconds.
6. Schedule a **refresh** and know the difference between Import and DirectQuery.

Hands-on: how to actually do it

Data model assumed

Star schema from Chapter 07: - `Fact_GL` (Date, AccountKey, CostCentreKey, Amount, Scenario) — Scenario = "Actual" or "Budget" - `Dim_Account` (AccountKey, Account, PLGroup, PLGroupSort, SignConvention) - `Dim_Date`, `Dim_CostCentre` (CostCentreKey, CostCentre, Region, Manager, ManagerEmail)

1. Core P&L measures (DAX)

```
Actual =
CALCULATE ( SUM ( Fact_GL[Amount] ), Fact_GL[Scenario] = "Actual" )

Budget =
CALCULATE ( SUM ( Fact_GL[Amount] ), Fact_GL[Scenario] = "Budget" )

-- Revenue and cost stored with natural signs (revenue +, expense −)
Revenue =
CALCULATE ( [Actual], Dim_Account[PLGroup] = "Revenue" )

Gross Profit =
CALCULATE ( [Actual],
    Dim_Account[PLGroup] IN { "Revenue", "COGS" } )

EBITDA =
CALCULATE ( [Actual],
    NOT Dim_Account[PLGroup] IN { "Depreciation", "Interest", "Tax" } )
```

Build the P&L statement itself with a **Matrix visual**: `Dim_Account[PLGroup]` on rows (sorted by `PLGroupSort`), `[Actual]` and `[Budget]` on values. This gives you a real statement layout instead of a bar chart.

2. Budget-vs-Actual variance

```
Variance = [Actual] - [Budget]

Variance % =
DIVIDE ( [Actual] - [Budget], [Budget] )   -- DIVIDE avoids /0 error

-- Sign-aware flag: cost overrun is adverse, revenue beat is favourable
Var Status =
VAR IsRevenue =
    SELECTEDVALUE ( Dim_Account[SignConvention] ) = "Income"
VAR Fav = IF ( IsRevenue, [Variance] ≥ 0, [Variance] ≤ 0 )
RETURN IF ( Fav, "Favourable", "Adverse" )
```

Conditional formatting on the Variance column: Format → Cell elements → Background color → *Rules* → if `Var Status` = "Adverse" then red, else green. Now the CFO sees the problem lines instantly.

3. KPI measures & cards

```
Gross Margin % = DIVIDE ( [Gross Profit], [Revenue] )

EBITDA Margin % = DIVIDE ( [EBITDA], [Revenue] )

-- Month-over-month growth using the date dimension
Revenue MoM % =
VAR Prev =
    CALCULATE ( [Revenue], DATEADD ( Dim_Date[Date], -1, MONTH ) )
RETURN DIVIDE ( [Revenue] - Prev, Prev )

Budget Achievement % = DIVIDE ( [Actual], [Budget] )
```

Drop each into a **Card** or **KPI visual**. For the KPI visual: Value = `[Revenue]`, Target = `[Budget]`, Trend axis = `Dim_Date[Month]`.

4. Drill-through

1. Add a new page, rename it `Transaction Detail`.
2. In the **Visualizations → Drill through** well, drag the field you want to drill *by*, e.g. `Dim_Account[Account]` (and/or `Dim_CostCentre[CostCentre]`).
3. Power BI auto-adds a "back" button. Put a Table visual on the page: Date, Account, Cost Centre, Amount, Voucher No.
4. On the summary P&L page, **right-click any total → Drill through → Transaction Detail.** The detail page opens filtered to that account.
5. Toggle **"Keep all filters" = On** so region/month context carries over.

5. Row-level security (RLS)

```
Modeling tab → Manage roles → Create
```

Role: **Region_Manager**

```
-- Filter Dim_CostCentre so each user sees only their region
[Region] =
    LOOKUPVALUE (
        UserMap[Region],
        UserMap[Email], USERPRINCIPALNAME ()
    )
```

Or the simpler "manager sees own cost centres" pattern:

```
-- Applied on Dim_CostCentre
Dim_CostCentre[ManagerEmail] = USERPRINCIPALNAME ()
```

Test before publishing: **Modeling** → **View as** → **tick Region_Manager**, optionally enter a test email under "Other user." The whole report re-filters. After **Publish**, in the Power BI Service go to the dataset → **Security** → add the Azure AD users/groups to the role. Without that last step, RLS does nothing in the Service.

> One `USERPRINCIPALNAME()` returns the logged-in user's email — that is what makes one report serve 50 managers.

Worked example / mini-project

Scenario: "Bharat Consumer Products Pvt Ltd" — FY 2025-26, three regions. You are handed a GL export. Reproduce this in ₹ lakhs.

`Fact_GL` (sample, Actual scenario, April):

Date	Account	PLGroup	CostCentre	Region	Scenario	Amount (₹ L)
2025-04-30	Product Sales	Revenue	CC-North	North	Actual	420
2025-04-30	Product Sales	Revenue	CC-South	South	Actual	510
2025-04-30	Raw Material	COGS	CC-North	North	Actual	−250
2025-04-30	Raw Material	COGS	CC-South	South	Actual	−290
2025-04-30	Salaries	Employee	CC-Head	HO	Actual	−95
2025-04-30	Freight	Distribution	CC-North	North	Actual	−38

Budget rows exist with Scenario = "Budget" (e.g. North Sales budget 400, South 480).

Build the P&L matrix (South region slicer applied):

P&L line	Actual	Budget	Variance	Var %	Status
Revenue	510	480	+30	+6.3%	Favourable
COGS	−290	−270	−20	+7.4%	Adverse
Gross Profit	**220**	**210**	**+10**	**+4.8%**	Favourable
Distribution	−22	−20	−2	+10%	Adverse
EBITDA (region)	**198**	**190**	**+8**	**+4.2%**	Favourable

KPI cards: Gross Margin % = 220/510 = **43.1%**; Budget Achievement (Rev) = 510/480 = **106.3%**.

Drill-through demo: the Regional Head clicks the COGS −290, drills through, and sees the Raw Material vouchers making up that number — spots that one supplier invoice caused the ₹20 L overrun.

RLS demo: publish with role `Region_Manager` on `ManagerEmail = USERPRINCIPALNAME()`. The South manager logs in → the entire report shows only 510 revenue, never North's 420. The Finance Head is in no role → sees the consolidated ₹930 revenue.

Reproduce it: paste the two tables into Excel, load via Get Data → Excel, mark `Dim_Date`, write the six measures above, and you have a working pack in ~30 minutes.

How it's tested

Interview questions: - "How do you build a P&L in Power BI without hard-coding rows for each line item?" (Answer: measures + Account dimension with a sort/group column.) - "Actuals came in over budget on marketing spend — how does your report show that as *adverse* while a revenue beat shows *favourable*?" (Sign convention logic.) - "Difference between a slicer filter and RLS?" (Slicer = user choice, removable; RLS = enforced at dataset, user cannot bypass.) - "Import vs DirectQuery — which for a monthly MIS pack?" (Import; faster, scheduled refresh.) - "What does `USERPRINCIPALNAME()` return and why does it matter for RLS?"

Practical/assessment test: Very common — a 45–60 min take-home or on-site. Typical brief: "*Here's a GL extract and a budget file. Build a P&L with Actual/Budget/Variance, add a Gross Margin KPI card, enable drill-through to transactions, and create an RLS role so a region manager sees only their region. Publish and share a test link.*" They grade: correct variance signs, no `/0` errors, working drill-through, and — the pass/fail line — whether RLS actually restricts data when they "View as role."

Common mistakes & how pros avoid them

Mistake	Fix
Hard-coding P&L rows as separate measures per line	Use one Account dimension + PLGroup + sort column; drive rows dynamically
`Variance % = Var / Budget` throwing errors on zero budget	Always use `DIVIDE()`
Variance sign wrong (cost overrun shown green)	Store a `SignConvention` and build a sign-aware `Var Status`
Building 6 separate files, one per region	One dataset + RLS
Forgetting to add users to the role in the Service	RLS in Desktop alone does nothing; assign members under dataset → Security
Totals in matrix look wrong ("the total isn't the sum of rows")	That's measure context — verify with a test filter, don't "fix" by summing columns
Drill-through page not filtered	Ensure the drill-by field is in the Drill through well and "Keep all filters" is On
Refresh fails after month-end	Use a parameterised folder/date path in Power Query, not a fixed filename

Learn-it roadmap & resources

Time to proficient: ~3–4 weeks part-time if you already did Chapter 07's modelling.

Week	Focus
1	P&L matrix + core measures; conditional formatting
2	Budget-vs-Actual, DIVIDE, sign-aware variance, KPI cards
3	Drill-through, bookmarks, tooltips, a clean CFO-ready layout
4	RLS (static + dynamic), publish to Service, scheduled refresh

Resources: - Microsoft Learn — "Power BI data analyst" learning path (free). - SQLBI (Marco Russo / Alberto Ferrari) — free articles/YouTube on variance, RLS, time intelligence. The gold standard for DAX. - Book: *The Definitive Guide to DAX* (Russo & Ferrari) for depth. - **Certification:** Microsoft **PL-300: Power BI Data Analyst Associate** — the one recruiters recognise; covers modelling, DAX, RLS, and publishing. ~₹4,800 exam fee in India, strongly worth listing on your CV.

Quick-reference

```
Actual   = CALCULATE(SUM(Fact_GL[Amount]), Fact_GL[Scenario]="Actual")
Budget   = CALCULATE(SUM(Fact_GL[Amount]), Fact_GL[Scenario]="Budget")
Variance = [Actual] - [Budget]
Var %    = DIVIDE([Actual]-[Budget], [Budget])
GM %     = DIVIDE([Gross Profit], [Revenue])
MoM %    = DIVIDE([Revenue]-CALCULATE([Revenue],DATEADD(Dim_Date[Date],-1,MONTH)),
                  CALCULATE([Revenue],DATEADD(Dim_Date[Date],-1,MONTH)))
RLS      = Dim_CostCentre[ManagerEmail] = USERPRINCIPALNAME()
```

Task	Click-path
P&L layout	Matrix visual → PLGroup on Rows, Actual/Budget on Values
Red/green variance	Format → Cell elements → Background color → Rules
Drill-through	Detail page → drag field to *Drill through* well → right-click total on summary
Create RLS role	Modeling → Manage roles → new role → DAX filter
Test RLS	Modeling → View as → tick role
Activate RLS live	Service → dataset → Security → add users to role
Refresh	Service → dataset → Schedule refresh (Import mode)

Sign rule: Income favourable when Actual ≥ Budget; Expense favourable when Actual ≤ Budget. Always `DIVIDE()`, never `/`. One dataset + RLS beats many files, every time.

Tableau & Choosing Your BI Tool

What it is & where it's used

Tableau is a drag-and-drop data-visualization and dashboarding tool. You connect it to a data source (Excel, CSV, SQL Server, Snowflake, Google Sheets), drag fields onto "shelves," and it draws charts. String a few charts together and you have an interactive dashboard that a CFO can filter by region, month, or product without touching a formula.

In finance/accounts roles, BI tools live wherever numbers need to be *seen* repeatedly by non-analysts:

Role	What they build in Tableau/Power BI
FP&A analyst	Budget vs actual dashboard, refreshed monthly
Revenue/AR analyst	Ageing dashboard, DSO trend, collection heatmap
Cost/plant controller	Cost-per-unit, variance waterfall by cost centre
Treasury	Cash-position dashboard across bank accounts
Tax/compliance	GST liability vs ITC trend, filing-status tracker
MIS executive	The monthly management deck, auto-refreshed

The pattern is identical everywhere: raw data → clean model → chart → published dashboard that others self-serve. Tableau is the market-leading tool for the "chart + dashboard" half. Power BI is its main rival. Excel is the third leg — and still the one you'll actually use most days.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you to *analyse* a company. It does not teach you to build a thing that 40 people open every Monday. The gap is **repeatable, self-service reporting**:

- College output = a one-off answer in a slide. Industry output = a *living dashboard* that refreshes itself and lets the viewer drill down without emailing you.
- Nobody in a B-school case study asks "how will this refresh next month?" or "can the regional head filter to just South zone?" Every real MIS job asks exactly that.
- Excel charts break when data grows or columns move. BI tools separate the *data model* from the *view*, so the report survives new months of data. That architectural idea — model once, visualise many — is the actual skill, and it's never taught.

Companies pay for BI because manual monthly reporting is a tax: a junior spends three days rebuilding the same deck. A good dashboard turns three days into a five-minute refresh. That saving is the job.

What "proficient" looks like

The bar an employer tests for — what a job-ready person does unaided:

- **Connect and shape:** pull data from Excel/SQL, fix data types, create a clean join or relationship, without needing pre-cleaned data.

- **Calculated fields:** write a profit-margin, YoY-growth, or running-total calc from scratch.
- **The right chart:** knows a trend = line, a part-to-whole = bar (not pie), a variance = waterfall or bar with reference line. Doesn't decorate.
- **Interactivity:** adds filters, parameters, and drill-downs so the viewer answers their own follow-up.
- **Publish & control access:** pushes to Tableau Server/Cloud or Power BI Service, sets a refresh schedule, and shares to the right people (not "everyone in company").
- **Tool judgment:** can say *why* this report is Tableau and that one is just an Excel PivotTable — and not over-engineer.

Hands-on: how to actually do it

1. Connecting data (Tableau Public / Desktop) `Connect → Microsoft Excel → select file → drag sheet to canvas`. Tableau splits fields into **Dimensions** (text/date — the "by what") and **Measures** (numbers — the "how much"). Sales *by* Region *over* Month: drag `Order Date` to Columns, `Sales` to Rows, `Region` to Colour.

2. A calculated field (`Analysis → Create Calculated Field`):

```
// Profit Margin %
SUM([Profit]) / SUM([Sales])
```

```
// YoY Sales Growth (table calc alternative)
(SUM([Sales]) - LOOKUP(SUM([Sales]), -1)) / ABS(LOOKUP(SUM([Sales]), -1))
```

```
// Bucketing AR ageing
IF [Days Overdue] ≤ 30 THEN "0-30"
ELSEIF [Days Overdue] ≤ 60 THEN "31-60"
ELSEIF [Days Overdue] ≤ 90 THEN "61-90"
ELSE "90+" END
```

3. LOD expression (Tableau's signature feature — compute at a different grain than the view):

```
// Sales per customer, regardless of what's on the view
{ FIXED [Customer Name] : SUM([Sales]) }
```

4. The Power BI / DAX equivalent (same logic, different language). In Power BI you write measures:

```
Total Sales = SUM(Sales[Amount])

Profit Margin % =
DIVIDE(SUM(Sales[Profit]), SUM(Sales[Amount]))

YoY Growth % =
VAR CurYr = [Total Sales]
VAR PrevYr = CALCULATE([Total Sales], DATEADD('Date'[Date], -1, YEAR))
RETURN DIVIDE(CurYr - PrevYr, PrevYr)
```

5. The Excel equivalent (for when a dashboard is overkill). A PivotTable + slicer *is* a mini-BI dashboard. To pull a value into a management sheet:

```
=XLOOKUP(A2, Data!$A:$A, Data!$D:$D, "Not found")

=SUMIFS(Sales[Amount], Sales[Region], "South", Sales[Month], "Apr")
```

6. Publishing (Tableau): `Server → Publish Workbook → sign in to Tableau Cloud → choose Project (folder) → set Permissions → set Data Source refresh schedule (e.g. daily 7am)`. Viewers open a URL; no Tableau install needed.

7. Publishing (Power BI): `Home → Publish → pick Workspace`. Then in the browser: `Workspace → Dataset → Schedule refresh → set gateway + time`. Share via `App` or add users to the workspace.

Worked example / mini-project

Build: an Accounts Receivable ageing dashboard for an Indian mid-size firm.

Sample data (`invoices.csv`), reproduce with ~200 rows like:

Invoice	Customer	Region	Invoice Date	Due Date	Amount (₹)	Paid?
INV-1001	Reliance Retail	West	2026-04-05	2026-05-05	4,50,000	No
INV-1002	Infosys	South	2026-05-12	2026-06-11	2,20,000	No
INV-1003	Tata Steel	East	2026-03-01	2026-03-31	8,75,000	No

Steps:

1. Connect the CSV in Tableau. Create `Days Overdue = DATEDIFF('day', [Due Date], TODAY())`.
2. Create the ageing bucket calc from section 4 above.
3. **Chart 1 (bar):** `Ageing Bucket` on Columns, `SUM(Amount)` on Rows. Instantly shows ₹ stuck in 90+.
4. **Chart 2 (heatmap):** `Region` on Rows, `Ageing Bucket` on Columns, `SUM(Amount)` on Colour. Red = West's 90+ bucket is your problem child.
5. **KPI:** `Total Outstanding = SUM([Amount])` filtered to `Paid? = "No"`; add `DSO ≈ (AR / Credit Sales) × 365` as a text tile.
6. Add a **Region filter** and a **Customer** quick filter. Combine all four onto one Dashboard sheet.

7. Publish to Tableau Cloud, schedule daily refresh, share to the collections team's email group only.

Outcome: the collections head opens one URL, filters to "West / 90+", and sees ₹8.75L from Tata Steel is the single biggest overdue — a decision that used to need a half-day Excel pull.

How it's tested

Interviews mix concept questions with a **live build test**.

Typical questions: - "When would you use Tableau over an Excel PivotTable?" (Answer: recurring, multi-user, needs scheduled refresh and interactivity; Excel for one-off analysis or heavy ad-hoc modelling.) - "Difference between a filter and a parameter?" (Filter limits data shown; parameter is a user-input value you plug into calcs/what-ifs.) - "What's an LOD expression / why FIXED?" (Compute at a grain independent of the view.) - "Bar vs pie — when?" (Almost always bar; pie only for 2–3 slices summing to 100%.) - "Star schema vs one flat table — why care?" (Fact + dimension tables; faster, cleaner measures, avoids duplication.)

The practical test (60–90 min, take-home or live): "Here's a messy sales/AR CSV. Build a dashboard showing [trend + breakdown + one KPI], make it filterable by region, and tell us the top insight." They score: correct chart choice, working calculated field, interactivity, cleanliness (no chartjunk), and whether you *found the insight*, not just drew boxes.

Common mistakes & how pros avoid them

- **Pie charts and 3D everything.** Pros default to bars and lines; colour carries one meaning, not five.
- **Building on a flat, un-modelled table.** Learn a simple star schema (one fact table, dimension tables for Date/Region/Customer). Makes measures correct and fast.
- **Over-engineering.** If two people need it once, it's an Excel PivotTable, not a Tableau server workbook. Pros match tool to lifespan.
- **No refresh plan.** A dashboard nobody scheduled dies in month two. Set the schedule *before* you share.
- **Sharing to "everyone."** Finance data is sensitive; set row/folder-level permissions.
- **Hard-coding dates/values in calcs.** Use `TODAY()`, parameters, and relative-date filters so it survives next month.
- **Ignoring performance.** Extracts (not live) for large data; filter at source; don't drag 2M rows into Tableau Public.

Learn-it roadmap & resources

Realistic time-to-proficiency: **3–5 weeks** part-time to job-ready if you already know Excel PivotTables.

Week	Focus
1	Excel PivotTables + charts mastery (the foundation)
2	Tableau Public: connect, dimensions/measures, basic charts
3	Calculated fields, LOD, filters, parameters, dashboards
4	Publishing, refresh, permissions; build 2 portfolio dashboards
5	Learn Power BI + DAX basics so you're tool-agnostic

Resources: - **Tableau Public** — free desktop tool + free portfolio hosting. Build here and put the link on your CV. - Tableau's official free training videos + "Makeover Monday" community datasets for practice. - **Power BI Desktop** — free download; the Microsoft Learn "PL-300" path is the best free curriculum. - Certifications: **Tableau Desktop Specialist** (~US$100) or **Microsoft PL-300 (Power BI Data Analyst)** — PL-300 is more recognised in Indian corporates because most run Microsoft 365.

Pick Power BI first if you're India-targeting and cost-sensitive: it's cheaper for employers, bundled with Office 365, and dominates Indian mid-market finance teams. Add Tableau for MNCs and analytics-heavy shops.

Quick-reference

Need	Excel	Tableau	Power BI
One-off analysis	✅ Best	Overkill	Overkill
Recurring multi-user dashboard	Weak	✅	✅
Scheduled auto-refresh	No	✅	✅
Cost for employer	Owned	₹₹₹	₹ (in O365)
Language for calcs	Formulas	Calc fields	DAX
Ad-hoc modelling / what-if	✅ Best	Weak	Medium

Chart cheat-sheet: trend → line · comparison → bar · part-to-whole → stacked bar (not pie) · variance → waterfall · relationship → scatter · distribution → histogram · concentration → heatmap.

Tableau essentials: Dimensions = "by what", Measures = "how much" · Filter = limit data · Parameter = user input · `{FIXED [X] : SUM([Y])}` = LOD · Extract = fast snapshot, Live = real-time.

Publish flow: clean model → build sheets → assemble dashboard → set filters → publish to Cloud/Service → schedule refresh → set permissions → share URL.

Decision rule: *Will 3+ people open this repeatedly?* → BI tool. *Just me, just now?* → Excel PivotTable.

Mini-project: An End-to-End Finance MIS Dashboard

What it is & where it's used

An **MIS (Management Information System) dashboard** is the one-page monthly picture that finance hands to a CEO/CFO/promoter: revenue vs target, gross margin, cash position, receivables ageing, top customers, expense heads, and where the business is bleeding. It sits on top of raw General Ledger (GL) exports and sales registers and turns thousands of transaction rows into 8-10 numbers a decision-maker actually reads.

This is the single most common "real" deliverable in Indian finance jobs. It's owned by **MIS Analysts, FP&A Analysts, Business Finance/Finance Business Partners, Financial Analysts, and Accounts Managers** in every SME, startup, and mid-cap. If you can build one unaided, you are immediately useful on day one — most freshers cannot.

The chapter builds the **full pipeline**: raw GL + sales CSV → a clean data model (star schema) → a Power BI / Excel dashboard → a monthly-reproducible process. Do it once here and you own a portfolio piece.

The gap: why companies want this (and college didn't teach it)

College teaches you to *read* a P&L and Balance Sheet that someone else prepared. Industry needs you to **manufacture** the P&L from a messy 40,000-row Tally/SAP dump — every month, on the 3rd working day, with zero errors, and explain the variances.

The specific gaps this closes:

College taught	Job needs
Interpret a finished P&L	Build P&L from a raw GL trial-balance export
Ratios in isolation	Ratios trended month-on-month with drill-down
"Prepare a statement" (once)	A *repeatable* process that runs every month unattended
One clean textbook dataset	Reconciling GL total to sales register, catching duplicates
Static answers	A live dashboard the CFO can filter by branch/product/month

The unstated skill is **data plumbing**: mapping a ledger's chart-of-accounts to reporting heads, joining sales to a calendar, and building it so next month is a one-click refresh, not a rebuild.

What "proficient" looks like

A job-ready person can, given a raw GL CSV and a sales register, unaided:

- Build a **cost-centre/account-head mapping table** so GL accounts roll up into P&L reporting lines.
- Construct a **star schema**: fact tables (GL, Sales) + dimension tables (Calendar, Account, Customer, Product).
- Write measures that produce **Revenue, COGS, Gross Margin %, EBITDA, MoM and YoY growth, MTD/YTD** correctly.
- Produce a **receivables ageing** bucket (0-30/31-60/61-90/90+).

- Tie the dashboard revenue back to the GL to the rupee (reconciliation).
- Refresh next month by **dropping in a new file** — no formula rewrites.
- Explain the top 3 variances in plain English.

Hands-on: how to actually do it

Raw inputs. Two files exported from Tally/ERP:

`gl.csv` — `Date, Voucher, AccountCode, AccountName, CostCentre, Debit, Credit` `sales.csv` — `Date, Invoice, CustomerCode, Customer, ProductCode, Product, Qty, Amount, GSTAmount`

And a mapping file you build by hand: `map_accounts.csv` — `AccountCode, ReportLine, ReportGroup` (e.g. `4001 → Sales → Revenue`, `5001 → Raw Material → COGS`, `6002 → Salaries → OpEx`).

Step 1 — Clean & shape in Python (repeatable)

```
import pandas as pd

gl   = pd.read_csv("gl.csv", parse_dates=["Date"])
mp   = pd.read_csv("map_accounts.csv")
sal  = pd.read_csv("sales.csv", parse_dates=["Date"])

# GL: signed amount (Dr +, Cr -) then map to report lines
gl["Amount"] = gl["Debit"].fillna(0) - gl["Credit"].fillna(0)
gl = gl.merge(mp, on="AccountCode", how="left")

# Catch unmapped accounts BEFORE they silently drop from the P&L
missing = gl[gl["ReportLine"].isna()]["AccountName"].unique()
assert len(missing) == 0, f"Unmapped accounts: {missing}"

gl["Month"] = gl["Date"].dt.to_period("M").astype(str)

# Monthly P&L pivot (income lines are credit-heavy so flip sign for reporting)
pnl = gl.groupby(["Month","ReportGroup","ReportLine"])["Amount"].sum().reset_index()
pnl.to_csv("fact_gl.csv", index=False)
sal.to_csv("fact_sales.csv", index=False)
```

Step 2 — Reconcile (never skip this)

```
gl_sales   = -gl.loc[gl["ReportLine"]=="Sales","Amount"].sum()   # credit → positive
reg_sales  = sal["Amount"].sum()
print("GL:", gl_sales, "Register:", reg_sales, "Diff:", gl_sales - reg_sales)
```

If the diff isn't zero (or a known GST/rounding gap), stop and investigate before dashboarding.

Step 3 — SQL alternative for the same rollup

```
SELECT  strftime('%Y-%m', g.Date)          AS Month,
        m.ReportGroup,
        m.ReportLine,
        SUM(g.Debit - g.Credit)            AS Amount
FROM    gl g
JOIN    map_accounts m ON g.AccountCode = m.AccountCode
GROUP BY Month, m.ReportGroup, m.ReportLine
ORDER BY Month;
```

Receivables ageing straight from open invoices:

```
SELECT CASE
         WHEN julianday('now') - julianday(Date) <= 30  THEN '0-30'
         WHEN julianday('now') - julianday(Date) <= 60  THEN '31-60'
         WHEN julianday('now') - julianday(Date) <= 90  THEN '61-90'
         ELSE '90+' END                    AS Bucket,
       SUM(Amount)                         AS Outstanding
FROM sales WHERE Paid = 0 GROUP BY Bucket;
```

Step 4 — Model it in Power BI (star schema)

Load `fact_gl`, `fact_sales`, and dimensions. Build a **Calendar** table and DAX measures:

```
Revenue     = CALCULATE( -SUM(fact_gl[Amount]), fact_gl[ReportLine]="Sales" )
COGS        = CALCULATE(  SUM(fact_gl[Amount]), fact_gl[ReportGroup]="COGS" )
Gross Margin   = [Revenue] - [COGS]
Gross Margin % = DIVIDE([Gross Margin], [Revenue])

Revenue LM  = CALCULATE([Revenue], DATEADD(Calendar[Date], -1, MONTH))
MoM %       = DIVIDE([Revenue] - [Revenue LM], [Revenue LM])
Revenue YTD = TOTALYTD([Revenue], Calendar[Date])
```

Step 5 — The Excel-only version (no Power BI)

Same result with `SUMIFS` off a helper table:

```
=SUMIFS(fact_gl[Amount], fact_gl[ReportLine],"Sales", fact_gl[Month],$B$1)*-1
=XLOOKUP([@AccountCode], map[AccountCode], map[ReportLine], "UNMAPPED")
=LET(rev, B5, cogs, B6, (rev-cogs)/rev)          // Gross Margin %
```

Use a PivotTable on `fact_gl` (Rows = ReportGroup/ReportLine, Columns = Month, Values = Sum of Amount) as the P&L engine, then a clean dashboard sheet referencing it with `GETPIVOTDATA`.

Worked example / mini-project

Fictional co: Surya Traders Pvt Ltd, April 2026. Raw GL rolls up to:

ReportGroup	ReportLine	Apr (₹)
Revenue	Sales	82,50,000
COGS	Raw Material	44,10,000
COGS	Freight Inward	3,20,000
OpEx	Salaries	11,40,000
OpEx	Rent	2,50,000
OpEx	Power & Fuel	1,80,000
Other	Interest	1,95,000

Derived dashboard KPIs:

Metric	Formula	Value
Revenue	—	₹82,50,000
COGS	44.10L + 3.20L	₹47,30,000
Gross Margin	Rev − COGS	₹35,20,000
Gross Margin %	35.20 / 82.50	**42.7%**
OpEx	11.40+2.50+1.80	₹15,70,000
EBITDA	GM − OpEx	₹19,50,000
EBITDA %	19.50 / 82.50	**23.6%**
PBT	EBITDA − Interest	₹17,55,000

Reconciliation: sales register total = ₹82,50,000 → matches GL Sales line to the rupee. Green tick.

Receivables ageing (from sales register, unpaid):

Bucket	₹	%
0-30	9,80,000	55%
31-60	4,60,000	26%
61-90	2,10,000	12%
90+	1,30,000	7%

MoM: March revenue was ₹78,00,000 → MoM = (82.50−78.00)/78.00 = **+5.8%**. Dashboard shows the KPI card in green with the arrow.

The one-page dashboard then has: 4 KPI cards (Revenue, GM%, EBITDA, PBT) with MoM arrows, a 12-month revenue+margin trend line, a top-10-customers bar, an expense-head donut, and the ageing bar. Filters:

Month, CostCentre. Next month: replace `gl.csv` and `sales.csv`, rerun the script / hit Refresh — done.

How it's tested

Practical assessment (the real gate): You're handed a raw GL CSV and told "*Build me a monthly P&L and a gross-margin trend by 5 pm.*" They watch whether you (a) map accounts, (b) reconcile, (c) get the sign convention right, (d) make it refreshable.

Common interview questions: - "Walk me through how you'd build an MIS from a Tally export." (They want: export → map → model → reconcile → visualize → refresh.) - "Your dashboard revenue is ₹2L higher than the sales register. What do you check first?" (Duplicate voucher, credit note not netted, GST mixed into amount, unmapped account.) - "Difference between MTD and YTD, and how you'd compute both." - "How do you make this reproducible next month without rebuilding?" - "What's the difference between gross margin and EBITDA margin?"

Timed Excel/SQL screen: SUMIFS across a criteria set, XLOOKUP with a fallback, a GROUP BY rollup, an ageing CASE statement. 30-45 minutes.

Common mistakes & how pros avoid them

Mistake	Fix
Unmapped GL accounts silently dropped from P&L	`assert` / `XLOOKUP(…,"UNMAPPED")` check every refresh
Wrong Dr/Cr sign → income shows negative	Fix sign convention once in the model, flip income for display
GST amount counted as revenue	Keep taxable value and GST in separate columns; report net
Hard-coded month → breaks next cycle	Use a Calendar dimension + slicer, never type "April" in formulas
No reconciliation → confident but wrong numbers	Always tie dashboard revenue to the GL and sales register
Rebuilding from scratch monthly	Parameterise the file path; make refresh one click
Averaging percentages (avg of GM%s ≠ total GM%)	Compute ratios from summed numerator/denominator (DIVIDE)
Credit notes / returns ignored	Net them before totalling revenue

Learn-it roadmap & resources

Time to proficiency: ~3-4 weeks of focused evenings if you already know basic Excel.

- **Week 1** — Excel: SUMIFS, XLOOKUP, PivotTables, Power Query (get-and-transform). Build the P&L from CSV.
- **Week 2** — Data modelling: star schema, Calendar tables, DAX basics. Rebuild in Power BI (free desktop).
- **Week 3** — SQL: SELECT/JOIN/GROUP BY/CASE. Redo the rollup in SQLite/DB Browser (free).
- **Week 4** — Polish: reconciliation, ageing, variance commentary; write a one-page README so it's reproducible.

Resources: Power BI Desktop (free); DB Browser for SQLite (free); Microsoft Learn "Power BI" path (free); ICAI/company Tally exports for realistic data. **Certifications that carry weight:** Microsoft **PL-300 (Power BI Data Analyst)** and any structured SQL course. Build one real dashboard end-to-end and put the `.pbix` + screenshots in a portfolio — that beats any certificate in interviews.

Quick-reference

Need	Tool / snippet
Map account → report line	`=XLOOKUP(code, map[Code], map[Line], "UNMAPPED")`
Monthly P&L rollup	`SUMIFS(Amount, Line, "Sales", Month, B1)`
SQL rollup	`SUM(Debit-Credit) … GROUP BY Month, ReportLine`
Revenue (DAX)	`CALCULATE(-SUM(Amount), Line="Sales")`
Gross Margin %	`DIVIDE([Revenue]-[COGS],[Revenue])`
MoM %	`DIVIDE([Rev]-[Rev LM],[Rev LM])`
YTD	`TOTALYTD([Revenue], Calendar[Date])`
Ageing bucket	`CASE WHEN days≤30 THEN '0-30' …`
Reconcile	GL Sales total == sales register total == dashboard Revenue

Golden rule: map → model → **reconcile** → visualise → make it one-click refreshable. If it doesn't tie to the rupee, it isn't done.

AI & automation in finance

What it is & where it's used

"AI & automation in finance" means using three overlapping tool families to do finance work faster and with fewer errors:

1. **LLMs / Copilots** — ChatGPT, Claude, Microsoft 365 Copilot, Gemini. You give plain-English instructions ("prompts") and get back formulas, SQL, draft emails, variance commentary, reconciliations logic, GST notice replies, DCF explanations.
2. **RPA (Robotic Process Automation)** — bots that click through screens and move data the way a human would: Power Automate, UiPath, or even Excel VBA / Office Scripts. Used for repetitive, rule-based tasks — downloading bank statements, keying invoices into Tally, pulling GSTR-2B every month.
3. **Built-in AI features** — Copilot inside Excel, Power BI Copilot (writes DAX and narratives), Tally's AI add-ons, GSTN's e-invoice/IRP automation.

Where it shows up by role:

Role	AI/automation use
Accounts / AP-AR executive	Auto-key invoices, 3-way match bots, draft vendor follow-up emails
GST / tax associate	Reconcile GSTR-2B vs purchase register, draft replies to notices, summarise circulars
FP&A / MIS analyst	Copilot writes DAX/formulas, generates variance commentary, builds monthly deck
Audit / internal audit	Sample selection, exception testing, summarise policies vs. transactions
Treasury / controller	Bank-recon bots, cash-flow forecasting prompts

The gap: why companies want this (and college didn't teach it)

Your MBA/CA syllabus teaches **concepts** — accounting standards, cost of capital, GST law. It does not teach that in a real month-end you will re-key the same numbers 40 times, or that a junior who can make a bot download 30 bank statements in 3 minutes is worth 5x one who does it by hand.

The specific gap:

- College assumes **one clean dataset**. Industry gives you a messy Tally export, a bank PDF, and a GSTR-2B JSON that must be joined — under time pressure.
- College grades **the right answer**. Employers pay for **the right answer produced repeatably and fast**, with an audit trail.
- Nobody teaches **how to instruct an AI safely** — what to paste, what NOT to paste (client PAN, salary data), and how to verify AI output before it hits a filing.

The finance-specific risk employers fear: an analyst who blindly trusts a hallucinated number in a board deck, or pastes confidential data into a public chatbot. The person who understands **both the tool and its limits** is the one who gets hired and promoted.

What "proficient" looks like

A job-ready person can, unaided:

- Write a **prompt that produces a working Excel formula, SQL query, or DAX measure** on the first or second try, then verify it against a known-good number.
- Use Copilot in Excel/Power BI to **draft variance commentary**, then edit it for accuracy (not ship it raw).
- Build a **Power Automate flow** that moves files/data on a schedule without manual clicks.
- Record an **Excel macro / Office Script** for a repetitive cleanup and re-run it monthly.
- Know **data-governance red lines**: never paste PII, client-identifiable, or price-sensitive data into a public LLM; use the company-approved enterprise tool.
- **Sanity-check every AI output** — recompute totals, tie to source, flag anything the model "confidently" made up.

The bar is not "can code an AI model." It is "can use AI as a fast, cheap junior analyst — and catch its mistakes."

Hands-on: how to actually do it

1. Prompt patterns that work for finance

Use this **structure** every time: *Role → Context → Task → Format → Constraints*.

```
Role: You are an FP&A analyst.
Context: I have a table with columns Month, Budget, Actual (INR lakhs).
Task: Write an Excel formula for the % variance in column D,
      and flag "REVIEW" if the unfavourable variance exceeds 10%.
Format: Give the formula only, then one line explaining it.
Constraints: Assume data starts in row 2. No macros.
```

Reusable prompt patterns:

Pattern	Use it for	Example opener
Explain-then-do	Learning while producing	"Explain what a 3-way match is, then give the SQL to find mismatches."
Formula generator	Excel/DAX/SQL	"Write an XLOOKUP that returns HSN code from Sheet2, exact match."
Reconciliation logic	GSTR-2B vs books	"Given two tables by GSTIN+invoice no, list rows in A not in B."
Draft & polish	Emails, commentary	"Draft a polite payment-reminder email; overdue Rs 2,45,000, 45 days."
Summarise-to-decision	Circulars, policies	"Summarise this GST circular into 5 action points for a trader."
Red-team / check	Catch your own errors	"Here's my DCF. What assumptions look aggressive?"

2. Getting real formulas from an LLM

Ask, then paste this into Excel:

```
=IFERROR(XLOOKUP(A2, Vendors[GSTIN], Vendors[VendorName], "Not found"), "Error")
```

Variance flag (from the prompt above):

```
=IF((C2-B2)/B2 < -0.10, "REVIEW", TEXT((C2-B2)/B2, "0.0%"))
```

3. SQL the LLM writes for you — GSTR-2B vs purchase register

```sql
-- Invoices in books but MISSING in 2B (ITC at risk)
SELECT pr.gstin, pr.invoice_no, pr.taxable_value, pr.igst
FROM purchase_register pr
LEFT JOIN gstr2b b
       ON pr.gstin = b.gstin AND pr.invoice_no = b.invoice_no
WHERE b.invoice_no IS NULL;
```

4. Python snippet for a repetitive reconciliation

```python
import pandas as pd

books = pd.read_excel("purchase_register.xlsx")
b2b   = pd.read_excel("gstr2b.xlsx")

merged = books.merge(b2b, on=["gstin", "invoice_no"],
                     how="outer", indicator=True)

only_books = merged[merged["_merge"] == "left_only"]   # ITC at risk
only_2b    = merged[merged["_merge"] == "right_only"]   # vendor filed, you didn't book
print(f"Missing in 2B: {len(only_books)} | Missing in books: {len(only_2b)}")
only_books.to_excel("itc_at_risk.xlsx", index=False)
```

5. DAX from Power BI Copilot (verify before trusting)

```
Variance % =
DIVIDE(
    SUM('P&L'[Actual]) - SUM('P&L'[Budget]),
    SUM('P&L'[Budget])
)
```

6. RPA basics — Power Automate (no code)

A first flow every finance junior should build:

1. Go to **make.powerautomate.com** → **Create** → **Scheduled cloud flow**.
2. Trigger: run **1st of every month, 9 AM**.
3. Action: **Outlook** → **Get attachments** where subject contains "Bank Statement".
4. Action: **OneDrive** → **Create file** in `/Recon/{Month}/`.
5. Action: **Send me an email** "3 statements filed."

For desktop clicking (e.g., keying into Tally), use **Power Automate Desktop** → record clicks → replace the manual keying loop.

Worked example / mini-project

Goal: Automate the monthly ITC reconciliation for a small trading firm, "Sharma Traders."

Inputs (reproduce with dummy data):

gstin	invoice_no	taxable_value (Rs)	igst (Rs)	source
27ABCDE1234F1Z5	INV-101	1,00,000	18,000	books
29PQRST5678K2Z1	INV-102	50,000	9,000	books
27ABCDE1234F1Z5	INV-101	1,00,000	18,000	2B
24LMNOP9012Q3Z9	INV-777	75,000	13,500	2B

Step 1 — Prompt the LLM:

```
Role: GST analyst. I have purchase_register and gstr2b tables
(gstin, invoice_no, taxable_value, igst). Write Python to output:
(a) invoices in books not in 2B, (b) in 2B not in books,
(c) total IGST at risk. Save to Excel.
```

Step 2 — Run the Python from section 4. Result:

- INV-102 → in books, **not in 2B** → Rs 9,000 IGST at risk (chase vendor to file).
- INV-777 → in 2B, **not in books** → Rs 13,500 (book it or query the vendor).

Step 3 — Copilot drafts the commentary: "For June, Rs 9,000 ITC is blocked pending vendor filing of INV-102; one unrecorded inward supply (INV-777, Rs 13,500 IGST) requires booking."

Step 4 — You verify: recompute `SUMIF` on IGST, tie totals to Tally's GSTR-2 report, confirm the two exceptions by eye. Then file.

Step 5 — Automate next month: save the Python + a Power Automate flow that drops both files into a folder and emails you the exception count. Month-end task drops from ~2 hours to ~10 minutes.

How it's tested

Interview questions:

- "How would you use ChatGPT/Copilot in your day, and what would you *never* paste into it?" (They test data-governance awareness.)

- "The AI gives you a formula that returns a wrong total. What's your process?" (Verification mindset.)
- "Walk me through automating a task you did manually." (Do you think in repeatable processes?)
- "What are LLM hallucinations and why do they matter for a filing?"

Practical / assessment tests companies give:

- **Timed Copilot/Excel test:** "Here's a messy 5,000-row export. Use any AI help to reconcile and produce a variance summary in 30 minutes." They watch *how* you prompt and *whether* you verify.
- **Prompt-craft screen:** given a business ask, write the prompt(s) you'd use.
- **RPA task:** "Build a flow that emails a reminder when an invoice is >30 days overdue."
- **Judgment case:** "This AI-generated board number looks off — find the error." (The error is planted.)

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Pasting client PAN/GSTIN/salary into a public chatbot	Use enterprise/approved tools; anonymise; paste structure, not sensitive values
Shipping AI output unchecked	Always recompute totals, tie to source, spot-check 3-5 rows
Trusting a hallucinated citation (fake section/circular no.)	Verify every legal reference against the actual GST portal / bare act
Vague prompts ("analyse this")	Use Role-Context-Task-Format-Constraints; give a sample of the data
Automating a broken process	Fix and document the manual process first, *then* automate
No audit trail	Keep the prompt, the source file, and a note of what you verified
Over-automating one-off tasks	Automate only recurring, rule-based work (the 80/20)

Learn-it roadmap & resources

Time to job-ready: ~4-6 weeks, part-time.

Week	Focus
1	Prompt patterns; use ChatGPT/Copilot to generate Excel formulas daily
2	LLM for SQL + reconciliation logic; learn to verify every output
3	Power Automate — build 2-3 scheduled/desktop flows
4	Copilot in Excel & Power BI (DAX + narratives)
5-6	End-to-end mini-project (the ITC recon above); learn data-governance rules

Resources:

- **Free:** Microsoft Learn — Power Automate & Copilot paths; Google's "Prompting Essentials"; Anthropic/OpenAI prompt guides; YouTube (Leila Gharani for Excel+AI).

- **Paid:** Microsoft 365 Copilot licence (if employer provides); UiPath Academy (free RPA certs); Coursera "Generative AI for Finance."
- **Certifications worth it:** Microsoft **PL-900** (Power Platform), **UiPath RPA Associate**, Microsoft **PL-300** (Power BI, includes Copilot).

Staying relevant: treat AI as a **force multiplier, not a replacement**. The finance jobs that survive belong to people who pair domain judgment (accounting standards, GST law, materiality) with tool fluency. Learn one new automation each month; keep a personal "prompt library" of what works.

Quick-reference

Prompt skeleton: `Role → Context (with data sample) → Task → Format → Constraints`

Key formulas / snippets:

Need	Snippet
Lookup with error handling	`=IFERROR(XLOOKUP(A2, tbl[key], tbl[val], "NA"), "Err")`
Variance flag	`=IF((Actual-Budget)/Budget < -0.1, "REVIEW", "OK")`
Missing in 2B (SQL)	`LEFT JOIN … WHERE b.invoice_no IS NULL`
Recon (Python)	`books.merge(b2b, on=[…], how="outer", indicator=True)`
Variance % (DAX)	`DIVIDE(SUM(Actual)-SUM(Budget), SUM(Budget))`

Red lines — never paste into a public LLM: PAN, Aadhaar, GSTIN-linked client data, salary/HR data, price-sensitive/unpublished financials.

Verify-before-ship checklist: recompute totals · tie to source · spot-check 3-5 rows · verify every cited section/circular · keep prompt + source as audit trail.

Automate only if: recurring + rule-based + stable process. One-off or judgment-heavy → do it manually.

Data literacy & storytelling

What it is & where it's used

Data literacy is the ability to read, interpret, question, and communicate numbers so they change a decision. Storytelling is the packaging: turning a correct-but-inert analysis into a narrative a busy CFO, partner, or promoter acts on in 90 seconds.

This is the last-mile skill that separates an analyst from a "spreadsheet operator." You can build a perfect variance model, but if the deck buries the one number that matters under 14 pie charts, nobody moves. Every finance role touches this:

Role	Where storytelling shows up
FP&A analyst	Monthly MIS deck, budget-vs-actual review, board pack
Audit (CA firm)	Summary of audit findings, ICFR observations to audit committee
Tax / GST	Reconciliation summary (GSTR-2B vs books) for management sign-off
Investment/equity research	One-page thesis, target-price bridge
Controllership	Flash results, cash-flow story, working-capital review
Consulting / TP	Client-facing "so what" slides

The output is almost always one of three things: a **one-page executive summary**, a **single chart that carries an argument**, or a **verbal 60-second answer** in a review meeting.

The gap: why companies want this (and college didn't teach it)

College trains you to *produce* answers — compute NPV, pass journal entries, prepare a cash-flow statement. It grades completeness. Industry rewards *compression and consequence*: what do I do differently because of this number?

The specific gaps employers complain about:

- **No "so what."** Fresh hires present *what happened* ("revenue is ₹42 Cr") but not *why it matters* ("revenue is ₹42 Cr, 8% below plan, driven entirely by the North zone, and it puts the Q4 incentive pool at risk").
- **Data dump, not argument.** A 30-tab workbook emailed with "PFA" instead of a 3-bullet takeaway.
- **Chart illiteracy.** Pie charts with 11 slices, dual-axis charts that mislead, 3-D bars, rainbow colours with no meaning.
- **No audience calibration.** Same detail for the analyst and the board. Executives want the decision; they'll ask for detail.
- **Burying the lede.** The headline finding on slide 23 instead of slide 1.

A CA syllabus never grades "did the audit committee understand the risk in 2 minutes?" — but that is exactly the billable skill.

What "proficient" looks like

A job-ready person can, unaided:

1. Open any dataset and state the **top 3 findings in one sentence each**, ranked by decision impact.
2. Write a **BLUF (Bottom Line Up Front)** executive summary: conclusion first, evidence second.
3. Pick the **right chart for the question** without thinking — comparison → bar, trend → line, part-to-whole → stacked bar (rarely pie), correlation → scatter, distribution → histogram.
4. Build a **variance/bridge (waterfall) chart** that explains a movement from A to B.
5. Strip **chart-junk**: no gridline clutter, no 3-D, no redundant legends, direct labels, one accent colour to highlight the point.
6. Tailor the same analysis into three lengths: **60-second verbal, one-page, full appendix.**
7. Defend every number ("where did ₹8 Cr come from?") and state the **action** the number implies.

Hands-on: how to actually do it

1. The BLUF structure (memorise this)

```
[Conclusion + recommendation]  ← one sentence, put it FIRST
   Because [3 supporting facts, ranked]
   Which means [the decision / risk / rupee impact]
   I recommend [specific action + owner + date]
```

Bad: "Please find the sales analysis attached." Good: "**We will miss the ₹500 Cr annual target by ~₹30 Cr unless North zone recovers.** North is 18% below plan (₹22 Cr gap), driven by two lost distributors; West and South are on track. Recommend the CSO review North distributor terms by 15th."

2. Chart choice — decision table

Your question	Chart	Avoid
A vs B vs C (compare)	Horizontal **bar**, sorted	Pie
Change over time	**Line**	Area stacks
Part-to-whole (2–4 parts)	**Stacked bar / 100% bar**	Pie > 4 slices
What moved the total A→B	**Waterfall (bridge)**	Two side-by-side bars
Relationship of two metrics	**Scatter**	Dual-axis line
Actual vs target	**Bullet chart / bar + target line**	Gauge
Many categories, one metric	Sorted **bar**, top-N + "Others"	Rainbow columns

3. Excel: build a variance bridge (waterfall)

Native waterfall (Excel 2016+): select the data, **Insert** → **Charts** → **Waterfall**. Then double-click the total column → check **"Set as Total"**.

To compute the bridge components for Plan → Actual:

```
Start (Plan):        =Plan
Volume effect:       =(Act_Vol − Plan_Vol) * Plan_Price
Price effect:        =(Act_Price − Plan_Price) * Act_Vol
Mix / other:         =Actual − Plan − Volume_effect − Price_effect
End (Actual):        =Start + SUM(effects)   ← must reconcile
```

Highlight the worst driver in a contrasting colour (right-click that point → Fill → dark red). Everything else stays grey. **One accent colour = one message.**

4. Excel: turn a table into a headline

```
="Revenue "&TEXT([@Actual],"₹#,##0,,\ \C\r")&", "&
 TEXT(([@Actual]-[@Plan])/[@Plan],"0%")&
 IF([@Actual]<[@Plan]," below plan"," above plan")
```

This auto-writes: `Revenue ₹42 Cr, −8% below plan`. Put it as a dynamic title so the chart title *is* the takeaway, not "Chart 1".

5. Direct labelling instead of legends

Legends force the eye to ping-pong. In a line chart, delete the legend and add a text box at the end of each line with the series name in the line's colour. Fewer than 4 series → always direct-label.

6. Python: a clean, chart-junk-free plot

```
import matplotlib.pyplot as plt

zones  = ['North', 'West', 'South', 'East']
gap    = [-22, 3, 5, -1]          # ₹ Cr vs plan
colors = ['#c0392b' if g < -5 else '#bdc3c7' for g in gap]

fig, ax = plt.subplots(figsize=(7, 3.5))
ax.barh(zones, gap, color=colors)
ax.axvline(0, color='#333', lw=0.8)
ax.set_title('North drags the plan by ₹22 Cr', loc='left', fontsize=13, weight='bold')
for s in ['top', 'right', 'bottom']:      # kill chart-junk
    ax.spines[s].set_visible(False)
ax.tick_params(length=0)
for y, g in enumerate(gap):               # direct data labels
    ax.text(g, y, f' {g:+d}', va='center',
            ha='left' if g > 0 else 'right')
ax.set_xticks([])                         # numbers on bars, no axis
plt.tight_layout(); plt.savefig('zone_gap.png', dpi=150)
```

The title states the conclusion; grey/red does the arguing; no gridlines, no legend, no axis noise.

7. Power BI / DAX: a KPI that tells the story

```
Rev vs Plan % =
VAR act  = SUM(Sales[Actual])
VAR plan = SUM(Sales[Plan])
RETURN DIVIDE(act - plan, plan)

Rev Status =
SWITCH(TRUE(),
    [Rev vs Plan %] < -0.05, "🔴 Behind",
    [Rev vs Plan %] <  0,    "🟡 Watch",
    "🟢 On track")
```

Drive conditional formatting off `Rev Status` so the reader sees red *before* they read a single number.

Worked example / mini-project

Scenario. You are FP&A at an Indian FMCG distributor. FY revenue = ₹470 Cr against a ₹500 Cr plan. The MD wants "why" in one page tomorrow.

Raw data (₹ Cr):

Zone	Plan	Actual	Gap
North	120	98	-22
West	150	153	+3
South	130	135	+5
East	100	84	-16
Total	**500**	**470**	**-30**

Step 1 — find the story. The ₹30 Cr miss is *not* broad — West and South beat plan. It's concentrated in North (-22) and East (-16), offset by +8 elsewhere. That's the lede.

Step 2 — one-line BLUF.

> **We closed FY at ₹470 Cr, ₹30 Cr (6%) short — but the miss is entirely North + East (-₹38 Cr); West & South over-delivered by ₹8 Cr.**

Step 3 — one chart. Horizontal bar of gap-by-zone, North & East in red, others grey, title = "Two zones caused the entire ₹30 Cr miss" (the Python snippet above renders exactly this).

Step 4 — the "so what" + action.

Finding	Rupee impact	Recommended action	Owner
North lost 2 distributors	-₹22 Cr	Renegotiate margins, re-appoint by Q1	Zonal Head N
East supply outages	-₹16 Cr	Add secondary depot	Supply Chain
West/South momentum	+₹8 Cr	Protect; replicate scheme	CSO

Step 5 — the one-pager layout:

```
┌────────────────────────────────────────────────┐
│ FY Revenue: ₹470 Cr  (-₹30 Cr / -6% vs plan)  │  ← BLUF banner
│                                                 │
│ [ zone-gap bar chart, North & East in red ]     │  ← ONE chart
│                                                 │
│ • 2 zones = 100% of miss                        │  ← 3 bullets
│ • West & South +₹8 Cr, on track                 │
│ • Recovery plan → +₹25 Cr addressable in Q1     │
│                                                 │
│ Ask: approve North distributor margin revision  │  ← the decision
└────────────────────────────────────────────────┘
```

Full zone/SKU/month detail goes in the appendix — referenced, not shown. The MD reads the banner, glances at one chart, and approves an action. That is data storytelling.

How it's tested

Interviews and assessments target the last mile explicitly:

Verbal / whiteboard questions - "You have 30 seconds and the CFO's attention. Give me the headline of this P&L." (tests BLUF instinct) - "This chart is a pie with 9 slices — what's wrong and how would you fix it?" - "Revenue is up 5% and profit is down 10%. Tell me the story." (margin/mix reasoning + narrative) - "What's the one number on this dashboard you'd escalate?"

Practical assessments - **Timed deck test:** given a messy 5-tab workbook, produce a 1-slide summary in 45 minutes. Graded on: is the conclusion first? is there one clear chart? is there a recommended action? - **Chart critique:** they hand you a deliberately bad chart (3-D, dual-axis, rainbow) and ask you to redo it. - **Case presentation:** analyse quarterly results and present to a panel; they interrupt with "so what?" and "what would you do?" - **Email test:** "Write the covering email a manager would actually read" — tests compression.

Scoring is rarely about spreadsheet mechanics; it's *did the reader know the decision without asking a follow-up question.*

Common mistakes & how pros avoid them

Mistake	Fix pros use
Conclusion on the last slide	BLUF — decision on slide 1, banner, or subject line
Pie charts / 3-D / rainbow	Sorted bars; one accent colour on the point that matters
Dual-axis to fake a correlation	Two panels or a scatter; never trick the eye
Chart title says "Revenue by Zone"	Title = the takeaway: "North caused the miss"
Legend the reader must decode	Direct-label series at the line's end
Presenting data, not the "so what"	Every number followed by "...which means..."
Same detail for board & analyst	Layer: 1-line → 1-page → appendix
Precision theatre (₹4,72,38,914)	Round to ₹4.7 Cr — precision ≠ credibility
Truncated y-axis exaggerating change	Start bars at zero; note breaks explicitly
No recommended action	End with action + owner + date

The professional's tell: they can delete 80% of the deck and the message survives.

Learn-it roadmap & resources

Realistic time-to-proficiency: 4–8 weeks of deliberate practice on top of existing Excel/BI skills. It's a *rewiring of habits*, not a new tool.

Week	Focus
1	BLUF structure; rewrite 5 old emails/decks conclusion-first
2	Chart choice table; remake 10 bad charts you find online
3	Build 3 variance/waterfall bridges in Excel from real data
4	One-page executive summaries — do 5, get feedback
5–6	Power BI storytelling: KPI cards, conditional formatting, drill-through
7–8	Mock panel presentations; practise the "so what?" reflex

Resources - *Storytelling with Data* — Cole Nussbaumer Knaflic (the standard; free blog + monthly challenge at storytellingwithdata.com). - *The Pyramid Principle* — Barbara Minto (BLUF / top-down structuring). - *Show Me the Numbers* — Stephen Few (chart design, anti-chart-junk). - Free: Google "Data Studio / Looker Studio" tutorials; matplotlib + seaborn galleries; the SWD blog exercises. - Certification: **Microsoft PL-300 (Power BI Data Analyst)** — includes a reporting/visual-design section; strong signal for FP&A/analyst roles in India (₹ market rate for the exam ~US$165). - Practice reps: take any listed Indian company's quarterly investor presentation and rewrite one slide better.

Quick-reference

BLUF template: Conclusion → because (3 facts) → which means (₹ impact) → I recommend (action, owner, date).

Chart picker: compare → sorted bar · trend → line · part-to-whole → stacked/100% bar (not pie) · movement A→B → waterfall · relationship → scatter · vs target → bullet.

Anti-chart-junk checklist: ☐ conclusion in the title ☐ one accent colour ☐ no 3-D ☐ no gridline clutter ☐ direct labels, no legend if <4 series ☐ y-axis starts at zero ☐ rounded numbers ☐ ≤1 core chart per message.

The 3 layers: 60-sec verbal → 1-page summary → full appendix.

The one test that matters: *Can the reader state the decision without asking a follow-up?* If no, cut and re-lead.

Excel dynamic takeaway title:

```
=TEXT([@Actual],"₹#,##0,,\ \C\r")&" ("&TEXT(([@Actual]-[@Plan])/[@Plan],"+0%;-0%")&" vs plan)"
```

Rounding rule: report to 2 significant figures for narrative (₹4.7 Cr), keep full precision in the appendix.

The data/SQL interview & test

What it is & where it's used

The SQL interview is the single most common practical filter between you and a data-touching finance job. Once a job description says "SQL", "data extraction", "pull your own numbers", or "self-serve analytics", assume there is a live or take-home SQL test — an MBA and a CA Intermediate do not exempt you from it.

Where it shows up:

Role	What the SQL screen looks like
FP&A / Finance analyst	3–5 questions on a sales/GL table, timed 30–45 min
Financial / Business analyst (fintech, e-comm)	Live shared editor, 45–60 min, joins + window functions
Revenue / Billing analyst	Take-home: reconcile invoices vs. payments
Data analyst (finance vertical)	HackerRank/Stratascratch style, 60–90 min
Audit analytics / Internal audit	Case: find duplicate vendors, split POs

The bar is not "can you write a `SELECT *`". It is "can you turn a vague business question into a correct query, unaided, under time pressure, and explain it."

The gap: why companies want this (and college didn't teach it)

College teaches you to *read* financial statements someone else prepared. The job is to *produce* the numbers from raw transactional tables — and those tables have nulls, duplicates, wrong data types, and 40 million rows. An MBA case study hands you a clean Exhibit 4; the real ERP hands you `sales_line_item` with `amount` stored as text and `INV-0001` sometimes typed `inv 0001`.

The specific gaps a SQL screen exposes:

- **Aggregation logic.** You know GROUP BY conceptually, but freeze on "revenue per customer *for their first month only*."
- **Joins under ambiguity.** LEFT vs INNER changes your revenue number. Getting this wrong = a wrong board deck.
- **Window functions.** Running totals, month-over-month growth, "top 3 per region", rank — never covered in a finance degree, asked in ~60% of screens.
- **Data hygiene instinct.** Pros check for duplicates and nulls *before* trusting a total. Freshers report the first number they get.

Companies test SQL because it is the cheapest reliable proxy for "can this person get the right number without hand-holding."

What "proficient" looks like

The concrete, job-ready bar. You can, unaided:

1. Filter and aggregate: `WHERE`, `GROUP BY`, `HAVING`, `COUNT/SUM/AVG`.

2. Join 2–4 tables and *know* why you chose INNER vs LEFT.
3. Use `CASE WHEN` for bucketing and conditional sums.
4. Write window functions: `ROW_NUMBER`, `RANK`, `SUM() OVER`, `LAG/LEAD`.
5. Structure a query with CTEs (`WITH`) instead of nested subqueries.
6. Handle nulls (`COALESCE`), dedupe, and cast types.
7. Compute finance staples: MoM growth, running total, top-N-per-group, cohort/retention, churn.
8. Read a business question and translate it — the actual skill being tested.

If you can do 1–6 fluently and reason through 7–8 out loud, you pass most finance-adjacent screens.

Hands-on: how to actually do it

Assume two tables:

```
customers(customer_id, name, city, signup_date)
orders(order_id, customer_id, order_date, amount, status)
```

Filter + aggregate — revenue per city, paid orders only:

```
SELECT c.city,
       SUM(o.amount)      AS total_revenue,
       COUNT(*)           AS num_orders
FROM   orders o
JOIN   customers c ON c.customer_id = o.customer_id
WHERE  o.status = 'PAID'
GROUP BY c.city
HAVING SUM(o.amount) > 100000
ORDER BY total_revenue DESC;
```

Conditional aggregation (pivot without a pivot) — paid vs refunded per month:

```
SELECT DATE_TRUNC('month', order_date) AS mth,
       SUM(CASE WHEN status='PAID'     THEN amount ELSE 0 END) AS paid,
       SUM(CASE WHEN status='REFUNDED' THEN amount ELSE 0 END) AS refunded
FROM   orders
GROUP BY 1
ORDER BY 1;
```

LEFT JOIN to find the gap — customers who never ordered:

```
SELECT c.customer_id, c.name
FROM   customers c
LEFT JOIN orders o ON o.customer_id = c.customer_id
WHERE  o.order_id IS NULL;
```

Window function — running total and month-over-month growth:

```
WITH monthly AS (
  SELECT DATE_TRUNC('month', order_date) AS mth,
         SUM(amount) AS revenue
  FROM   orders
  WHERE  status = 'PAID'
  GROUP BY 1
)
SELECT mth,
       revenue,
       SUM(revenue) OVER (ORDER BY mth)                 AS running_total,
       revenue - LAG(revenue) OVER (ORDER BY mth)       AS mom_change,
       ROUND(100.0 * (revenue - LAG(revenue) OVER (ORDER BY mth))
             / LAG(revenue) OVER (ORDER BY mth), 1)      AS mom_pct
FROM   monthly
ORDER BY mth;
```

Top-N per group — top 2 customers by spend in each city:

```
WITH ranked AS (
  SELECT c.city, c.name,
         SUM(o.amount) AS spend,
         ROW_NUMBER() OVER (PARTITION BY c.city
                            ORDER BY SUM(o.amount) DESC) AS rn
  FROM   orders o
  JOIN   customers c ON c.customer_id = o.customer_id
  GROUP BY c.city, c.name
)
SELECT city, name, spend FROM ranked WHERE rn <= 2;
```

Data hygiene — the check pros run first:

```
-- duplicate orders?
SELECT order_id, COUNT(*) FROM orders GROUP BY order_id HAVING COUNT(*) > 1;
-- nulls hiding in the money column?
SELECT COUNT(*) FROM orders WHERE amount IS NULL;
```

Worked example / mini-project

Case (India e-commerce, reproduce this locally): "Give me GST-inclusive net revenue by month for FY2024-25, the MoM growth, and flag any month where refunds exceeded 5% of gross."

Seed data (paste into SQLite / Postgres):

```sql
CREATE TABLE orders(order_id INT, customer_id INT, order_date DATE,
                    amount NUMERIC, status TEXT);
INSERT INTO orders VALUES
(1,101,'2024-04-05', 50000,'PAID'),
(2,102,'2024-04-18', 30000,'PAID'),
(3,103,'2024-04-22', 12000,'REFUNDED'),
(4,101,'2024-05-10', 80000,'PAID'),
(5,104,'2024-05-15', 20000,'PAID'),
(6,105,'2024-05-27', 60000,'REFUNDED'),
(7,102,'2024-06-02', 90000,'PAID'),
(8,103,'2024-06-19',  5000,'REFUNDED');
```

Amounts are GST-inclusive at 18%. Solution:

```sql
WITH m AS (
  SELECT DATE_TRUNC('month', order_date) AS mth,
         SUM(CASE WHEN status='PAID' THEN amount ELSE 0 END)     AS gross_paid,
         SUM(CASE WHEN status='REFUNDED' THEN amount ELSE 0 END) AS refunds
  FROM orders
  GROUP BY 1
)
SELECT mth,
       gross_paid,
       ROUND(gross_paid / 1.18, 0)                        AS net_ex_gst,
       ROUND(gross_paid - gross_paid/1.18, 0)             AS gst_component,
       ROUND(gross_paid - LAG(gross_paid) OVER (ORDER BY mth), 0) AS mom_change,
       CASE WHEN refunds > 0.05 * (gross_paid + refunds)
            THEN 'FLAG' ELSE 'OK' END                     AS refund_flag
FROM m
ORDER BY mth;
```

Expected reading: April gross ₹80,000 (net ₹67,797, GST ₹12,203), May ₹100,000 (MoM +₹20,000) but refunds ₹60,000 vs gross ₹100,000 → **FLAG** (>5%). Being able to *narrate* that last line — "May looks great on revenue but refund rate is 37%, so I'd investigate before reporting growth" — is what gets you the offer.

How it's tested

Format you'll actually face:

Type	Detail
Live shared editor	Zoom + DB Fiddle / CoderPad; 3–4 questions, 45 min, they watch you think
Timed platform	HackerRank / StrataScratch / DataLemur; auto-graded, 60–90 min
Take-home	A CSV + "answer these 5 business questions", 24–48 hr
Verbal-only	Whiteboard/no-run; they judge structure, not exact syntax

Questions that recur:

- "Second-highest salary / order value." (Tests window functions or subquery.)
- "Customers who ordered in Jan but not Feb." (Anti-join / `NOT IN` / `EXCEPT`.)
- "Difference between `WHERE` and `HAVING`; `INNER` vs `LEFT JOIN`." (Verbal.)
- "MoM revenue growth %." (Window + `LAG`.)
- "Top 3 products per category." (`ROW_NUMBER` + `PARTITION BY`.)
- "Deduplicate this table keeping the latest row." (`ROW_NUMBER()` then `WHERE rn=1`.)
- "How would you check this number is right?" (The hygiene answer — most freshers miss it.)

How they grade: correctness first, then whether you clarified assumptions, then readability (CTEs > nested subqueries), then that you sanity-checked the output.

Common mistakes & how pros avoid them

Mistake	Fix
Diving into SQL before understanding the question	Restate it: "So 'active customer' means ordered in last 30 days?" — always clarify first
Using `INNER JOIN` when the question needs unmatched rows	If the ask is "who *didn't*", it's a `LEFT JOIN … IS NULL` or `EXCEPT`
`COUNT(*)` vs `COUNT(column)` confusion	`COUNT(col)` ignores nulls; know which you mean
Filtering on an aggregate in `WHERE`	Aggregate conditions go in `HAVING`, not `WHERE`
Nulls silently breaking sums/joins	`COALESCE(amount,0)`; check `IS NULL` early
Wrong grain → double counting after a join	Aggregate before joining, or check for fan-out with a row count
Reporting the first number without checking	Run the dedupe + null check first; state "I verified no dupes"
Nested subquery spaghetti	Use `WITH` CTEs — readable and graders reward it
Integer division (`1/2 = 0`)	Multiply by `100.0` or cast for percentages

Learn-it roadmap & resources

Realistic time-to-proficiency from zero, part-time: **4–6 weeks** to pass most finance screens.

Week	Focus
1	SELECT/WHERE/GROUP BY/ORDER BY/HAVING — SQLBolt (free, ~4 hrs)
2	Joins (all types), CASE, subqueries — Mode SQL Tutorial (free)
3	Window functions + CTEs — the part that actually differentiates you
4	Drill interview questions — DataLemur (free tier), StrataScratch
5–6	Timed mocks + one take-home; build the mini-project above in SQLite

Resources:

- **Free:** SQLBolt, Mode Analytics SQL Tutorial, DataLemur (Nick Singh, finance-heavy), PostgreSQL Tutorial, LeetCode Database (50 problems).
- **Practice envs:** DB Fiddle, sqliteonline.com — zero install, paste and run.
- **Paid (optional):** StrataScratch (~$30/mo, real company questions), "Ace the Data Science Interview" book.
- **Certification:** not required and rarely asked for in finance roles — a GitHub repo with 3 solved case queries beats a certificate. If you want one, Google Data Analytics (Coursera) or Microsoft DP-900 signal basics.

Install SQLite locally (`sqlite3` , ships free) and reproduce every query above — running beats reading.

Quick-reference

```
-- Skeleton
SELECT col, AGG(x) FROM t
JOIN t2 ON t.id=t2.id
WHERE  filter_rows
GROUP BY col
HAVING AGG(x) > n              -- filter on aggregate
ORDER BY col LIMIT k;

-- Window
ROW_NUMBER() OVER (PARTITION BY g ORDER BY x DESC)  -- top-N per group
SUM(x)     OVER (ORDER BY d)                        -- running total
LAG(x)     OVER (ORDER BY d)                        -- prev row → MoM
RANK() / DENSE_RANK()                               -- ties handling

-- Patterns
COALESCE(x,0)                       -- null-safe
LEFT JOIN … WHERE b.id IS NULL      -- anti-join ("didn't")
A EXCEPT B                          -- rows in A not in B
100.0 * a / b                       -- avoid integer division
WITH cte AS (…) SELECT … FROM cte   -- readable > nested
```

Question type	Weapon
Top-N per group	`ROW_NUMBER() … PARTITION BY`
MoM / running total	`LAG` / `SUM() OVER`
"Who didn't..."	LEFT JOIN ... IS NULL / `EXCEPT`
Nth highest	window or correlated subquery
Pivot	`SUM(CASE WHEN …)`
Dedupe	`ROW_NUMBER()` , keep `rn=1`

Golden rule: clarify the question → check the data → write the CTE → sanity-check the number → explain it out loud.

Cheat-sheet: SQL, DAX & pandas

What it is & where it's used

This chapter is a dense, printable reference for the three languages that do the heavy lifting in a modern finance-analytics job: **SQL** (pull and shape data from a database), **DAX** (build measures inside Power BI / Excel Power Pivot), and **pandas** (Python data-wrangling). You use SQL to answer "how much did we bill client X in FY25?" against a warehouse; DAX to make a P&L dashboard that recalculates as a CFO clicks filters; pandas to reconcile a 200k-row bank statement against the ledger in 30 seconds.

Roles that touch these daily: **FP&A analyst**, **finance/business analyst**, **management-reporting (MIS) executive**, **revenue/GL accountant in a shared-service centre**, **credit / risk analyst**, and **audit-analytics** teams at the Big 4. In Indian GCCs (Deloitte USI, EY GDS, Accenture, Genpact, TresVista) a job ad that says "hands-on with Power BI and SQL" is testing exactly the muscle memory below.

The gap: why companies want this (and college didn't teach it)

An MBA-Finance or CA course teaches you *what* a P&L, a reconciliation, or a variance is. It does not teach you to compute it over 500,000 rows without Excel choking. The gap is **mechanical fluency at scale**:

- College: `VLOOKUP` two sheets. Job: `LEFT JOIN` a 4-million-row invoice table to a customer master and not create duplicates.
- College: a static ratio in a template. Job: a `CALCULATE(SUM( … ), FILTER( … ))` measure that stays correct when the user slices by region *and* month.
- College: retype numbers. Job: a pandas script that runs the same reconciliation every morning, unattended.

Employers pay for the person who can turn a vague ask ("why did margin drop in Q3?") into a query, a measure and a chart in an hour. That is a skill nobody grades in an exam, so it screens candidates hard.

What "proficient" looks like

The concrete bar an interviewer expects you to clear **unaided**:

- **SQL**: write a `JOIN` + `GROUP BY` + `HAVING` query; use a window function (`SUM() OVER` , `ROW_NUMBER()`) for running totals and de-duplication; explain the difference between `WHERE` and `HAVING` , and `INNER` vs `LEFT` join, without notes.
- **DAX**: know that a measure recomputes in *filter context*; write `CALCULATE` with a filter; build a YoY / YTD measure using `SAMEPERIODLASTYEAR` and `TOTALYTD` ; understand `SUM` (a column) vs `SUMX` (row-by-row).
- **pandas**: load a CSV/Excel, `merge` two frames, `groupby().agg()` , `pivot_table` , handle NaNs, and export back to Excel — cleanly, in under 20 lines.

Hands-on: how to actually do it

SQL — the clauses you use every day

Logical execution order (memorise this — it explains 90% of bugs): `FROM → JOIN → WHERE → GROUP BY → HAVING → SELECT → ORDER BY → LIMIT`.

```
-- Revenue by customer for FY25 (Apr-24 to Mar-25), > 10 lakh only
SELECT c.customer_name,
       SUM(i.amount)                        AS total_rev,
       COUNT(*)                             AS invoice_count
FROM   invoices i
JOIN   customers c ON c.customer_id = i.customer_id   -- INNER: only matched rows
WHERE  i.invoice_date BETWEEN '2024-04-01' AND '2025-03-31'
GROUP  BY c.customer_name
HAVING SUM(i.amount) > 1000000              -- filter AFTER aggregation
ORDER  BY total_rev DESC
LIMIT  20;
```

```
-- Window functions: running total + latest row per customer
SELECT customer_id, invoice_date, amount,
       SUM(amount) OVER (PARTITION BY customer_id ORDER BY invoice_date) AS running_total,
       ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY invoice_date DESC) AS rn
FROM   invoices;          -- keep WHERE rn = 1 in an outer query to get the latest invoice
```

```
-- Categorise ageing (AR buckets) with CASE
SELECT invoice_id, amount,
       CASE WHEN DATEDIFF(CURRENT_DATE, due_date) ≤ 0   THEN 'Not due'
            WHEN DATEDIFF(CURRENT_DATE, due_date) ≤ 30  THEN '0-30'
            WHEN DATEDIFF(CURRENT_DATE, due_date) ≤ 90  THEN '31-90'
            ELSE '90+' END AS ageing_bucket
FROM   invoices WHERE status = 'OPEN';
```

`COALESCE(col, 0)` swaps NULLs for 0; `LEFT JOIN` keeps unmatched left rows (use it to find invoices with **no** payment: `WHERE p.payment_id IS NULL`).

DAX — measures for a Power BI P&L

```
Total Sales     = SUM(Sales[Amount])
Total Cost      = SUM(Sales[Cost])
Gross Margin %  = DIVIDE([Total Sales] - [Total Cost], [Total Sales])   -- DIVIDE = safe /0

-- Filter context override: sales only for the North region, ignoring page slicers
North Sales     = CALCULATE([Total Sales], Sales[Region] = "North")

-- Row-by-row: value = qty * price computed per row, then summed
Revenue         = SUMX(Sales, Sales[Qty] * Sales[Price])

-- Time intelligence (needs a marked Date table)
Sales YTD       = TOTALYTD([Total Sales], 'Date'[Date], "31-03")   -- Indian FY year-end
Sales LY        = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY %           = DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

`SUM` aggregates one column; `SUMX` iterates a table and evaluates an expression per row — use `SUMX` whenever the math is `A * B` at row level. `CALCULATE` is the one function to truly master: it *modifies filter context*.

pandas — reconciliation and reshaping

```
import pandas as pd

ledger = pd.read_excel("ledger.xlsx", sheet_name="GL")
bank   = pd.read_csv("bank_stmt.csv", parse_dates=["txn_date"])

# Clean + merge on a key
recon = ledger.merge(bank, on="utr_no", how="outer", indicator=True)
unmatched = recon[recon["_merge"] != "both"]          # breaks to investigate

# Group & aggregate (the SQL GROUP BY equivalent)
summary = (ledger
           .groupby("cost_centre")
           .agg(total=("amount", "sum"), count=("amount", "size"))
           .reset_index())

# Pivot to a matrix (months across columns)
pt = ledger.pivot_table(index="account", columns="month",
                        values="amount", aggfunc="sum", fill_value=0)

ledger["amount"] = ledger["amount"].fillna(0)          # handle blanks
summary.to_excel("mis_summary.xlsx", index=False)      # ship it
```

Worked example / mini-project

Task: monthly MIS — revenue by region with a YoY view — from a raw sales table.

Sample `sales` table (₹ lakh):

sale_date	region	amount
2024-05-10	South	12.5
2024-05-22	North	8.0
2025-05-08	South	15.0
2025-05-19	North	9.5

Step 1 — SQL to pull the monthly base:

```
SELECT region,
       DATE_FORMAT(sale_date, '%Y-%m') AS mth,
       SUM(amount)                     AS rev
FROM   sales
GROUP  BY region, DATE_FORMAT(sale_date, '%Y-%m')
ORDER  BY region, mth;
```

Step 2 — load into Power BI, build measures:

```
Rev     = SUM(sales[amount])
Rev LY  = CALCULATE([Rev], SAMEPERIODLASTYEAR('Date'[sale_date]))
YoY %   = DIVIDE([Rev] - [Rev LY], [Rev LY])
```

For May: South ₹15.0L vs ₹12.5L LY → **+20%**; North ₹9.5L vs ₹8.0L → **+18.75%**. Drop a matrix visual (rows = region, values = Rev, Rev LY, YoY %) and a Date slicer.

Step 3 — same answer in pandas as a sanity check:

```
df["yr"] = df["sale_date"].dt.year
piv = df.pivot_table(index="region", columns="yr", values="amount", aggfunc="sum")
piv["yoy_%"] = (piv[2025] - piv[2024]) / piv[2024] * 100
print(piv.round(2))
```

Three tools, one number — that cross-check *is* the job.

How it's tested

- **SQL screen (30–45 min, HackerRank / live editor):** you're given 2–3 tables and asked "top 5 customers by revenue", "second-highest salary" (window function), or "invoices with no payment" (`LEFT JOIN … IS NULL`). Interviewers love `WHERE` vs `HAVING` and `INNER` vs `LEFT` .

- **Power BI / DAX case:** a take-home dataset — "build a P&L dashboard with YoY and a region slicer." They check whether your measures survive slicing (filter-context awareness) and whether you used `DIVIDE` not `/`.
- **pandas / Python round:** "read these two files, reconcile, output the breaks" or a `groupby` + `pivot_table` puzzle.
- **Verbal:** "Explain filter context." "SUM vs SUMX?" "How do you avoid a fan-out / duplicate join?"

Common mistakes & how pros avoid them

Mistake	Fix
Duplicated rows after a join (fan-out)	Join on a **unique** key; check row count before/after; aggregate before joining
`WHERE` used to filter an aggregate	Aggregates go in `HAVING`, not `WHERE`
Dividing by zero blows up DAX	Always `DIVIDE(a, b)` — returns blank on /0
Using `SUM` where math is row-level	Use `SUMX(table, qty*price)`
No Date table → time intelligence fails	Add a marked `Date` table, join it, use `TOTALYTD` / `SAMEPERIODLASTYEAR`
`merge` silently drops rows	Use `how="outer", indicator=True`, inspect `_merge`
Chained-assignment / SettingWithCopy warning	Use `.loc[]` or `.copy()`
Trusting NULL == NULL in SQL	Test with `IS NULL`; `COALESCE` before comparing

Learn-it roadmap & resources

Realistic time to interview-ready (2 hrs/day):

- **SQL:** 3–4 weeks. Free: SQLBolt, Mode Analytics SQL tutorial, LeetCode Database (Easy/Medium), StrataScratch.
- **DAX / Power BI:** 4–6 weeks. Free: Microsoft Learn "Power BI" path, SQLBI.com (Marco Russo), "Learn DAX" YouTube. Cert (optional but strong on a resume): **PL-300 Microsoft Power BI Data Analyst** (~US$165 / ~₹4,800).
- **pandas:** 3–4 weeks. Free: Kaggle "Pandas" + "Python" micro-courses, the official 10-minutes-to-pandas guide.

Practice on real data: your bank statement, a GST sales register, or public datasets (data.gov.in). Rebuild one MIS report end-to-end across all three tools — that single project teaches more than 20 tutorials.

Quick-reference

SQL

```
SELECT col, AGG(x) FROM t JOIN u ON t.k=u.k
WHERE ...  GROUP BY col  HAVING AGG(x)>n  ORDER BY col DESC  LIMIT n;
Joins: INNER (matched) | LEFT (keep left) | anti-join: LEFT + IS NULL
Window: SUM()/AVG() OVER(PARTITION BY .. ORDER BY ..), ROW_NUMBER(), RANK(), LAG()/LEAD()
NULLs: COALESCE(c,0) | IS NULL | NULLIF(a,b)
Dates: BETWEEN, DATEDIFF(), DATE_FORMAT()
```

DAX

```
SUM / AVERAGE / COUNTROWS(col)          -- aggregate one column
SUMX(tbl, expr)                         -- row-by-row then sum
CALCULATE(expr, filter1, filter2)       -- modify filter context (the key one)
DIVIDE(num, den[, 0])                   -- safe division
FILTER(tbl, cond) | ALL() | ALLEXCEPT() -- filter helpers
SAMEPERIODLASTYEAR / TOTALYTD / DATESYTD-- time intelligence
RELATED() | RELATEDTABLE()              -- cross-table lookup
```

pandas

```
pd.read_csv / read_excel / to_excel
df.merge(o, on="k", how="left"/"outer", indicator=True)
df.groupby("c").agg(t=("x","sum"), n=("x","size"))
df.pivot_table(index=, columns=, values=, aggfunc="sum", fill_value=0)
df.loc[mask, "col"] = v   |  df["c"].fillna(0)  |  df.drop_duplicates()
df.assign(newcol=...)  |  df.sort_values("c", ascending=False)
```

Mental map: SQL `GROUP BY` = pandas `groupby` = DAX aggregate in filter context. SQL `JOIN` = pandas `merge` = DAX relationship. Same idea, three dialects.

Practical Accounting & ERP

Practical, Hands-On, Job-Ready — India-first + Global

Real formulas, queries, entries & tools — closing the gap between what an MBA teaches and what finance employers pay for

Contents

1. **Theory vs the job: what accounting roles actually do**
2. **The accounting cycle in a real company**
3. **Journal entries you will actually pass**
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15. **Cheat-sheet: entries, close checklist & shortcuts**

Theory vs the job: what accounting roles actually do

What it is & where it's used

An accounting job is not "posting debits and credits." It is *operating a system of record* — usually an ERP (TallyPrime, SAP, Oracle NetSuite, Zoho Books, Microsoft Dynamics) — under deadlines, so that money moves correctly, taxes are filed on time, and management can trust the numbers. The theory you learned (double-entry, AS/Ind AS, ratio analysis) is the *grammar*. The job is *writing paragraphs* in that grammar every day, fast, without errors, and reconciling when it breaks.

Three roles you will actually target, and what they own:

Role	Owns day-to-day	ERP touchpoints	Typical India CTC (0-4 yrs)
Accounts Executive / Accountant	Booking invoices, bank entries, GST/TDS data, vendor payments, ledger scrutiny	Tally/Zoho voucher entry, GST portal, bank statements	₹3-6 LPA
Assistant Manager (AM) – Finance	Month-end close, reconciliations, MIS, GST returns, review of juniors' entries	SAP FICO / NetSuite, Excel MIS, GSTN, TRACES	₹8-16 LPA
Controller / Finance Manager	Books integrity, statutory audit, board MIS, controls, cash-flow, ERP process design	Full ERP + BI (Power BI), consolidation	₹18-40 LPA+

Every one of these lives inside a **close calendar** (the monthly rhythm of shutting the books) and a **compliance calendar** (GSTR-1 by the 11th, GSTR-3B by the 20th, TDS payment by the 7th). Miss those and the job goes wrong regardless of how well you know accounting standards.

The gap: why companies want this (and college didn't teach it)

Your MBA/CA syllabus optimized for *exam answers*: "define materiality," "compute the current ratio," "pass the rectification entry." Employers optimize for *throughput and reliability*. The gap is specific:

College taught	The job needs
Prepare a Trial Balance from given balances	*Extract* the TB from Tally/SAP, spot the ₹ that doesn't tie, and fix it before close
The GST concept (input credit, output tax)	File GSTR-3B on the portal, reconcile GSTR-2B vs purchase register, claim exactly the eligible ITC
Journal entries in a neat notebook	400 vendor invoices booked with correct TDS section, HSN, cost centre, and GL code
"Reconciliation is comparing two statements"	A 3-way bank rec at month-end where ₹18,750 is stuck and you must find *which* entry
Ratios from a printed balance sheet	Build a self-refreshing MIS in Excel/Power BI that the CFO reads on the 3rd

Nobody in college made you close books against a clock, chase a mismatched paise, or defend an ITC number to an auditor. That is the entire gap — and it is closeable in weeks, not years, because it is *procedural* skill, not new theory.

What "proficient" looks like

The concrete bar an interviewer or a 3-month probation tests for. A job-ready person can, **unaided**:

- Book any voucher in Tally/ERP with the correct ledger, tax, TDS section, and cost centre — and know *why* each leg hits that account.
- Reconcile a bank statement to the books and explain every open item (uncleared cheque, bank charge, direct debit).
- Pull a GSTR-2B, match it against the purchase register in Excel, and produce a clean "eligible ITC" figure with a reason for every mismatch.
- Do a month-end close: accruals, prepaid amortization, depreciation, provisions, and hand over a tied-out TB.
- Build an MIS in Excel using `SUMIFS`, `XLOOKUP`, and a PivotTable that updates when you paste new data.
- Read a GL and answer "why is the telephone expense up 40% this month?" in five minutes.

If you can do those six things without asking anyone, you are worth ₹6-10 LPA in India today.

Hands-on: how to actually do it

1. The entries that are 80% of the job. Real Dr/Cr, India context.

Purchase of raw material ₹1,00,000 + 18% GST from a registered vendor:

Account	Dr (₹)	Cr (₹)
Purchases A/c	1,00,000	
Input CGST A/c	9,000	
Input SGST A/c	9,000	
To Vendor (Sundry Creditor)		1,18,000

Professional fees ₹50,000 to a consultant, TDS u/s 194J @ 10%:

Account	Dr (₹)	Cr (₹)
Professional Fees A/c	50,000	
To TDS Payable (194J)		5,000
To Consultant (Creditor)		45,000

Month-end prepaid insurance (₹24,000 paid for the year, book 1 month):

Account	Dr (₹)	Cr (₹)
Insurance Expense A/c	2,000	
To Prepaid Insurance A/c		2,000

2. TallyPrime click-path (voucher entry). `Gateway of Tally → Vouchers → F9 (Purchase)` → set *Party A/c name* → select *Purchase ledger* → *GST ledgers auto-populate from the ledger's tax setup* → enter qty/rate → `Ctrl+A` to save. For TDS, the expense ledger must have *"Is TDS applicable = Yes"* and the correct *Nature of Payment* set, so Tally deducts automatically.

3. GST portal path (GSTR-3B). `gst.gov.in → Login → Returns Dashboard → select period → GSTR-3B → Prepare Online` → fill 3.1 (outward tax), 4 (eligible ITC from 2B) → `Save` → `Proceed to Payment` → create challan (PMT-06) → `File with DSC/EVC`.

4. The reconciliation in Excel. Match purchase register to GSTR-2B on invoice number + GSTIN:

```
=XLOOKUP(A2&B2, GSTR2B!$A:$A & GSTR2B!$B:$B, GSTR2B!$D:$D, "NOT IN 2B")
```

Flag tax mismatches:

```
=IF(ABS(C2-VLOOKUP(A2,GSTR2B!$A:$D,4,0))>1,"MISMATCH","OK")
```

Sum eligible ITC by GST rate:

```
=SUMIFS(ITC_Amount, Rate, 18, Status, "OK")
```

5. SQL, when the data lives in a database (NetSuite/SAP export, or a startup on Postgres). Ledger movement for one account:

```
SELECT gl_account, SUM(debit) - SUM(credit) AS net_movement
FROM journal_lines
WHERE posting_date BETWEEN '2026-06-01' AND '2026-06-30'
  AND gl_account = '5100-Telephone'
GROUP BY gl_account;
```

Top 10 vendors by spend this quarter:

```
SELECT vendor_name, SUM(amount) AS total_spend
FROM ap_invoices
WHERE invoice_date >= '2026-04-01'
GROUP BY vendor_name
ORDER BY total_spend DESC
LIMIT 10;
```

6. Python for the reconciliation nobody wants to do by hand:

```python
import pandas as pd
books = pd.read_excel("purchase_register.xlsx")
portal = pd.read_excel("gstr2b.xlsx")
merged = books.merge(portal, on="invoice_no", how="outer",
                     suffixes=("_books", "_2b"), indicator=True)
only_books = merged[merged["_merge"] == "left_only"]   # ITC claimed but not in 2B → risk
only_2b    = merged[merged["_merge"] == "right_only"]   # in 2B but not booked → missed ITC
print(only_books[["invoice_no", "tax_books"]])
```

7. Power BI / DAX measure for MIS:

```
Gross Margin % =
DIVIDE(
    SUM(Sales[Revenue]) - SUM(Sales[COGS]),
    SUM(Sales[Revenue])
)
```

Worked example / mini-project

Close June 2026 books for "Kaveri Traders Pvt Ltd" and produce the MIS. Reproduce this in Excel + Tally.

Given for June: Sales ₹42,00,000; Purchases ₹28,00,000; Salaries ₹6,00,000 (TDS 192 ₹40,000); Rent ₹1,50,000 (TDS 194I 10% = ₹15,000); Insurance ₹24,000 paid 1 Apr for 12 months; Machinery ₹12,00,000 (dep @ 15% WDV, book 1 month).

Step 1 — Accruals & adjustments.

Entry	Dr (₹)	Cr (₹)
Insurance Expense / To Prepaid	2,000	2,000
Depreciation / To Acc. Depreciation (12,00,000×15%÷12)	15,000	15,000
Rent / To TDS 194I / To Landlord	1,50,000	15,000 + 1,35,000

Step 2 — Provisional P&L (MIS):

Line	₹
Revenue	42,00,000
Less: Purchases (COGS)	28,00,000
Gross Profit	**14,00,000**
Salaries	6,00,000
Rent	1,50,000
Insurance	2,000
Depreciation	15,000
Net Profit	**6,33,000**

Step 3 — Compliance figures to file: GST output (say 18%) ₹7,56,000; ITC from purchases ₹5,04,000; net GST payable ₹2,52,000 (GSTR-3B by 20 Jul). TDS payable = 40,000 + 15,000 = ₹55,000 (deposit by 7 Jul, challan 281).

Step 4 — MIS one-liner in Excel (from a transactions sheet):

```
=SUMIFS(Amount, Type, "Revenue", Month, "Jun") - SUMIFS(Amount, Type, "Expense", Month, "Jun")
```

Deliverable: a tied-out TB, a P&L that a manager reads in 30 seconds, and two compliance numbers ready to file. *That* is a month-end close.

How it's tested

Interview questions (verbal): - "Walk me through the entry for a purchase with GST and TDS." (They watch whether you know TDS is on the base, not on GST.) - "GSTR-2B shows ₹5 lakh ITC, your books show ₹5.4 lakh. What do you do?" - "What's the difference between provision and accrual? Give a real example." - "Bank balance per books is ₹2,00,000, per statement ₹1,85,000. Reconcile."

Practical assessments (this is what filters people): - **Timed Excel test (30-45 min):** given a raw transaction dump, build a PivotTable, write `SUMIFS` / `XLOOKUP`, and produce a summary. No internet. - **"Close these books" case:** a messy trial balance with 5 missing adjustments; find and pass them. - **Reconciliation screen:** two files (bank statement + ledger), identify every unmatched line. - **SQL screen (for FinTech/analyst roles):** join AP invoices to payments, find unpaid > 60 days. - **Tally live test:** book 10 vouchers correctly with tax and cost centres in front of them.

Common mistakes & how pros avoid them

Mistake	Why it happens	How pros avoid it
Deducting TDS on the GST-inclusive amount	Confusing base vs total	TDS is always on the *taxable value*, not GST
Claiming full ITC without checking 2B	Trusting the invoice	Reconcile 2B *before* filing 3B, every month
Hard-coding numbers in the MIS	Speed under pressure	Formulas + references only; never type a total
Forgetting reversing entries for accruals	Not thinking about next month	Pass accrual + auto-reverse on the 1st
`VLOOKUP` breaking on inserted columns	Column-index fragility	Use `XLOOKUP / INDEX-MATCH`
Posting to the wrong cost centre / GL	Autopilot	Ledger scrutiny before close; review top 10 variances
Round-off / paise mismatches ignored	"It's only ₹2"	Reconcile to zero; ₹2 today is ₹2 lakh mislabeled tomorrow

Learn-it roadmap & resources

Realistic time-to-proficiency from an MBA/CA-Inter base: **6-10 weeks of deliberate practice.**

Week	Focus	Resource
1-2	Tally end-to-end + real vouchers	TallyPrime (free 7-day licence), Tally Education, YouTube: FinTaxPro
2-3	GST returns + reconciliation	GST portal sandbox, ICAI GST material, ClearTax blogs
3-4	TDS, month-end close, accruals	TRACES portal, ICAI practical guides
4-6	Excel for finance	Excel: `SUMIFS/XLOOKUP/INDEX-MATCH`, PivotTables — Chandoo.org, Corporate Finance Institute (CFI)
6-8	Power BI / SQL for MIS	Microsoft Learn (free), Mode SQL tutorial, DataCamp
8-10	ERP exposure	SAP FICO fundamentals (openSAP free), NetSuite/Zoho trial

Certifications worth listing: CFI's *Financial Modeling & Valuation Analyst (FMVA)*, Microsoft PL-300 (Power BI), TallyPrime certification, and — highest signal in India — your CA articleship/CA Inter itself. Pair one tool cert with demonstrable practice files.

Quick-reference

Item	Value / formula
GSTR-1 due	11th of next month
GSTR-3B due	20th of next month
TDS deposit due	7th of next month
TDS 194J (professional)	10%
TDS 194I (rent, plant/machinery / land-building)	2% / 10%
TDS 194C (contractor, non-individual)	2%
TDS 192 (salary)	slab-based
Standard GST slabs	0 / 5 / 12 / 18 / 28%
Match invoice to 2B	`=XLOOKUP(inv&gstin, 2B!inv&gstin, 2B!tax, "NOT IN 2B")`
Conditional sum	`=SUMIFS(sum_range, crit_range, criteria)`
Net GST payable	Output tax − Eligible ITC
Depreciation (WDV, monthly)	`Opening WDV × rate ÷ 12`
Bank rec identity	Book bal ± uncleared items = Bank bal
Ledger movement (SQL)	`SUM(debit) − SUM(credit) GROUP BY account`
Gross Margin % (DAX)	`DIVIDE(Revenue − COGS, Revenue)`

The one line to remember: the exam rewarded *knowing* the entry; the job rewards *booking 400 of them correctly, reconciling to zero, and filing on time*. Close that gap and you are job-ready.

The accounting cycle in a real company

What it is & where it's used

The accounting cycle is the assembly line that turns a shoebox of receipts into a signed balance sheet. In a real company it runs every single day and closes hard every month. The flow never changes:

Source document → Voucher → Journal/Ledger → Trial Balance → Adjustments → Financial Statements → Close.

Every finance role lives somewhere on this line. An **Accounts Payable executive** creates purchase vouchers from vendor invoices. An **AR/billing analyst** raises sales invoices (the source doc for the *customer*). A **staff accountant** posts journals and reconciles ledgers. A **GL/R2R (Record-to-Report) analyst** owns the trial balance and month-end close. A **financial reporting** person builds the P&L and Balance Sheet. In India this happens in **TallyPrime, Zoho Books, or SAP/Oracle**; the artefacts (invoice, voucher, ledger, TB) are identical across all of them. If you can trace one rupee from a supplier invoice all the way to "Trade Payables" on the balance sheet, you understand the job.

The gap: why companies want this (and college didn't teach it)

College teaches you to *pass* journal entries from a textbook that already tells you "Purchased goods for ₹50,000." It never shows you the actual **tax invoice** that the ₹50,000 came from, why it has a GSTIN, an HSN code, a place-of-supply, and CGST+SGST split — and how one wrong GSTIN kills your Input Tax Credit. It teaches debit-credit rules but not *voucher types*, *bill-wise tracking*, *reconciliation*, or what "the books don't tie" means at 9pm on the 5th of the month.

The industry gap is this: **college gives you the entry, the job gives you the evidence and makes you tie it out.** Employers pay for the person who can take a messy pile of real artefacts and produce a clean, reconciled trial balance that a CA can sign. That's a *process* skill, not a theory skill — and it's exactly what an MBA/CA-Inter syllabus underweights.

What "proficient" looks like

A job-ready person can, unaided:

- Read a **GST tax invoice** and identify GSTIN, HSN/SAC, taxable value, CGST/SGST/IGST, and place of supply — and know whether ITC is available.
- Pick the **correct voucher type** in Tally (Purchase, Sales, Payment, Receipt, Contra, Journal) — this is the #1 thing juniors get wrong.
- Post entries with **bill-wise details** so payables/receivables can be aged.
- Extract a **trial balance**, spot that it doesn't tie, and find the difference (transposition, one-sided posting, wrong side).
- Pass **month-end adjustments**: depreciation, prepaid, accruals, provisions, closing stock.
- Roll the adjusted TB into a **P&L and Balance Sheet** and prove the BS balances.
- Reconcile **bank, GSTR-2B vs purchase register, and vendor ledgers**.

Hands-on: how to actually do it

Step 1 — Source document to journal. A vendor invoice: goods ₹1,00,000 + 18% GST (intra-state) = ₹18,000 (₹9,000 CGST + ₹9,000 SGST), total ₹1,18,000.

Account	Dr (₹)	Cr (₹)
Purchases	1,00,000	
Input CGST	9,000	
Input SGST	9,000	
To Trade Payables — ABC Traders		1,18,000

Step 2 — In TallyPrime (the actual click-path):

```
Gateway of Tally → Vouchers → F9 (Purchase)
  Party A/c name: ABC Traders  → (bill-wise: New Ref, Inv-042)
  Purchase ledger: Purchases @18% (set GST details, HSN)
  GST auto-computes CGST + SGST
  Ctrl+A to accept
```

Rule of thumb for voucher type: **money out of bank = Payment (F5); money in = Receipt (F6); bank↔cash or bank↔bank = Contra (F4); credit purchase = F9; credit sale = F8; anything with no cash/bank movement (depreciation, provisions) = Journal (F7).**

Step 3 — Trial balance in Excel. Say you exported ledger closing balances. Prove it ties:

```
=SUMIF(C:C,"Dr",D:D) - SUMIF(C:C,"Cr",D:D)   → must equal 0
```

Find a suspicious difference that's divisible by 9 (a transposition error, e.g. 5,400 keyed as 4,500):

```
=IF(MOD(ABS(TotalDr-TotalCr),9)=0,"Likely transposition","Check one-sided posting")
```

Step 4 — Pull the vendor sub-ledger with SQL (any ERP with a DB behind it):

```sql
SELECT v.party_name,
       SUM(CASE WHEN j.dc = 'D' THEN j.amount ELSE 0 END) AS debit,
       SUM(CASE WHEN j.dc = 'C' THEN j.amount ELSE 0 END) AS credit,
       SUM(CASE WHEN j.dc = 'C' THEN j.amount ELSE -j.amount END) AS balance
FROM journal_lines j
JOIN parties v ON v.id = j.party_id
WHERE j.ledger = 'Trade Payables'
  AND j.txn_date <= '2026-03-31'
GROUP BY v.party_name
HAVING balance <> 0
ORDER BY balance DESC;
```

Step 5 — Reconcile GSTR-2B vs purchase register in Python (the ITC check every AP team runs):

```
import pandas as pd
pr   = pd.read_excel("purchase_register.xlsx")   # your books
b2b  = pd.read_excel("gstr2b.xlsx")              # from GST portal

merged = pr.merge(b2b, on=["gstin", "invoice_no"],
                  how="outer", indicator=True, suffixes=("_books", "_2b"))

print(merged.loc[merged["_merge"]=="left_only"])   # in books, NOT in 2B → ITC at risk
print(merged.loc[merged["_merge"]=="right_only"])  # in 2B, NOT in books → missed entry

merged["tax_diff"] = merged["tax_books"].fillna(0) - merged["tax_2b"].fillna(0)
print(merged.loc[merged["tax_diff"].abs() > 1])    # value mismatches
```

Worked example / mini-project

Reproduce a one-month close for "Nashik Auto Parts Pvt Ltd." Opening: Capital ₹5,00,000, Bank ₹5,00,000. March transactions:

Date	Event	Voucher	Entry
02	Buy stock ₹1,00,000 +18% GST on credit (ABC)	Purchase F9	Dr Purchases 1,00,000 / Dr Input CGST 9,000 / Dr Input SGST 9,000 / Cr ABC 1,18,000
10	Sell goods ₹1,50,000 +18% GST on credit (XYZ)	Sales F8	Dr XYZ 1,77,000 / Cr Sales 1,50,000 / Cr Output CGST 13,500 / Cr Output SGST 13,500
15	Pay rent ₹20,000 by bank	Payment F5	Dr Rent 20,000 / Cr Bank 20,000
20	Receive ₹1,77,000 from XYZ	Receipt F6	Dr Bank 1,77,000 / Cr XYZ 1,77,000
28	Pay ABC ₹1,18,000	Payment F5	Dr ABC 1,18,000 / Cr Bank 1,18,000

Month-end adjustments (Journal F7): - Depreciation on ₹1,20,000 machinery @15% p.a. for 1 month = **₹1,500** → Dr Depreciation / Cr Machinery. - Closing stock ₹40,000 → shown in P&L (credit) and BS (asset). - GST payable = Output ₹27,000 − Input ₹18,000 = **₹9,000** net liability.

Trial Balance (extract), then the statements:

P&L: Sales 1,50,000 + Closing stock 40,000 − Purchases 1,00,000 − Rent 20,000 − Depreciation 1,500 = **Net profit ₹68,500.**

Balance Sheet:

Liabilities	₹	Assets	₹
Capital 5,00,000 + Profit 68,500	5,68,500	Bank (5,00,000−20,000+1,77,000−1,18,000)	5,39,000
GST payable	9,000	Closing stock	40,000
		Machinery (1,20,000−1,500)	1,18,500
Wait — Machinery wasn't bought this month; treat opening PPE ₹1,20,000 funded from capital			
Total	**5,77,500**	**Total**	**5,77,500**

The point of the exercise: pass every entry in Tally, export the TB, and confirm **Assets = Liabilities + Equity to the rupee.** If it's off, walk back through the ledgers.

How it's tested

Interview questions: - "Walk me from a supplier invoice to the balance sheet." (Trace the artefact chain.) - "Which voucher type for paying salary? For depreciation?" (Payment; Journal.) - "Your trial balance is off by ₹900 — what do you check first?" (Divisible by 9 → transposition.) - "What's GSTR-2B and why reconcile it?" (ITC eligibility.) - "Golden rules of accounting?" (Debit the receiver / what comes in / expenses & losses.)

Practical assessments companies actually give: 1. A **timed Tally test**: "Here are 15 vouchers, post them and give me the P&L in 30 minutes." 2. An **Excel case**: raw ledger dump → build the TB, find the ₹X difference, produce statements. 3. A **reconciliation screen**: two files (books vs bank / books vs 2B), find the mismatches — often live in Excel with `VLOOKUP/XLOOKUP`.

```
=XLOOKUP(A2, GST2B!B:B, GST2B!E:E, "MISSING IN 2B")
```

Common mistakes & how pros avoid them

Mistake	Fix
Using Journal (F7) for a bank payment	Any cash/bank movement gets Payment/Receipt/Contra, never Journal
Booking gross invoice to Purchases (GST included)	Split tax to Input CGST/SGST; only net goes to Purchases
Not enabling bill-wise details	Turn on bill-wise → payables/receivables become ageable
Claiming ITC not in GSTR-2B	Reconcile 2B monthly *before* filing GSTR-3B
Forcing the TB to tie with a "Suspense" plug and forgetting it	Investigate the difference; suspense is temporary, not a fix
Ignoring the sign of closing stock	Appears twice: P&L credit AND balance-sheet asset
Posting to the wrong period	Lock periods after close; date-stamp every voucher

Pros reconcile *as they go*, keep a **close checklist**, and never trust a TB that ties without also checking that individual sub-ledgers (AP/AR/bank) match their supporting schedules.

Learn-it roadmap & resources

Time to proficiency: 6–10 weeks of daily practice from a working debit-credit base.

- **Weeks 1–2:** Golden rules, voucher types, pass 50 entries by hand.
- **Weeks 3–5:** TallyPrime end-to-end — company creation, GST setup, all voucher types, export TB and statements. (Free: *Tally Education* self-learning; YouTube *FinTaxPro*, *CA Rahul Malodia*.)
- **Weeks 6–8:** Excel reconciliations (XLOOKUP, SUMIF, pivot the ledger), one full month-end close.
- **Weeks 9–10:** GST portal — file a practice GSTR-3B, reconcile 2B; touch SAP FICO concepts.

Certifications that carry weight: Tally's **TallyPrime with GST** certificate; **SAP FICO** (for MNC GL roles); your **CA-Inter** itself is the strongest signal. Free portal practice: log into `gst.gov.in` with a sandbox/test GSTIN.

Quick-reference

Item	Detail
Cycle	Source doc → Voucher → Ledger → TB → Adjustments → Financials → Close
Golden rules	Personal: Dr receiver, Cr giver · Real: Dr in, Cr out · Nominal: Dr expenses/losses, Cr income/gains
Tally voucher keys	F4 Contra · F5 Payment · F6 Receipt · F7 Journal · F8 Sales · F9 Purchase
TB tie test	Σ Debits = Σ Credits; diff ÷ 9 → transposition; diff even → one-sided
GST intra-state	CGST + SGST (9%+9%); inter-state → IGST 18%
ITC check	Purchase register must match GSTR-2B before GSTR-3B
Accounting equation	Assets = Liabilities + Equity (must tie to the rupee)
Excel workhorses	`=XLOOKUP()` `=SUMIF()` `=MOD(diff,9)` pivot tables
Key adjustments	Depreciation, prepaid, accruals, provisions, closing stock
Depreciation (SLM/month)	Cost × rate ÷ 12 → Dr Depreciation / Cr Asset

Journal entries you will actually pass

What it is & where it's used

A journal entry is the atom of accounting: for every transaction you record which account to **Debit (Dr)** and which to **Credit (Cr)**, in equal amounts. Everything downstream — the trial balance, GST returns, the P&L, the balance sheet, the audit — is just journal entries aggregated. If you can pass the right entry for a real business event, you can do 70% of an accounts executive's job.

This is the core skill of every hands-on finance/accounts role: **Accounts Executive, Accounts Payable/Receivable, GST/Tax executive, Article assistant, Financial Analyst (who must read the entries auditors passed), and Controllers** who review them. In practice you pass these in **TallyPrime, Zoho Books, SAP, Oracle NetSuite, or QuickBooks** — but the tool only asks you the same question: *which head is Dr, which is Cr, and for how much?*

The gap: why companies want this (and college didn't teach it)

College teaches you the rule ("debit the receiver, credit the giver") and makes you pass textbook entries with clean numbers. Industry throws messy, tax-laden, multi-leg reality at you:

- A single sales invoice has **five lines** — party, sales, CGST, SGST, and possibly TCS or a round-off — not the two-line entry from your textbook.
- Real entries carry **input tax credit (ITC)** logic, **TDS deduction**, **reverse charge (RCM)**, **accruals at month-end**, and **prepaid amortisation** that colleges skip entirely.
- Nobody in a job says "pass a journal entry." They say *"book this vendor bill,"* *"the electricity bill for March hasn't come — provide for it,"* *"amortise the annual insurance,"* *"run depreciation for the quarter."* You must translate business language into Dr/Cr yourself.

The gap employers pay to close: **turning a real document (an invoice, a payslip, a bank statement, a rent agreement) into the correct, tax-compliant entry — first try, without a template.**

What "proficient" looks like

A job-ready person can, unaided:

1. Look at any **GST invoice** and book it with the correct CGST/SGST vs IGST split (intra- vs inter-state).
2. Know when to **debit an asset vs an expense**, and when tax is an asset (ITC) vs a cost (blocked credit / RCM).
3. Pass **month-end adjustment entries** — accruals, prepaids, depreciation, provisions — so the P&L reflects the *period*, not just cash movement.
4. Handle **TDS on both sides** (deducted by us on expenses; deducted by customers on our income).
5. Self-check: every entry's **total Dr = total Cr**, and the resulting balance makes intuitive sense (an asset stays debit, income stays credit).

Hands-on: how to actually do it

Golden rule set (modern, balance-sheet approach): **Assets & Expenses increase on Debit; Liabilities, Equity & Income increase on Credit.** Reverse to decrease.

1. Sales with GST (intra-state, ₹1,00,000 + 18% GST)

Account	Dr (₹)	Cr (₹)
Debtor / Bank A/c	1,18,000	
To Sales A/c		1,00,000
To Output CGST @9%		9,000
To Output SGST @9%		9,000

Inter-state? Replace CGST+SGST with a single **Output IGST @18% ₹18,000**.

2. Purchase with GST (ITC claim)

Account	Dr (₹)	Cr (₹)
Purchases A/c	50,000	
Input CGST @9%	4,500	
Input SGST @9%	4,500	
To Creditor A/c		59,000

Input GST is a **current asset** (receivable from government), not an expense.

3. Expense with TDS (professional fees ₹1,00,000, TDS 10% u/s 194J)

Account	Dr (₹)	Cr (₹)
Professional Fees A/c	1,00,000	
Input CGST @9%	9,000	
Input SGST @9%	9,000	
To Vendor A/c		1,08,000
To TDS Payable (194J)		10,000

TDS is computed on the **taxable value (₹1,00,000), not on GST**. You pay the vendor ₹1,08,000 and deposit ₹10,000 with the government.

4. Payroll (gross ₹5,00,000; PF ₹30,000 employee; TDS ₹40,000; net paid ₹4,30,000)

Account	Dr (₹)	Cr (₹)
Salaries & Wages A/c	5,00,000	
To Salary Payable A/c		4,30,000
To PF Payable (employee)		30,000
To TDS Payable (192)		40,000

Employer PF/ESI is a **separate** entry: `Dr PF Employer Contribution (expense) / Cr PF Payable`.

5. Accrual (electricity for March, bill not yet received, est. ₹25,000)

Account	Dr (₹)	Cr (₹)
Electricity Expense A/c	25,000	
To Outstanding Expenses A/c		25,000

Reverse next month when the actual bill arrives, then book the real invoice.

6. Prepaid (annual insurance ₹1,20,000 paid 1 Apr; monthly charge ₹10,000)

At payment:

Account	Dr (₹)	Cr (₹)
Prepaid Insurance A/c	1,20,000	
To Bank A/c		1,20,000

Every month-end:

Account	Dr (₹)	Cr (₹)
Insurance Expense A/c	10,000	
To Prepaid Insurance A/c		10,000

7. Depreciation (machine ₹6,00,000, SLM, 10% p.a. → ₹5,000/month)

Account	Dr (₹)	Cr (₹)
Depreciation A/c	5,000	
To Accumulated Depreciation A/c		5,000

8. Provision (audit fees payable, estimated ₹75,000)

Account	Dr (₹)	Cr (₹)
Audit Fees A/c	75,000	
To Provision for Audit Fees A/c		75,000

TallyPrime click-path (Sales voucher)

`Gateway of Tally → Vouchers → F8 (Sales) → Party A/c name → Sales ledger → select Item/amount → GST ledgers auto-calc if rates are set in the stock item → Ctrl+A to save.` For a pure JV use **F7 (Journal)**.

Worked example / mini-project

Reproduce March books for "Kaleru Traders" (Telangana, intra-state).

Transactions: 1. Sold goods ₹2,00,000 + 18% GST on credit to a Hyderabad customer. 2. Bought raw material ₹80,000 + 18% GST on credit from a local vendor. 3. Paid March rent ₹40,000 + 18% GST; landlord deducts nothing, we deduct TDS 10% u/s 194I. 4. Ran payroll: gross ₹1,00,000, TDS ₹8,000, net ₹92,000. 5. Month-end: depreciation ₹5,000; provide ₹15,000 for unbilled electricity.

Entries:

#	Account	Dr (₹)	Cr (₹)
1	Debtors	2,36,000	
	To Sales		2,00,000
	To Output CGST / SGST		18,000 / 18,000
2	Purchases	80,000	
	Input CGST / SGST	7,200 / 7,200	
	To Creditors		94,400
3	Rent	40,000	
	Input CGST / SGST	3,600 / 3,600	
	To Landlord		43,200
	To TDS Payable (194I)		4,000
4	Salaries	1,00,000	
	To Salary Payable		92,000
	To TDS Payable (192)		8,000
5a	Depreciation	5,000	
	To Accumulated Dep.		5,000
5b	Electricity	15,000	
	To Outstanding Expenses		15,000

Net GST payable check: Output ₹36,000 − Input (₹14,400 + ₹7,200) = **₹14,400 payable** in GSTR-3B. This single reconciliation is what a GST executive computes every month.

How it's tested

Interview questions: - "Pass the entry for a credit sale of ₹1,00,000 with 18% GST, inter-state." (Expect: IGST, not CGST/SGST.) - "Insurance paid for the full year — how do you treat it?" (Prepaid + monthly amortisation.) - "Difference between provision and accrual?" (Provision = estimated known liability; accrual = expense incurred, bill awaited.) - "Is input GST an expense?" (No — a current asset / ITC.) - "Bill received but goods not yet delivered — book it?" (Match to period; goods-in-transit / accrual logic.)

Practical assessment: Firms hand you **5–10 vouchers or a stack of invoices and a 30-minute timer** in Tally or on paper, and ask you to book them and produce a trial balance that ties. Some give a "close these books" case: raw transactions + instructions to pass month-end adjustments and hand back a P&L. Big-4 article interviews probe the *why* behind each Dr/Cr.

Common mistakes & how pros avoid them

Mistake	Fix
Charging CGST/SGST on an inter-state sale	Check place of supply first: same state → C+S; different state → IGST.
Deducting TDS on the GST portion	TDS is on the **taxable value only**.
Treating input GST as an expense	Book it as **Input CGST/SGST/IGST (asset)** so ITC flows to GSTR-3B.
Capitalising a repair / expensing an asset	Asset = future benefit > 1 yr; repair = restore current condition.
Forgetting to reverse accruals next period	Pass a reversing JV on day 1 of the new month.
Provision & actual bill both hitting expense (double count)	Reverse the provision when the real invoice is booked.
Entry doesn't balance	Always verify **ΣDr = ΣCr** before saving.

Learn-it roadmap & resources

Time to proficiency: 4–6 weeks of daily practice (you already have the theory from CA Inter).

- **Week 1–2:** Master the modern golden rules; hand-write the 8 entry types above until automatic.
- **Week 3:** TallyPrime — do the free **Tally Education** course or ICAI's Tally module; book 50+ real invoices.
- **Week 4:** GST layer — practise CGST/SGST/IGST/RCM and reconcile a month to GSTR-3B on the GST portal (sandbox).
- **Ongoing:** Pull a real company's financials and reverse-engineer what entries produced each line.

Resources: ICAI Accounting & Taxation study material (free), TallyPrime GST self-learning (free), Zoho Books "Accountant" free plan for sandbox practice, and Cleartax/TaxGuru articles for TDS/GST rate updates.
Certifications that signal competence: Tally ACE/Professional, and your CA Inter itself.

Quick-reference

Event	Debit	Credit
Credit sale (intra)	Debtor	Sales + Output CGST + Output SGST
Credit sale (inter)	Debtor	Sales + Output IGST
Purchase	Purchases + Input CGST/SGST	Creditor
Expense + TDS	Expense + Input GST	Vendor + TDS Payable
Payroll	Salaries	Salary Payable + PF + TDS
Accrual	Expense	Outstanding Expenses
Prepaid (pay) / (charge)	Prepaid Asset / Expense	Bank / Prepaid Asset
Depreciation	Depreciation	Accumulated Depreciation
Provision	Expense	Provision A/c

Rules: Assets & Expenses ↑ = Dr · Liabilities, Equity, Income ↑ = Cr · TDS on taxable value only · Input GST = asset · Same state → CGST+SGST, different state → IGST · Always ΣDr = ΣCr.

Common rates: GST 5/12/18/28% · TDS 194J (professional) 10% · 194I (rent) 10% · 194C (contractor) 1–2% · 192 (salary) at slab.

TallyPrime Hands-On

What it is & where it's used

TallyPrime is the accounting and compliance backbone of Indian SMEs. Roughly 2 million-plus businesses run their books on Tally, which means when a hiring manager at a mid-size firm, a CA practice, or a trading/manufacturing company reads "Tally" on your resume, they mean: *can you actually pass entries, generate GST returns, and pull a Trial Balance without hand-holding?*

Roles that require it, day one:

- **Accounts Executive / Accounts Assistant** — voucher entry, bank reconciliation, ledger scrutiny.
- **Accountant / Sr. Accountant** — month-end close, GST/TDS working, MIS reports.
- **Audit article / audit assistant** — you *receive* client Tally data and vouch it.
- **Tax / GST executive** — GSTR-1 and GSTR-3B reconciliation straight out of Tally.
- **Finance analyst at an SME** — Tally is the source system before data hits Excel.

Globally, the equivalents are QuickBooks (US/UK SMEs), Xero (ANZ/UK), Zoho Books, and SAP B1 for larger shops. The *muscle* you build in Tally — double-entry discipline, master setup, tax mapping, reconciliation — ports directly to all of them.

The gap: why companies want this (and college didn't teach it)

An MBA or B.Com teaches you *what* a journal entry is. It does not teach you that in Tally you press **F9** for a Purchase, that a "ledger" must sit under the correct "group" or your Balance Sheet mis-classifies, or that a wrong GST rate at ledger level silently corrupts your GSTR-1.

The specific gaps employers pay to close:

College teaches	The job needs
Debit the receiver, credit the giver	Which *voucher type* (F5/F6/F7/F8/F9) records it, and the exact click-path
"GST is 18%"	Setting CGST/SGST/IGST ledgers, HSN codes, and tax-rate at stock/ledger level so returns auto-compute
Bank reconciliation concept	Matching Tally to a bank statement, setting the *bank date*, clearing 40 uncleared entries
Trial Balance definition	Drilling from Balance Sheet → group → ledger → voucher to find a ₹500 mismatch

Companies want someone who can be handed a laptop with TallyPrime open and be productive in an hour, not someone who needs a week of "where's the sales entry?"

What "proficient" looks like

A job-ready person can, unaided:

1. Create a company with correct financial year, GSTIN, and features enabled.
2. Build the group/ledger structure and stock items with tax rates.

3. Pass all five core vouchers (Sales, Purchase, Payment, Receipt, Journal) with GST auto-calculating.
4. Reconcile a bank ledger against a statement and explain every uncleared item.
5. Generate and *interpret* Balance Sheet, P&L, Trial Balance, Day Book, GSTR-1, GSTR-3B, and the Ledger.
6. Fix a mismatched Trial Balance by drill-down.

Speed benchmark: a proficient user passes a standard sales invoice in under 30 seconds and closes a month's bank rec in under 20 minutes.

Hands-on: how to actually do it

> Navigation note: `Gateway of Tally` is your home. `Alt+G` (Go To) is the universal search — type any report/master name. `Ctrl+A` saves any screen. `Esc` backs out.

1. Company setup

```
Open TallyPrime → Alt+K (Company) → Create Company
  Company Name : Sharma Traders
  Financial year beginning : 1-Apr-2025
  Books beginning from      : 1-Apr-2025
  Set/Alter GST details     : Yes  → State: Telangana, GSTIN: 36ABCPS1234F1Z5, Reg type: Regul
  Base currency symbol      : ₹
Ctrl+A to save
```

Enable features once: **F11 (Features)** → set *Maintain Accounts = Yes*, *Enable Goods and Services Tax (GST) = Yes*, *Maintain Inventory = Yes*.

2. Masters — groups and ledgers

Every ledger MUST sit under a group; the group decides where it lands in the financials.

```
Gateway of Tally → Create (or Alt+G → "Create Ledger")
```

Core ledgers to create:

Ledger	Under (Group)	Notes
Sales – GST 18%	Sales Accounts	Set GST rate 18%, HSN
Purchase – GST 18%	Purchase Accounts	Set GST rate 18%
CGST	Duties & Taxes	Type: GST, Tax type: Central
SGST	Duties & Taxes	Type: GST, Tax type: State
IGST	Duties & Taxes	Type: GST, Tax type: Integrated
HDFC Bank	Bank Accounts	Enter A/c no. + IFSC
Cash	Cash-in-Hand	Auto-exists
ABC Enterprises (customer)	Sundry Debtors	Set GSTIN + state
XYZ Suppliers (vendor)	Sundry Creditors	Set GSTIN + state
Rent	Indirect Expenses	—

3. GST setup

```
F11 → GST = Yes → set company GSTIN & state.
At ledger level (Sales/Purchase): Set/alter GST details = Yes → Taxability: Taxable, Integrate
Stock item level: same GST-details screen, plus HSN/SAC code.
```

Rule Tally applies automatically: buyer state = seller state → **CGST + SGST**; different states → **IGST**. Set the state correctly on party ledgers or Tally picks the wrong tax.

4. Voucher entry — the five you use daily

Voucher	Key	Records
Payment	F5	Money going out (pay vendor, expenses)
Receipt	F6	Money coming in (customer pays)
Contra	F4	Bank↔Cash transfers
Purchase	F9	Buying goods/services
Sales	F8	Selling goods/services
Journal	F7	Adjustments, provisions, depreciation

Sales invoice (F8) — sell 10 units @ ₹1,000, intra-state 18% GST:

```
Gateway of Tally → Vouchers → F8 (Sales)
  Party A/c name : ABC Enterprises
  Sales ledger   : Sales – GST 18%
  Item : Product-A   Qty 10   Rate 1000   → Amount 10,000
  CGST  → 900   (auto)
  SGST  → 900   (auto)
  Total : 11,800
Ctrl+A to save
```

The equivalent journal Tally posts behind the screen:

Account	Dr (₹)	Cr (₹)
ABC Enterprises (Debtor)	11,800	
Sales – GST 18%		10,000
Output CGST		900
Output SGST		900

Receipt (F6) — customer pays ₹11,800 into HDFC:

Account	Dr (₹)	Cr (₹)
HDFC Bank	11,800	
ABC Enterprises		11,800

Payment (F5) — pay rent ₹20,000 from bank:

Account	Dr (₹)	Cr (₹)
Rent	20,000	
HDFC Bank		20,000

Journal (F7) — provide depreciation ₹5,000:

Account	Dr (₹)	Cr (₹)
Depreciation	5,000	
Accumulated Depreciation		5,000

5. Bank reconciliation

```
Gateway of Tally → Alt+G → "Bank Reconciliation" → select HDFC Bank
  (or: Banking → Bank Reconciliation)
For each entry, in the "Bank Date" column enter the date it actually cleared per the bank stat
Entries with a bank date = reconciled. Blank bank date = still uncleared.
Bottom of screen shows: Balance as per Company books vs Balance as per Bank → difference must
Ctrl+A to save.
```

6. Key reports and their click-paths

Report	Click-path
Balance Sheet	Gateway → Balance Sheet (or `Alt+G` → "Balance Sheet")
Profit & Loss	Gateway → Profit & Loss A/c
Trial Balance	Gateway → Display More Reports → Trial Balance
Day Book (all vouchers of a day)	Gateway → Day Book (`Alt+G` → "Day Book")
Ledger (a party's full statement)	Display More Reports → Account Books → Ledger
GSTR-1 (outward)	Gateway → Display More Reports → GST Reports → GSTR-1
GSTR-3B (summary)	GST Reports → GSTR-3B
Stock Summary	Gateway → Stock Summary
Outstanding receivables	Display More Reports → Statements of Accounts → Outstanding → Receivables

Universal tricks: **F2** changes the report date/period, **Alt+F2** sets a date range, **Ctrl+F12** filters, and **Alt+E** exports to Excel/PDF. On any figure, press **Enter to drill down** all the way to the voucher.

Worked example / mini-project

Reproduce a mini month for **Sharma Traders**, April 2025. Create the company and ledgers above, then enter:

1. **02-Apr — Purchase (F9):** 50 units Product-A @ ₹700 from XYZ Suppliers, GST 18%. Value ₹35,000 + ₹6,300 CGST/SGST split (₹3,150 each) = ₹41,300.
2. **05-Apr — Payment (F5):** Pay XYZ ₹41,300 from HDFC Bank.
3. **10-Apr — Sales (F8):** 10 units @ ₹1,000 to ABC Enterprises, GST 18% → ₹11,800.
4. **15-Apr — Sales (F8):** 20 units @ ₹1,000 to ABC → ₹23,600.
5. **20-Apr — Receipt (F6):** ABC pays ₹35,400 into HDFC.
6. **30-Apr — Payment (F5):** Rent ₹20,000 from HDFC.
7. **30-Apr — Journal (F7):** Depreciation ₹5,000.

Now verify:

- **Trial Balance** (Gateway → Display More Reports → Trial Balance): total Dr = total Cr. If not, drill in.

- **Output GST payable** = ₹6,120 (on ₹34,000 sales); **Input GST** = ₹6,300 (on purchase). Net ITC carried forward ₹180. Check via **GSTR-3B**.
- **HDFC Bank ledger** movement: −41,300 −20,000 +35,400 = closing outflow ₹25,900 from opening. Reconcile assuming the ₹20,000 rent cheque hasn't cleared — set bank dates on the other two, leave rent's blank, and confirm the reconciliation difference = ₹20,000.
- **Stock Summary:** 50 in − 30 out = **20 units** Product-A on hand @ ₹700 = ₹14,000 closing stock.
- **P&L:** Gross Profit = Sales ₹34,000 − COGS (30 × ₹700 = ₹21,000) = ₹13,000; less Rent ₹20,000 and Depreciation ₹5,000 → net loss for the mini-month.

If every number ties, you've done a real close.

How it's tested

Interview questions:

- Which voucher for a credit sale vs a cash sale? (F8; cash = F8 with Cash as party.)
- Difference between a group and a ledger?
- How does Tally decide CGST+SGST vs IGST?
- Where do you set the GST rate — company, ledger, or stock item? (Any/all; most specific wins.)
- My Trial Balance doesn't match — how do you find the error? (Drill Balance Sheet → group → ledger → voucher.)
- What's the entry for depreciation? For a bad debt write-off?

Practical test (the real screen): Companies give a laptop with TallyPrime and say:

> "Here are 8 vouchers and a bank statement. Enter them, reconcile the HDFC bank account, and show me the GSTR-1 and Balance Sheet."

Or hand you a company where the Trial Balance is off by ₹X and ask you to find it. They watch whether you *use shortcuts* (F8/F9, Enter-to-drill, Alt+E) or hunt through menus — that tells them your real hours on the tool.

Common mistakes & how pros avoid them

Mistake	Consequence	Pro habit
Ledger under wrong group (e.g. Rent under Sundry Creditors)	Mis-stated Balance Sheet	Verify group at creation; check Trial Balance grouping
Wrong/blank state on party ledger	GST computes IGST vs CGST/SGST wrong	Always fill GSTIN + state on parties
Using Journal (F7) for everything	Loses inventory + GST auto-calc	Use F8/F9 for trade, F7 only for adjustments
Forgetting HSN/SAC	GSTR-1 rejects/errors on portal	Set HSN at stock/ledger level upfront
Not setting bank date in rec	Reconciliation never balances	Enter bank date only for cleared items
Editing a saved company's financial year mid-way	Data corruption	Set FY correctly at creation
Not backing up	Total data loss	Alt+K → Backup weekly; keep company data folder copied

Learn-it roadmap & resources

Time to proficiency: 3–4 weeks of daily practice (1–2 hrs) to be job-ready for an Accounts Executive role; 2 months to be fully independent on GST returns and close.

Week	Focus
1	Install, company setup, groups/ledgers, F5/F6/F8/F9
2	GST setup, stock items, inventory vouchers, HSN
3	Bank rec, all reports, drill-down, exports
4	Full mini-close + GSTR-1/3B reconciliation

Resources:

- **TallyPrime Educational mode** — free download from tallysolutions.com; runs full software (only restricts some dates) — practice unlimited.
- **Tally official YouTube channel & Help (F1 → Help)** — free, authoritative click-paths.
- **Certification:** *TallyEducation "TallyPrime with GST" / Tally ACE & PRO* certificates — inexpensive, recognized by Indian SMEs and add a resume line.
- Practice with any B.Com "Tally practical" workbook — do the entries, don't just read them.

Quick-reference

Voucher shortcuts (from Voucher screen):

Key	Voucher
F4	Contra
F5	Payment
F6	Receipt
F7	Journal
F8	Sales
F9	Purchase

Navigation & actions:

Key	Action
Alt+G	Go To (universal search)
Alt+K	Company menu
F11	Features
Ctrl+A	Save/accept
Enter	Drill down
F2 / Alt+F2	Change date / period
Alt+E	Export (Excel/PDF)
Esc	Go back

GST logic: same state → CGST + SGST (½ each); different state → IGST (full rate). Common rates: 5%, 12%, 18%, 28%.

Report paths: Gateway of Tally → Balance Sheet / Profit & Loss / Day Book / Stock Summary directly; Trial Balance, Ledger, GST Reports, Outstanding via **Display More Reports**.

Golden rule: every ledger under the right group → correct financials; correct state on parties → correct GST; bank date only on cleared items → clean reconciliation.

SAP FICO / S/4HANA for Finance

What it is & where it's used

SAP is the accounting engine that runs most large companies on earth. **FICO** is the finance half of SAP: **FI (Financial Accounting)** produces the statutory books — General Ledger, Accounts Payable, Accounts Receivable, Asset Accounting, Bank — that feed the balance sheet and P&L. **CO (Controlling)** is the management-accounting side — cost centres, profit centres, internal orders, product costing, profitability analysis (CO-PA). **S/4HANA** is SAP's current-generation product (the successor to ECC 6.0) built on the in-memory HANA database, where FI and CO are merged into a single table called the **Universal Journal (ACDOCA)** — one line item now carries both the GL account and the cost object, so there's no more month-end reconciliation between FI and CO.

Where it's used: every Nifty-50 company, most listed mid-caps, and — critically for an India-based candidate — the **Global Capability Centres (GCCs)** and Big-4 / captive shared-service centres in Bengaluru, Hyderabad, Pune, Gurgaon and Chennai. Roles that require it: **R2R (Record-to-Report) analyst, P2P (AP) associate, O2C (AR) associate, GL accountant, SAP FICO consultant, financial analyst, internal auditor, and tax analyst.** If a job description says "hands-on SAP" and pays a premium over a Tally job, this is the skill they mean.

The gap: why companies want this (and college didn't teach it)

An MBA or CA course teaches you *what* a journal entry is and *how* to prepare a trial balance by hand. It never teaches you that in a real company you **never touch the trial balance directly** — you post a document, the system derives the GL impact, and controls stop you from doing anything stupid. Colleges teach Tally at best, where one person does everything on one screen. SAP is the opposite: it is **built for control and scale**, where AP, AR, GL and treasury are separate teams, every posting leaves an audit trail, and a document once posted **cannot be deleted** (only reversed).

The specific gap employers pay to close:

- You think in **T-accounts**; SAP thinks in **document types, posting keys, and GL account determination**.
- You've never seen **automated postings** (tax, WHT/TDS, exchange differences, GR/IR clearing) that the system generates for you.
- You don't know **org structure** — that a "company" in SAP is a *Company Code*, and one instance can run 40 legal entities across 12 countries simultaneously.
- You've never done a **period-end close** on a clock with SLA penalties.

A fresher who can navigate SAP, post a vendor invoice, run FBL3N, and explain the P2P cycle is instantly worth more than one who only knows theory.

What "proficient" looks like

The concrete bar an R2R/AP/AR hire is tested against:

1. **Navigate by T-code** without hunting through menus — type `/nFB60`, `Enter`, you're on the vendor-invoice screen.

2. **Post the four core documents unaided**: GL journal (FB50), vendor invoice (FB60), customer invoice (FB70), general posting with both debit and credit lines (F-02).
3. **Read a GL account** using FBL3N and explain every line — distinguish open items from cleared items, drill into the source document (FB03).
4. **Explain the org structure**: Client → Company Code → Chart of Accounts → Business Area / Profit Centre / Cost Centre.
5. **Explain the end-to-end cycles**: P2P (PR → PO → GR → Invoice → Payment) and O2C (Order → Delivery → Billing → Receipt), and where FI hooks in.
6. **Clear open items** (F-32 customer, F-44 vendor, F.13 auto-clearing) and understand **GR/IR** reconciliation.
7. **Know what changed in S/4HANA**: Universal Journal, Business Partner (replaces separate customer/vendor masters), New Asset Accounting, Fiori apps.

Hands-on: how to actually do it

Org structure (memorise this hierarchy):

```
Client (e.g. 100)            ← top of the SAP box; shared master data
  └ Company Code (e.g. 1000, "ACME India Pvt Ltd")  ← a legal entity, produces B/S + P&L
       ├ Chart of Accounts (e.g. YCOA)   ← the GL account master list
       ├ Business Area / Segment
       ├ Profit Centre     (CO — internal P&L slice)
       └ Cost Centre       (CO — where costs land: HR, IT, Finance)
```

The core T-codes (type into the command box, prefix with /n to switch):

T-code	What it does	Tally equivalent
FB50	Post GL journal (GL accounts only)	Journal Voucher (F7)
FB60	Post vendor (AP) invoice	Purchase Voucher (F9)
FB70	Post customer (AR) invoice	Sales Voucher (F8)
F-02	General posting, manual Dr/Cr, any account	Journal with full control
FB03	Display any posted document	Drill into voucher
FBL3N	GL account line-item display	Ledger display
FBL1N	Vendor line items	Party ledger (creditor)
FBL5N	Customer line items	Party ledger (debtor)
F-53 / F110	Manual / automatic outgoing payment	Payment Voucher (F5)
FB08	Reverse a document	Delete/alter voucher
F-32 / F-44	Clear customer / vendor open items	Bill-wise adjustment

Posting keys (the two-digit code that tells SAP debit-or-credit and account type — this trips up every fresher):

Posting key	Meaning
40	GL account — Debit
50	GL account — Credit
01	Customer — Debit (invoice)
31	Vendor — Credit (invoice)
25	Vendor — Debit (payment)

Posting vs Tally — the mental shift. In Tally you *are* the ledger. In SAP you post a **document** with a header (date, type, company code) and line items; the system derives the GL hit, adds tax and TDS lines automatically, assigns a **document number** from a range, and freezes it. You can't delete — only **reverse (FB08)**, which creates an equal-and-opposite document. Every field is validated against master data before it saves.

Example: post a vendor invoice via FB60 (click-path). 1. `/nFB60` → set **Company Code** 1000. 2. **Vendor** = 100234, **Invoice date**, **Posting date**, **Amount** ₹1,18,000, **Currency** INR. 3. **Tax** tab: tax code `V0` /GST code; tick *Calculate tax* so SAP splits the ₹18,000 GST. 4. Line item: GL account 400100 (Office Expenses), amount ₹1,00,000, **Cost Centre** IN-ADMIN. 5. Check the **simulate** button (traffic light green) → **Post**. Document 1900000123 created.

The entry SAP builds behind that one screen:

Posting key	Account	Dr (₹)	Cr (₹)
40	Office Expenses (400100)	1,00,000	
40	Input CGST (154010)	9,000	
40	Input SGST (154020)	9,000	
31	Vendor — ACME Supplies (100234)		1,18,000

Worked example / mini-project

Scenario: You are a GL analyst at a Bengaluru GCC. Reproduce a mini month-end for Company Code 1000 (April 2026).

Step 1 — Rent accrual (FB50). Rent of ₹5,00,000 for April is unpaid at month-end.

PK	Account	Dr (₹)	Cr (₹)
40	Rent Expense (401200)	5,00,000	
50	Accrued Expenses (250100)		5,00,000

Step 2 — Customer invoice (FB70). Bill client TechCorp ₹11,80,000 incl. 18% GST.

PK	Account	Dr (₹)	Cr (₹)
01	Customer — TechCorp (200500)	11,80,000	
50	Consulting Revenue (300100)		10,00,000
50	Output CGST (254010)		90,000
50	Output SGST (254020)		90,000

Step 3 — Receipt & clearing (F-28, then F-32). TechCorp pays ₹11,80,000. Post the incoming payment, then clear the open invoice against the receipt so the customer line nets to zero.

Step 4 — Verify with FBL3N. Run FBL3N on GL 300100 (Consulting Revenue) for 01.04.2026–30.04.2026, status *All items*. You should see the ₹10,00,000 credit. Drill in (double-click → FB03) to confirm it traces to your FB70 document.

Step 5 — Reversal test. Say Step 1 rent should have been ₹4,00,000. Do **not** edit — reverse document via **FB08** (reversal reason 01), then re-post the correct amount. Now FBL3N on 401200 shows three lines (original +5,00,000, reversal −5,00,000, correct +4,00,000) — a clean audit trail no auditor can object to.

This five-step flow *is* what an R2R analyst does 200 times a month.

How it's tested

Interview questions (verbal screen): - "Walk me through the P2P cycle and where FI gets hit." (Expected: PR → PO → GR posts *Dr Inventory / Cr GR-IR*; Invoice posts *Dr GR-IR / Cr Vendor*; Payment posts *Dr Vendor / Cr Bank*.) - "Difference between FB50, FB60 and F-02?" (FB50 = GL only; FB60 = vendor sub-ledger; F-02 = fully manual, any account/posting key.) - "What is GR/IR and why does it need clearing?" - "Cost centre vs profit centre vs internal order?" - "What changed in S/4HANA vs ECC?" (Universal Journal ACDOCA, Business Partner, New Asset Accounting, Fiori, no FI-CO reconciliation.) - "How do you correct a wrongly posted document?" (Reverse via FB08 — never delete.)

Practical / assessment tests: - A **live-system or screenshot exercise**: "Here's a vendor bill — post it in FB60 and show the document number and the resulting entry." - A **journalising test**: given 8–10 business events, write the Dr/Cr with correct posting keys. - A **"read this FBL3N" screen**: identify open vs cleared items and explain a suspense balance. - A **close case**: "These four accruals are pending and the sub-ledger doesn't tie to GL — what do you do first?"

Common mistakes & how pros avoid them

Mistake	Why it hurts	Pro habit
Trying to *delete* a posted document	Impossible; flags you as a non-user	Always **reverse (FB08)**
Confusing Company Code with Client	Wrong-entity posting	Company Code = legal entity; check it first, every time
Wrong posting key (40 vs 50)	Entry lands on the wrong side	Memorise 40/50/01/31/25 cold
Ignoring **document date vs posting date**	Wrong period / GST return mismatch	Posting date drives the accounting period
Manually keying tax/TDS lines	SAP auto-derives them; manual = imbalance	Use tax codes, tick *Calculate tax*
Posting to a **cost element without a cost object**	Hard error in CO	Always attach cost centre/order on P&L accounts
Confusing open-item vs cleared	Reconciliation chaos	Learn F-32/F-44 clearing early
Leaving items in **GR/IR** unresolved	Audit finding at year-end	Run **MR11 / F.13** regularly

Learn-it roadmap & resources

Realistic time-to-proficiency: 4–8 weeks part-time to interview-ready as an *end user* (post, display, clear, explain cycles). 4–6 months for a **consultant-level** understanding (configuration, org structure setup, integration).

Week	Focus
1	Navigation, org structure, GL master data, FB50
2	AP: FB60, F-53, vendor line items (FBL1N)
3	AR: FB70, F-28, clearing (F-32)
4	Month-end: accruals, reversals (FB08), FBL3N reporting
5–6	CO basics: cost centres, profit centres, CO-PA concept
7–8	P2P & O2C end-to-end; S/4HANA differences; Fiori

Resources: - **SAP Learning Hub / learning.sap.com** — free "SAP S/4HANA for Financial Accounting Associates" learning journey (aligns with cert **C_TS4FI**). - **openSAP** free courses on S/4HANA Finance. - Practice access: a **free trial/IDES sandbox** or an institute-provided server (a real login beats any video). - Books: *SAP S/4HANA Finance: An Introduction* (SAP Press); *Configuring SAP S/4HANA Finance*. - **Certification worth doing:** SAP Certified Associate — SAP S/4HANA Cloud Public Edition, Finance (`C_TS4FI` / `C_S4CFI`). For an *accountant* role, hands-on beats the cert; for a *consultant* role, the cert matters.

Quick-reference

Org hierarchy: Client → Company Code (legal entity) → Chart of Accounts → Business Area / Profit Centre / Cost Centre.

Must-know T-codes: FB50 (GL post) · FB60 (vendor inv) · FB70 (customer inv) · F-02 (general/manual) · FB03 (display doc) · FBL3N (GL items) · FBL1N (vendor) · FBL5N (customer) · FB08 (reverse) · F-32/F-44 (clear cust/vendor) · F110 (auto payment run) · F.13 (auto-clear) · MR11 (GR/IR clear).

Posting keys: 40 = GL Dr · 50 = GL Cr · 01 = Customer Dr · 31 = Vendor Cr · 25 = Vendor Dr.

P2P entries: GR → *Dr Inventory / Cr GR-IR* · Invoice → *Dr GR-IR / Cr Vendor* · Payment → *Dr Vendor / Cr Bank*.

O2C entries: Billing → *Dr Customer / Cr Revenue + Output GST* · Receipt → *Dr Bank / Cr Customer*.

S/4HANA one-liners: Universal Journal = **ACDOCA** (FI + CO in one table) · **Business Partner** replaces separate customer/vendor masters · **New Asset Accounting** · **Fiori** UI · no FI-CO reconciliation.

Golden rule: Never delete — **reverse (FB08)**. Every posting leaves a trail.

Reconciliations That Matter

Reconciliation is the daily discipline of proving that two independent records of the same money agree — and explaining every rupee that doesn't. It is the single most common task juniors are handed on day one, and the fastest way to lose a manager's trust if you get it wrong.

What it is & where it's used

A reconciliation compares two versions of "truth" and isolates the differences (called *reconciling items*), each of which must have a documented reason and, if it's a real error, a journal entry. The four that actually matter in an Indian finance job:

Recon	Compares	Owned by	Consequence if wrong
Bank (BRS)	Cash-book / ledger vs bank statement	Accounts / Treasury	Cash misstated, fraud undetected
Vendor (AP)	Your ledger vs vendor's Statement of Account (SOA)	Accounts Payable	Double payment, disputed dues
Customer (AR)	Your ledger vs customer's SOA	Accounts Receivable	Bad-debt surprises, wrong collections
Inter-company (ICO)	Entity A's books vs Entity B's books	Group / consolidation	Consolidation doesn't tie, audit qualification
GST 2B recon	Purchase register vs GSTR-2B	Tax / Indirect Tax	ITC lost, notices under Sec 16

Roles that live and die by this: **Accounts Executive, AP/AR Analyst, R2R (Record-to-Report) associate in a GCC/Shared Service, GST/Indirect-tax executive, Treasury analyst, and audit staff** doing substantive testing.

The gap: why companies want this (and college didn't teach it)

MBA and CA-Inter teach you *what* a BRS is and make you draw the T-format statement. Employers don't care about the T-format — the bank feed and the ERP already exist. What they pay for is the **matching engine in your head and in Excel**: taking 4,000 bank lines and 3,800 book lines, matching them at speed, and confidently saying "these 12 items are genuine timing differences, these 3 are our posting errors, this one is a fraud." College gives you 20 clean transactions; the job gives you messy exports with different date formats, truncated narrations, part-payments, and TDS/bank-charge noise. Nobody teaches the **GSTR-2B reconciliation** at all, yet it directly controls how much Input Tax Credit a company can claim — real cash. That gap (clean-theory vs messy-volume-with-money-at-stake) is exactly what this chapter closes.

What "proficient" looks like

A job-ready person can, unaided:

- Take two raw exports and reconcile them in Excel using `XLOOKUP` / `SUMIFS` / `Power Query` within an hour, not a day.
- Explain *every* reconciling item as timing, error, or omission — and pass the "so what's the journal?" test.

- Categorise GST 2B mismatches into *match / mismatch / in-books-not-2B / in-2B-not-books* and know the ITC action for each.
- Build a **repeatable** recon (Power Query refresh, not manual re-keying every month).
- Know when a difference is a red flag (fraud, GST notice) versus routine.

Hands-on: how to actually do it

Bank reconciliation (BRS)

The modern BRS is a **two-way tick-mark**: match book lines to bank lines; whatever is left on each side is a reconciling item.

Match on a helper key (amount + date window). In Excel, flag whether each book entry appears in the bank statement:

```
' Book sheet, col F = is this cleared in the bank?
=IF(COUNTIFS(Bank[Amount], D2, Bank[Date], "≤"&TODAY()) > 0, "Cleared", "Uncleared")

' Better: exact two-way match with XLOOKUP on a composite key
' Build key = Amount&"|"&Narration-token in both sheets, then:
=IFERROR(XLOOKUP(G2, Bank[Key], Bank[Date], "NOT IN BANK"), "NOT IN BANK")
```

The reconciliation itself:

```
Balance as per Cash Book (Dr)                     12,45,000
Add: Cheques issued but not yet presented        + 2,30,000
Less: Cheques deposited but not yet cleared      -  1,10,000
Less: Bank charges not in books                  -     1,850
Add: Interest credited by bank not in books      +     3,200
Less: Direct debit (EMI) not in books            -    45,000
= Balance as per Bank Statement                   13,21,350
```

Every item that is *our* omission (bank charges, interest, direct debits) needs a journal:

Item	Dr	Cr	Amount
Bank charges	Bank Charges A/c	Bank A/c	1,850
Interest credited	Bank A/c	Interest Income A/c	3,200
EMI auto-debit	Loan A/c	Bank A/c	45,000

Cheques not presented / not cleared need **no** entry — they are pure timing.

Vendor / Customer reconciliation

You get the vendor's **SOA (PDF/Excel)** and match it to your ledger. Load both into **Power Query** and do an anti-join to find one-sided items:

1. Data → Get Data → From File (vendor SOA) and From File (your ledger export from Tally/SAP).
2. On invoice number, `Merge Queries` → **Left Anti** → gives invoices in *your* books not in vendor's.
3. Repeat **Right Anti** for vendor's-not-yours.
4. Net balance check: `= SUM(YourDebits) - SUM(YourCredits)` vs vendor's closing.

Typical causes of a vendor difference and the fix:

Difference	Reason	Action
Vendor shows invoice you don't	GRN/invoice not booked	Book the purchase (or dispute)
You show payment vendor doesn't	Cheque in transit / wrong UTR	Share UTR, no entry
TDS deducted	Vendor booked gross	Reconcile with TDS deducted; educate vendor
Debit/credit note timing	Return not yet in vendor books	Confirm CN reference

Inter-company reconciliation

Entity A's "due from B" must equal Entity B's "due to A", sign-flipped. Use a shared **ICO matrix**. In SQL against a group ledger:

```
SELECT a.doc_id, a.entity AS from_entity, b.entity AS to_entity,
       a.amount AS a_side, b.amount AS b_side,
       a.amount + b.amount AS difference
FROM ico_ledger a
JOIN ico_ledger b
  ON a.doc_id = b.doc_id
 AND a.counterparty = b.entity
WHERE ABS(a.amount + b.amount) > 1        -- non-zero net = mismatch
ORDER BY ABS(a.amount + b.amount) DESC;
```

Differences are usually **FX rate** (each entity used a different rate), **cut-off** (goods in transit at period-end), or a **missed booking**. The elimination entry at consolidation only works if the mismatch is zero, so this must clear before close.

GST 2B reconciliation

GSTR-2B is the auto-drafted ITC statement. You may claim ITC **only** for invoices appearing in 2B (Sec 16(2)(aa)). Match your **purchase register** to 2B on **GSTIN + Invoice No + Taxable Value + Tax**.

Python is ideal because 2B is a downloadable JSON/Excel and volumes are large:

```
import pandas as pd

books = pd.read_excel("purchase_register.xlsx")
gst2b = pd.read_excel("GSTR2B.xlsx")

# Normalise keys — invoice numbers are the usual pain point
for df in (books, gst2b):
    df["key"] = (df["GSTIN"].str.strip()
                 + "|" + df["InvNo"].str.upper().str.replace(r"\W", "", regex=True)
                 + "|" + df["TaxableValue"].round(0).astype(int).astype(str))

merged = books.merge(gst2b, on="key", how="outer",
                     suffixes=("_bk", "_2b"), indicator=True)

status = {"both": "Matched",
          "left_only": "In books, NOT in 2B (hold ITC)",
          "right_only": "In 2B, NOT in books (book it)"}
merged["Recon"] = merged["_merge"].map(status)

# Tax-value mismatch even when both present
m = merged["_merge"] == "both"
merged.loc[m & (merged["Tax_bk"].round() != merged["Tax_2b"].round()),
           "Recon"] = "Value mismatch — probe"

merged.to_excel("2B_recon.xlsx", index=False)
```

Action per bucket: **Matched** → claim; **In-books-not-in-2B** → do *not* claim yet, follow up with vendor to file GSTR-1; **In-2B-not-in-books** → book the purchase; **Value mismatch** → check rate/rounding, raise with vendor.

Worked example / mini-project

Scenario: Month-end GST 2B recon for Acme Traders Pvt Ltd, March 2026.

Purchase register (books):

Vendor GSTIN	Inv No	Taxable	GST
29ABCDE1234F1Z5	INV-101	1,00,000	18,000
27PQRSX5678L1Z2	5023	50,000	9,000
06LMNOP9012K1Z8	A/778	2,00,000	36,000

GSTR-2B (portal):

Supplier GSTIN	Inv No	Taxable	GST
29ABCDE1234F1Z5	INV101	1,00,000	18,000
27PQRSX5678L1Z2	5023	50,000	4,500
33ZZZZZ0000Z1Z0	90	80,000	14,400

Run the Python above (invoice normaliser strips the "-" so INV-101 = INV101). Result:

Recon status	Invoice	ITC action
Matched	INV-101	Claim 18,000
Value mismatch	5023	Books 9,000 vs 2B 4,500 — vendor filed at 9% wrongly; hold 4,500, chase
In books, not in 2B	A/778	Vendor hasn't filed GSTR-1; **do not claim 36,000** this month
In 2B, not in books	90	14,400 in 2B but no purchase booked — investigate; possibly wrong GSTIN mapped to us

Eligible ITC to claim in GSTR-3B = ₹18,000 + ₹4,500 = **₹22,500**, versus ₹63,000 the books naively suggest. That ₹40,500 gap is exactly why this recon exists — claiming the full amount invites a Sec 16 notice with interest.

How it's tested

Interview questions: - "A cheque you issued hasn't cleared — does it need a journal entry?" (No, timing.) - "Bank statement shows a credit you can't identify. What do you do?" (Park in suspense, investigate, never assume income.) - "Why can't you claim ITC that isn't in 2B?" (Sec 16(2)(aa).) - "Inter-company balance doesn't tie by ₹1,200 — likely cause?" (FX rate or cut-off timing.)

Practical tests companies give: - A **timed Excel test**: two sheets, "reconcile and list differences in 30 minutes" — they watch whether you reach for `XLOOKUP` / `COUNTIFS` /Power Query or start eyeballing. - A **GST case**: raw purchase register + 2B Excel, "how much ITC can we claim?" — the trap is the value-mismatch and not-in-2B rows. - A **close simulation** in a GCC: "here's a trial balance with an unreconciled bank line, clear it" — they score whether your journals are correct and balanced.

Common mistakes & how pros avoid them

Mistake	Why it hurts	Pro habit
Matching on amount alone	Two ₹50,000 payments collide	Match on composite key (party + inv + amount)
Passing entries for timing items	Overstates/understates cash	Only *errors and omissions* get journals
Dumping unknowns into a "misc" bucket	Hides fraud	Everything gets a named reason or goes to suspense with a ticket
Claiming full ITC ignoring 2B	Interest + penalty under GST	Claim only matched value
Re-keying the recon every month	Slow, error-prone	Build once in Power Query, refresh monthly
Ignoring rounding	100s of false mismatches	Round to nearest rupee before comparing
Not aging the differences	Old items rot	Track "days open" on every reconciling item

Learn-it roadmap & resources

Time to proficiency: 4–6 weeks of deliberate practice alongside a job or dummy data.

- **Week 1–2:** Master BRS logic + Excel matching (`XLOOKUP` , `SUMIFS` , `COUNTIFS`). Free: ExcelIsFun / Chandoo lookup playlists.
- **Week 3:** Power Query merge/anti-join for AP/AR recon. Free: Microsoft Power Query docs; "Get & Transform" tutorials.
- **Week 4:** GST 2B recon — read Sec 16 & Rule 36; practice on the **GST portal** (Returns → GSTR-2B → download Excel). Free: GSTN help + ICAI GST background material.
- **Week 5–6:** Automate one recon end-to-end in Python (pandas) or an ERP (TallyPrime bank reconciliation: *Gateway → Banking → Bank Reconciliation*).

Certifications that signal this skill: **ICAI GST certificate course**, **Microsoft PL-300 (Power BI/Query)**, or any R2R/AP-AR training from a GCC. None are mandatory — a clean reconciled workbook you built beats a certificate in interviews.

Quick-reference

Need	Formula / step
Is book line in bank?	`=COUNTIFS(Bank[Amt],D2,Bank[Date],"≤"&TODAY())>0`
Two-way match key	`=A2&"|"&TEXT(ROUND(B2,0),"0")` both sheets, then `XLOOKUP`
One-sided items	Power Query → Merge → **Left/Right Anti**
ICO net difference	`a.amount + b.amount` (should = 0)
2B match key	`GSTIN | InvNo(normalised) | TaxableValue`
Bank charge JE	Dr Bank Charges / Cr Bank
Bank interest JE	Dr Bank / Cr Interest Income
Auto-debit EMI JE	Dr Loan / Cr Bank

Reconciling-item rule of thumb: *Timing → no entry. Error/omission → journal. Unknown → suspense + investigate, never income.*

GST 2B buckets: Matched → claim · In-books-not-2B → hold, chase vendor · In-2B-not-books → book it · Value mismatch → probe rate/rounding.

Legal anchors: BRS is management practice (no statute) · GST ITC gated by **Sec 16(2)(aa)** and **Rule 36** · TDS reconciliation ties to **Form 26AS / AIS**.

Month-end close: the checklist & journals

What it is & where it's used

Month-end close (MEC) is the disciplined process of finalising a company's books for a period so the numbers can be trusted, reported, and locked. It's the difference between a ledger that *has* transactions and financials that are *complete, accurate, and cut off correctly*. Every accrual, prepaid, provision, depreciation run, reconciliation, and inter-company match happens in a fixed window — typically **Working Day 1 (WD1) to WD5** — ending with a signed-off Trial Balance and a flux (variance) commentary sent to management.

Roles that live and die by the close:

Role	What they own in the close
Accounts Executive / AP-AR	Cut-off, accruals of unbilled expenses, vendor/customer recons
GL Accountant / Financial Analyst	Journals, depreciation, provisions, prepaid amortisation, TB
Assistant Manager – Finance	Flux analysis, review, sign-off, schedules
FP&A	Actual vs budget variance, board pack
Statutory/Internal Audit	Tests cut-off, provisions, and the close controls

In India this maps directly to the monthly cycle around **GSTR-1 (11th), GSTR-3B (20th), TDS payment (7th)**, and quarterly Ind AS/Schedule III reporting. Globally the same skeleton drives US GAAP/IFRS closes on a "5-day close" or "faster close" mandate.

The gap: why companies want this (and college didn't teach it)

College teaches you to pass a *single* adjusting entry in an exam — "provide depreciation @15% WDV" — with the numbers handed to you. It never teaches the **operational reality**:

- Nobody hands you the numbers. You *derive* the accrual from open POs, GRNs, and a rate card.
- Everything is **time-boxed**. The CFO wants a P&L by WD3, not "whenever it's ready."
- **Cut-off** is a control, not a concept — an invoice dated 30-Sep that hits on 3-Oct must still land in September.
- The books must **tie out**: bank recon, GST 2B vs books, AP sub-ledger to GL control account.
- You must **explain movements** (flux), not just produce them. "Why is repairs up 40% MoM?" is the actual job.

Employers pay for someone who can take a messy set of sub-ledgers and produce a defensible TB *on a deadline, unaided*. That reliability — checklist-driven, self-reviewing — is what an MBA syllabus skips entirely.

What "proficient" looks like

A job-ready person can, without supervision:

1. Own a **close calendar** and drive owners to hit WD deadlines.
2. Compute and post **accruals, prepaids, provisions, and depreciation** with a supporting schedule for each.

3. Apply **cut-off** correctly across revenue, expenses, and inventory.
4. Reconcile **sub-ledger to GL** (AP, AR, fixed assets, bank) and clear differences.
5. Produce a **flux analysis** with real commentary (drivers, not restated numbers).
6. Deliver a **reusable checklist** with maker/checker sign-off and an audit trail.
7. Reverse accruals cleanly the next period and avoid double-counting.

Hands-on: how to actually do it

1. The close calendar

Build it once, reuse forever. A minimal WD-based calendar:

WD	Task	Owner
WD1	Freeze sub-ledgers (AP/AR/inventory), post cut-off accruals	AP/AR
WD1-2	Bank recon, GST 2B vs books, TDS reconciliation	GL
WD2	Prepaid amortisation, depreciation run, provisions	GL
WD3	Inter-company match, draft TB, first flux	AM
WD4	Review adjustments, finalise schedules	AM
WD5	Lock period, MIS pack + flux commentary to CFO	FP&A

2. The core journals

Accrual — unbilled expense (services received in Sep, invoice not in). Say ₹1,80,000 of consultancy, GRN raised, no bill:

Account	Dr	Cr
Professional fees (P&L)	1,80,000	
Accrued expenses (Provision for expenses)		1,80,000

Reverse on 1-Oct so the actual invoice doesn't double-count:

Account	Dr	Cr
Accrued expenses	1,80,000	
Professional fees		1,80,000

Prepaid amortisation — annual insurance ₹1,20,000 paid 1-Apr, monthly charge:

```
Monthly = 1,20,000 / 12 = 10,000
```

Account	Dr	Cr
Insurance expense (P&L)	10,000	
Prepaid insurance (BS)		10,000

Depreciation — SLM, asset ₹6,00,000, life 5 yrs, no residual:

```
Monthly dep = 6,00,000 / (5*12) = 10,000
```

Account	Dr	Cr
Depreciation expense	10,000	
Accumulated depreciation		10,000

Provision — doubtful debts (Expected Credit Loss, Ind AS 109), 2% on ₹40,00,000 receivables:

Account	Dr	Cr
Impairment loss / Bad debt expense	80,000	
Provision for doubtful debts		80,000

3. Prepaid & accrual schedules in Excel

Auto-compute the monthly prepaid charge and remaining balance:

```
Monthly amort:  =IF(EOMONTH($D2,0)≥E$1, $C2/$B2, 0)
   where C2 = total prepaid, B2 = months, D2 = start date, E$1 = period-end date

Closing balance: =$C2 - SUMPRODUCT(($A$1:E$1≥$D2)*($A$1:E$1≤EOMONTH($D2,$B2-1))*($C2/$B2))
```

Pull the current period's accrual from an open-PO extract:

```
=SUMIFS(GRN[Value], GRN[Billed], "No", GRN[GRN_Date], "≤"&EOMONTH(TODAY(),0))
```

4. Cut-off test (SQL)

Catch invoices booked in the wrong period — doc date in Sep but posting date in Oct:

```
SELECT doc_no, vendor, doc_date, posting_date, amount
FROM ap_invoices
WHERE doc_date  ≤ '2026-09-30'
  AND posting_date > '2026-09-30'
  AND posting_date ≤ '2026-10-05';   -- likely September cut-off items
```

5. Sub-ledger to GL tie-out (SQL)

```
SELECT g.control_balance,
       s.subledger_balance,
       g.control_balance - s.subledger_balance AS difference
FROM (SELECT SUM(amount) control_balance FROM gl WHERE account='AP_CONTROL') g
CROSS JOIN (SELECT SUM(open_amount) subledger_balance FROM ap_open_items) s;
-- difference MUST be 0
```

6. Flux analysis (Python)

```
import pandas as pd
tb = pd.read_excel("TB.xlsx")  # cols: Account, Sep, Aug
tb["MoM_abs"] = tb["Sep"] - tb["Aug"]
tb["MoM_pct"] = (tb["MoM_abs"] / tb["Aug"].replace(0, pd.NA)) * 100
# flag anything moving > 10% AND > 50k for commentary
flux = tb[(tb["MoM_pct"].abs() > 10) & (tb["MoM_abs"].abs() > 50000)]
print(flux.sort_values("MoM_abs", key=abs, ascending=False))
```

7. TallyPrime click-path (recurring journals)

`Gateway of Tally > Vouchers > F7 (Journal)` → Dr expense, Cr provision → in the narration tag the reversal. For depreciation, keep an **Accounting Voucher** template and `Ctrl+D` won't help — instead duplicate last month's voucher via `Alt+2` from the Day Book and edit the date. Verify posting under `Display > Trial Balance > F5 (detailed)`.

Worked example / mini-project

Scenario: Close September 2026 for *Meridian Traders Pvt Ltd*. Draft TB shows PBT of ₹12,00,000 before adjustments. Post these:

#	Adjustment	Basis	Amount
1	Electricity accrual (bill due Oct)	Meter reading	₹45,000
2	Prepaid insurance charge	₹1,20,000 / 12	₹10,000
3	Depreciation — plant	SLM ₹6,00,000 / 60	₹10,000
4	Provision — doubtful debts	2% × ₹40,00,000	₹80,000
5	Interest accrual on loan	₹50,00,000 × 9% / 12	₹37,500

Impact on PBT:

```
Adjusted PBT = 12,00,000 - 45,000 - 10,000 - 10,000 - 80,000 - 37,500
             = 10,17,500
```

Flux commentary (Sep vs Aug), driver-based:

Account	Aug	Sep	MoM %	Commentary
Power & fuel	38,000	45,000	+18%	Higher production run + summer load
Provision – debtors	60,000	80,000	+33%	Debtor book grew ₹8L; ECL rate held at 2%
Interest cost	0	37,500	new	New ₹50L WC loan drawn 1-Sep

Deliverable: signed TB + these five schedules + flux note. On 1-Oct, reverse adjustments #1 and #5 (accruals), keep #2–#4 (they're period charges, not reversing accruals).

How it's tested

Interview questions - "Walk me through your month-end close, day by day." (Tests you actually did it.) - "Difference between accrual, provision, and reserve?" (Accrual = known amount, timing gap; provision = estimated liability; reserve = appropriation of profit.) - "An invoice dated 29-Sep arrives on 4-Oct — which period?" (September; cut-off follows the event, not the receipt.) - "How do you make sure an accrual isn't double-counted next month?" (Auto-reverse on WD1.) - "Your AP sub-ledger is ₹2,000 off the control account — what do you do?"

Practical tests companies give - A **timed Excel case**: raw sub-ledger dump → build prepaid & accrual schedules and a corrected TB in 45–60 min. - A **"close these books" case study**: 8–10 adjustments described in words; you must journalise, produce adjusted P&L, and write 3-line flux. - A **SQL/data screen** for analyst roles: write the cut-off and tie-out queries above. - A **reconciliation drill**: bank statement vs cash book, find and classify the differences.

Common mistakes & how pros avoid them

Mistake	Fix
Forgetting to reverse accruals → double expense	Post accruals as **auto-reversing** journals dated WD1 next month
Cut-off by invoice-received date, not event date	Apply the "goods/services received" test; run the SQL cut-off query
No supporting schedule behind a journal	One journal = one schedule; auditor's first ask
Flux that restates numbers ("up because it went up")	Commentary must name a **driver**: volume, rate, one-off, timing
Provisions that swing wildly with no policy	Fix a documented rate/methodology (e.g. ECL matrix) and hold it
Closing before sub-ledgers tie to GL	Tie-out difference must be **zero** before you lock the period
Posting to a period you've already locked	Lock the period in the ERP; late items go to the next open period with a note
Depreciation on the wrong pro-rata basis	Confirm policy: full-month, mid-month, or days-based; be consistent

Learn-it roadmap & resources

Time to proficiency: ~6–8 weeks of deliberate practice if you already know debits/credits.

Week	Focus
1–2	Journals cold: accrual, prepaid, provision, depreciation, and their reversals
3	Cut-off + reconciliations (bank, AP/AR, GST 2B vs books)
4	Build a close calendar + reusable checklist in Excel
5	Flux analysis — write real commentary on 3 months of dummy TBs
6–8	Run a full mock close end-to-end, timed, in Tally + Excel

Resources - *ICAI* Ind AS material — provisions (Ind AS 37), ECL (Ind AS 109), PP&E (Ind AS 16). - TallyPrime free trial — practise recurring journals and the Day Book workflow. - Corporate Finance Institute (CFI) — free "Month-End Close" and financial-accounting lessons. - CPA/ACCA record-to-report (R2R) notes for the globally-portable framing. - Practice data: generate a dummy sub-ledger in Excel and close it three months running.

Certifications that signal this: CA Intermediate (Accounting + Advanced Accounting), ACCA FA/FR, or any employer's internal R2R certification. For BPO/GCC "record-to-report" roles, the close checklist *is* the job description.

Quick-reference

Reversing vs non-reversing - Reverse next period: accruals (unbilled expense, interest, salary accrual). - Do NOT reverse: prepaid amortisation, depreciation, provisions (they're period charges/estimates).

Standard journals (Dr / Cr)

Entry	Dr	Cr
Accrual	Expense	Accrued liabilities
Prepaid charge	Expense	Prepaid asset
Depreciation	Depreciation exp	Accumulated dep
Provision (ECL)	Impairment loss	Provision
Interest accrual	Interest exp	Interest payable

Key formulas

```
Monthly prepaid = Total / No. of months
Monthly SLM dep = (Cost − Residual) / (Life in years * 12)
WDV dep         = Opening WDV * rate / 12
MoM flux %      = (Current − Prior) / Prior * 100
Tie-out diff    = GL control − Sub-ledger  (must = 0)
```

Close discipline - Freeze sub-ledgers → post adjustments → tie out → flux → lock. - One journal = one schedule = one owner. - Cut-off follows the **event date**, not the receipt date. - Flux commentary names a **driver**, never restates the number.

Ind AS in practice

What it is & where it's used

Ind AS (Indian Accounting Standards) are India's IFRS-converged standards. They are **mandatory** for listed companies and large unlisted companies (net worth ≥ ₹250 crore), plus their holding/subsidiary/associate/JV entities. Everyone else runs on the older **AS** (Accounting Standards) or the simpler Ind AS-lite. So the first job skill is knowing *which* framework the entity is on.

Four standards do 80% of the real work on the job:

Standard	Topic	Where it bites
Ind AS 115	Revenue from contracts with customers	Every sale, every SaaS subscription, every construction/AMC contract
Ind AS 116	Leases	Office rent, vehicles, equipment, plant land
Ind AS 109	Financial instruments	Receivables (ECL provisioning), investments, borrowings, derivatives
Ind AS 16	Property, plant & equipment	Capitalisation, componentisation, depreciation

Roles that need this: **financial reporting / R2R analysts**, statutory-audit associates (CA articles/qualified), controllers, FP&A when building models off Ind AS numbers, and anyone preparing or auditing consolidated financials.

The gap: why companies want this (and college didn't teach it)

MBA and even CA-Inter teaches *what* the standard says. The job is *applying* it to a messy contract nobody wrote with accounting in mind. College hands you "recognise revenue at fair value"; the employer hands you a ₹1.8 crore master service agreement with a discount, a free 3-month support add-on, and milestone billing, and asks: *how much revenue this quarter, and pass the entries.*

The specific gaps employers pay to close: - **Unbundling contracts** into performance obligations and allocating price — pure judgement, no textbook number. - **Building the Ind AS 116 ROU asset + lease liability amortisation schedule** in Excel from a rent agreement. This is a near-universal interview test. - **ECL (Expected Credit Loss) provision matrices** under Ind AS 109 — replacing the old "provide when overdue 180 days" rule with a forward-looking model. - Knowing the **AS → Ind AS journal differences** because most Indian firms carry both a tax book (AS/ICDS) and a reporting book (Ind AS).

What "proficient" looks like

A job-ready person can, **unaided**:

1. Read a contract and split it into performance obligations, allocate the transaction price, and state the recognition pattern (point-in-time vs over-time).
2. Build a lease amortisation schedule in Excel that ties: opening liability × rate = interest, less payment = closing; and ROU depreciated straight-line.

3. Build an ECL provision matrix by ageing bucket and compute the loss allowance.
4. Pass the full set of Dr/Cr entries for each, including deferred tax where the book/tax base differs.
5. Explain *why* — e.g. why a security deposit gets discounted, why a lease liability uses the incremental borrowing rate.

Hands-on: how to actually do it

Ind AS 115 — the 5-step model

1. Identify the contract → 2. Identify performance obligations (POs) → 3. Determine transaction price → 4. Allocate price to POs (by standalone selling price, SSP) → 5. Recognise revenue as each PO is satisfied.

Allocation in Excel — software licence ₹10,00,000 + 1-yr support billed together for ₹11,00,000; SSPs are ₹10,00,000 and ₹2,00,000.

```
Total SSP          =10,00,000+2,00,000           → 12,00,000
Licence allocated  =1100000*(1000000/1200000)    → 9,16,667  (point-in-time)
Support allocated  =1100000*(200000/1200000)     → 1,83,333  (over 12 months)
Monthly support rev =183333/12                    → 15,278
```

Journal at contract start (support unearned):

Particulars	Dr (₹)	Cr (₹)
Bank / Trade receivable	11,00,000	
To Revenue – Licence		9,16,667
To Contract liability (deferred support)		1,83,333

Each month: Dr Contract liability 15,278 / Cr Revenue – Support 15,278.

Ind AS 116 — lease liability + ROU (lessee)

Lease liability = **PV of future lease payments** at the incremental borrowing rate (IBR).

```
=PV(rate_per_period, n_periods, -payment, 0, type)
=-PV(0.10, 5, 300000, 0, 0)          'annual ₹3,00,000 rent, 5 yrs, 10% IBR, arrears
```

ROU asset = lease liability + initial direct costs + restoration provision − incentives.

Initial entry:

Particulars	Dr (₹)	Cr (₹)
Right-of-use asset	11,37,236	
To Lease liability		11,37,236

Then each period split payment into interest and principal (see worked example).

Ind AS 109 — ECL simplified approach for receivables

Provision matrix: apply a historical loss rate (adjusted for forward-looking factors) to each ageing bucket.

```
ECL_bucket = Gross_exposure * loss_rate
Total_ECL  = SUMPRODUCT(exposure_range, loss_rate_range)
```

Impairment entry: Dr Impairment loss (P&L) / Cr Allowance for ECL.

Security deposit (interest-free) under Ind AS 109 — a ₹5,00,000 deposit refundable in 3 years, discounted at 10%:

```
=5,00,000/(1.10)^3   → 3,75,657   'financial asset at amortised cost
Difference 1,24,343  → prepaid rent (ROU/expense), unwound as interest income
```

Ind AS 16 — componentisation & depreciation

Capitalise cost + directly attributable costs (freight, install, testing) − trade discounts; borrowing costs if a qualifying asset (Ind AS 23). Depreciate each significant **component** with a different useful life separately.

```
Depreciation p.a. = (Cost − Residual) / Useful_life           'SLM
=SLN(cost, salvage, life)
=DB(cost, salvage, life, period)                               'WDV / declining balance
```

Worked example / mini-project

Build a 5-year Ind AS 116 lease schedule. Office rent ₹3,00,000/year in arrears, 5 years, IBR 10%.

Liability at inception `=-PV(0.10,5,300000)` = **₹11,37,236**. ROU = same.

Yr	Opening liab	Interest @10%	Payment	Closing liab	ROU dep (SLM)
1	11,37,236	1,13,724	3,00,000	9,50,960	2,27,447
2	9,50,960	95,096	3,00,000	7,46,056	2,27,447
3	7,46,056	74,606	3,00,000	5,20,661	2,27,447
4	5,20,661	52,066	3,00,000	2,72,727	2,27,447
5	2,72,727	27,273	3,00,000	~0	2,27,447

Year-1 entries:

Particulars	Dr (₹)	Cr (₹)
Interest expense (finance cost)	1,13,724	
Lease liability	1,86,276	
To Bank		3,00,000
Depreciation – ROU asset	2,27,447	
To Accumulated depreciation – ROU		2,27,447

Compare to old AS 19 operating lease: total P&L would be a flat ₹3,00,000 rent. Under Ind AS 116, Year-1 charge = 1,13,724 + 2,27,447 = **₹3,41,171** — front-loaded. That front-loading is the single most-asked interview point. Reproduce this in Excel; the closing balance hitting ~0 in year 5 is your check.

Reproduce the ECL piece with this matrix:

Ageing	Exposure (₹)	Loss rate	ECL (₹)
Not due	20,00,000	0.5%	10,000
1–90 days	8,00,000	2%	16,000
91–180	3,00,000	10%	30,000
>180	1,00,000	50%	50,000
Total	**32,00,000**		**1,06,000**

`=SUMPRODUCT(B2:B5, C2:C5)` → 1,06,000. Entry: Dr Impairment loss 1,06,000 / Cr Allowance for ECL 1,06,000.

How it's tested

Interview questions: - Walk me through the 5-step revenue model with an example. When is revenue *over time* vs *point in time*? - What's the day-1 entry for a lease? Why does an operating lease now sit on the balance sheet? - Difference between ECL and the old incurred-loss model? What's the "simplified approach"? - Which costs get capitalised into PPE? Is training cost capitalised? (No.) - AS vs Ind AS revenue difference — what changed for a bundled software sale?

Practical tests companies give: - **Timed Excel (30–45 min):** "Here's a rent agreement — build the ROU + lease liability schedule and the year-1 journals." Extremely common for reporting/audit roles. - **Case:** a contract with two deliverables and a discount — allocate the price and give quarterly revenue. - **Provision matrix:** ageing given, compute ECL with SUMPRODUCT. - Big-4 audit assessments give an accounting-treatment memo: "conclude with reference to the standard paragraph."

Common mistakes & how pros avoid them

Mistake	Fix
Recognising bundled revenue upfront	Split into POs, defer service portion as contract liability
Using the wrong lease discount rate	Use IBR (or rate implicit if determinable); document the source
Straight-lining lease expense under Ind AS 116	Only AS 19 straight-lines; Ind AS 116 splits interest + depreciation
Forgetting short-term/low-value lease exemption	≤12 months or low-value → expense straight-line, no ROU
Provisioning ECL only when overdue	ECL is forward-looking; even *not-due* balances carry a small rate
Capitalising training/admin/launch costs in PPE	Only directly attributable costs to bring asset to working condition
Ignoring the deferred tax on Ind AS adjustments	Book base ≠ tax base → create DTA/DTL (Ind AS 12)
Discounting the security deposit but forgetting to unwind	Recognise interest income each year; amortise the prepaid rent

Learn-it roadmap & resources

Time to proficiency: 6–8 weeks part-time to handle these four standards confidently in Excel and journals.

- **Weeks 1–2:** Ind AS 115 + 116 — the two highest-yield. Build one lease schedule and one revenue-allocation sheet from scratch.
- **Weeks 3–4:** Ind AS 109 (ECL, amortised cost, security deposits) + Ind AS 16.
- **Weeks 5–6:** Deferred tax linkage (Ind AS 12), consolidation basics, first-time adoption (Ind AS 101).
- **Weeks 7–8:** Real annual reports — read the "significant accounting policies" note of a listed company and rebuild one disclosure.

Resources: - **ICAI** free Ind AS study material + the bare standards (mca.gov.in) — authoritative and free. - **EY / KPMG / Deloitte / PwC** "Ind AS pocket guides" and illustrative disclosures — free PDFs, industry-grade. - Company annual reports (Infosys, Reliance) — free, best worked examples of real disclosures. - **Certification:** ICAI's Certificate Course on Ind AS; **Dip IFR (ACCA)** is the globally-portable equivalent — highly regarded and transferable to IFRS roles abroad.

Quick-reference

Item	Formula / rule
Revenue (5 steps)	Contract → POs → Price → Allocate (by SSP) → Recognise
Price allocation	`=Total*(SSP_item/Total_SSP)`
Lease liability	`=-PV(IBR, n, payment, 0, type)`
ROU asset	Liability + IDC + restoration − incentives
Lease interest	Opening liability × IBR
ECL	`=SUMPRODUCT(exposure, loss_rate)`
Deposit at amortised cost	`=Deposit/(1+r)^n`
PPE depreciation	`=SLN(cost,salvage,life)` or `=DB(…)` for WDV
Lease exemption	≤12 months OR low-value → straight-line expense
Applicability	Listed OR net worth ≥ ₹250 cr (+ group entities)

Golden rule: Ind AS front-loads and puts things *on* the balance sheet (leases, ECL, discounted deposits). If your answer looks like flat AS-19 rent or "provide only when overdue," you're on the wrong standard.

Financial Statement Preparation & Schedule III

What it is & where it's used

Financial statement preparation is the act of converting a raw **trial balance (TB)** — a flat list of ledger balances — into the two statutory face documents every Indian company must file: the **Balance Sheet** and the **Statement of Profit and Loss**, plus the **Notes to Accounts** that explain every line. The mandated format is **Schedule III of the Companies Act, 2013** (Division I for entities on AS, Division II for Ind AS, Division III for NBFCs).

This is the single most common deliverable in an accounts department. It is used by:

- **Accounts/finance executives** doing the monthly and annual "close".
- **Audit article assistants / audit associates** who redraw a client's TB into Schedule III during statutory audit.
- **Consulting/ROC compliance teams** preparing statements for AOC-4 filing.
- **FP&A and controllership** who start every management pack from a Schedule III shell.

If you can take a messy TB and hand back a compliant Balance Sheet, P&L, and notes without supervision, you are doing the core job an accountant is paid for.

The gap: why companies want this (and college didn't teach it)

An MBA (and even much of CA theory) teaches you the *T-format* balance sheet — the horizontal "Liabilities on left, Assets on right" layout from the 1956 Act. **That format is illegal for companies today.** Schedule III mandates the **vertical format** with a fixed line-item sequence, mandatory sub-note groupings, and prior-year comparatives.

The specific gaps employers see in freshers:

What college teaches	What the job needs
T-format, no notes	Vertical Schedule III + 20-odd notes
"Sundry debtors"	"Trade Receivables", split by ageing & security
Lump everything as "Reserves"	Reserves & Surplus note with movement
Ignore current/non-current split	Every asset/liability classified by 12-month rule
One year only	Two-year comparatives, regrouped

College stops at the TB. The employer's work *starts* at the TB. Schedule III is procedural knowledge — grouping rules and a rigid template — that is simply never drilled academically.

What "proficient" looks like

A job-ready person can, **unaided**, do all of this:

1. Take a 150-line TB in Excel and map each ledger to its Schedule III line and note number.
2. Apply the **current vs non-current** test correctly (the 12-month / operating-cycle rule).

3. Build **Notes 1–2** (Share Capital, Reserves & Surplus with movement) and Notes for PPE, Trade Receivables (with the mandatory **ageing schedule**), Trade Payables ageing, Borrowings, and each expense head.
4. Ensure the Balance Sheet **ties** (Total Equity & Liabilities = Total Assets) and that P&L profit flows into Reserves.
5. Handle appropriations: **provision for tax, proposed dividend, transfer to reserves**.
6. Deliver with **prior-year comparatives** and note cross-references, ready for audit.
7. Know the FY2021 amendment additions: **ageing schedules, ratios disclosure, promoter shareholding, CSR, Ind AS vs AS differences**.

Hands-on: how to actually do it

Step 0 — the extended trial balance in Excel

Lay the TB out with a mapping column. This is the workhorse.

Col	Header	Content
A	Ledger	e.g. "Bank – HDFC CC"
B	Debit	4,50,000
C	Credit	
D	SIII_Head	"Cash & Cash Equivalents"
E	Note_No	12
F	Nature	Asset / Liab / Inc / Exp

Check the TB balances first:

```
=SUM(B:B)-SUM(C:C)        → must be 0
```

Pull a note total from the mapping (the key formula):

```
=SUMIFS($B:$B,$E:$E,12) - SUMIFS($C:$C,$E:$E,12)
```

Sum debits minus credits for every ledger tagged to Note 12. Do this for each note number and the statement builds itself.

Map ledgers to heads with a lookup table (build a master `tbl_Map` of ledger→head):

```
=XLOOKUP(A2, tbl_Map[Ledger], tbl_Map[SIII_Head], "**UNMAPPED**")
```

Filter for `**UNMAPPED**` to catch anything you forgot — an unmapped ledger is the #1 cause of a balance sheet that doesn't tie.

Step 1 — the current / non-current test

A liability is **current** if it is due within 12 months or the operating cycle; else non-current. Same logic for assets.

```
=IF(Due_Date - Reporting_Date ≤ 365, "Current", "Non-current")
```

Classic split: current maturities of long-term debt → **Other Current Liabilities**; the rest → **Long-term Borrowings**.

Step 2 — the mandatory ageing schedules (post-2021 amendment)

Trade Receivables must be bucketed. In Excel:

```
=SUMIFS(Amt, Days, "≥0", Days, "<181")          'Less than 6 months
=SUMIFS(Amt, Days, ">180", Days, "<366")        '6 months – 1 year
=SUMIFS(Amt, Days, ">365", Days, "<731")        '1–2 years
```

where `Days = TODAY() - Invoice_Date`. The buckets required: **< 6m, 6m–1y, 1–2y, 2–3y, > 3y**, split into *Undisputed – considered good / doubtful* and *Disputed*.

Step 3 — the closing journal entries

Before statements, pass the closing/appropriation entries:

#	Particulars	Dr	Cr
1	Depreciation Expense A/c ... Dr	8,00,000	
	To Accumulated Depreciation A/c		8,00,000
2	Profit & Loss A/c ... Dr	12,00,000	
	To Provision for Tax A/c		12,00,000
3	Surplus in P&L (Retained Earnings) ... Dr	5,00,000	
	To General Reserve A/c		5,00,000
4	All Revenue A/cs ... Dr / All Expense A/cs ... Cr	x	x
	To/By Profit & Loss A/c (closing)		

Step 4 — TallyPrime shortcut

Tally already outputs Schedule III if ledgers are grouped correctly: `Gateway of Tally → Balance Sheet → F12 (Configure) → "Method of showing Balance Sheet: Vertical" → Yes`. Then `Alt+F5` for detailed, `Ctrl+B` to change the schedule basis. But you still verify the grouping — Tally trusts your ledger's parent group.

Worked example / mini-project

Sunrise Traders Pvt Ltd — FY2024-25 trial balance (₹):

Ledger	Dr	Cr
Equity Share Capital (1,00,000 × ₹10)		10,00,000
General Reserve (opening)		4,00,000
Surplus in P&L (opening)		1,50,000
Term Loan – SBI (repay 2028)		6,00,000
Trade Payables		3,20,000
Plant & Machinery (gross)	18,00,000	
Accumulated Depreciation		5,00,000
Inventory	4,10,000	
Trade Receivables	5,60,000	
Cash & Bank	2,40,000	
Revenue from Operations		40,00,000
Purchases	22,00,000	
Employee Benefits Expense	6,50,000	
Depreciation	1,80,000	
Other Expenses	3,30,000	
Totals	**63,70,000**	**69,70,000**

Difference ₹6,00,000 = profit not yet closed. **Statement of P&L:**

Particulars	Note	2024-25 (₹)
Revenue from Operations	16	40,00,000
Total Income		**40,00,000**
Purchases of Stock-in-Trade	18	22,00,000
Employee Benefits Expense	19	6,50,000
Depreciation & Amortisation		1,80,000
Other Expenses	20	3,30,000
Total Expenses		**33,60,000**
Profit before Tax		**6,40,000**
Less: Tax @ 25%		1,60,000
Profit for the year		**4,80,000**

Appropriate: transfer ₹1,00,000 to General Reserve. Closing Surplus = 1,50,000 + 4,80,000 − 1,00,000 = **₹5,30,000**. General Reserve = 4,00,000 + 1,00,000 = **₹5,00,000**.

Balance Sheet as at 31 Mar 2025:

Particulars	Note	₹
I. EQUITY AND LIABILITIES		
(1) Shareholders' Funds		
(a) Share Capital	1	10,00,000
(b) Reserves & Surplus	2	10,30,000
(2) Non-current Liabilities		
Long-term Borrowings	3	6,00,000
(3) Current Liabilities		
Trade Payables	4	3,20,000
Provision for Tax	5	1,60,000
TOTAL		**31,10,000**
II. ASSETS		
(1) Non-current Assets		
PPE (18,00,000 − 6,80,000)	6	11,20,000
(2) Current Assets		
Inventories	7	4,10,000
Trade Receivables	8	5,60,000
Cash & Cash Equivalents	9	2,40,000
TOTAL		**31,10,000**

Note accumulated depreciation = 5,00,000 opening + 1,80,000 current = 6,80,000. Both sides tie at **₹31,10,000**.

Note 2 – Reserves & Surplus (show the movement):

	₹
General Reserve (4,00,000 + 1,00,000)	5,00,000
Surplus: Opening 1,50,000 + Profit 4,80,000 − Transfer 1,00,000	5,30,000
Total	**10,30,000**

Reproduce this in a workbook: TB on sheet 1, mapping column, `SUMIFS` per note, statements auto-populate.

How it's tested

Interview questions: - "Difference between T-format and Schedule III vertical format?" - "How do you classify current vs non-current — give the rule." - "Where does proposed dividend go now vs pre-2016?" (Now: disclosed in notes, provided only when declared.) - "Walk me from trial balance to balance sheet." -

"What new disclosures did the March 2021 Schedule III amendment add?" (Ageing schedules, ratios, promoter holding, CSR, shortfall utilisation.)

Practical/assessment tests: - A **timed Excel test (60–90 min):** here's a TB, produce Balance Sheet + P&L + Notes 1 and 2. They check if it ties and if groupings are right. - **"Regroup this" case:** a competitor-style messy TB with wrong heads; fix the classification. - **Audit-firm test:** given a client TB and last year's signed accounts, prepare current-year statements *with comparatives* and flag reclassifications. - Sometimes a **Tally practical:** post given entries, group ledgers, generate the vertical balance sheet.

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Balance sheet doesn't tie	Reconcile via the `SUM(Dr)-SUM(Cr)=0` check *before* building; hunt `**UNMAPPED**` ledgers
Netting off — showing TR net of advances	Never net; show gross assets and gross liabilities separately
Forgetting current maturities of long-term debt	Split loan repayment schedule; reclassify the next-12-months portion to current
Missing ageing schedules / ratios (post-2021)	Keep a checklist of the 8 new mandatory disclosures
Depreciation shown as reduction in gross PPE	Keep gross block and accumulated depreciation as separate note columns
Prior-year figures not regrouped	Regroup comparatives to match current classification and add a note
Wrong reserve movement	Always present Reserves & Surplus as opening + additions − deductions = closing
Rounding chaos	Fix one unit (₹ lakhs/₹ full) across all statements; state it in the header

Pros build **one linked template** and reuse it — they never rekey. They also keep a **lead schedule** per note tying back to the TB, which is exactly what auditors ask for.

Learn-it roadmap & resources

Time to proficiency: 3–5 weeks of deliberate practice if you already know debits/credits.

- **Week 1:** Read Schedule III Division I text (it's short — MCA website, free). Memorise the line-item order.
- **Week 2:** Rebuild 3 real published company balance sheets from their notes backwards into a TB, then forward again.
- **Week 3:** Do 5 timed TB→statements exercises in Excel with `SUMIFS` mapping.
- **Week 4:** Add the 2021 amendment disclosures (ageing, ratios). Do the same in TallyPrime.
- **Week 5:** Practise comparatives and appropriations.

Resources: - **Schedule III, Companies Act 2013** — the primary source, free on mca.gov.in. - **ICAI "Guidance Note on Division I & II of Schedule III"** — free PDF, the definitive reference. - **Taxmann / Bharat's "Company Balance Sheet & P&L"** — worked formats (paid). - **TallyPrime learning (Tally**

Education) for the ERP path. - **Certification:** CA Intermediate (Advanced Accounting) covers this fully; ICAI's Certificate Course on Financial Reporting; any "Company Final Accounts" MOOC.

Quick-reference

Schedule III Balance Sheet order (Division I):

```
EQUITY & LIABILITIES
  1 Shareholders' Funds → Share Capital | Reserves & Surplus | Money pending allotment
  2 Non-current Liabilities → Long-term Borrowings | Deferred Tax Liab | Other | LT Provisions
  3 Current Liabilities → ST Borrowings | Trade Payables | Other CL | ST Provisions
ASSETS
  1 Non-current Assets → PPE | Intangibles | Non-current Investments | LT L&A | Other
  2 Current Assets → Inventories | Current Investments | Trade Receivables | Cash & Cash Eq |
```

P&L order: Revenue from Operations → Other Income → Total Income → (Cost of Materials, Purchases, Change in Inventory, Employee Benefits, Finance Costs, Depreciation, Other Expenses) → PBT → Tax → PAT → EPS.

Item	Rule / value
Current classification	Due / realised within 12 months or operating cycle
TB check	`=SUM(Dr)-SUM(Cr)` must = 0
Note total formula	`=SUMIFS(Dr,Note,n)-SUMIFS(Cr,Note,n)`
Ageing buckets	<6m, 6m-1y, 1-2y, 2-3y, >3y
2021 new disclosures	Ageing, ratios, promoter holding, CSR, title deeds, shortfall use
Proposed dividend	Not provided; disclosed in notes until declared
Format	Vertical only (T-format illegal for companies)
Filing form	AOC-4 to ROC

Consolidation basics in practice

What it is & where it's used

Consolidation is combining a parent company and its subsidiaries into **one set of financial statements**, as if the group were a single economic entity. You don't just add the two balance sheets — you **eliminate** the internal relationships (the parent's investment, inter-company loans, inter-company sales) so the group statement shows only what the group owns and owes *to the outside world*.

In India this is governed by **Ind AS 110** (Consolidated Financial Statements) for listed / large companies and **AS 21** for others; globally it's **IFRS 10**. A company must consolidate when it **controls** another entity — practically, when it holds **more than 50% of voting power** (or controls the board / relevant activities).

Who does this at work:

Role	Consolidation task
Group / Corporate accountant	Prepares the quarterly/annual consolidated financials
Audit associate (Big 4 / mid-tier)	Tests elimination entries, goodwill, NCI
FP&A analyst	Builds the consolidation model in Excel
M&A / valuation analyst	Computes goodwill on acquisition (PPA)
Financial reporting (controllership)	Owns the consol close each period

Even a "single company" job touches this the moment the firm buys a stake, sets up a subsidiary, or spins off a division.

The gap: why companies want this (and college didn't teach it)

College teaches the **theory** — "eliminate the investment against equity, recognise goodwill and minority interest" — usually with a clean single-line example. It almost never makes you build a real **consolidation worksheet** with columns, elimination journals, and a proof that it balances. So freshers can *recite* AS 21 but freeze when handed two trial balances and told "give me the consol by 6 pm."

The specific gaps employers see:

- Candidates confuse **elimination entries** (removing internals) with **regular journals** (they don't hit any single company's books — they live only in the consol worksheet).
- They can't compute **goodwill** correctly — they net the *whole* subsidiary equity instead of only the **parent's share**, and forget the **NCI on net assets**.
- They don't know that **pre-acquisition** profits get capitalised (part of goodwill) while **post-acquisition** profits flow to consolidated reserves.
- They can't split profit between the **parent** and **Non-Controlling Interest (NCI / minority interest)**.

Closing this gap = being the person who can actually *produce* the number, not just describe it.

What "proficient" looks like

A job-ready person, handed a parent + subsidiary trial balance, can unaided:

1. Set up a **consolidation worksheet** (Parent | Subsidiary | Eliminations Dr | Eliminations Cr | Consolidated).
2. Compute **goodwill** = Cost of investment − Parent's share of net assets at acquisition.
3. Compute **NCI** = NCI % × net assets (at acquisition, then updated for post-acq movements).
4. Pass the **four core eliminations**: investment vs equity, inter-co debtor/creditor, inter-co sales/purchases, and **unrealised profit on stock**.
5. Split the year's profit into **owners of parent** and **NCI**, and prove the consol balances.
6. Explain *why* each entry exists in one sentence.

Hands-on: how to actually do it

The four eliminations you must know

1. Cancel the parent's investment against the subsidiary's equity at acquisition (this is where goodwill and NCI are born):

Particulars	Dr (₹)	Cr (₹)
Share capital (subsidiary)	XXX	
Reserves at acquisition (subsidiary)	XXX	
Goodwill (balancing, if positive)	XXX	
To Investment in subsidiary (parent)		XXX
To Non-Controlling Interest		XXX

2. Cancel inter-company balances (parent owes/lent to subsidiary):

Particulars	Dr (₹)	Cr (₹)
Inter-company creditor	XXX	
To Inter-company debtor		XXX

3. Cancel inter-company sales & purchases (so group revenue isn't inflated):

Particulars	Dr (₹)	Cr (₹)
Sales (inter-company)	XXX	
To Purchases / COGS (inter-company)		XXX

4. Remove unrealised profit on closing stock (goods still held within the group):

Particulars	Dr (₹)	Cr (₹)
Consolidated P&L (COGS)	XXX	
To Inventory		XXX

The Excel formulas that actually build the worksheet

Assume columns: `C` =Parent, `D` =Subsidiary, `E` =Elim Dr, `F` =Elim Cr. Consolidated column `G` :

```
# Assets (Dr nature): Parent + Sub + Elim Dr − Elim Cr
=C4 + D4 + E4 − F4

# Liabilities/Equity/Income (Cr nature): Parent + Sub − Elim Dr + Elim Cr
=C10 + D10 − E10 + F10

# Goodwill = Cost of investment − Parent% × Net assets at acquisition
=Cost_Invest − (Parent_pct * NA_acq)

# NCI at reporting date = NCI% × (Net assets at acquisition + post-acq reserves)
=NCI_pct * (NA_acq + Post_acq_reserves)

# Unrealised profit on stock = intra-group closing stock × margin
=IntraGroup_Stock * (GP_pct)         # if margin is on SALES
=IntraGroup_Stock * (Markup/(1+Markup))   # if profit is a % on COST

# Proof: total Elim Dr must equal total Elim Cr
=SUM(E:E)−SUM(F:F)      # must return 0
```

Use `SUMIF` to pull a full trial balance into the worksheet by account code:

```
=SUMIF(Parent_TB[Code], $B4, Parent_TB[Amount])
=SUMIF(Sub_TB[Code],    $B4, Sub_TB[Amount])
```

Python check (reproducible, audit-friendly)

```
cost_invest   = 1_800_000
parent_pct    = 0.80
sc_sub        = 1_000_000     # share capital of subsidiary
res_acq       = 500_000       # reserves at acquisition
na_acq        = sc_sub + res_acq

goodwill = cost_invest − parent_pct * na_acq
nci_acq  = (1 − parent_pct) * na_acq
print(f"Goodwill: Rs {goodwill:,.0f}")   # 1,800,000 − 0.8*1,500,000
print(f"NCI @acq: Rs {nci_acq:,.0f}")
```

Worked example / mini-project

Facts. On 1 Apr 2025, **Parent Ltd** buys **80%** of **Sub Ltd** for **₹18,00,000**. At that date Sub's share capital = **₹10,00,000**, reserves = **₹5,00,000** (net assets = ₹15,00,000). By 31 Mar 2026 Sub's reserves have grown to

₹8,00,000 (so **₹3,00,000 post-acquisition** profit). During the year Parent sold goods worth **₹2,00,000** to Sub at a **25% margin on cost**; **₹1,00,000** of it is still in Sub's closing stock. Parent's own reserves = ₹40,00,000. Inter-company: Sub owes Parent ₹1,50,000.

Step 1 — Goodwill:

```
Goodwill = 18,00,000 − (80% × 15,00,000)
         = 18,00,000 − 12,00,000 = ₹6,00,000
```

Step 2 — NCI at year-end:

```
NCI = 20% × (Net assets at reporting date)
    = 20% × (10,00,000 + 8,00,000) = 20% × 18,00,000 = ₹3,60,000
```

Step 3 — Unrealised profit on stock (25% on cost → margin = 25/125 = 20% of price):

```
Stock in group = ₹1,00,000 → profit element = 1,00,000 × 25/125 = ₹20,000
```

This ₹20,000 is deducted from consolidated inventory *and* from the profit-maker (Parent), so from consolidated reserves.

Step 4 — Consolidated reserves (owners of parent):

```
Parent's own reserves            40,00,000
+ Parent's share of post-acq (80% × 3,00,000)   2,40,000
− Unrealised profit                   (20,000)
= Consolidated reserves          ₹42,20,000
```

Step 5 — The elimination journals:

#	Entry	Dr (₹)	Cr (₹)
1	Share capital (Sub)	10,00,000	
	Reserves at acq (Sub)	5,00,000	
	Goodwill	6,00,000	
	To Investment in Sub		18,00,000
	To NCI (20% × 15,00,000)		3,00,000
2	Inter-co payable (Sub)	1,50,000	
	To Inter-co receivable (Parent)		1,50,000
3	Sales	2,00,000	
	To COGS		2,00,000
4	COGS (unrealised profit)	20,000	
	To Inventory		20,000

Post-acq NCI top-up: NCI is credited a further **20% × 3,00,000 = ₹60,000** (₹3,00,000 + ₹60,000 = **₹3,60,000**, matching Step 2).

Proof: goodwill ₹6,00,000 + consolidated reserves ₹42,20,000 + NCI ₹3,60,000 all tie back; the worksheet's Elim Dr = Elim Cr. That's a complete, reproducible consolidation you can rebuild in a fresh Excel tab in under 30 minutes.

How it's tested

Interview questions (conceptual): - "Why do we eliminate the investment against equity?" (Avoid double-counting — the investment *is* the net assets.) - "Pre-acquisition vs post-acquisition profit — what's the accounting difference?" (Pre → goodwill/NCI; post → consolidated reserves.) - "Goodwill formula? What if it's negative?" (Negative = capital reserve / bargain purchase, credited to P&L under Ind AS 103.) - "How is NCI measured?" (Proportionate net assets, or fair value / full-goodwill method.) - "What's unrealised profit and why remove it?"

Practical tests companies give: - A **timed Excel case**: two trial balances + a fact sheet → build the consol worksheet and give goodwill, NCI, consolidated reserves in 45–60 min. - A **"spot the missing elimination"** task: a consol that doesn't balance; find the un-cancelled inter-co entry. - **PPA (purchase price allocation)** exercise for M&A roles: fair-value the net assets, then derive goodwill.

Common mistakes & how pros avoid them

Mistake	The fix
Using **whole** subsidiary equity for goodwill	Use only **parent's %** of net assets at acquisition
Netting goodwill against **year-end** reserves	Goodwill uses reserves **at the acquisition date** only
Forgetting the **post-acquisition NCI top-up**	NCI = NCI% × *reporting-date* net assets, not acquisition-date
Margin **on cost vs on sales** confusion	25% on cost = 20% on price; use `markup/(1+markup)`
Consolidating a **≤50% associate** by adding lines	Associates use **equity method** (one-line), not full consolidation
Eliminations leaking into a single company's ledger	Eliminations live **only** in the consol worksheet — never posted to Tally of either entity
Consol worksheet doesn't balance	Add a **Elim Dr − Elim Cr = 0** check cell before you trust any number

Pros keep a **one-line "why"** next to every elimination and a **reconciliation of NCI and goodwill** as a standing tab — auditors ask for exactly that.

Learn-it roadmap & resources

Time to proficiency: ~2–3 weeks of focused practice if you already know basic financial statements.

Week	Focus
1	AS 21 / Ind AS 110 concepts; goodwill & NCI by hand on paper
2	Build 3–4 consol worksheets in Excel from scratch (add inter-co, stock, dividends)
3	Add complications: mid-year acquisition, fair-value adjustments, associates (equity method)

Resources: - **ICAI** Ind AS 110 / AS 21 study material and illustrations (free, India-specific). - CA Inter / CA Final **Advanced Accounting** consolidation chapters — the best structured drills for Indian candidates. - IFRS Foundation's **IFRS 10 / IAS 28** basics for the global framing. - Practice by taking any two public companies' standalone statements and attempting a mini-consol. - **Certifications that signal this skill:** CA, ACCA (FR & SBR papers), CPA (FAR), CFA (Financial Reporting).

Quick-reference

```
Consolidate when:  Control (usually >50% voting power)   → Ind AS 110 / AS 21 / IFRS 10
Associate (20–50%):  Equity method (one line), NOT full consol
```

Item	Formula
Goodwill	Cost of investment − Parent% × Net assets **at acquisition**
Negative goodwill	Bargain purchase → Capital reserve / P&L (Ind AS 103)
NCI (proportionate)	NCI% × Net assets **at reporting date**
Consolidated reserves	Parent reserves + Parent% × post-acq profit − unrealised profit
Unrealised profit (margin on sales)	Intra-group stock × GP%
Unrealised profit (markup on cost)	Intra-group stock × markup/(1+markup)

The four eliminations: (1) Investment ↔ Equity (birth of goodwill + NCI); (2) Inter-co debtor ↔ creditor; (3) Inter-co sales ↔ purchases; (4) Unrealised profit on stock.

Golden checks: Elim Dr = Elim Cr · Goodwill uses acquisition-date reserves · NCI uses reporting-date net assets · Eliminations never touch either company's own ledger.

MIS Reporting

What it is & where it's used

MIS (Management Information System) reporting is the recurring, decision-ready summary of a company's financial and operational performance — usually a **monthly MIS pack** that goes to the CFO, MD, board, investors, or a lender. It is not the statutory financials (those follow Schedule III / Ind AS and go out annually). MIS is *internal, fast, and opinionated*: it tells management "here is what happened, here is why, here is what to do."

A typical monthly pack contains: P&L actual vs budget vs last month, cash position and 13-week cash flow, receivables (AR) and payables (AP) ageing, and a KPI dashboard (gross margin %, EBITDA %, DSO, DPO, headcount cost, revenue per employee). It closes with **commentary** — the paragraph that explains variances.

Roles that live and die by MIS: **FP&A analyst, management accountant, finance manager, financial controller, business finance / commercial finance partner, and startup "finance person #1."** In an FMCG or manufacturing SME the accountant who "does the books in Tally" is also expected to produce the MIS. In a funded startup, the founder wants an MIS pack by the 7th of every month or the investor calls get awkward. This is one of the most *hireable* practical skills in Indian finance because every company above ~₹5 crore turnover needs it and very few juniors can build one.

The gap: why companies want this (and college didn't teach it)

College teaches you to *prepare* a P&L and balance sheet from a trial balance. It never teaches you to *interpret* one for a non-finance MD, or to build the same P&L **five ways** (by month, by product, by branch, actual vs budget, YoY). The MBA case method discusses strategy at 30,000 feet; the CA syllabus drills accounting standards. Neither hands you a blank Excel and says "the month closed yesterday, the MD wants the pack tomorrow at 9am, go."

The specific gaps MIS closes:

- **From "recording" to "reporting."** Employers already have Tally posting the entries. They pay you to turn that data into a one-page story.
- **Variance thinking.** Nobody in college asks "revenue is ₹8L below budget — is it price, volume, or mix?" That decomposition is the whole job.
- **Speed and templating.** Real MIS is the *same* template every month. The skill is building it once so it refreshes in 20 minutes, not rebuilding it for 3 days.
- **Commentary.** A number without a "so what" is useless to management. The gap is writing 5 crisp bullet points a CEO actually reads.

What "proficient" looks like

A job-ready person can, unaided:

1. Pull a trial balance / ledger export from Tally or an ERP and map every GL to an MIS line (revenue, COGS, opex buckets) using a **mapping table**, not manual sorting.

2. Build a **P&L that auto-refreshes** with actual vs budget vs prior month and computes variances in ₹ and %.
3. Produce **AR/AP ageing** (0-30 / 31-60 / 61-90 / 90+) from an open-item invoice list.
4. Compute the standard KPIs — Gross Margin %, EBITDA %, DSO, DPO, CCC, Current Ratio, Revenue/Employee — and know what each *means*.
5. Build a **13-week rolling cash flow** so management sees the runway, not just the bank balance.
6. Write **commentary** that attributes each material variance to a driver (price/volume/mix/timing/one-off).
7. Do all of the above on a locked template that refreshes when new data is pasted in — turnaround measured in hours, not days.

Hands-on: how to actually do it

Step 1 — Export the data

From **TallyPrime**: `Gateway of Tally → Display More Reports → Trial Balance` → press `Alt+E` (Export) → format **Excel (Spreadsheet)** → set "Show Ledger-wise" and full year. For ledgers/ageing: `Display More Reports → Statements of Accounts → Outstandings → Receivables` → `Alt+E`. This gives you a bill-by-bill open items list with due dates.

Step 2 — Map GLs to MIS lines (never sort by hand)

Keep a `Mapping` sheet: column A = Ledger name, column B = MIS bucket. Then in your data sheet:

```
=XLOOKUP([@Ledger], Mapping[Ledger], Mapping[MIS_Bucket], "UNMAPPED")
```

Guard against new ledgers slipping through:

```
=IF(COUNTIF(Mapping[Ledger],[@Ledger])=0, "⚠ NEW LEDGER", "OK")
```

Step 3 — Build the P&L with SUMIFS

Actual by bucket, then variance vs budget:

```
Actual:      =SUMIFS(Data[Amount], Data[MIS_Bucket], $A5, Data[Month], "Jun")
Var ₹:       =[@Actual]-[@Budget]
Var %:       =IFERROR([@Actual]/[@Budget]-1, "")
Margin %:    =[@GrossProfit]/[@Revenue]
```

Step 4 — AR/AP ageing buckets

Given an invoice list with `Due_Date` and `Outstanding`:

```
Days overdue: =MAX(0, TODAY()-[@Due_Date])
Bucket:       =IFS([@Days]≤30,"0-30", [@Days]≤60,"31-60", [@Days]≤90,"61-90", TRUE,"90+")
```

Then a summary pivot, or:

```
=SUMIFS(AR[Outstanding], AR[Customer], $A2, AR[Bucket], B$1)
```

Step 5 — The KPI block

```
DSO    = (Trade Receivables / Credit Sales) * Days_in_period
       = (Receivables / Revenue) * 30
DPO    = (Trade Payables / Purchases) * 30
CCC    = DSO + DIO - DPO
EBITDA% = EBITDA / Revenue
```

Optional — SQL if the data sits in a database

```
SELECT mis_bucket,
       SUM(CASE WHEN period = '2026-06' THEN amount END) AS actual_jun,
       SUM(CASE WHEN period = '2026-05' THEN amount END) AS actual_may
FROM gl_entries g
JOIN gl_mapping m ON g.ledger = m.ledger
GROUP BY mis_bucket
ORDER BY mis_bucket;
```

AR ageing in SQL:

```
SELECT customer,
  SUM(CASE WHEN CURRENT_DATE-due_date<=30 THEN outstanding ELSE 0 END) AS b_0_30,
  SUM(CASE WHEN CURRENT_DATE-due_date BETWEEN 31 AND 60 THEN outstanding ELSE 0 END) AS b_31_6
  SUM(CASE WHEN CURRENT_DATE-due_date BETWEEN 61 AND 90 THEN outstanding ELSE 0 END) AS b_61_9
  SUM(CASE WHEN CURRENT_DATE-due_date>90 THEN outstanding ELSE 0 END) AS b_90_plus
FROM open_invoices
GROUP BY customer;
```

Optional — Power BI / DAX for a live dashboard

```
Revenue       = SUM(Fact[Amount])
GM %          = DIVIDE([Gross Profit], [Revenue])
Rev vs Bud % = DIVIDE([Revenue] - [Budget Revenue], [Budget Revenue])
DSO           = DIVIDE([Trade Receivables], [Revenue]) * 30
YoY Rev %     = DIVIDE([Revenue] - CALCULATE([Revenue], DATEADD('Cal'[Date],-1,YEAR)),
                      CALCULATE([Revenue], DATEADD('Cal'[Date],-1,YEAR)))
```

The month-end adjusting entries behind the numbers

MIS is only right if provisions are booked. Typical month-end Dr/Cr:

Entry	Dr	Cr
Accrue electricity bill not yet received	Power & Fuel Exp	Provision for Expenses
Depreciation for the month	Depreciation	Accumulated Depreciation
Prepaid insurance monthly charge	Insurance Exp	Prepaid Insurance
Salaries earned, paid next month	Salaries	Salary Payable

Worked example / mini-project

Company: Vitesse Components Pvt Ltd, an auto-parts SME. June 2026 close.

P&L (₹ lakh)	Actual	Budget	Var ₹	Var %	Prior Mo
Revenue	210	230	(20)	-8.7%	205
COGS	138	147	9	—	131
Gross Profit	**72**	**83**	**(11)**	**-13.3%**	**74**
GM %	34.3%	36.1%	-1.8pp		36.1%
Employee cost	28	27	(1)		27
Other opex	19	20	1		19
EBITDA	**25**	**36**	**(11)**	**-30.6%**	**28**
EBITDA %	11.9%	15.7%			13.7%

AR ageing (₹ lakh):

Customer	0-30	31-60	61-90	90+	Total
Mahindra Tier-1	42	18	0	0	60
Local OEM	12	9	7	14	42
Total	**54**	**27**	**7**	**14**	**102**

KPIs: DSO = (102/210)×30 = **14.6 days ✓** but ₹14L is 90+ overdue — a red flag. DPO ≈ 32 days. GM slipped because a steel price hike raised COGS while a volume dip lost fixed-cost absorption.

Commentary (what management actually reads):

- Revenue ₹210L, **8.7% below budget** — driven by a 9% volume shortfall at Local OEM; price and mix held.
- **GM fell 1.8pp to 34.3%** — ₹6L from steel cost inflation (not passed to customers), ₹5L from weaker fixed-cost absorption on lower volume.
- **EBITDA ₹25L vs ₹36L budget**; the entire ₹11L miss is gross-margin, not overhead — opex is on plan.
- **₹14L receivables >90 days at Local OEM** — same customer driving the volume drop. Recommend credit hold and a collection call before July dispatch.
- **Action:** file a price-revision request with Local OEM to recover steel inflation; target ₹4L/month margin recovery from August.

Reproduce it: paste a Tally TB into `Data`, map GLs, and every table above refreshes with SUMIFS.

How it's tested

Interview questions: - "Walk me through what goes into a monthly MIS pack." - "Revenue is up but EBITDA is down — how do you explain that to the MD?" - "How do you decompose a revenue variance into price, volume, and mix?" - "What's the difference between MIS and statutory reporting?" - "DSO jumped from 45 to 60 days — what do you investigate?"

Practical / assessment tests companies give: - A **timed Excel test** (45-90 min): here's a raw TB and last month's template — produce the P&L, ageing, and 3 commentary bullets. - A **"broken template"** test: SUMIFS returns wrong totals because of an unmapped ledger; find and fix it. - A **case:** "Books close today, MD wants the pack by 9am — what's your checklist?" - Sometimes a **SQL screen** (build the ageing query) or a **Power BI take-home** (one dashboard page from a CSV).

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Hard-coding numbers / manual sorting each month	Locked template driven by SUMIFS + XLOOKUP mapping; only paste raw data
New ledger silently dropped from P&L	An "UNMAPPED / ⚠ NEW LEDGER" check cell that must read zero before publishing
P&L doesn't tie to Tally	Reconcile MIS revenue & PAT to the TB every month — a control-total cell
Commentary just restates numbers ("revenue down 8%")	Attribute to a *driver* — price/volume/mix/timing/one-off — and add an action
Ignoring cash — only P&L	Always include cash position + 13-week flow; profit ≠ liquidity
MIS on the 20th	Automate so it lands by working day 5-7; late MIS is useless
Mixing accruals in one month, cash in another	Book month-end provisions (depreciation, accruals, prepaids) before pulling
One giant P&L, no segmentation	Slice by product/branch/customer — that's where insight hides

Learn-it roadmap & resources

Time to proficiency: ~6-8 weeks part-time if you already know accounting. Week 1-2: Excel (SUMIFS, XLOOKUP, IFS, pivots, tables). Week 3-4: build a full P&L + ageing template from a real Tally export. Week 5-6: KPIs, variance decomposition, commentary writing. Week 7-8: Power BI / SQL layer for a live dashboard.

Resources: - *Excel:* ExcelJet (free formula reference), Chandoo.org, Leila Gharani (YouTube). - *FP&A / MIS:* CFI's FP&A and Financial Modeling courses (paid); "Financial Intelligence" by Berman & Knight (interpretation). - *Power BI:* Microsoft Learn "Power BI in a Day" (free), Enterprise DNA. - *SQL:* Mode SQL Tutorial (free), SQLBolt. - *India-specific:* TallyPrime documentation on exports and outstandings; practice on any SME's TB.

Certifications: Microsoft PL-300 (Power BI Data Analyst), CFI's FMVA (includes FP&A/dashboard modules), Tally certifications. None are mandatory — a **portfolio MIS pack** you built beats any certificate in an interview.

Quick-reference

Item	Formula / step
Map GL to bucket	`=XLOOKUP(Ledger, Map[L], Map[Bucket], "UNMAPPED")`
Actual by bucket	`=SUMIFS(Data[Amt], Data[Bucket], $A5, Data[Month], "Jun")`
Variance %	`=IFERROR(Actual/Budget-1,"")`
Ageing bucket	`=IFS(D≤30,"0-30",D≤60,"31-60",D≤90,"61-90",TRUE,"90+")`
Gross Margin %	Gross Profit / Revenue
EBITDA %	EBITDA / Revenue
DSO	(Receivables / Revenue) × Days
DPO	(Payables / Purchases) × Days
CCC	DSO + DIO − DPO
Tally TB export	Trial Balance → `Alt+E` → Excel
Tally AR ageing	Outstandings → Receivables → `Alt+E`
Control check	MIS PAT must tie to Tally TB; UNMAPPED count = 0
Delivery target	Working day 5-7, same template every month

Pack contents, in order: P&L (actual/budget/prior) → EBITDA bridge → cash position + 13-week flow → AR/AP ageing → KPI dashboard → commentary (5 bullets, each = variance + driver + action).

Working with Auditors

What it is & where it's used

Every company of any size gets audited. In India a statutory audit under the Companies Act, 2013 is mandatory for **every** company (private, public, one-person) regardless of turnover. On top of that a **tax audit** under Section 44AB of the Income-tax Act kicks in when business turnover crosses ₹1 crore (₹10 crore if cash receipts and payments are each ≤ 5%) or professional receipts cross ₹50 lakh. Add GST audit/reconciliation (GSTR-9C), internal audit, and — for listed or PE-backed firms — quarterly limited reviews and often a US-GAAP/IFRS group audit.

"Working with auditors" is the recurring job of the person who owns the books: the **accountant, finance executive, financial controller, or FP&A analyst**. You are the auditor's counterparty. You hand over the trial balance, build the supporting schedules, answer queries, produce documents from the **PBC list** (Prepared By Client), and defend your numbers. Do it well and the audit closes in three weeks with a clean report. Do it badly and it drags for three months, throws up adjustments that dent your P&L, and burns your reputation with management. This is a career skill that quietly separates a "junior who does entries" from a "controller who runs the close."

The gap: why companies want this (and college didn't teach it)

Your B.Com/MBA taught you *audit theory from the auditor's chair* — sampling, materiality, SA 500, assertions, the audit risk model. What no classroom teaches is the **auditee's craft**: how to keep books that survive scrutiny, how to build a schedule that ties to the GL to the paisa, how to respond to a query without opening ten new ones, and how to project-manage a team of CAs who are sitting in your office asking for the 47th document.

Employers feel this gap acutely because audit season is expensive. A messy auditee means overtime, delayed financials, delayed board meetings, and delayed loan renewals (banks want audited statements). When a hiring manager asks "*Have you handled audits?*" they mean: can you produce a fixed-asset register that reconciles to the GL, explain a variance without panicking, and close the audit on schedule? That skill is learned only by doing — which is exactly why it commands a premium.

What "proficient" looks like

A job-ready person can, unaided:

- Produce a **clean trial balance** that ties to the GL, with no suspense balances and no "misc" dumping grounds.
- Build every schedule on the standard PBC list from the accounting system, each **reconciling to the TB line** to zero.
- Anticipate the **common audit queries** and keep the backup ready *before* it's asked.
- Draft a **lead schedule** (a one-page summary per FS caption that rolls up to the number in the financial statements).
- Handle the **BRS, GST reconciliation (GSTR-1 vs 3B vs books), TDS reconciliation (26AS vs books), and inter-company reconciliation**.

- Respond to a query in writing with the entry, the document, and the rationale — closing it, not extending it.
- Run the audit as a **project**: a PBC tracker, a query log, and a status call cadence.

Hands-on: how to actually do it

1. Build an audit-ready trial balance

Export the TB from Tally/ERP, then sanity-check it. Two rules: **debits = credits**, and **no orphan/suspense balances**.

```
' TB balances?
=SUM(D2:D500)-SUM(E2:E500)          ' Dr total minus Cr total → must be 0

' Flag suspense / clearing accounts that should be nil at year-end
=IF(AND(ISNUMBER(SEARCH("suspense",A2)),ABS(F2)>0),"CLEAR ME","")
```

2. The lead schedule (ties GL → FS)

For each financial-statement line, a lead schedule groups the underlying ledgers. Use `SUMIFS` against your TB dump so it auto-ties:

```
' Trade Receivables lead = sum of all ledgers mapped to "Debtors"
=SUMIFS(TB[ClosingBal], TB[FSGroup], "Trade Receivables")

' Tie-out check against the balance sheet figure
=ROUND(SUMIFS(TB[ClosingBal],TB[FSGroup],"Trade Receivables")-BS_TradeReceivables,0)  ' must b
```

3. Debtors ageing (the #1 requested schedule)

```
' Bucket each invoice by days outstanding from invoice date
=LOOKUP((TODAY()-[@InvoiceDate]),{0,31,61,91,181},{"0-30","31-60","61-90","91-180",">180"})
```

If your data is in SQL, generate the ageing straight from the ledger:

```
SELECT customer,
       SUM(CASE WHEN dso ≤30 THEN amount ELSE 0 END) AS "0-30",
       SUM(CASE WHEN dso BETWEEN 31 AND 90 THEN amount ELSE 0 END) AS "31-90",
       SUM(CASE WHEN dso BETWEEN 91 AND 180 THEN amount ELSE 0 END) AS "91-180",
       SUM(CASE WHEN dso > 180 THEN amount ELSE 0 END) AS ">180"
FROM (SELECT customer, amount,
             DATEDIFF(CURRENT_DATE, invoice_date) AS dso
      FROM ar_open_items) t
GROUP BY customer
ORDER BY 5 DESC;
```

4. Bank reconciliation (auditors always vouch this)

```
import pandas as pd
books = pd.read_csv("bank_ledger.csv")     # your GL bank ledger
stmt  = pd.read_csv("bank_statement.csv")   # bank's statement

# Un-reconciled: entries in books not matched by (date, amount) in the statement
merged = books.merge(stmt, on=["value_date","amount"], how="left", indicator=True)
unpresented = merged[merged["_merge"]=="left_only"]      # cheques issued, not cleared
print("Book balance     :", books.amount.sum())
print("Un-reconciled amt:", unpresented.amount.sum())
```

5. GST reconciliation (GSTR-1 vs 3B vs books)

On the GST portal: **Login → Returns Dashboard → select FY → GSTR-9 → Download GSTR-1/3B summary (PDF/Excel)**. Then in Excel:

```
' Difference between books' output tax and GSTR-3B tax paid - should be ~0
=Books_OutputGST - GSTR3B_TaxPaid
```

6. TDS reconciliation (26AS / AIS vs books)

Download Form 26AS from the income-tax portal (**Login → e-File → Income Tax Returns → View Form 26AS → TRACES**). Match TDS credited per 26AS to TDS receivable in books; the delta usually means a customer deducted but didn't file, or a timing mismatch.

7. A typical audit adjustment (JV) you'll pass

When the auditor finds an unrecorded expense (say, ₹1,20,000 audit fee not provided for):

Account	Dr (₹)	Cr (₹)
Legal & Professional Charges (P&L)	1,20,000	
Provision for Audit Fees (Current Liab.)		1,20,000

Narration: Being provision for FY 2025-26 statutory audit fee not recorded, adjustment per audit.

Worked example / mini-project

Scenario: You're the accountant at *Nimbus Traders Pvt Ltd*, FY 2025-26. Turnover ₹8.4 crore. The statutory auditor's team arrives Monday. Build the audit file.

Step 1 — TB and lead schedules. Export the TB. It shows Trade Receivables ₹92,40,000. Your debtors ageing (built with the `SUMIFS` / `LOOKUP` above) sums to ₹92,40,000 — **it ties**. Good. The `>180 days` bucket is ₹6,10,000 across two customers.

Step 2 — Anticipate the query. The auditor will ask "*Is provision for doubtful debts needed on the ₹6.1 lakh?*" One customer (₹4,10,000) paid ₹4,00,000 in April 2026 — attach the bank credit as **subsequent-receipt evidence**, no provision needed. The other (₹2,00,000) is disputed — you provide 100%.

Account	Dr (₹)	Cr (₹)
Provision for Doubtful Debts (P&L)	2,00,000	
Allowance for Doubtful Debts (contra-asset)		2,00,000

Step 3 — Bank & GST. BRS shows ₹35,000 of unpresented cheques — documented. GSTR-3B tax paid = ₹14,90,000; books output GST = ₹15,02,000. Delta ₹12,000 → a March sales invoice was booked but reported in the April return. Note it in the reco; no adjustment needed (timing).

Step 4 — Fixed assets. FAR shows additions of ₹4,50,000 (a machine). Depreciation per Schedule II (WDV, useful life 15 yrs) for 9 months = you compute and tie to the depreciation ledger.

Step 5 — Deliverable. A single workbook: `Nimbus_Audit_FY26.xlsx` with tabs — *TB, Lead Schedules, Debtors Ageing, Creditors Ageing, BRS, GST Reco, TDS Reco, FAR, Query Log*. Every schedule footer ties to the FS to zero. That workbook *is* the difference between a 2-week and a 2-month audit.

How it's tested

Interview questions: - "Walk me through how you prepare for a statutory audit." (They want: PBC list → schedules → tie-outs → query handling.) - "What is a PBC list? Name 8 items on it." - "How do you reconcile GSTR-1, GSTR-3B and books? What causes differences?" - "A customer is 200 days overdue but paid after year-end. Do you provide for it?" (No — subsequent receipt is audit evidence.) - "What's the entry for an unrecorded expense the auditor finds?" - "Difference between statutory audit, tax audit, and internal audit?"

Practical/assessment tests: - A **timed Excel case**: given a raw TB dump and a ledger extract, build a debtors ageing and a lead schedule that ties — in 45 minutes. - A **"here's a messy BRS, reconcile it"** exercise with planted un-presented cheques and a bank charge not in books. - A **query-response drill**: they hand you five auditor queries and grade whether your written answers *close* them (entry + document + rationale) or invite follow-ups.

Common mistakes & how pros avoid them

Mistake	What pros do
Schedules don't tie to the TB	Every schedule footer has a `=schedule − TB line` cell that must show 0 before sending.
"Suspense" / "misc" ledgers with balances	Clear all suspense to nil before the audit; auditors treat suspense as a red flag.
Answering queries verbally	Maintain a **written query log**: query, response, document ref, status, owner, date.
Giving auditors raw system access with no map	Provide a clean export + a "where things are" index; control the narrative.
Passing adjustments only in the audit file, not in books	Post every agreed adjustment in the ERP so next year opens clean.
Re-doing schedules from scratch each request	Build **formula-driven** schedules off a single TB dump so a re-export refreshes everything.
Missing subsequent events	Review April–May bank statements and post-year-end invoices before finalising.

Learn-it roadmap & resources

Time to proficiency: ~2–3 months of focused effort, or one full audit cycle on the job.

Week	Focus
1	TB hygiene, lead schedules, tie-outs in Excel (SUMIFS, ROUND checks)
2	Debtors/creditors ageing, BRS, FAR + Schedule II depreciation
3	GST reco (GSTR-1/3B/2B/9C), TDS reco (26AS/AIS), inter-co
4	Build a full mock audit file end-to-end; run a query-log drill
5–8	Shadow a real audit; own the PBC tracker

Resources: - ICAI Standards on Auditing (free PDFs) — read SA 500 (evidence), SA 560 (subsequent events), SA 580 (representations) from the *auditee's* angle. - GST portal help + GSTR-9C reconciliation format (free, cbic-gst.gov.in). - TallyPrime "Audit & Compliance" and any ERP's standard schedules export. - **Certifications:** CA Intermediate (Audit paper) is the gold standard in India; for tooling, an advanced-Excel course pays off fastest.

Quick-reference

Standard PBC list (keep these ready):

#	Item
1	Trial balance + general ledger dump
2	Lead schedules per FS caption
3	Debtors & creditors ageing
4	Bank statements + BRS (all accounts)
5	Fixed-asset register + depreciation working
6	GST reco (GSTR-1/3B/2B, GSTR-9C)
7	TDS reco (26AS/AIS vs books) + challans
8	Inventory listing + valuation basis
9	Loan statements, sanction letters, interest working
10	Payroll register, PF/ESI/PT challans
11	Related-party transactions list
12	Board/AGM minutes, statutory registers

Key thresholds & checks:

Item	Value / Rule
Tax audit (Sec 44AB) — business	Turnover > ₹1 cr (₹10 cr if cash ≤ 5%)
Tax audit — profession	Receipts > ₹50 lakh
Statutory audit	All companies, no threshold
GSTR-9C reconciliation	Aggregate turnover > ₹5 cr
TB tie-out	Dr total − Cr total = 0
Schedule tie-out	`Schedule − FS line = 0`

Golden rules: tie every schedule to zero · clear all suspense · answer queries in writing with entry + document + rationale · post agreed adjustments in the ERP · review subsequent events before sign-off.

AP, AR & Fixed-Asset Registers

What it is & where it's used

Three sub-ledgers hang off every general ledger, and in a real accounts team they are usually three different desks:

- **Accounts Payable (AP)** — money you owe vendors. Governed by the **Purchase-to-Pay (P2P)** cycle: PR → PO → GRN → 3-way match → vendor invoice → payment.
- **Accounts Receivable (AR)** — money customers owe you. Governed by the **Order-to-Cash (O2C)** cycle: sales order → dispatch → invoice → collection → cash application.
- **Fixed-Asset Register (FAR)** — the list of every capitalised asset, its cost, location, and depreciation, feeding the depreciation schedule that hits P&L every month.

These are the entry-level and mid-level engine rooms of finance. **AP executive, AR / credit-control analyst, R2R (record-to-report) associate, fixed-asset accountant, and GL accountant** roles live here. In a Big-4 or GCC (Global Capability Centre) in Bengaluru/Hyderabad/Pune, "P2P analyst" and "O2C analyst" are literally two of the largest hiring pools. If you can run these three sub-ledgers and reconcile them to the GL, you are employable in accounts on day one.

The gap: why companies want this (and college didn't teach it)

College teaches you the *journal entry* for a purchase and the *formula* for depreciation. It never shows you:

- That an invoice **cannot be posted** until it three-way-matches a PO and a GRN — so 40% of an AP job is chasing mismatches, not posting.
- **Ageing** — the single most-watched AP/AR metric — and how a bucketed ageing report is built and read.
- That depreciation in practice is driven by a **register with hundreds of line items**, block-of-assets rules under the Income-tax Act, and Schedule II useful lives under the Companies Act — two *different* depreciation numbers on the same asset.
- **GST input tax credit (ITC)** blocking, **TDS** deduction on vendor payments, and **e-invoice / IRN** validation — India-specific gates that decide whether a payable is even legal to pay.

Employers pay for the person who can take a messy vendor ledger and produce a clean, reconciled, aged, tax-correct sub-ledger. That is a *process* skill, not a formula.

What "proficient" looks like

A job-ready person can, unaided:

1. Take a raw invoice register + PO register + GRN register and run a **3-way match** in Excel, flagging quantity/price/tax mismatches.
2. Build a **31/60/90/90+ ageing report** for AP or AR using formulas, and calculate **DSO / DPO**.
3. Post the full P2P and O2C journal entries **including GST and TDS**, and reconcile the sub-ledger control account to the GL.
4. Maintain a **fixed-asset register** and generate a **depreciation schedule** under both Companies Act Schedule II (SLM/WDV) and Income-tax block-WDV.

5. Do it in **TallyPrime** (India SME reality) and understand the same flow in an ERP (SAP FICO / Oracle / Zoho / NetSuite).

Hands-on: how to actually do it

1. Three-way match (Excel)

Given three sheets — `PO`, `GRN`, `Invoice`, keyed on PO number — pull PO and GRN data next to each invoice line:

```
=XLOOKUP([@PO_No], PO[PO_No], PO[PO_Qty], "PO missing")
=XLOOKUP([@PO_No], GRN[PO_No], GRN[Recv_Qty], "GRN missing")
```

Then a match-status flag:

```
=IF(AND([@Inv_Qty]≤[@GRN_Qty], ABS([@Inv_Rate]-[@PO_Rate])<0.01),
   "MATCH", "HOLD-"&TEXTJOIN("/",TRUE,
     IF([@Inv_Qty]>[@GRN_Qty],"QTY",""),
     IF(ABS([@Inv_Rate]-[@PO_Rate])≥0.01,"PRICE","")))
```

Anything not `MATCH` goes on hold — you do not pay it.

2. Ageing report (Excel)

With an invoice `Due_Date` and a snapshot `AsOf` date:

```
=IF([@Balance]=0,"", $AsOf - [@Due_Date])        // days overdue
```

Bucket it:

```
=IFS([@Days]≤0,"Not due",[@Days]≤30,"0-30",
     [@Days]≤60,"31-60",[@Days]≤90,"61-90",TRUE,"90+")
```

Summarise the outstanding by bucket per party:

```
=SUMIFS(Inv[Balance], Inv[Party],[@Party], Inv[Bucket],"90+")
```

DSO (days sales outstanding) and **DPO**:

```
DSO = (Total AR / Credit Sales in period) * Days in period
DPO = (Total AP / Credit Purchases in period) * Days in period
```

3. Ageing / register in SQL

```
SELECT party_name,
  SUM(CASE WHEN days<=30            THEN balance END) AS d0_30,
  SUM(CASE WHEN days BETWEEN 31 AND 60 THEN balance END) AS d31_60,
  SUM(CASE WHEN days BETWEEN 61 AND 90 THEN balance END) AS d61_90,
  SUM(CASE WHEN days>90             THEN balance END) AS d90_plus,
  SUM(balance) AS total_due
FROM (
  SELECT party_name, balance,
         DATEDIFF(CURRENT_DATE, due_date) AS days
  FROM ar_open_items WHERE balance <> 0
) t
GROUP BY party_name
ORDER BY total_due DESC;
```

4. Depreciation schedule in Python

```
import pandas as pd
far = pd.DataFrame({
    "asset":["Laptop-01","Plant-A","Delivery Van"],
    "cost":[80000, 2500000, 900000],
    "life_yrs":[3, 15, 8],          # Schedule II useful life
    "method":["SLM","WDV","WDV"]})

def dep(row):
    if row.method == "SLM":
        return round(row.cost / row.life_yrs, 2)          # no residual for simplicity
    rate = 1 - (0.05) ** (1/row.life_yrs)                  # WDV rate, 5% residual
    return round(row.cost * rate, 2)

far["annual_dep"] = far.apply(dep, axis=1)
print(far)
```

The WDV rate formula is the same one Schedule II implies: `rate = 1 - (residual/cost)^(1/life)`.

5. Journal entries (India, with GST & TDS)

Purchase of ₹1,00,000 goods + 18% GST, TDS 194C @1% on a contractor bill:

Account	Dr	Cr
Purchases / Expense A/c	1,00,000	
Input CGST @9%	9,000	
Input SGST @9%	9,000	
To Vendor A/c		1,17,000

On payment with TDS ₹1,000:

Account	Dr	Cr
Vendor A/c	1,17,000	
To TDS Payable (194C)		1,000
To Bank		1,16,000

Sale of ₹2,00,000 + 18% IGST (interstate):

Account	Dr	Cr
Customer A/c	2,36,000	
To Sales A/c		2,00,000
To Output IGST		36,000

Capitalising the ₹9,00,000 van + monthly depreciation:

Account	Dr	Cr
Fixed Asset – Vehicles	9,00,000	
Input CGST/SGST	81,000	
To Vendor		9,81,000
Depreciation A/c (monthly)	9,375	
To Accumulated Depreciation		9,375

6. TallyPrime click-path

- **P2P:** Gateway → Vouchers → F9 Purchase → select party → stock item → GST auto-computes from ledger rates. For 3-way in Tally: enable *Purchase Order* (F11 → Enable PO) → Order → Receipt Note (GRN) → Purchase against them.
- **AR ageing:** Gateway → Display More Reports → Statements of Accounts → Ageing Analysis → set bucket ranges (0-30/31-60/...).
- **FAR:** create asset ledger under *Fixed Assets* group; book depreciation via Journal (F7) at month-end.

Worked example / mini-project

Reproduce this. A trading firm, month of June 2026.

Vendor open items as on 30-Jun-2026:

Vendor	Invoice	Inv Date	Due Date	Amount (₹)
Sharma Traders	P-101	05-Apr	05-May	1,20,000
Sharma Traders	P-140	20-May	19-Jun	80,000
Global Supply	P-155	02-Jun	02-Jul	2,50,000

Ageing as on 30-Jun: P-101 is 56 days overdue → **31-60**; P-140 is 11 days → **0-30**; P-155 not yet due. AP ageing:

Vendor	0-30	31-60	Not due	Total
Sharma Traders	80,000	1,20,000	–	2,00,000
Global Supply	–	–	2,50,000	2,50,000

DPO if June credit purchases were ₹15,00,000: `(4,50,000 / 15,00,000) × 30 = 9 days`.

Fixed asset added 15-Jun-2026: Plant ₹12,00,000, Schedule II life 15 yrs, SLM, 5% residual. - Annual dep = (12,00,000 − 60,000) / 15 = ₹76,000. - June (pro-rata 16 days of 30) = 76,000 × 16/365 ≈ **₹3,331**. - Income-tax side: Plant & Machinery block, WDV @15%. Since put to use < 180 days in FY, **half depreciation** = 12,00,000 × 15% × ½ = ₹90,000 for the year. Two different numbers — that's the point.

Build the ageing sheet with the formulas above and tie the sub-ledger total (₹4,50,000) back to the AP control account in the GL. If it doesn't tie, you have an unposted invoice or a duplicate.

How it's tested

- **Timed Excel test (30–45 min):** "Here's an invoice dump and a PO dump — build an ageing report and flag mismatches." They want XLOOKUP/SUMIFS/IFS, a pivot, and correct buckets.
- **SQL screen:** write the ageing CASE-WHEN query; compute DSO.
- **Case / "close these books":** given trial balance + a list of pending items, post accruals, book depreciation, reconcile AP/AR sub-ledger to GL.
- **Interview questions:** "Walk me through P2P." "What is a 3-way match and why?" "GRN posted, invoice not received at month-end — what entry?" (Answer: **GR/IR / goods-received-not-invoiced accrual**.) "Difference between Companies Act and Income-tax depreciation?" "How do you treat capital WIP?" "What is DSO and how do you reduce it?" "Blocked ITC under GST — give examples."

Common mistakes & how pros avoid them

- **Paying an unmatched invoice.** Pros never release payment on a HOLD line; the match is the control.
- **Expensing a capital item** (or capitalising a repair). Rule of thumb: if it extends life or capacity → capitalise.
- **One depreciation number.** Pros keep the **Companies Act book and the Income-tax block-WDV in parallel**, and reconcile the deferred-tax difference.
- **Ignoring GR/IR at close.** Received-not-invoiced and invoiced-not-received both need accruals or the P&L is wrong.

- **Claiming blocked ITC** (motor vehicles, personal use, works contract) — reverses later with interest.
- **Ageing off invoice date, not due date.** Overdue is measured from the *due* date.
- **Sub-ledger not tied to GL.** Always reconcile the control account monthly — the number employers trust is the reconciled one.

Learn-it roadmap & resources

Stage	Time	Do this
Excel for AP/AR	1 wk	XLOOKUP, SUMIFS, IFS, pivots on a sample ledger
P2P & O2C process	1 wk	Map both cycles end-to-end; learn GR/IR, 3-way match
TallyPrime	2 wks	Book purchases/sales with GST, PO/GRN, ageing, FAR
GST + TDS + Depreciation	2 wks	ITC rules, TDS sections (194C/194J/194Q), Schedule II vs blocks
ERP exposure	ongoing	SAP FICO or Oracle basics; free NetSuite/Zoho trials

Resources: ICAI study material (Accounting + Taxation) — you already have it; TallyPrime free educational mode; GSTN portal help; SAP FICO intro on Udemy; the *Corporate Finance Institute* AP/AR primers. **Certifications:** Tally certification, SAP FICO, or a GCC's internal R2R/P2P badge. Realistic time to job-ready: **6–8 weeks** of focused practice.

Quick-reference

Item	Key thing
P2P order	PR → PO → GRN → 3-way match → invoice → pay
O2C order	SO → dispatch → invoice → collect → cash apply
3-way match	PO ≈ GRN ≈ Invoice (qty & price)
Ageing buckets	0-30 / 31-60 / 61-90 / 90+ from **due date**
DSO	AR ÷ Credit Sales × Days
DPO	AP ÷ Credit Purchases × Days
Accrual at close	GR/IR (goods recd not invoiced)
SLM dep	(Cost − Residual) ÷ Life
WDV rate	1 − (Residual/Cost)^(1/Life)
IT depreciation	Block WDV; ½ rate if used <180 days
TDS sections	194C (contract 1/2%), 194J (prof 10%), 194Q (purchase 0.1%)
Blocked ITC	Motor vehicles, personal, works contract
Excel core	`XLOOKUP`, `SUMIFS`, `IFS`, `DATEDIF`, PivotTable

Master these three sub-ledgers and their reconciliations, and you can walk into any AP/AR/R2R desk in an Indian GCC or SME and be productive in a week.

The accounting/controllership interview & practical test

What it is & where it's used

Every accounts, finance, tax, audit and controllership role in India ends with two filters: a **structured interview** (technical + situational) and a **practical test** — a timed hands-on assessment where you actually pass entries, build a reconciliation, or "close the books" on a mock trial balance. This is standard at Big 4 (Deloitte, PwC, EY, KPMG), captive GCCs (JPMorgan, Amazon, Genpact, Accenture), mid-market CFO teams, and startups hiring an R2R (Record-to-Report), AP/AR, or GL accountant.

The interview tests whether you *understand* debit-credit logic, accruals, and month-end. The practical test verifies you can *execute* it in Excel and Tally/SAP under time pressure. Roles that hit this: GL Accountant, R2R Analyst, AP/AR Executive, Financial Analyst, Assistant Manager–Finance, Tax Associate, and Audit Associate.

The gap: why companies want this (and college didn't teach it)

An MBA or CA-Inter teaches you *why* a provision is created and how AS/Ind AS classifies it. It rarely makes you sit down and **book 40 entries against a bank statement in 45 minutes** or explain why a suspense account won't clear. Colleges grade theory papers; employers pay for a clean, tied-out trial balance by Working Day 3 (WD+3).

The specific gaps this chapter closes:

- **Speed + accuracy on entries** — knowing the entry vs. keying it correctly with the right sign, ledger, and cost centre.
- **Reconciliation muscle** — bank recon, GSTR-2B vs. books, AP/AR sub-ledger vs. GL, intercompany.
- **Month-end sequencing** — you can't accrue before you cut off; you can't close before recons pass.
- **Explaining the "why"** — interviewers probe "*why did you debit that?*" College never made you defend an entry out loud.

What "proficient" looks like

A job-ready candidate can, unaided:

1. Pass any routine entry (accrual, prepaid, depreciation, provision, forex, TDS, GST) with correct Dr/Cr, narration, and cost centre.
2. Build a **bank reconciliation** from a statement + cashbook and explain every reconciling item.
3. Reconcile **GSTR-2B to purchase register** and quantify ITC to claim/hold.
4. Take an **unadjusted trial balance → post adjusting entries → produce an adjusted TB → P&L and Balance Sheet** that balances.
5. Explain the **month-end close calendar** (WD-2 to WD+5) and what happens each day.
6. Answer "*accruals vs. provisions,*" "*deferred revenue,*" "*why does depreciation hit P&L but not cash*" crisply.

Hands-on: how to actually do it

The entries they will ask you to pass

Scenario	Journal Entry (Dr / Cr)
Accrue Dec electricity bill Rs.50,000 not yet received	Dr Electricity Expense 50,000 / Cr Outstanding Expenses (Accrued Liab) 50,000
Prepaid insurance Rs.1,20,000 for 12 months, book monthly	On payment: Dr Prepaid Insurance 1,20,000 / Cr Bank 1,20,000. Monthly: Dr Insurance Exp 10,000 / Cr Prepaid Insurance 10,000
Depreciation on machinery (WDV) Rs.80,000	Dr Depreciation 80,000 / Cr Accumulated Depreciation 80,000
Provision for doubtful debts Rs.25,000	Dr Bad Debt Provision (P&L) 25,000 / Cr Provision for Doubtful Debts 25,000
Salary Rs.5,00,000, TDS u/s 192 Rs.40,000, PF Rs.30,000	Dr Salaries 5,00,000 / Cr TDS Payable 40,000 / Cr PF Payable 30,000 / Cr Salary Payable 4,30,000
Purchase Rs.1,00,000 + 18% GST from registered vendor	Dr Purchases 1,00,000 / Dr Input CGST 9,000 / Dr Input SGST 9,000 / Cr Vendor 1,18,000
Vendor invoice, TDS 194J @10% on Rs.1,00,000	Dr Professional Fees 1,00,000 / Cr TDS Payable 10,000 / Cr Vendor 90,000
Deferred revenue: received Rs.1,20,000 annual SaaS upfront	On receipt: Dr Bank 1,20,000 / Cr Deferred Revenue 1,20,000. Monthly: Dr Deferred Revenue 10,000 / Cr Revenue 10,000
Forex payable USD 10,000 @ 83 booked, settled @ 85	Dr Import Purchase 8,30,000 / Cr Vendor 8,30,000. On settlement: Dr Vendor 8,30,000 / Dr Forex Loss 20,000 / Cr Bank 8,50,000

Golden rule to recite: **Debit what comes in / expenses & assets; Credit what goes out / income & liabilities.** Modern form: *assets & expenses increase on debit; liabilities, equity & income increase on credit.*

Bank reconciliation in Excel

Match cashbook to bank statement. Flag unmatched items:

```
=IF(COUNTIFS(Bank[Amt],[@Amt],Bank[Date],[@Date])>0,"Matched","Open")
```

Reconciliation math:

```
Balance as per Cashbook               1,00,000
Add: Cheques issued not yet presented   +15,000
Less: Cheques deposited not yet cleared  -8,000
Less: Bank charges not in books          -1,200
Add: Interest credited not in books      +2,000
= Balance as per Bank Statement        1,07,800
```

GSTR-2B vs. purchase register (ITC reconciliation)

```
Books ITC per invoice    :  =SUMIFS(PR[GST],PR[GSTIN],[@GSTIN],PR[Inv],[@Inv])
2B ITC per invoice       :  =SUMIFS(GSTR2B[GST],GSTR2B[GSTIN],[@GSTIN],GSTR2B[Inv],[@Inv])
Status                   :  =IF(ROUND(Books-2B,0)=0,"OK",IF(Books>2B,"Not in 2B - HOLD ITC","Ve
```

Only ITC appearing in 2B is claimable — books-only invoices get held.

Post adjusting entries to a trial balance with SQL

```
-- Adjusted trial balance from a journal_lines table
SELECT a.account_name,
       SUM(jl.debit)  AS total_dr,
       SUM(jl.credit) AS total_cr,
       SUM(jl.debit - jl.credit) AS net
FROM journal_lines jl
JOIN accounts a ON a.id = jl.account_id
WHERE jl.period = '2026-03'
GROUP BY a.account_name
ORDER BY a.account_name;

-- Sanity check: total debits MUST equal total credits
SELECT SUM(debit) AS dr, SUM(credit) AS cr,
       SUM(debit) - SUM(credit) AS diff
FROM journal_lines WHERE period = '2026-03';   -- diff must be 0
```

Tally / GST portal click-paths

- **Pass a journal in TallyPrime:** Gateway of Tally → Vouchers → press **F7** (Journal) → Dr ledger → amount → Cr ledger → amount → narration → **Ctrl+A** to save.
- **Bank recon in Tally:** Banking → Bank Reconciliation → select bank ledger → enter "Bank Date" against each voucher → unreconciled balance shows at bottom.
- **Download GSTR-2B:** gst.gov.in → login → Returns Dashboard → select period → **GSTR-2B** → Download (Excel).

Worked example / mini-project — "Close these books" (Rs.)

You get this unadjusted trial balance for **Meghna Traders Pvt Ltd, March 2026**:

Ledger	Dr	Cr
Cash & Bank	4,50,000	
Debtors	6,00,000	
Machinery	10,00,000	
Purchases	25,00,000	
Salaries	8,00,000	
Sales		42,00,000
Creditors		5,50,000
Share Capital		8,00,000
Rent	1,50,000	
Total	**55,50,000**	**55,50,000**

Adjustments to book (WD+2):

1. Depreciation @10% on machinery → Dr Depreciation 1,00,000 / Cr Accum. Dep 1,00,000
2. Salary for March Rs.72,000 unpaid → Dr Salaries 72,000 / Cr Salary Payable 72,000
3. Rent Rs.30,000 prepaid (Apr) → Dr Prepaid Rent 30,000 / Cr Rent 30,000
4. Provision for doubtful debts @2% of debtors → Dr Bad Debt Prov 12,000 / Cr Provision 12,000

Adjusted P&L:

Item	Rs.
Sales	42,00,000
Less: Purchases	(25,00,000)
Less: Salaries (8,00,000+72,000)	(8,72,000)
Less: Rent (1,50,000−30,000)	(1,20,000)
Less: Depreciation	(1,00,000)
Less: Bad debt provision	(12,000)
Net Profit	**5,96,000**

Balance Sheet ties: Equity 8,00,000 + Profit 5,96,000 = 13,96,000; Liabilities = Creditors 5,50,000 + Salary Payable 72,000 + Provision 12,000 + Accum Dep 1,00,000. Assets = Bank 4,50,000 + Debtors 6,00,000 + Machinery 10,00,000 + Prepaid Rent 30,000. **Both sides = 20,80,000.** Books closed.

How it's tested

Interview questions (rapid-fire): - Difference between accrual and provision? (Accrual = known amount, incurred; provision = estimated liability.) - Deferred revenue — asset or liability, and why? (Liability — you owe service.) - Why does depreciation reduce profit but not cash? (Non-cash allocation of past capex.) - Three golden rules / modern accounting equation. - What is a contra entry? A suspense account? Why would a TB not tie? - Walk me through your month-end close. - Prepaid vs. outstanding expense — where does each sit on the balance sheet?

Practical assessments companies actually give: - **Timed Excel test (30–45 min):** given a bank statement + cashbook, build the BRR; or given raw data, use SUMIFS/VLOOKUP/pivot to produce a summary. - **"Pass these entries":** 8–12 scenarios on paper or in Tally, graded on Dr/Cr, amount, and narration. - **"Close these books" case** (like the worked example above): unadjusted TB + adjustments → adjusted TB + financials. - **Recon screen:** GSTR-2B vs. purchase register or AP sub-ledger vs. GL; quantify and explain the gap. - **SQL/BI screen** (GCC/analyst roles): write a GROUP BY to produce a TB or aging.

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Reversing Dr/Cr on accruals/provisions	Say the entry aloud: "expense up = debit; liability up = credit."
Forgetting the reversal in next period	Book accruals as auto-reversing (JV type in ERP).
Claiming full ITC ignoring 2B	Claim only 2B-matched ITC; park the rest in "ITC on hold."
TB doesn't tie, chasing randomly	Diff ÷ 2 = a wrong-sign posting; diff ÷ 9 = transposition.
Netting TDS wrong on vendor payment	TDS credited to *payable*, not netted into expense.
No narration / no cost centre	Every entry needs a narration + dimension; auditors reject blanks.
Closing before recons pass	Recons are a *gate*, not a *report* — fix breaks first.

Learn-it roadmap & resources

Phase	Time	Focus
1	1 week	Drill 50 journal entries until reflexive; recite golden rules.
2	1 week	Bank recon + GST recon in Excel (SUMIFS, XLOOKUP, pivots).
3	1 week	Full close cycle: unadjusted TB → adjustments → financials.
4	Ongoing	Mock interviews; explain every entry out loud.

Resources: ICAI Accounting/Cost study material (free); TallyPrime education mode (free); GST portal help section; Zoho Books / QuickBooks free trials for close practice; ExcelJet for lookup/SUMIFS; NISM & ICAI for credibility. Certifications that signal readiness: **CA Inter**, **Tally Certification**, Microsoft **Excel Associate (MO-201)**, and any GCC-run R2R bootcamp.

Time to practical proficiency: **3–4 focused weeks** if you already know the theory — the bottleneck is speed, not concepts.

Quick-reference

Item	Rule / Formula
Modern rule	Assets & Expenses ↑ = Dr; Liab, Equity & Income ↑ = Cr
Accrual entry	Dr Expense / Cr Outstanding Liability
Prepaid entry	Dr Prepaid Asset / Cr Bank; then Dr Expense / Cr Prepaid
Deferred revenue	Dr Bank / Cr Deferred Rev; then Dr Deferred Rev / Cr Revenue
Depreciation	Dr Depreciation / Cr Accumulated Depreciation
GST purchase	Dr Purchase + Dr Input CGST + Dr Input SGST / Cr Vendor
TDS on payment	Dr Expense / Cr TDS Payable / Cr Vendor (net)
BRR direction	Cashbook + unpresented cheques − uncleared deposits ± bank items = Bank
TB won't tie	Diff÷2 = sign error; Diff÷9 = transposition
ITC test	Claim only GSTR-2B-matched invoices
Close calendar	WD-2 cutoff → WD+1 accruals → WD+2 recons → WD+3 TB → WD+5 report
Tally journal	F7 → Dr / Cr → narration → Ctrl+A
Adjusted TB (SQL)	`SELECT account, SUM(debit-credit) GROUP BY account` (diff must be 0)

Cheat-sheet: entries, close checklist & shortcuts

What it is & where it's used

This chapter is the laminated card you tape to your monitor. It collects the journal entries you post 50 times a month, the month-end close checklist that stops you forgetting the depreciation run, and the keyboard shortcuts that separate a 4-hour close from a 40-minute one in TallyPrime and SAP.

Every accounts, tax, and FP&A role touches this:

Role	Uses this for
Accounts Payable/Receivable exec	Repeatable vendor/customer entries, TDS, GST
GL / R2R accountant	Month-end close, accruals, provisions, reconciliations
Tax executive	GST output/input, RCM, TDS entries
FP&A analyst	Reading the TB fast, month-over-month checks
Startup finance / founder's office	Owning the whole close solo in Tally/Zoho

The value is speed with accuracy. Nobody promotes the person who *knows* the accrual entry; they promote the one who posts all 40 of them correctly before the 3rd working day.

The gap: why companies want this (and college didn't teach it)

College teaches you the *logic* of double-entry — "debit the receiver." It never makes you post 200 entries under a deadline, never shows you a real month-end close calendar, and never times you on Tally. The industry gap is muscle memory plus process:

- **Volume & repetition:** Real jobs are the same 25 entry-types, endlessly. You must post them without thinking.
- **The close is a *process*, not an entry:** Books don't just "close" — someone runs a 30-line checklist in a fixed order (accruals before depreciation before tax before TB freeze).
- **Tool fluency:** An MBA never opens Tally. Employers assume you can do `Alt+G > Voucher`, `Ctrl+A` to save, and F-key navigation blind.
- **Statutory muscle:** TDS sections, GST heads (CGST/SGST/IGST), RCM — these are daily entries, not exam theory.

Close this gap and you are useful on day one instead of day ninety.

What "proficient" looks like

A job-ready person can, unaided:

- Post any of the 25 common entries correctly with the right TDS section / GST head.
- Run a full month-end close against a checklist and produce a clean trial balance.
- Navigate TallyPrime entirely by keyboard, and do basic SAP FI transactions (FB50, F-02, FBL3N).
- Explain *why* an accrual reverses next month and *why* prepaid sits as an asset.

- Spot a suspense-account balance or a mismatched control account and fix it before the manager sees it.

Hands-on: how to actually do it

1. The 25 entries you post constantly

#	Transaction	Dr	Cr
1	Credit purchase of goods	Purchases + Input CGST + Input SGST	Creditor (Vendor)
2	Credit sale of goods	Debtor (Customer)	Sales + Output CGST + Output SGST
3	Interstate sale	Debtor	Sales + Output IGST
4	Payment to vendor	Creditor	Bank
5	Receipt from customer	Bank	Debtor
6	Professional fee bill (TDS 194J @10%)	Professional Fees	Vendor; TDS Payable (194J)
7	Rent paid (TDS 194I @10%)	Rent	Bank; TDS Payable (194I)
8	Contractor bill (TDS 194C @2%)	Contract Exp	Vendor; TDS Payable (194C)
9	Salary paid	Salaries	Bank; TDS Payable (192); PF Payable; ESI Payable
10	Depreciation	Depreciation	Accumulated Depreciation
11	Expense accrual (month-end)	Expense	Accrued/Outstanding Expenses
12	Reversal of accrual (next month)	Accrued Expenses	Expense
13	Prepaid at payment	Prepaid Expense	Bank
14	Prepaid amortisation	Expense	Prepaid Expense
15	Provision for doubtful debts	Bad Debt Expense	Provision for Doubtful Debts
16	Bank charges	Bank Charges	Bank
17	Interest accrued on FD	Accrued Interest	Interest Income
18	RCM on freight/GTA	Input IGST (RCM)	RCM Payable
19	GST payment (setoff)	Output CGST/SGST/IGST	Input GST + Electronic Cash Ledger
20	TDS deposit	TDS Payable	Bank
21	Purchase of fixed asset	Fixed Asset + Input GST	Vendor/Bank
22	Cash withdrawal (contra)	Cash	Bank
23	Owner capital introduced	Bank	Capital
24	Forex gain/loss on settlement	Debtor/Bank (loss: Forex Loss)	Forex Gain / Debtor
25	Year-end P&L transfer	Sales/Income	Trading & P&L

2. TallyPrime keyboard shortcuts (v3.x)

```
Alt+G      Go To (jump to any report/voucher)
F1         Help / switch company
Ctrl+A     Accept & save the current screen (the money shortcut)
Ctrl+Q     Quit without saving
F4         Contra        F5  Payment      F6  Receipt
F7         Journal       F8  Sales        F9  Purchase
Alt+F7     Stock Journal
Ctrl+F9    Debit Note    Ctrl+F8  Credit Note
Alt+C      Create master on the fly (ledger/item)
Alt+D      Delete voucher / line
Alt+X      Cancel voucher
Alt+2      Duplicate a voucher (from Day Book)
Ctrl+Enter Drill into / edit a master mid-entry
Alt+F1     Detailed view in reports
F2         Change period      Alt+F2  Change period range
F12        Configure current screen
D (Gateway)Day Book
```

3. SAP FI T-codes you must know

```
FB50    Enter G/L account document (fast entry)
F-02    General posting (with posting keys 40 Dr / 50 Cr)
FB60    Vendor invoice        FB70   Customer invoice
F-53    Outgoing payment      F-28   Incoming payment
FBL1N   Vendor line items     FBL5N  Customer line items
FBL3N   G/L account line items
FS10N   G/L balance display
F.01    Financial statements   S_ALR_87012277  GL balances report
FB08    Reverse a document     F-44   Vendor clearing
```

4. The month-end close checklist (order matters)

```
[ ] 1.  Cut-off: all invoices/bills for the month entered
[ ] 2.  Bank reconciliation — every account tied to statement
[ ] 3.  Cash count vs cash ledger
[ ] 4.  AR ageing reviewed; AP ageing reviewed
[ ] 5.  Accruals for unbilled expenses posted
[ ] 6.  Prepaid amortisation for the month posted
[ ] 7.  Depreciation run
[ ] 8.  Inventory valuation / stock reconciliation
[ ] 9.  Inter-company & control account reconciliation
[ ] 10. Provisions (doubtful debts, bonus, gratuity) reviewed
[ ] 11. GST: GSTR-2B vs books input reco; output liability tied out
[ ] 12. TDS: deducted vs payable reconciled, challan ready
[ ] 13. Payroll reconciled (salary, PF, ESI, PT)
[ ] 14. Suspense account = ZERO
[ ] 15. Trial balance drawn; P&L and BS reviewed vs last month
[ ] 16. Variance/flux commentary for FP&A
[ ] 17. Lock/freeze the period
```

Worked example / mini-project

Reproduce a mini month-end for **Nova Traders Pvt Ltd** (Karnataka), month of June.

Transactions during June:

1. Credit sale to a Karnataka customer, ₹1,00,000 + 18% GST.

```
Dr  Debtors — Sharma & Co        1,18,000
    Cr  Sales                        1,00,000
    Cr  Output CGST (9%)                 9,000
    Cr  Output SGST (9%)                 9,000
```

2. Professional fee bill from an auditor ₹50,000, TDS 194J @10%.

```
Dr  Professional Fees              50,000
    Cr  ABC & Associates (net)          45,000
    Cr  TDS Payable — 194J               5,000
```

Month-end adjustments (close checklist steps 5–7):

3. Electricity bill for June not yet received, estimated ₹12,000 (accrual):

```
Dr  Electricity Expense            12,000
    Cr  Outstanding Expenses            12,000
```

4. Insurance ₹24,000 paid 1-June for the year → ₹2,000/month amortisation:

```
Dr  Insurance Expense              2,000
    Cr  Prepaid Insurance              2,000
```

5. Depreciation on equipment (WDV ₹6,00,000 @ 15% p.a. = ₹7,500/month):

```
Dr  Depreciation                   7,500
    Cr  Accumulated Depreciation       7,500
```

GST tie-out (step 11): Output = ₹18,000. Suppose Input from GSTR-2B = ₹6,300. Net payable in cash = ₹11,700.

```
Dr  Output CGST                    9,000
Dr  Output SGST                    9,000
    Cr  Input CGST/SGST                 6,300
    Cr  Electronic Cash Ledger         11,700
```

Quick TB sanity check in Excel — paste ledger balances, then verify it balances:

```
=IF(ROUND(SUM(Debit)-SUM(Credit),2)=0,"BALANCED","OUT BY "&SUM(Debit)-SUM(Credit))
```

Month-over-month flux for FP&A commentary:

```
=IFERROR((Jun-May)/May, "n/a")   ' format as %; flag any line moving >10%
```

In Tally: `Alt+G > Trial Balance > Alt+F1` (detailed) confirms suspense = 0 and totals match.

How it's tested

Interview questions:

- "Pass the entry for a professional fee bill of ₹50,000 with TDS." (they want section + net vendor amount)
- "What reverses at the start of next month and why?" (accruals)
- "Walk me through your month-end close, in order."
- "CGST vs IGST — when do you charge which?"
- "Where does a prepaid expense sit and why is it an asset?"

Practical assessments companies actually give:

- **Timed Tally test:** "Here are 15 vouchers — enter them in 20 minutes and produce the TB." Graded on speed, correct heads, and a zero suspense.
- **"Close these books" case:** a messy trial balance with missing accruals/depreciation; you must clean it and explain adjustments.
- **Excel screen:** bank reconciliation from two datasets, or a TB that won't balance — find the plug.

- **SAP navigation:** "Post a vendor invoice in FB60 and display it in FBL1N."

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Forgetting to reverse last month's accrual → double-count	Post accrual + reversal as a *pair* immediately; or use auto-reversing journals
Wrong GST head (CGST/SGST vs IGST)	Rule: same state = C+S, different state = I. Check place of supply, not billing address
Ignoring suspense account balance	Suspense must be ZERO before freezing — it's the first thing a reviewer checks
TDS on gross vs net confusion	TDS is on the base amount; vendor gets base − TDS; deposit TDS by the 7th
Closing before bank reco	Bank reco is step 2, not optional. Unreconciled = books not closed
Editing a posted voucher instead of reversing	In SAP use FB08; keep an audit trail — never silently overwrite
Skipping cut-off	An invoice dated 30-June entered in July inflates next month. Enforce a hard cut-off date

Learn-it roadmap & resources

Time to proficiency: 6–10 weeks of daily practice to be job-ready on entries + close; Tally fluency in ~3 weeks of real use.

Week	Focus
1–2	Master the 25 entries cold; drill TDS sections & GST heads
3–4	TallyPrime end-to-end: masters, vouchers, GST, reports — by keyboard only
5–6	Run a full mock month-end close against the checklist
7–8	Bank reco + GSTR-2B reco in Excel; TB troubleshooting
9–10	SAP FI basics (if targeting large corporates)

Resources:

- **Tally**: TallyPrime free educational version + Tally Education's official "TallyEssential" certification (paid, recognised in India).
- **GST/TDS**: CBIC and Income-tax portal help sections; ICAI study material (free, you already have it).
- **SAP FI**: free openSAP courses; any SAP FICO end-user PDF for T-code drills.
- **Excel/recons**: practise with a downloaded bank statement + ledger dump.
- **Certifications worth it in India:** TallyEssential/TallyProfessional; add SAP FICO only if targeting MNC ERP roles.

Quick-reference

GST rule: same state → CGST + SGST; different state → IGST. Setoff order: IGST first, then CGST/SGST.

Statutory	Rate / Section	Due
TDS Professional	194J @ 10%	7th next month
TDS Rent (land/bldg)	194I @ 10%	7th
TDS Contractor	194C @ 1%/2%	7th
TDS Salary	192 (slab)	7th
GST return	GSTR-1 / 3B	11th / 20th
PF / ESI	—	15th

Tally survival keys: `Alt+G` go-to · `Ctrl+A` save · `F5` payment · `F6` receipt · `F7` journal · `F8` sales · `F9` purchase · `Alt+C` create master · `Alt+2` duplicate.

SAP survival codes: `FB50` GL post · `FB60` vendor invoice · `FBL3N` GL line items · `FS10N` balance · `FB08` reverse.

Close order: cut-off → bank reco → AR/AP → accruals → prepaids → depreciation → inventory → provisions → GST/TDS/payroll → suspense=0 → TB → freeze.

Excel balance check: `=IF(ROUND(SUM(Debit)-SUM(Credit),2)=0,"BALANCED","OUT")`

Practical Taxation & Compliance

Practical, Hands-On, Job-Ready — India-first + Global

Real formulas, queries, entries & tools — closing the gap between what an MBA teaches and what finance employers pay for

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Tax on the job vs textbooks

What it is & where it's used

A tax/compliance role is not "knowing the Income-tax Act." It is running a **recurring machine** that takes raw business transactions, classifies them correctly, computes the right tax, files the right return before a legal deadline, pays the money, and keeps proof — every single month, forever. The Act is the rulebook; the job is operating the assembly line.

Where this sits in real org charts:

Role	What tax/compliance work looks like day-to-day
Accounts Executive (SME / startup)	Files GSTR-1 and GSTR-3B, deposits TDS, reconciles GSTR-2B vs purchase register in Excel
Tax Analyst (mid/large corp)	Owns a compliance calendar, prepares TDS returns (26Q/24Q), advance-tax workings, litigation trackers
Big-4 / consulting associate	Same mechanics but across many clients; automates recons in Excel/Power Query
AP/AR & Controller-track	Ensures every vendor invoice has correct GST + TDS *before* payment, so books close clean
Global finance (US/EU/APAC)	Sales-tax/VAT filings, withholding, statutory reporting — different rates, identical *rhythm*

The common thread across geographies: **deadlines are the product.** A brilliant computation filed one day late is a failure; a mediocre computation filed on time and later revised is normal.

The gap: why companies want this (and college didn't teach it)

Your MBA and CA-Inter taught you **how tax is computed** — slab rates, Section 40(a)(ia) disallowance, input tax credit eligibility, the theory of "supply." That knowledge is necessary and assumed. Nobody pays you for it, because it's in every textbook and now in every LLM.

What college did **not** teach, and what the company is actually buying:

- **The calendar mindset.** Textbooks are organized by *concept* (chapter on TDS, chapter on GST). Work is organized by *date* (7th = TDS deposit, 11th = GSTR-1, 20th = GSTR-3B). The unit of work is a deadline, not a topic.
- **Reconciliation, not computation.** 80% of the job is matching two lists that should agree but don't — books vs GST portal, TDS deducted vs Form 26AS, vendor invoice vs GSTR-2B. Textbooks never show two lists.
- **Documentation & audit trail.** The answer isn't "₹1,80,000 TDS." It's "₹1,80,000, here is the challan CIN, here is the working, here is who approved it."
- **Tooling.** Real tax runs on Excel, TallyPrime/SAP, the GST and TRACES portals, and increasingly Power Query/SQL — none of which appear in a syllabus.

Employers see thousands of candidates who can quote a section and zero who can hand back a clean reconciliation by 5 PM. That scarcity is your paycheck.

What "proficient" looks like

The concrete bar. A job-ready person can, **unaided**:

1. Take a month's sales and purchase data in Excel and produce a GSTR-3B summary (output tax, eligible ITC, net payable) with formulas, not by hand.
2. Reconcile the purchase register against GSTR-2B and produce a list of mismatched invoices with the *reason* (missing, wrong GSTIN, wrong amount, wrong period).
3. Compute TDS on a batch of vendor invoices, pick the correct section (194C vs 194J vs 194Q), and generate the deposit figure per section.
4. Read a compliance calendar and, on any given day, say what is due and whether it's done.
5. Pass a payment for release only after confirming GST is claimable and TDS is deducted — i.e., protect the company from disallowance.

If you can do these five, you clear the practical test for almost any India accounts/tax role.

Hands-on: how to actually do it

1. Build the compliance calendar (the core artifact)

This is the single most job-defining table. Memorize its shape.

Due date	Compliance	Form	Period	Applies to
7th	TDS/TCS deposit	Challan ITNS-281	Prev. month	All deductors (7th, but 30 Apr for March)
11th	GST outward supplies	GSTR-1	Prev. month	Monthly filers
13th	GST outward (QRMP)	IFF	Prev. month	Quarterly filers
20th	GST summary + payment	GSTR-3B	Prev. month	Monthly filers
25th	GST payment (QRMP)	PMT-06	Prev. month	Quarterly filers
15th (Jun/Sep/Dec/Mar)	Advance tax	Challan 280	Quarter	Companies & taxpayers
31 Jul (Q1)/31 Oct/31 Jan/31 May	TDS return	24Q/26Q	Quarter	Deductors

Excel formula to flag what's due and overdue against today's date:

```
=IF([@Done]="Yes","✓ Done",
   IF([@DueDate]<TODAY(),"⚠ OVERDUE",
     IF([@DueDate]≤TODAY()+3,"DUE SOON","")))
```

Days remaining, with weekend awareness:

```
=NETWORKDAYS(TODAY(),[@DueDate])-1
```

2. TDS section picker (stop guessing)

```
=XLOOKUP([@Nature], Sections[Nature], Sections[Section], "REVIEW")
```

TDS amount with threshold logic (194J professional, threshold ₹30,000, rate 10%):

```
=IF([@YTDPayment]+[@Invoice] > 30000, ROUND([@Invoice]*10%, 0), 0)
```

3. GST output tax and net payable

```
Output tax   =SUMIFS(Sales[TaxAmt], Sales[Type],"Taxable")
Eligible ITC =SUMIFS(Purch[TaxAmt], Purch[ITCEligible],"Yes")
Net payable  =MAX(OutputTax-EligibleITC, 0)
```

4. GSTR-2B reconciliation (the interview favourite) — Power Query merge

Load `PurchaseRegister` and `GSTR2B` as tables, then in Power Query:

```
= Table.NestedJoin(PurchaseRegister, {"GSTIN","InvNo"},
                   GSTR2B, {"GSTIN","InvNo"},
                   "Match", JoinKind.LeftOuter)
```

Expand, then classify:

```
=IF([@Match]="","Not in 2B - hold ITC",
   IF(ROUND([@BookTax],0)<>ROUND([@2BTax],0),"Value mismatch","Matched"))
```

5. The journal entries you'll actually pass

Vendor bill with GST input and TDS deducted (professional fee ₹1,00,000 + 18% GST, TDS @10%):

Account	Dr (₹)	Cr (₹)
Professional Fees (expense)	1,00,000	
Input CGST	9,000	
Input SGST	9,000	
To Vendor A/c		1,08,000
To TDS Payable (194J)		10,000

On depositing TDS:

Account	Dr (₹)	Cr (₹)
TDS Payable (194J)	10,000	
To Bank		10,000

6. TallyPrime / portal click-paths

- **Record GST purchase:** Gateway of Tally → Vouchers → F9 (Purchase) → select party → item → GST ledgers auto-populate if configured.
- **File GSTR-1:** Gateway → Display More Reports → GST Reports → GSTR-1 → export JSON → gst.gov.in → Returns Dashboard → select period → GSTR-1 → upload JSON → Submit → File with DSC/EVC.
- **Deposit TDS:** incometax.gov.in → e-Pay Tax → select 281 → Company/Non-company, section code → pay → save challan CIN.

Worked example / mini-project

Scenario: You run compliance for *Nova Traders Pvt Ltd*, June 2026.

Sales (all intra-state, 18%):

Invoice	Taxable (₹)	GST (₹)
S-101	5,00,000	90,000
S-102	3,00,000	54,000
Total	**8,00,000**	**1,44,000**

Purchases:

Invoice	Taxable (₹)	GST (₹)	In GSTR-2B?
P-201 (goods)	4,00,000	72,000	Yes
P-202 (services)	1,00,000	18,000	No

Step 1 — ITC: Only invoices reflected in GSTR-2B are claimable (Rule 36(4)/Sec 16). P-202 is not in 2B → hold ₹18,000. Eligible ITC = **₹72,000**.

Step 2 — GSTR-3B net GST: Output 1,44,000 − ITC 72,000 = **₹72,000 payable** by 20 Jul via GSTR-3B; pay via PMT-06/cash ledger.

Step 3 — TDS: P-202 is a professional service, ₹1,00,000 > ₹30,000 threshold → 194J @10% = **₹10,000 TDS** to deposit by 7 Jul (Challan 281). Vendor is paid ₹1,08,000 − ₹10,000 = ₹98,000.

Step 4 — Calendar check on 5 Jul:

Due	Task	Status
7 Jul	Deposit TDS ₹10,000	DUE SOON
11 Jul	File GSTR-1 (₹8,00,000 sales)	Pending
20 Jul	GSTR-3B, pay ₹72,000	Pending

Reproduce this in one Excel sheet: a Sales tab, a Purchase tab with an "In2B" column, and a Calendar tab using the flag formula above. That workbook *is* a portfolio piece.

How it's tested

Interview questions (verbal): - "Walk me through the GST monthly compliance cycle and every due date." - "A vendor invoice is in your books but not in GSTR-2B. What do you do?" - "Difference between 194C, 194J and 194Q — and which rate?" - "What happens if you deduct TDS but pay the vendor without depositing it?" (Answer: 30% expense disallowance u/s 40(a)(ia) until deposited.)

Practical/assessment tests (what actually decides it): - **Timed Excel test (30–45 min):** Given a raw sales+purchase dump, produce GSTR-3B output tax, eligible ITC, and net payable using formulas. They watch whether you use `SUMIFS` / `XLOOKUP` or manually add — manual = reject. - **The reconciliation case:** Two files (purchase register + GSTR-2B) that don't tie. Deliver a mismatch list with reasons. This is the single most common practical screen. - **"Close the month" case:** A trial balance and a stack of pending items; book the TDS/GST provisions and hand back a clean set.

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Claiming ITC not in GSTR-2B	Reconcile 2B **before** filing 3B; hold unmatched ITC, don't claim on faith
Paying vendor before deducting TDS	Gate: no payment released until TDS field is filled — protects against 40(a)(ia)
Treating tax as a computation, not a calendar	Maintain a live calendar; the deadline is the deliverable
No documentation	Save every challan CIN, ARN, and working; "trust me" fails an audit
Hard-coding rates in cells	Keep a rate/section master table and `XLOOKUP` into it — one place to update
Manual copy-paste recon	Use Power Query merge; it's repeatable and leaves an audit trail
Missing the March exception	TDS for March is due **30 April**, not 7 April

Learn-it roadmap & resources

Realistic time-to-proficiency: 8–12 weeks of deliberate practice alongside study.

- **Weeks 1–2:** Build the compliance calendar from scratch in Excel. Learn `SUMIFS`, `XLOOKUP`, `NETWORKDAYS`, `IF`.
- **Weeks 3–4:** GST mechanics hands-on — free-play in the GST portal sandbox, file dummy GSTR-1/3B. Read CBIC's GST flyers.
- **Weeks 5–6:** TDS — sections, thresholds, TRACES portal, Form 26AS reading. Deposit a test challan on the income-tax portal.
- **Weeks 7–8:** Power Query reconciliation; automate the 2B match.
- **Weeks 9–12:** Do the mini-project monthly with fresh numbers; add TallyPrime.

Resources: - Free: CBIC GST portal help, income-tax e-filing portal guides, ClearTax/TaxGuru blogs, ICAI study material (you already have it). - Paid/high-ROI: TallyPrime with GST certificate; ClearTax or Zoho Books trial for hands-on filing; Microsoft Excel/Power Query courses. - Certifications that signal job-readiness: **TallyPrime GST**, ICAI's practical GST/TDS courses, and your CA-Inter itself.

Quick-reference

Due dates (monthly filer): TDS deposit 7th · GSTR-1 11th · GSTR-3B 20th · TDS return end of month after quarter · Advance tax 15 Jun/Sep/Dec/Mar.

Key TDS sections:

Section	Nature	Rate	Threshold (₹)
194C	Contractor	1% (ind/HUF) / 2%	30,000 single / 1,00,000 aggregate
194J	Professional/technical	10% (2% technical)	30,000
194Q	Purchase of goods	0.1%	50,00,000
194H	Commission/brokerage	5%	15,000

Core formulas:

```
=XLOOKUP(nature, Sections[Nature], Sections[Section], "REVIEW")
=SUMIFS(Sales[Tax], Sales[Type],"Taxable")
=MAX(OutputTax-EligibleITC,0)
=IF([@Done]="Yes","✓",IF([@Due]<TODAY(),"OVERDUE",""))
```

Golden rules: ITC only if in GSTR-2B · Deduct TDS before paying vendor · The deadline is the deliverable · Keep the challan/ARN as proof.

GST in Practice I: Registration, Invoicing, E-Invoice & E-Way Bill

What it is & where it's used

Every business in India with taxable turnover above the threshold lives inside the GST machine: it registers, raises tax invoices, pushes those invoices to the government's Invoice Registration Portal (IRP) to get an IRN and QR code, and generates an e-way bill before goods move. These four steps are the daily plumbing of the finance-tax function. Get one wrong and the buyer loses input tax credit (ITC), the truck gets detained, or a notice lands.

Roles that do this hands-on:

- **Accounts/AP-AR executives** — raise sales invoices, book purchases, match ITC.
- **GST/Indirect-tax analysts** — registration amendments, e-invoicing exceptions, reconciliations.
- **Finance controllers & CA-firm article/audit staff** — sign off that the invoice-to-e-way-bill chain is compliant.
- **ERP/Tally implementers** — configure the IRP/e-way-bill API integration.

If you can register an entity, cut a compliant tax invoice, generate the IRN, and produce the e-way bill without asking anyone — you are doing what a company actually pays a tax executive for.

The gap: why companies want this (and college didn't teach it)

An MBA or CA-Inter syllabus teaches you *what* Section 22 says about registration thresholds and *what* Rule 46 lists as mandatory invoice fields. It does **not** put you on the GST portal at `gst.gov.in`, hand you a login, and say "register this Private Limited company by end of day." The gap is procedural muscle memory:

College teaches	Job needs
"Threshold is ₹40L/₹20L"	Which state, goods vs services, which ARN to track
"Invoice must have GSTIN, HSN, tax rate"	The exact 16-char invoice numbering rule, place-of-supply logic for IGST vs CGST+SGST
"E-invoicing is mandatory over a limit"	The ₹5 crore AATO trigger, generating IRN in Tally/IRP, handling a rejected payload
"E-way bill for movement of goods"	₹50,000 rule, Part-A vs Part-B, validity by distance, extending an expired bill

Employers pay for the person who has *clicked the buttons*, seen the error `2172: Duplicate IRN`, and knows the fix. That is unteachable from a textbook.

What "proficient" looks like

A job-ready person can, unaided:

1. **Register** a new GSTIN end-to-end (TRN → Part-B → ARN → GSTIN) and read the 15-digit GSTIN structure.
2. Cut a **tax invoice** that survives an audit: correct series, HSN/SAC, place of supply, IGST vs CGST+SGST split, reverse-charge flag.

3. Generate an **e-invoice** (IRN + signed QR) from the ERP and reconcile IRNs against the GSTR-1 auto-population.
4. Generate and **manage an e-way bill**: Part-A + Part-B, pick the right transport mode, extend/cancel, and know the distance-to-validity table.
5. Know **who is exempt** from e-invoice (SEZ units as suppliers exempt, banks, transporters) and e-way bill (goods < ₹50k, non-motorised transport).

Hands-on: how to actually do it

A. GST Registration (portal click-path)

```
gst.gov.in → Services → Registration → New Registration
Part-A: Select "Taxpayer" → State → Legal Name (as per PAN) → PAN → email + mobile
        → OTP verification → TRN (Temporary Reference Number) generated
Part-B (login with TRN): 10 tabs
   Business Details → Promoters/Partners → Authorized Signatory
   → Principal Place of Business → Additional Places → Goods & Services (HSN/SAC)
   → Bank Accounts (can add post-registration) → State Specific → Verification
Submit with DSC (mandatory for Companies/LLP) or EVC/e-Sign
→ ARN generated → track at Services > Track Application Status
→ GSTIN + registration certificate (Form REG-06) in ~7 working days
```

Decode the GSTIN `29ABCDE1234F1Z5`:

Chars	Meaning
1-2	State code (29 = Karnataka)
3-12	PAN of entity
13	Entity number of same PAN in state
14	`Z` by default
15	Check-digit

Excel to validate GSTIN length + embedded PAN:

```
=IF(AND(LEN(A2)=15, MID(A2,3,10)=B2), "Valid structure", "Check GSTIN")
```

(A2 = GSTIN, B2 = PAN)

B. Tax invoice — mandatory contents (Rule 46)

A compliant B2B tax invoice must carry: supplier name/address/**GSTIN**; consecutive **invoice number** (≤16 chars, alphanumeric, unique per FY); date; recipient GSTIN + name; **place of supply** (for inter-state); **HSN/SAC**; description; qty; taxable value; **tax rate & amount split into CGST/SGST or IGST**; reverse-charge flag; signature/DSC.

IGST vs CGST+SGST decision (the logic that trips freshers):

```
' C2 = supplier state code, D2 = place-of-supply state code, E2 = taxable value, F2 = rate%
IGST  =IF(C2<>D2, E2*F2, 0)
CGST  =IF(C2=D2, E2*F2/2, 0)
SGST  =IF(C2=D2, E2*F2/2, 0)
```

Journal entry — outward B2B sale, ₹1,00,000 + 18% IGST (inter-state):

Account	Dr (₹)	Cr (₹)
Debtor / Customer A	1,18,000	
Sales		1,00,000
Output IGST		18,000

Intra-state (18% = 9% CGST + 9% SGST):

Account	Dr (₹)	Cr (₹)
Debtor	1,18,000	
Sales		1,00,000
Output CGST		9,000
Output SGST		9,000

C. E-invoicing (IRN + QR)

Applicable when **Aggregate Annual Turnover (AATO) > ₹5 crore** in any FY since 2017-18. You do **not** create the invoice on the IRP — your ERP generates the JSON, posts it to the IRP, and gets back the **IRN** (64-char hash) + **signed QR** + digitally signed invoice.

TallyPrime path:

```
Gateway of Tally → Alter → GST Details → set "e-Invoicing applicable: Yes"
Voucher entry (F8 Sales) → Save → prompt "Provide GST details?" → Yes
Exchange → Send for e-Invoicing → IRP login (via GSP/API) → IRN + QR fetched back
Print → invoice now shows IRN + QR
```

IRP portal (manual / bulk): `einvoice1.gst.gov.in` → Login → e-Invoice → Bulk Upload → upload JSON via offline tool → download IRN file.

Python to build a minimal e-invoice payload (concept for API integration):

```python
import requests, json

payload = {
    "Version": "1.1",
    "TranDtls": {"TaxSch": "GST", "SupTyp": "B2B", "RegRev": "N"},
    "DocDtls": {"Typ": "INV", "No": "INV/2026/0042", "Dt": "03/07/2026"},
    "SellerDtls": {"Gstin": "29ABCDE1234F1Z5", "LglNm": "Acme Pvt Ltd",
                   "Addr1": "MG Road", "Loc": "Bengaluru", "Pin": 560001, "Stcd": "29"},
    "BuyerDtls": {"Gstin": "27PQRS5678K1Z2", "LglNm": "Beta LLP", "Pos": "27",
                  "Addr1": "Andheri", "Loc": "Mumbai", "Pin": 400053, "Stcd": "27"},
    "ItemList": [{"SlNo": "1", "PrdDesc": "Steel bracket", "HsnCd": "7308",
                  "Qty": 100, "Unit": "NOS", "UnitPrice": 1000, "TotAmt": 100000,
                  "AssAmt": 100000, "GstRt": 18, "IgstAmt": 18000, "TotItemVal": 118000}],
    "ValDtls": {"AssVal": 100000, "IgstVal": 18000, "TotInvVal": 118000}
}

headers = {"Authorization": auth_token, "Content-Type": "application/json",
           "user_name": user, "Gstin": "29ABCDE1234F1Z5"}
r = requests.post("https://einv-apisandbox.nic.in/eivital/v1.04/invoice",
                  headers=headers, data=json.dumps(payload))
print(r.json()["Data"]["Irn"], r.json()["Data"]["SignedQRCode"])
```

D. E-way bill

Required when **consignment value > ₹50,000** and goods move (inter-state, or intra-state where the state mandates). Two parts:

- **Part-A**: GSTIN of supplier/recipient, invoice no/date, HSN, value, place-of-dispatch/delivery pincodes.
- **Part-B**: transporter ID, vehicle number / transport doc number.

Portal path `ewaybillgst.gov.in`:

```
Login → e-Way Bill → Generate New
Transaction Type: Outward → Sub-type: Supply
Doc type: Tax Invoice → No + Date
From (auto from GSTIN) → To (buyer GSTIN + pincode)
Item: HSN, qty, taxable value, tax rate → system computes value
Part-B: Mode = Road → Vehicle No (KA01AB1234) → Approx distance (km)
Submit → 12-digit EWB number + validity generated
```

Validity table (regular cargo):

Distance	Validity
Up to 200 km	1 day
Every additional 200 km (or part)	+1 day

Expiring bill → **Extend** under e-Way Bill > Extend Validity (only within 8 hrs before/after expiry).

Worked example / mini-project

Scenario: Acme Pvt Ltd (GSTIN 29..., Karnataka, AATO ₹12 cr) sells 100 steel brackets @ ₹1,000 to Beta LLP (Mumbai) on 03-Jul-2026. Distance 980 km by road, vehicle KA01AB1234.

Reproduce the full chain:

1. **Invoice value:** 100 × ₹1,000 = ₹1,00,000 taxable. Inter-state → **IGST 18% = ₹18,000**. Invoice total **₹1,18,000**. Invoice no `INV/2026/0042`.
2. **E-invoice:** AATO > ₹5 cr → mandatory. Post the JSON above to IRP → receive IRN + QR. The IRN auto-populates GSTR-1, so you do **not** re-key this invoice in GSTR-1.
3. **E-way bill:** value ₹1,18,000 > ₹50,000 → mandatory. Part-A from invoice; Part-B vehicle KA01AB1234. Distance 980 km → validity = ceil(980/200) = **5 days**.
4. **Books (Acme, seller):**

Account	Dr (₹)	Cr (₹)
Beta LLP (Debtor)	1,18,000	
Sales – Steel brackets		1,00,000
Output IGST		18,000

5. **Buyer (Beta LLP) claims ITC** only because Acme's IRN flowed to GSTR-1 → GSTR-2B. Beta's entry:

Account	Dr (₹)	Cr (₹)
Purchases	1,00,000	
Input IGST	18,000	
Acme Pvt Ltd (Creditor)		1,18,000

Reconciliation check in Excel — match IRN invoices to GSTR-2B:

```
=IF(COUNTIF(GSTR2B[InvNo], [@InvNo])>0, "Matched", "MISSING in 2B")
```

How it's tested

Interview questions: - "Turnover ₹6 cr — is e-invoicing mandatory? From when?" (Yes, > ₹5 cr.) - "Karnataka seller, Karnataka buyer — CGST+SGST or IGST?" (CGST+SGST; place of supply = Karnataka.) - "Consignment ₹45,000 — e-way bill needed?" (No, below ₹50k — unless state rule or handicraft exception.) - "Buyer says no ITC showing — where do you look?" (Did seller generate IRN / file GSTR-1 → check GSTR-2B.) - "Vehicle broke down mid-transit, bill expiring — what do you do?" (Extend validity; update Part-B with new vehicle.)

Practical assessment companies give: - A timed task: "Here's a PO — raise a compliant tax invoice in Tally/Excel with correct tax split." - "Register this dummy entity on the GST portal sandbox and screenshot the ARN." - "Given this invoice JSON that IRP rejected with error 2172 — fix it." (Duplicate IRN → invoice already registered; don't resubmit.) - A reconciliation screen: match a purchase register against a GSTR-2B extract and flag missing ITC.

Common mistakes & how pros avoid them

Mistake	Fix / pro habit
Invoice number > 16 chars or resets mid-year	Keep series ≤16 alphanumeric, unique per FY; document the series scheme
Charging IGST for intra-state (or vice versa)	Always derive from **place of supply**, not billing address
Generating e-way bill but skipping e-invoice (or vice versa)	Both are separate obligations; build both into the ERP save flow
Resubmitting a rejected IRN payload blindly	Read the error code — 2172 = duplicate, don't retry; 2150 = invalid GSTIN
Letting e-way bill expire in transit	Track validity; extend within the 8-hr window before/after expiry
Assuming ITC is automatic	ITC depends on **seller** filing GSTR-1; reconcile against GSTR-2B monthly
Wrong HSN digits (4 vs 6 vs 8)	AATO > ₹5 cr needs **6-digit HSN**; below, 4-digit for B2B

Learn-it roadmap & resources

Time to proficiency: ~3-4 weeks of hands-on practice alongside a real or dummy books set.

Week	Focus
1	Register a dummy entity on GST portal; decode GSTINs; Rule 46 invoice fields
2	Cut 20 invoices in TallyPrime with correct tax splits and journal entries
3	E-invoicing sandbox (`einv-apisandbox.nic.in`) — generate IRNs, handle rejects
4	E-way bills end-to-end; monthly GSTR-2B reconciliation in Excel

Resources: - GSTN official user manuals — `tutorial.gst.gov.in` (free, authoritative). - E-invoice & e-way bill sandbox portals for API practice. - ICAI GST background material (free for students). - TallyPrime GST self-learning modules; ClearTax / IRIS blogs for edge cases. - **Certification:** ICAI Certificate Course on GST; NACIN GST practitioner path; MSME GST practitioner courses.

Quick-reference

Item	Rule of thumb
Registration threshold	₹40L goods / ₹20L services (₹20L/₹10L special-category states)
GSTIN	15 char: 2 state + 10 PAN + 1 entity + Z + check-digit
E-invoice trigger	AATO > ₹5 crore in any FY since 2017-18
E-invoice output	64-char IRN + signed QR (from IRP, not manually created)
E-way bill trigger	Consignment value > ₹50,000
EWB validity	1 day / 200 km, +1 day per extra 200 km
Intra-state tax	CGST + SGST (equal split)
Inter-state tax	IGST (= CGST + SGST rate)
ITC source	GSTR-2B (depends on seller's GSTR-1)
Key portals	gst.gov.in, einvoice1.gst.gov.in, ewaybillgst.gov.in
Common error 2172	Duplicate IRN — invoice already registered, do not resubmit
Invoice number	≤16 chars, alphanumeric, unique per FY (Rule 46)
HSN digits	6-digit if AATO > ₹5 cr, else 4-digit (B2B)

GST in Practice II: Returns & ITC

What it is & where it's used

Every registered business in India files GST returns monthly or quarterly. Three documents do the heavy lifting:

- **GSTR-1** — the outward supply return. You declare every sales invoice, credit/debit note, and export. This is what *feeds your customers' ITC*.
- **GSTR-3B** — the summary self-assessment return. You compute output tax, claim eligible ITC, and *pay the net tax*. This is where money actually moves.
- **GSTR-2B** — an auto-drafted, static ITC statement generated by the portal on the 14th. It tells you the *maximum ITC you're legally allowed* to claim, based on what your suppliers filed.

The job is: file GSTR-1 accurately, reconcile purchase books against GSTR-2B, claim only eligible and matched ITC, set it off correctly (IGST → CGST → SGST rules), and pay the balance in cash.

Who does this: accounts executives, GST/indirect-tax analysts, AP/AR teams, tax consultants at CA firms, controllers, and anyone in a finance role at a company with a GSTIN. In interviews for accounts/tax roles in India, "walk me through 3B vs 2B reconciliation" is a near-guaranteed question.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you *what* GST is — dual tax, value-added chain, seamless credit. It never makes you sit with a purchase register and a 2B and find the ₹18,000 of ITC you can't claim because a vendor forgot to file. That gap is expensive: over-claimed ITC triggers interest at 18% p.a. under Section 50, plus penalties. Under-claimed ITC is working capital left on the table.

Companies pay for the person who can: 1. Turn a messy sales ledger into a valid GSTR-1 JSON without errors. 2. Reconcile books vs 2B and produce a *vendor follow-up list*. 3. Apply the set-off order correctly so the electronic cash ledger payment is exactly right — not a rupee more.

College gives theory; the employer needs someone who won't cause a notice.

What "proficient" looks like

A job-ready person can, unaided:

- Prepare GSTR-1 from an invoice dump, split correctly into B2B, B2C (large/small), exports, credit notes, and HSN summary.
- Download GSTR-2B, run a reconciliation in Excel that flags **matched / mismatched / missing-in-books / missing-in-2B** lines.
- Decide ITC eligibility: block Section 17(5) items (cars, personal use, club membership, works contract for immovable property, CSR), apply Rule 42/43 reversals for exempt supplies.
- Apply the set-off hierarchy and compute the exact cash payable.
- Pass the correct journal entries and file 3B so the cash ledger, credit ledger, and books all tie.

Hands-on: how to actually do it

Step 1 — GSTR-1 structure

GSTR-1 is filed table-wise. The tables you touch most:

Table	What goes in it
4A/4B	B2B invoices (registered buyers)
5A	B2C Large (inter-state, invoice > ₹1 lakh)
6A	Exports
7	B2C Others (consolidated, rate-wise)
9B	Credit/Debit notes
12	HSN-wise summary

Portal path: `Returns Dashboard → select period → GSTR-1 → Prepare Online` (or upload JSON via Offline Tool). Filing is by the **11th** of next month (monthly) or **13th** for QRMP quarterly.

Step 2 — ITC set-off order (Section 49A + Rule 88A)

This is the single most-tested computation. IGST credit must be **fully exhausted first**, and can be used against any head. CGST credit cannot touch SGST and vice-versa.

```
Order of utilisation:
1. IGST credit  → pay IGST, then CGST, then SGST (in that order)
2. CGST credit  → pay CGST, then IGST
3. SGST credit  → pay SGST, then IGST
Rule: CGST credit can NEVER pay SGST liability (and vice-versa)
```

Step 3 — GSTR-2B reconciliation in Excel

Load purchase register (books) on one sheet, GSTR-2B download on another. Match on GSTIN + Invoice No.

```
' Match flag: does this books-invoice appear in 2B?
=IF(ISNUMBER(MATCH(A2&"|"&B2, TwoB!$A$2:$A$5000, 0)), "In 2B", "Missing in 2B")

' Tax difference between books and 2B (using XLOOKUP)
=LET(
   key, A2&"|"&B2,
   b2b_tax, XLOOKUP(key, TwoB_Key, TwoB_TotalTax, 0),
   books_tax, F2,
   books_tax - b2b_tax)
```

Build a status column with SUMIFS to reconcile totals:

```
' Total ITC available per 2B for the period
=SUMIFS(TwoB!$H:$H, TwoB!$D:$D, "≥"&DATE(2025,4,1), TwoB!$D:$D, "≤"&DATE(2025,4,30))
```

Step 4 — Python reconciliation (for volume)

```python
import pandas as pd

books = pd.read_excel("purchase_register.xlsx")
twob  = pd.read_excel("gstr2b.xlsx")

for df in (books, twob):
    df["key"] = df["gstin"].str.strip() + "|" + df["invoice_no"].str.strip()

merged = books.merge(twob, on="key", how="outer",
                     suffixes=("_books", "_2b"), indicator=True)

def status(r):
    if r["_merge"] == "left_only":  return "Missing in 2B (chase vendor)"
    if r["_merge"] == "right_only": return "In 2B, not in books (book it)"
    if abs(r["tax_books"] - r["tax_2b"]) > 1: return "Tax mismatch"
    return "Matched"

merged["status"] = merged.apply(status, axis=1)
eligible_itc = merged.loc[merged.status == "Matched", "tax_2b"].sum()
print("Claimable ITC:", round(eligible_itc, 2))
merged.to_excel("recon_output.xlsx", index=False)
```

Step 5 — Journal entries

Recording a purchase with ITC (intra-state, 18% = 9% CGST + 9% SGST):

Account	Dr	Cr
Purchases / Expense A/c	1,00,000	
Input CGST A/c	9,000	
Input SGST A/c	9,000	
To Creditor / Bank		1,18,000

Set-off and payment at month end:

Account	Dr	Cr
Output CGST A/c	15,000	
Output SGST A/c	15,000	
To Input CGST A/c		9,000
To Input SGST A/c		9,000
To Electronic Cash Ledger (Bank)		12,000

Worked example / mini-project

Scenario — "Nord Traders Pvt Ltd", April 2025. Reproduce this end to end.

Output tax (from GSTR-1):

Supply	Taxable value	IGST	CGST	SGST
Inter-state B2B	10,00,000	1,80,000	–	–
Intra-state B2B	8,00,000	–	72,000	72,000
Total output		**1,80,000**	**72,000**	**72,000**

ITC per books: IGST ₹1,20,000, CGST ₹90,000, SGST ₹90,000 — total ₹3,00,000.

But 2B reconciliation flags: - One vendor (invoice for ₹50,000 + ₹9,000 CGST + ₹9,000 SGST) **not in 2B** → not filed by supplier. Reverse ₹18,000. - A ₹15,000 IGST invoice is for a **motor car** → blocked u/s 17(5). Reverse ₹15,000.

Eligible ITC: IGST = 1,20,000 − 15,000 = **1,05,000**; CGST = 90,000 − 9,000 = **81,000**; SGST = 90,000 − 9,000 = **81,000**.

Set-off (IGST first):

Head	Output	IGST credit used	Own-head credit	Cash
IGST	1,80,000	1,05,000	–	75,000
CGST	72,000	–	72,000 (of 81k)	0
SGST	72,000	–	72,000 (of 81k)	0

IGST credit (₹1,05,000) is fully consumed against IGST liability, leaving **₹75,000 IGST payable in cash**. CGST and SGST are each covered by own-head credit (₹72,000 used, ₹9,000 each carried forward).

Cash to pay via challan (PMT-06): ₹75,000. Credit carried forward: CGST ₹9,000 + SGST ₹9,000.

3B filing path: `GSTR-3B → Table 3.1 output → Table 4 ITC (4A auto from 2B, 4B reversals for the ₹15k car + ₹18k unfiled) → Table 6.1 payment → Create Challan → pay → File with DSC/EVC.`

How it's tested

Interview questions: - "Explain the IGST-first set-off rule. Can CGST credit pay SGST?" (No.) - "A vendor's invoice is in your books but not in 2B — do you claim ITC?" (No — Rule 36(4)/Section 16(2)(aa): ITC only if it appears in 2B.) - "Name five Section 17(5) blocked credits." - "Difference between 2A and 2B?" (2A is dynamic/real-time; 2B is static, generated on the 14th — 2B is the basis for claim.) - "Due dates for GSTR-1 and 3B under QRMP?"

Practical assessment: companies hand you a purchase register + a 2B extract and ask you to (a) reconcile in Excel within 30–45 minutes, (b) list vendors to follow up, (c) state the exact claimable ITC and cash payable. Consulting firms give a "prepare this 3B" case with blocked credits hidden in the data to see if you catch them.

Common mistakes & how pros avoid them

Mistake	Fix
Claiming ITC not in 2B	Claim only matched-in-2B lines; keep a "provisional/chase" bucket separately
Ignoring Section 17(5) blocks	Maintain a tagged expense-code list (car, food, club, CSR) auto-flagged
CGST↔SGST cross set-off	Lock a set-off template; never let one head pay the other
GSTR-1 vs 3B mismatch	Reconcile 3B Table 3.1 total to GSTR-1 before filing — a common notice trigger
Matching only on invoice no.	Always match on GSTIN + invoice no. (invoice numbers repeat across vendors)
Forgetting Rule 42/43 reversal	If you have exempt supplies, reverse common-credit proportionately
Late filing interest	Pay within due date; interest u/s 50 is 18% p.a. on cash-paid portion

Pros reconcile *before* filing, not after a notice. They keep an audit trail: books → 2B → adjustments → 3B, each tying to the next.

Learn-it roadmap & resources

Time to proficiency: 4–6 weeks of hands-on practice if you already know GST basics.

- Week 1–2: GSTR-1 structure, file 2–3 dummy returns on the GST portal sandbox / offline tool.
- Week 3: Build a reusable Excel 2B reconciliation template (XLOOKUP + SUMIFS + status column).
- Week 4: Master set-off logic and blocked credits; do 10 worked computations.
- Week 5–6: Do a full month-end close for a sample company end-to-end.

Resources: - GST portal (gst.gov.in) — free Returns Offline Tool and help manuals. - CBIC GST Flyers + Section 16, 17(5), 49, Rule 36(4)/42/43 (free, authoritative). - ICAI Indirect Tax study material (free for students). - TallyPrime / Zoho Books / ClearTax GST — practice the automated recon flow. - Certifications: ICAI Certificate Course on GST; GSTP (Goods & Services Tax Practitioner) exam via NACIN.

Quick-reference

Item	Detail
GSTR-1 due	11th (monthly) / 13th (QRMP quarterly)
GSTR-3B due	20th (monthly); 22nd/24th staggered (QRMP)
GSTR-2B generated	14th of next month (static)
ITC condition	Must appear in 2B (Sec 16(2)(aa)) + invoice + receipt of goods/services + tax paid + 180-day payment rule
Set-off order	IGST first (any head) → CGST (CGST then IGST) → SGST (SGST then IGST); no CGST↔SGST
Blocked ITC (17(5))	Motor vehicles ≤13 seats, food/beverage, club, health/life insurance (unless mandated), works contract for immovable property, personal use, CSR, goods lost/stolen/gifted
Interest (Sec 50)	18% p.a. on late/short cash payment; 24% on excess ITC utilised
ITC time limit	Earlier of 30th Nov following FY-end or annual return date
Key Excel	`=XLOOKUP(gstin&"|"&inv, 2b_key, 2b_tax, 0)` ; `=SUMIFS()` for period totals
Payment challan	Form PMT-06 → Electronic Cash Ledger → offset in 3B Table 6.1

GST in practice III: Annual Return & Notices

What it is & where it's used

The GST **annual return (GSTR-9)** and the **reconciliation statement (GSTR-9C)** are the year-end "close" of the GST cycle. Every month or quarter you filed GSTR-1 (outward supplies) and GSTR-3B (summary + tax payment); the annual return rolls all of that into one consolidated filing per GSTIN per financial year, reconciled against your books.

Beyond the return, GST is now a **notice-heavy regime**. The department runs automated matching (GSTR-1 vs 3B, 3B vs 2B, e-way bills vs sales, e-invoice vs GSTR-1) and throws out scrutiny notices — **ASMT-10** (scrutiny discrepancy), **DRC-01A / DRC-01** (demand), **DRC-03** (voluntary payment), plus ADT-01 for departmental audit. Knowing how to read, reconcile and *reply* to these is a core, billable skill.

Roles that need this: GST/indirect-tax executives, accounts payable/receivable leads doing month-end and year-end, tax analysts in a Big-4 or mid-tier firm, finance managers signing off the annual return, and anyone in a consulting practice handling client compliance. In an audit/consulting firm, "can you draft an ASMT-10 reply and finalise 9C" is a direct hiring criterion.

The gap: why companies want this (and college didn't teach it)

College teaches GST *law* — sections, definitions, the constitutional backstory. It does **not** teach you to:

- Open a GSTR-9 offline utility, map ledger balances into 17 tables, and explain a `Table 8` ITC difference.
- Reconcile **turnover as per books vs turnover as per GSTR-9** and defend the delta (unbilled revenue, credit notes, schedule-III items, cross-charge).
- Read a system-generated ASMT-10 that says "ITC in 3B exceeds 2B by ₹4,32,118" and produce a line-item reconciliation instead of panicking.
- Decide *when to pay via DRC-03* and *when to contest*, and draft the covering reply.

Employers pay for the person who closes the loop: books → returns → reconciliation → notice reply → resolution. That end-to-end confidence is the gap.

What "proficient" looks like

A job-ready person can, unaided:

1. **File GSTR-9** end-to-end: pull auto-populated data, correct Tables 4–9, 10–14, and 17–18, and reconcile ITC (Table 6 vs Table 8).
2. **Prepare GSTR-9C**: reconcile turnover (Table 5), tax paid (Table 9), ITC (Table 12–14), and quantify any additional liability payable through DRC-03.
3. **Build a reconciliation working** in Excel that ties GSTR-1, 3B, 2B and books, with a clear "reason for difference" column.
4. **Respond to ASMT-10 within 30 days** with a reasoned reply (ASMT-11) plus annexures.
5. **Handle a DRC-01A/01**: agree-and-pay via DRC-03, or contest via DRC-06, knowing the limitation and pre-deposit rules.

Hands-on: how to actually do it

GSTR-9: portal click-path

```
GST Portal → Login → Services → Returns → Annual Return
→ Select FY → GSTR-9 "Prepare Online"
→ Download: (a) GSTR-9 System Computed Summary (PDF)
            (b) GSTR-1 Summary, (c) GSTR-3B Summary
→ Fill Tables 4–19 → Compute Liability → Pay any diff via DRC-03
→ File with DSC/EVC
```

Key tables and where numbers come from:

Table	Content	Source
4	Outward taxable supplies (B2B, B2C, exports, RCM)	GSTR-1
5	Exempt / nil / non-GST outward	GSTR-1
6	ITC availed (inputs / capital goods / services / RCM)	GSTR-3B
7	ITC reversed (Rule 42/43, 37, ineligible)	Books
8	ITC reconciliation: 2A/2B vs availed	Auto + books
9	Tax paid	GSTR-3B
10–14	Prior-year adjustments in next-year returns	GSTR-1/3B of Apr–Oct next FY
17–18	HSN summary outward / inward	Books

The core Excel reconciliation (3B vs 2B ITC — the #1 notice trigger)

Lay out one row per GSTIN of the supplier. Use `SUMIFS` to net books vs portal, then a reason column.

```
' Books ITC per supplier (from purchase register)
=SUMIFS(Books[ITC], Books[GSTIN], A2)

' 2B ITC per supplier (from downloaded GSTR-2B)
=SUMIFS(GSTR2B[ITC], GSTR2B[GSTIN], A2)

' Difference
=D2 - E2                     ' Books minus 2B

' Auto-tag the reason
=IFS( ABS(F2)<1, "Matched",
      F2>0,  "In books, not in 2B (supplier not filed / timing)",
      F2<0,  "In 2B, not in books (missed invoice / RCM)" )
```

To match a single invoice both ways, `XLOOKUP` on a concatenated key:

```
=XLOOKUP(A2&B2, GSTR2B[GSTIN]&GSTR2B[InvNo], GSTR2B[TaxableValue], "NOT IN 2B")
```

SQL: pull the mismatch straight from your ERP

```sql
SELECT  b.supplier_gstin,
        SUM(b.igst + b.cgst + b.sgst)          AS itc_books,
        COALESCE(SUM(g.itc_total), 0)          AS itc_2b,
        SUM(b.igst + b.cgst + b.sgst)
          - COALESCE(SUM(g.itc_total), 0)      AS diff
FROM    purchase_register b
LEFT JOIN gstr2b g
       ON  b.supplier_gstin = g.supplier_gstin
       AND b.invoice_no     = g.invoice_no
       AND b.fin_year        = '2024-25'
GROUP BY b.supplier_gstin
HAVING  ABS(SUM(b.igst + b.cgst + b.sgst) - COALESCE(SUM(g.itc_total),0)) > 1
ORDER BY ABS(diff) DESC;
```

Python: turnover recon for GSTR-9C Table 5

```python
import pandas as pd

books = pd.read_excel("trial_balance_revenue.xlsx")   # ledger-wise revenue
gstr9 = 24_50_00_000        # outward turnover per GSTR-9 Table 4/5

recon = pd.DataFrame([
    ("Turnover as per audited financials", books["amount"].sum()),
    ("Add: Unbilled revenue at year-end",  -12_00_000),  # not yet in GST
    ("Less: Credit notes not in books",     -4_50_000),
    ("Add: Schedule III / non-GST supply",  -8_00_000),
    ("Less: Cross-charge to branches",      15_00_000),
], columns=["particulars", "amount"])

recon.loc[len(recon)] = ("Turnover as per GSTR-9", recon["amount"].sum())
recon["check_vs_9"] = recon["amount"].iloc[-1] - gstr9   # should be ~0
print(recon.to_string(index=False))
```

Journal entry when 9C throws up extra liability

You pay short-paid GST through **DRC-03** (cash only, no ITC). Book it as:

Account	Dr	Cr
GST Expense / Prior-period tax (P&L)	₹50,000	
Interest on GST (P&L)	₹9,000	
To Bank (DRC-03 challan)		₹59,000

If it relates to reversed ITC:

Account	Dr	Cr
ITC Reversed – Expense	₹50,000	
To Input CGST/SGST/IGST		₹50,000

Responding to notices — the mechanics

Notice	Meaning	Your reply form	Deadline
ASMT-10	Scrutiny discrepancy	ASMT-11 (+ annexures)	30 days
DRC-01A	Pre-show-cause intimation	Part B of DRC-01A / DRC-03	~15 days
DRC-01	Show cause notice (demand)	DRC-06	30 days
DRC-03	*You* pay voluntarily	— (challan)	Anytime
ADT-01	Departmental audit notice	Produce records	As stated

Portal path to reply: `Services → User Services → View Additional Notices/Orders → View → Reply`. Attach a signed PDF reconciliation as annexure.

Worked example / mini-project

Scenario — Acme Traders Pvt Ltd, FY 2024-25. Reconcile and respond to an ASMT-10.

Data:

Item	GSTR-3B (₹)	GSTR-2B (₹)	Books (₹)
ITC availed	42,00,000	39,50,000	41,80,000

The system issues **ASMT-10**: "ITC availed in GSTR-3B (₹42,00,000) exceeds ITC in GSTR-2B (₹39,50,000) by ₹2,50,000. Explain."

Step 1 — Build the bridge.

Reconciliation item	₹
ITC as per 2B	39,50,000
Add: RCM ITC (self-invoiced, not in 2B)	1,20,000
Add: Invoices filed by supplier in Apr-2025 2B (timing)	90,000
Add: Import IGST (from ICEGATE, not vendor 2B)	60,000
Explained ITC	**42,20,000**
ITC in 3B	42,00,000
Excess claimed to reverse	Nil (books ₹41,80,000; ₹20,000 already reversed)

Step 2 — Conclusion. The ₹2,50,000 gap is fully explained by RCM, import IGST and timing — all legitimate. Nothing to pay. A ₹20,000 ineligible item was already reversed in March-2025 3B.

Step 3 — File ASMT-11 attaching the above table plus RCM self-invoices, bill of entry, and the April 2B extract. Closing line: "*In view of the above reconciliation, the difference is fully explained and no further tax is payable. We request closure of proceedings.*"

Step 4 — If ₹20,000 were NOT already reversed: pay via DRC-03 (`Services → User Services → My Applications → Intimation of Voluntary Payment DRC-03 → Cause = Scrutiny`), ₹20,000 tax + interest @18% p.a., and reference the ARN in the ASMT-11 reply.

How it's tested

Interview questions

- "Turnover per books is ₹50 Cr but GSTR-9 shows ₹48.5 Cr. Give me five reasons for the difference." (Unbilled revenue, credit notes, Schedule-III, cross-charge, financial-year cut-off / advances.)
- "Difference between Table 6 and Table 8 of GSTR-9?"
- "You get an ASMT-10 for 3B-vs-2B ITC excess. Walk me through your reply."
- "When do you use DRC-03 vs DRC-06?"
- "Is GSTR-9C still audited by a CA?" (Now self-certified since FY 2020-21; no mandatory CA audit, but a reconciliation statement is filed if turnover > ₹5 Cr.)

Practical assessment (very common) — you're handed three files (GSTR-1 summary, 3B summary, purchase register) and asked to:

1. Produce a 3B-vs-2B ITC reconciliation with a reason column (timed, ~45 min in Excel).
2. Draft a one-page ASMT-11 reply from a sample notice.
3. Fill Table 8 of GSTR-9 and state the eligible ITC.

Firms grade on whether your reconciliation *ties* and whether your reasons are defensible.

Common mistakes & how pros avoid them

- **Treating GSTR-9 as auto-filed.** The portal pre-fills, but *you* must correct Tables 6–8 and 17. Pros always download and re-derive.

- **Ignoring next-year adjustments (Tables 10–14).** Amendments and ITC of last FY claimed in Apr–Oct of the next FY must be reported here — most people miss it and create a false mismatch.
- **Reversing ITC on RCM/import just because it's not in 2B.** RCM and import IGST legitimately don't appear in 2B — never reverse them to "match".
- **Missing the 30-day ASMT-10 window.** No reply → best-judgment assessment (ASMT-13). Diarise every notice the day it lands.
- **Paying disputed demand via DRC-03 out of fear.** If you have a defensible position, contest via DRC-06; a DRC-03 payment can be read as admission.
- **Not reconciling e-invoice / e-way bill to GSTR-1.** The department does — a sales-vs-eway gap is a fresh notice waiting to happen.
- **Interest not computed.** Any short-paid tax carries 18% p.a.; forgetting it invites a follow-up demand.

Learn-it roadmap & resources

Time to proficiency: ~6–8 weeks part-time if you already know monthly GST.

Week	Focus
1–2	Master 3B-vs-2B and 1-vs-3B reconciliation in Excel
3	GSTR-9 tables end-to-end on the offline utility
4	GSTR-9C turnover, tax and ITC reconciliation
5	Notices: ASMT-10/11, DRC family, draft sample replies
6	Departmental audit (ADT-01), records to keep, limitation
7–8	Full mock: file a dummy 9/9C and answer a sample notice

Resources

- GST Portal help + downloadable GSTR-9/9C **offline utilities** (free, hands-on practice).
- **CBIC** notifications & the CGST Act s.44 (annual return), s.61 (scrutiny), s.65 (audit), s.73/74 (demands).
- ICAI *Technical Guide on Annual Return & Reconciliation Statement* (free PDF) — the practitioner's bible.
- ClearTax / TaxGuru walkthroughs for portal screens; Taxmann GST reckoner for rates.
- **Certification:** ICAI Certificate Course on GST, or the government **GST Practitioner (GSTP)** enrolment for practice.

Quick-reference

Form	Purpose
GSTR-9	Annual return (turnover > ₹2 Cr; optional below)
GSTR-9C	Reconciliation statement, self-certified (turnover > ₹5 Cr)
ASMT-10 / 11	Scrutiny notice / reply
ASMT-13	Best-judgment assessment (if no reply)
DRC-01A	Pre-SCN intimation
DRC-01 / 06	Show cause notice / reply
DRC-03	Voluntary / scrutiny payment (cash only)
ADT-01 / 02	Audit notice / findings

Key deadlines: GSTR-9 & 9C — **31 Dec** of next FY. ASMT-10 reply — **30 days**. DRC-01 reply — **30 days**.

Interest: 18% p.a. on short-paid tax; ITC-related interest under s.50.

Limitation for demands: s.73 (non-fraud) — 3 years; s.74 (fraud) — 5 years from annual-return due date.

Golden rules of reconciliation

```
Books   ┐
GSTR-1 ─┼─→ must tie ←─┬─ GSTR-3B
2B     ─┘              └─ e-invoice / e-way bill
RCM & Import IGST → in books, NOT in 2B → never reverse to match
Every difference needs a named reason. No unexplained delta.
```

TDS/TCS in practice

What it is & where it's used

TDS (Tax Deducted at Source) means the *payer* withholds a slice of tax before paying a vendor, employee, landlord or contractor, and deposits it with the government against the payee's PAN. **TCS (Tax Collected at Source)** is the mirror image — the *seller* collects a small extra amount from the buyer on certain sales (scrap, motor vehicles > ₹10 lakh, some goods sales) and remits it.

It is the government's cash-flow engine and its biggest data-matching tool: every rupee deducted lands in the payee's **Form 26AS / AIS**, and if it doesn't match, someone gets a notice.

Roles that live and breathe this:

Role	TDS/TCS work
Accounts Payable executive	deduct correct section/rate at invoice booking
Payroll executive	monthly TDS on salary (192), Form 16 in May
Tax/compliance associate	monthly challans, quarterly 24Q/26Q, corrections
Statutory auditor / articled clerk	vouch deductions, verify 26Q vs books, 40(a)(ia) disallowance
Vendor master / procurement	206AB/206CCA PAN screening before onboarding

If you touch a payment run in India, you touch TDS.

The gap: why companies want this (and college didn't teach it)

An MBA teaches that "tax is deducted at source." It does **not** teach that you have to:

- pick the *right section* from a 40-row rate chart at the moment of booking,
- apply the **higher rate under 206AB** when a vendor hasn't filed returns,
- deposit by the **7th of next month** (30 April for March) or eat 1.5%/month interest,
- file **26Q** with the exact challan-deduction mapping or the return gets rejected,
- and know that a wrong deduction triggers a **30% expense disallowance** under section 40(a)(ia).

Colleges test *concepts*; employers test *whether the challan is right and filed on time*. A single missed deposit means interest, a ₹200/day late-filing fee under 234E, and a partner's phone call. That operational, deadline-driven, penalty-aware muscle is exactly what companies pay for and what no classroom simulates.

What "proficient" looks like

A job-ready person can, unaided:

1. Read an invoice and say **"194J, 10%, ₹X"** or **"194C, 1%/2%, ₹Y"** in seconds.
2. Apply **threshold logic** (e.g. 194C single ₹30,000 / annual ₹1,00,000; 194J ₹30,000; 194Q ₹50 lakh) and **206AB** higher-rate override.
3. Pass the correct **journal entry** and reconcile the TDS payable ledger to the challan.

4. Generate a **challan (ITNS 281 / e-Pay Tax)**, download the CSI file, and validate the return in **RPU/TRACES**.
5. Explain the interplay of **194Q (buyer deducts) vs 206C(1H) (seller collects)** and who wins.
6. Produce **Form 16A** from TRACES and **Form 16** (Part A + Part B) for payroll.

Hands-on: how to actually do it

Key sections cheat-rates (FY 2025-26, resident, PAN available)

Section	Nature of payment	Threshold	Rate
192	Salary	Basic exemption	Slab (avg)
194C	Contractor / sub-contractor	₹30,000 single / ₹1,00,000 p.a.	1% (Ind/HUF), 2% (others)
194J	Professional / technical fees	₹30,000	10% (2% for technical/call-centre)
194H	Commission / brokerage	₹20,000	2%
194I	Rent — plant/machinery / land-building	₹2,40,000 p.a.	2% / 10%
194Q	Purchase of goods (buyer T/O > ₹10 cr)	₹50,00,000	0.1% on excess
206C(1H)	Sale of goods (seller T/O > ₹10 cr)	₹50,00,000	0.1% on excess
206C(1)	Scrap	—	1%
No PAN	any	—	**20% (206AA)**

206AB / 206CCA — the "non-filer" penalty

If the payee is a **"specified person"** (didn't file ITR for the relevant previous year *and* aggregate TDS/TCS ≥ ₹50,000 in that year), deduct at the **higher of**: twice the normal rate, or 5%. Check status on the **Reporting Portal → Compliance Check for 206AB** (bulk PAN upload). Never onboard a big vendor without this screen.

Excel: pick section, apply threshold, compute TDS

```
' Rate lookup from a RateTable (Section | Rate)
=XLOOKUP([@Section], RateTable[Section], RateTable[Rate])

' Apply annual threshold: only deduct once cumulative crosses limit
=IF([@CumInvoiceValue]>[@Threshold], [@InvoiceValue]*[@Rate], 0)

' 194Q – 0.1% only on purchases ABOVE 50 lakh (cumulative from same seller)
=MAX(0, MIN([@CumPurchase],[@CurrPurchase]) - MAX(5000000,[@CumPurchase]-[@CurrPurchase]))*0.0

' 206AB override: higher of 2x rate or 5%
=IF([@IsSpecifiedPerson]="Y", MAX([@Rate]*2, 5%), [@Rate])
```

Journal entries

Booking a ₹1,00,000 professional-fee invoice (194J @10%):

Account	Dr	Cr
Professional Fees (expense)	1,00,000	
To Vendor A/c		90,000
To TDS Payable – 194J		10,000

On deposit (7th of next month):

Account	Dr	Cr
TDS Payable – 194J	10,000	
To Bank		10,000

TallyPrime click-path

`Gateway of Tally → Vouchers → F9 Purchase` → select TDS nature of payment on the ledger → Tally auto-computes → `Alt+G → TDS Reports → Challan Reconciliation` → `Ctrl+P` for ITNS 281.

Deposit & return calendar

Task	Due date
Deposit TDS (Apr–Feb)	**7th** of next month
Deposit TDS (March)	**30 April**
26Q/27Q (non-salary)	Q1 31-Jul, Q2 31-Oct, Q3 31-Jan, Q4 **31-May**
24Q (salary)	same quarterly dates
Form 16A (from TRACES)	15 days after return due date
Form 16 (salary)	**15 June**

Deposit via **income tax portal → e-Pay Tax → Proceed (TDS/TCS) → ITNS 281**. Download the **CSI file** from the portal — the RPU validation needs it.

Worked example / mini-project

Vermilion Analytics Pvt Ltd, June 2025 payments:

Vendor	Service	Section	Amount (₹)	Note
CA Rao & Co	Audit fee	194J	2,00,000	PAN valid
BuildRight (partnership)	Office renovation	194C	5,00,000	contractor
Mehta (proprietor)	Freelance design	194J	40,000	**non-filer (206AB)**
Landlord (building)	Office rent	194I(b)	3,00,000	annual

Computation

```
194J  Rao & Co : 2,00,000 × 10%           = 20,000
194C  BuildRight (firm, 2%) : 5,00,000 × 2% = 10,000
194J  Mehta, 206AB → max(10%×2, 5%)=20% :
      40,000 × 20%                          =  8,000
194I  Rent (building 10%) : 3,00,000 × 10% = 30,000
────────────────────────────────────────────────
Total TDS for June                          = 68,000
```

Deposit **₹68,000 by 7 July 2025** (four challans or one, split by section in the return). If deposited on 20 July instead: `Interest u/s 201 = 68,000 × 1.5% × 2 months (deducted June, paid July → part-months) = ₹2,040`.

Return (26Q, Q1) mapping — each deductee row links to the challan by BSR code + date + serial. Validate in **RPU** → generate `.fvu` → upload on TRACES/income-tax portal → download **Form 16A** for each vendor two weeks later.

Reproduce it: build the four-row table in Excel, add the `206AB` override formula, sum column, and you have the exact number a compliance associate files.

How it's tested

Interview questions - "Vendor raises a ₹28,000 194C invoice, then a ₹15,000 one — do you deduct? On what?" (Yes, on ₹43,000 total, since annual ₹1,00,000 not crossed but single-bill ₹30,000 crossed on second? — trap: 194C single limit ₹30,000, aggregate ₹1,00,000; neither crossed individually but aggregate matters — expected answer: deduct once ₹1,00,000 aggregate crosses.) - "194Q vs 206C(1H) — same transaction, both above ₹50L, who acts?" (194Q buyer prevails; seller stops collecting.) - "What is section 40(a)(ia)?" (30% disallowance of expense if TDS not deducted/paid.) - "Client didn't deduct 206AB higher rate — consequences?" (short deduction, interest, disallowance.)

Practical test companies give: - A timed sheet of 10 invoices → identify section, rate, TDS amount, and total challan (30 min in Excel). - "Here's a 26Q .fvu error log — fix the challan-deduction mismatch." - Payroll case: compute annual salary TDS, spread over 12 months, produce Form 16 Part B.

Common mistakes & how pros avoid them

Mistake	Fix
Deducting on **GST component** of invoice	Deduct on **taxable value only** (excl. GST), if GST shown separately
Wrong Ind/HUF vs firm rate in 194C (1% vs 2%)	Check payee constitution in vendor master
Missing **206AB** higher rate	Run bulk Compliance Check every quarter; flag in vendor master
Depositing March TDS by 7 April	March deadline is **30 April** — don't overpay interest, don't miss
Challan section ≠ return section	Reconcile TDS payable ledger to challan before filing
Quoting no PAN → 20% forgotten	Block payment until PAN captured
Late return	₹200/day fee u/s **234E** (capped at TDS amount) — file even if late

Pros keep a **TDS control sheet**: section-wise payable ledger reconciled to challans each month, and a vendor master with PAN + 206AB status refreshed quarterly.

Learn-it roadmap & resources

Time to proficiency: 3–4 weeks of hands-on, if you already know accounting basics.

Week	Focus
1	Section/rate chart, thresholds, journal entries
2	e-Pay Tax challan, CSI file, TallyPrime TDS
3	RPU + TRACES: file a dummy 26Q, generate Form 16A
4	24Q + Form 16, 206AB compliance check, corrections

Resources (free): Income Tax India e-filing portal (practice login), TRACES "e-Tutorials", NSDL/Protean **RPU & FVU** utilities (download free), CBDT rate chart PDF. **Paid:** ICAI BoS TDS module, ClearTDS / Winman TDS software (free trials), any "TDS return filing" Udemy course (₹500). **Certification:** none required, but ICAI Intermediate (Taxation) and a "GST & TDS Practitioner" course strengthen the CV.

Quick-reference

```
DEPOSIT: 7th next month | MARCH: 30 April
RETURNS: 26Q/24Q – 31 Jul / 31 Oct / 31 Jan / 31 May
FORM 16: 15 June | FORM 16A: 15 days after return

RATES (PAN available, FY25-26):
 192  salary        slab
 194C contractor    1% ind/huf, 2% others   (30k/1L)
 194J prof fees     10% (2% technical)       (30k)
 194H commission    2%                        (20k)
 194I rent          2% P&M / 10% land-bldg    (2.4L)
 194Q buy goods     0.1% >50L (buyer T/O>10cr)
 206C(1H) sell gds  0.1% >50L (seller side)
 206C(1) scrap      1%
 NO PAN (206AA)     20%
 206AB non-filer    higher of 2×rate or 5%

INTEREST 201: 1%/mo (not deducted) or 1.5%/mo (not paid)
LATE FEE 234E: ₹200/day (max = TDS)
DISALLOWANCE 40(a)(ia): 30% of expense
```

Key formula: `TDS = Taxable value (excl. GST) × applicable rate`, applied once the cumulative payment crosses the section threshold, with a 206AB override for non-filers.

Income-tax computation & finalisation

What it is & where it's used

Income-tax computation is the process of converting an entity's book profit (or an individual's gross earnings) into **taxable income**, applying the correct slab/rate, adjusting for deductions and reliefs, and arriving at the **final tax payable or refundable** for the year. Finalisation is the year-end ritual — reconciling advance tax, TDS, self-assessment tax, and the return — so the books, Form 26AS/AIS, and the ITR all agree.

Roles that live and die on this skill:

Role	What they touch
Tax associate (Big 4 / CA firm)	Computing income for 50-200 clients per season, ITR filing
Accounts / finance executive (industry)	Advance-tax working, provision for tax, deferred tax
Financial controller	Sign-off on current-tax and MAT provision at year-end close
Payroll / HR-finance	TDS on salary (Sec 192), Form 16
Anyone doing a company statutory audit	Tax provision, MAT, tax reconciliation in notes

If you can take a trial balance and produce a defensible tax computation sheet, you are immediately useful in March-September (the Indian filing crunch).

The gap: why companies want this (and college didn't teach it)

MBA and even CA theory teach you the *heads of income* and the *slab rates*. What they don't teach is the **mechanical, spreadsheet-driven finalisation** that a real client engagement demands:

- Books show *depreciation as per Companies Act*; tax needs *depreciation as per Income-tax Act (WDV, block of assets)*. Nobody teaches you to run both.
- You must **add back** disallowances (Sec 40(a)(ia), 43B, 37) that never appear in a textbook profit.
- You must decide **old vs new regime** (115BAC for individuals, 115BAA for companies) — a real Rupee decision, not a definition.
- You must reconcile tax paid to **Form 26AS / AIS** before filing, or the return gets a mismatch notice.
- **MAT (Sec 115JB)** vs normal tax — computing both and paying the higher — is a pure-practice skill.

Colleges test "what is Section 80C." Employers test "here is the trial balance, give me the computation, the advance-tax challans, and the provision entry by 6 pm."

What "proficient" looks like

A job-ready person can, **unaided**:

1. Build a **computation of total income** sheet in Excel from a trial balance — five heads, adjustments, deductions, tax, cess, interest.
2. Run **tax-depreciation** on the block-of-assets method and reconcile it to book depreciation.
3. Compute **MAT under 115JB** and compare with normal tax; know MAT credit u/s 115JAA.

4. Prepare an **advance-tax schedule** (15%/45%/75%/100% cut-offs) and compute interest u/s 234B/234C.
5. Decide **115BAA / 115BAC** with a side-by-side comparison.
6. Pass the **provision-for-tax and deferred-tax** journal entries and tie them to the computation.
7. Reconcile the computation to **Form 26AS/AIS** and file the correct **ITR** (ITR-6 company, ITR-3/2 individual).

Hands-on: how to actually do it

Step 1 — Company computation skeleton (FY 2024-25 / AY 2025-26)

Start from **net profit as per Statement of P&L**, then adjust:

```
Net profit as per books (before tax)            xxxx
Add:  Depreciation as per Companies Act          xxx
      Disallowances u/s 40(a)(i)/(ia)            xxx
      43B items unpaid (GST, PF, bonus, interest) xxx
      Provisions / donations debited             xxx
      Fines & penalties, personal expenses       xxx
Less: Depreciation as per Income-tax Act        (xxx)
      Income taxed under other heads / exempt    (xxx)
      Deductions Chapter VI-A (80G, 80JJAA…)     (xxx)
= Total Income (round off to nearest ₹10, Sec 288A)
```

Excel — put the add-backs in a signed column so a single SUM works:

```
' B = amount, C = +1 add / -1 less
=B2 (net profit)
Total Income  =B2 + SUMPRODUCT(add_amounts, add_signs)
Round off     =MROUND(TotalIncome,10)
```

Step 2 — Tax rate logic (nested IF is fine, choose readable)

Company, opting **115BAA** (22% + 10% surcharge + 4% cess = **25.168%**), no MAT:

```
=ROUND(TotalIncome * 0.25168, 0)
```

Company **not** opting 115BAA — normal rate 25% (turnover ≤ ₹400 cr) or 30%, plus surcharge slab, plus 4% cess. Surcharge helper:

```
=IF(TotalIncome>100000000, 0.12, IF(TotalIncome>10000000, 0.07, 0))
```

Individual — old vs new regime side by side (AY 2025-26 new-regime slabs):

```
=LET(ti, TaxableIncome,
  MAX(0,
   (MIN(ti,700000)-300000)*0.05
 + (MIN(ti,1000000)-700000)*0.10
 + (MIN(ti,1200000)-1000000)*0.15
 + (MIN(ti,1500000)-1200000)*0.20
 + (ti-1500000)*0.30 ))
```

Add 4% cess: `=Tax*1.04`. Apply **87A rebate** if new-regime total income ≤ ₹7,00,000 → tax = 0.

Step 3 — Tax depreciation (block of assets, WDV)

```
Opening WDV  + Additions − Sale proceeds = Closing base
Dep = base × block rate   (½ rate if asset used < 180 days)
```

Block	Rate
Plant & machinery (general)	15%
Furniture & fittings	10%
Computers & software	40%
Buildings (non-residential)	10%
Motor cars	15%

```
Full-rate dep  = (OpenWDV + Add180plus − Sale) * Rate
Half-rate dep  = Add_below180 * Rate/2
```

Step 4 — MAT (Sec 115JB)

Book profit × **15% + surcharge + cess**. Pay **higher of** normal tax and MAT. Excess MAT over normal tax is **MAT credit** carried 15 years.

```
NormalTax = …
MAT       = ROUND(BookProfit*0.15*1.04*(1+surcharge),0)
TaxPayable = MAX(NormalTax, MAT)
MATCredit  = MAX(0, MAT − NormalTax)
```

Note: a company under **115BAA is exempt from MAT** — that's a big reason to opt in.

Step 5 — Journal entries at finalisation

Date	Particulars	Dr (₹)	Cr (₹)
31-Mar	Advance income-tax A/c … Dr	20,00,000	
	To Bank		20,00,000
31-Mar	Income-tax expense (P&L) … Dr	26,50,000	
	To Provision for tax		26,50,000
31-Mar	Deferred tax expense … Dr	1,20,000	
	To Deferred tax liability		1,20,000
Next yr	Provision for tax … Dr	26,50,000	
	To Advance tax + TDS		24,00,000
	To Bank (self-assessment 140A)		2,50,000

Step 6 — Advance-tax & 234C interest

Due date	Cumulative %	234C trigger if paid <
15 Jun	15%	12%
15 Sep	45%	36%
15 Dec	75%	75%
15 Mar	100%	100%

234B interest = **1% p.m.** on shortfall (if advance tax < 90% of assessed) from April to payment. 234C = 1% p.m. on each instalment shortfall.

Worked example / mini-project

Alpha Traders Pvt Ltd, FY 2024-25. Reproduce this in Excel.

Line	₹
Net profit as per books	1,00,00,000
Add: Depreciation (Companies Act)	18,00,000
Add: GST unpaid at year-end (43B)	3,00,000
Add: Donation to CM Relief debited	2,00,000
Add: Penalty for late GST filing	50,000
Less: Depreciation (Income-tax Act)	(22,00,000)
Business income	**1,01,50,000**
Less: 80G (50% of ₹2,00,000)	(1,00,000)
Total income	**1,00,50,000**

Turnover ₹380 cr → base rate **25%**. Not opting 115BAA.

```
Tax @25%                        = 25,12,500
Surcharge @7% (TI > ₹1 cr)      =  1,75,875
                                  _________
                                  26,88,375
Cess @4%                        =  1,07,535
Total normal tax                = 27,95,910
```

MAT check: book profit ≈ ₹1,05,50,000 → MAT @15% × 1.04 × 1.07 = ₹17,60,900 < normal tax. **Pay ₹27,95,910.**

If instead Alpha opts **115BAA**: 1,00,50,000 × 25.168% = **₹25,29,384** — a genuine ₹2.66 lakh saving, no surcharge slab, no MAT. That comparison *is* the deliverable a manager wants.

Advance tax paid ₹24,00,000; TDS credit (26AS) ₹1,50,000; balance ₹2,45,910 → pay as **self-assessment tax (Sec 140A)** via challan before filing **ITR-6**.

How it's tested

Interview questions - "Company book profit is ₹50 lakh, tax depreciation is higher than book — walk me through the computation." - "When would a company still pay MAT under the old regime but not under 115BAA?" - "Difference between 234B and 234C interest?" - "Advance tax first instalment date and percentage?" - "Deferred tax vs current tax — which timing differences create DTL?"

Practical assessments - **Timed Excel test (60-90 min):** given a trial balance + fixed-asset schedule, produce the computation sheet, tax, MAT, and advance-tax working. - **"Finalise these books" case:** pass provision, advance-tax adjustment, and deferred-tax entries; tie the ITR figure to the ledger. - **26AS reconciliation:** they hand you a 26AS PDF and books; find the TDS mismatch.

Common mistakes & how pros avoid them

Mistake	Fix
Using book depreciation for tax	Always maintain a separate **block-of-assets** register; never copy Companies-Act dep.
Forgetting 43B — claiming unpaid GST/PF/bonus	Add back anything statutory **unpaid by the ITR due date**.
Ignoring MAT because "we have profit"	Compute **both** every year; pay the higher.
Missing the ½-rate rule on assets used < 180 days	Tag every addition with its put-to-use date.
Not reconciling to **AIS/26AS** before filing	Download 26AS + AIS first; match TDS line by line.
Applying old surcharge slabs to a 115BAA company	115BAA has a **flat 10% surcharge**, no slab.
Forgetting 234C on quarterly shortfall	Model the four cut-offs in the advance-tax sheet.
Rounding wrong	Round **total income** to nearest ₹10 (288A), tax to nearest ₹1.

Pros keep a **standard computation template** and a **checklist of add-backs** so nothing is missed under season pressure.

Learn-it roadmap & resources

Phase	Time	What to do
Basics	2 weeks	Five heads, slabs, both regimes — CA Inter Taxation module
Computation	3 weeks	Build 10 company + 10 individual computations in Excel from scratch
Depreciation & MAT	1 week	Block-of-assets register; MAT under 115JB
Filing	1 week	File a dummy ITR-3 and ITR-6 on the income-tax portal
Provision & deferred tax	1 week	AS 22 / Ind AS 12 entries, tie to computation

Resources - Income-tax portal (incometax.gov.in) — free ITR utilities, Form 26AS, AIS - ICAI Study Material — Taxation (authoritative, free) - CBDT rate cards / Finance Act each year (surcharge, slabs change annually) - Cleartax / Taxmann computation guides for worked examples - **Certification:** CA / CMA; or a practical GST+ITR filing course (Cleartax, Udemy) for non-CA roles

Realistic time-to-proficiency: **6-8 weeks** of daily practice to independently finalise a straightforward company.

Quick-reference

Company rates (AY 2025-26)

Basis	Rate (+cess 4%)
115BAA	22% + 10% surcharge = **25.168%** (no MAT)
Turnover ≤ ₹400 cr	25% + surcharge slab
Others	30% + surcharge slab
MAT (115JB)	15% of book profit

Company surcharge: 7% (TI > ₹1 cr), 12% (TI > ₹10 cr).

Individual new-regime slabs (AY 25-26): Nil ≤3L; 5% 3-7L; 10% 7-10L; 15% 10-12L; 20% 12-15L; 30% >15L. **87A rebate** up to ₹7L → tax nil.

Advance tax: 15% Jun · 45% Sep · 75% Dec · 100% Mar.

Interest: 234A (late filing), 234B (< 90% paid), 234C (instalment shortfall) — all **1% p.m.**

Key sections: 40(a)(ia) TDS disallowance · 43B statutory dues on payment · 115JB MAT · 115JAA MAT credit (15 yrs) · 115BAA company concessional · 115BAC individual new regime · 140A self-assessment · 288A round-off.

Forms: ITR-6 (company) · ITR-3 (business individual) · ITR-2 (no business) · Form 26AS + AIS (reconcile before filing).

ITR Filing & Tax Audit

What it is & where it's used

Income Tax Return (ITR) filing is the annual act of reporting a taxpayer's income, deductions, and tax paid to the Income Tax Department, and squaring off the balance (refund or demand). Tax audit under **Section 44AB** is the statutory examination of a business/profession's books by a Chartered Accountant, reported in **Form 3CA/3CB + 3CD**, that must be completed before the ITR of an audited assessee is filed.

Every finance function touches this:

Role	What they do with ITR/tax audit
Accounts executive	Assembles the trial balance, ledgers, TDS/26AS reconciliation feeding the return
Tax associate (CA firm / Big 4)	Prepares 3CD annexures, drafts returns for clients, uploads on portal
Finance manager / controller	Signs off on tax provision, coordinates the statutory + tax audit close
Payroll / HR finance	Issues Form 16, ensures salary TDS matches employee ITRs
Founder / freelancer	Files own ITR-3/ITR-4, decides presumptive vs. audit

If you are hired into any accounts or tax seat in India, July–September (the return + audit season) is the busiest, highest-visibility part of your year. Knowing the mechanics cold makes you immediately useful.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you what taxable income *is*. It does not teach you:

- **Which of the seven ITR forms** actually applies to a given fact pattern, and why using the wrong one gets the return treated as defective (Section 139(9)).
- How to **reconcile Form 26AS, AIS (Annual Information Statement), and TIS** against the books before you touch a return — the single most common real-world task.
- The **exact turnover thresholds** that trigger tax audit, and the trap in the 5% cash-transaction rule.
- How to fill **Clause 21 (disallowances), Clause 26 (Section 43B), Clause 34 (TDS compliance)** of Form 3CD — the clauses that consume 80% of audit time.
- The **portal click-path**: DSC registration, JSON utility, UDIN, CA-assignee acceptance workflow.

Colleges test theory in a vacuum. Employers pay for someone who can take a messy Tally backup and produce a filed, defensible return without supervision. That end-to-end capability is the gap.

What "proficient" looks like

A job-ready person can, unaided:

1. Pick the **correct ITR form** for salary, capital gains, business, presumptive, or company income.
2. Download and **reconcile 26AS + AIS** to the books, flagging every mismatch with a reason.
3. Determine **tax-audit applicability** in under two minutes given turnover, profession, and cash ratios.
4. Populate **Form 3CD clause-by-clause** from a trial balance and supporting schedules.

5. Compute **total income, tax, interest under 234A/B/C**, and know whether it's a refund or payable.
6. Generate **UDIN**, get the audit report accepted by the assessee, and file the ITR before the due date.

Hands-on: how to actually do it

Step 1 — Pick the ITR form

Form	Who files it
ITR-1 (Sahaj)	Resident individual, income ≤ ₹50L: salary, one house property, other sources
ITR-2	Individual/HUF with capital gains, >1 house, foreign assets — **no business income**
ITR-3	Individual/HUF with **business/professional income** (incl. F&O, intraday)
ITR-4 (Sugam)	Presumptive income u/s 44AD/44ADA/44AE, turnover ≤ ₹2 crore
ITR-5	Firms, LLPs, AOP, BOI
ITR-6	Companies (other than those claiming 11 exemption)
ITR-7	Trusts, political parties, u/s 139(4A)–(4D)

Step 2 — Reconcile 26AS / AIS to books (the daily task)

Pull TDS from 26AS and match to your books. In Excel:

```
=XLOOKUP(A2, TDS_26AS[TAN], TDS_26AS[TaxDeducted], 0)          // TDS as per 26AS for each par
=B2 - C2                                                        // Books TDS minus 26AS TDS
=IF(ABS(D2)<1, "Match", IF(D2>0,"Short in 26AS","Excess in 26AS"))
```

Reconcile receipts against AIS SFT entries:

```
import pandas as pd
books = pd.read_excel("sales_ledger.xlsx")          # party, invoice, amount
ais   = pd.read_excel("ais_export.xlsx")            # party, reported_amount

recon = books.merge(ais, on="party", how="outer", suffixes=("_books","_ais"))
recon["diff"] = recon["amount"].fillna(0) - recon["reported_amount"].fillna(0)
recon["flag"] = recon["diff"].apply(lambda d: "OK" if abs(d) < 1 else "INVESTIGATE")
recon.to_excel("ais_recon.xlsx", index=False)
```

Step 3 — Test tax-audit applicability (Section 44AB)

```
def tax_audit_required(turnover, cash_receipts_pct, cash_payments_pct,
                       is_profession, gross_receipts_prof,
                       opted_presumptive=False, declared_below_presumptive=False):
    if is_profession:
        if gross_receipts_prof > 50_00_000:          # 44AB(b): ₹50 lakh
            return "Audit required (profession)"
    else:
        # Business: ₹1 cr default; ₹10 cr if cash ≤5% both ways
        if cash_receipts_pct ≤ 5 and cash_payments_pct ≤ 5:
            limit = 10_00_00_000
        else:
            limit = 1_00_00_000
        if turnover > limit:
            return f"Audit required (business, limit ₹{limit:,})"
    if declared_below_presumptive and turnover > 0:  # 44AB(e)
        return "Audit required (declared below presumptive)"
    return "No audit required"

print(tax_audit_required(9_50_00_000, 3, 4, False, 0))   # → No audit (₹10cr limit)
print(tax_audit_required(1_20_00_000, 40, 30, False, 0)) # → Audit required (₹1cr)
```

Step 4 — Choose the audit report form

- **Form 3CA** — accounts already audited under another law (Companies Act, LLP Act). CA just annexes 3CD.
- **Form 3CB** — accounts NOT audited under any other law (proprietor, most partnerships). CA gives an opinion + annexes 3CD.
- **Form 3CD** — the 44-clause statement of particulars, annexed to *both* 3CA and 3CB.

Step 5 — Key Form 3CD clauses (where time goes)

Clause	What it captures
12	Presumptive income (44AD/44ADA etc.)
16	Amounts not credited to P&L
21(a)	Personal/capital/penalty expenses disallowed
21(b)	Disallowance u/s 40(a) — TDS not deducted/paid
26	Section 43B — statutory dues paid vs. outstanding
27	CENVAT/GST credit, prior-period items
31	Loans/deposits u/s 269SS/269T (cash > ₹20,000)
34	TDS/TCS compliance — deducted, deposited, return filed

Step 6 — Journal for the tax provision

Particulars	Dr (₹)	Cr (₹)
Income Tax Expense (P&L)	3,60,000	
To Provision for Tax		3,60,000
(Being provision for current tax created)		
Advance Tax Paid A/c	2,80,000	
TDS Receivable A/c	60,000	
To Bank		2,80,000
Provision for Tax	3,60,000	
To Advance Tax Paid		2,80,000
To TDS Receivable		60,000
To Provision — Balance Payable		20,000

Worked example / mini-project

Client: Sharma Traders (proprietor, wholesale). FY 2025-26.

Item	Amount (₹)
Turnover (all digital)	3,40,00,000
Cash receipts	4,20,000 (1.2%)
Cash payments	6,80,000 (2.0%)
Net profit as per books	24,50,000
Depreciation (books)	3,10,000
Depreciation (Income Tax Act)	3,85,000
Interest to bank paid before due date	1,20,000
GST payable outstanding on 30 Sep (43B)	90,000

1. Audit applicability: cash both ≤ 5% → limit is ₹10 cr. Turnover ₹3.4 cr < ₹10 cr → **no 44AB audit** unless declaring below 44AD's 6%/8% presumptive. Profit ₹24.5L is 7.2% of turnover, above 6% → **no audit**. Suppose the client instead declares only ₹18L (5.3%) → **audit triggered under 44AB(e)**.

2. Assume audit applies. Compute business income:

```
Net profit per books                    24,50,000
Add: Book depreciation                  +3,10,000
Less: Tax depreciation                  -3,85,000
Add: GST unpaid u/s 43B (Clause 26)      +90,000
────────────────────────────────────────────────
Business income                         24,65,000
```

3. Tax (individual, new regime, FY25-26 slabs):

```
Income                 24,65,000
Tax:
  0-4L    : 0
  4-8L    : 5% x 4L      = 20,000
  8-12L   : 10% x 4L     = 40,000
  12-16L  : 15% x 4L     = 60,000
  16-20L  : 20% x 4L     = 80,000
  20-24.65L:25% x 4.65L  = 1,16,250
  Total tax             = 3,16,250
  +4% cess              = 12,650
  Total liability       = 3,28,900
```

4. Interest 234B/C applies if advance tax < 90% paid — compute and add. **5. Reconcile TDS credit from 26AS, adjust, arrive at payable/refund. 6.** CA generates **UDIN**, uploads 3CB-3CD, client accepts, then ITR-3 is filed.

Reproduce this in a spreadsheet: one tab for book-to-tax adjustments, one for the slab calc using nested `IF` / `SUMPRODUCT`, one for the 3CD Clause 26 schedule.

How it's tested

Interview questions: - "Turnover is ₹8 crore, 100% digital — is tax audit required?" (No, ₹10 cr limit.) - "Client is a doctor with ₹55L receipts — which form, is audit needed?" (ITR-3; audit yes, > ₹50L.) - "Difference between 3CA and 3CB?" - "What is Clause 34 and why does it matter?" (TDS compliance; drives 40(a)(ia) disallowance.) - "What is UDIN and when is it generated?"

Practical assessment: Firms hand you a **Tally backup + 26AS PDF** and a laptop, and ask you to (a) prepare the tax-audit computation, (b) fill a blank 3CD in the utility, and (c) tell them the tax payable — timed, 60–90 minutes. Big 4 tax teams also give a **book-to-tax reconciliation case** in Excel. Expect a short "spot the disallowance" test on a sample P&L.

Common mistakes & how pros avoid them

Mistake	Fix
Wrong ITR form (ITR-1 with capital gains) → defective u/s 139(9)	Map income sources to the form table first
Ignoring AIS/TIS, filing only on books	Always download AIS; reconcile before filing
Filing ITR before uploading & accepting the audit report	Sequence: 3CB-3CD → UDIN → assessee acceptance → then ITR
Missing UDIN within 60 days	Generate UDIN at report signing, not later
Treating the ₹10 cr limit as automatic	It needs BOTH cash receipts and payments ≤ 5%
Forgetting 43B add-back for unpaid statutory dues	Reconcile GST/PF/ESI paid vs. outstanding at year-end
Not reporting 40(a)(ia) TDS defaults in Clause 34/21(b)	Cross-check TDS returns to expense ledgers

Learn-it roadmap & resources

Time to job-ready: 8–12 weeks of focused practice alongside CA Inter tax study.

- **Weeks 1–2:** ITR form selection, heads of income, new vs. old regime slabs.
- **Weeks 3–4:** 26AS/AIS reconciliation in Excel; file 3–4 dummy ITR-1/ITR-2 on the portal's offline JSON utility.
- **Weeks 5–8:** Section 44AB applicability drills; fill 3CD clause-by-clause on sample data.
- **Weeks 9–12:** Full mock — Tally backup to filed ITR-3 with 3CB-3CD.

Resources (free): Income Tax portal (incometax.gov.in) — offline utilities, FAQs; ICAI *Guidance Note on Tax Audit u/s 44AB*; department's YouTube how-to-file videos; ClearTax and Tax Guru blogs for worked examples. **Paid:** any CA-firm articleship (the real training ground), Udemy "Income Tax Return Filing" practical courses. **Certification:** the **CA qualification** itself authorizes signing tax audits; CA Inter + articleship experience is what employers screen for.

Quick-reference

Trigger	Threshold
Business tax audit (default)	Turnover > ₹1 crore
Business tax audit (cash ≤ 5% both ways)	Turnover > ₹10 crore
Profession tax audit	Gross receipts > ₹50 lakh
Presumptive business (44AD)	Turnover ≤ ₹2 crore, deem 6% (digital)/8% (cash)
Presumptive profession (44ADA)	Receipts ≤ ₹75 lakh, deem 50%
269SS/269T cash loan limit	₹20,000

Form map: 3CA = accounts audited elsewhere · 3CB = not audited elsewhere · 3CD = 44-clause annexure (always).

Filing sequence: Books close → 26AS/AIS reconcile → 3CB-3CD prepared → **UDIN** → CA uploads → assessee accepts → **ITR filed**.

Key clauses: 21(b) = TDS disallowance · 26 = 43B dues · 31 = cash loans · 34 = TDS compliance.

Due dates (typical): Non-audit ITR — 31 Jul · Tax audit report — 30 Sep · Audited ITR — 31 Oct.

Interest: 234A (late filing) · 234B (< 90% advance tax) · 234C (deferment of instalments) — all @ 1% p.m.

Tax Tools & Portals

What it is & where it's used

Indian tax compliance runs on four government portals and a handful of accounting/tax software packages. You don't "know tax" for a job — you *operate* these tools:

Tool	Owner	What it does
GSTN portal (gst.gov.in)	Goods & Services Tax Network	File GSTR-1/3B/9, generate e-invoices & e-way bills, claim ITC, reconcile GSTR-2B
TRACES (tdscpc.gov.in)	Income Tax Dept	TDS returns' aftermath — download Form 16/16A, justification reports, correct TDS defaults, Form 26AS
Income-tax e-filing (incometax.gov.in)	Income Tax Dept	File ITR, respond to notices, e-verify, pay tax, view AIS/TIS
ClearTax / Zoho Books / TallyPrime	Private	Bulk GST filing, invoicing, books of accounts, reconciliation at scale

Roles that live in these tools daily: GST executive, TDS/accounts executive, statutory compliance analyst, audit associate (Big 4 and mid-tier), tax consultant, and every SME accountant. If a JD says "hands-on GST & TDS compliance," it means these exact screens.

The gap: why companies want this (and college didn't teach it)

An MBA/CA-theory background teaches you *sections* — Section 16 ITC conditions, Section 194J TDS on professional fees, Rule 36(4). Employers don't pay you to recite Section 16; they pay you to **click the right buttons, catch a ₹40,000 ITC mismatch, and file before the 20th** without breaking anything.

The gap is procedural and unforgiving: - College says "reconcile purchases with GSTR-2B." No one showed you the actual GSTR-2B JSON, how to VLOOKUP it against your purchase register, or what to do with an invoice the vendor never uploaded. - College says "deduct TDS." No one made you generate a challan, map it in an FVU file, and download a corrected Form 16A after a short-deduction default. - Portals change constantly (new IMS on GSTN, AIS on e-filing). Textbooks are always 2 years stale.

Companies want someone who has *touched the portal*, because the cost of a wrong click is real: late fees, interest, blocked ITC, and notices.

What "proficient" looks like

A job-ready person can, unaided:

1. Log into GSTN, upload a GSTR-1 via offline tool JSON, file GSTR-3B, and know the offset order (IGST → CGST/SGST).
2. Pull GSTR-2B, reconcile it against the purchase register, and produce a "match / mismatch / not-in-2B" report.
3. Generate an e-invoice IRN and an e-way bill, and cancel one within the time window.
4. On TRACES: download Form 16A, read a Justification Report, and file a TDS correction for a wrong PAN.

5. On e-filing: reconcile Form 26AS + AIS with books, file an ITR, and respond to a 143(1) intimation.
6. In Tally/Zoho: pass a GST-compliant sales entry, run GSTR-1 report, and export the return.

The bar is **speed + zero rework**. A senior can file a month's returns for 20 GSTINs; a junior does 2–3 cleanly and reconciles the rest.

Hands-on: how to actually do it

GSTN portal — file GSTR-3B (click-path)

```
Login → Returns Dashboard → select FY & month → GSTR-3B "Prepare Online"
→ 3.1 Outward supplies: enter taxable value, IGST/CGST/SGST
→ 4  Eligible ITC: auto-drafts from 2B; edit 4(A)(5) All other ITC
→ 5  Exempt/nil → 6.1 Payment of tax
→ "Make Payment/Create Challan" if cash needed → File with EVC/DSC
```

Offset rule the portal enforces: **IGST credit first fully utilised, then CGST, then SGST**, cross-utilisation IGST↔CGST/SGST allowed, but CGST credit can never pay SGST.

GSTR-2B reconciliation in Excel

Purchase register in one sheet, downloaded 2B in another. Match on **GSTIN + Invoice No + Taxable value**:

```
=XLOOKUP(A2&"|"&B2, Reg!$A:$A&"|"&Reg!$B:$B, Reg!$D:$D, "NOT IN BOOKS")
```

Flag ITC differences:

```
=IF(ABS(D2-E2)≤1,"Match",IF(E2="NOT IN 2B","ITC not available yet","MISMATCH ₹"&D2-E2))
```

Rounding tolerance of ₹1 avoids false mismatches from paise rounding.

e-Way bill validity (know the formula)

Normal cargo: **1 day per 200 km** (part thereof = +1 day). 640 km → ceil(640/200) = 4 days.

```
=CEILING(distance_km/200,1)  ' days of validity
```

TRACES — download Form 16A

```
Login (deductor) → Downloads → Form 16A → enter FY, Quarter, Form type (26Q)
→ submit request → Requested Downloads → download .zip
→ open in TRACES PDF Generation Utility → outputs password-protected PDF
```

Form 16A PDF password = **TAN in caps** (e.g., `BLRA12345B`) unless changed.

Income-tax e-filing — pay TDS challan (Minor head 200)

```
e-Pay Tax → New Payment → TDS/TCS (Challan 281) → AY → Minor head 200 (regular)
or 400 (demand) → fill section (e.g., 94C) → Net banking → download CIN challan
```

Journal entries — TDS on ₹1,00,000 professional fee (194J, 10%)

Date	Particulars	Dr (₹)	Cr (₹)
Invoice	Professional Fees A/c Dr	1,00,000	
	To Consultant A/c		90,000
	To TDS Payable (194J) A/c		10,000
Payment	Consultant A/c Dr	90,000	
	To Bank A/c		90,000
Deposit	TDS Payable A/c Dr	10,000	
	To Bank A/c		10,000

TallyPrime — GST sales entry click-path

```
Gateway of Tally → Vouchers → F8 Sales → party name → sales ledger (with GST rate)
→ item + qty + rate → CGST/SGST ledgers auto-calc → Ctrl+A save
Reports: Display More Reports → GST Reports → GSTR-1 → export JSON for portal
```

Worked example / mini-project

Scenario: You run March compliance for *Acme Traders Pvt Ltd*, Karnataka (intra-state).

Sales (outward): ₹10,00,000 taxable, GST 18% → CGST ₹90,000 + SGST ₹90,000. Purchases per books: ₹6,00,000 taxable, ITC CGST ₹54,000 + SGST ₹54,000. Downloaded **GSTR-2B** shows only ₹5,00,000 (ITC ₹45,000 + ₹45,000) — one vendor (₹1,00,000, ITC ₹9,000+₹9,000) hasn't filed.

Step 1 — Reconcile. Your Excel match flags that vendor invoice as "NOT IN 2B." Per Section 16(2)(aa)/Rule 36(4), you can claim ITC **only** to the extent in 2B.

Step 2 — Claimable ITC: CGST ₹45,000 + SGST ₹45,000 (not ₹54,000).

Step 3 — Net cash payable in GSTR-3B:

Head	Output tax	ITC (2B)	Cash payable
CGST	₹90,000	₹45,000	₹45,000
SGST	₹90,000	₹45,000	₹45,000
Total	₹1,80,000	₹90,000	**₹90,000**

Step 4 — Challan: Create PMT-06 for ₹90,000, pay, then file GSTR-3B.

Step 5 — Follow-up: Email the defaulting vendor to file their GSTR-1. When it appears in a later month's 2B, claim the ₹18,000. Record it in an "ITC pending" tracker so it isn't forgotten (it lapses if not claimed by Nov 30 of next FY).

Reproduce this in a spreadsheet — it's exactly what a GST executive does 12 times a year per client.

How it's tested

Interview questions: - "Walk me through filing GSTR-3B. What's the ITC set-off order?" - "Vendor didn't upload an invoice — can I claim ITC? Until when?" - "Difference between GSTR-2A and 2B?" (2A is dynamic/live; 2B is static, generated 14th, and is the basis for ITC.) - "TDS on ₹50,000 rent to an individual — which section, rate?" (194I, 10% on land/building.) - "What's in Form 26AS vs AIS?"

Practical assessments companies actually give: - A **timed reconciliation test**: here's a purchase register and a GSTR-2B export — produce the mismatch report in Excel in 30 minutes. - A **Tally case**: "Enter these 10 vouchers and generate GSTR-1." They check your GST ledgers and HSN. - A **login demo**: they hand you a sandbox/their portal and say "show me where you'd download Form 16A" or "file this nil GSTR-3B." - **TDS challan mapping**: given payments, compute TDS, section, and due dates.

Common mistakes & how pros avoid them

Mistake	Consequence	Pro habit
Claiming full book ITC, ignoring 2B	Notice, ITC reversal + 24% interest	Reconcile 2B *every* month, claim only what's reflected
Filing 3B before reconciling 2B	Locked-in error (3B can't be revised)	2B recon → then 3B, in that order
Wrong TDS section/rate	Short-deduction default on TRACES	Keep a section-rate cheat-sheet; verify PAN status
Missing e-invoice for B2B (turnover > ₹5 cr)	Invoice invalid, ITC denied to buyer	Auto-generate IRN at billing in Tally/Zoho
Filing GSTR-1 after 3B	3B ITC/liability mismatch	Always GSTR-1 first, then 3B
Forgetting DSC vs EVC	Companies/LLPs *must* use DSC	Keep DSC token drivers installed & tested
Ignoring AIS before ITR	Mismatch notice u/s 143(1)	Reconcile AIS + 26AS + books first

Pros also **never file on the last day** — portals crash on the 20th. They keep a filing calendar and file 2–3 days early.

Learn-it roadmap & resources

Time to job-ready: 6–10 weeks of hands-on practice alongside theory.

Week	Focus
1–2	GST basics + GSTN portal navigation; file a dummy nil GSTR-3B/1
3–4	GSTR-2B reconciliation in Excel; e-invoice + e-way bill
5–6	TDS: sections, challans, TRACES, Form 16/16A
7–8	TallyPrime GST + Zoho Books; end-to-end monthly cycle
9–10	Income-tax e-filing: ITR, 26AS/AIS, notices

Resources: - **Free:** GSTN's own tutorials (tutorial.gst.gov.in), TRACES e-tutorials, Income Tax Dept YouTube, CBIC flyers. Practice on GSTN's offline tools (free download). - **Paid:** ClearTax GST certification, Tally's TallyPrime with GST course (TallyEducation), Udemy "GST + TDS practical" courses (₹500–1,500 on sale). - **Certifications that signal proficiency:** Tally ACE/Professional, ClearTax certifications, ICAI's GST certificate course (if eligible).

Best practice: get a **free trial ClearTax/Zoho account** and a GSTN test login, and actually file dummy returns. Nothing beats muscle memory on the real screens.

Quick-reference

GST return due dates

Return	Frequency	Due
GSTR-1	Monthly / QRMP	11th / 13th
GSTR-3B	Monthly / QRMP	20th / 22nd–24th
GSTR-9 (annual)	Yearly	31 Dec

Common TDS sections

Section	Payment	Rate
192	Salary	Slab
194C	Contractor	1% (indiv) / 2% (co.)
194J	Professional/technical	10% / 2%
194I	Rent (land/bldg)	10%
194H	Commission	2%
194Q	Purchase of goods >₹50L	0.1%

Key portal facts - GSTR-2B generated on the **14th**; basis for ITC. GSTR-2A is dynamic. - ITC set-off: **IGST first**, then CGST, then SGST (no CGST↔SGST). - e-Way bill validity: **1 day / 200 km**. - Form 16A password = **TAN (caps)**. - TDS deposit due: **7th of next month** (April–Feb); March by **30 April**. - Company/LLP e-filing: **DSC mandatory**; others EVC allowed. - ITC lapse deadline: **30 Nov** of next FY.

Payroll & compliance basics

What it is & where it's used

Payroll is the monthly process of converting an employee's CTC into a **net-pay number in their bank account**, while simultaneously deducting and depositing the right taxes and social-security contributions with the government — TDS, PF, ESI, Professional Tax. Get it wrong and you don't just annoy staff; you trigger interest, penalties, and a personal liability on the "principal officer" for TDS defaults.

Roles that live in this:

- **Payroll executive / HR-Ops** — runs the monthly cycle, files challans.
- **Accounts / Finance associate** — books the payroll journal, reconciles the salary-payable and statutory-dues ledgers.
- **Tax associate** — computes TDS on salary (Sec 192), issues Form 16, files Form 24Q.
- **Consultant / articleship (CA)** — verifies client payroll, catches PF/ESI wage-base errors in audit.

Even if you never "do payroll," every finance person books the salary journal and answers the auditor's question: "*Does statutory dues payable tie to what was actually deposited?*"

The gap: why companies want this (and college didn't teach it)

MBA and CA-Inter teach you that "salary is an expense" and that "Section 192 requires TDS." What they don't teach:

- The difference between **CTC, gross, and take-home** — three numbers that confuse every fresher and every new joiner you'll have to explain it to.
- That PF is on **Basic + DA** (capped at ₹15,000 wage), ESI is on **gross ≤ ₹21,000**, and PT is a **slab by state** — three different wage bases, three different rules.
- The **old vs new tax regime** choice and how it changes monthly TDS.
- That TDS on salary is deposited by the **7th**, PF by the **15th**, ESI by the **15th**, and the quarterly **24Q** return is what actually populates the employee's Form 26AS/AIS.

Colleges test definitions; employers test whether you can produce a payroll register, a challan, and a Form 16 that reconciles. That reconciliation muscle is the gap.

What "proficient" looks like

A job-ready person can, unaided:

- Build a **salary structure** from a CTC number, splitting Basic/HRA/allowances correctly.
- Compute **monthly TDS** under both regimes by annualising, applying deductions, and dividing tax by remaining months.
- Calculate **PF (12%+12%), ESI (0.75%+3.25%), PT** on the correct wage bases.
- Produce a **payroll register**, pass the **journal entry**, and reconcile statutory-dues ledgers to challans paid.
- Know the **due dates** and the **forms** (24Q, ECR, ESI return, Form 16 Part A + B).

Hands-on: how to actually do it

1. Salary structure from CTC (Excel)

A common ₹9,00,000 CTC split. Layout with formulas (Basic = 40% of CTC is a typical policy):

Component	Formula (cell refs)	Monthly	Annual
Basic	`=ROUND(CTC*0.40/12,0)`	30,000	3,60,000
HRA	`=ROUND(Basic*0.50,0)` (metro)	15,000	1,80,000
Special allowance	balancing figure	20,500	2,46,000
Employer PF	`=ROUND(MIN(Basic,15000)*0.12,0)`	1,800	21,600
CTC	`=SUM( … )`	**75,000**	**9,00,000**

Gross = Basic + HRA + Special = ₹65,500/month. Employer PF is part of CTC but **not** part of gross.

2. Statutory deductions (the three wage bases)

```
Employee PF   =ROUND(MIN(Basic_DA,15000)*0.12,0)        ' 1,800
ESI (if gross≤21000) =ROUND(Gross*0.0075,0)             ' 0 here, gross>21k
PT (Karnataka) =IF(Gross≥25000,200,0)                   ' 200
```

ESI applies only when gross ≤ ₹21,000 (₹25,000 for disabled). Employer ESI = 3.25% of gross.

3. Monthly TDS on salary — the annualise-and-divide method

```
Step 1  Projected annual gross          = monthly gross × 12  (+ bonus already known)
Step 2  Less: exemptions (HRA, std ded ₹50k)
Step 3  Less: Chapter VI-A (80C, 80D…) — only if OLD regime
Step 4  = Taxable income → apply slab   → gross tax
Step 5  + 4% cess                       → total tax liability
Step 6  Monthly TDS = (total tax − TDS already deducted) / months remaining
```

Excel slab tax (new regime FY 2024-25) using a helper approach:

```
=ROUND(
  MAX(0,MIN(TI,700000)-300000)*0.05
 +MAX(0,MIN(TI,1000000)-700000)*0.10
 +MAX(0,MIN(TI,1200000)-1000000)*0.15
 +MAX(0,MIN(TI,1500000)-1200000)*0.20
 +MAX(0,TI-1500000)*0.30 ,0)
```

Then `Tax_with_cess = Tax*1.04`, and apply the **87A rebate** (nil tax if taxable ≤ ₹7,00,000 under new regime).

4. The payroll journal entry

For total gross ₹6,55,000, employee PF ₹18,000, TDS ₹40,000, ESI ee ₹0, PT ₹2,000, plus employer PF ₹18,000:

Account	Dr	Cr
Salaries & Wages A/c	6,55,000	
Employer PF Contribution A/c	18,000	
To TDS Payable (Sec 192)		40,000
To PF Payable (ee 18,000 + er 18,000)		36,000
To PT Payable		2,000
To Salary Payable (net)		5,95,000

On payment of net salary: `Salary Payable Dr / To Bank`. On depositing each statutory due: `PF Payable Dr / To Bank`, etc. The payable ledgers should hit **zero** after deposit — that is your reconciliation.

5. Portal click-paths (compliance run)

- **TDS challan (ITNS 281):** incometax.gov.in → e-Pay Tax → New Payment → *TDS on Salary (192)* → Assessment Year → fill tax/interest → pay → save the **CIN/BSR + challan no.** By the **7th**.
- **PF (ECR):** unifiedportal-emp.epfindia.gov.in → Payments → ECR Upload → upload text file → verify → Prepare Challan → Pay. By the **15th**.
- **ESI:** esic.in → File Monthly Contribution → enter wages → Submit → Pay online. By the **15th**.
- **Form 24Q (quarterly):** prepare via RPU utility → validate with FVU → upload on TRACES/e-filing → download **Form 16 Part A** from TRACES after processing.

Worked example / mini-project

Rohan, ₹9,00,000 CTC, joins 1 Apr, new regime, no 80C. Reproduce his year:

- Monthly gross = ₹65,500 → annual gross = ₹7,86,000.
- Less standard deduction ₹50,000 → **taxable = ₹7,36,000**.
- Slab tax (new): 3–7L @5% = ₹20,000; 7–7.36L @10% = ₹3,600 → ₹23,600.
- 87A rebate: taxable > ₹7,00,000, so **no rebate**. Tax = ₹23,600.
- Cess 4% = ₹944 → **annual tax = ₹24,544**.
- **Monthly TDS = 24,544 / 12 = ₹2,045** (round; adjust in March).

Monthly deductions: PF ₹1,800 + PT ₹200 + TDS ₹2,045 = ₹4,045. **Take-home = 65,500 − 4,045 = ₹61,455.**

Now build the full 12-row payroll register in Excel with columns *Basic, HRA, Special, Gross, PF, PT, TDS, Net*, sum the TDS column (₹24,540) and confirm it matches the four 24Q quarterly returns and Rohan's Form 16 Part B. If they tie, your compliance run is clean.

How it's tested

Interview questions: - "Difference between CTC, gross and net?" (you must draw the three wage bases) - "What's the TDS due date and what happens if you miss it?" (7th; 1.5%/month interest + Sec 40(a)(ia) disallowance) - "PF is calculated on what?" (Basic + DA, capped ₹15,000) - "Old vs new regime — which gives lower TDS for someone with ₹1.5L 80C and ₹2L home-loan interest?"

Practical assessments: - A **timed Excel test**: "Here's a CTC, build the salary structure and compute monthly TDS both regimes." (45 min) - A **reconciliation case**: "Statutory dues payable shows ₹1,10,000; challans total ₹95,000 — find the ₹15,000 gap." (usually an employer-PF line not deposited or an admin-charge omission) - "**Prepare Form 16 Part B** from this payroll register."

Common mistakes & how pros avoid them

Mistake	Fix
PF on gross instead of Basic+DA	Wage base = Basic+DA, `MIN( ... ,15000)` cap
Forgetting **employer** PF/ESI in CTC and in the journal	Employer contributions are a cost + a payable, not just employee side
ESI applied above ₹21,000 gross	ESI ceases once gross crosses ₹21,000 (continues till contribution-period end)
Depositing TDS but not filing 24Q	No 24Q → nothing in employee's 26AS → Form 16 Part A can't generate
Rebate 87A confusion	New regime: nil tax ≤ ₹7L taxable; ₹1 over means full slab tax (marginal relief applies)
Standard deduction missed	₹50,000 (₹75,000 new regime FY24-25) available to all salaried
Statutory dues not reconciled to challans	Every payable ledger must zero out post-deposit each month

Pros keep a **compliance calendar** (7th TDS, 15th PF/ESI, quarter-end 24Q) and never close the month until payable ledgers reconcile to the CIN/challan numbers.

Learn-it roadmap & resources

Time to proficiency: 3–4 weeks if you already know basic accounting.

- Week 1 — Salary structuring + the three wage bases. Build 5 dummy structures in Excel.
- Week 2 — TDS on salary both regimes; reproduce the worked example, then vary 80C/HRA.
- Week 3 — Run one full mock cycle: register → journal → challans → 24Q → Form 16.
- Week 4 — Do it inside **TallyPrime payroll** and on the **EPFO/ESIC demo** flows.

Resources: - **Free:** ClearTax and TaxAdda salary/TDS guides; EPFO & ESIC official portals; income-tax e-filing help section; RPU utility docs (Protean/NSDL). - **Paid:** TallyPrime payroll module; Udemy "Indian Payroll & Statutory Compliance" courses; greytHR / Zoho Payroll free trials to see a real SaaS payroll run. - **Certification:** CA-Inter Taxation already covers Sec 192; NISM has none here, but greytHR Academy and Tally certifications signal payroll-tool fluency.

Quick-reference

Item	Rule / Rate	Due date	Form
TDS on salary	Sec 192, avg rate, annualise ÷ months	7th next month	Challan 281 → 24Q (qtrly)
PF	12% ee + 12% er on Basic+DA (cap ₹15,000)	15th	ECR
ESI	0.75% ee + 3.25% er on gross ≤ ₹21,000	15th	Monthly contribution
Professional Tax	State slab (KA/MH ≈ ₹200/mo max ₹2,500/yr)	varies by state	State PT return
Form 16	Part A (TRACES) + Part B (employer)	by 15 Jun	Form 16
Standard deduction	₹50,000 old / ₹75,000 new (FY24-25)	—	—
87A rebate (new)	Nil tax if taxable ≤ ₹7,00,000	—	—
Late TDS interest	1% (not deducted) / 1.5% (not paid) per month	—	—

Golden checks: gross = Basic+HRA+allowances (no employer PF); take-home = gross − PF − PT − TDS − ESI(ee); every statutory-payable ledger must zero out after deposit; TDS column total = sum of four 24Q returns = Form 16 Part B.

Handling tax notices & assessments

What it is & where it's used

A tax notice is a formal communication from the Income Tax Department asking you to confirm something, produce evidence, or explain a mismatch. Handling them means: reading the notice correctly, identifying the section it was issued under, gathering the right documents, drafting a reply, and filing the response on the e-filing / e-proceedings portal before the deadline — without escalating a routine query into a full-blown assessment or penalty.

This is one of the most billed, most feared, and least-taught skills in Indian finance. It shows up everywhere:

Role	How they touch notices
Accounts / Finance executive	Receives 143(1) intimations for the company, reconciles the mismatch, books the demand/refund
Tax associate (CA firm / Big 4)	Drafts replies to 143(2), 142(1), 148 for dozens of clients
Payroll / TDS team	Handles TRACES defaults, 200A intimations, short-deduction notices
Finance manager / controller	Signs off on representation, manages assessment exposure and provisions
Founder / SME owner	Personally liable — needs someone who can read a notice and not panic

The moment a notice lands in the inbox, everyone looks for the one person who "knows how to handle it." That person gets promoted and paid.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you the *theory* of assessment — sections, timelines, appellate hierarchy — as exam facts. It never shows you an actual notice PDF, never makes you log into the portal, never makes you draft a reply under a 15-day clock. The gap is entirely operational:

- College teaches "Section 143(1) is a summary assessment." It never shows you that the intimation is an auto-generated CPC document where **Column A (as filed)** and **Column B (as computed)** sit side by side, and your whole job is to explain the delta.
- College never teaches the **e-Proceedings** tab, the difference between a mere *intimation* and an *adverse* notice, or that ignoring a 143(1) refund-adjustment proposal for 30 days = deemed acceptance.
- Faceless assessment (from 2020) changed everything to written submissions uploaded online. There is no "going to the officer's chamber" anymore — your **drafting quality is the representation.** MBAs are never taught to write a submission.

Employers pay for the person who can turn a scary PDF into a calm, evidenced, on-time reply. That is a learnable, mechanical skill — and it's the gap this chapter closes.

What "proficient" looks like

A job-ready person, handed a notice, can unaided:

1. **Identify the section and sub-type** within 60 seconds (intimation vs scrutiny vs reassessment).
2. State the **response deadline and consequence of missing it.**
3. Pull the **matching data** — ITR, Form 26AS, AIS/TIS, Form 16/16A, books, bank statements — and build a reconciliation.
4. Draft a **point-wise reply** that answers each query with a document reference (Annexure-1, -2...).
5. File it via **e-Proceedings**, attach a consolidated PDF, and download the acknowledgement.
6. Know when to **agree and pay**, when to **contest**, and when to escalate to a CA/counsel.

Hands-on: how to actually do it

Step 1 — Decode the section

Section	What it is	Trigger	Typical deadline	Consequence of ignoring
143(1)	Intimation (summary processing by CPC)	Arithmetic error, TDS/tax mismatch, disallowance u/s 143(1)(a)	30 days to respond to adjustment proposal	Adjustment auto-confirmed; demand raised
142(1)	Inquiry before assessment	Non-filing, or AO wants documents/info	As stated (often 15 days)	Best-judgment assessment u/s 144; penalty u/s 272A
143(2)	Scrutiny notice	Return picked for detailed scrutiny (CASS/manual)	Must be **served within 3 months from end of FY in which return filed**; reply as scheduled	Assessment proceeds; adverse additions
148	Income escaping assessment (reassessment)	AO has "information" income escaped; preceded by 148A show-cause	As stated (30 days typical)	Reassessment, tax + interest + penalty
156	Notice of demand	Follows any order creating a demand	30 days to pay	1% p.m. interest u/s 220(2); recovery
245	Adjustment of refund against demand	Old demand outstanding	30 days to respond	Refund adjusted

Step 2 — Log in and locate it

Portal click-path (incometax.gov.in):

```
Login → Pending Actions → e-Proceedings
→ select the AY → View Notices
→ read the notice PDF + "Response" button
→ Submit Response → upload PDF (max ~5 MB/attachment)
→ e-Verify (Aadhaar OTP / DSC) → download Acknowledgement (Transaction ID)
```

For 143(1) mismatches specifically: `Pending Actions → Response to Outstanding Demand` or the **"Agree / Disagree"** grid inside the intimation.

Step 3 — Build the reconciliation (Excel)

The core analytical task in 90% of notices is: *the department's number ≠ your number — explain the gap.* A 143(1)(a) TDS mismatch is a classic:

```
' Match income booked vs AIS/26AS by TAN/party
=XLOOKUP(A2, AIS!$A:$A, AIS!$C:$C, "NOT IN AIS")

' Gap column
=Books_Amount - AIS_Amount

' Flag material differences
=IF(ABS(D2)>1000, "EXPLAIN", "OK")

' TDS credit reconciliation
=SUMIF('26AS'!B:B, A2, '26AS'!D:D)   ' TDS per 26AS for this TAN
```

Put it in a clean table you can paste into the reply and attach:

Party / TAN	As per 26AS/AIS (₹)	As per Books/ITR (₹)	Difference (₹)	Reason
ABC Ltd / MUMA1234B	5,00,000	5,00,000	0	Matched
XYZ Bank / DELX9876C	82,400	0	82,400	FD interest — accrued, offered in AY 2024-25

Step 4 — Draft the reply (reusable skeleton)

```
To: The Assessing Officer / CPC / NaFAC
PAN: ABCDE1234F   |   AY: 2024-25
Notice u/s 143(1)(a) dated 12-06-2024, DIN: <DIN>

Sub: Response to proposed adjustment - reg.

1. This is with reference to the captioned intimation proposing an
   adjustment of ₹82,400 on account of interest income appearing in
   Form 26AS but allegedly not offered to tax.

2. The assessee submits that the said interest was offered under
   "Income from Other Sources" in Schedule OS, Row 3 of the return
   (refer Annexure-1: ITR-V / computation extract).

3. The apparent mismatch arises solely due to TDS being reflected in
   26AS in AY 2024-25 while the income was offered on accrual basis.
   Reconciliation enclosed as Annexure-2.

4. It is therefore prayed that the proposed adjustment be dropped and
   the return be processed as filed.

Enclosures: Annexure-1 (Computation), Annexure-2 (Reconciliation),
Annexure-3 (Bank interest certificate)
```

Merge everything into **one indexed PDF** (reply + annexures) before uploading.

Worked example / mini-project

Scenario. Rao Traders Pvt Ltd files ITR for AY 2024-25 declaring total income ₹42,00,000, tax paid ₹10,92,000. On 12 Jun 2024 CPC issues a **143(1)(a)** intimation:

- Adds ₹1,20,000 — "contract receipts in 26AS not reconciled with turnover."
- Disallows ₹35,000 — "late deposit of employee PF, u/s 36(1)(va)."
- Proposed demand: ₹48,050 (tax + interest u/s 234).

Your workflow:

1. **Read** → It's an *intimation with adjustment proposal*, not scrutiny. 30-day clock.
2. **Reconcile the ₹1,20,000.** Pull 26AS contract receipts (Section 194C) vs revenue ledger:

```
=SUMIF('26AS'!C:C, "194C", '26AS'!E:E)      ' 26AS 194C receipts
=SUMIFS(Sales!Amt, Sales!Party, "Bharat Infra")
```

Finding: the ₹1,20,000 is a **March advance** appearing in 26AS (deductor booked it) but recognised as revenue in the *next* year per AS-9/Ind AS 115. **Contest** — it's timing, not escapement.

3. **The ₹35,000 PF.** Check challan dates vs due date. EPF for Feb 2024 (due 15 Mar) was actually deposited **22 Mar**. Post *Checkmate Services (SC, 2022)*, employee-contribution deposited after the PF-Act due date is **not deductible. Agree** — do not fight a lost cause.
4. **Respond on portal:** Disagree on ₹1,20,000 (with reconciliation Annexure), Agree on ₹35,000.
5. **Revised demand** after acceptance: tax on ₹35,000 @ ~26% ≈ ₹9,100 + interest ≈ ₹1,500 = **₹10,600.** Pay via **e-Pay Tax** → **Challan (Minor Head 400 – tax on regular assessment)**.
6. **Journal entries:**

Particulars	Dr (₹)	Cr (₹)
Income Tax Expense (prior period) A/c	10,600	
To Provision for Tax / Bank A/c		10,600
(Being demand u/s 143(1) accepted for PF disallowance, AY 24-25)		

The ₹1,20,000 timing item: **no entry** — it reverses naturally as it's taxed next year. Download the acknowledgement (Transaction ID), file it. Done in under two hours.

How it's tested

Interview questions: - "What's the difference between 143(1) and 143(2)?" (Summary/CPC vs detailed scrutiny/AO.) - "You get a 143(1)(a) with a TDS mismatch — walk me through your first four steps." - "Client got a 148. What do you check before replying?" (Whether 148A procedure was followed, validity of reopening, time limits, the "information" basis.) - "Deadline for responding to a 143(1) adjustment? Consequence of missing it?" (30 days; deemed acceptance.) - "Employee PF deposited late — deductible?" (No, post-Checkmate.)

Practical assessments companies actually give: - **The "here's a notice PDF, draft the reply" test** — a sample intimation and a folder of documents; produce the reconciliation + reply in 60–90 min. - **A mismatch reconciliation in Excel** — 26AS vs books, find and explain every gap. - **A portal walkthrough** — "show me where you'd file the response" (screen-share). - **Case judgment** — a mixed notice where some items should be agreed and some contested; they test whether you know *when to fight*.

Common mistakes & how pros avoid them

Mistake	Why it hurts	Pro habit
Treating every 143(1) as an emergency	Most are refunds/no-action intimations	Read the last page — "no action required" vs adjustment proposal
Missing the 30-day window	Adjustment auto-confirmed, demand raised	Diary every DIN with deadline the day it arrives
Fighting genuinely lost items (late PF, bogus expense)	Invites penalty, wastes goodwill	Concede clean losers, contest strong points only
Replying without annexures / DIN reference	Reply treated as unsubstantiated	Every claim → an annexure; quote DIN & AY on page 1
Uploading 12 separate files	Officer can't follow; looks amateur	One indexed, bookmarked PDF
Not verifying the notice is genuine	Fraud/phishing PDFs exist	Verify DIN on portal: `Authenticate Notice/Order Issued by ITD`
Ignoring AIS/TIS	Half of all mismatches originate there	Reconcile books → AIS/TIS → 26AS every time

Learn-it roadmap & resources

Time to proficiency: 4–6 weeks of deliberate practice alongside a job.

Week	Focus
1	Memorise the section table; read 5 real anonymised notices
2	Master the portal: e-Proceedings, Response to Demand, Authenticate DIN, e-Pay Tax
3	Build 3 reconciliations (26AS/AIS vs books) in Excel end-to-end
4	Draft 3 replies from templates; get them reviewed by a CA
5–6	Shadow live notices; learn agree-vs-contest judgment

Resources: - **Income Tax portal help section** + the "e-Proceedings" user manual (free). - **income-tax.gov.in** → **AIS** and **TRACES** (for TDS/26AS) — practice logins. - Bare Act: **Sections 139–158** (skim), free on incometaxindia.gov.in. - **Taxmann / Taxsutra** blogs for landmark cases (Checkmate, Ashish Agarwal on 148). - ICAI's **Background Material on Faceless Assessment** (free PDF). - Certification: **ICAI Certificate Course on Faceless Assessment**, or any GST+IT practitioner course; the CA Inter Taxation paper itself covers the statutory base.

Quick-reference

Section cheat-sheet - `143(1)` — CPC intimation → **30 days** to respond to adjustment - `142(1)` — inquiry / call for docs → reply or face 144 - `143(2)` — scrutiny → served within **3 months** of FY-end of filing - `148`

— reassessment → preceded by **148A** show-cause - `156` — demand notice → **pay in 30 days**, else 1% p.m. u/s 220(2) - `245` — refund adjusted against old demand

Portal paths - File reply: `Pending Actions → e-Proceedings → Submit Response` - Pay demand: `e-Pay Tax → Minor Head 400` - Verify notice: `Authenticate Notice/Order Issued by ITD` (use DIN)

Excel workhorses - `=XLOOKUP(party, AIS_range, amt_range, "NOT IN AIS")` - `=SUMIF('26AS'!TAN, party, '26AS'!TDS)` - `=IF(ABS(Books-AIS)>1000,"EXPLAIN","OK")`

Golden rules 1. Identify the section first. 2. Note the deadline the day it arrives. 3. Reconcile before you draft. 4. Concede losers, contest winners. 5. One indexed PDF, every claim annexed, DIN on page 1. 6. Always download the acknowledgement.

International tax & transfer pricing basics

What it is & where it's used

Cross-border money movement triggers a second layer of tax rules on top of domestic law. Three things dominate day-to-day practice:

1. **Withholding tax (WHT) on foreign payments** — when an Indian company pays a non-resident (software licence, consultancy, interest, royalty, dividend), it must deduct tax at source under **Section 195** before remitting, and file **Form 15CA/15CB**.
2. **DTAA (Double Taxation Avoidance Agreement)** — treaties India signs with ~95 countries that cap those WHT rates and decide *which* country gets to tax.
3. **Transfer pricing (TP)** — when two related entities (an Indian arm and its US/UAE/Singapore parent) trade with each other, the price must be **arm's length** (what unrelated parties would charge), documented, and reported.

Who does this: International tax analysts and TP associates in Big 4 and MNC in-house teams; finance/AP staff at any GCC (Global Capability Centre) or exporter who process foreign vendor invoices; treasury; and CA-firm associates. If you are targeting a GCC in Bengaluru/Hyderabad/Pune or an MNC shared-services role, this is a core, screened skill.

The gap: why companies want this (and college didn't teach it)

An MBA teaches "double taxation is bad, treaties exist." Employers pay for the *mechanics*: **which rate applies, which form to file, and by when.** Every foreign vendor payment a GCC makes stops at the finance desk asking: *do we withhold, how much, and is a 15CB certificate needed?* Get it wrong and the company either over-deducts (vendor screams) or under-deducts (30% disallowance under **Section 40(a)(i)** plus interest and penalty).

The specific gaps: - Nobody teaches the **195 → DTAA → 15CA/15CB decision tree**. - College never mentions **Tax Residency Certificate (TRC)** and **Form 10F** — the two documents without which you *cannot* apply the lower treaty rate. - TP is taught as economics; industry wants **benchmarking, the five methods, Form 3CEB, and a Local File** — deadlines and documentation, not theory. - **PE (Permanent Establishment)**, **BEPS**, **GAAR**, and now **Pillar Two 15% global minimum tax** are interview-live but classroom-absent.

What "proficient" looks like

A job-ready person can, unaided:

- Take a foreign vendor invoice and determine **taxability, section, DTAA article, and rate** — and explain *why*.
- Fill **Form 15CA (Part A/B/C/D)** and know when a CA's **Form 15CB** is mandatory.
- Apply a **treaty rate** correctly, insisting on **TRC + Form 10F + no-PE declaration** on file.
- Gross-up when the contract is **"net of tax."**
- Explain the **five TP methods** and pick the right one; read a **benchmarking study** and its arm's-length range (inter-quartile).

- Know the **Form 3CEB / Master File (3CEAA) / CbCR (3CEAD)** thresholds and due dates.

Hands-on: how to actually do it

Step 1 — The Section 195 decision tree

```
Is the payee a non-resident?  ── No ──> use normal TDS (194J etc.)
        │ Yes
Is the income taxable in India (Sec 5/9 – source rule)?  ── No ──> no WHT (but 15CA Part D)
        │ Yes
Rate = LOWER of  (a) Income-tax Act rate  and  (b) DTAA rate
        │  (DTAA only if payee gives TRC + Form 10F + no-PE declaration)
Deduct, deposit by 7th of next month, file 15CA (+15CB if remittance is taxable & > ₹5 lakh/yr
```

Step 2 — Common Act vs treaty rates (add surcharge + 4% cess to Act rates)

Payment	Act rate (Sec)	Typical DTAA rate
Royalty / FTS (fees for technical services)	20% (115A)	10–15%
Interest	20% (194LC etc.)	10–15%
Dividend	20% (115A)	5–15%

Step 3 — WHT calculation in Excel

```
# B2 = invoice (foreign, gross), B3 = Act rate, B4 = DTAA rate
# WHT rate actually applied:
=MIN(B3,B4)
# WHT amount (treaty, no surcharge/cess on treaty rate):
=B2*MIN(B3,B4)
# If contract is "NET OF TAX" – gross up:
=B2/(1-MIN(B3,B4))          # grossed-up amount
=B2/(1-MIN(B3,B4))-B2       # tax borne by payer
```

Worked gross-up: vendor must *receive* $10,000 net, treaty rate 10% → gross = 10000/(1−0.10) = **$11,111.11**, WHT = **$1,111.11**.

Step 4 — Journal entries (₹, royalty of ₹10,00,000, 10% treaty WHT, payer bears nothing)

Date	Particulars	Dr (₹)	Cr (₹)
Invoice booked	Royalty expense A/c ...... Dr	10,00,000	
	To Foreign vendor A/c		9,00,000
	To TDS payable (195) A/c		1,00,000
Remittance	Foreign vendor A/c ...... Dr	9,00,000	
	To Bank A/c		9,00,000
Deposit TDS (by 7th)	TDS payable (195) A/c ... Dr	1,00,000	
	To Bank A/c		1,00,000

Step 5 — File Form 15CA / 15CB (income-tax portal)

```
incometax.gov.in → login → e-File → Income Tax Forms → File Now
→ Form 15CA
   Part A : taxable remittance ≤ ₹5 lakh in FY
   Part B : taxable, > ₹5 lakh, order/certificate u/s 195(2) obtained
   Part C : taxable, > ₹5 lakh — REQUIRES CA's Form 15CB first
   Part D : NOT taxable in India (no WHT)
→ For Part C: CA fills & e-verifies Form 15CB (DIN, amount, rate, DTAA article)
→ Acknowledgement number goes to the bank (AD) with Form A2 to release funds.
```

Step 6 — The five transfer-pricing methods

Method	One-line logic	Best when
CUP (Comparable Uncontrolled Price)	Compare the exact price to a third-party price	Commodities, loans, royalties with public comparables
RPM (Resale Price)	Work back from resale margin	Distributor buys & resells with no value-add
CPM (Cost Plus)	Cost + arm's-length markup	Contract manufacturer / service provider on cost
TNMM (Transactional Net Margin)	Compare *net* operating margin (OP/cost, OP/sales)	Default for GCC/IT/ITeS captives — most used in India
PSM (Profit Split)	Split combined profit by contribution	Highly integrated, unique intangibles

TNMM in Excel — the arm's-length range from a comparable set:

```
# C2:C11 = OP/OC margins of 10 comparables; B2 = tested party's OP/OC
=QUARTILE.INC(C2:C11,1)     # lower quartile (arm's-length floor)
=MEDIAN(C2:C11)             # median
=QUARTILE.INC(C2:C11,3)     # upper quartile (ceiling)
=IF(AND(B2≥QUARTILE.INC(C2:C11,1),B2≤QUARTILE.INC(C2:C11,3)),"Arm's length","ADJUST to media
```

Step 7 — TP filing thresholds

Form	What	Trigger / due date
Form 3CEB	Accountant's report on all int'l & specified domestic transactions	Any int'l related-party txn — due **31 Oct**
Master File (3CEAA)	Group-level info	Consolidated group revenue > ₹500 cr **and** int'l txn > ₹50 cr — **30 Nov**
CbCR (3CEAD)	Country-by-country report	Group revenue > **€750 mn** (~₹6,750 cr)

Worked example / mini-project

Setup: *Nimbus Analytics India Pvt Ltd* is a captive (GCC) doing software development for its US parent, billed **cost + 15%**. Total operating cost FY25-26 = **₹40,00,000**. During the year it also pays **£20,000** to a UK consultant for technical services (contract is net of tax).

Part 1 — Transfer price (CPM/TNMM): - Revenue billed to parent = 40,00,000 × 1.15 = **₹46,00,000**; Operating profit = ₹6,00,000; **OP/OC = 15.0%**. - Benchmarking set (10 comparable IT captives), OP/OC margins: 11, 12, 12.5, 13, 14, **14.5, 15, 16, 18, 20**. - Lower quartile ≈ **12.63%**, median **14.25%**, upper quartile **16.5%**. - Tested margin 15% lies inside 12.63%–16.5% → **arm's length. No adjustment. Report in Form 3CEB.**

Part 2 — WHT on the UK payment (FTS): - India–UK DTAA, FTS/royalty rate = **10%** (10F + TRC + no-PE on file). - Net of tax → gross up: 20,000 / (1 − 0.10) = **£22,222.22**; WHT = **£2,222.22**. - At ₹105/£: remit £20,000 (₹21,00,000), deposit WHT ₹2,33,333 by the 7th, file **Form 15CB (CA)** then **15CA Part C** (taxable, > ₹5 lakh), give ack + Form A2 to the bank.

Reproduce it: drop the 10 margins in a column, run the `QUARTILE.INC` formulas, and the gross-up formula for £20,000 — you have re-derived both the TP conclusion and the WHT.

How it's tested

Interview questions: - "Indian co pays a US company for cloud software — is it royalty/FTS? Do you withhold, at what rate, which form?" - "What is a PE and why does the parent care about creating one in India?" - "TRC missing — can you still apply the DTAA rate?" (No — TRC + Form 10F are mandatory.) - "Which TP method for a captive IT services centre, and why?" (TNMM.) - "What is the inter-quartile range and what happens if the tested margin falls below it?" - "One-line on BEPS / Pillar Two 15% minimum tax."

Practical assessments: - **Timed Excel/case:** given an invoice + a mini benchmarking table, compute the WHT, decide the 15CA part, and state whether the TP margin is arm's length. - **Form-fill drill:** map five vendor invoices to 15CA Part A/B/C/D. - **Grossing-up test** on a net-of-tax contract (the classic trip-up). - **Read-a-TP-study:** hand you a benchmarking report, ask you to defend the comparables and the range.

Common mistakes & how pros avoid them

Mistake	Fix
Applying DTAA rate with no **TRC + Form 10F** on file	No documents → use the higher Act rate; collect docs before remitting
Adding **surcharge + cess** on top of the treaty rate	Treaty rate is the ceiling — apply it flat
Ignoring **gross-up** on "net of tax" contracts	Always check who bears the tax; divide by (1 − rate)
Filing 15CA **Part D** for a taxable remittance	Part D is only for non-taxable; taxable > ₹5L needs Part C + 15CB
Choosing comparables with different functions for TNMM	Match FAR (functions, assets, risks); reject product/geography mismatches
Missing **3CEB (31 Oct)** — ₹1 lakh penalty	Calendar every related-party client separately
Forgetting **equalisation levy / TDS 194-O** on digital/e-com	Screen digital-service payments separately

Learn-it roadmap & resources

Time to job-ready: 8–12 weeks part-time.

- **Weeks 1–3:** Sections 5, 9, 195; WHT rates; do 20 15CA/15CB mappings. *ICAI International Taxation module* (free PDF) + the income-tax portal sandbox.
- **Weeks 4–6:** DTAA reading — pull the **India–US, India–UK, India–UAE, India–Singapore** treaties from incometaxindia.gov.in; practise finding the royalty/FTS/interest article and rate.
- **Weeks 7–9:** TP — the five methods, TNMM benchmarking, Rule 10B/10D; build one benchmarking sheet end-to-end.
- **Weeks 10–12:** BEPS Action Plans, PE, GAAR, **Pillar Two**; read the **OECD Transfer Pricing Guidelines** (free, skim) and a real Form 3CEB.

Resources: ICAI study material (free); OECD TP Guidelines & Model Tax Convention (free); Taxmann/Vinod Singhania *International Taxation* (paid); Big-4 alert emails (free). **Certifications:** ICAI's Diploma in International Taxation; the **ADIT (Advanced Diploma in International Taxation)** by CIOT UK — the gold standard for MNC/GCC international-tax roles.

Quick-reference

Item	Value
WHT on foreign payment section	**195**
Disallowance if not deducted	**40(a)(i)** — 100%
TDS deposit due	**7th of next month**
Treaty rate needs	**TRC + Form 10F + no-PE declaration**
15CA Part A	taxable ≤ ₹5L/FY
15CA Part C	taxable > ₹5L → needs **15CB**
15CA Part D	not taxable
Gross-up (net of tax)	`= amount / (1 - rate)`
Arm's-length range	`QUARTILE.INC(range,1)` → `QUARTILE.INC(range,3)`
Default captive method	**TNMM**
Form 3CEB due	**31 Oct**
Master File / CbCR forms	**3CEAA / 3CEAD**
Global minimum tax	**Pillar Two — 15%**

Golden rule: *Rate = lower of Act and DTAA — but only if TRC + 10F are on file; when in doubt, withhold higher and let the vendor claim credit.*

Tax-tech & automation

What it is & where it's used

"Tax-tech" is the stack of software, scripts and glue that turns a manual, deadline-driven compliance job into a repeatable, auditable pipeline. In practice it means three things:

1. **Reconciliation engines** — matching GSTR-2B vs your purchase register, TDS 26AS/AIS vs books, bank vs cash book, vendor ledger vs their ledger.
2. **Return automation** — pushing invoice data from your ERP/accounting system to the GST portal, TRACES, or the income-tax utility via APIs or JSON, instead of keying it in.
3. **AI/ML assist** — document extraction (OCR of invoices), anomaly flags ("this HSN usually attracts 18%, you booked 5%"), and drafting notice replies.

Who needs it: GST practitioners, indirect-tax analysts in industry, TDS/payroll teams, statutory-audit associates running substantive testing, FP&A analysts who own the tax provision, and anyone in a **GSP/ASP** (GST Suvidha Provider / Application Service Provider) or Big-4 managed-services team. If your job title contains "tax", "compliance", "audit", "AR/AP", or "finance analyst", part of your value is now measured in *how few hours a month you spend keying and matching*.

The gap: why companies want this (and college didn't teach it)

MBA and CA-Inter teach you **what** GSTR-2B is and the *law* behind Section 16(2)(aa) ITC matching. They do not teach you what to do when you have a 40,000-row purchase register and a 38,000-row 2B that don't tie out, due tomorrow.

The industry reality:

College taught	Job actually needs
"Reconcile ITC as per rules"	Match 40k rows in 20 min with a repeatable macro
Definition of TDS sections	Bulk-validate 26AS vs books, isolate mismatches
Manual return filing walkthrough	Generate portal JSON from ERP, no re-keying
"Maintain documentation"	Version-controlled, audit-trailed, re-runnable

Companies pay for tax-tech because manual work **doesn't scale, isn't auditable, and breaks at month-end**. A person who can hand back a reconciled file with the mismatches *categorised and explained* is worth 3x one who "will get to it." This chapter closes the gap between knowing the rule and shipping the reconciled file.

What "proficient" looks like

A job-ready person can, **unaided**:

- Take two exports (register + 2B) and produce a four-bucket reconciliation — *matched, mismatch-in-value, in-books-not-in-2B, in-2B-not-in-books* — in Excel or Python.

- Write a `XLOOKUP` / `SUMIFS` recon and know its failure modes (duplicate GSTINs, trailing spaces, invoice-number formatting).
- Build the same recon in SQL when the data is 500k rows and won't fit Excel.
- Use a pivot / Power Query to refresh the recon monthly with one click.
- Read and generate the **GSTR-1 JSON schema**, understand the offline utility, and know where APIs (via a GSP) replace it.
- Explain what should stay manual (judgement, notice replies) vs automated (matching, arithmetic).

You are *not* expected to build the AI model. You are expected to **use** the tools and **verify** their output — the tool proposes, you dispose.

Hands-on: how to actually do it

1. The GST ITC reconciliation in Excel

Assume `Books` sheet and `GSTR2B` sheet, each with columns: `GSTIN`, `InvoiceNo`, `Taxable`, `IGST`, `CGST`, `SGST`. First build a clean match key (never match on invoice number alone — normalise it):

```
' Match key: GSTIN + cleaned invoice no (upper, no spaces/special chars)
=UPPER(SUBSTITUTE(TRIM([@GSTIN])&"|"&[@InvoiceNo]," ",""))
```

Pull the 2B taxable value against each books row:

```
=XLOOKUP([@MatchKey], GSTR2B[MatchKey], GSTR2B[Taxable], "NOT IN 2B", 0)
```

Bucket the result:

```
=IF(D2="NOT IN 2B","In books, not in 2B",
   IF(ROUND(D2-[@Taxable],0)=0,"Matched",
   "Value mismatch: "&TEXT(D2-[@Taxable],"₹#,##0")))
```

Reverse-check (rows in 2B missing from books) with a `COUNTIFS` on the 2B sheet:

```
=IF(COUNTIFS(Books[MatchKey],[@MatchKey])=0,"In 2B, not in books","OK")
```

Summarise with a pivot, or:

```
=SUMIFS(Books[IGST], Books[Bucket], "Matched")   ' safe-to-claim ITC
```

2. Same recon in SQL (when Excel dies)

```
SELECT
    COALESCE(b.match_key, g.match_key)               AS match_key,
    b.taxable                                        AS books_taxable,
    g.taxable                                        AS gstr2b_taxable,
    CASE
        WHEN b.match_key IS NULL THEN 'In 2B, not in books'
        WHEN g.match_key IS NULL THEN 'In books, not in 2B'
        WHEN ROUND(b.taxable - g.taxable, 0) = 0 THEN 'Matched'
        ELSE 'Value mismatch'
    END                                              AS bucket,
    ROUND(COALESCE(b.taxable,0) - COALESCE(g.taxable,0), 2) AS diff
FROM books b
FULL OUTER JOIN gstr2b g ON b.match_key = g.match_key
ORDER BY bucket, diff DESC;
```

A `FULL OUTER JOIN` on the normalised key is the entire recon — both "missing" buckets fall out for free. (In MySQL, which lacks FULL OUTER JOIN, emulate with `LEFT JOIN … UNION … RIGHT JOIN`.)

3. Automating it in Python (monthly, re-runnable)

```
import pandas as pd

def clean_key(df):
    df["key"] = (df["GSTIN"].str.strip().str.upper() + "|" +
                 df["InvoiceNo"].str.replace(r"[^A-Za-z0-9]", "", regex=True))
    return df

books = clean_key(pd.read_excel("books.xlsx"))
b2b   = clean_key(pd.read_excel("gstr2b.xlsx"))

m = books.merge(b2b, on="key", how="outer",
                suffixes=("_bk", "_2b"), indicator=True)

def bucket(r):
    if r["_merge"] == "left_only":  return "In books, not in 2B"
    if r["_merge"] == "right_only": return "In 2B, not in books"
    return "Matched" if round(r.Taxable_bk - r.Taxable_2b) == 0 else "Value mismatch"

m["bucket"] = m.apply(bucket, axis=1)
m.to_excel("recon_output.xlsx", index=False)
print(m["bucket"].value_counts())
```

Run it every month by dropping two files in a folder — 30 seconds vs a day.

4. AI-assist: invoice extraction + anomaly flag

OCR a scanned invoice to structured fields (Tesseract free, or cloud Document AI), then apply a rules layer the "AI" can't be trusted to get right alone:

```
# after OCR gives fields dict: {"hsn":"8471", "rate":5, "taxable":100000}
EXPECTED = {"8471": 18}          # HSN 8471 (computers) → 18%
flag = "REVIEW: rate looks wrong" if EXPECTED.get(fields["hsn"]) ≠ fields["rate"] else "ok"
```

The model reads; **your rules table decides**. That is the professional pattern for AI in compliance.

5. Portal / Tally click-paths

- **GSTR-1 (offline utility):** GST portal → Downloads → Offline Tools → *Returns Offline Tool* → paste/import Excel → *Generate JSON* → upload on portal → *Preview* → *Submit* → *File with EVC/DSC*.
- **GSTR-2B download:** portal → Returns Dashboard → select period → GSTR-2B → *Download* (Excel/JSON) — this is your recon input.
- **TallyPrime GSTR-1 export:** Gateway of Tally → Display More Reports → GST Reports → GSTR-1 → `Alt+E` (Export) → JSON → upload to portal.
- **API route:** enterprises skip the portal and file through a **GSP** (ClearTax/Cygnet/Zoho) via authenticated API — data flows ERP → GSP → GSTN with an audit trail and no manual JSON.

Worked example / mini-project

Scenario: Acme Traders Pvt Ltd, March 2026. You must finalise ITC before filing GSTR-3B.

Purchase register (books): 5 invoices. GSTR-2B (as downloaded): 5 lines. Amounts in ₹.

Books

GSTIN	InvoiceNo	Taxable	IGST
27AAAC…1Z5	INV-001	1,00,000	18,000
27AAAC…1Z5	inv 002	50,000	9,000
29BBBC…4Z1	445	2,00,000	36,000
24CCCC…7Z9	A-99	80,000	14,400
07DDDD…2Z3	771	60,000	10,800

GSTR-2B

GSTIN	InvoiceNo	Taxable	IGST
27AAAC…1Z5	INV001	1,00,000	18,000
27AAAC…1Z5	INV002	50,000	9,000
29BBBC…4Z1	445	1,80,000	32,400
07DDDD…2Z3	771	60,000	10,800

After `clean_key` normalises invoice numbers, the recon produces:

Bucket	Invoice	Diff (Taxable)	Action
Matched	INV-001, inv 002, 771	0	Claim ITC fully
Value mismatch	445	₹20,000	Vendor under-reported; claim ₹32,400 only, follow up
In books, not in 2B	A-99 (₹80,000)	—	ITC **not eligible** yet (Sec 16(2)(aa)); park in "ITC not available"

ITC decision: Eligible now = 18,000 + 9,000 + 32,400 + 10,800 = **₹70,200**. Held back = ₹14,400 (A-99) + ₹3,600 (445 shortfall) = ₹18,000, tracked in a register until it appears in a later 2B.

Journal for the eligible ITC:

Account	Dr (₹)	Cr (₹)
Purchases	3,90,000	
Input IGST (eligible)	70,200	
ITC on hold (not in 2B)	18,000	
Trade Payables		4,78,200

That single reconciled file + register + entry is exactly the deliverable an employer means by "close the GST."

How it's tested

Interview questions - "Your purchase register shows ₹5L ITC but 2B shows ₹4.6L. Walk me through what you do." (Answer: four-bucket recon, hold ineligible under 16(2)(aa), track in a register, chase vendors.) - "How do you match when invoice numbers are formatted differently?" (Normalise: TRIM, UPPER, strip special chars; match on GSTIN+cleaned no, not amount.) - "When would you move from Excel to SQL/Python?" (Row count > ~100k, monthly repeatability, audit trail.)

Practical assessments companies give - A **timed Excel test (30–45 min):** two messy sheets, "reconcile and give me the mismatch summary." They watch for XLOOKUP/Power Query, key normalisation, and whether you catch duplicates. - A **SQL screen:** "write the join that shows only mismatches" — they want FULL OUTER JOIN and a CASE bucket. - A **case study:** raw GSTR-2B + register files emailed to you, produce a recon and an ITC eligibility note. Grading is on *correct buckets and the eligibility judgement*, not fancy formulas.

Common mistakes & how pros avoid them

Mistake	Why it bites	Pro habit
Matching on invoice number alone	Same number across vendors → false matches	Always GSTIN + normalised invoice key
Ignoring trailing spaces / case	`INV 001` ≠ `INV001` → false mismatches	`TRIM`, `UPPER`, `SUBSTITUTE`/regex first
Trusting VLOOKUP with duplicates	Returns first hit, hides duplicates	`COUNTIFS` to detect dupes before matching
Claiming ITC not in 2B	Sec 16(2)(aa) violation → interest + penalty	Park in "ITC on hold" register
Overwriting last month's file	No audit trail	Dated files / Git / Power Query source refresh
Blindly trusting AI/OCR output	Silent wrong HSN or rate	Rules layer + human review of exceptions
Hard-coding, no re-run	Breaks next month	Parameterise (Power Query, function, script)

Learn-it roadmap & resources

Time to job-ready: 8–10 weeks, part-time.

Weeks	Focus	Resource (free unless noted)
1–2	Excel recon: XLOOKUP, SUMIFS, COUNTIFS, pivots	ExcelJet, Chandoo.org
3–4	Power Query (refreshable recons)	Microsoft Learn: Power Query
5–6	SQL joins on real files	SQLBolt, Mode SQL Tutorial
7–8	Python + pandas merges	Kaggle "Pandas" micro-course
9–10	GST portal/JSON + a GSP trial	GSTN offline utility docs; ClearTax/Zoho free tier

Certifications that signal this: ICAI's *Certificate Course on GST*; Microsoft **PL-300 (Power BI)** for the analytics angle; any GSP vendor's product certification (ClearTax/Cygnet) if you target managed-services roles. None are mandatory — a **portfolio recon file** on GitHub beats a certificate in interviews.

Quick-reference

Task	Tool	Command / step
Match key	Excel	`=UPPER(SUBSTITUTE(TRIM(GSTIN)&"|"&Inv," ",""))`
Pull value	Excel	`=XLOOKUP(key, rng_key, rng_val, "NOT FOUND")`
Detect duplicate	Excel	`=COUNTIFS(rng, key)>1`
Full recon	SQL	`FULL OUTER JOIN … ON key` + `CASE` bucket
Missing rows	SQL	`WHERE b.key IS NULL OR g.key IS NULL`
Monthly recon	Python	`df1.merge(df2, on="key", how="outer", indicator=True)`
Refreshable recon	Power Query	Data → Get Data → merge queries → Refresh
GSTR-1 file	Portal	Offline Tool → Generate JSON → upload → File (EVC/DSC)
GSTR-2B	Portal	Returns Dashboard → 2B → Download
Tally export	TallyPrime	GST Reports → GSTR-1 → `Alt+E` → JSON
API filing	GSP	ERP → GSP (ClearTax/Cygnet) → GSTN

Golden rules: normalise before you match · never claim ITC absent from 2B · the tool proposes, you dispose · every recon must re-run next month unchanged.

The Tax Role Interview & Practical Test

What it is & where it's used

The tax role interview is a two-part gate: a **conceptual conversation** ("explain input tax credit blocking") and a **practical test** where you actually *do the work* — compute a GST liability, calculate TDS on a vendor bill, or reconcile GSTR-2B against the purchase register in Excel. This is where 60% of MBA-Finance candidates lose the offer: they can define ITC but freeze when handed a messy purchase dump and told "reconcile this, you have 30 minutes."

Roles that run this gate: **Tax Executive / Analyst** (Big 4 compliance teams, mid-tier CA firms), **Accounts Payable with TDS ownership**, **GST Compliance Associate**, **Finance Executive** at any company above ~₹5 crore turnover, and offshore **India-tax delivery centres** (EY GDS, Deloitte USI, PwC AC). Globally, the equivalent is a VAT/sales-tax analyst or an indirect-tax associate — same muscle, different rate tables.

The gap: why companies want this (and college didn't teach it)

College teaches you the *Income Tax Act sections* and the *CGST Act charging provisions*. It never makes you file a single GSTR-3B or match a single invoice. The gap is **execution under real-world mess**:

College taught	The job actually needs
"TDS u/s 194J is 10%"	Vendor bill has GST — do you deduct TDS on the base or the gross?
"ITC is available on inputs"	2B shows the invoice but your vendor filed late — claim now or defer?
Definition of RCM	Booking the RCM self-invoice and the offsetting ITC in Tally
"File GSTR-3B monthly"	Reconciling 2B vs books, finding the 12 mismatches, and drafting the follow-up email

Employers pay for someone who closes a period **without supervision** and **without penalty-triggering errors**. That skill is invisible on a marksheet, so they test it live.

What "proficient" looks like

The concrete bar a job-ready candidate clears unaided:

- Given a vendor invoice, compute **GST (correct place-of-supply → CGST+SGST vs IGST)** and **TDS on the base value excluding GST**, and state both due dates.
- Take a raw purchase register + a downloaded **GSTR-2B JSON/Excel** and produce a reconciliation flagging: matched, in-books-not-in-2B, in-2B-not-in-books, and value/GSTIN mismatches.
- Pass the **220-day / rule-37 / Section 16(4)** ITC eligibility tests from memory.
- Write the **journal entries** for a purchase with ITC, an RCM transaction, and a TDS deduction.
- Know the **return calendar** cold: GSTR-1 (11th), GSTR-3B (20th), TDS payment (7th), TDS return (quarterly, 31st of next month).

Hands-on: how to actually do it

1. Compute GST + TDS on a vendor bill

A professional-services vendor in the same state bills you **₹1,00,000 + 18% GST**.

```
Base value        = 1,00,000
GST @ 18% (9+9)   =   18,000   → CGST 9,000 + SGST 9,000 (intra-state)
Invoice total     = 1,18,000

TDS u/s 194J @10% = 10,000   ← on BASE (₹1,00,000), NOT on ₹1,18,000
Net payable       = 1,18,000 − 10,000 = 1,08,000
```

The single most-tested trap: **TDS is deducted on the value excluding GST** when GST is shown separately (CBDT Circular 23/2017). Interstate? The ₹18,000 becomes IGST instead of CGST+SGST.

2. Excel formulas the test uses

Place-of-supply logic (auto-split CGST/SGST vs IGST):

```
=IF([@SupplierState]=[@BuyerState], [@Taxable]*[@Rate]/2, 0)   → CGST
=IF([@SupplierState]=[@BuyerState], [@Taxable]*[@Rate]/2, 0)   → SGST
=IF([@SupplierState]<>[@BuyerState], [@Taxable]*[@Rate], 0)    → IGST
```

TDS with threshold and section rate lookup:

```
=IF([@BaseValue]>=VLOOKUP([@Section],ThresholdTbl,2,0),
    ROUND([@BaseValue]*VLOOKUP([@Section],RateTbl,2,0),0), 0)
```

3. Reconcile GSTR-2B vs purchase register (the core test)

Match on a composite key of **GSTIN + Invoice No.**, then compare tax values:

```
Key:        =[@SupplierGSTIN]&"|"&[@InvoiceNo]
2B Taxable: =XLOOKUP([@Key], TwoB[Key], TwoB[Taxable], "NOT IN 2B")
Status:     =IFS(
   [@[2B Taxable]]="NOT IN 2B","In books, not in 2B",
   ISNA(XLOOKUP([@Key],Books[Key],Books[Key])),"In 2B, not in books",
   ROUND([@BookTaxable],0)<>ROUND([@[2B Taxable]],0),"Value mismatch",
   TRUE,"Matched")
```

4. Journal entries

Transaction	Dr	Cr
Purchase with ITC	Purchases ₹1,00,000; CGST ITC ₹9,000; SGST ITC ₹9,000	Vendor ₹1,18,000
TDS on the above (194J)	Vendor ₹10,000	TDS Payable ₹10,000
RCM (freight ₹10,000) — self-invoice	RCM Expense ₹10,000; IGST ITC ₹1,800	Vendor ₹10,000; IGST RCM Payable ₹1,800

5. GST-portal click-path (2B download)

`Services → Returns → Returns Dashboard → select period → GSTR-2B → View / Download → Download Excel`. This is what you'd narrate if asked "walk me through pulling 2B."

Worked example / mini-project

Reconcile a month for "Acme Traders" (Maharashtra). Purchase register (books) vs GSTR-2B:

Supplier	GSTIN	Inv No	Books Taxable	Books GST	2B Taxable	2B GST
Alpha Ltd	27AAA...1Z5	INV-101	50,000	9,000	50,000	9,000
Beta Co	27BBB...2Z4	INV-205	30,000	5,400	30,000	5,400
Gamma Inc	24GGG...3Z3	INV-88	20,000	3,600	—	—
Delta LLP	27DDD...4Z2	INV-410	40,000	7,200	40,000	7,600
Zeta Pvt	27ZZZ...5Z1	—	—	—	15,000	2,700

Run the `Status` formula:

- **INV-101, INV-205** → *Matched* (claim ITC ₹9,000 + ₹5,400).
- **Gamma INV-88** → *In books, not in 2B* — vendor hasn't filed. **Defer** ITC of ₹3,600 (Sec 16(2)(aa): no 2B, no credit).
- **Delta INV-410** → *Value mismatch* — books GST ₹7,200 vs 2B ₹7,600. Either your invoice-value entry is wrong or vendor over-reported. Investigate; claim only the reconciled amount.
- **Zeta** → *In 2B, not in books* — vendor filed against your GSTIN but you have no bill. Could be a wrong GSTIN by the vendor or a missed booking. Do **not** claim until traced.

Eligible ITC this month = ₹14,400 (only matched lines). The deliverable is this table plus a one-line follow-up email: "*Gamma & Delta — please confirm GSTR-1 filing / invoice value for INV-88 and INV-410.*"

How it's tested

Interview questions (verbal):

1. "Vendor bills ₹1,00,000 + 18% GST for professional fees — how much TDS, on what value?" (Answer: ₹10,000 on the base.)
2. "Invoice is in your books but not in 2B — can you take ITC?" (No — Sec 16(2)(aa).)

3. "Difference between GSTR-2A and 2B?" (2A is dynamic/live; 2B is static, generated 14th, and is the ITC basis.)
4. "What's blocked ITC under Sec 17(5)?" (Motor vehicles, personal consumption, works contract for immovable property, etc.)
5. "RCM on which common expenses?" (GTA/freight, legal fees from advocates, director sitting fees, import of services.)
6. "Due dates for GSTR-1, 3B, TDS payment, TDS return?"

Practical/assessment test (hands-on):

- A **timed 30-45 min Excel** with a purchase dump and a 2B extract: "reconcile and tell me eligible ITC." (Exactly the mini-project above.)
- "Here's a vendor invoice — compute GST and TDS, give me the net payable and journal entry."
- A **TallyPrime** live task: pass a purchase voucher with ITC and a TDS deduction.
- A **case**: "Vendor filed GSTR-1 late in the next month — how does your reconciliation and ITC treatment change?"

Common mistakes & how pros avoid them

Mistake	Fix
Deducting TDS on GST-inclusive amount	Deduct on base value when GST is shown separately (Circular 23/2017)
Claiming ITC because it's in *books*	ITC follows **2B**, not books — Sec 16(2)(aa)
Matching on invoice number alone	Use **GSTIN + Invoice No.** — invoice numbers repeat across vendors
Ignoring rounding	`ROUND( ... ,0)` both sides before comparing; paise differences flag false mismatches
Forgetting **Sec 16(4)** time limit	ITC lapses after 30 Nov of next FY / annual return date
Treating 2A and 2B as interchangeable	2B is the static, auditable ITC basis; quote it in the interview
Missing RCM self-invoice	RCM creates a liability *and* an offsetting ITC — book both

Pros also **keep the reconciliation reproducible**: a single key column, `XLOOKUP` not `VLOOKUP` (returns "NOT IN 2B" cleanly), and a status column that speaks in review language ("In books, not in 2B") rather than TRUE/FALSE.

Learn-it roadmap & resources

Time to interview-ready: 3-5 weeks if you already know the concepts.

- **Week 1** — Rebuild the GST computation + place-of-supply logic in Excel; memorise the return calendar and Sec 17(5)/16 rules.
- **Week 2** — Build the 2B reconciliation template from scratch; run it on 3 sample datasets.
- **Week 3** — TDS: section rates, thresholds, Form 26Q, and journal entries; practise 10 vendor-bill computations.

- **Week 4** — TallyPrime vouchers (purchase-with-ITC, TDS, RCM); mock the timed Excel test twice.

Resources: - GST portal (`gst.gov.in`) — file NIL returns on a test/own registration to see the real UI. - CBIC's *GST Flyers* and Circular 23/2017 (TDS-on-GST) — free, authoritative. - ICAI Study Material (Taxation) — you already have this via CA Inter. - ClearTax / TaxGuru blogs for reconciliation walk-throughs. - **Certification:** ICAI's *Certificate Course on GST*, or a Tally-with-GST certificate — useful signalling for non-CA roles.

Quick-reference

Item	Value
TDS base	Value **excluding** GST (if GST shown separately)
GST intra-state	CGST + SGST (split rate)
GST inter-state	IGST (full rate)
ITC basis	**GSTR-2B** (static, gen. 14th)
ITC condition	Sec 16(2)(aa) — must appear in 2B
ITC time limit	Sec 16(4) — 30 Nov of next FY
Blocked ITC	Sec 17(5)
GSTR-1 due	11th of next month
GSTR-3B due	20th of next month
TDS payment	7th of next month
TDS return (26Q)	Quarterly, 31st of month after quarter
Recon key	`GSTIN & "|" & InvoiceNo`
Recon function	`XLOOKUP(key, 2B[key], 2B[val], "NOT IN 2B")`
Common TDS sections	194C (1/2%), 194J (10%), 194I (10% rent), 194H (5%)
Common RCM	GTA freight, advocate fees, director fees, import of service

Cheat-sheet: due-dates, rates & forms

What it is & where it's used

Every finance/accounts/tax job in India runs on a compliance calendar. Miss a GST return by a day and interest plus late fees start ticking; miss a TDS deposit and you lose the expense deduction. The person who *knows the dates cold* — the 7th, 11th, 15th, 20th of every month — is the one the manager trusts with the filings.

This chapter is the reference sheet you keep pinned. It's used by:

- **Accounts executives / AP-AR** — TDS deduction, GST input reconciliation.
- **Tax associates (Big 4, boutique CA firms)** — return filing, advance tax, TDS returns.
- **Finance analysts / controllers** — cash-flow planning around statutory outflows.
- **Founders / SME accountants** — the one-person compliance team.

Interviewers *love* this because it can't be bluffed. You either know that TDS is deposited by the **7th** and the GSTR-3B is by the **20th**, or you don't.

> Rates below reflect **FY 2025-26 / AY 2026-27**. Always re-verify against the latest Finance Act — rates change every Budget.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you *what* TDS is (a mechanism to collect tax at source). It never teaches you that Section 194J (professional fees) is 10%, 194C (contractor) is 1%/2%, 194Q (goods purchase) is 0.1%, or that all of them get deposited by the 7th of the next month via **Challan ITNS-281**. College gives you the concept; the employer pays for the **execution under a deadline**.

The gap is precisely this: colleges teach *principles*, jobs run on *dates, forms, sections and thresholds*. A fresher who says "I'll deduct TDS on this consultant bill at 10% under 194J, deposit by the 7th, and file 26Q for the quarter" is instantly worth more than one who defines TDS beautifully but can't file it.

What "proficient" looks like

A job-ready person can, *without Googling*:

1. Look at a purchase/expense invoice and state the correct **TDS section, rate, and threshold**.
2. Name the **due date** for TDS deposit, TDS return, GSTR-1, GSTR-3B, advance tax, and ROC filings.
3. Compute **interest and late fee** for a delayed filing.
4. Pick the **right form** (26Q vs 24Q vs 27Q; GSTR-1 vs 3B vs 9; ITR-1 vs 3 vs 6).
5. Read a GST portal / TRACES dashboard and know what's pending.

The bar isn't memorising all of the Income-tax Act — it's the **20 high-frequency dates, rates and forms** that recur every single month.

Hands-on: how to actually do it

1. The monthly compliance rhythm (burn this into memory)

Date	Obligation	Form / Challan
7th	Deposit TDS/TCS for previous month	Challan ITNS-281
11th	File GSTR-1 (outward supplies, monthly filers)	GSTR-1
13th	GSTR-1 (QRMP quarterly) / IFF (optional)	GSTR-1 / IFF
15th	PF & ESI deposit; Advance tax (quarterly)	ECR / Challan 280
20th	File & pay GSTR-3B (monthly filers)	GSTR-3B
22nd/24th	GSTR-3B (QRMP, state-wise)	GSTR-3B
30th/31st	TDS return (quarterly), ROC filings	26Q/24Q, MGT-7/AOC-4

2. TDS rate & section quick lookup (most-used)

Section	Nature of payment	Rate	Threshold (FY25-26)
192	Salary	Slab	Basic exemption
194A	Interest (non-bank)	10%	₹5,000
194C	Contractor (individual/HUF)	1%	₹30,000 single / ₹1,00,000 aggregate
194C	Contractor (others)	2%	same
194H	Commission / brokerage	2%	₹20,000
194I	Rent — plant & machinery	2%	₹2,40,000
194I	Rent — land/building	10%	₹2,40,000
194J	Professional / technical fees	10% / 2%	₹30,000
194Q	Purchase of goods	0.1%	₹50,00,000
206C(1H)	TCS on sale of goods	0.1%	₹50,00,000

> No PAN → TDS at **20%** (or twice the rate, whichever is higher).

3. Interest & late-fee formulas (put these in Excel)

```
' TDS interest — late DEDUCTION: 1% per month (or part) from due-to-deduct to actual deduction
=TDS_Amount * 1% * Months_Delay

' TDS interest — late DEPOSIT: 1.5% per month (or part) from deduction to deposit
=TDS_Amount * 1.5% * ROUNDUP((Deposit_Date - Deduction_Date)/30, 0)

' GST late fee (3B/1): Rs 50/day (Rs 20/day nil), capped
=MIN(Days_Late * 50, Cap)

' GST interest on late tax payment: 18% p.a. on net tax
=Net_Tax * 18% * Days_Late / 365
```

4. Which form? (decision cheat-sheet)

```
TDS returns:   Salary → 24Q   |  Resident non-salary → 26Q  |  Non-resident → 27Q  |  TCS → 27
GST returns:   Sales → GSTR-1 |  Summary+pay → GSTR-3B       |  Annual → GSTR-9 (>Rs 2 cr) + 9
Income Tax:    Individual (no biz) → ITR-1/2 | Individual+biz → ITR-3 | Firm/LLP → ITR-5 | Com
ROC (company): Annual return → MGT-7 | Financials → AOC-4 | Director KYC → DIR-3 KYC
```

5. Income-tax slabs — New Regime (default, AY 2026-27)

Income slab	Rate
Up to ₹4,00,000	Nil
₹4,00,001 – ₹8,00,000	5%
₹8,00,001 – ₹12,00,000	10%
₹12,00,001 – ₹16,00,000	15%
₹16,00,001 – ₹20,00,000	20%
₹20,00,001 – ₹24,00,000	25%
Above ₹24,00,000	30%

Rebate u/s 87A: nil tax up to ₹12,00,000 taxable income. Standard deduction ₹75,000 (salary). Verify against the current Finance Act each year.

Worked example / mini-project

Scenario: Acme Traders Pvt Ltd, June 2026. Build the month's compliance sheet.

Transactions: - Paid consultant ₹80,000 for audit support (194J). - Paid contractor (company) ₹1,50,000 for fit-out (194C). - Office rent to landlord ₹60,000/month (194I building). - June sales ₹18,00,000 + GST 18%; input

GST available ₹1,40,000.

Step 1 — TDS:

Payment	Section	Rate	TDS
Consultant ₹80,000	194J	10%	₹8,000
Contractor ₹1,50,000	194C	2%	₹3,000
Rent ₹60,000	194I(b)	10%	₹6,000
Total TDS			**₹17,000**

Deposit ₹17,000 via ITNS-281 by **7 July 2026**. File 26Q by **31 July 2026** (Q1).

Step 2 — GST: - Output GST = 18,00,000 × 18% = ₹3,24,000 - Input credit = ₹1,40,000 - Net GST payable = **₹1,84,000** - File GSTR-1 by **11 July**, GSTR-3B + pay ₹1,84,000 by **20 July 2026**.

Step 3 — What if 3B is filed on 5 August (15 days late)? - Late fee = 15 × ₹50 = ₹750 - Interest = 1,84,000 × 18% × 15/365 = **₹1,361** - Total avoidable cost = **₹2,111** — the entire point of knowing the calendar.

Journal entry for the consultant bill + TDS:

Particulars	Dr	Cr
Professional Fees A/c	80,000	
To TDS Payable (194J)		8,000
To Consultant (Bank/Payable)		72,000

How it's tested

Interview questions: - "By when do you deposit TDS deducted in June?" → 7 July. - "TDS rate and section on a ₹1 lakh consultancy bill?" → 194J, 10%, ₹10,000. - "Difference between GSTR-1 and GSTR-3B?" → 1 = outward supply detail; 3B = summary + tax payment. - "Company files which ITR?" → ITR-6. - "Interest for depositing TDS one month late?" → 1.5% per month.

Practical assessments companies give: 1. **Timed compliance-calendar test** — a table of transactions; you fill section, rate, TDS, due date under 20 minutes. 2. **Excel late-fee model** — build interest/late-fee formulas for a set of delayed filings. 3. **"File this in Tally"** — pass TDS entries, generate 26Q, reconcile GSTR-2B vs purchase register. 4. **GST recon case** — spot the invoices in books but missing from GSTR-2B and quantify blocked ITC.

Common mistakes & how pros avoid them

Mistake	Fix
Deducting TDS at wrong rate/section	Keep the section table taped to your monitor; verify threshold *and* rate
Missing the "part of a month" rule	Interest is per month **or part** — 31 days = 2 months, always ROUNDUP
Forgetting the aggregate threshold (194C ₹1 L/yr)	Track year-to-date payments per vendor, not just single bills
Filing 3B without reconciling 2B	Reconcile ITC monthly; claim only what appears in GSTR-2B
No-PAN vendor deducted at normal rate	20% flat when PAN absent — always collect PAN first
Depositing TDS but not filing the return	Deposit ≠ return; 26Q/24Q are separate quarterly obligations
Using old FY rates	Re-check every rate after the Union Budget

Pros run a **shared calendar with reminders 2 days before each due date**, keep a vendor master with PAN + default TDS section, and reconcile GST every month — never at year-end.

Learn-it roadmap & resources

Time to proficiency: 3–4 weeks of hands-on filing to internalise the calendar; ~2 months to be trusted end-to-end.

- **Week 1** — Memorise the monthly rhythm (7/11/15/20) and top-10 TDS sections.
- **Week 2** — Practice GSTR-1 & 3B on the GST portal sandbox; do 2B reconciliation.
- **Week 3** — File a mock 26Q on TRACES; build the interest/late-fee Excel model.
- **Week 4** — Do a full month's close for a dummy company (like the example above).

Resources: - **Free:** income-tax portal (incometax.gov.in), GST portal (gst.gov.in), CBIC & CBDT circulars, ClearTax/TaxGuru blogs, ICAI study material (Taxation). - **Paid:** ClearTax / Zoho Books hands-on, Udemy "GST & TDS Practical", any local CA-firm articleship (best learning). - **Certification:** CA Intermediate (Taxation paper), ICAI GST certificate course, GST Practitioner exam.

Quick-reference

```
MONTHLY:  7th TDS deposit | 11th GSTR-1 | 15th PF/ESI/adv tax | 20th GSTR-3B
QUARTERLY: TDS return by end of month after quarter (Q4 salary 24Q → 31 May)
ADV TAX:  15 Jun 15% | 15 Sep 45% | 15 Dec 75% | 15 Mar 100%
ANNUAL:   ITR (non-audit) 31 Jul | audit 31 Oct | GSTR-9 31 Dec | AOC-4/MGT-7 within 30/60 day
```

Quick TDS	Sec	Rate
Salary	192	Slab
Contractor	194C	1%/2%
Commission	194H	2%
Rent (building)	194I	10%
Professional	194J	10%
Goods purchase	194Q	0.1%

Interest/Fee	Rate
TDS late deduction	1% / month
TDS late deposit	1.5% / month
GST late tax	18% p.a.
GST late fee	₹50/day (₹20 nil), capped
Income-tax 234B/C	1% / month

Forms: 24Q (salary TDS) · 26Q (non-salary) · 27Q (NR) · 27EQ (TCS) · GSTR-1/3B/9/9C · ITR-1/3/5/6 · ITNS-281 (TDS challan) · Challan 280 (income tax).

Practical FP&A & Corporate Finance

Practical, Hands-On, Job-Ready — India-first + Global

Real formulas, queries, entries & tools — closing the gap between what an MBA teaches and what finance employers pay for

Contents

1. **What FP&A & Corporate Finance Actually Do**
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What FP&A & Corporate Finance Actually Do

What it is & where it's used

FP&A (Financial Planning & Analysis) is the team that turns a company's raw accounting data into forward-looking decisions: how much will we earn next quarter, can we afford this hire, why did margin drop 2%, should we approve this ₹40 lakh capex. **Corporate finance** is the broader umbrella — capital structure, funding, treasury, valuation, M&A — of which FP&A is the operational, monthly-heartbeat piece.

Where it lives, by role:

Role	What the FP&A/corp-fin work looks like
FP&A Analyst	Builds/updates the monthly forecast, variance decks, budget templates
Business Finance Partner	Sits with a Sales/Ops head, defends their P&L, challenges spend
Treasury Analyst	Cash-flow forecast, banking, working-capital, FX exposure
Corp Dev / IB Analyst	Valuation models (DCF, comps), deal support, board packs
Startup Finance (India)	All of the above at once — MIS, runway, investor reporting, GST
Controller / Accounts	Owns the actuals that FP&A forecasts against

In India this shows up as "MIS Analyst", "Business Finance", "FP&A", or simply "Finance Manager" in a startup. The tool stack is near-universal: **Excel** (the real ERP of finance), a **BI layer** (Power BI/Tableau), the **ERP/ledger** (Tally, SAP, NetSuite, Zoho), and increasingly **SQL/Python** to get data out without begging IT.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you WACC, CAPM, and the Modigliani-Miller theorem. Your CA course taught you AS/Ind-AS, journal entries, and how to *audit* a set of books. Neither taught you the actual job, which is **80% "get the number, explain the number, project the number" and 20% theory.**

The concrete gaps:

- **College is backward-looking; FP&A is forward-looking.** You were graded on preparing/auditing past financials. Employers pay you to *forecast* the next 12 months and be roughly right.
- **College gives clean data; the job gives garbage.** No textbook problem has duplicate cost centres, a ₹ in a text field, or three spellings of "Reliance". Real data cleaning is 40% of the job.
- **College values the exact answer; the job values the fast, defensible, on-time answer.** A board deck at 90% accuracy on Monday beats 100% on Thursday.
- **Nobody taught you Excel as a modelling language** — dynamic arrays, `INDEX/MATCH`, `SUMIFS`, scenario toggles, clean model architecture. CA and MBA both assume you'll "pick it up."
- **Nobody taught you to talk to a database.** The data you need lives in an ERP behind a `SELECT` statement, not in a nicely formatted PDF.

Close these five and you are hireable. The rest of this chapter (and book) is how.

What "proficient" looks like

The bar an employer tests for — what a job-ready person does **without hand-holding**:

- Given a raw GL/trial-balance export, build a **3-statement-linked** or at least a clean P&L model in under a day.
- Write a **variance bridge** (Budget → Actual, explaining each ₹ of the gap) and narrate it in three sentences a CFO understands.
- Pull their own data: a `SUMIFS` / `XLOOKUP` in Excel, a `GROUP BY` in SQL, a `groupby` in pandas.
- Build a **rolling 12-month forecast** with driver-based assumptions (volume × price, headcount × cost), not a flat +10%.
- Produce a **cash-flow / runway** view and know the difference between profit and cash.
- Turn all of it into a **1-page board/MIS deck** with a clear story, not a data dump.
- Know the **calendar**: what's due on working-day 1, 3, 5 of the month.

The FP&A monthly calendar (memorise this — interviewers ask):

Working day	Activity
WD 1–3	Close: accruals booked, actuals locked by Controller
WD 3–4	FP&A pulls actuals, refreshes model, builds variance vs budget/forecast
WD 4–5	Commentary written, MIS/board deck built
WD 5–6	Business reviews with department heads
WD 7–10	Re-forecast next quarter, update rolling forecast
Quarterly	Board pack, re-forecast; Annually: the budget (Aug–Nov cycle)

Hands-on: how to actually do it

1. Get actuals against budget — the workhorse `SUMIFS`. Given a transaction table (`Data` sheet: Cost Centre in A, Account in B, Month in C, Amount in D):

```
=SUMIFS(Data!$D:$D, Data!$B:$B, $A2, Data!$C:$C, F$1)
```

This sums Amount where Account = the row label and Month = the column header — the entire skeleton of an MIS pulls this way.

2. Look up master data — `XLOOKUP` (or `INDEX/MATCH` for legacy Excel):

```
=XLOOKUP(A2, Master!$A:$A, Master!$C:$C, "NOT MAPPED")
=INDEX(Master!$C:$C, MATCH(A2, Master!$A:$A, 0))
```

3. The variance bridge column:

```
Variance      =Actual - Budget
Variance %    =IFERROR((Actual-Budget)/Budget, 0)
Fav/Unfav     =IF(AND(N("expense"),Actual>Budget),"Unfav","Fav")
```

4. Pull the raw data yourself with SQL (don't wait for IT):

```
SELECT cost_centre,
       account_name,
       DATE_TRUNC('month', txn_date) AS period,
       SUM(amount) AS actual
FROM   gl_transactions
WHERE  txn_date >= '2026-04-01'          -- FY26-27 (Indian FY)
GROUP  BY cost_centre, account_name, period
ORDER  BY period, cost_centre;
```

5. Clean and aggregate in Python when Excel chokes on volume:

```
import pandas as pd
gl = pd.read_csv("gl_export.csv")
gl["amount"] = pd.to_numeric(gl["amount"], errors="coerce").fillna(0)
gl["month"]  = pd.to_datetime(gl["txn_date"]).dt.to_period("M")

pivot = (gl.groupby(["cost_centre", "month"])["amount"]
           .sum()
           .unstack(fill_value=0))
pivot["FY_total"] = pivot.sum(axis=1)
pivot.to_excel("actuals_by_cc.xlsx")
```

6. A driver-based revenue line (DAX for Power BI MIS):

```
Revenue Fcst = SUMX(VALUES(Product[SKU]),
    [Avg Price] * [Forecast Units])

Var to Budget % =
DIVIDE([Actual] - [Budget], [Budget], 0)
```

7. The accrual journal FP&A insists Controllers book so the number is complete (electricity bill not yet received, ₹2,00,000):

Account	Dr	Cr
Power & Fuel (Expense, P&L)	₹2,00,000	
Accrued Expenses (Liability, BS)		₹2,00,000

Reversed next month when the actual invoice lands. Understanding this is why FP&A and accounting can't be separated.

Worked example / mini-project

Reproduce this: a monthly MIS for "Nimbus Retail Pvt Ltd" (India), month = Apr-2026.

Budget vs Actual, ₹ lakh:

Line	Budget	Actual	Var (₹)	Var %	Comment
Revenue	500.0	472.0	(28.0)	-5.6%	2 stores opened late
COGS	300.0	288.9	11.1	fav	in line with volume
Gross Profit	**200.0**	**183.1**	**(16.9)**	**-8.5%**	GM 38.8% vs 40.0%
Employee cost	70.0	74.0	(4.0)	-5.7%	3 early hires
Rent	25.0	25.0	0.0	—	fixed
Marketing	30.0	41.0	(11.0)	-37%	pulled forward launch spend
Other opex	20.0	19.5	0.5	fav	
EBITDA	**55.0**	**23.6**	**(31.4)**	**-57%**	see bridge

The EBITDA bridge (the deliverable a CFO actually reads):

```
Budget EBITDA                    55.0
  Revenue shortfall (volume)    (11.2)   ← 28 x 40% GM
  Gross margin slip (1.2pt)      (5.7)
  Extra headcount                (4.0)
  Marketing pull-forward        (11.0)
  Other opex saving               0.5
Actual EBITDA                    23.6
```

Three-sentence narrative: "EBITDA missed by ₹31.4L, but ~₹11L is marketing we deliberately pulled forward and ₹4L is hiring ahead of Q2 — both timing, not structural. The real concern is the ₹17L gross-profit miss from two delayed store openings. Recovery expected from June once stores stabilise." That narrative — not the spreadsheet — is what gets you promoted.

Build it: dump the 8 lines in Excel, add `Var` / `Var %` formulas above, build the bridge as a waterfall chart. Thirty minutes of work; it *is* the job.

How it's tested

Interview questions: - "Walk me through the three financial statements and how they connect." (cash is the linking answer) - "Budget vs Actual vs Forecast — what's the difference?" (annual-fixed vs actual vs latest-estimate) - "You forecast ₹100 revenue, actual is ₹80. How do you investigate?" (decompose: volume vs price vs mix, by segment) - "Difference between profit and cash?" (working capital, non-cash items, capex) - "What is a rolling forecast and why use it?"

Practical tests (increasingly the real filter): - **Timed Excel test (30–60 min):** given a raw transaction dump, build a P&L pivot, add variance columns, `XLOOKUP` a mapping table, and write two bullets of commentary. Graded on speed, correct absolute referencing, and no hardcoded numbers. - **Case / "close these books":** an accrual + prepaid adjustment set, produce adjusted EBITDA. - **SQL screen:** "write a query for revenue by region by month." A `GROUP BY` with a date filter. - **Take-home model:** build a 12-month driver-based forecast from assumptions and present a 1-slide recommendation.

Common mistakes & how pros avoid them

Mistake	What pros do
Hardcoding numbers inside formulas	Every input is a labelled, coloured (blue) cell; formulas are black
Flat "+10%" forecasts	Driver-based: units × price, heads × cost, so you can defend each ₹
Reporting variance without a *why*	Every variance gets a one-line cause: price/volume/mix/timing
Confusing profit with cash	Always keep a cash/runway view beside the P&L
Circular refs / broken 3-statement links	Iterative calc on; isolate the interest-on-debt circularity
Dumping a 40-row table on the CFO	One page, one chart, three sentences, one recommendation
Waiting for IT to give data	Learn `SUMIFS`, SQL `GROUP BY`, pandas `groupby` and self-serve
Model with no version/date	Filename `MIS_Apr26_v3_2026-05-06.xlsx`; never overwrite last close

Learn-it roadmap & resources

Realistic time-to-proficiency: 3–6 months of deliberate practice alongside a job or CA prep.

Weeks	Focus	Milestone
1–3	Excel modelling: `SUMIFS`, `XLOOKUP`, `INDEX/MATCH`, pivots, dynamic arrays	Build the Nimbus MIS unaided
4–6	3-statement model + variance bridge	Link IS→BS→CFS, no plugs
7–9	SQL basics: `SELECT/WHERE/GROUP BY/JOIN`	Pull actuals from a sample DB
10–12	Power BI/DAX + a driver-based rolling forecast	1-page live dashboard
13–20	Python/pandas for cleaning; a DCF/valuation	Portfolio of 3 models

Resources: - **Free:** Corporate Finance Institute (CFI) free courses; Aswath Damodaran's NYU valuation lectures (YouTube, gold standard); Microsoft/Kaggle SQL & pandas tracks; ICAI study material for the accounting base. - **Paid/cert:** CFI's **FMVA** (financial modelling), Wall Street Prep / Breaking Into Wall Street for bankers, Coursera Excel-to-BI specialisations. In India, your **CA / CFA** is the credibility layer; FMVA/BIWS is the *skills* layer employers actually test. - **Practice data:** any company's published annual report — rebuild their P&L and forecast a year forward.

Quick-reference

Need	Tool / formula
Actual by account & month	`=SUMIFS(amt, acct, $A2, month, F$1)`
Map master data	`=XLOOKUP(key, lookup, return, "NA")`
Legacy lookup	`=INDEX(ret, MATCH(key, look, 0))`
Variance % (safe)	`=IFERROR((Act-Bud)/Bud, 0)`
Pull DB data	`SELECT dim, SUM(amt) … GROUP BY dim`
Clean/aggregate big data	`df.groupby("x")["amt"].sum()`
BI variance	`DIVIDE([Actual]-[Budget], [Budget], 0)`
Accrual entry	Dr Expense / Cr Accrued liability
Indian FY	1 Apr – 31 Mar; label FY26-27
Close cadence	Actuals WD3, variance WD4, deck WD5
Model hygiene	Blue = input, black = formula, never hardcode
The deliverable	1 page, 1 chart, 3 sentences, 1 recommendation

Budgeting in practice

What it is & where it's used

A budget is the company's financial plan for the next year (usually the fiscal year, in India **1 April to 31 March**) expressed in numbers: how much revenue each product/region will bring, what each department is allowed to spend, and what profit falls out. In practice, "budgeting" is the **annual process** of collecting numbers from every cost-centre owner, challenging them, consolidating them into a single P&L (and often cash flow and balance sheet), getting board sign-off, and then loading the approved figures into the accounting/ERP system so that **actual vs budget (variance)** reporting can run every month.

Roles that live and die by this:

Role	What they do in budgeting
FP&A Analyst	Builds the template, chases cost-centre owners, consolidates, runs variance
Finance Business Partner	Sits with a department head (Sales, Marketing, Ops) to build that unit's budget
Management Accountant / Cost Accountant	Standard costs, overhead absorption rates, cost-centre allocation
Financial Controller	Owns the calendar, reviews, presents to CFO
Startup Finance / "Founder's office"	Runway, burn, hiring plan — the same discipline, lighter tooling

Budgeting sits next to **forecasting** (which re-estimates mid-year) and **planning** (the strategic 3-5 year view). The budget is the *committed baseline* you get judged against.

The gap: why companies want this (and college didn't teach it)

An MBA teaches you the *master budget* diagram — sales budget → production budget → materials/labour/overhead budgets → cash budget → budgeted P&L. Cost accounting (CA Inter Paper 4) teaches flexed budgets and variance formulas. Both are correct and both are **useless on day one of the job**, because the real work is:

- **Process and people, not formulas.** 80% of budgeting is chasing 15 cost-centre owners for their numbers, handling the ones who pad, and reconciling versions.
- **A working template.** College never makes you build a driver-based Excel model that 12 departments feed into and that rolls up cleanly.
- **Top-down targets vs bottom-up build.** Textbooks present one clean number. Reality: the CEO says "grow EBITDA 20%" (top-down) while cost-centres submit numbers that imply a *loss* (bottom-up). Closing that gap **is** the job.
- **Version control and audit trail.** V1, V2, "final", "final_CFO_v3". Nobody teaches you how not to lose the plot.

Employers pay for the person who can turn a messy set of departmental asks into one board-ready, defendable number — fast, and without breaking the model.

What "proficient" looks like

A job-ready person can, unaided:

1. Build a **cost-centre budget template** in Excel that a non-finance manager can fill in, with input cells clearly separated from formula cells.
2. Build **revenue bottom-up from drivers** (units × price, or headcount × productivity), not a single typed number.
3. **Consolidate** many cost-centre tabs into one P&L using structured references / `SUMIFS`, so adding a department doesn't break anything.
4. Run a **top-down reconciliation**: show the CEO's target, the bottom-up submission, and the gap, then run scenarios to close it.
5. Produce **budget vs actual variance** with % and a favourable/adverse flag.
6. Manage a **budget calendar** and know who owes what by when.

Hands-on: how to actually do it

1. The cost-centre input template (one tab per department)

Keep **inputs blue, formulas black** — a real convention. Give each owner a driver, not a blank cell.

```
        A                    B         C        D        E        F        G
1  Cost Centre: Marketing   Owner: R.Sharma   CC Code: 5200
2  Line item         Driver           Rate     Qty      Apr ...  Total FY  Notes
3  Salaries          headcount        ₹90,000  6        =C3*D3   =E3*12
4  Digital ads       monthly spend    ₹250,000 1        =C4*D4   =E4*12
5  Events            per event        ₹400,000 8        =C5*D5   =E5*12
```

Monthly phasing (seasonality) instead of flat ÷12 — spread an annual number across months using a weight row:

```
=$F$6 * INDEX($Weights, MATCH(H$1, $MonthHdr, 0))
```

2. Build revenue bottom-up (driver-based)

Never type a revenue number. Model it:

```
Revenue = Units_sold * Avg_price
Units_sold (SaaS) = Opening_customers + New − Churned
=SUMPRODUCT(units_row, price_row)     ' mix of products
```

Growth-rate build for a month row (B = prior month, MoM growth in $C$1):

```
=B2*(1+$C$1)
```

3. Consolidate cost centres into one P&L

Put every cost-centre line into one long **flat table** (`tblBudget`) with columns `CC_Code | Account | Month | Amount`. Then the P&L is just `SUMIFS`:

```
=SUMIFS(tblBudget[Amount],
        tblBudget[Account], $A5,
        tblBudget[Month],   B$4)
```

Adding a 16th department = paste more rows, formulas don't change. This is why pros use a flat table, not 15 differently-shaped tabs.

Cross-check the roll-up two ways (belt and braces):

```
=SUMIFS( … )                 ' by account
=SUMPRODUCT((tblBudget[CC_Code]=code)*tblBudget[Amount])   ' by cost centre
' the two grand totals MUST tie
```

4. Top-down vs bottom-up reconciliation

	Amount (₹ Cr)
CEO target EBITDA (top-down)	12.0
Bottom-up submission EBITDA	8.4
Gap to close	**3.6**

Close the gap with a lever table and `Goal Seek` (Data → What-If Analysis → Goal Seek: *Set cell* = EBITDA, *To value* = 12, *By changing* = a cost-cut % or price-uplift cell).

5. Variance (budget vs actual), the monthly output

```
Variance   =Actual − Budget
Variance % =(Actual−Budget)/Budget
Flag (cost)=IF(Actual≤Budget,"Favourable","Adverse")
Flag (rev) =IF(Actual≥Budget,"Favourable","Adverse")
```

6. Same thing in SQL (when data lives in a warehouse)

```
SELECT  cc.cost_centre,
        a.account,
        SUM(CASE WHEN f.scenario='BUDGET' THEN f.amount END) AS budget,
        SUM(CASE WHEN f.scenario='ACTUAL' THEN f.amount END) AS actual,
        SUM(CASE WHEN f.scenario='ACTUAL' THEN f.amount END)
      - SUM(CASE WHEN f.scenario='BUDGET' THEN f.amount END) AS variance
FROM    fact_gl f
JOIN    dim_cost_centre cc ON cc.cc_key = f.cc_key
JOIN    dim_account     a  ON a.acc_key = f.acc_key
WHERE   f.fiscal_year = 2027
GROUP BY cc.cost_centre, a.account
ORDER BY variance;
```

7. Python — phase an annual budget across 12 months with seasonality

```
import pandas as pd
weights = pd.Series([0.06,0.06,0.07,0.08,0.08,0.09,
                     0.09,0.09,0.09,0.09,0.10,0.10])   # sums to 1.0
annual = 24_000_000                                    # ₹2.4 Cr
monthly = (annual * weights).round(0)
print(monthly)
```

8. Power BI DAX for the dashboard

```
Budget     = CALCULATE(SUM(Fact[Amount]), Fact[Scenario]="Budget")
Actual     = CALCULATE(SUM(Fact[Amount]), Fact[Scenario]="Actual")
Variance   = [Actual] - [Budget]
Variance % = DIVIDE([Variance], [Budget])
```

Worked example / mini-project

Company: *Nirmaan Interiors Pvt Ltd*, a Pune SME. Build FY2027-28 budget. Reproduce this.

Step 1 — Revenue (bottom-up). Two streams:

Stream	Driver	Value	Annual revenue
Residential projects	40 projects × ₹6,00,000		₹2,40,00,000
Commercial fit-outs	8 projects × ₹18,00,000		₹1,44,00,000
Total revenue			**₹3,84,00,000**

Step 2 — Direct costs (COGS ~55% of revenue).

Item	Basis	₹
Materials	40% of revenue	1,53,60,000
Site labour (contract)	15% of revenue	57,60,000
Gross profit		**1,72,80,000 (45%)**

Step 3 — Cost-centre budgets (overheads).

CC	Line	Monthly ₹	Annual ₹
Admin (5100)	Salaries (5 staff)	3,00,000	36,00,000
Admin	Office rent	1,20,000	14,40,000
Marketing (5200)	Digital + events	1,50,000	18,00,000
Design (5300)	Software + salaries	4,00,000	48,00,000
Total overheads			**1,16,40,000**

Step 4 — Budgeted P&L.

	₹
Revenue	3,84,00,000
COGS	(2,11,20,000)
Gross profit	1,72,80,000
Overheads	(1,16,40,000)
EBITDA	**56,40,000 (14.7%)**

Step 5 — Top-down challenge. CEO wants EBITDA ₹70,00,000. Gap = ₹13,60,000. Levers modelled: +5% price on commercial (₹7,20,000), cut events budget 30% (₹3,00,000), one design hire deferred 6 months (₹3,60,000) → closes ₹13,80,000. Re-run: **EBITDA ₹70,20,000**. Present both versions plus the bridge.

Step 6 — Month 1 variance. April actual overheads ₹10,20,000 vs budget ₹9,70,000 → variance ₹50,000 **Adverse (+5.2%)** — flag to Admin owner.

How it's tested

Interview questions - "Walk me through how you'd build a company's annual budget from scratch." (They want *process* + *calendar*, not formulas.) - "Top-down vs bottom-up — which do you use and why?" (Answer: both — bottom-up build, top-down target, reconcile the gap.) - "A cost-centre owner submits a number ₹20L above last year with no justification. What do you do?" - "How do you handle version control across 15 departments?" - "Difference between a fixed and a flexed budget?" (Flexed = re-cast at actual activity level for fair variance.)

Practical assessments - **Timed Excel test (45-60 min):** given a raw cost-centre extract, build a consolidated budget with `SUMIFS`, add a driver-based revenue build, and produce a variance column. They watch for input/formula separation, no hard-coding, and a tie-out. - **Case:** "Here's the CEO's EBITDA target and the bottom-up submission — close the gap and justify each lever" (often with Goal Seek). - **Modelling test:** build a 3-scenario (base/best/worst) budget with a scenario switch (`CHOOSE` / `INDEX`).

Common mistakes & how pros avoid them

Mistake	How pros avoid it
Hard-coding numbers into formulas	Every assumption in a labelled blue input cell
÷12 flat phasing	Seasonality weight row that sums to 100%
15 differently-shaped tabs that won't roll up	One flat table + `SUMIFS` ; template locked for owners
Typing a revenue number	Driver-based build (units × price, headcount × output)
No top-down check	Always show target vs bottom-up vs gap bridge
Version chaos ("final_v7")	Dated filenames + a change-log tab; single source of truth
Padding / sandbagging by owners	Challenge against prior-year actuals and per-unit ratios
Budget ignored after April	Monthly budget-vs-actual pack with variance commentary
No cash budget, only P&L	Add a cash/working-capital line — profit ≠ cash

Learn-it roadmap & resources

Time to proficiency: 6-10 weeks part-time if you already know Excel basics.

Weeks	Focus
1-2	Excel for FP&A: `SUMIFS` , `INDEX/MATCH` , `XLOOKUP` , tables, data validation
3-4	Build a driver-based revenue model + cost-centre template
5-6	Consolidation, top-down/bottom-up reconciliation, Goal Seek, scenarios
7-8	Variance reporting + a Power BI / DAX dashboard
9-10	End-to-end mock budget for a fictional company (portfolio piece)

Resources - CA Inter Cost & Management Accounting — *Budgets & Budgetary Control* chapter (free ICAI module) for the theory backbone. - CFI *FP&A / Budgeting & Forecasting* course (paid) — job-shaped. - Corporate Finance modelling on YouTube (free): "driver-based budgeting", "budget vs actual dashboard Power BI". - Practise on real templates: build the *Nirmaan Interiors* example above yourself.

Certifications: CMA (India) or CMA (US) both weight budgeting heavily; CFI **FMVA** is the practical badge recruiters recognise for FP&A.

Quick-reference

Need	Formula / step
Consolidate cost centres	`=SUMIFS(tbl[Amount], tbl[Account],A5, tbl[Month],B$4)`
Revenue from mix	`=SUMPRODUCT(units, price)`
MoM growth	`=Prior*(1+growth%)`
Phase annual → months	`annual * weight_row` (weights sum to 1)
Variance	`=Actual-Budget`
Variance %	`=(Actual-Budget)/Budget`
Cost flag	`=IF(Actual≤Budget,"Fav","Adv")`
Close top-down gap	Data → What-If → **Goal Seek**
Scenario switch	`=CHOOSE(switch, base, best, worst)`
DAX variance	`Variance = [Actual]-[Budget]`

Golden rules: FY = Apr-Mar · inputs blue, formulas black · never hard-code · build revenue from drivers · one flat table, not 15 tabs · always reconcile top-down vs bottom-up · budget is worthless without monthly variance.

Typical budget calendar (SME/mid-market):

When	Milestone
Nov	CFO issues assumptions + top-down targets, template out
Early Dec	Cost-centre owners submit bottom-up (V1)
Mid Dec	FP&A consolidates, first gap-to-target review
Late Dec	Iterations V2/V3, lever decisions
Jan	CFO / board review and sign-off
Feb-Mar	Load approved budget into ERP; lock baseline
Apr	Year starts; first budget-vs-actual pack

Forecasting & Rolling Forecasts

What it is & where it's used

A forecast is your best current estimate of where the numbers will land — revenue, costs, cash, headcount — given what you know today. It is different from a **budget** (a fixed annual target, locked once) and from **actuals** (what happened). FP&A lives in the gap between them: budget vs actual vs forecast.

A **rolling forecast** drops the "we plan once a year in October and pray" model. Instead you always look 12–18 months ahead, re-forecasting every month or quarter, so the horizon "rolls" forward and never shrinks to zero in December.

Where it shows up on the job:

- **FP&A analyst / manager** — owns the monthly re-forecast and the board deck.
- **Business finance / finance business partner** — forecasts a specific P&L (a region, a product line, a plant).
- **Startup finance / founder's office** — cash runway and revenue forecasts for investors.
- **Treasury** — 13-week cash-flow forecast.
- **Corporate development / IB** — the operating model behind a DCF is a forecast.

If a job description says "variance analysis," "driver-based planning," "cash flow forecasting," or "management reporting," this chapter is the core skill.

The gap: why companies want this (and college didn't teach it)

College hands you a completed cash-flow statement and asks you to compute a ratio. Employers hand you a blank sheet and last year's messy actuals and say "tell me what Q3 looks like, and defend it." That is a completely different muscle.

Specifically, the gaps:

College taught	The job needs
One "correct" answer	A defensible *estimate* with assumptions written down
Static full-year budget	A living forecast updated monthly
Revenue as a single number	Revenue broken into **drivers** (units × price, or leads × conversion × ARPU)
Ignore what already happened	**Run-rate** the year-to-date actuals into the remaining months
Model never checked again	**Forecast accuracy** measured and improved each cycle

The commercial reason companies pay for this: a forecast that is wrong by 15% every quarter causes the wrong hiring, the wrong inventory buy, and a missed loan covenant. A finance person who forecasts within ±3–5% and *explains the miss* is directly protecting cash and credibility.

What "proficient" looks like

A job-ready person can, unaided:

1. Take 6 months of actuals + a budget and produce a full-year forecast in Excel with a clean **actuals | forecast** split (bold vertical line at the current month).
2. Choose the right method per line: **run-rate** for stable overheads, **driver-based** for revenue and variable cost.
3. Build a **rolling 12-month** version where changing "current month" re-slices actual vs forecast automatically.
4. Track **forecast accuracy** (MAPE / bias) and show whether they consistently over- or under-call.
5. Write a 5-bullet assumptions box a non-finance manager can challenge.
6. Explain a variance in business language: "Revenue is ₹4L below forecast because conversion dropped from 3.0% to 2.4%, not because traffic fell."

Hands-on: how to actually do it

1. Run-rate (simplest, for stable lines)

Run-rate = annualize what's happened so far. If YTD actual through month `n` is known:

```
' YTD actual in B2, current month number in B3 (e.g. 6)
=B2 / B3 * 12                          ' straight run-rate for the full year
=B2 + (B2 / B3) * (12 - B3)            ' actual-to-date + run-rate for remaining months
```

Use run-rate for rent, salaries, subscriptions, insurance — costs that don't swing with volume.

2. Driver-based (for revenue and variable cost)

Break the number into the levers a manager actually controls. For a D2C business:

```
Revenue = Sessions × Conversion% × Average Order Value
```

```
' Sessions C2, Conversion C3, AOV C4
=C2 * C3 * C4
```

For variable cost tie it to the same driver:

```
' Units D2, cost per unit D3
=D2 * D3
```

The power: to model a scenario you change *one driver*, not 40 output cells.

3. The actuals-vs-forecast switch (the heart of a rolling model)

Put a single cell `CurrentMonth` (say = 6). Each month column knows if it's actual or forecast:

```
' Row of month numbers in C1:N1 (1..12), CurrentMonth in $B$1
' Actual row C2:N2, Forecast-logic below
=IF(C$1 ≤ $B$1, C_actual, C_forecast)
```

A cleaner pattern using named cells and SWITCH-style logic:

```
=IF(C$1≤$B$1, INDEX(Actuals,C$1), INDEX(ForecastDrivers,C$1))
```

Then a live "Full Year" column blends both:

```
' Actuals for elapsed months + forecast for the rest
=SUMPRODUCT((MonthNums≤CurrentMonth)*ActualRow)
 + SUMPRODUCT((MonthNums>CurrentMonth)*ForecastRow)
```

4. Forecast accuracy (MAPE and bias)

Once actuals land, score last cycle's forecast:

```
' Actual A2:A13, Forecast F2:F13
' MAPE  (lower = better; <5% is strong for revenue)
=AVERAGE(ABS(A2:A13 - F2:F13) / A2:A13)     ' enter as array / dynamic array
' Bias  (positive = you over-forecast on average)
=AVERAGE((F2:F13 - A2:A13) / A2:A13)
```

5. SQL to pull the actuals your forecast rests on

```
SELECT DATE_TRUNC('month', txn_date) AS month,
       SUM(amount)                   AS revenue,
       COUNT(DISTINCT order_id)      AS orders
FROM sales
WHERE txn_date ≥ '2025-04-01'      -- Indian FY start
GROUP BY 1
ORDER BY 1;
```

6. Python for a quick trend / seasonal forecast

```
import pandas as pd

df = pd.read_csv("monthly_revenue.csv", parse_dates=["month"])
df = df.set_index("month").asfreq("MS")

# 3-month moving-average run-rate
df["run_rate"] = df["revenue"].rolling(3).mean()

# simple seasonal index (this month vs 12-month avg)
df["seasonal_idx"] = df["revenue"] / df["revenue"].rolling(12).mean()
print(df.tail())
```

7. DAX (Power BI) — actual vs forecast in one measure

```
Actual or Forecast =
IF (
    MAX ( 'Calendar'[MonthNum] ) ≤ [Current Month],
    [Actual Revenue],
    [Forecast Revenue]
)

MAPE =
AVERAGEX (
    VALUES ( 'Calendar'[MonthNum] ),
    DIVIDE ( ABS ( [Actual Revenue] - [Forecast Revenue] ), [Actual Revenue] )
)
```

Worked example / mini-project

Business: a Bengaluru D2C skincare brand. FY26 (Apr–Mar). You have actuals Apr–Sep; forecast Oct–Mar.

Actuals (₹, revenue):

Month	Sessions	Conv%	AOV (₹)	Revenue (₹)
Apr	80,000	2.8%	950	21,28,000
May	85,000	2.9%	940	23,17,100
Jun	82,000	2.7%	960	21,25,440
Jul	90,000	3.0%	970	26,19,000
Aug	95,000	3.1%	980	29,58,100
Sep	98,000	3.0%	990	29,10,600

YTD actual revenue = **₹1,50,58,240**.

Driver forecast Oct–Mar — assumptions written down: - Sessions grow 3% MoM (festive push Oct–Nov, +8% those two months). - Conversion holds at 3.0%. - AOV rises ₹10/month.

Month	Sessions	Conv%	AOV (₹)	Forecast Rev (₹)
Oct	1,05,840 (+8%)	3.0%	1,000	31,75,200
Nov	1,14,307 (+8%)	3.0%	1,010	34,63,500
Dec	1,17,736 (+3%)	3.0%	1,020	36,02,720
Jan	1,21,268	3.0%	1,030	37,47,180
Feb	1,24,906	3.0%	1,040	38,97,070
Mar	1,28,653	3.0%	1,050	40,52,570

Forecast H2 = **₹2,19,38,240**. **Full-year forecast = ₹3,69,96,480.**

Now the accuracy check. Say last year you forecast Sep at ₹27,50,000 and actual came in ₹29,10,600:

```
Error% = (27,50,000 − 29,10,600) / 29,10,600 = −5.5%  (you under-forecast)
```

Do this for all six actual months, average the absolute values → your MAPE. If it's 4.2%, you tell the board: "Historical forecast accuracy is ~96%; treat the ₹3.70 Cr as ±₹15L."

Reproduce it: put months across columns, `CurrentMonth=6`, drivers in a block, and one `IF(month≤6, actual, driver_forecast)` row. Add a MAPE cell. That single tab is a legitimate portfolio piece.

How it's tested

Interview questions: - "Walk me through building a revenue forecast for a business you know." (They want *drivers*, not "I'd grow it 10%.") - "Difference between budget, forecast, and actual?" - "Your forecast was off by 12% — how do you find out why?" (Bridge it by driver.) - "Run-rate vs driver-based — when each?" - "What's a rolling forecast and why bother?"

Practical assessments (very common in FP&A hiring): - **Timed Excel case (45–60 min):** given raw actuals + a budget, build a full-year forecast with an actuals/forecast split and a variance column. Scored on structure, correct `SUMPRODUCT` / `IF` logic, and a written assumptions box. - **"Break this forecast" exercise:** they hand you a model and ask what's wrong (hardcoded numbers inside formulas, no assumptions tab, circular references). - **Variance bridge:** turn a ₹4L revenue miss into a volume/price/mix waterfall. - **Case presentation:** 3–5 slides — forecast, key drivers, risks, ask. Communication is graded as heavily as the math.

Common mistakes & how pros avoid them

Mistake	Fix
Hardcoding numbers inside formulas (`=C2*1.1`)	Every assumption lives in its own labelled cell; formulas only reference cells
No assumptions tab	One box: growth %, conversion, AOV, headcount plan — dated and initialled
Forecasting the total, not the drivers	Always decompose: units × price, or funnel steps
Ignoring YTD actuals — re-forecasting from budget	Anchor on actuals; only the *remaining* months are forecast
Straight-lining seasonal businesses	Apply a seasonal index; festive/quarter-end spikes are real in India
Never scoring past accuracy	Track MAPE + bias each cycle; a persistent bias means fix the method
Sandbagging (deliberately low) or hockey-sticks	Bias metric exposes it; pros forecast the honest number and flag risk separately
One number, no range	Give base / best / worst, or "±X% based on historical accuracy"

Pros also keep a **version log** — snapshot each month's forecast so you can see how your Dec call for full-year revenue drifted from Aug to Nov. That drift *is* the story the CFO wants.

Learn-it roadmap & resources

Time to job-ready: 4–8 weeks of deliberate practice if your Excel is already decent.

Week	Focus
1	Excel mechanics: `INDEX` , `SUMPRODUCT` , `IF` , `XLOOKUP` , dynamic arrays; build a run-rate
2	Driver-based revenue model + assumptions tab
3	Rolling 12-month structure with the `CurrentMonth` switch
4	Variance bridge + MAPE/bias tracking
5–6	Full mini-project (like above) end to end; write the assumptions & risks
7–8	Power BI/DAX or Python version; present a 3-slide case out loud

Resources: - *Free:* Corporate Finance Institute (CFI) FP&A free lessons; Microsoft's Excel forecast docs; `statsmodels` / `pandas` docs for Python; any real company's investor deck (reverse-engineer their drivers). - *Paid / high-signal:* CFI **FMVA** certification (financial modeling + FP&A modules), Wall Street Prep FP&A course, Rob Marlow / Leila Gharani Excel on YouTube (free but excellent). - *Cert worth naming on a resume in India:* FMVA (CFI). Your CA Intermediate + MBA already cover the accounting depth — the modeling cert closes the "can you build it in Excel" question.

Build one public forecast model (dummy or a listed company's segment) and put it on GitHub/LinkedIn. That artifact beats any line on a CV.

Quick-reference

Need	Formula / step
Straight run-rate (full year)	`=YTD / MonthNo * 12`
Actual + run-rate remainder	`=YTD + (YTD/MonthNo)*(12-MonthNo)`
Driver revenue	`=Sessions * Conv% * AOV`
Actual/forecast switch	`=IF(Month≤CurrentMonth, Actual, Forecast)`
Blended full year	`=SUMPRODUCT((M≤Cur)*Act)+SUMPRODUCT((M>Cur)*Fcst)`
MAPE	`=AVERAGE(ABS(Act-Fcst)/Act)` (array)
Bias	`=AVERAGE((Fcst-Act)/Act)`
Variance %	`=(Actual-Forecast)/Forecast`

Rules of thumb: revenue MAPE < 5% is strong; run-rate stable overheads, driver-model everything volume-linked; anchor on actuals and only forecast the remaining months; every assumption in its own cell; snapshot each forecast version; always give a range, never a false single point.

Variance Analysis & Bridges

What it is & where it's used

Variance analysis is the discipline of explaining *why* actual results differ from plan (budget), forecast, or prior period — and doing it in a way that points a manager to a decision. A "bridge" is the structured walk from one number to another (Budget Revenue → Actual Revenue, or LY EBITDA → TY EBITDA) broken into named, additive drivers: price, volume, mix, FX, rate, efficiency, one-offs.

If you only remember one thing: **a variance is useless until it's decomposed into a driver someone owns.** "Revenue is ₹40 lakh below plan" is a symptom. "Volume was fine but realised price fell ₹18/unit because we over-discounted the North region" is an action.

Where it's used and who's tested on it:

Role	How variance analysis shows up
FP&A analyst	Monthly Budget-vs-Actual (BvA) deck, driver bridges, commentary
Business finance partner	Explaining a plant/region/product-line miss to the GM
Controller / R2R	Flux analysis at close (BS & P&L movement explanations)
Cost accountant (CA)	Standard costing variances (MPV, MUV, LRV, LEV, overhead)
Corporate strategy / IR	Bridging EBITDA/EPS growth for board and investor decks

The gap: why companies want this (and college didn't teach it)

College teaches the *formulas* — material price variance = (SP − AP) × AQ — as isolated exam sums. Industry needs the opposite skill: taking a messy actuals file plus a budget file, reconciling them to the last rupee, decomposing the gap into 4–6 drivers that **sum exactly** to the total variance, and writing three sentences a busy GM will act on.

The specific gaps:

- **Reconciliation, not calculation.** Real files don't tie. Actuals sit in the ERP at transaction grain; budget sits in a flat Excel by product-month. You must join them and prove the bridge foots.
- **Mix.** Almost no MBA course properly separates *volume* from *mix*. It's the driver managers most often get wrong, and the one interviewers probe.
- **Materiality & narrative.** A 40-line variance table is not analysis. Pros surface the 3 drivers that explain 80% of the gap and stay silent on noise.
- **Sign discipline.** Favourable/unfavourable (F/U) is not "positive/negative" — a cost overspend is a negative number but you must label it U. Getting this wrong in a board deck is career-limiting.

What "proficient" looks like

A job-ready person, handed a budget file and an actuals extract, can unaided:

1. Reconcile both to the same grain and confirm totals tie to the GL.
2. Build a **price/volume/mix bridge** where the components sum exactly to (Actual − Budget).

3. Produce a **waterfall chart** in Excel or Power BI showing the walk.
4. Correctly label every variance F or U with the right sign convention.
5. Apply a materiality threshold (e.g. > ₹5 lakh or > 5%) and comment on only those.
6. Write commentary in the form **[Driver]** → **[₹ impact, F/U]** → **[root cause]** → **[action/owner]**.
7. Do it monthly on a repeatable, refreshable model — not a one-off hack.

Hands-on: how to actually do it

The decomposition formulas (memorise these)

For a single product, comparing Budget (B) to Actual (A):

```
Total Rev variance = (Aqty × Aprice) − (Bqty × Bprice)

Price variance  = (Aprice − Bprice) × Aqty
Volume variance = (Aqty − Bqty) × Bprice
```

Price + Volume = Total. That's the two-way split. Now add **mix** for a multi-product portfolio (the version employers actually want):

```
Volume variance = (Total Aunits − Total Bunits) × Budget avg price   ← pure size
Mix variance    = Σ [ (Actual mix% − Budget mix%) × Total Aunits × Bprice_i ]
Price variance  = Σ [ (Aprice_i − Bprice_i) × Aqty_i ]
```

Rate/efficiency (cost side, standard costing):

```
Material Price Var (MPV)  = (Std price − Act price) × Act qty
Material Usage Var (MUV)  = (Std qty − Act qty) × Std price
Labour Rate Var (LRV)     = (Std rate − Act rate) × Act hrs
Labour Efficiency (LEV)   = (Std hrs − Act hrs) × Std rate
```

Excel — building the bridge

Assume `Actuals` and `Budget` tables keyed by Product. Pull budget values next to actuals:

```
=XLOOKUP([@Product], Budget[Product], Budget[Qty])          'budget qty
=XLOOKUP([@Product], Budget[Product], Budget[Price])        'budget price
```

Per-row driver columns:

```
Price var   =([@ActPrice]-[@BudPrice])*[@ActQty]
Volume var  =([@ActQty]-[@BudQty])*[@BudPrice]
Check       =[@PriceVar]+[@VolVar]-([@ActQty]*[@ActPrice]-[@BudQty]*[@BudPrice])   'must be 0
```

Portfolio volume vs mix (SUMPRODUCT):

```
BudAvgPrice =SUMPRODUCT(Budget[Qty],Budget[Price])/SUM(Budget[Qty])
VolumeVar   =(SUM(Actuals[ActQty])-SUM(Budget[Qty]))*BudAvgPrice
MixVar      =SUMPRODUCT((Actuals[ActQty]/SUM(Actuals[ActQty])
             -Budget[Qty]/SUM(Budget[Qty]))*SUM(Actuals[ActQty])*Budget[Price])
```

F/U flag:

```
=IF([@TotalVar]≥0,"F","U")     'revenue: positive = Favourable
=IF([@CostVar]≤0,"F","U")      'costs:   spend below budget = Favourable
```

The waterfall chart (native, Excel 2016+)

Select the bridge series (Start, +Price, +Volume, +Mix, End) → **Insert** → **Waterfall**. Right-click the Start and End columns → **Set as Total**. Colour F green, U red. Done — no float-column hacks.

DAX (Power BI) measures

```
Price Var =
SUMX( VALUES(Sales[Product]),
    ([Actual Price] - [Budget Price]) * [Actual Qty] )

Volume Var =
( SUM(Actuals[Qty]) - SUM(Budget[Qty]) ) * [Budget Avg Price]

Total Var = [Actual Revenue] - [Budget Revenue]
Mix Var   = [Total Var] - [Price Var] - [Volume Var]   // plug so it foots
```

SQL — reconcile actuals to budget at source

```
SELECT  b.product,
        b.qty              AS bud_qty,
        b.price            AS bud_price,
        a.qty              AS act_qty,
        a.price            AS act_price,
        (a.price-b.price)*a.qty                       AS price_var,
        (a.qty -b.qty )*b.price                       AS volume_var,
        a.qty*a.price - b.qty*b.price                 AS total_var
FROM        budget  b
LEFT JOIN   (SELECT product, SUM(qty) qty,
                    SUM(qty*price)/SUM(qty) price
             FROM invoices WHERE period='2026-06'
             GROUP BY product) a  ON a.product=b.product
ORDER BY ABS(a.qty*a.price - b.qty*b.price) DESC;      -- biggest miss first
```

Python — reusable bridge function

```
import pandas as pd

def revenue_bridge(bud, act):
    m = bud.merge(act, on="product", suffixes=("_b","_a"))
    m["price_var"]  = (m.price_a - m.price_b) * m.qty_a
    m["volume_var"] = (m.qty_a  - m.qty_b ) * m.price_b
    tot_var = (m.qty_a*m.price_a - m.qty_b*m.price_b).sum()
    bud_avg = (bud.qty*bud.price).sum()/bud.qty.sum()
    vol = (act.qty.sum()-bud.qty.sum())*bud_avg
    mix = tot_var - m["price_var"].sum() - vol
    return {"price": m.price_var.sum(), "volume": vol,
            "mix": mix, "total": tot_var}
```

Worked example / mini-project

Setup: A mid-size Indian FMCG unit sells two SKUs. June 2026 budget vs actual:

SKU	Bud Qty	Bud Price ₹	Act Qty	Act Price ₹
Premium	10,000	250	9,000	260
Value	20,000	100	26,000	92

Budget revenue = 10,000×250 + 20,000×100 = ₹45,00,000 Actual revenue = 9,000×260 + 26,000×92 = ₹23,40,000 + ₹23,92,000 = ₹47,32,000 **Total variance = +₹2,32,000 (F)**

Now decompose:

Price = (260−250)×9,000 + (92−100)×26,000 = +90,000 − 2,08,000 = **−₹1,18,000 (U)**

Portfolio volume — budget avg price = 45,00,000 / 30,000 = ₹150. Total units: bud 30,000, act 35,000. Volume = (35,000−30,000)×150 = **+₹7,50,000 (F)**

Mix = Total − Price − Volume = 2,32,000 − (−1,18,000) − 7,50,000 = **−₹4,00,000 (U)**

Bridge (foots to +2,32,000):

Driver	₹	F/U
Budget revenue	45,00,000	
Price	−1,18,000	U
Volume	+7,50,000	F
Mix	−4,00,000	U
Actual revenue	**47,32,000**	**F**

Commentary a GM will act on:

> Revenue beat plan by ₹2.3L (F), but the beat is low quality. Volume drove +₹7.5L as total units ran 17% ahead. This was more than offset by an adverse mix of −₹4.0L (growth came from Value SKU, up 30%, while Premium fell 10%) and adverse price of −₹1.2L (Value discounted ₹8/unit to move stock). **Action:** Sales to defend Premium volume and roll back Value discounting — current trajectory dilutes gross margin even as top line grows. Owner: Regional Sales Head.

Reproduce it: drop the four-row table into Excel, add the driver columns above, insert a Waterfall chart.

How it's tested

Interview questions: - "Revenue is up but margin is down — walk me through how you'd find why." (They want price/mix separation.) - "Difference between volume and mix variance?" (The classic filter.) - "This cost variance is +₹2L — favourable or unfavourable?" (Sign discipline trap.) - "Your bridge doesn't foot. Where do you look first?" (Grain mismatch, missing SKUs, FX.)

Practical assessments: - **Timed Excel test (45–60 min):** given `budget.xlsx` + `actuals.xlsx`, build a BvA with price/volume/mix bridge, a waterfall, and 3 bullets of commentary. - **Flux/close case:** "Here's this month's P&L vs last month — explain every line moving > ₹5L." (Controller roles.) - **Standard costing sum:** compute MPV/MUV/LRV/LEV and reconcile to total cost variance (CA/cost-accountant roles). - **Take-home:** a data set where the naive two-way split hides a mix story — they check if you catch it.

Common mistakes & how pros avoid them

Mistake	Fix
Bridge doesn't foot to total	Add a `check = sum(drivers) - total` column; never ship until 0
Confusing volume with mix	Volume = *size* at budget avg price; mix = *composition* shift. Compute volume first, plug mix
Wrong F/U sign on costs	Costs: under-spend = F. Revenue: over = F. Label explicitly, don't rely on sign
Commenting on everything	Set a materiality gate (₹ or %); comment on the top 3 drivers only
Cause with no action	Every comment ends in an action + owner, not just "due to lower volume"
Comparing apples to oranges	Reconcile grain/currency/period first; strip FX into its own bridge column
Double-counting price & mix	Price on Actual qty, mix as the plug — don't derive both independently

Learn-it roadmap & resources

Time to proficiency: 4–6 weeks part-time for someone with your accounting base.

- **Week 1:** Nail the formulas — two-way (price/volume), then three-way (add mix). Do 10 hand sums. Standard costing variances overlap directly with CA Cost material.
- **Week 2:** Rebuild the worked example in Excel; master XLOOKUP, SUMPRODUCT, native Waterfall.
- **Week 3:** Automate — the Python function above, or a Power BI model with the DAX measures. Learn to refresh, not rebuild.
- **Week 4:** Write commentary. Take any bridge and force yourself to 3 action-oriented bullets under a materiality gate.

Resources: - CA Inter Cost & Management Accounting — the standard costing chapter *is* variance analysis (free, you already have it). - CFI's *FP&A* and *Budgeting & Forecasting* courses (paid) — bridge and commentary templates. - Leila Gharani / "Excel Off The Grid" YouTube — waterfall and dynamic bridges (free). - Practice data: Kaggle "retail sales" sets; build BvA on them. - **Certification signal:** CMA (US) or CIMA cover this deeply; for your target roles, a strong Excel bridge in your portfolio beats another cert.

Quick-reference

```
Two-way:
  Price var  = (Aprice − Bprice) × Aqty
  Volume var = (Aqty − Bqty) × Bprice

Three-way (portfolio):
  Volume = (ΣAqty − ΣBqty) × Budget avg price
  Mix    = Total − Price − Volume        (plug, must foot)

Cost / standard costing:
  MPV = (Sp−Ap)×Aq   MUV = (Sq−Aq)×Sp
  LRV = (Sr−Ar)×Ah   LEV = (Sh−Ah)×Sr
```

Item	Rule
F/U — revenue	Actual > Budget = F
F/U — cost	Actual < Budget = F
Foot check	Σ drivers − total = 0 (mandatory)
Materiality	comment only if > ₹5L or > 5%
Waterfall	Insert → Waterfall → Set first/last as Total
Commentary	Driver → ₹ impact (F/U) → root cause → action + owner
Volume vs mix	Volume = size @ budget avg price; Mix = composition shift

Management reporting & the board deck

What it is & where it's used

Management reporting is the monthly ritual where Finance turns the trial balance into a story the leadership team can act on. Two artefacts dominate:

1. **The monthly reporting pack (MIS pack)** — a 15-40 page internal document circulated 5-10 working days after month-end. It carries the P&L vs budget, cash, working capital, department spend, and KPI dashboards.
2. **The board deck** — a tighter 12-20 slide summary presented to the board of directors / investors every month or quarter. It is the pack, distilled and narrated.

Roles that live in this work: **FP&A Analyst/Manager, Management Accountant, Business Finance Partner, Financial Controller, Head of Finance/CFO**, and in startups the **Finance lead** who does all of it alone. If a job description says "prepare monthly MIS," "budget vs actual variance analysis," "board reporting," or "flash reporting," this chapter is that job.

The gap: why companies want this (and college didn't teach it)

College teaches you to *prepare* financial statements per Ind AS / Schedule III. It does not teach you to **explain** them to a CEO who has 90 seconds. The gap is threefold:

- **From accuracy to insight.** Anyone can show that Gross Margin fell to 38%. The paid skill is stating *why* (input cost up 4%, discounting up 2%, mix shift toward low-margin SKUs) and *what to do*.
- **From statutory to managerial.** The board doesn't care about Schedule III line ordering. They want revenue by segment, contribution margin, cash runway, and CAC/LTV — cuts that never appear in the audited financials.
- **From data to narrative.** MBAs build DCFs; employers need someone who can build a clean waterfall, write a three-bullet commentary, and not blow up the CFO in front of the board. Nobody grades "commentary writing" in a degree, but it's the entire value of the role.

What "proficient" looks like

A job-ready person can, **unaided**, take a trial balance and a budget file and produce by Day 5:

- A **budget-vs-actual P&L** with variance columns (₹ and %) and a favourable/adverse flag.
- A **bridge/waterfall** explaining the movement in profit (or revenue) from budget to actual, or MoM.
- A **cash and working-capital** view: closing cash, burn, runway, DSO/DPO/DIO.
- A **one-page executive summary**: 3-5 bullets, each a number + cause + action.
- A **board deck** that opens with the punchline, not the appendix.
- Commentary that survives the "so what?" test on every slide.

They know the reporting calendar, tie every number back to the GL, and never present a variance they can't explain.

Hands-on: how to actually do it

1. The variance table (Excel)

Assume actuals in column C, budget in D. Variance and flag:

```
Variance ₹:   =C2-D2
Variance %:   =IFERROR((C2-D2)/ABS(D2),"n/a")
Flag:         =IF(N2=0,"–",IF(ISNUMBER(SEARCH("expense",$A2)), IF(C2<D2,"Fav","Adv"), IF(C2>D2
```

For revenue/profit rows, actual > budget is Favourable; for cost rows it flips — hence the `SEARCH("expense" … )` switch, or keep a sign column.

Pull actuals from the GL dump with a keyed lookup instead of manual typing:

```
=XLOOKUP($A2, GL!$B:$B, GL!$H:$H, 0)          'exact GL account match
=SUMIFS(GL!$H:$H, GL!$C:$C, $A2, GL!$E:$E, "≥"&$F$1, GL!$E:$E, "≤"&$G$1)   'sum by cost cent
```

Materiality filter — only comment on what matters. A common rule: variance is reportable if `ABS(₹) > 500000 OR ABS(%) > 10%`.

```
=IF(OR(ABS(E2)>500000, ABS(F2)>0.10), "COMMENT", "")
```

2. The profit bridge (waterfall)

Insert → Chart → Waterfall in Excel 365. Data laid out as:

Step	Value (₹ lakh)
Budget EBITDA	120
Volume	+18
Price	+9
Input cost	-14
Employee cost	-6
Actual EBITDA	127

Mark "Budget EBITDA" and "Actual EBITDA" as **Totals** (double-click the point → Set as Total) so they anchor to the axis.

3. Rolling KPIs (Power BI / DAX)

```
Revenue MTD = CALCULATE([Revenue], DATESMTD('Date'[Date]))
Revenue YTD = CALCULATE([Revenue], DATESYTD('Date'[Date]))
Var to Budget % = DIVIDE([Revenue] - [Budget], [Budget])
Gross Margin % = DIVIDE([Revenue] - [COGS], [Revenue])
Cash Runway (months) = DIVIDE([Closing Cash], AVERAGEX(DATESINPERIOD('Date'[Date], MAX('Date'[
```

4. Pull the trial balance from Tally

TallyPrime → **Gateway of Tally** → **Display More Reports** → **Trial Balance** → **F12 (Configure)** set "Show Closing Balances" = Yes → **Alt+E (Export)** → **Excel (Spreadsheet)**. For segment cuts, enable **Cost Centres** first (F11 → Maintain Cost Centres = Yes) and export the **Cost Centre Breakup**.

5. Automating the merge (Python)

```
import pandas as pd

actual = pd.read_excel("tb_jun26.xlsx")          # GL account, Amount
budget = pd.read_excel("budget_fy27.xlsx")        # GL account, Jun

rep = actual.merge(budget, on="GL account", how="left", suffixes=("_act","_bud"))
rep["Var_INR"] = rep["Amount"] - rep["Jun"]
rep["Var_pct"] = rep["Var_INR"] / rep["Jun"].abs()
rep["Flag"]    = rep["Var_pct"].apply(lambda x: "Adv" if x < -0.10 else ("Fav" if x > 0.10 els
rep.to_excel("variance_jun26.xlsx", index=False)
```

Worked example / mini-project

Company: Zephyr Foods Pvt Ltd, a D2C snacks brand. June FY27 close.

Reporting P&L (₹ lakh):

Line	Actual	Budget	Var ₹	Var %	Flag
Net Revenue	480	500	-20	-4%	Adv
COGS	298	300	-2	fav cost	Fav
Gross Profit	**182**	**200**	**-18**	**-9%**	Adv
GM %	37.9%	40.0%	-2.1pp		Adv
Marketing	96	85	+11	+13%	Adv
Employee cost	54	55	-1		Fav
Other opex	32	30	+2		Adv
EBITDA	**0**	**30**	**-30**		Adv

Cash: Opening ₹620L, closing ₹548L, net burn ₹72L → runway = 548 / (avg 3-mo burn ₹68L) = **8.1 months**. DSO 42 days (up from 35), DIO 61 days.

Executive summary (this is the deliverable that earns the salary):

- **Revenue ₹480L, 4% below budget** — offline distributor sell-through slowed in Tier-2 (-₹28L); e-com grew +₹8L, partly offsetting.
- **Gross margin fell to 37.9% (-2.1pp)** — heavier discounting on the ₹99 SKU during the Amazon sale drove it; input costs were actually favourable.
- **EBITDA at breakeven vs ₹30L budget** — the miss is 60% margin, 40% a ₹11L marketing overspend on a campaign pulled forward from July.
- **Runway 8.1 months, DSO up 7 days** — one distributor is ₹40L overdue; collections escalated, targeting closure by 15 Jul.
- **Ask:** approve reallocating July marketing (already spent in June) and tighten discount-approval to protect Q2 margin.

Board deck order (12 slides): (1) Executive summary, (2) EBITDA bridge budget→actual, (3) Revenue by channel, (4) Gross margin trend + driver, (5) Opex, (6) Cash & runway, (7) Working capital / AR ageing, (8) KPI scorecard (CAC, LTV, repeat rate), (9) Key risks, (10) Decisions requested, (11) Next-month outlook, (12) Appendix.

How it's tested

Interview questions: - "Walk me through how you'd build a monthly MIS pack from a trial balance." - "Gross margin dropped 200bps this month. How do you find out why?" - "What's the difference between a statutory P&L and a management P&L?" - "How do you decide what goes on a board slide vs the appendix?" - "What's your reporting calendar look like around close?"

Practical assessments: - **Timed Excel/case (60-90 min):** given a raw TB and a budget tab, build the variance P&L, a waterfall, and write a 5-bullet summary. Graded on formula cleanliness (no hardcoding), correct fav/adverse logic, and whether commentary explains *causes*. - **"Fix the deck" test:** they hand you a bloated 40-slide deck and ask you to cut it to 12 and rewrite the opening slide. - **Live commentary drill:** here's the variance table — give me three bullets in two minutes.

Common mistakes & how pros avoid them

Mistake	What pros do
Reporting *what* happened, never *why*	Every material variance gets a one-line cause and action
Commentary restates the number ("Revenue was ₹480L")	State cause + quantify + implication
Data doesn't tie to the GL	Reconcile the pack total to the TB before circulating — always
40-slide deck, punchline on slide 31	Answer-first: executive summary is slide 1
Commenting on every line	Apply a materiality threshold; stay silent on noise
Hardcoded numbers, broken next month	Formula-driven, refreshable from a single data tab
Inconsistent numbers across slides	One source-of-truth data model feeds every slide
Presenting a variance you can't explain	Never take a number to the board you haven't traced
Colours/signs flipped (adverse shown green)	Lock a convention: adverse = red, favourable = green, consistently

Learn-it roadmap & resources

Time to proficiency: 6-10 weeks of deliberate practice if you already know Excel and basic accounting.

- **Weeks 1-2:** Master variance tables, SUMIFS/XLOOKUP, and fav/adverse logic. Rebuild a real company's segment P&L from its annual report.
- **Weeks 3-4:** Waterfalls, bridges, and writing commentary. Draft a 5-bullet exec summary daily on any dataset.
- **Weeks 5-6:** Build a KPI dashboard in Power BI or Excel. Learn the close calendar and reconciliation discipline.
- **Weeks 7-10:** Assemble a full mock board deck end-to-end and get it critiqued.

Resources: - CFI **FP&A** and **Financial Modeling (FMVA)** courses (paid) — strong on reporting structure. - Free: **Corporate Finance Institute** blog on variance analysis; **Microsoft Learn** for DAX; the "Financial Times of variance commentary" — practice by writing your own on public quarterly results. - Books: *Financial Planning & Analysis and Performance Management* (Alexander). - India: ICAI SFM/Cost material for cost-variance mechanics; study any listed company's investor deck as a board-deck template.

Quick-reference

Item	Formula / Step
Variance ₹	`=Actual-Budget`
Variance %	`=IFERROR((Act-Bud)/ABS(Bud),"n/a")`
Pull actual from GL	`=SUMIFS(GL!Amt, GL!Acct, key, GL!Date, "≥"&start, GL!Date,"≤"&end)`
Materiality flag	`=IF(OR(ABS(var)>500000,ABS(pct)>0.10),"COMMENT","")`
Runway (months)	`Closing cash / avg monthly net burn`
DSO	`AR / Revenue × days`
DPO	`AP / COGS × days`
DIO	`Inventory / COGS × days`
GM % (DAX)	`DIVIDE([Rev]-[COGS],[Rev])`
Tally TB export	Display More Reports → Trial Balance → Alt+E → Excel
Deck rule	Answer-first; exec summary slide 1; appendix last
Commentary formula	Number + Cause + Implication + Action
Colour convention	Adverse = red, Favourable = green
Close cadence	Pack by working day 5; board deck by day 7-10

The one line to remember: the board doesn't pay you to report the number — they pay you to explain the number and tell them what to do about it.

The driver-based operating model

What it is & where it's used

A **driver-based operating model** is a P&L (and often a mini balance sheet / cash view) where the numbers are *outputs* of operational assumptions, not typed-in guesses. Instead of "Revenue next month = ₹1.2 Cr because gut feel", you build:

> Revenue = **Units sold × Price**, where Units sold = **Sales reps × Deals per rep × Win rate**, and Price = **List price × (1 − discount %)**.

Change the number of reps or the win rate, and the whole P&L moves — consistently, traceably. The "drivers" are the handful of real-world levers (headcount, traffic, conversion, price, utilisation, occupancy, throughput) that management can actually pull.

Where it's used, by role:

Role	How they use it
FP&A analyst	Monthly forecast, annual operating plan (AOP), budget-vs-actual bridges
Corporate finance / strategy	Scenario planning, board decks, "what if we hire 10 reps"
Startup finance / founder's office	Investor model, runway, unit economics, fundraising deck
Business finance partner	Sitting with the Sales/Ops head, translating their plan into rupees
Investment/PE analyst	Bottoms-up revenue build for a target company

This is the single most-hired-for skill in FP&A. "Can you build a driver-based model?" is asked in almost every FP&A interview in India and globally.

The gap: why companies want this (and college didn't teach it)

MBA finance teaches you to *analyse* a given P&L: ratios, CAPM, NPV of a fixed cash-flow stream. It rarely teaches you to *construct* the P&L from operations. CA/Inter teaches you to *record and report* what already happened (accounting), and *cost* per unit — but not to project forward off business drivers under uncertainty.

The gap employers pay to close:

1. **Forward vs backward.** Accounting looks back; FP&A builds the number *before* it happens.
2. **Drivers vs line items.** College says "assume revenue grows 10%". Industry asks "*why* 10% — is it price or volume, and which lever created it?" A flat growth % is unusable in a board meeting because no one can act on it.
3. **Structure & auditability.** A real model separates **inputs (blue), calculations (black), outputs**. Nobody hard-codes a number inside a formula. Colleges never grade you on model hygiene.
4. **Linkage.** Revenue drives COGS drives headcount drives cash. College treats these as separate chapters; the job treats them as one connected sheet.

Close this gap and you're immediately more useful than someone who only knows Excel functions.

What "proficient" looks like

A job-ready person can, unaided:

- Take a business ("D2C skincare brand", "SaaS", "QSR restaurant chain") and **name its 5–8 real drivers** in under two minutes.
- Build a **12-month driver-based P&L** in Excel with a clean input block, monthly calc engine, and summary — properly colour-coded and formula-linked (zero hard-codes in calc cells).
- Wire a **scenario switch** (Base / Upside / Downside) that flips the whole model from one cell.
- Explain a **variance**: "Revenue missed by ₹40 lakh — ₹30L was volume (fewer units), ₹10L was price (higher discounting)."
- Sanity-check outputs with **ratios and reasonableness** (gross margin %, revenue per head, is EBITDA margin plausible for this industry?).
- Do it **fast** — a timed case in 60–90 minutes.

Hands-on: how to actually do it

Step 1 — Lay out the structure

Three zones. Use fill colours: **blue font = input**, **black = formula**, **green = link from another sheet**.

```
[INPUTS]        drivers & assumptions (editable)
[CALC ENGINE]   monthly build: Jan..Dec across columns
[OUTPUT]        P&L summary, KPIs, scenario toggle
```

Step 2 — Build the driver revenue block (SaaS example)

Put months across columns (C:N). Assumptions down the left.

```
                    Jan      Feb      Mar  ...
New leads           2,000    2,100    2,205     ← input, grows 5%/mo
Lead→trial %        10%      10%      10%       ← input
Trial→paid %        25%      25%      25%       ← input
New customers       =C_leads*C_l2t*C_t2p
Churn %             3%
Beginning custs     0        =prev End
Ending custs        =Begin + New - Begin*Churn%
ARPU (₹/mo)         1,500
MRR (₹)             =Ending custs * ARPU
```

Excel for New customers in C6:

```
=C2*C3*C4
```

Ending customers in C8 (Feb onward references prior End):

```
=C7 + C6 - C7*C5
```

MRR:

```
=C8*C9
```

Step 3 — Drive costs *off* the same engine

Costs should be formulas of drivers, not standalone lines.

```
COGS / hosting       = MRR * 0.15                         'gross margin driver
Sales headcount      = ROUNDUP(New_customers / 20, 0)     '1 rep closes 20/mo
Sales salary cost    = Sales_headcount * 80000            '₹80k/head/mo
CAC spend            = New_leads * 50                      '₹50 cost per lead
```

Step 4 — Use lookups so structure stays clean

Pull an assumption by name instead of hard-referencing a cell:

```
=XLOOKUP("ARPU", Assumptions[Driver], Assumptions[Value])
```

Older Excel / Google Sheets:

```
=INDEX(Assumptions!$C:$C, MATCH("ARPU", Assumptions!$B:$B, 0))
```

Step 5 — Add a scenario switch

Cell **B1** holds `Base` / `Upside` / `Downside`. Store three columns of assumptions and pick with:

```
=CHOOSE(MATCH($B$1,{"Base";"Upside";"Downside"},0), C_base, C_up, C_down)
```

Or with a data table / `SWITCH`:

```
=SWITCH($B$1,"Base",5%,"Upside",8%,"Downside",2%)   'monthly lead growth
```

Step 6 (optional) — Same model in Python for a large forecast

```
import pandas as pd
m = pd.date_range("2026-04-01", periods=12, freq="MS")
d = pd.DataFrame(index=m)
d["leads"] = 2000 * (1.05 ** range(12))
d["new_cust"] = d.leads * 0.10 * 0.25
cust, beg = [], 0
for n in d.new_cust:
    end = beg + n - beg*0.03
    cust.append(end); beg = end
d["end_cust"] = cust
d["MRR"] = d.end_cust * 1500
d["COGS"] = d.MRR * 0.15
d["gross_profit"] = d.MRR - d.COGS
print(d.round(0))
```

Variance analysis (price vs volume bridge)

```
Volume variance = (Actual units − Budget units) × Budget price
Price variance  = (Actual price − Budget price) × Actual units
```

Excel:

```
=(Act_units-Bud_units)*Bud_price   'volume effect
=(Act_price-Bud_price)*Act_units   'price effect
```

Worked example / mini-project — QSR restaurant chain (India)

Build a monthly P&L for a cloud-kitchen brand. Reproduce this.

Drivers (inputs):

Driver	Value
Orders/day/kitchen	120
Kitchens live	4 (add 1 every quarter)
Average order value (AOV)	₹350
Days/month	30
Food cost %	32% of revenue
Packaging/order	₹18
Delivery commission	22% of revenue
Staff/kitchen	6 @ ₹18,000/mo
Rent/kitchen	₹60,000/mo

Calc (Month 1, 4 kitchens):

```
Orders/mo   = 120 × 30 × 4              = 14,400
Revenue     = 14,400 × ₹350             = ₹50,40,000
Food cost   = 32% × Revenue             = ₹16,12,800
Packaging   = 14,400 × ₹18              = ₹2,59,200
Commission  = 22% × Revenue             = ₹11,08,800
Staff       = 6 × 4 × ₹18,000           = ₹4,32,000
Rent        = 4 × ₹60,000               = ₹2,40,000
────────────────────────────────────────────────────────
EBITDA      = 50.40L − 16.13 − 2.59 − 11.09 − 4.32 − 2.40
            = ₹13,87,200   (27.5% margin)
```

Excel layout — Revenue row across 12 months, kitchens stepping up:

```
Kitchens (C):  =IF(MOD(COLUMN()-COLUMN($C$1),3)=0, B_kitchens+1, B_kitchens)
Orders   (C):  =120*30*C_kitchens
Revenue  (C):  =C_orders*350
Food     (C):  =C_revenue*0.32
EBITDA   (C):  =C_revenue-C_food-C_pack-C_comm-C_staff-C_rent
EBITDA % (C):  =C_ebitda/C_revenue
```

Now the payoff: change **Orders/day from 120 to 100** in *one* input cell — EBITDA margin drops to ~20% and you can tell the founder exactly why. Change AOV to ₹400 and watch margin expand. That live sensitivity *is* the deliverable.

Extension: add a scenario toggle (Base 120 / Upside 150 / Downside 90 orders/day) and a 12-month EBITDA chart.

How it's tested

Interview questions: - "Walk me through how you'd build a revenue model for [our business]." - "What are the 3 biggest drivers of this P&L? Which would you stress-test?" - "Revenue is up 15% but EBITDA is flat — what happened?" (tests driver thinking) - "How do you separate price effect from volume effect?" - "How do you keep a model auditable when three people edit it?"

Practical assessments (common in India + globally): 1. **Timed Excel case (60–90 min):** "Here's a business description and some assumptions. Build a 12-month driver-based P&L with a scenario switch." Graded on structure, formula linkage (no hard-codes), correctness, and colour-coding. 2. **Take-home model:** Build an investor/operating model over 2–3 days; present it. 3. **Live "break my model":** They change an input and watch whether your P&L flows correctly and whether you can explain the movement. 4. **Case interview (consulting/strategy):** Verbal driver tree — "estimate annual revenue of a Domino's outlet" (seats/orders/AOV).

Prep tip: keep a **blank driver-model template** you can rebuild from memory in 30 minutes.

Common mistakes & how pros avoid them

Mistake	Why it hurts	Pro fix
Hard-coding numbers inside calc formulas	Nobody can trace or flex it	Every number lives once, in a blue input cell; calcs only reference
Growth % with no driver behind it	"10% because 10%" — unactionable	Decompose into price × volume, or leads × conversion
Costs typed independently of revenue	Model breaks when volume changes	Drive variable costs *off* revenue/units
No input/output separation	Reviewers can't audit	Blue/black/green colour convention, dedicated Inputs tab
Over-engineering (200 drivers)	Slow, fragile, unreadable	5–8 drivers that matter; rest are ratios
Circular references (interest ↔ cash)	#REF errors, breaks	Iterative calc on, or a copy-paste "circularity switch"
No sanity check	Ships a 60% EBITDA margin for a QSR	Benchmark margins/ratios before sending
Balance sheet doesn't balance	Instant credibility loss	Build a check row: `Assets − (L+E) = 0`

Learn-it roadmap & resources

Time to proficiency: 6–10 weeks part-time if you already know Excel basics. Realistic milestones:

- **Week 1–2:** Master `INDEX/MATCH` / `XLOOKUP` , `CHOOSE` , `SUMIFS` , absolute vs relative refs, colour convention. Build a single-page revenue driver block.
- **Week 3–4:** Full 12-month P&L for one business type; add variable-cost linkage.
- **Week 5–6:** Scenario toggle, sensitivity/data tables, variance (price/volume) bridge.

- **Week 7–8:** 3-statement linkage (P&L → BS → CF) and a check row; then rebuild from a blank sheet under a timer.

Resources: - *Free:* CFI's free Excel/financial-modeling articles; Aswath Damodaran's model spreadsheets (NYU, free); Corporate Finance Institute YouTube; the "Breaking Into Wall Street" free tutorials. - *Paid (India-relevant):* CFI **FMVA** certification (~US$500–800, globally recognised); WallStreetPrep / BIWS financial modeling course; Udemy "Financial Modeling for FP&A" courses (₹500–2,000 on sale). - *Practice:* rebuild the operating models of listed companies (Zomato, Nykaa, DMart) from their annual reports — reverse-engineer the drivers.

Certifications that signal this skill: FMVA (CFI), CFA (concepts, not modeling per se), and simply a strong take-home model on your GitHub/portfolio — for FP&A roles a good sample model beats a certificate.

Quick-reference

Driver trees (memorise): | Business | Revenue = | |---|---| | SaaS | Customers × ARPU; Customers = Leads × Conv% − Churn | | Retail/QSR | Stores × Orders/day × Days × AOV | | E-commerce | Traffic × Conversion% × AOV | | Services/consulting | Billable heads × Utilisation% × Hours × Rate | | Manufacturing | Capacity × Utilisation% × Price/unit |

Key formulas:

```
Pick assumption:   =XLOOKUP(key, names, values)
Scenario switch:   =CHOOSE(MATCH($B$1,{"Base";"Up";"Down"},0), a,b,c)
                   =SWITCH($B$1,"Base",x,"Up",y,"Down",z)
Volume variance:   =(Act_qty-Bud_qty)*Bud_price
Price variance:    =(Act_price-Bud_price)*Act_qty
Gross margin %:    =(Revenue-COGS)/Revenue
Revenue/head:      =Revenue/Headcount
BS check row:      =Total_assets-(Total_liab+Total_equity)   'must = 0
```

Colour convention: blue = input · black = formula · green = link · never hard-code inside a calc.

Reasonableness benchmarks (India, rough): SaaS gross margin 70–85% · QSR EBITDA 15–25% · retail net margin 3–6% · IT services EBITDA 18–25%.

Golden rule: if you can't point to the *driver* behind a number, it isn't a model — it's a guess.

Scenario & sensitivity planning

What it is & where it's used

Scenario and sensitivity planning is the discipline of asking **"what if the world is different from my base plan?"** and answering it with numbers, not adjectives. Two distinct techniques hide under this umbrella:

- **Sensitivity analysis** — change *one* driver at a time (price, volume, GST rate, USD/INR, interest rate) and watch what happens to profit, cash, or valuation. Answers: "How fragile is my number?"
- **Scenario analysis** — change *a bundle* of drivers together into a coherent story (Base / Best / Worst), because in real life a demand crash also drags price and forex with it. Answers: "What range of outcomes am I actually staring at?"

Where it shows up on the job: - **FP&A analyst** — building the annual operating plan (AOP) with upside/downside cases the CFO presents to the board. - **Corporate finance / M&A** — DCF valuations always carry a sensitivity table on WACC and terminal growth. - **Credit & lending** — banks stress-test a borrower's DSCR under a downside case before sanctioning. - **Treasury** — forex and interest-rate sensitivity on covenants and cash runway. - **Investment / equity research** — bull/base/bear target prices. - **Startup finance** — runway under fast-burn vs slow-burn cases for the next fundraise.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you to compute *one* NPV, *one* IRR, *one* EPS. Real business runs on **ranges and triggers**, not point estimates. The board never asks "what's the number?" — it asks "what happens if China dumps, if the rupee hits 90, if we lose our largest client?"

The specific gaps: - College builds **static** models. Industry needs a model where you change an input cell and everything downstream recalculates cleanly — which requires you to actually *link* formulas instead of hardcoding. - College treats assumptions as given. Industry pays you to **isolate the 3-4 drivers that matter** out of 200 line items, and to know which one, if wrong, sinks the plan. - College stops at the answer. Industry wants the **decision**: at what volume do we break even, at what forex rate do we breach our loan covenant, how much can price fall before this deal is NPV-negative.

A person who can build a clean driver-based model with a working scenario switch and a two-variable data table is worth 2-3x a person who can only reproduce a textbook NPV.

What "proficient" looks like

A job-ready person can, unaided:

1. **Structure a model so every assumption lives in one place** (a colour-coded input block) and flows through formulas — no hardcoded numbers buried in calculations.
2. Build a **scenario switch** (one cell drives Base/Best/Worst) using `CHOOSE` or `INDEX`, so the whole P&L flips with one dropdown.
3. Run a **one-variable and two-variable Data Table** (Excel's `What-If Analysis`) and read a **tornado chart** to rank drivers by impact.
4. State the **decision-useful output**: break-even point, margin of safety, the threshold value of a driver (e.g. "NPV goes negative below ₹92/unit"), and the probability-weighted expected value.

5. Explain **which scenario is realistic vs stress** — Best/Worst are planning bookends, not forecasts, and shouldn't be simple ±10% on everything.

Hands-on: how to actually do it

1. The scenario switch (the heart of it)

Put a single control cell, say `C1`, holding 1, 2, or 3 (Base/Best/Worst). Lay assumptions in a grid, then pull the active one with `CHOOSE`:

```
                B          C(Base)    D(Best)    E(Worst)
Row5  Volume    =CHOOSE($C$1,C5,D5,E5)   1,00,000  1,20,000   75,000
Row6  Price     =CHOOSE($C$1,C6,D6,E6)   500       540        460
Row7  Var cost  =CHOOSE($C$1,C7,D7,E7)   300       290        320
```

Cell `B5` formula: `=CHOOSE($C$1, C5, D5, E5)` — flip `C1` and the entire model recalculates. `INDEX` works too: `=INDEX(C5:E5, $C$1)`.

Make `C1` a dropdown: **Data → Data Validation → List → 1,2,3**, or map to text with `=MATCH(F1, {"Base","Best","Worst"},0)`.

2. One-variable sensitivity (Data Table)

Layout: put your output formula (say EBITDA in `H1`) at top-right, list the driver values down a column, then select the block and **Data → What-If Analysis → Data Table → Column input cell = the price cell**.

```
        =H1(EBITDA formula)
460      2,10,00,000
480      3,00,00,000
500      3,90,00,000   ← base
520      4,80,00,000
540      5,70,00,000
```

3. Two-variable sensitivity (the classic valuation table)

Corner cell = output (NPV). Rows = terminal growth, columns = WACC. Select the whole grid → Data Table → **Row input = WACC cell, Column input = growth cell.**

```
NPV      10%      11%      12%      13%
2.0%   1,240    1,050      900      770
2.5%   1,380    1,160      985      840
3.0%   1,560    1,290    1,085      920
```

4. Threshold / break-even with Goal Seek

Data → What-If → Goal Seek → Set NPV cell To 0 by changing Price cell. It solves for the price where NPV = 0 — your decision trigger.

Break-even volume by formula:

```
=Fixed_Cost / (Price - Variable_Cost_per_unit)
Margin of Safety % = (Actual_Sales - BEP_Sales) / Actual_Sales
```

5. Python — Monte Carlo when ranges beat three cases

```python
import numpy as np
rng = np.random.default_rng(42)
n = 100_000
volume = rng.normal(100_000, 15_000, n)          # demand uncertainty
price  = rng.triangular(460, 500, 540, n)         # min, mode, max
varcost= rng.triangular(290, 300, 320, n)
fixed  = 2_00_00_000
ebitda = volume*(price-varcost) - fixed
print(f"Mean EBITDA: Rs {ebitda.mean():,.0f}")
print(f"P(loss): {(ebitda<0).mean():.1%}")
print(f"5th pctile (worst case): Rs {np.percentile(ebitda,5):,.0f}")
```

6. Python — one-line tornado (driver ranking)

```python
base = {"price":500,"vol":100000,"vc":300,"fixed":20000000}
def ebitda(p): return p["vol"]*(p["price"]-p["vc"])-p["fixed"]
b = ebitda(base)
for k in ["price","vol","vc"]:
    lo, hi = dict(base), dict(base)
    lo[k]*=0.9; hi[k]*=1.1
    print(f"{k:6} swing: Rs {abs(ebitda(hi)-ebitda(lo)):,.0f}")
```

Sort the swings descending and you have a tornado chart's data.

Worked example / mini-project

Company: a mid-size auto-component maker, "Bharat Forgings Ltd." Build a one-year EBITDA plan with three cases and find the decision triggers.

Base assumptions:

Driver	Base	Best	Worst
Volume (units)	1,00,000	1,20,000	75,000
Price (₹/unit)	500	540	460
Variable cost (₹/unit)	300	290	320
Fixed cost (₹)	2,00,00,000	2,00,00,000	2,10,00,000

Compute EBITDA per case — `Volume × (Price − VarCost) − Fixed`:

Case	Contribution/unit	EBITDA
Base	₹200	1,00,000 × 200 − 2,00,00,000 = **₹0**
Best	₹250	1,20,000 × 250 − 2,00,00,000 = **₹1,00,00,000**
Worst	₹140	75,000 × 140 − 2,10,00,000 = **−₹99,00,000**

The Base case sits *exactly at break-even* — a red flag that leaps out the moment you build the table. This is the value of the exercise.

Probability-weighting (management assigns 60% Base, 15% Best, 25% Worst):

```
Expected EBITDA = 0.60(0) + 0.15(1,00,00,000) + 0.25(-99,00,000)
                = 0 + 15,00,000 − 24,75,000 = -Rs 9,75,000
```

Expected value is *negative* even though the base looks fine — the fat downside tail dominates. Decision: renegotiate the fixed-cost base or lock in a minimum-volume contract before committing.

Break-even trigger (Goal Seek on Base): to reach EBITDA of ₹50 lakh, solve required price: `Price = (Fixed + Target)/Volume + VarCost = (2,00,00,000 + 50,00,000)/1,00,000 + 300 = ₹550`. So management needs a **10% price rise** or the plan under-delivers.

Reproduce it: put the four drivers in rows with a `CHOOSE` switch in `C1`, drop a two-variable Data Table of Price (rows) × Volume (columns) against EBITDA, and add a tornado using the Python snippet above.

How it's tested

Interview questions: - "Walk me through how you'd build a Base/Best/Worst case for this business." - "What's the difference between sensitivity and scenario analysis?" - "Your DCF says NPV is ₹500 cr. Which two inputs would you sensitise and why?" - "Volume falls 20% — is that best handled by a scenario or a sensitivity?" - "What's margin of safety and why does the board care?"

Practical/assessment tests you will actually get: - **Timed Excel test (45-60 min):** given a raw P&L, build a driver-based model with a working scenario dropdown and a two-variable data table. Graders check: are inputs separated from formulas, does flipping the switch recalc everything, is there any hardcoded number inside a calculation. - **Case study:** "Here's a factory investment. Should we build it?" — expected deliverable is NPV with a sensitivity table and a written recommendation naming the trigger variable. - **Live model audit:** they hand you someone's model and ask "what breaks this?" — you're expected to find the assumption that swings the answer most.

Common mistakes & how pros avoid them

Mistake	Why it's wrong	Pro fix
Best/Worst = base ±10% on every line	Drivers don't move independently; this is lazy	Build coherent *stories* — a demand shock also hits price and utilisation
Hardcoding numbers inside formulas	Scenario switch can't reach them; model lies	Every assumption in a coloured input cell, formulas reference cells only
Sensitising 15 variables	Noise, no signal	Tornado-rank first, sensitise the top 3-4
Symmetric up/down assumptions	Downside is usually fatter (costs spike faster than they fall)	Use asymmetric ranges; check the loss tail
Presenting three numbers, no decision	Board wants the "so what"	Always state the trigger/threshold and the recommended action
Forgetting cash & covenants	A profitable worst case can still breach DSCR and default	Stress the balance sheet and covenants, not just EBITDA
Circular references from interest-on-debt	Model shows `#REF` /circular warning	Enable iterative calc deliberately, or use a copy-paste interest toggle

Learn-it roadmap & resources

Time to proficiency: 4-6 weeks part-time if you already know Excel basics.

- **Week 1-2:** Master `CHOOSE` , `INDEX/MATCH` , Data Validation dropdowns, and the What-If Analysis menu (Data Table, Goal Seek, Scenario Manager). Rebuild any past P&L as a driver-based model.
- **Week 3:** One- and two-variable data tables + tornado charts. Do 5 DCF sensitivity tables.
- **Week 4:** Break-even, margin of safety, probability-weighting, expected value.
- **Week 5-6:** Monte Carlo in Python (`numpy`), and stress-testing balance-sheet covenants.

Resources: - *Free:* Corporate Finance Institute (CFI) free scenario-analysis lessons; ASimpleModel.com (excellent free model-building series); Aswath Damodaran's NYU valuation lectures on YouTube (the gold standard for sensitivity in DCF). - *Paid:* CFI's **FMVA** certification (has a dedicated scenario & sensitivity module) — most recognised for FP&A/corp-fin roles. Wall Street Prep / Breaking Into Wall Street modelling courses. - *India-specific:* For CA students, the Cost & FM papers already cover marginal costing, break-even and margin of safety — connect that theory directly to the Excel build.

Quick-reference

Task	Tool / Formula
Scenario switch	`=CHOOSE($C$1, base, best, worst)` or `=INDEX(range,$C$1)`
Dropdown control	Data → Data Validation → List
1-var sensitivity	Data → What-If → Data Table → Column input cell
2-var sensitivity	Corner=output; Data Table → Row input + Column input
Solve for a threshold	Data → What-If → Goal Seek → Set cell To 0 by changing
Break-even volume	`=Fixed / (Price − VarCost/unit)`
Margin of safety	`=(Actual − BEP) / Actual`
Contribution/unit	`=Price − Variable cost/unit`
Expected value	$=\Sigma(\text{Probability}_i \times \text{Outcome}_i)$
Monte Carlo	Python `numpy.random` → percentiles + P(loss)
Driver ranking	Tornado chart: ±10% each driver, sort by swing

Golden rules: inputs separate from formulas · never hardcode inside a calc · rank drivers before sensitising · state the trigger and the decision · stress cash & covenants, not just profit.

Unit Economics, KPIs & Metrics

What it is & where it's used

Unit economics is the profit-and-loss of a *single unit* — one customer, one order, one subscription, one delivery — stripped of company-wide noise. If you can't make money on one unit, scaling only makes you lose money faster. **KPIs and metrics** are the small set of numbers that tell you whether that unit is healthy and whether the business is on track.

This is the language of modern FP&A. It shows up in:

- **FP&A / business finance** — building the operating model, board decks, and "burn multiple" narratives.
- **Startup / SaaS / D2C finance** — CAC, LTV, payback, cohort retention, contribution margin per order.
- **Strategy & investor relations** — same metrics translated for VCs and lenders.
- **Corporate FP&A in traditional firms** — contribution margin by SKU/region, gross margin bridges.

Anyone touching a **board deck, a fundraise model, or a "why are we burning cash" review** lives in this chapter. In India, every funded startup (Zomato, Meesho, Nykaa, a Series-A SaaS firm) reports these numbers — and every finance hire is expected to compute and defend them.

The gap: why companies want this (and college didn't teach it)

Your MBA taught you the **income statement top-down**: Revenue → COGS → Gross Profit → EBITDA. That's an *accounting* view. It tells you the company made ₹X, not whether the *next customer* is worth acquiring.

The industry works **bottom-up and forward-looking**:

College taught	Industry needs
Gross margin % for the whole firm	Contribution margin **per unit**, after variable selling/logistics
Historical P&L	LTV — *future* value of a customer you just acquired
"Marketing is an expense"	CAC as an **investment** with a payback period
Revenue growth	Retention, cohorts, and *quality* of that growth
One blended number	Metrics **by business model** (SaaS ≠ D2C ≠ marketplace)

The gap: colleges teach financial *accounting* and static ratio analysis. Employers need you to model the *economics of growth* — connect a marketing spend today to a cash return 14 months later, and prove the business gets *more* efficient as it scales, not less. Nobody teaches cohort analysis or the CAC payback formula in a classroom.

What "proficient" looks like

A job-ready person can, unaided:

1. Build a **contribution margin** waterfall for a unit and name every variable cost line.
2. Compute **CAC** correctly (fully-loaded, not just ad spend) and **LTV** three different ways.
3. State the **LTV:CAC** and **payback** rules of thumb and explain *why* they matter for cash.

4. Build a **cohort retention table** in Excel/SQL and read revenue retention (NRR/GRR) off it.
5. Pick the **right KPIs for the business model** in front of them and defend the choice.
6. Spot when a metric is being **gamed** (blended CAC hiding paid CAC, LTV using revenue not margin).

The bar is not memorising formulas — it's judgement about *which* number is honest.

Hands-on: how to actually do it

Contribution margin (per unit)

Contribution margin = Revenue − **variable** costs. Fixed costs (rent, salaries, tools) are excluded.

```
Contribution Margin (CM1) = Net Revenue − COGS − variable selling/logistics
CM per unit               = CM1 / units
CM %                      = CM1 / Net Revenue
Breakeven units           = Fixed Costs / CM per unit
```

Excel for a D2C order:

```
Net revenue      =  B2 - B3                         'AOV minus discount
COGS             =  B4
Contribution     =  B2 - B3 - B4 - B5 - B6 - B7     'less shipping, payment gateway, returns p
CM %             =  (B2-B3-B4-B5-B6-B7) / (B2-B3)
Breakeven units  =  Fixed_Costs / Contribution_per_unit
```

CAC — Customer Acquisition Cost

```
CAC = (Total Sales + Marketing spend, fully loaded) / New customers acquired
```

Fully loaded = ad spend + salaries of the sales/marketing team + tools + agency fees + attributed discounts/coupons. **Blended CAC** divides by *all* new customers (incl. organic). **Paid CAC** divides by only paid-channel customers — always higher, always the honest one.

```
Blended CAC = 12,00,000 / 4,000 = ₹300
Paid CAC    = 12,00,000 / 2,000 = ₹600   ← the number a VC will ask for
```

LTV — Lifetime Value (three ways)

```
1. Simple:        LTV = ARPU × Gross Margin% × Avg customer lifespan (in periods)
2. Churn-based:   LTV = (ARPU × Gross Margin%) / Monthly churn rate
3. Discounted:    LTV = Σ  [ margin_t / (1+r)^t ]   over expected life
```

Rule: **always use margin, never revenue.** LTV on revenue overstates value by (1 − margin).

```
ARPU (monthly)       = ₹500
Gross margin         = 70%   →  contribution ARPU = ₹350
Monthly churn        = 5%    →  avg life = 1/0.05 = 20 months
LTV (churn method)   = 350 / 0.05 = ₹7,000
LTV:CAC              = 7,000 / 600 = 11.7x
```

Payback period

```
CAC Payback (months) = CAC / (monthly contribution margin per customer)
                     = 600 / 350 = 1.7 months
```

SQL — cohort retention

```
WITH first_order AS (
  SELECT customer_id,
         DATE_TRUNC('month', MIN(order_date)) AS cohort_month
  FROM orders
  GROUP BY customer_id
),
activity AS (
  SELECT o.customer_id,
         f.cohort_month,
         DATE_TRUNC('month', o.order_date) AS active_month,
         SUM(o.net_revenue) AS revenue
  FROM orders o
  JOIN first_order f ON o.customer_id = f.customer_id
  GROUP BY 1,2,3
)
SELECT cohort_month,
       DATEDIFF('month', cohort_month, active_month) AS month_index,
       COUNT(DISTINCT customer_id) AS active_customers,
       SUM(revenue) AS cohort_revenue
FROM activity
GROUP BY 1,2
ORDER BY 1,2;
```

Python — churn and LTV from the cohort table

```
import pandas as pd

df = pd.read_csv("cohorts.csv")  # cohort_month, month_index, active_customers
pivot = df.pivot(index="cohort_month", columns="month_index",
                 values="active_customers")

# Retention % = each month vs month 0
retention = pivot.div(pivot[0], axis=0)

# Monthly churn implied by month 1 retention
churn = 1 - retention[1].mean()
arpu, gm = 500, 0.70
ltv = (arpu * gm) / churn
print(f"Avg monthly churn: {churn:.1%}  |  LTV: Rs {ltv:,.0f}")
```

DAX — Net Revenue Retention in Power BI

```
NRR :=
VAR StartRev = CALCULATE([Revenue], DATEADD('Date'[Date], -12, MONTH))
RETURN DIVIDE([Revenue], StartRev)   -- >100% = expansion beats churn
```

Worked example / mini-project

"Kirana Cloud" — a B2B SaaS billing app for Indian retail shops. Reproduce this in Excel.

Inputs (monthly, per customer):

Item	Value
Subscription (ARPU)	₹1,500
Gross margin	80% → contribution ARPU ₹1,200
Monthly logo churn	4%
Fully-loaded S&M / month	₹8,00,000
New paying customers / month	100

Step-by-step:

```
CAC             = 8,00,000 / 100              = ₹8,000
Avg lifespan    = 1 / 0.04                    = 25 months
LTV             = 1,200 × 25                   = ₹30,000
LTV:CAC         = 30,000 / 8,000              = 3.75x      ✅ (>3 healthy)
CAC payback     = 8,000 / 1,200               = 6.7 months ✅ (<12 healthy)
Annual GRR      = (1 − 0.04)^12               = 61%        ⚠️ churn too high
```

The catch a good analyst spots: LTV:CAC looks fine (3.75x), but **61% gross retention means 39% of customers gone within a year** — the LTV is propped up by a long theoretical tail. Fix churn before scaling spend. This is exactly the kind of "the headline metric hides the problem" insight FP&A gets hired for.

Now add **NRR**: if surviving customers upgrade seats and expansion adds 8% while churn removes 4%, NRR = 104% — the business grows revenue even with zero new logos. That single number reframes the entire investment case.

Build it: one sheet with the input block, one CAC/LTV/payback block, and a 12-column cohort retention grid using `=B0*(1-churn)^col`. Add conditional formatting (green >100% NRR, red <90% GRR).

How it's tested

Interview questions:

- "Walk me through CAC and LTV. What's a good LTV:CAC ratio and why 3x?"
- "Difference between blended and paid CAC? Which would a founder quote to look good?"
- "Company has LTV:CAC of 5x but is burning cash. How?" (Answer: long payback — cash out now, LTV realised over years.)
- "What KPIs matter for a marketplace vs a SaaS vs a D2C brand?"
- "GRR vs NRR — which can exceed 100% and why?"

Practical / take-home tests:

1. **Timed Excel case (45–60 min):** raw orders table given → build contribution margin, CAC, LTV, payback, and a cohort retention grid with conditional formatting.
2. **SQL screen:** "Write a cohort retention query from this `orders` table" (the query above is the expected answer).
3. **Metrics-audit case:** a deck with inflated numbers → "find what's wrong." Expected catches: LTV on revenue not margin, blended CAC hiding paid CAC, churn understated by counting only voluntary churn.
4. **Board-deck build:** given a P&L, produce the 6 KPIs that matter and a one-line narrative each.

Common mistakes & how pros avoid them

Mistake	Fix
LTV computed on **revenue**, not gross margin	Always multiply ARPU by GM% first
Quoting **blended CAC** to hide expensive paid channels	Report paid CAC separately; segment by channel
Ignoring **CAC payback** — a 5x LTV:CAC with 30-month payback still kills cash	Show payback alongside the ratio
Putting **fixed costs** into contribution margin	CM uses variable costs only
Assuming churn is constant	Early cohorts churn faster; use the cohort curve, not one rate
Using **one KPI set** for every business	Match KPIs to model (see cheat-sheet)
Counting a cancelled-then-resubscribed user as retained	Define churn rules explicitly and document them
LTV with **infinite lifespan** on high churn	Cap horizon (e.g., 36 months) or discount future margin

Pros write down every definition (a "metrics dictionary") so nobody argues about what "active customer" means at month-end.

Learn-it roadmap & resources

Time to proficiency: 3–5 weeks part-time.

- **Week 1** — Contribution margin, CAC, LTV, payback. Rebuild the worked example from scratch in Excel.
- **Week 2** — Cohort analysis in Excel, then reproduce it in SQL.
- **Week 3** — Business-model KPIs; read 3 real investor decks (Zomato, Nykaa, a SaaS S-1) and recompute their metrics.
- **Week 4–5** — Python/DAX automation of a cohort report; build a KPI dashboard.

Resources (mostly free):

- *For Entrepreneurs* (David Skok) — the canonical SaaS metrics essays. Free.
- a16z & Bessemer "Cloud metrics" primers — free, investor-grade definitions.
- Zomato / Nykaa investor presentations (India context) — free on their IR pages.
- *Financial Intelligence* (Berman & Knight) — margin thinking. Paid.
- Practice data: Kaggle "Online Retail" and "Telco Churn" datasets — build cohorts on real rows.
- **Certification:** none required specifically, but **FMVA (CFI)** and **Wall Street Prep** cover this inside their FP&A tracks. For most India roles, a strong Excel + SQL portfolio project beats a certificate.

Quick-reference

Formulas:

Metric	Formula	Healthy
Contribution margin	Revenue − variable costs	> 0, ideally 30%+
CAC (paid)	Fully-loaded S&M / paid customers	model-dependent
LTV	(ARPU × GM%) / churn	—
LTV:CAC	LTV / CAC	**3–5x**
CAC payback	CAC / monthly contribution/customer	**< 12 months**
GRR	1 − revenue churn	> 90% (SaaS)
NRR	(Start + expansion − churn) / Start	**> 100%**
Rule of 40 (SaaS)	Growth% + Profit margin%	≥ 40
Burn multiple	Net burn / Net new ARR	< 1 great, > 2 poor

KPIs by business model:

Model	The metrics that matter
SaaS	MRR/ARR, NRR, GRR, logo churn, CAC payback, Rule of 40, magic number
D2C / e-commerce	AOV, contribution margin/order, repeat rate, RTO %, blended vs paid CAC, cohort repeat
Marketplace	GMV, take rate, contribution/order, buyer & seller retention, liquidity
Lending / fintech	NIM, CAC, GNPA, credit cost, LTV net of losses, collection efficiency
Subscription media	ARPU, churn, engagement (DAU/MAU), content cost per sub

The one-liner to remember: *If contribution margin per unit isn't positive, no amount of scale, funding, or growth fixes it.*

Business Partnering & Commercial Finance

What it is & where it's used

Business partnering is the shift from finance *reporting on* the business to finance *sitting inside* the business — helping sales, operations, marketing and product make better commercial decisions with money as the lens. A "commercial finance" or "finance business partner" (FBP) is the person a Sales Director calls before signing a ₹4 crore deal at 12% discount, or the one the plant head asks whether the new SKU actually makes money after freight and returns.

You are not the scorekeeper anymore. You are the co-pilot who says: "*That discount kills your margin — here's a rebate structure that protects it and still wins the deal.*"

Where it shows up in roles:

Role	What partnering looks like
Finance Business Partner / Commercial Finance	Deal desk, pricing approvals, regional P&L ownership
FP&A Manager	Challenging budget assumptions with cost/ops teams
Revenue / Sales Finance	Discount governance, incentive design, quota setting
Cost Controller (manufacturing)	Product costing, make-vs-buy, capex justification
Startup Finance lead	Unit economics, CAC/LTV, board narrative

The gap: why companies want this (and college didn't teach it)

Your MBA taught NPV, contribution margin and Porter's five forces. It did **not** teach you how to walk into a room where a Sales VP outranks you, disagree with a discount, and leave with the room on your side. College graded you on the *right answer*. Business partnering pays you for the *adopted answer* — analysis that a non-finance person actually acts on.

The specific gaps employers keep hiring to close:

- **Translation gap** — you can compute EBITDA margin; can you tell a plant manager "every 1% scrap you cut adds ₹18 lakh to our bottom line"?
- **Influence-without-authority gap** — you don't manage sales, yet you must change their behaviour.
- **Commercial-context gap** — a "good" number isn't good if it ignores that the customer is a strategic logo, or that the competitor just cut price.
- **So-what gap** — freshers deliver a variance report; partners deliver *the one decision the report implies*.

CA/MBA curricula reward technical correctness in isolation. Companies lose crores to margin leakage, bad discounts and unmanaged price/cost gaps — they pay partners to plug that.

What "proficient" looks like

A job-ready business partner can, unaided:

- Build a **deal / pricing model** that shows gross margin, contribution and net margin *after* discounts, rebates, freight, credit cost and GST — and give a clear go/no-go.

- Run a **price-volume-mix (PVM) bridge** explaining why revenue or margin moved, and attribute it to who owns each lever.
- Sit in a commercial meeting, listen, and reframe a request ("we need 15% off") into a structured alternative that protects margin.
- Say **no with a number** and **yes with a condition** — never a flat block.
- Own a **regional/product P&L**: forecast it, explain the miss, and drive one corrective action.
- Design or vet a **sales incentive** so reps aren't paid to destroy margin.
- Deliver a one-slide **"CFO-ready"** recommendation: situation → options → recommendation → risk.

The bar is *behaviour change*, not analysis volume. Proficiency = "sales asks finance first."

Hands-on: how to actually do it

1. The deal margin model (the single most-used tool)

Build a clean waterfall from list price to net margin. Copy-usable Excel:

```
List price (per unit)                  =B2
Discount %                             =B3
Net price                              =B2*(1-B3)
Volume (units)                         =B5
Gross revenue                          =B4*B5
COGS/unit                              =B7
Gross margin ₹                         =(B4-B7)*B5
Freight/unit                           =B9
Rebate % (on net)                      =B10
Credit cost (net price * days/365 * WACC)  =B4*(B12/365)*$B$14
Contribution ₹                         =B8-(B9*B5)-(B4*B10*B5)-(credit)
Contribution %                         =B15/B6
```

The margin-preservation formula every partner memorises — **how much extra volume a discount must generate to break even**:

```
Break-even volume uplift % = Discount% / (GM% - Discount%)
Excel: =disc/(gm - disc)
```

At 40% gross margin, a 10% price cut needs volume to rise **33%** just to stand still — put *that* in front of sales and the conversation changes.

2. Price-Volume-Mix bridge (explaining the movement)

```
Price effect  = (Price_curr - Price_prior) * Volume_prior
Volume effect = (Volume_curr - Volume_prior) * Price_prior
Mix effect    = Σ(VolShare_curr - VolShare_prior) * (Margin_sku - Margin_total_prior) * TotalV
```

Excel for a two-period table:

```
=(C_priceCurr − C_pricePrior)*C_volPrior        ' price
=(C_volCurr − C_volPrior)*C_pricePrior           ' volume
=SUMPRODUCT( … )                                  ' mix
```

3. SQL — pull the numbers partners actually get asked for

Margin leakage by customer (who is over-discounting):

```sql
SELECT customer_name,
       SUM(list_price*qty)                          AS gross_rev,
       SUM(net_price*qty)                           AS net_rev,
       1 - SUM(net_price*qty)/SUM(list_price*qty)  AS realized_disc,
       SUM((net_price-cogs)*qty)/SUM(net_price*qty) AS net_margin_pct
FROM sales_lines
WHERE invoice_date ≥ '2026-04-01'
GROUP BY customer_name
HAVING 1 - SUM(net_price*qty)/SUM(list_price*qty) > 0.15
ORDER BY realized_disc DESC;
```

4. Python — quick deal sensitivity for a meeting

```python
import numpy as np, pandas as pd

def deal_margin(list_p, disc, cogs, vol, freight=0, rebate=0):
    net = list_p*(1-disc)
    contrib = (net - cogs - freight - net*rebate)*vol
    return contrib, contrib/(net*vol)

for d in np.arange(0, 0.25, 0.05):
    c, pct = deal_margin(1000, d, 600, 500, freight=40, rebate=0.02)
    print(f"disc {d:.0%}: contribution Rs{c:,.0f}  margin {pct:.1%}")
```

5. DAX — a live margin measure for the dashboard sales sees

```
Net Margin % =
DIVIDE(
    SUMX(Sales, Sales[NetPrice]*Sales[Qty]) - SUMX(Sales, Sales[COGS]*Sales[Qty]),
    SUMX(Sales, Sales[NetPrice]*Sales[Qty])
)

Discount Leakage Rs =
SUMX(Sales, (Sales[ListPrice]-Sales[NetPrice])*Sales[Qty])
```

6. The influence script (the real skill)

Structure every pushback as **Data → So-what → Option → Ask**:

> "This deal at 15% off lands us at 6% net margin vs our 14% floor (data). At this margin we're funding the customer, not earning (so-what). Two options: hold price with a 60-day payment term sweetener, or 15% off tied to a 20% volume commit (option). Which do you want me to model for tomorrow's call? (ask)"

Worked example / mini-project

Scenario: You are FBP for a mid-size FMCG distributor in Pune. Sales wants to close a modern-trade chain at **15% discount** on a product with list ₹1,000, COGS ₹620. Current business is 4,000 units/quarter at 5% discount. Sales promises "big volume."

Step 1 — current state

Metric	Value
List	₹1,000
Net price (5% off)	₹950
COGS	₹620
Contribution/unit	₹330
Volume	4,000
Contribution ₹	₹13,20,000
GM % (on net)	34.7%

Step 2 — proposed deal at 15%

Net price = ₹850. Contribution/unit = ₹850 − ₹620 = ₹230.

Break-even volume to keep ₹13,20,000 contribution: `13,20,000 / 230 = 5,740 units` → a **43.5% volume jump** just to hold flat.

Extra discount vs GM check: incremental discount = 10 points on a 34.7% margin base → `0.10/(0.347-0.10)` ≈ **40% uplift required**. Consistent.

Step 3 — add freight ₹30/unit + 2% rebate + 45-day credit at 12% WACC

Credit cost/unit = 850 × 45/365 × 12% ≈ ₹12.6. Rebate = ₹17. Net contribution/unit = 230 − 30 − 17 − 12.6 = **₹170.4**. Break-even now = `13,20,000/170.4 = 7,746 units` → **94% volume growth needed**. The "big volume" promise is nowhere near.

Step 4 — the recommendation slide

> *Situation:* 15% ask needs ~94% volume growth to be margin-neutral. Sales forecasts +25%. *Options:* (a) 8% off + ₹30 freight to customer's DC = margin-neutral at +20% volume. (b) 15% off only above 6,000 units (tiered rebate). (c) Walk. *Recommendation:* Option (a) — protects the ₹13.2L base, wins the logo, achievable volume. *Risk:* Competitor may match; revisit in Q2.

You just turned a margin-destroying deal into a defensible one — *that* is business partnering.

How it's tested

Interview questions:

- "A sales head wants a 20% discount you think is wrong. Walk me through what you do." (They want structure + influence, not a flat "I'd refuse.")
- "How much extra volume justifies a 10% price cut at 40% margin?" (Answer: 33%.)
- "Explain a P&L miss to a non-finance CEO in 30 seconds."
- "Difference between gross, contribution and net margin — and which one do you use for a one-off deal?" (Contribution / incremental.)
- "Tell me about a time you changed a business decision with data." (Behavioural — have a STAR story ready.)

Practical assessments companies actually give:

- **Timed deal-model build (45–60 min):** raw price/cost/discount data in Excel → build margin waterfall + break-even + a go/no-go recommendation cell. They check formula correctness *and* whether you wrote a clear recommendation.
- **Case presentation:** "Here's a losing product line. Present 3 options to the leadership team in 10 minutes." Judged on commercial judgement and communication.
- **Roleplay:** an interviewer plays a pushy sales director; you must hold margin while keeping the relationship.
- **PVM bridge test:** two-period revenue/margin data → decompose and explain.

Common mistakes & how pros avoid them

Mistake	What pros do
Saying a flat "no"	Say "no with a number, yes with a condition" — always offer a structured alternative
Leading with gross margin on a one-off deal	Use **contribution / incremental** margin for deals; gross for portfolio
Ignoring below-the-line costs (freight, rebate, credit, returns)	Model *net* landed margin — that's where deals silently bleed
Drowning the room in a 12-tab model	One slide: situation → options → recommendation → risk
Being the "business prevention department"	Be the enabler who finds the *how*, not just the *no*
Owning the analysis but not the follow-through	Track whether the decision was acted on and what it delivered
Forgetting GST/credit period in Indian deals	Build 45–90 day credit cost and GST cash-flow timing into the model
Sandbagging with sales' own optimistic volume	Stress-test volume; show break-even so the burden shifts to them

Learn-it roadmap & resources

Time to proficiency: 6–12 months on the job; 8–10 focused weeks to become interview-ready if you already know Excel and P&L basics.

Phase	Weeks	Focus
1	1–2	Master margin math: gross vs contribution vs net, break-even, PVM
2	3–4	Build 3 deal models from scratch; add freight/rebate/credit layers
3	5–6	Storytelling: turn every model into a 1-slide recommendation
4	7–8	Influence & roleplay: practice the Data→So-what→Option→Ask script
5	ongoing	Own a mock regional P&L; forecast, explain variance, recommend

Resources:

- *Books:* "Financial Intelligence" (Berman & Knight) — translation skills; "The Finance Business Partner" (Ian Wallis); "Made to Stick" (Heath) for communication.
- *Free:* CFI's FP&A / Business Partnering articles; Mike Pearl / "The FP&A Guy" (Paul Barnhurst) LinkedIn + podcast; Excel margin-model templates on Wall Street Prep blog.
- *Paid/cert:* CFI **FMVA** (modeling backbone), AICPA **CGMA** (India-relevant management-accounting + partnering), your **CA** cost & management accounting papers map directly.
- *Practice:* rebuild a listed FMCG/pharma company's segment P&L from its annual report and write the "what I'd tell the CEO" note.

Quick-reference

```
Break-even volume uplift %  = Discount% / (GM% - Discount%)
Contribution/unit           = Net price - variable cost (COGS+freight+rebate+credit)
Net price                   = List * (1 - Disc%)
Credit cost/unit            = Net price * (credit days/365) * WACC
Realized discount %         = 1 - (Net rev / Gross rev)
PVM: Price = ΔPrice*Vol_prior ; Volume = ΔVol*Price_prior ; Mix = mix-shift*margin gap
```

Margin type	Use it for
Gross margin	Portfolio / product profitability
Contribution margin	One-off deals, incremental decisions, make-vs-buy
Net margin	Full customer/deal profitability after all costs

The three phrases that make you a partner: 1. "No with a number, yes with a condition." 2. "So what does this mean for *your* decision?" 3. "Here are three options — I recommend option B, and here's the risk."

Recommendation slide skeleton: Situation → Options → Recommendation → Risk. One slide. Every time.

Capex, project appraisal & business cases

What it is & where it's used

Capex (capital expenditure) is money spent to acquire or upgrade long-lived assets — a new plant, a fleet of delivery vans, an ERP rollout, a warehouse, machinery. Unlike opex (rent, salaries, electricity — expensed in the year), capex sits on the balance sheet and depreciates over years. **Project appraisal** is the discipline of deciding *which* capex is worth doing, and a **business case** is the document that carries that recommendation to the people who sign the cheque.

Every rupee of capex is a bet: pay cash now, receive uncertain cash later. The job is to convert that bet into three numbers — **NPV, IRR, payback** — plus a written argument, and defend it.

Where this skill lives: - **FP&A analyst** — builds the model, runs sensitivities, owns the capex tracker. - **Corporate finance / business finance partner** — writes the business case, presents to the investment committee. - **Treasury** — checks it against the WACC and funding plan. - **Internal audit / controllership** — runs the **post-investment review** (did we get what we promised?). - Startups, PE portfolio finance, project finance (infra, renewables) — the entire deal *is* an appraisal.

The gap: why companies want this (and college didn't teach it)

College teaches you to compute NPV from a clean table of cash flows handed to you. Industry never hands you the cash flows — **you build them**, and 90% of the skill is in the assumptions, not the discounting.

Specific gaps a fresh MBA/CA has: - **You discount profit instead of cash.** Textbooks blur PAT and cash flow. In practice you must add back depreciation, adjust for working capital, and handle tax on *cash* terms. - **You ignore the tax shield on depreciation.** In India, depreciation is a WDV block under the Income Tax Act — it changes the cash flows and nobody in college models it. - **You treat sunk costs and allocated overheads as relevant.** They aren't. Only *incremental* cash flows count. - **You never touch terminal value, working-capital release, or salvage.** - **You've never written the one-page ask** that a CFO actually reads, or been forced to say "here's what kills this project." - **Post-investment review** — closing the loop — is not taught at all, yet it's what auditors and boards demand.

What "proficient" looks like

A job-ready person can, unaided:

1. Build a **10-year DCF for a project** in Excel from raw assumptions — revenue driver, cost ramp, capex phasing, working capital, tax, depreciation shield, salvage/terminal value.
2. Compute **NPV, IRR, discounted payback, and profitability index** with the correct sign convention and timing (year-0 outflow).
3. Choose the **right discount rate** (WACC or a hurdle rate) and explain *why*.
4. Run a **sensitivity / scenario / tornado** analysis and name the 2-3 variables that actually swing the decision.
5. Write a **one-page business case**: problem, options, recommendation, financials, risks, ask.
6. Design and run a **post-investment review** 12-18 months later against the original baseline.

7. Know the **decision rules and their traps** — IRR on non-conventional cash flows, mutually exclusive projects, capital rationing.

Hands-on: how to actually do it

The cash-flow engine (this is the whole game)

Free cash flow to the project each year:

```
FCF = (Revenue − Cash Costs − Depreciation) × (1 − tax)
      + Depreciation                           ← add back non-cash
      − Capex
      − ΔWorking Capital
```

The `Depreciation × tax` piece is the **tax shield** — it's why depreciation belongs in the model even though it's non-cash.

Core Excel formulas (copy-usable)

Lay years across columns. Put the initial outlay in **Year 0**.

```
NPV (correct way — year-0 outflow is NOT discounted):
=B10 + NPV(rate, C10:L10)          ' B10 = Year-0 cash flow (negative), C10:L10 = Years 1-10

IRR:
=IRR(B10:L10)

XIRR / XNPV (when dates are irregular — use these in real deals):
=XNPV(rate, B10:L10, B1:L1)         ' B1:L1 = actual dates
=XIRR(B10:L10, B1:L1)

Discounted payback — cumulative discounted CF crosses zero:
Discount factor:   =1/(1+$B$2)^C$1
Discounted CF:     =C10*C_DF
Cumulative:        =B_cum + C_disc
Payback year (approx): =MATCH(TRUE, cum_range>0, 0)

Profitability Index:
=NPV(rate,C10:L10) / -B10           ' PV of inflows / initial outlay
```

Common trap: `=NPV(rate, B10:L10)` is *wrong* if B10 is the Year-0 outflow — Excel's NPV assumes the first value is one period away. Always keep Year 0 outside the NPV() and add it.

Depreciation & tax shield — India (WDV block method)

Income-tax depreciation is on the **block of assets** at WDV rates (e.g. plant & machinery general block = 15%). Book depreciation (Companies Act, Schedule II, often SLM over useful life) differs — model the

income-tax one for cash tax.

```
WDV depreciation, Year n:
=Opening_WDV * 15%
Closing_WDV = Opening_WDV - Dep         ' carries to next year's opening
```

Also model the **additional depreciation** (20% extra in year of purchase for new plant in manufacturing, where still applicable) if relevant.

Sensitivity in Excel

Use a **Data Table** (What-If Analysis → Data Table). Put NPV in a corner cell `=NPV_cell`, discount rates down the left, a swing variable across the top. For a one-way tornado, flex each input ±10% and chart the NPV delta.

Python — same appraisal, reproducible

```
import numpy_financial as npf
import numpy as np

cf = [-500, 120, 140, 160, 180, 200]   # Year 0..5, in ₹ lakh
rate = 0.13                             # WACC 13%

npv = cf[0] + npf.npv(rate, cf[1:])     # or npf.npv(rate, cf) if cf[0] is t=0-discounted conv
irr = npf.irr(cf)
disc = [c/(1+rate)**t for t, c in enumerate(cf)]
cum = np.cumsum(disc)
payback = next(t for t, v in enumerate(cum) if v > 0)

print(f"NPV ₹{npv:,.1f}L | IRR {irr:.1%} | Disc. payback ~Yr {payback}")
```

Note: `npf.npv` treats the first element as t=0 (already un-discounted), so pass the whole `cf` list directly — different convention from Excel. Verify once with a hand calc.

Journal entries — when capex actually hits the books

Event	Dr	Cr	Amount
Purchase machine on credit	Plant & Machinery	Vendor payable	₹50,00,000
GST input on capital goods	Input CGST/SGST (ITC)	Vendor payable	₹9,00,000
Installation cost (capitalise)	Plant & Machinery	Bank	₹2,00,000
Year-end depreciation (Schedule II)	Depreciation expense	Accumulated depreciation	₹5,20,000
Salvage on disposal	Bank / Accum. dep.	Plant & Machinery / Gain	at exit

ITC on capital goods is fully available (not amortised over 60 months post-2017, except for the common-credit reversal rule under Rule 43 if used partly for exempt supplies).

Worked example / mini-project

Project: A mid-size Indian manufacturer evaluates a new packaging line.

Assumptions: - Capex ₹500 lakh, installed Year 0. Life 5 years, salvage ₹50 lakh. - Incremental revenue ₹400 lakh/yr; cash costs 55% of revenue. - Working capital = 15% of revenue, built in Year 0, released in Year 5. - Tax 25%. WACC 13%. Depreciation for cash tax: WDV 15%.

Working-capital block = 15% × 400 = ₹60 lakh outflow at Year 0, released Year 5.

₹ lakh	Yr 0	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5
Revenue		400	400	400	400	400
Cash costs (55%)		−220	−220	−220	−220	−220
WDV Dep @15%		−75	−64	−54	−46	−39
PBT		105	116	126	134	141
Tax @25%		−26	−29	−31	−34	−35
PAT		79	87	94	101	106
+ Dep (add back)		75	64	54	46	39
− Capex	−500					
− ΔWC	−60					+60
+ Salvage (net)						+50
FCF	**−560**	**154**	**151**	**149**	**147**	**255**

Now the numbers (rate 13%):

```
NPV  = −560 + 154/1.13 + 151/1.13² + 149/1.13³ + 147/1.13⁴ + 255/1.13⁵
     ≈ −560 + 136 + 118 + 103 + 90 + 138  ≈  ₹125 lakh   → POSITIVE, accept
IRR  ≈ 21%   (> 13% WACC → accept)
Discounted payback ≈ 4.3 years
PI   = (560+125)/560 ≈ 1.22   (>1 → accept)
```

One-page business-case skeleton you'd attach: 1. **Problem/opportunity** — current line at 95% capacity; turning away ₹400L of orders. 2. **Options** — (a) do nothing, (b) outsource, (c) new line. Recommend (c). 3. **Financials** — NPV ₹125L, IRR 21%, payback 4.3 yrs, PI 1.22. 4. **Risks & sensitivity** — NPV turns negative if revenue falls >18% or costs exceed 63%. Volume is the swing variable. 5. **The ask** — approve ₹560L capex+WC; funded from internal accruals; go-live Q3.

How it's tested

Interview questions: - "Walk me through building free cash flow for a project from scratch." (They want dep add-back, WC, tax shield, capex phasing.) - "NPV vs IRR — which do you trust for two mutually exclusive projects, and why?" (NPV; IRR can mislead on scale and reinvestment.) - "A project has two IRRs — what happened?" (Non-conventional cash flows / sign changes; use NPV or MIRR.) - "What discount rate did you use and why?" (WACC for average-risk; risk-adjusted hurdle otherwise.) - "Payback ignores what?" (Time value and cash flows after payback — that's why it's a screen, not a decision rule.)

Practical tests companies give: - A **timed 30-45 min Excel case**: raw assumptions in one tab, "return NPV/IRR/payback and a recommendation." They check the year-0 handling, the tax shield, and whether your formulas are dynamic (change an input, does NPV update). - A **business-case writing test**: given a scenario, produce a one-pager with a recommendation. - A **"critique this model"** test: they hand you a flawed model and ask what's wrong (sunk cost included, NPV discounting year 0, hard-coded values).

Common mistakes & how pros avoid them

Mistake	Fix
Discounting Year-0 outflow inside `NPV()`	Keep Yr 0 outside: `=B10+NPV(rate,C10:L10)`
Including **sunk costs** (already-spent feasibility study)	Only incremental future cash flows are relevant
Allocating **corporate overhead** that won't change	Include only *incremental* overhead
Discounting **PAT instead of FCF**	Add back depreciation, adjust working capital
Ignoring the **depreciation tax shield**	Model it; it can flip a marginal NPV
Forgetting **working-capital release** and **salvage** at the end	Add both in the terminal year
Comparing projects of **different lives** on raw NPV	Use equivalent annual annuity (EAA)
Trusting **IRR** on non-conventional flows	Use NPV or MIRR
Nominal cash flows discounted at a **real** rate (or vice versa)	Match: nominal↔nominal, real↔real
No **post-investment review**	Bake the baseline into the approval so it can be audited

Post-investment review (PIR) — the pro move most people skip. 12-18 months after go-live, pull actual capex, actual revenue/cost, actual payback, and compare to the approved case. Document variances and *why*. This is what internal audit and the board look for, and it's how the FP&A team gets its next forecast right.

Learn-it roadmap & resources

Time to proficiency: 3-4 weeks of focused practice to pass an Excel appraisal test; 2-3 months to write business cases confidently.

- **Week 1** — DCF mechanics: build 5 project models by hand in Excel; drill NPV/IRR/XNPV/XIRR/PI.
- **Week 2** — Cash-flow building: incremental analysis, tax shield, WC, terminal value, EAA, MIRR.

- **Week 3** — Sensitivity, scenario, Data Tables, tornado charts; Python `numpy_financial`.
- **Week 4** — Write 3 one-page business cases; design a PIR template.

Resources: - CFA Level I / II "Corporate Issuers — Capital Budgeting" readings (free curriculum outlines; the clearest treatment). - ICAI CA Inter **Financial Management** — Capital Budgeting chapter (India tax + depreciation context, free study material). - Damodaran Online (NYU) — free capital budgeting lectures and spreadsheets. - CFI's **Financial Modeling & Valuation Analyst (FMVA)** — paid, hands-on, good for the Excel muscle. - `numpy-financial` docs for Python; Microsoft Excel Data Table / What-If docs.

Certifications that signal this skill: CFA, FMVA, CIMA/CMA, and for India the CA qualification itself.

Quick-reference

Metric	Formula / rule	Accept if
NPV	`=B0 + NPV(rate, Yr1:YrN)`	NPV > 0
IRR	`=IRR(range)` / `=XIRR(cf, dates)`	IRR > WACC/hurdle
Discounted payback	cumulative discounted CF crosses 0	≤ target years
Profitability Index	PV inflows ÷ initial outlay	PI > 1
MIRR	`=MIRR(cf, fin_rate, reinv_rate)`	> hurdle (use for weird flows)
EAA	`NPV × rate / (1-(1+rate)^-n)`	higher, for unequal lives

Free cash flow: `(Rev − CashCost − Dep) × (1−t) + Dep − Capex − ΔWC` **Tax shield:** `Depreciation × tax rate` **India WDV P&M general block:** 15%; **book dep:** Schedule II (SLM/WDV over useful life) **GST:** ITC on capital goods fully available (watch Rule 43 reversal for exempt use) **Decision hierarchy:** NPV rules > IRR; screen with payback; rank under rationing by PI. **Only relevant cash flows:** incremental, future, after-tax. Ignore sunk costs & unchanged overhead. **Always close the loop:** approve with a baseline → post-investment review at 12-18 months.

Working-capital & cash-flow management

What it is & where it's used

Working capital is the money trapped inside the operating engine of a business: cash tied up in **receivables** (customers who owe you), **inventory** (stock sitting in a warehouse), minus the free financing you get from **payables** (suppliers you haven't paid yet). Manage it well and you release crores of cash without raising a single rupee of debt. Manage it badly and a profitable company runs out of cash and dies — the classic "profitable but insolvent" trap.

The **cash-conversion cycle (CCC)** measures how many days a rupee stays trapped: `CCC = DIO + DSO − DPO`. The **13-week cash-flow forecast** is the tactical instrument treasurers and CFOs live by — a rolling week-by-week view of every rupee coming in and going out for the next quarter.

Roles that pay for this skill: - **FP&A / Treasury analyst** — builds and owns the 13-week model, flags cash crunches. - **Credit control / AR analyst** — attacks DSO, chases overdue invoices. - **Corporate finance / CFO office** — negotiates supplier terms (DPO), decides on invoice discounting or a cash-credit limit. - **Financial controllers** — reconcile the forecast to the actual bank balance every Monday.

The gap: why companies want this (and college didn't teach it)

College teaches you that "working capital = current assets − current liabilities" and moves on. Nobody shows you that this single ratio hides three independent levers, each owned by a different department, each negotiable, each worth real cash. Your MBA taught you accrual accounting — revenue when earned, expense when incurred. But cash doesn't move on accruals. A ₹1 crore sale booked in March that gets collected in July does nothing for June's payroll.

Employers pay because the gap is *cash timing*, not profit. A founder can read a P&L. What they can't do is answer "will we make payroll in week 6?" — and that answer requires a bottoms-up 13-week model that no textbook builds. This is the skill that gets a 26-year-old into the room where the CFO decides whether to draw down the working-capital loan.

What "proficient" looks like

A job-ready person can, unaided:

1. Compute DSO, DPO, DIO and the CCC from a trial balance or Tally export, and explain what each day is worth in rupees.
2. Build a rolling 13-week cash-flow forecast in Excel from AR/AP ageing, payroll, GST, and loan schedules — with a live bank-balance line that flags any week going negative.
3. Run a **collections waterfall**: take an AR ageing and predict when each bucket actually converts to cash based on historical collection %.
4. Recommend a concrete lever — "cut DSO by 8 days via 2/10 net-45 terms, releases ₹1.4 cr" — with the number to back it.
5. Reconcile last week's forecast to the actual bank statement and explain the variance (a "cash bridge").

Hands-on: how to actually do it

The three ratios (Excel)

Assume an annual model. Put revenue, COGS, and closing balances on a sheet:

```
DSO  = Accounts Receivable / Revenue        * 365
DPO  = Accounts Payable    / COGS (or Purchases) * 365
DIO  = Inventory           / COGS             * 365
CCC  = DIO + DSO − DPO
```

Excel, with cells named:

```
=(_AR/_Revenue)*365          ' DSO
=(_AP/_COGS)*365             ' DPO
=(_Inventory/_COGS)*365      ' DIO
=DIO+DSO−DPO                 ' CCC
```

Rupee value of one DSO day = `Revenue / 365`. If revenue is ₹120 cr, one day of DSO = ₹32.9 lakh of cash. Cut DSO by 10 days → release **₹3.29 cr**.

Collections waterfall from an AR ageing (Excel)

Historical collection profile: of a month's sales, 40% collects in the current month, 35% at 30 days, 20% at 60 days, 5% at 90. Lay sales across rows, apply the profile with a shifted `SUMPRODUCT`:

```
=SUMPRODUCT($B2:$B13, INDEX($Profile,ROW()-…))  ' shift each cohort by its lag
```

Simpler, most analysts use a diagonal drag of `=Sales_MonthN * Collect%_bucket` and sum each week's column.

DSO/DPO/DIO levers — what to actually pull

Lever	Tactic	Cash effect
DSO ↓	2/10 net-30 early-pay discount; auto-reminders at day −3, 0, +7; stop-supply at 60 dpd; invoice same-day not month-end	Pulls cash forward
DPO ↑	Renegotiate supplier terms 30→45; pay on due date not early; use dynamic discounting only if return beats your cost of capital	Pushes cash back
DIO ↓	ABC analysis, cut slow-movers, JIT on A-items, consignment stock	Frees inventory cash

Pulling the numbers from Tally

Ratios need AR, AP, Inventory, Revenue, COGS. In TallyPrime: - **AR ageing**: `Gateway of Tally → Display More Reports → Statements of Accounts → Ageing Analysis → Ledger` (or `Bills Receivable → F6: Ageing`). Set ageing periods 0-30 / 31-60 / 61-90 / 90+. - **AP ageing**: same path, `Bills Payable`. - **Closing**

stock: `Stock Summary`. Revenue/COGS: `Profit & Loss A/C`. - Export each with `Alt+E → Excel (Spreadsheet)` and pull into the model.

The 13-week forecast — Python skeleton

When AR/AP ageings are large, a Python roll is faster than dragging Excel:

```
import pandas as pd, numpy as np

weeks = pd.date_range("2026-07-06", periods=13, freq="W-MON")
cf = pd.DataFrame(index=weeks)

# Inflows
cf["collections"]   = [1.20,0.95,1.40,1.10,0.80,1.05,1.30,0.90,1.15,1.00,1.25,0.85,1.10]  # cr
cf["other_inflow"]  = 0.0

# Outflows (negative)
cf["payables"]      = [-0.70,-0.55,-0.90,-0.60,-0.75,-0.50,-0.80,-0.65,-0.70,-0.60,-0.85,-0.55
cf["payroll"]       = np.where(cf.index.day ≤ 7, -0.45, 0.0)   # 1st-of-month payroll
cf["gst_tds"]       = np.where(cf.index.day.isin(range(18,26)), -0.35, 0.0)  # 20th GST
cf["loan_emi"]      = -0.12
cf["capex"]         = 0.0

cf["net"]        = cf.drop(columns=[]).sum(axis=1)
opening          = 0.60   # cr, this Monday's bank balance
cf["closing"]    = opening + cf["net"].cumsum()
print(cf[["collections","payables","payroll","gst_tds","net","closing"]].round(2))

crunch = cf[cf["closing"] < 0]
if not crunch.empty:
    print("CASH CRUNCH in week(s):", list(crunch.index.date))
```

Cash-flow classification (accounting anchor)

The 13-week model is *direct-method* operating cash. Contrast with AS-3 / Ind AS 7 indirect method used in statutory accounts:

Activity	Examples
Operating	Collections, supplier payments, payroll, GST, TDS
Investing	Capex, asset sale, investments
Financing	Loan drawdown/EMI, equity, dividend

Journal entries behind the levers

Recording an early-payment discount **given** to a customer (2/10 net-30, ₹1,00,000 invoice, they pay in 10 days):

Account	Dr (₹)	Cr (₹)
Bank	98,000	
Discount Allowed	2,000	
To Debtors (Customer A)		1,00,000

Discount **received** by taking a supplier's early-pay offer:

Account	Dr (₹)	Cr (₹)
Creditors (Supplier B)	50,000	
To Bank		49,000
To Discount Received		1,000

Worked example / mini-project

Prakash Fabricators Pvt Ltd, FY26. Revenue ₹120 cr, COGS ₹90 cr. Closing AR ₹22 cr, Inventory ₹15 cr, AP ₹11 cr.

```
DSO = 22/120 * 365 = 66.9 days
DIO = 15/90  * 365 = 60.8 days
DPO = 11/90  * 365 = 44.6 days
CCC = 60.8 + 66.9 - 44.6 = 83.1 days
```

83 days of operating cash is locked up. **One DSO day = ₹120cr/365 = ₹32.9 lakh.**

The ask: the CFO wants to fund a ₹6 cr machine from internal cash, no new loan. Target the levers:

Lever	Move	Days	Cash released
DSO	67 → 55 (credit control + 2/10 terms)	−12	12 × 0.329 = ₹3.95 cr
DIO	61 → 52 (ABC + cut slow SKUs)	−9	9 × (90cr/365) = ₹2.22 cr
DPO	45 → 52 (renegotiate top 5 vendors)	+7	7 × (90cr/365) = ₹1.73 cr

Total released ≈ **₹7.9 cr** — machine funded, CCC drops from 83 to **55 days**.

13-week check. Feed the new (faster) collections profile into the Python model above. Week 5 previously dipped to −₹0.15 cr (a crunch) because of clustered GST + payroll. After pulling DSO forward, week-5 closing goes to +₹0.42 cr — the crunch disappears without drawing the cash-credit limit. That single chart — closing balance line staying above zero for 13 weeks — is the deliverable a CFO signs off on.

How it's tested

Interview questions - "A company is profitable but keeps running out of cash. Walk me through why." (Answer: CCC too long — cash trapped in AR/inventory; growth funds working capital.) - "What's the difference between the direct and indirect cash-flow method?" - "DSO went from 45 to 60 days QoQ. Give me three causes and how you'd investigate." (Deteriorating collections, a few large late accounts, revenue-recognition timing, channel-mix shift to slow payers.) - "How would you build a 13-week cash forecast from scratch? What are your inputs?" - "One DSO day — how much cash is that for a ₹500 cr revenue company?" (≈₹1.37 cr.)

Practical / assessment tests - **Timed Excel case (45–60 min):** given an AR ageing + AP ageing + payroll and GST schedule, build a 13-week forecast with a flagged negative-week and a one-paragraph recommendation. - **Ratio drill:** from a trial balance, compute DSO/DPO/DIO/CCC and the rupee value of a 10-day DSO cut. - **Collections-waterfall test:** apply a historical collection % profile to a sales forecast to project weekly cash inflow. - **Variance / bridge:** reconcile last week's forecast to the actual bank statement, explain the gap.

Common mistakes & how pros avoid them

- **Forecasting on invoice date, not collection date.** A ₹50 lakh March invoice is *not* March cash. Pros apply a collection lag profile.
- **Using annual DPO/DSO on a weekly model.** Ratios are diagnostic; the 13-week model needs actual dated line items (specific EMIs, the 20th GST payment, the 7th payroll).
- **Forgetting GST/TDS timing.** GST is due by the 20th, TDS by the 7th. These are large, clustered, non-negotiable outflows that cause the classic mid-month crunch.
- **Chasing DPO by simply paying suppliers late.** That destroys supplier relationships and forfeits discounts. Pros *negotiate* longer terms and only stretch to the agreed due date.
- **No rolling refresh.** A forecast built once and never updated is dead in a week. Roll it every Monday: drop the past week, add week 14, re-anchor opening cash to the actual bank balance.
- **Confusing profit with cash.** Depreciation, accruals, and prepaid items break the link. The 13-week model ignores accruals entirely — it's pure cash in/out.
- **Not reconciling to the bank.** If your model's opening balance doesn't tie to the bank statement to the rupee, nobody trusts the closing line.

Learn-it roadmap & resources

Time to proficiency: ~4–6 weeks part-time if you already know Excel and basic accounting. Week 1: ratios + CCC. Weeks 2–3: build one real 13-week model end-to-end. Week 4: collections waterfall + variance bridge. Weeks 5–6: automate the roll in Python/Power Query.

Resource	Type	Focus
CFI "Cash Flow Forecasting" course	Paid	13-week model, treasury view
Wall Street Prep / Breaking Into Wall Street WC modeling	Paid	Working-capital schedule in a 3-statement model
ICAI FM study material (working-capital management)	Free	India-context, exam rigour
ACCA / CIMA treasury notes	Free/Paid	Globally-portable practice
CTP (Certified Treasury Professional, AFP)	Cert	Treasury/cash-management gold standard
Excel `SUMPRODUCT` , `EOMONTH` , `WORKDAY` mastery	Free	The engine of every model

Build a real one for a friend's business or from a public annual report — that single artifact beats any certificate in an interview.

Quick-reference

```
DSO = AR / Revenue        * 365     One DSO day = Revenue / 365
DPO = AP / COGS           * 365     One DIO/DPO day = COGS / 365
DIO = Inventory / COGS    * 365
CCC = DIO + DSO − DPO               Lower = better
```

Lever	Direction	Effect on cash
DSO	↓	Releases cash (collect faster)
DPO	↑	Releases cash (pay later)
DIO	↓	Releases cash (less stock)

13-week model checklist: opening bank = actual balance · collections on *collection* date · payables on due date · payroll (7th) · GST (20th) · TDS (7th) · loan EMIs · capex · closing = opening + Σnet · flag any week < 0 · roll every Monday · reconcile to bank.

India timing anchors: GST payment by 20th · TDS deposit by 7th · Advance tax 15 Jun/Sep/Dec/Mar · typical B2B terms 30–90 days.

Valuation in practice (for decisions)

What it is & where it's used

Valuation "for decisions" is deliberately *not* the 200-tab IPO model. It is the fast, defensible number you build to answer a business question: *Should we buy this competitor? Is this vendor overpriced? What's our own company roughly worth for ESOP pricing or a funding round? Is Segment B destroying value?* The output is a **range plus a recommendation**, delivered in hours or a day, not weeks.

Two workhorses do 90% of the job:

- **DCF (Discounted Cash Flow)** — intrinsic value from projected free cash flows discounted at WACC. Used when you have a view on future cash flows and want to test whether the market/asking price is sane.
- **Comps (Comparable company / transaction multiples)** — value by analogy: apply peer EV/EBITDA, EV/Revenue, or P/E to the target. Faster, market-anchored, needs a peer set.

Roles that pay for this: **Corporate Development / M&A**, **FP&A** (business-case and buy-vs-build decisions), **investment banking / equity research**, **PE/VC associates**, **strategy/consulting**, and increasingly **finance business partners** who must sanity-check a capex or acquisition proposal in a meeting.

The gap: why companies want this (and college didn't teach it)

College teaches you the *Gordon Growth formula* and makes you discount a clean cash-flow stream to a single "correct" answer. Industry never has a clean stream. The gap is threefold:

1. **Speed and judgment over precision.** Employers want someone who, given a messy P&L, can produce a value range by end of day and *say which assumptions matter*. Textbooks reward exactness on made-up numbers.
2. **The plumbing MBA skips.** Converting equity value ↔ enterprise value (add net debt, minorities, subtract cash), mid-year convention, terminal value as 70–80% of value, calendarising to India's Apr–Mar FY, treating leases and ESOPs — none of this is in the lecture.
3. **Framing a decision, not a number.** A valuation that ends in "it's worth ₹142.7 cr" is useless. "It's worth ₹120–160 cr; below ₹135 cr it's accretive to our EPS, above ₹150 cr we're paying for synergies we haven't underwritten" — that gets you hired.

What "proficient" looks like

A job-ready person can, **unaided and under time pressure**:

- Build a 5-year DCF from a P&L in under 45 minutes, with a clean **FCFF → Enterprise Value → Equity Value → per-share** bridge.
- Compute **WACC** from a cap structure and defend every input (risk-free = 10-yr G-Sec, ERP, beta, cost of debt post-tax).
- Pull a **comps table**, pick the right multiple for the sector (EV/EBITDA for manufacturing, EV/Revenue or EV/GMV for early SaaS, P/B for banks), and reconcile it against the DCF.
- Run a **sensitivity table** (value vs WACC and terminal growth) and state the swing factors.

- Say **"precision doesn't matter here"** when it doesn't — e.g. a ₹5 cr build-vs-buy where the answer is obvious at any reasonable discount rate.

Hands-on: how to actually do it

1. The FCFF build (Excel)

Free Cash Flow to Firm each year:

```
FCFF = EBIT × (1 − tax) + D&A − Capex − ΔNet Working Capital
```

Layout with years across columns (B:F = FY26–FY30). Assume EBIT in row 10, D&A row 11, Capex row 12, ΔNWC row 13, tax rate in `$B$3`:

```
=B10*(1-$B$3) + B11 - B12 - B13        # FCFF for FY26, copy right
```

2. Discount factors and PV

Put WACC in `$B$2`. Use **mid-year convention** (cash arrives mid-period) for a tighter number:

```
# Period number in row 20: 1,2,3,4,5
# Mid-year discount factor:
=1/(1+$B$2)^(B20-0.5)
# PV of each year's FCFF:
=B14*B21          # FCFF row 14 × discount factor row 21
```

3. Terminal value (Gordon growth), then discount it

```
# Terminal value at end of year 5, g in $B$4:
=F14*(1+$B$4)/($B$2-$B$4)
# PV of TV (discount 5 full years, or 4.5 with mid-year):
=TV/(1+$B$2)^(F20-0.5)
```

Sanity check: **TV should be 60–80% of total EV**. Above 85% means your explicit forecast is doing no work — extend it.

4. EV → Equity → per share bridge

```
Enterprise Value      = SUM(PV of FCFF) + PV of TV
(−) Net Debt          = Total Debt − Cash & equivalents
(−) Minority Interest
(−) Preference capital
(+) Non-operating assets / investments
= Equity Value
÷ Diluted shares (incl. ESOP via treasury method)
= Value per share
```

5. WACC

```
# E = market cap, D = net debt, Re = cost of equity (CAPM), Rd = pre-tax cost of debt, Tc = ta
WACC = E/(E+D)*Re + D/(E+D)*Rd*(1-Tc)
# CAPM:
Re = Rf + Beta*ERP
# India today (illustrative): Rf ≈ 7.0% (10-yr G-Sec), ERP ≈ 6.5–7%, Beta from peers
=0.07 + 1.1*0.065     # → 14.15%
```

6. Comps table with a live multiple

```
# Peer EV/EBITDA in D2:D6; apply median to target EBITDA in $B$8:
=MEDIAN(D2:D6)*$B$8               # implied EV
# EV/Revenue quick screen for a target with revenue in $B$9:
=MEDIAN(E2:E6)*$B$9
```

7. Sensitivity (Data Table)

Put EV formula in a corner cell. Select the grid, `Data → What-If Analysis → Data Table`. Row input = terminal growth, Column input = WACC. Instant 2-D value matrix.

8. Python — same DCF, scriptable for screening many targets

```
import numpy as np

def dcf_ev(fcff, wacc, g, midyear=True):
    fcff = np.array(fcff, dtype=float)
    n = np.arange(1, len(fcff) + 1)
    adj = 0.5 if midyear else 0.0
    df = 1 / (1 + wacc) ** (n - adj)
    pv_explicit = (fcff * df).sum()
    tv = fcff[-1] * (1 + g) / (wacc - g)
    pv_tv = tv / (1 + wacc) ** (n[-1] - adj)
    return pv_explicit + pv_tv, pv_tv / (pv_explicit + pv_tv)

ev, tv_share = dcf_ev([120, 138, 158, 179, 200], wacc=0.14, g=0.05)
print(f"EV ₹{ev:,.0f} cr | TV is {tv_share:.0%} of value")
```

Worked example / mini-project

Decision: Your company (mid-size auto-components maker) is screening **TargetCo**, a bolt-on supplier. Asking price is **₹150 cr equity**. Should you bid?

Given (₹ cr): FY25 Revenue 100, EBIT 15, D&A 6, Capex 8, tax 25%, NWC growing ₹2 cr/yr. Growth 12% → tapering. Net debt ₹20 cr. Peers trade at **8× EV/EBITDA**. WACC 14%, terminal g 5%.

FCFF forecast:

Item (₹ cr)	FY26	FY27	FY28	FY29	FY30
EBIT	16.8	18.8	20.7	22.4	23.9
EBIT×(1−0.25)	12.6	14.1	15.5	16.8	17.9
+ D&A	6.7	7.3	7.8	8.2	8.5
− Capex	9.0	9.7	10.3	10.7	11.0
− ΔNWC	2.0	2.0	2.0	2.0	2.0
FCFF	**8.3**	**9.7**	**11.0**	**12.3**	**13.4**

DCF: PV of FCFF (mid-year, 14%) ≈ ₹37 cr. Terminal value = 13.4×1.05/(0.14−0.05) = **₹156 cr**; PV ≈ **₹85 cr**. TV is 70% of EV — healthy.

- **Enterprise Value ≈ ₹122 cr** → − net debt ₹20 cr → **Equity ≈ ₹102 cr**.

Comps cross-check: FY25 EBITDA = EBIT 15 + D&A 6 = ₹21 cr. 8× = **EV ₹168 cr** → − ₹20 cr = **Equity ₹148 cr**.

The decision (this is the deliverable):

> DCF says ₹102 cr equity, comps say ₹148 cr. The gap is the market pricing in growth our cash flows don't fully capture. The ₹150 cr ask sits at the *top* of the comps range and ~45% above intrinsic DCF. **Recommendation: counter at ₹110–125 cr.** Above ₹135 cr, the deal only works if we can underwrite ≥₹4 cr/yr of procurement/logistics synergy — which we have not validated. Precision beyond this range is wasted; the negotiation, not the third decimal, decides the outcome.

How it's tested

Interview questions - "Walk me from EBITDA to equity value per share." (The bridge — most common screen.) - "Terminal value is 90% of your DCF. Comfortable? What do you change?" - "EV/EBITDA vs P/E — when do you use which, and why is EV/EBITDA capital-structure neutral?" - "Your WACC drops 1% — roughly what happens to value?" (Should answer *directionally and materially*, ~15–25% for a growth name.) - "When would you *not* build a DCF?"

Practical tests - **Timed Excel case (45–90 min):** raw P&L + balance sheet handed over; build DCF + comps + a football-field chart and a one-line recommendation. Graded on the *bridge, WACC defensibility, and sensitivity* — not formatting. - **Take-home M&A screen:** "Here are 5 targets, pick the one to pursue." They test whether you triage with quick comps before over-modelling. - **LBO / accretion-dilution** for PE/IB: given purchase price and financing, is the deal EPS-accretive?

Common mistakes & how pros avoid them

Mistake	Fix
Discounting FCFF with cost of equity	FCFF → **WACC**; FCFE → cost of equity. Never mix.
Mixing EV and equity items	EV multiples ↔ EV numerator (EBITDA, EBIT, sales); equity multiples ↔ P/E, P/B. Don't apply EV/EBITDA to net income.
Terminal g ≥ nominal GDP	Cap g at ~4–5% India nominal-ish; a company can't outgrow the economy forever.
Terminal value 90%+ of EV	Extend explicit forecast or use an exit multiple as a check.
Forgetting the bridge	Always subtract net debt, minorities, prefs to get to equity.
False precision	Report a **range**; run sensitivity; state the 2 assumptions that move it.
Cherry-picked peers	Match size, growth, margins, geography; show median *and* range, not one flattering comp.
Double-counting synergies into standalone value	Value standalone first, then show synergy as a separate, clearly-flagged layer.

Learn-it roadmap & resources

Time to proficiency: 6–10 weeks part-time if you already know accounting.

- **Weeks 1–2:** FCFF/FCFE mechanics + the bridge. Build 3 DCFs from scratch, no template.
- **Weeks 3–4:** WACC, CAPM, beta unlevering/relevering; sensitivity tables.

- **Weeks 5–6:** Comps — build a peer set from screeners (Screener.in, Tijori for India), reconcile with DCF.
- **Weeks 7–8:** M&A screening, accretion/dilution, a light LBO.
- **Ongoing:** value 1 real listed company a week; check against the CMP and explain the gap.

Resources - *Damodaran Online* (NYU) — free spreadsheets, India ERP/beta data, the global gold standard. - Aswath Damodaran's YouTube valuation series (free). - Macabacus / Breaking Into Wall Street / Wall Street Prep — paid, template-heavy, interview-focused. - **Screener.in** and **Tijori** for Indian financials; NSE/BSE filings; RBI for the 10-yr G-Sec. - Your **CA Intermediate FM** already covers cost of capital, leverage, and capital budgeting — leverage it directly. - Certifications that signal this: **CFA Level II** (equity valuation), or a focused financial-modeling course (FMVA / WSP).

Quick-reference

Item	Formula / rule
FCFF	EBIT(1−t) + D&A − Capex − ΔNWC
FCFE	FCFF − Interest(1−t) + Net borrowing
WACC	E/V·Re + D/V·Rd(1−t)
CAPM (Re)	Rf + β·ERP
Terminal value	$FCFF_5$(1+g)/(WACC−g)
EV → Equity	EV − net debt − minorities − prefs + non-op assets
Discount (mid-year)	1/(1+WACC)^(n−0.5)
India Rf	10-yr G-Sec yield (~7%)
India ERP	~6.5–7%
Sector multiple	Mfg → EV/EBITDA; SaaS → EV/Revenue; Bank → P/B; mature → P/E
TV as % of EV	Aim 60–80%; >85% = extend forecast
Excel sensitivity	Data → What-If → Data Table (WACC × g)
Golden rule	Deliver a **range + a decision**, not a single number

The FP&A Tech Stack

What it is & where it's used

The FP&A tech stack is the set of tools a finance team uses to *collect actuals, build a plan, and report the variance between the two*. In 90% of Indian companies — from a ₹50 crore manufacturer to the finance shared-service centre of an MNC — that stack is still **Excel plus a source system** (Tally, SAP, Oracle NetSuite, or Zoho Books). As the company scales, a dedicated **planning tool** (Anaplan, Workday Adaptive Planning, Datarails, Oracle EPM Cloud, Board) is layered on top to kill the spreadsheet chaos.

You will touch this stack in almost every finance role:

Role	What they do with the stack
FP&A analyst	Builds the budget model, monthly forecast, variance decks
Business finance partner	Slices margin by product/region, runs scenarios
Financial reporting / controllership	Feeds actuals from the ERP into the plan
Corporate development / strategy	Long-range plans, M&A models
Startup finance (fractional CFO)	Runs the whole thing in one Excel/Google Sheets model

The uncomfortable truth: **the tool is a means, not the skill.** Employers pay for someone who can design a driver-based model and *decide* when a spreadsheet has outgrown its usefulness. This chapter teaches both.

The gap: why companies want this (and college didn't teach it)

An MBA or CA course teaches you to *read* financial statements and to *value* a business. It never makes you *build the machine that produces a forecast every month*. The gaps employers complain about:

1. **You know accounting, not modelling.** You can pass a consolidation question but can't build a three-statement model that flexes when volume changes.
2. **You've never seen "version hell."** College models are one-shot. Real FP&A runs Budget v1, v2, Reforecast Q2, Actuals — and someone always emails `Budget_FINAL_v3_USE_THIS.xlsx`. You've never felt that pain, so you don't respect the tools built to solve it.
3. **You don't know the data plumbing.** Actuals live in the ERP at transaction-level with 400 GL codes. The plan is at product-line level. Nobody taught you the *mapping* between them — the single hardest, most valuable FP&A skill.
4. **You've never justified a ₹40 lakh/year software purchase.** Knowing *when* to move off Excel and *how to say so to a CFO* is a promotion-grade skill.

Closing this gap is what turns a "reporting analyst" into an "FP&A lead."

What "proficient" looks like

A job-ready person can, unaided:

- Build a **driver-based P&L** in Excel where changing one assumption (say, price per unit) ripples correctly through revenue, COGS, GST, and EBITDA.

- Pull actuals from an ERP export and **reconcile them to the plan** using a mapping table, not manual retyping.
- Explain a **variance** in business terms: "EBITDA missed by ₹1.2 Cr — ₹80 L volume, ₹40 L price/mix."
- Articulate the **EPM value chain**: model → data integration → workflow → reporting.
- Give a crisp answer to "*When do we move off Excel onto Anaplan/Adaptive?*" with real triggers, not vibes.
- Use structured references, `SUMIFS`, `XLOOKUP`, and dynamic arrays — no hard-coded cell soup.

Hands-on: how to actually do it

Excel: the driver-based revenue block

Never hard-code revenue. Build it from drivers so it flexes.

```
Assumptions (named cells / a Assumptions tab):
  Price_Unit   = 1200      (₹ per unit)
  Vol_Growth   = 0.03      (3% MoM)
  Units_Jan    = 10000

Month row   :  Jan      Feb                 Mar  ...
Units (B5)  :  =Units_Jan   =B5*(1+Vol_Growth)   =C5*(1+Vol_Growth)
Revenue     :  =B5*Price_Unit
```

XLOOKUP for the GL-to-plan mapping (replaces VLOOKUP)

Your ERP export has GL codes; your plan needs line items. Map them once:

```
=XLOOKUP([@GLCode], MapTable[GLCode], MapTable[PlanLine], "UNMAPPED")
```

The `"UNMAPPED"` flag is deliberate — any new GL the ERP throws at you surfaces instead of silently dropping out of the P&L.

SUMIFS to aggregate actuals to the plan grain

```
=SUMIFS(GL[Amount], GL[PlanLine], $A2, GL[Month], B$1, GL[Entity], "IN01")
```

Variance and % variance, protected against divide-by-zero

```
Variance    =Actual - Budget
Var %       =IFERROR((Actual-Budget)/ABS(Budget), "n/m")
```

Python: pull the ERP CSV, map it, pivot to a plan-ready grid

When the export is 200k rows, Excel chokes. Pandas doesn't.

```python
import pandas as pd

gl  = pd.read_csv("gl_actuals_apr26.csv")           # TransID, GLCode, Amount, Month, Entity
mp  = pd.read_csv("gl_plan_map.csv")                # GLCode, PlanLine

df  = gl.merge(mp, on="GLCode", how="left")
unmapped = df[df["PlanLine"].isna()]["GLCode"].unique()
print("UNMAPPED GLs (fix the map!):", unmapped)

pivot = (df.dropna(subset=["PlanLine"])
           .pivot_table(index="PlanLine", columns="Month",
                        values="Amount", aggfunc="sum", fill_value=0))
pivot.to_excel("actuals_planready.xlsx")
```

SQL: the same aggregation straight from the warehouse

If actuals sit in a database, skip the export entirely:

```sql
SELECT m.plan_line,
       g.period,
       SUM(g.amount) AS actual_amt
FROM   gl_transactions g
JOIN   gl_plan_map    m ON g.gl_code = m.gl_code
WHERE  g.entity = 'IN01'
  AND  g.period BETWEEN '2026-04' AND '2026-06'
GROUP  BY m.plan_line, g.period
ORDER  BY m.plan_line, g.period;
```

DAX: a variance measure in Power BI / EPM reporting

```
Variance =
VAR bud = CALCULATE([Amount], 'Scenario'[Name] = "Budget")
VAR act = CALCULATE([Amount], 'Scenario'[Name] = "Actual")
RETURN act - bud

Variance % =
DIVIDE([Variance], CALCULATE([Amount], 'Scenario'[Name] = "Budget"))
```

What a planning tool (Adaptive / Anaplan) does that Excel can't

In Adaptive or Anaplan the same driver model becomes a **multi-dimensional cube**: `Account × Time × Entity × Product × Version`. You write one formula on a line item and it applies across every intersection. Key mechanics you'll use:

- **Versions**: Budget, Forecast, Actual are first-class — no `_v3_FINAL` files.

- **Data integration**: a scheduled connector loads ERP actuals nightly, auto-mapped.
- **Workflow**: each cost-centre owner enters their numbers; a submit/approve chain locks them.
- **Audit trail**: every cell change is logged with who/when.

Worked example / mini-project

Reproduce this. You are FP&A for a mid-size FMCG distributor. Build the April FY27 flash.

Assumptions:

Driver	Value
Opening units/month	50,000
Price per unit	₹250
Gross margin	32%
Fixed opex/month	₹28,00,000
GST rate (output)	18%

Budget P&L (April):

Line	Formula	Amount (₹)
Revenue	50,000 × 250	1,25,00,000
COGS	Revenue × 68%	85,00,000
Gross profit	Rev − COGS	40,00,000
Fixed opex	given	28,00,000
EBITDA	GP − opex	**12,00,000**

Now actuals come in from Tally: units sold **46,000**, realised price **₹258** (price hike stuck), COGS ₹82.1 L, opex ₹29.2 L.

Line	Budget	Actual	Variance	Driver
Units	50,000	46,000	−4,000	volume
Price	250	258	+8	price
Revenue	1,25,00,000	1,18,68,000	−6,32,000	see below
COGS	85,00,000	82,10,000	+2,90,000 (fav)	
Gross profit	40,00,000	36,58,000	−3,42,000	
Opex	28,00,000	29,20,000	−1,20,000	
EBITDA	12,00,000	7,38,000	**−4,62,000**	

Price-volume bridge on revenue (the answer a CFO wants):

```
Volume effect = (46,000 − 50,000) × ₹250        = −10,00,000
Price effect  = (₹258 − ₹250) × 46,000           = +3,68,000
Net revenue variance                              = −6,32,000  ✓
```

The story: "*Revenue missed ₹6.3 L — we lost ₹10 L on volume (4,000 fewer units) but recovered ₹3.7 L on the price increase. EBITDA fell ₹4.6 L, mostly volume-driven plus a ₹1.2 L opex overrun.*" That one sentence is the entire job.

Journal entry for the recognised April sales (net of the price realised):

Account	Dr (₹)	Cr (₹)
Trade Receivables	1,40,04,240	
Revenue		1,18,68,000
Output CGST		10,68,120
Output SGST		10,68,120

How it's tested

Interview questions:

- "Walk me through building a driver-based revenue forecast."
- "You get an ERP export at transaction level; the plan is at product level. How do you bridge them?" (They want to hear *mapping table*, not "I retype it.")
- "EBITDA missed budget. How do you find out why?" (Price-volume-mix bridge.)
- "When would you recommend moving off Excel to Anaplan or Adaptive?"

The practical test — this is what actually decides it:

- A **timed Excel case (45–90 min):** raw actuals + budget on two tabs. Build a variance report, a price-volume bridge, and a one-line commentary. They watch whether you use `SUMIFS` / `XLOOKUP` or manual filtering.
- A **"break my model" review:** they hand you a spreadsheet with a hard-coded total and ask you to find the error.
- Occasionally a **SQL/Power BI screen** for FP&A roles at data-mature firms: "aggregate these GL rows to plan lines and compute variance."

Good answer to the move-off-Excel question:

> "Move off Excel when three things co-occur: (1) the model breaks on version control — multiple people editing, `FINAL_v3` emails; (2) actuals load is a manual monthly copy-paste that eats two days; (3) we need bottom-up input from 10+ cost-centre owners with approval workflow. At that point a tool like Adaptive pays back — roughly ₹20–40 L/year, justified by analyst time saved and fewer errors."

Common mistakes & how pros avoid them

Mistake	What pros do
Hard-coding numbers inside formulas	All assumptions in one labelled Assumptions tab, referenced by name
`VLOOKUP` with hard column index	`XLOOKUP` / `INDEX-MATCH` — survives inserted columns
No "UNMAPPED" catch on GL mapping	Default flag so new GL codes surface, never silently drop
Manual monthly copy-paste of actuals	Structured tables + a repeatable pull (Power Query/Python/connector)
Reporting variance without the *why*	Always a price-volume-mix bridge and a one-line story
Buying Anaplan to fix a modelling problem	Fix the model design first; a tool amplifies good design, not bad
One giant tab	Separate Inputs / Calcs / Outputs; colour-code inputs (blue) vs formulas (black)
Divide-by-zero in Var %	Wrap in `IFERROR`, show "n/m"

Learn-it roadmap & resources

Realistic time-to-proficiency: 3–4 months of deliberate practice to clear an FP&A Excel test; 6–12 months on the job to be trusted with a planning tool.

Phase	Weeks	Focus
Excel modelling core	1–4	`SUMIFS`, `XLOOKUP`, dynamic arrays, structured refs, a 3-statement model
Driver-based planning	5–8	Build a budget + reforecast; price-volume-mix bridges
Data layer	9–12	Power Query, basic SQL, pandas for large actuals
EPM exposure	ongoing	Free Adaptive/Anaplan trials, model-builder tutorials

Resources:

- **Free:** Corporate Finance Institute free courses; Anaplan "Level 1 Model Builder" (free); Workday Adaptive product tutorials; Microsoft Learn (Power Query, DAX); `pandas` docs.
- **Paid / cert:** CFI **FMVA** (globally portable, strong in India), Anaplan **Certified Model Builder**, Adaptive Planning certification (usually employer-sponsored), Wall Street Prep for modelling drills.
- **Practice:** rebuild your own company's or a listed FMCG's segment P&L from its annual report as a driver model.

Quick-reference

Need	Formula / step
Lookup (modern)	`=XLOOKUP(key, lookup_col, return_col, "NA")`
Aggregate to grain	`=SUMIFS(amt, key_col, key, month_col, mth)`
Safe % variance	`=IFERROR((Act-Bud)/ABS(Bud),"n/m")`
Volume effect	`(Act_units − Bud_units) × Bud_price`
Price effect	`(Act_price − Bud_price) × Act_units`
SQL aggregate	`SELECT plan_line, period, SUM(amount) … GROUP BY plan_line, period`
DAX variance	`act − bud` via `CALCULATE([Amount], Scenario[Name]="Actual"/"Budget")`
Move-off-Excel triggers	version hell + manual actuals load + multi-owner input/workflow
EPM tools (India-relevant)	Anaplan, Workday Adaptive, Oracle EPM Cloud, Board, Datarails
EPM value chain	Model → Data integration → Workflow → Reporting/Audit
Output GST split	CGST 9% + SGST 9% (intra-state) or IGST 18% (inter-state)

The FP&A Interview & Case (with cheat-sheet)

What it is & where it's used

The FP&A interview is the final filter between "I did an MBA / cleared CA Inter" and a ₹6-18 LPA offer as an **FP&A Analyst, Business Finance Analyst, Financial Analyst, or Budgeting Analyst**. It has three moving parts, and every serious company (Amazon, Flipkart, Genpact, EY/Deloitte finance-transformation teams, GCCs like Wells Fargo/JPMC Hyderabad, plus mid-market CFO offices) tests all three:

1. **Conceptual Q&A** — do you understand the three statements, drivers, and variance?
2. **A live/take-home case** — build a mini-model, forecast, and explain the variance.
3. **A timed Excel/technical screen** — XLOOKUP, pivots, sometimes SQL or Power BI.

This chapter is the closing chapter: it assumes you've read the modelling, variance, and budgeting chapters, and now packages everything for the room. Roles that hire on this exact skill set: FP&A Analyst, Cost Analyst, Revenue Analyst, Corporate Finance Associate, and finance roles inside SaaS/D2C startups where "the analyst who owns the model" is a real job title.

The gap: why companies want this (and college didn't teach it)

College taught you to **calculate** ratios and **pass** journal entries against a given trial balance. It did not teach you to sit in front of a hiring manager and answer *"Revenue missed budget by ₹40 lakh — walk me through why, in 90 seconds."* That answer requires **price/volume/mix decomposition**, a **bridge**, and a **recommendation** — none of which appear in a textbook exam.

The specific gaps this chapter closes:

College teaches	Job actually tests
Compute NPV given cash flows	Build the cash flows yourself from assumptions
Define working capital	Forecast WC and explain the cash impact
"Variance = Actual − Budget"	*Why* the variance happened + what to do
Solve one clean problem	Handle a messy 5,000-row export under time pressure
Written exams	Talk while you build, defend your number

Employers pay for **judgement under ambiguity**, communicated crisply. The interview is engineered to detect exactly that.

What "proficient" looks like

A job-ready candidate, unaided, can:

- **Answer the three-statement linkage question cold** — how a ₹100 depreciation change flows through P&L, cash flow, and balance sheet (net income −₹75 assuming 25% tax, cash +₹25, retained earnings −₹75, PP&E −₹100... balances).
- **Build a driver-based revenue forecast** in Excel in under 15 minutes (units × price, or customers × ARPU).

- **Decompose a variance** into price, volume, and mix — not just report the delta.
- **Write an XLOOKUP / SUMIFS / pivot** without googling.
- **Read a SQL screen** — `GROUP BY`, `JOIN`, a window function.
- **State a recommendation** in one sentence: "Cut CAC by moving 20% of spend from Google to referral; that recovers ₹22L of the ₹40L miss."

The bar is not "knows everything." It's "can produce a defensible number and a decision from messy inputs, and explain it to a non-finance manager."

Hands-on: how to actually do it

The Excel core they will test

```
# Lookup a budget figure by cost centre (modern)
=XLOOKUP(A2, Budget[CostCentre], Budget[Amount], "Not found")

# Conditional sum - actuals for one department, one month
=SUMIFS(Actuals[Amt], Actuals[Dept], "Sales", Actuals[Month], "Jun")

# Variance % with divide-by-zero guard
=IFERROR((Actual-Budget)/Budget, 0)

# Two-way lookup (row = account, col = month) without a pivot
=INDEX(Data, MATCH($A2, Accounts, 0), MATCH(B$1, Months, 0))

# CAGR over n years
=(End/Start)^(1/n) - 1
```

Price / Volume / Mix — the question that separates analysts from clerks

```
Volume variance = (Actual Qty - Budget Qty) × Budget Price
Price variance   = (Actual Price - Budget Price) × Actual Qty
Mix variance     = shift in product mix at standard margins
Total = Volume + Price (+ Mix)
```

The SQL screen (revenue by month from a transactions table)

```
SELECT DATE_TRUNC('month', invoice_date) AS mth,
       SUM(amount)                       AS revenue,
       COUNT(DISTINCT customer_id)       AS active_customers,
       SUM(amount) / COUNT(DISTINCT customer_id) AS arpu
FROM   invoices
WHERE  invoice_date >= '2025-04-01'
GROUP  BY 1
ORDER  BY 1;

-- Month-over-month growth with a window function
SELECT mth, revenue,
       revenue - LAG(revenue) OVER (ORDER BY mth) AS mom_delta
FROM   monthly_rev;
```

Python (if the JD mentions it)

```
import pandas as pd
df = pd.read_csv("actuals.csv")
piv = df.pivot_table(index="account", columns="month",
                     values="amount", aggfunc="sum")
piv["variance"] = piv["Jun"] - piv["budget"]
piv["var_pct"]  = piv["variance"] / piv["budget"]
print(piv.sort_values("variance").head(10))   # 10 worst misses
```

DAX (Power BI screens)

```
Variance = SUM(Actuals[Amount]) - SUM(Budget[Amount])
Var % = DIVIDE([Variance], SUM(Budget[Amount]))
YTD Actual = TOTALYTD(SUM(Actuals[Amount]), 'Date'[Date])
```

Worked example / mini-project

The case (typical take-home, India SaaS): "Q1 FY26 revenue was budgeted at ₹5.00 Cr; actual came in at ₹4.60 Cr. Explain the ₹40L miss and recommend one action."

Given data:

Metric	Budget	Actual
New customers	500	460
ARPU (₹/customer)	₹1,00,000	₹1,00,000
Marketing spend	₹75L	₹80L
CAC (₹/customer)	₹15,000	₹17,391

Step 1 — bridge the revenue miss.

```
Revenue = Customers × ARPU
Volume variance = (460 − 500) × ₹1,00,000 = −₹40,00,000
Price/ARPU variance = (1,00,000 − 1,00,000) × 460 = 0
Total = −₹40L   → 100% a volume (customer-acquisition) problem
```

The miss is **not** pricing. ARPU held. We acquired 40 fewer customers.

Step 2 — why fewer customers? CAC rose from ₹15,000 to ₹17,391 (+16%) *while* spend rose ₹5L. So we spent **more** and got **fewer** — efficiency collapsed. Customers acquired = ₹80L ÷ ₹17,391 ≈ 460. Had CAC held at ₹15,000, ₹80L would have bought ~533 customers.

Step 3 — the one-sentence recommendation.

> "The ₹40L miss is entirely a volume problem driven by a 16% CAC spike, not pricing. Reallocate spend from the underperforming paid channel toward referral (historically ₹8,000 CAC); at that efficiency, the existing ₹80L budget recovers the 40-customer shortfall and ₹40L of revenue within one quarter."

Step 4 — the Excel that backs it:

```
Volume var =(460-500)*100000          → -4000000
CAC actual =80000000/460              → 17391
Cust if CAC held =80000000/15000      → 5333
```

That is a complete, defensible answer: bridge → root cause → decision → number. Reproduce it in a blank sheet in under 20 minutes and you clear most FP&A case rounds.

How it's tested

Round 1 — conceptual (verbal), the greatest hits:

- "Walk me through the three statements. Now change depreciation by ₹100 — trace it."
- "What's the difference between a budget, a forecast, and a plan?" (Budget = fixed target set once; forecast = updated view of reality; plan = the strategy behind the numbers.)
- "Revenue is up but cash is down — how?" (WC bloat: receivables/inventory up, or capex.)
- "What KPIs would you track for a D2C brand?" (CAC, LTV, LTV/CAC, contribution margin, repeat rate, cash conversion cycle.)

- "How do you build a revenue forecast?" (Driver-based: units × price, cohort, or run-rate — never a flat % without a driver.)

Round 2 — the case: a take-home model or a live "build a 3-year forecast from these assumptions," followed by defending every number.

Round 3 — the timed technical screen: 30-45 min. A messy CSV; deliver a pivot, an XLOOKUP-driven variance report, and often a chart. GCCs increasingly add a SQL screen and a Power BI task.

What they're really scoring: structure (did you bridge, not just subtract?), speed, accuracy under pressure, and whether you *ended with a decision*.

Common mistakes & how pros avoid them

Mistake	What pros do
Reporting a variance without explaining *why*	Always bridge: price / volume / mix, then root cause
Hardcoding numbers inside formulas	Every driver lives in a labelled input cell; formulas reference cells
Forecasting revenue as a flat "+10%"	Tie growth to a driver (customers, price, capacity)
Silent during the live case	Narrate: "I'm laying inputs top-left, calcs below, output right"
No recommendation	End every answer with one decision sentence + the ₹ impact
Circular refs / broken balance sheet	Know the depreciation and interest-on-debt linkages cold
Over-engineering the model	Deliver a clean, right answer fast; polish only if time remains
Guessing on Excel syntax	Drill XLOOKUP, SUMIFS, INDEX-MATCH, IFERROR until automatic

Learn-it roadmap & resources

Time to interview-ready: 6-10 weeks if you already have the MBA/CA base.

Weeks	Focus
1-2	Excel drills: XLOOKUP, SUMIFS, pivots, INDEX-MATCH — daily reps
3-4	Three-statement model from scratch, 5x, no template
5-6	Variance/bridge cases; price-volume-mix until reflexive
7-8	SQL basics (SELECT/JOIN/GROUP BY/window) + one Power BI dashboard
9-10	Mock interviews out loud; timed 30-min Excel cases

Resources - **Free:** CFI free Excel & FP&A intros; Corporate Finance Institute cheat-sheets; Ben Felix / Aswath Damodaran (valuation intuition); LeetCode/StrataScratch (easy SQL); Microsoft Learn (Power BI/DAX). - **Paid (high ROI):** CFI **FMVA** (~₹40-70k, globally recognised for FP&A), Wall Street Prep / Breaking Into Wall Street modelling courses. - **India-specific:** for controllership-flavoured roles your **CA Inter** costing + accounting is already a strong differentiator — lead with it.

Quick-reference

Core formulas

Metric	Formula
Variance	Actual − Budget
Variance %	(Actual − Budget) / Budget
Volume var	(Act Qty − Bud Qty) × Bud Price
Price var	(Act Price − Bud Price) × Act Qty
CAGR	(End/Start)^(1/n) − 1
Gross margin	(Revenue − COGS) / Revenue
Contribution margin	(Revenue − Variable cost) / Revenue
EBITDA	EBIT + Dep + Amort
LTV	ARPU × Gross margin × Avg lifetime
LTV/CAC	Healthy ≥ 3×
CAC	Total S&M spend / New customers
CCC	DSO + DIO − DPO
Working capital	Current assets − Current liabilities
Free cash flow	CFO − Capex

Excel one-liners

Task	Formula
Lookup	`=XLOOKUP(key, lookup, return, "NA")`
Conditional sum	`=SUMIFS(sum, crit1, val1, crit2, val2)`
Two-way lookup	`=INDEX(d, MATCH(r,rows,0), MATCH(c,cols,0))`
Guarded divide	`=IFERROR(a/b, 0)`
Count by criteria	`=COUNTIFS(rng, crit)`

The interview mantra: *Bridge the number → find the root cause → end with one decision and its ₹ impact.* Structure beats speed; a decision beats a correct-but-mute calculation.

Bloomberg, Capital IQ & Advanced Modeling — Hands-On

Hands-On Drills — Do the Task, Not Just Define It (India, 2026)

Step-by-step functions, T-codes, worked models and interview drills

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Terminal Orientation: Bloomberg & Capital IQ Function Drills

What you'll be able to do

Sit down at a Bloomberg terminal cold and drive it: pick the right asset class with the yellow keys, load a ticker, run the core functions an analyst uses every day (DES, FA, GP, EQS, RV, BICS, WACC, BETA), and read the screens fast enough to answer a "what's this company worth and how's it trading" question in five minutes. You'll also know where the same things live in S&P Capital IQ (CapIQ) so you can switch platforms without re-learning. By the end you can look up Infosys, read its description, financials and price chart, and pull a peer set — the raw material for every model in this guide.

The drill — step by step

1. The yellow keys (market sectors). The keyboard's yellow keys set the asset class so the terminal knows what "INFO" means. The ones you'll touch: `EQUITY` (stocks), `GOVT` (sovereign bonds), `CORP` (corporate bonds), `CMDTY` (commodities), `CRNCY` (FX), `INDEX`, `MMKT` (money markets), `MTGE`. The workflow is always: **type the name → press the yellow key → press `<GO>`.**

`<GO>` (the green key) executes. `<HELP>` once explains the current screen; `<HELP><HELP>` (twice) opens live chat with the Bloomberg help desk — genuinely useful, a real person answers.

2. Load a security. Type `INFO` then press `EQUITY`. The autocomplete/ticker-lookup drops a match list. Infosys trades in several places, so you disambiguate by exchange code:

What you type	Meaning
`INFO IN <EQUITY> <GO>`	Infosys, NSE India listing (`IN` = India)
`INFO US <EQUITY> <GO>`	Infosys ADR on NYSE
`INFY IN <EQUITY>`	also resolves (ticker vs. short name)

`IN` is the Bloomberg country/exchange code for India. If you don't know the ticker, type the company name and hit `EQUITY <GO>` — the security-finder (SECF) list appears; arrow down and Enter.

3. Now run the core functions. Once a security is loaded, you just type the mnemonic and `<GO>`; it applies to the loaded ticker. Drill them in this order:

- **`DES <GO>`** — Description. Business summary, sector, key execs, listing details, GICS/BICS classification, a snapshot of price and market cap. Your first stop on any name.
- **`FA <GO>`** — Financial Analysis. The workhorse. Tabs across the top: Income Statement (IS), Balance Sheet (BS), Cash Flow (CF), Ratios, Segments, Multiples. Change the period toggle to Annual/Quarterly and the currency (INR/USD). This is where you read 5 years of P&L.
- **`GP <GO>`** — Graph Price. Candlestick/line price chart. Change the range with the field boxes (`1D`, `1M`, `1Y`, `5Y`). `GIP` = intraday, `GPO` = price only.
- **`GPO`, `HP`** — `HP` (Historical Prices) is the tabular price history you'd export.

- `EQS <GO>` — Equity Screening. Build a peer/universe screen with criteria (country = India, BICS sub-industry = IT Consulting & Services, market cap > $5bn). Returns a list you can save — this seeds your comps set.
- `RV <GO>` — Relative Value. Auto-builds a comparable-companies grid for the loaded name (EV/EBITDA, P/E, growth) against a peer group Bloomberg guesses; you can edit the peers. This is a pre-baked comps table.
- `BICS` — Bloomberg Industry Classification. Used inside EQS/RV to define the peer universe (Bloomberg's answer to GICS).
- `WACC <GO>` — Weighted Average Cost of Capital. Bloomberg's computed cost of equity (via CAPM), cost of debt, capital weights, and blended WACC. Read it, don't trust it blindly — check the inputs.
- `BETA <GO>` — regression of the stock vs. its index (adjusted/raw beta, R^2). Set the index to NSE Nifty for an Indian name.
- `CN <GO>` — Company News. `TOP <GO>` = top market news.
- `PORT <GO>` — Portfolio & Risk Analytics (needs a saved portfolio).
- `FLDS <GO>` — field finder; searches every data field name (needed later for Excel BDP).

Navigation muscle memory: the **red** `<CANCEL>` stops a running query; `<MENU>` (or the left-arrow) goes back one screen; `PANEL` **/green arrows** cycle between the four screen panels; `1<GO>` … `4<GO>` open a function on a numbered panel.

4. Capital IQ (the cheaper, web-based cousin). CapIQ is browser-based, not a keyboard terminal. Layout is a left-nav of modules:

- **Companies** — search a company → a company profile with tabs: Tearsheet, Financials, Comparable Companies, Transactions, Estimates, Ownership.
- **Screening** — the EQS equivalent; build company/transaction screens with a criteria builder.
- **Comps** — pre-built comparable-company templates you clone.
- **Transactions** — M&A/precedent-deal database (used in Chapter 4).
- **Tearsheet** — a one-page company summary (description, financials, multiples) you can export to PDF/Excel — the standard "hand this to the MD" page.

Same job, different door: Bloomberg `DES` ≈ CapIQ company profile; `FA` ≈ CapIQ Financials tab; `RV` / `EQS` ≈ CapIQ Comps/Screening; deal work ≈ CapIQ Transactions.

Worked example — Infosys in five screens: 1. `INFO IN EQUITY <GO>` → loads. 2. `DES <GO>` → read: "Infosys Ltd, IT consulting, HQ Bengaluru, mkt cap ≈ ₹6.5 lakh cr / ~$78bn, ADR: INFY US." 3. `FA <GO>` → IS tab, Annual, INR → note FY revenue ≈ ₹1.53 lakh cr, EBIT margin ≈ 21%. 4. `GP <GO>`, set 5Y → see the multi-year price trend. 5. `RV <GO>` → peer grid vs. TCS, Wipro, HCLTech — read Infosys EV/EBITDA vs. peer median.

The output

Your "orientation output" is a filled scratch sheet — one row per function, so you can answer a snap valuation question:

Function	Field read	Infosys value (illustrative)
DES	Sector / mkt cap	IT Services / ~$78bn
FA (IS)	Revenue / EBIT margin	₹1.53 lakh cr / ~21%
FA (Ratios)	ROE	~30%
GP 5Y	Trend	up, cyclical dips
BETA	Adj. beta vs Nifty	~0.7
WACC	Blended WACC	~11–12%
RV	EV/EBITDA vs peer median	e.g. 18x vs 16x → slight premium

Checks & gotchas

- **Wrong yellow key = wrong or no security.** `INFO` alone means nothing; `INFO EQUITY` does. Always press the sector key.
- **Which listing?** ADR (`US`) vs. local (`IN`) have different currency, price and multiples. State which one you're using.
- **Currency & scale on FA.** Toggle INR vs. USD deliberately; Indian filings are often in ₹ crore — confirm the units box before you quote a number.
- **WACC/BETA are model outputs, not gospel.** Bloomberg's beta window and index choice drive the number — check the regression index is Nifty, not S&P 500, for an Indian name.
- **RV peers are auto-picked.** Bloomberg's default peer set will include names you'd reject; you must curate (Chapter 3).

Interview drill

Q: "Walk me through the first three things you do on a Bloomberg when handed a new ticker." A: "Load it with the right yellow key, run `DES` to understand the business, size and classification, then `FA` to read the last few years of income statement, margins and returns. `GP` for the price trend. That tells me what it does, how it earns and how the market's treating it before I touch a model."

Q: "What's the difference between EQS and RV?" A: " `EQS` is a screener — I define criteria and it returns a universe. `RV` is relative-value — it takes the loaded name and builds a comps grid against a peer group, showing valuation multiples side by side. EQS finds the peers; RV compares them. In CapIQ these are the Screening and Comparable Companies modules."

Q: "Bloomberg vs. Capital IQ — when would you reach for which?" A: "Bloomberg for live markets, fixed income, FX, real-time news and speed once you know the mnemonics. CapIQ for deep private-company data, M&A/transaction comps and clean Excel-exportable tearsheets. Most desks use both."

Practise free

You can rehearse 80% of this without a terminal: - **Bloomberg Market Concepts (BMC)** — Bloomberg's own ~8-hour paid-but-cheap e-course (often free via a college/library licence); teaches the mental model and mnemonics. - **Screener.in** — India's best free `FA` substitute: 10-year P&L, balance sheet, cash flow, ratios per company. Look up Infosys and read exactly what FA shows. - **Tijori Finance** — free tier gives segment and peer views (an RV-style comps grid). - **NSE India / BSE India** — official price history and corporate filings (your `HP` / `CN`). - **TIKR / stockanalysis.com** — free CapIQ-style tearsheets and global comps. Drill: open Screener.in, pull Infosys, TCS, Wipro, HCLTech side by side, and reproduce the RV table above by hand. That single exercise builds the reading speed the terminal only accelerates.

Pulling & Cleaning Data into Excel (BDP/BDH/BQL, CapIQ Plugin)

What you'll be able to do

Get data out of Bloomberg and Capital IQ into a spreadsheet **as live, refreshable formulas** — not copy-paste that rots the moment the price moves. You'll write `=BDP` for a single current value, `=BDH` for a time series, `=BDS` for bulk/list data, know when to reach for BQL, and pull the same financials from CapIQ with `=CIQ` formulas. You'll build a clean, model-ready "Data" tab: one place all raw pulls land, from which your model reads — the discipline that stops a model from silently breaking. Worked outcome: five years of Infosys financials sitting in a tab your DCF/comps can point at.

The drill — step by step

0. Install & log in. The Bloomberg add-in installs from the terminal (`WAPI` / `API<GO>` → "Excel Add-in") and needs you logged into Bloomberg on that machine. In Excel you'll see a **Bloomberg** ribbon tab. CapIQ's plugin installs separately and adds a **Capital IQ** ribbon (log in with your S&P credentials).

1. `=BDP` — Bloomberg Data Point (one static/current value). Syntax: `=BDP("security", "field")`.

```
=BDP("INFO IN Equity","PX_LAST")          → last price
=BDP("INFO IN Equity","CUR_MKT_CAP")      → market cap
=BDP("INFO IN Equity","BEST_PE_RATIO")    → forward P/E (BEST = consensus estimate)
=BDP("INFO IN Equity","CRNCY")            → INR
```

The trick is knowing the **field mnemonic**. Find them with `FLDS <GO>` on the terminal (searchable field dictionary) or the FieldSearch button on the ribbon. `PX_LAST`, `CUR_MKT_CAP`, `SALES_REV_TURN`, `EBITDA`, `IS_EPS`, `BS_TOT_ASSET`, `TRAIL_12M_ …`, `BEST_ …` (estimates) are the ones you'll reuse.

2. `=BDH` — Bloomberg Data History (a time series). Syntax: `=BDH("security","field(s)","start","end", [optional args])`.

```
=BDH("INFO IN Equity","PX_LAST","1/1/2021","12/31/2025","Per=M")
```

`Per=M` (monthly), `Per=Q`, `Per=Y` set frequency; `Fill=P` carries prices over non-trading days; `Days=W` restricts to weekdays. BDH spills a two-column (date, value) array **downward** — leave blank cells below the formula or it errors with `#N/A Requesting Data` colliding into your labels.

3. `=BDS` — Bloomberg Data Set (bulk / list data that returns many rows/cols). Used for things that aren't a single number: index members, dividend history, the full list of a company's segments or peers.

```
=BDS("NIFTY Index","INDX_MEMBERS")      → all Nifty constituents
=BDS("INFO IN Equity","DVD_HIST")       → dividend history table
```

4. BQL — Bloomberg Query Language (the modern, powerful way). BQL does filtering, aggregation and universe screening in one formula, returning a table. It's what you graduate to when BDP/BDH get clumsy.

Conceptually: `get( … )` fields `for( … )` a universe `with( … )` parameters.

```
=BQL("members('NIFTY Index')","name, px_last, pe_ratio")
```

returns every Nifty member with its name, price and P/E in one spill. You don't need mastery on day one — know it exists and that it replaces stacking 50 BDP calls.

5. CapIQ Excel plugin — `=CIQ` formulas. Same idea, S&P's data. `=CIQ("IQ ticker / identifier","mnemonic", [period])`.

```
=CIQ("INFO:","IQ_TOTAL_REV","IQ_FY-4")     → revenue, 4 years ago
=CIQ("INFO:","IQ_EBITDA","IQ_FY")          → latest FY EBITDA
=CIQ("INFO:","IQ_EBITDA","IQ_FY-1")        → prior FY
```

`IQ_FY`, `IQ_FY-1…-4` walk back fiscal years; `IQ_FQ` for quarters; `IQ_NTM` / `IQ_FY+1` for forward estimates. CapIQ mnemonics (`IQ_TOTAL_REV`, `IQ_EBIT`, `IQ_NI`, `IQ_TOTAL_DEBT`, `IQ_CASH_ST_INVEST`, `IQ_TEV` = enterprise value) come from the plugin's formula builder — use it, don't memorise.

6. Build the clean Data tab. Discipline that separates a pro model from a mess: - One tab named `Data_Raw`. Formulas only reference it; the model never calls BDP directly in the middle of a calc. - **Columns = periods (FY-4…FY), rows = line items.** Put the entity ticker in one cell (`B1`) and reference it: `=CIQ($B$1,"IQ_TOTAL_REV","IQ_FY-4")`. Change one cell → whole tab re-points to a new company. - Immediately below each live-formula block, **paste-special → values** into a mirror block if you need a point-in-time snapshot (see gotchas). - Add a **units row** and a **currency cell**; convert to a single reporting currency explicitly.

Worked example — Infosys 5-yr revenue & EBITDA, model-ready:

Row / `IQ_FY-`	FY-4	FY-3	FY-2	FY-1	FY
Formula (Revenue)	`=CIQ($B$1,"IQ_TOTAL_REV","IQ_FY-4")`	… `-3`	… `-2`	… `-1`	… `IQ_FY`
Revenue (₹ cr)	100,472	121,641	146,767	153,670	162,990
EBITDA (₹ cr)	26,000	30,700	34,900	36,100	38,900
EBITDA margin	25.9%	25.2%	23.8%	23.5%	23.9%

(Numbers illustrative — the point is every cell is a formula keyed to `$B$1`.)

The output

A single `Data_Raw` tab: ticker cell up top, a currency/units header, one clean block per statement (IS, BS, CF), periods across, live formulas that refresh on open, and a values-mirror for the version you actually valued off. Your comps and DCF tabs contain **zero** BDP/CIQ calls — they read from `Data_Raw` by cell reference. Hand it to a colleague, they change `$B$1` from `INFO:` to `TCS:` and the whole book reprices.

Checks & gotchas

- **`#N/A Requesting Data` / `#N/A Invalid Field`** — data still loading (wait/refresh) vs. a wrong mnemonic (fix via FLDS/formula builder). Don't build on top of a cell that's still `Requesting`.
- **Point-in-time vs. restated.** Live formulas pull *today's* view of history — companies restate, and consensus estimates drift daily. If your valuation must be reproducible, paste-special values and date-stamp the snapshot. This is the single most common "my model changed overnight" bug.
- **Currency & scale.** CapIQ may return USD while Bloomberg FA showed INR; ₹ crore vs. ₹ mn vs. absolute. Force one currency, one scale, and label it. A 100x error hides here.
- **Fiscal-year misalignment.** `IQ_FY` for an Indian March year-end ≠ a US December year-end peer — you'll calendarise in Chapter 3.
- **Blank-cell collisions.** BDH/BDS spill arrays; leave room or they overwrite (or error against) adjacent cells.
- **Broken on a machine without a login.** These formulas only resolve where the add-in + entitlement live. Email the workbook and the recipient sees `#NAME?` unless they too have the plugin — send the values-mirror.

Interview drill

Q: "Difference between BDP, BDH and BDS?" A: "BDP returns one current/static value for one field. BDH returns a historical time series with a frequency you set. BDS returns bulk or list-type data — index members, dividend schedules — that isn't a single scalar. One point, one history, one set."

Q: "How do you make sure a Bloomberg-linked model is reproducible?" A: "Isolate all live pulls in a Data tab, then paste-special the values you actually valued off and date-stamp them, because live formulas re-pull restated financials and shifting consensus every time the book opens. The audit version is static; the live version is for updating."

Q: "Why keep a separate Data tab instead of calling BDP inside the model?" A: "Auditability and portability — one place to check every input, one cell to re-point the whole model to a new ticker, and the calc engine keeps working even if the data connection drops."

Practise free

The formula *thinking* transfers completely without a licence: - **Python `yfinance`** — `yf.Ticker("INFY.NS").financials` and `.history(period="5y")` reproduce BDH/CIQ pulls for Indian tickers (`.NS` = NSE). Build the exact `Data_Raw` layout in pandas and export to Excel. - **Screener.in export** — the "Export to Excel" button on any company gives you 10 years of financials in one sheet; treat it as your CapIQ dump and build the clean tab on top. - **`nsepy` / NSE bhavcopy** — free historical prices for the BDH drill. - **`=STOCKHISTORY()`** in Microsoft 365 Excel — a genuine free BDH analogue for prices: `=STOCKHISTORY("INFY","1/1/2021",,2,0,0,1)`. Drill: replicate the Infosys revenue/EBITDA table above from a Screener.in export, then rebuild it in pandas so you understand what the plugin does under the hood.

Building a Comparable Companies (Comps) Analysis

What you'll be able to do

Build a trading-comps table from scratch: choose a defensible peer set, pull the raw inputs (market cap, net debt → enterprise value, revenue, EBITDA, EBIT, net income, shares, EPS), compute the multiples (EV/EBITDA, EV/Sales, EV/EBIT, P/E, PEG), calendarise so mismatched fiscal years are comparable, strip one-offs so the numbers are clean, take the median, and apply it to your target to imply a value and a football-field range. Worked end-to-end on Indian IT services with Infosys as the target.

The drill — step by step

1. Pick the peer set — this is 50% of the work. Comparable means *comparable*: same industry, similar business model, similar size, geography and margin profile. Use `EQS` /BICS on Bloomberg or CapIQ Screening: BICS "IT Consulting & Services", country India, market cap band. For Infosys the honest peer set is TCS, Wipro, HCLTech, Tech Mahindra, LTIMindtree. You could add Accenture/Cognizant as global reference points, but flag them — different scale and currency. **Write down your inclusion rule**; you will be asked to defend every name.

2. Pull the raw inputs into a clean grid (from your `Data_Raw` tab, Chapter 2). Per company you need:

- Share price and diluted shares → **Equity value (market cap)**
- Total debt + preferred + minority interest − cash & equivalents → **Net debt**
- **Enterprise Value (EV) = Equity value + Net debt** (+ minorities + preferred)
- Revenue, EBITDA, EBIT, Net income, EPS — both **LTM** (last twelve months) and **forward** (NTM / FY+1 consensus, `BEST_` / `IQ_NTM`)

3. Compute the multiples. The iron rule: **numerator and denominator must belong to the same claimants.**

Multiple	Formula	Whose metric
EV/EBITDA	EV ÷ EBITDA	capital-structure-neutral, pre-D&A
EV/EBIT	EV ÷ EBIT	after D&A, still pre-financing
EV/Sales	EV ÷ Revenue	for low/negative-margin names
P/E	Price ÷ EPS (or Equity ÷ Net income)	equity holders only
PEG	P/E ÷ EPS growth %	P/E normalised for growth

EV pairs with pre-interest metrics (EBITDA, EBIT, Sales) because EV represents *all* capital providers. **P/E** pairs with net income/EPS because both are *after* interest — equity-only. Never put EV over net income, or price over EBITDA.

4. Calendarise. Infosys, TCS, Wipro run **March year-ends**; Accenture runs **August**, Cognizant **December**. Comparing raw FY figures mixes periods. Fix by converting everyone to LTM, or calendarising to a common calendar year: e.g. CY = (months of FYn in the calendar year × FYn) + (remaining months × FYn+1), weighted.

For a purely Indian peer set with aligned March ends this is mostly moot — which is itself a reason to keep the set domestic.

5. Adjust for one-offs (clean the metric). Strip non-recurring items so multiples reflect underlying earnings: remove litigation settlements, restructuring, one-time visa/impairment charges, gains on asset sales; normalise a tax rate if a period had a one-off tax credit. Document each adjustment in a footnote row. A peer that took a big restructuring charge will show a distorted EBIT — un-distort it or its multiple lies.

6. Summarise — median, not just mean. Compute **min, 25th percentile, median, mean, 75th, max** for each multiple down the peer column. **Lead with the median** — it's robust to one outlier; the mean gets dragged by a single 40x name.

7. Imply the target's value. Apply the peer median multiple to the target's own metric:

- Implied EV = peer median EV/EBITDA × target EBITDA
- Implied Equity = Implied EV − target net debt
- Implied price = Implied equity ÷ target diluted shares
- Cross-check with P/E: Implied price = peer median P/E × target EPS

8. Football field. Plot each methodology's implied per-share range (25th–75th percentile of each multiple) as a horizontal bar; the overlap is your defensible value range, next to the current price.

Worked example — Infosys target, illustrative LTM figures:

Company	EV (₹ cr)	EBITDA (₹ cr)	EV/EBITDA	Net income (₹ cr)	P/E
TCS	14,20,000	62,000	22.9x	46,000	30.9x
Wipro	2,60,000	22,000	11.8x	13,100	19.8x
HCLTech	3,90,000	34,000	11.5x	15,700	24.8x
Tech Mahindra	1,30,000	9,500	13.7x	4,300	30.2x
LTIMindtree	1,60,000	8,200	19.5x	4,600	34.8x
Median			**13.7x**		**30.2x**

Apply to Infosys (EBITDA ≈ ₹38,900 cr, net debt ≈ −₹30,000 cr net cash, ~415 cr shares, EPS ≈ ₹66): - Implied EV = 13.7 × 38,900 ≈ **₹5,32,930 cr** - Implied equity = 5,32,930 + 30,000 (net cash adds back) ≈ **₹5,62,930 cr** - Implied price = 5,62,930 cr ÷ 415 cr ≈ **₹1,357/sh** - P/E cross-check: 30.2 × 66 ≈ **₹1,993/sh** → the spread tells you which multiple the market is really paying and that Infosys screens richer on P/E than EV/EBITDA vs. this set.

The output

A one-page comps sheet: peer rows with EV build-up and each multiple; a stats block (min/25th/median/mean/75th/max) per multiple; a target-implied-value block showing EV → equity → per-share for EV/EBITDA and P/E; and a football-field chart with the implied ranges against the live price. Footnotes list peer-selection rationale and every one-off adjustment.

Checks & gotchas

- **EV must be consistent** — always subtract cash and add debt, minorities and preferred the same way for every peer, or the ranking is noise.
- **Net cash companies** (most Indian IT) have EV < market cap; adding back net cash *raises* implied equity. Get the sign right.
- **LTM vs. forward mismatch** — never compare one peer's forward multiple to another's trailing. Pick one basis for the whole table.
- **Diluted, not basic shares** — include options/RSUs; treasury-stock method.
- **Outliers** — a high-growth or newly-listed peer can carry a 40x multiple; that's why you lead with the median and eyeball the range.
- **Currency** — don't mix an Accenture USD EV/EBITDA into an INR table without noting the multiple is currency-neutral but the size context isn't.
- **Garbage peers = garbage output.** The most common mistake is a lazy peer set. Comps are only as good as comparability.

Interview drill

Q: "Why EV/EBITDA over P/E?" A: "EV/EBITDA is capital-structure-neutral and pre-D&A, so it compares operating value across companies with different debt loads and depreciation policies — cleaner for cross-company comparison and for buyers who'll refinance the target. P/E is distorted by leverage and tax. I'd use both and explain divergences."

Q: "Your target trades at a discount to the peer median EV/EBITDA — buy signal?" A: "Not automatically. A discount can be justified — slower growth, lower margins, weaker returns on capital, governance or client-concentration risk. I'd check whether the discount is explained by fundamentals before calling it cheap. Comps tell you *where* it trades, not *why*."

Q: "How do you handle mismatched fiscal year-ends in a comps set?" A: "Calendarise — convert everyone to LTM or to a common calendar year by weighting the overlapping fiscal periods — so I'm comparing the same window. For an all-Indian March-end set it's usually moot, which is one reason I prefer a domestic peer group."

Practise free

- **Screener.in / TIKR / stockanalysis.com** — pull EV, EBITDA, net income, P/E for TCS, Wipro, HCLTech, Tech Mahindra, LTIMindtree and Infosys and build the exact grid above by hand.
- **Tijori Finance** — free peer-comparison views that mirror an RV screen.
- **Python** — `yfinance` `.info` gives `enterpriseValue`, `enterpriseToEbitda`, `trailingPE`; loop over the peer tickers, `pd.DataFrame`, compute median, imply Infosys value. This is the single best rep: it forces you to build EV yourself and see where net cash flips the sign. Drill target: reproduce the median EV/EBITDA and the implied Infosys price, then write the two-line rationale for why Infosys deserves a premium or discount to it.

Building a Precedent Transactions Analysis

What you'll be able to do

Build a precedent-transactions (deal comps) valuation: source relevant M&A deals from CapIQ Transactions or Mergermarket, screen them to a defensible set, pull the transaction value and target financials at the time of each deal, compute transaction multiples (EV/EBITDA, EV/Sales, EV/EBIT), read the control premium implied, adjust for timing/synergies, and apply the median to your target to imply an acquisition value. You'll know exactly why precedent multiples sit *above* trading comps and when to trust them. Worked on an Indian IT-services target.

The drill — step by step

1. Understand what you're building. Trading comps (Chapter 3) value a company as an independent public entity. Precedent transactions value it as an *acquisition target* — what real buyers actually paid to gain control. Deal multiples embed a **control premium** (typically 20–40% over the undisturbed price) and often **synergies**, so they run higher. Use precedents when the question is "what would someone pay to buy this?"

2. Source the deals. In **CapIQ → Transactions** (or **Mergermarket**), screen the M&A database on:

- **Industry** of the target (BICS/GICS = IT Services)
- **Geography** (India targets; add global IT-services deals as reference, flagged)
- **Date** — last 3–5 years only; older deals reflect a different rate/valuation regime
- **Deal size** band comparable to your target
- **Stake** — control deals (>50%, ideally 100%); exclude minority stakes, they don't carry a control premium
- **Deal status** — completed (or announced), not rumoured/withdrawn
- **Consideration** — note cash vs. stock (stock deals can inflate headline value)

Bloomberg equivalent: `MA <GO>` (M&A / deal search) and `EVTS`. Each deal record gives announced date, target, acquirer, deal value, stake, and often the target's LTM financials at announcement.

3. Pull the inputs per deal. For each transaction you need the **Transaction Enterprise Value (TEV)** — offer equity value + target net debt assumed — and the **target's LTM financials at the announcement date** (revenue, EBITDA, EBIT). Critical: use the financials as they were *at the time of the deal*, not today's. CapIQ Transactions usually stores these; otherwise pull the target's LTM as of the announcement quarter.

4. Compute transaction multiples.

Multiple	Formula
EV/EBITDA	TEV ÷ target LTM EBITDA at announcement
EV/Sales	TEV ÷ target LTM revenue
EV/EBIT	TEV ÷ target LTM EBIT

Same claimant-consistency rule as trading comps — TEV over pre-interest metrics.

5. Read the control premium. Where the target was public, premium = (offer price per share ÷ undisturbed price ~1 day/1 month before announcement) − 1. Record it; the median premium is itself a data point ("deals in this space cleared at ~25–30% premia").

6. Adjust for timing and synergies. - **Timing / regime:** a 2021 deal struck at peak multiples over-states today's fair value in a higher-rate 2026 environment. Weight recent deals more; note the vintage. - **Synergies:** strategic buyers pay up for cost/revenue synergies embedded in the price. That's why a *strategic* precedent multiple can overshoot standalone value — flag strategic vs. financial-sponsor deals; sponsor deals are cleaner "no-synergy" reads. - **Deal structure:** all-stock deals at a frothy acquirer price inflate headline TEV; note consideration mix.

7. Summarise and apply. Take **min/25th/median/75th/max** of each transaction multiple; **lead with the median**. Apply to the target's *current* LTM metric:

- Implied TEV = median deal EV/EBITDA × target LTM EBITDA
- Implied equity = Implied TEV − target net debt (+ net cash)
- Implied price = Implied equity ÷ diluted shares

Worked example — a mid-cap Indian IT-services target, illustrative deals:

Announced	Target	Acquirer	TEV (₹ cr)	LTM EBITDA (₹ cr)	EV/EBITDA	Premium
2022	Mid-cap IT co A	Global SI	12,000	750	16.0x	32%
2023	Digital eng. co B	PE sponsor	6,500	480	13.5x	26%
2021	ER&D services C	Strategic	9,000	500	18.0x	40%
2024	BPM/IT co D	Global SI	7,200	540	13.3x	24%
Median					**14.75x**	**29%**

Apply the median 14.75x to a target with LTM EBITDA ≈ ₹900 cr and net cash ≈ ₹500 cr, ~50 cr diluted shares: - Implied TEV = 14.75 × 900 ≈ **₹13,275 cr** - Implied equity = 13,275 + 500 ≈ **₹13,775 cr** - Implied price = 13,775 cr ÷ 50 cr ≈ **₹276/sh**

Note this sits above where trading comps (say median 11–12x) would land — that gap is roughly the control premium plus synergy value.

The output

A one-page deal-comps sheet: one row per transaction (date, target, acquirer, stake, consideration, TEV, target LTM metric, each multiple, premium); a stats block leading with the median; a "vintage" column so recency is visible; and an implied-value block (TEV → equity → per-share) applied to your target's current metrics. Footnotes flag strategic-vs-sponsor and any all-stock deals. Usually shown as the *top* bar of the football field, above trading comps.

Checks & gotchas

- **Point-in-time financials.** Use the target's EBITDA *as of the deal announcement*, not its latest — the multiple is a ratio of the price paid to what the buyer bought then. Mixing today's EBITDA with an old price is the classic error.
- **Stale deals.** Pre-2021 multiples reflect near-zero rates; don't average a 2019 deal against 2025 without weighting for regime.

- **Strategic vs. financial buyer.** Strategic multiples carry synergies; sponsor deals don't. Blending them without a flag over-states standalone value.
- **Minority vs. control.** A 15% stake purchase has no control premium — exclude it or you'll understate.
- **Consideration mix.** All-stock deals can inflate TEV if the acquirer's stock was rich at signing.
- **Thin data.** M&A in a niche can give you only 3–4 usable deals — say so; a precedent set of two is a range, not a valuation.
- **Deal value ≠ equity value.** TEV includes assumed net debt; don't quote it as the equity cheque.

Interview drill

Q: "Why do precedent transactions usually give a higher value than trading comps?" A: "Because deal prices include a control premium — buyers pay above the undisturbed market price to acquire control — and often synergies a strategic buyer expects to realise. Trading comps value the company as-is in the public market; precedents value it as an acquisition. The gap is roughly premium plus synergy."

Q: "Which financials do you use for the target in a precedent multiple — current or at the time of the deal?" A: "At the time of the deal. The multiple is price-paid over what the buyer was buying, so I use the target's LTM figures as of the announcement date. Using today's financials against an old price gives a meaningless ratio."

Q: "When are precedent transactions less reliable than comps?" A: "When deals are stale — struck in a very different rate or valuation regime — when the sample is tiny, or when the set mixes strategic and sponsor buyers with very different synergy assumptions. Deal data is also lumpy and lags; trading comps update daily. I'd caveat the range accordingly."

Practise free

No paid deal database? Reconstruct precedents from public sources: - **BSE/NSE announcements & SEBI SAST filings** — open-offer documents disclose offer price, stake and undisturbed price → compute the control premium directly. - **Company press releases / investor decks** — acquirers publish deal value and target revenue/EBITDA; capture TEV and the metric. - **VCCEdge / Tracxn (free tiers), news archives (Mint, ET, Business Standard, Reuters)** — India M&A deal terms. - **Mergermarket / CapIQ Transactions** if you have campus/library access. Drill: pick five real Indian IT/BPO acquisitions from the last four years, dig the TEV and target LTM EBITDA out of the press releases, build the exact table above, take the median, and imply a value for a listed mid-cap target — then compare it to where trading comps put the same company and explain the gap.

Building a Full DCF from Terminal Data

What you'll be able to do

Pull a company's historicals off a terminal (Bloomberg/CapIQ or a free proxy), forecast unlevered free cash flow (FCFF) for five years, build a WACC bottom-up using CAPM, layer in a terminal value two ways (Gordon growth and exit multiple), discount everything, walk enterprise value down to a per-share equity value, and pressure-test it with a 2-way sensitivity grid and a football-field chart. You will carry one real-ish company — call it **Bharat Consumer Ltd (BCL)** — all the way to a rupee-precise fair value, and you'll know how to spot when your own model is quietly cheating.

The drill — step by step

Step 1 — Pull the raw financials. On Bloomberg, `BCL IN Equity <GO>`, then `FA <GO>` (Financial Analysis). Tab to Income Statement and Cash Flow, set periodicity Annual, and export to Excel with the red `Export` button (or `=BDH("BCL IN Equity","SALES_REV_TURN","2021","2025")` in the Excel add-in). On CapIQ it's the "Financials" template. Free proxy: Screener.in export, or Tijori/annual report PDFs. Grab five years so you have growth rates and margins to anchor forecasts.

BCL FY25 actuals (₹ cr): Revenue 12,000, EBITDA 2,160 (18.0%), D&A 360, EBIT 1,800, tax rate 25.17% (India new regime + surcharge/cess), Capex 420, and change in working capital that consumed ₹150 cr as sales grew.

Step 2 — Forecast the revenue driver. Don't forecast 15 lines; forecast the driver and let margins do the rest. BCL grew revenue ~11% historically; I taper it: 11% → 10% → 9% → 8% → 7%. Hold EBITDA margin flat at 18% (state this assumption explicitly — margin expansion is where models lie). D&A ≈ 3% of sales, capex ≈ 3.5% of sales, working capital drag ≈ 1.25% of incremental sales.

Step 3 — Build FCFF, year by year. The formula, memorise it:

```
FCFF = EBIT × (1 − tax) + D&A − Capex − ΔNWC
```

Year 1 (FY26): Revenue = 12,000 × 1.11 = 13,320. EBITDA = 2,397.6. D&A = 399.6. EBIT = 1,998. NOPAT = 1,998 × (1−0.2517) = 1,495.1. + D&A 399.6 − Capex (3.5%×13,320 = 466.2) − ΔNWC (1.25% × 1,320 = 16.5) = **FCFF ₹1,412.0 cr**.

Roll the same engine forward:

₹ cr	FY26	FY27	FY28	FY29	FY30
Revenue	13,320	14,652	15,971	17,249	18,456
EBIT	1,998.0	2,197.8	2,395.7	2,587.3	2,768.4
NOPAT	1,495.1	1,644.6	1,792.7	1,936.1	2,071.6
+ D&A	399.6	439.6	479.1	517.5	553.7
− Capex	466.2	512.8	559.0	603.7	646.0
− ΔNWC	16.5	16.7	16.5	16.0	15.1
FCFF	**1,412.0**	**1,554.7**	**1,696.3**	**1,833.9**	**1,964.2**

Step 4 — Build WACC from CAPM. Risk-free: `GIND10YR Index <GO>` → 10-yr G-sec ≈ **6.9%**. Beta: on Bloomberg `BCL IN Equity BETA <GO>` gives raw and adjusted beta; use adjusted (Blume) ≈ **0.85** (or lever an industry beta). Equity risk premium for India: Damodaran's mature-market + country risk ≈ **7.5–8.0%**; use 7.5%.

```
Cost of equity = Rf + β × ERP = 6.9% + 0.85 × 7.5% = 13.28%
```

Cost of debt: BCL's coupon / rating spread ≈ 8.5% pre-tax, after-tax = 8.5% × (1−0.2517) = 6.36%. Capital weights at market: equity ₹28,000 cr, debt ₹4,000 cr → E/V = 87.5%, D/V = 12.5%.

```
WACC = 0.875 × 13.28% + 0.125 × 6.36% = 11.62% + 0.795% = 12.42%
```

Round to **12.4%**.

Step 5 — Terminal value, two ways. Gordon: pick g = 5.0% (below nominal GDP ~10.5%, sanity intact).

```
TV = FCFF_FY30 × (1+g) / (WACC − g) = 1,964.2 × 1.05 / (0.124 − 0.05)
   = 2,062.4 / 0.074 = ₹27,870 cr
```

Exit multiple cross-check: apply 12× EV/EBITDA to FY30 EBITDA (18% × 18,456 = 3,322): TV = ₹39,864 cr. The two disagree — that's the point; the Gordon TV implies a cheaper business. I'll run the DCF on Gordon and show exit as a bound.

Step 6 — Discount. Discount factor = 1/(1.124)^t (mid-year convention optional; here year-end).

	FY26	FY27	FY28	FY29	FY30	TV
CF	1,412.0	1,554.7	1,696.3	1,833.9	1,964.2	27,870
DF @12.4%	0.8897	0.7916	0.7043	0.6266	0.5575	0.5575
PV	1,256.3	1,230.7	1,194.7	1,149.1	1,095.1	15,538.5

Sum of PV(explicit FCFF) = ₹5,925.9 cr. PV(TV) = ₹15,538.5 cr. **Enterprise Value = ₹21,464 cr.**

Step 7 — EV-to-equity bridge.

```
Equity value = EV − Net debt − Minority interest − Preferred + Investments/associates
```

BCL: EV 21,464 − Net debt (debt 4,000 − cash 900 = 3,100) − minority 200 + associate investments 300 = **₹18,464 cr**. Shares outstanding = 100 cr. **Fair value = ₹184.6 / share.** Current price ₹280 → the DCF says overvalued ~34%; you'd flag that, not force the assumptions to fit.

The output

A one-screen model block: the FCFF build (table above), a WACC box (Rf 6.9%, β 0.85, ERP 7.5%, Ke 13.28%, Kd(at) 6.36%, weights 87.5/12.5, WACC 12.4%), the discounting strip, and the bridge:

```
EV                       21,464
 − Net debt              (3,100)
 − Minority                (200)
 + Associates               300
= Equity value           18,464
÷ Shares (cr)               100
= Value / share        ₹ 184.6
```

TV as % of EV = 15,538.5 / 21,464 = **72.4%**. Implied exit EV/EBITDA on the Gordon TV = 27,870 / 3,322 = **8.4×** — internally consistent and *below* the 12× comps multiple, so the DCF is conservative.

Checks & gotchas

- **TV should be 60–80% of EV.** 72% is normal. Above ~85% means your explicit window is too short or g too high.
- **Implied terminal growth vs implied exit multiple** must both be sane. Back out each from the other and eyeball them.
- **g < WACC always**, and g ≤ long-run nominal GDP. g of 8% in a 12.4% WACC world quietly doubles your TV.
- **Consistency of nominal terms**: nominal cash flows → nominal WACC (India, don't mix a real Rf with nominal growth).
- **Tax rate**: use the marginal/statutory rate for NOPAT, not the accounting effective rate distorted by one-offs.
- **Don't double-count**: FCFF is pre-financing, so discount at WACC and *never* subtract interest in the cash flow. Subtract debt only in the bridge.
- **Mid-year convention** lifts value ~6% (half a period less discounting); pick one and disclose it.
- **Circularity**: market-value weights depend on the equity value you're solving for. Iterate once or fix weights at target capital structure.

Interview drill

Q: Why FCFF discounted at WACC and not FCFE at cost of equity? FCFF is capital-structure-neutral — it's the cash to all providers, so you discount at the blended cost of all capital (WACC) and get enterprise value.

FCFE is post-debt cash to equity, discounted at Ke, giving equity value directly. FCFF is preferred when leverage is changing because WACC is more stable than a moving Ke; you avoid re-forecasting the debt schedule inside the cash flow.

Q: Your terminal value is 72% of EV — is the model useless? No, that's structurally normal because a going concern's value is dominated by cash flows beyond any five-year window. What matters is that the terminal assumptions are defensible: g of 5% is below nominal GDP, and the implied exit multiple of 8.4× is below trading comps of 12×, so I'm not smuggling in optimism through the terminal — if anything I'm conservative.

Q: Move ERP from 7.5% to 8.5% — direction and rough magnitude? Ke rises ~0.85%, WACC rises ~0.74%, so the denominator (WACC − g) widens and value falls sharply because TV is convex in the spread. A move from 12.4% to 13.1% WACC roughly cuts fair value from ₹185 toward ₹160 — a ~13% hit from a 1-point ERP change, which is exactly why the sensitivity table matters.

Practise free

You don't need a live terminal. Pull five years of financials from **Screener.in** (free export to Excel) or the company's annual report; use **GIND10YR** substitute from RBI/worldgovernmentbonds.com for the risk-free rate; get **beta** by regressing 2 years of weekly stock returns against Nifty in Excel (`=SLOPE(stock_returns, index_returns)`) — that literally *is* what the terminal's BETA function does. Take **ERP** from Damodaran's free country-risk-premium spreadsheet (updated each January). Build the whole FCFF → WACC → TV → bridge in one Excel tab, then add a two-input `Data Table` (Data ▸ What-If ▸ Data Table) with WACC across the top and g down the side to reproduce the sensitivity grid and football field for free.

Building an LBO Model (Sources/Uses, Debt Schedule, Returns)

What you'll be able to do

Take a target, set an entry enterprise value off an EBITDA multiple, build a Sources & Uses table that funds the deal, layer a realistic debt stack (Term Loan A/B + a revolver + optionally mezzanine), forecast EBITDA and free cash flow, run a cash sweep that pays debt down year by year, exit at a multiple, and compute the two numbers a PE interviewer actually cares about — **IRR** and **MOIC** — then stress them on leverage and exit. You'll carry one target, **Deccan Components Ltd (DCL)**, from a ₹1,000 cr entry to a five-year return, to the rupee.

The drill — step by step

Step 1 — Set the entry. DCL LTM EBITDA = ₹150 cr. Entry multiple = 8.0× → **Entry EV = ₹1,200 cr**. Assume no existing debt acquired (cash-free, debt-free deal), so equity purchase price = EV = 1,200. This is the "purchase enterprise value" line.

Step 2 — Decide leverage. Sponsors lever to what cash flow can service. Total debt = 4.5× EBITDA = ₹675 cr, split:

Tranche	× EBITDA	₹ cr	Rate	Amort
Term Loan A	2.0×	300	9.5%	10%/yr mandatory
Term Loan B	2.0×	300	10.5%	1%/yr, bullet
Revolver	—	0 drawn	9.0%	as needed
Total debt	**4.5×**	**675**		

Step 3 — Sources & Uses. Uses = what you must pay for. Sources = how you fund it. They must balance; the sponsor equity is the plug.

```
USES                          SOURCES
Purchase EV         1,200     Term Loan A          300
Transaction fees       30     Term Loan B          300
Financing fees         15     Revolver               0
                             Sponsor equity (plug) 645
TOTAL USES          1,245     TOTAL SOURCES      1,245
```

Sponsor equity = 1,245 − 675 = **₹570… wait, plug to balance: 1,245 − 600 (TLA+TLB) = ₹645 cr.** (Note the ₹45 cr of fees are funded by equity — that's why equity is 645, not 570.) Equity as % of total capitalisation = 645/1,245 = 51.8%.

Step 4 — Project EBITDA and FCF. DCL grows revenue 8% with 25% EBITDA margin holding. Build the cash available for debt paydown:

```
Cash for sweep = EBITDA − cash interest − cash taxes − capex − ΔNWC − mandatory amort
```

Year 1: EBITDA 150 × 1.08 = 162. Interest: TLA 9.5%×300 = 28.5, TLB 10.5%×300 = 31.5 → 60.0. D&A 30, EBIT 132, less interest 60 = PBT 72, tax @25% = 18, so cash taxes 18. Capex = 6% of sales (sales ≈ 648) = 38.9. ΔNWC = 8. Mandatory amort TLA 10%×300 = 30, TLB 1%×300 = 3.

FCF before sweep = 162 − 60 − 18 − 38.9 − 8 = 37.1. Less mandatory amort 33 = **₹4.1 cr free for the cash sweep** in Year 1 (early years are tight — that's realistic).

Step 5 — Debt schedule with cash sweep. Each year: opening balance − mandatory amort − optional sweep = closing balance. Sweep hits the most expensive prepayable tranche first (usually TLB after TLA mandatory). Interest is computed on the average or opening balance (be consistent). Roll five years:

₹ cr	Y1	Y2	Y3	Y4	Y5
EBITDA	162.0	175.0	189.0	204.1	220.5
Cash interest	60.0	56.5	52.2	47.0	40.8
Cash taxes	18.0	21.4	25.1	29.2	33.7
Capex	38.9	42.0	45.4	49.0	52.9
ΔNWC	8.0	8.6	9.3	10.0	10.8
FCF pre-amort	37.1	46.5	57.0	68.9	82.3
Total debt paid	37.1	46.5	57.0	68.9	82.3
Debt closing	**637.9**	**591.4**	**534.4**	**465.5**	**383.2**

By Year 5, ₹675 cr of debt is down to **₹383.2 cr** — the deleveraging engine that drives PE returns.

Step 6 — Exit and returns. Exit at Year 5, EBITDA = ₹220.5 cr. Exit multiple — conservatively assume *multiple contraction* to 7.5× (you never underwrite to expansion):

```
Exit EV = 7.5 × 220.5 = 1,653.8
− Net debt at exit      (383.2)  [assume minimal cash]
= Exit equity value    1,270.6
```

MOIC = exit equity / entry equity = 1,270.6 / 645 = **1.97×**. IRR = MOIC^(1/5) − 1 = 1.97^0.2 − 1 = **14.5%**. Below the ~20% PE hurdle — so at 8.0× entry / 4.5× leverage / 7.5× exit, this deal is marginal. That's a *finding*, and knowing it beats a model that always says "great deal."

The output

The finished LBO is four linked blocks on one screen: Sources & Uses (balances at 1,245), the debt schedule (above), the FCF build, and a returns box:

```
Entry equity            645.0
Exit EV (7.5×)        1,653.8
Exit net debt          (383.2)
Exit equity          1,270.6
MOIC                    1.97×
IRR (5-yr)              14.5%
```

Return attribution (the "bridge" PE loves): of the ₹625.6 cr equity gain, **deleveraging** contributed 675 − 383 = ₹291.8 cr, **EBITDA growth** at constant multiple contributed (220.5−150)×7.5 = ₹528.8 cr, and **multiple change** contributed (7.5−8.0)×150 = −₹75 cr. That decomposition is the answer to "where do your returns come from?"

Checks & gotchas

- **Sources must equal Uses** to the rupee, every time, including fees. Fees are a Use funded by equity — forgetting them overstates returns.
- **The revolver is the backstop**: if FCF goes negative, you *draw* the revolver, you don't create negative debt paydown. Model a minimum cash balance.
- **Interest is circular** (interest depends on debt, debt depends on sweep, sweep depends on interest). Enable iterative calc or use a beginning-balance interest convention to break it cleanly.
- **Never underwrite multiple expansion.** Assume flat or contracting exit; if the deal only works on expansion, it doesn't work.
- **Cash taxes ≠ book taxes** once you have amortising fees and interest shields — keep a separate tax build.
- **MOIC and IRR must be consistent**: IRR ≈ MOIC^(1/years) − 1 only with a single entry and single exit and no interim dividends; a dividend recap changes both.
- **Sanity**: entry leverage 4.5× with EBITDA margin 25% and rising rates — check the interest coverage (EBITDA/interest) stays above ~2.5× or the deal is un-financeable.

Interview drill

Q: What are the three drivers of LBO returns? Deleveraging (FCF pays down debt, so equity grows even at a flat multiple), EBITDA growth (revenue growth plus margin expansion), and multiple arbitrage (buy low, sell high — but you never *underwrite* to this; you assume flat or lower exit). In my DCL model, deleveraging added ₹292 cr and EBITDA growth ₹529 cr, while the multiple contraction *cost* ₹75 cr — showing returns can survive a worse exit multiple if the operating story delivers.

Q: Why does more leverage boost IRR but raise risk? Higher debt shrinks the equity cheque, so the same absolute equity gain is a larger percentage return — that's the IRR lift. But it also raises fixed cash interest, thinning the FCF cushion; a small EBITDA miss can breach covenants or force a revolver draw, and equity is first-loss. Leverage amplifies both the upside and the probability of a zero.

Q: Walk me from entry EV to sponsor equity. Entry EV = entry multiple × LTM EBITDA. Add transaction and financing fees to get total uses. Fund it with the debt tranches you can raise given leverage capacity; sponsor equity is the plug that makes sources equal uses. So equity = total uses − total debt drawn, and it explicitly absorbs the fees.

Practise free

No LBO software needed — it's all Excel. Pull a mid-cap's EBITDA from **Screener.in**, pick an entry multiple from **comparable transactions** (news of recent PE deals, or trading comps as a floor). Build Sources & Uses, then a five-row debt schedule with a `=MIN(cash_available, debt_outstanding)` sweep and iterative calculation turned on (File ▸ Options ▸ Formulas ▸ Enable iterative). Compute IRR with `=RATE` or `=(exit/entry)^(1/n)-1`, and MOIC as a simple ratio. Add a `Data Table` sensitising IRR against entry leverage (rows) and exit multiple (columns) to reproduce the classic PE sensitivity grid. Rehearse the return-attribution bridge by hand — that verbal decomposition is what actually gets tested in PE interviews.

Writing the Equity Research Note / Initiation

What you'll be able to do

Turn a finished model into a sell-side **initiation report** that a desk would actually publish: a crisp one-line thesis, a rating and price target that *fall out of the model* (not the other way round), a business and industry section grounded in channel checks, an explicit forecast block, a blended DCF-plus-comps valuation, a risk register, catalysts, and a one-page summary — written in SEBI-compliant, non-promotional prose. You'll produce a short worked excerpt for **Bharat Consumer Ltd (BCL)** and know exactly what an Equirus/Nuvama/street-side analyst JD is asking for.

The drill — step by step

Step 1 — Write the thesis in one sentence, first. If you can't state why the stock is mispriced in a line, you don't have a note. BCL example:

> *"We initiate on BCL with a REDUCE and a ₹185 target (34% downside): the street is extrapolating 15% volume growth into a category that is structurally slowing to high-single-digits, and current multiples price in margin expansion the company has repeatedly guided against."*

Note the structure: rating, target, downside %, and the *variant perception* — what you see that the consensus doesn't.

Step 2 — Set rating and target from the model, transparently. The target must be reproducible. Blend your DCF (from chapter 5, ₹185) with a comps cross-check:

Method	Value/share	Weight
DCF (WACC 12.4%, g 5%)	₹185	60%
EV/EBITDA comps (11× FY27E)	₹196	25%
P/E comps (28× FY27E EPS)	₹178	15%
Blended target	**₹187**	100%

Round to **₹185**. Map to a rating band the desk uses (e.g. BUY >15% upside, ADD 5–15%, REDUCE −5 to −15%, SELL <−15%). At ₹280 vs ₹185, that's −34% → **REDUCE** (or SELL, per house bands).

Step 3 — Business & industry, grounded in channel checks. Don't rewrite the annual report. Explain the *unit economics* and the *industry structure*, then evidence it. Channel checks = the primary work: call 8–10 distributors, visit modern-trade shelves, check e-commerce pricing, talk to two ex-employees, read the last four earnings-call transcripts for a change in tone. Write it as evidence: "*Our checks across 12 distributors in 3 states indicate secondary sales decelerating to ~7% YoY, versus management's 12–14% primary-sales run-rate — implying channel inventory build.*" That specificity is the moat of a good note.

Step 4 — Forecasts, shown explicitly. Analysts publish their numbers so clients can disagree with a specific line. Present three years:

₹ cr	FY25A	FY26E	FY27E	FY28E
Revenue	12,000	13,320	14,652	15,971
growth %	—	11.0	10.0	9.0
EBITDA	2,160	2,398	2,637	2,875
margin %	18.0	18.0	18.0	18.0
PAT	1,290	1,433	1,578	1,720
EPS (₹)	12.9	14.3	15.8	17.2

State the two or three assumptions that matter (flat margin, tapering volume) and where you differ from consensus ("*our FY27 EPS is 6% below Bloomberg consensus of ₹16.8*").

Step 5 — Valuation section. Show the DCF assumptions box, the sensitivity grid (WACC × g), and the comps table so the target is auditable. Include a **football field** — DCF range, EV/EBITDA range, P/E range, 52-week range — with the current price line, so the reader sees where ₹280 sits versus every method.

Step 6 — Risks and catalysts. Risks are two-sided and specific: *upside risk* — faster rural recovery lifts volumes to 12%, taking the DCF to ₹215; *downside* — raw-material inflation compresses margin 150bps to ₹160. Catalysts with dates: Q1 results (Jul), monsoon data (Aug), a rumoured price hike. A note without dated catalysts is an opinion, not a trade.

Step 7 — One-pager and compliance. The front page is the whole note in one screen: rating, target, price, up/downside, market cap, the thesis in three bullets, a mini financial summary table, and the football field. Then the **SEBI RA disclosures**: registration number, analyst certification ("*views accurately reflect my personal views; no part of compensation was tied to the recommendation*"), holdings disclosure, and no forward-looking guarantee language. Prose must be **non-promotional**: no "must buy", no "guaranteed", no target framed as certainty — SEBI (Research Analysts) Regulations require balanced, evidence-based language.

The output

A worked excerpt of the initiation:

BCL IN | REDUCE | TP ₹185 (−34%) | CMP ₹280 | Mkt cap ₹28,000 cr

Thesis. We initiate coverage of Bharat Consumer with a REDUCE rating and a ₹185 12-month target. The market is pricing sustained mid-teens volume growth into a category our channel work shows decelerating to ~7%. At 28× FY27E EPS, the stock discounts margin expansion management has explicitly guided against.

Why now. (1) Secondary-sales checks across 12 distributors point to channel inventory build; (2) consensus FY27 EPS of ₹16.8 sits 6% above our ₹15.8; (3) the risk-reward is skewed — our bull case is ₹215 (+8% from a rural recovery) against a bear case of ₹160 (−15% on margin compression).

Valuation. DCF (WACC 12.4%, terminal g 5.0%) yields ₹185; an 11× FY27E EV/EBITDA cross-check gives ₹196. We weight the DCF 60% given the long-duration cash-flow profile. Football field below shows CMP above every method's central estimate.

Key risks to our call. Faster-than-expected rural volume recovery; benign input costs; a value-accretive acquisition.

RA Reg. No. INHxxxxxxxxx. Analyst certification and disclosures on p.12.

Checks & gotchas

- **The target must tie to the model.** If a reader rebuilds your DCF and can't get near ₹185, the note has no credibility. Keep the assumptions box in the report.
- **Rating consistency**: a REDUCE with a target 34% below price is right; a BUY with a target *below* the current price is an instant credibility kill — check the arithmetic.
- **Two-sided risks.** A note that only lists downside for a SELL (or only upside for a BUY) reads as a pitch, not research. SEBI-compliant means balanced.
- **No promissory language.** "Will", "guaranteed", "multibagger" are red flags for both compliance and quality.
- **Consensus context**: always state where you sit vs consensus — that's your variant perception, the reason the note exists.
- **Dated catalysts.** "Catalyst: better performance" is empty. Name the event and the month.
- **Disclosures are mandatory**, not optional garnish — RA registration, certification, holdings, and conflicts.

Interview drill

Q: What makes a research note different from a valuation model? The model produces a number; the note produces a *decision with a variant perception*. Its job is to identify where the market is wrong and why, evidence it with primary work like channel checks, and frame a risk-reward with catalysts — the target price is an output of that argument, not the argument itself. A great note could be right on the call even if the exact target is off by 5%.

Q: How do you blend DCF and comps into one target? They answer different questions — DCF is intrinsic and long-duration-sensitive, comps are relative and reflect current market sentiment. I weight toward DCF for stable cash-generative businesses and toward comps for cyclical or sentiment-driven names, then present both plus a football field so the client sees the range, not a false-precision single number. For BCL I used 60% DCF given its long, predictable cash flows.

Q: How do SEBI RA norms shape how you write? They require the recommendation to reflect genuine analysis, compensation not to be tied to the specific call, and full disclosure of holdings and conflicts, plus registration and analyst certification. Practically it forces balanced, evidence-based, non-promotional prose — two-sided risks, no guarantees — which, conveniently, is also what makes research credible to a professional reader.

Practise free

You don't need a Bloomberg terminal or a research license. Pick a listed company, build the model from **Screener.in** data (chapter 5's method), and do real **channel checks** for free — visit shops, check Amazon/Blinkit pricing, read the last four **earnings-call transcripts** (free on the company IR site or Screener's "Documents"), and note where management tone shifts. Read three or four **published initiations** (many are free on brokerage sites, Trendlyne, or research aggregators) purely to internalise the *structure and register* — thesis-first, evidence-dense, non-promotional. Then write a 2-page note with a one-pager and a football field (`Data Table` + a stacked bar in Excel). Study the SEBI (Research Analysts) Regulations disclosure list — it's public — and put a compliant disclosure block on your practice note so the habit is automatic.

The Modeling Test & Terminal Interview Drills + Cheat-Sheet

What you'll be able to do

Walk into a timed modelling test (the 60–90 minute build-a-3-statement-and-DCF-from-a-blank-sheet exercise) and know exactly what the reviewer is watching, how to structure for speed, which keyboard shortcuts save you ten minutes, and how to keep the model auditable. You'll also handle the rapid-fire terminal function quiz and the standing valuation Q&A, and you'll have a one-page cheat-sheet of every formula and function you need under time pressure.

The drill — step by step

Step 1 — Before you type, set the frame (2 min). Read the prompt, note the deliverable (usually: 3-statement + DCF, one or two scenarios, an answer to "is it cheap?"). Lay out tabs: `Assumptions`, `Model` (IS/BS/CF stacked), `DCF`, `Output`. Colour convention immediately — **blue = hardcoded input, black = formula, green = link to another sheet**. Reviewers grade this in the first 30 seconds.

Step 2 — Build the income statement top-down (10 min). One driver: revenue growth. Everything else as a % of revenue or a margin. Wire tax off PBT. Don't hardcode any calculated cell — if a number can be computed, compute it.

Step 3 — Balance sheet and the cash-flow bridge (20 min). Build working-capital via **days**: DSO, DIO, DPO drive receivables, inventory, payables. PP&E roll-forward: opening + capex − depreciation. Debt and equity from a simple schedule. Then the **cash flow statement links everything**: CFO from net income + D&A ± ΔNWC, CFI = −capex, CFF = debt/equity flows. Ending cash flows back to the balance sheet.

Step 4 — Make it balance (5 min). The whole test is really "does your balance sheet balance?" Build a check row: `Assets − Liabilities − Equity = 0` in every column, conditionally formatted red if non-zero. If it doesn't tie, 90% of the time it's the cash flow statement not fully feeding ending cash, or a working-capital sign error.

Step 5 — DCF on top (15 min). FCFF = EBIT×(1−t) + D&A − capex − ΔNWC (chapter 5). WACC box, terminal value (Gordon), discount, EV-to-equity bridge, per-share. Add a `Data Table` sensitivity (WACC × g).

Step 6 — Checks and a one-line answer (5 min). Turn on every check row. Then, in a cell or out loud, state the conclusion: "*Fair value ₹185 vs price ₹280 — overvalued, DCF-implied exit multiple 8.4× below comps 12×.*" The reviewer wants a *view*, not just a number.

What they're actually watching

Dimension	Pass	Fail
Structure	Inputs separated, colour-coded, one driver	Hardcodes buried in formulas
Formulas	No plugs, everything links	Balancing figure typed in by hand
Checks	Balance check + cash tie, visible	No checks, silent errors
Speed	Shortcuts, no mouse for navigation	Formatting the sheet for 20 min
Judgement	States a view with a sanity check	Delivers a number with no context

Speed shortcuts (no-mouse modelling)

- `F2` edit cell, `F4` toggle absolute refs (`$A$1`), `Ctrl+;` today's date.
- `Alt+=` autosum. `Ctrl+Shift+arrow` select to edge. `Ctrl+D` / `Ctrl+R` fill down/right.
- `Alt,H,O,I` autofit column. `Alt,A,W,T` open Data Table. `Ctrl+[` trace precedent.
- `F9` recalc; `Ctrl+`` (grave) show all formulas for a fast audit.
- Name your WACC and tax cells so formulas read `=EBIT*(1-tax)` not `=D12*(1-$B$4)`.

Terminal / CapIQ function quiz (rapid fire)

Task	Bloomberg	CapIQ / free proxy
Price history to Excel	`=BDH( … )`	`=CIQ( … )` / Screener export
Current field (P/E, mkt cap)	`=BDP("TICK","PE_RATIO")`	`=CIQ("IQ_PE_EXCL")`
Company financials screen	`FA <GO>`	Financials template
Beta	`BETA <GO>`	`SLOPE()` regression (free)
Comparable companies	`RV <GO>`	Comps template
Relative valuation / football	`EQRV <GO>`	build manually
Debt / capital structure	`DDIS <GO>`	annual report
Estimates / consensus	`EEO <GO>` / `EE <GO>`	Trendlyne
Supply chain	`SPLC <GO>`	annual report
News run	`CN <GO>`	Google News

The output

A finished test artefact is: a colour-coded 3-statement model that balances in every column (visible check row of zeros), a DCF tab with a WACC box and a WACC×g sensitivity grid, and a one-line verdict. It looks disciplined — inputs in blue on their own tab, no orphaned hardcodes, checks green. And the one-pager below, which you can reconstruct from memory in the first two minutes of any test as your scaffold.

The one-page cheat-sheet

Free cash flow

```
FCFF = EBIT × (1 − t) + D&A − Capex − ΔNWC     → discount at WACC → EV
FCFE = FCFF − Interest×(1−t) + Net borrowing    → discount at Ke  → Equity
```

WACC & CAPM

```
Ke   = Rf + β × ERP
Kd   = pre-tax cost of debt × (1 − t)
WACC = (E/V)×Ke + (D/V)×Kd
```

Terminal value

```
Gordon:  TV = FCFF_n × (1+g) / (WACC − g)      [g < WACC, g ≤ nominal GDP]
Exit:    TV = EBITDA_n × exit multiple
Implied exit multiple = Gordon TV / EBITDA_n   (cross-check)
```

Bridge

```
Equity = EV − Net debt − Minority − Preferred + Associates
Per share = Equity / diluted shares
```

Multiples

```
EV/EBITDA, EV/EBIT, EV/Sales   (capital-structure neutral → use EV)
P/E, P/B, P/FCFE               (equity multiples → use price)
PEG = P/E ÷ growth
```

LBO

```
Sources = Uses (equity is the plug, fees are a Use)
MOIC = exit equity / entry equity
IRR  ≈ MOIC^(1/years) − 1      (single entry/exit, no interim CF)
Returns = deleveraging + EBITDA growth + multiple change
```

Best practice: blue input / black formula / green link · one driver, margins do the rest · no hardcoded plugs · balance-check row = 0 · TV = 60–80% of EV · nominal CF with nominal WACC · state a view.

Checks & gotchas

- **Balance-sheet imbalance** almost always = cash flow statement not fully wired or a working-capital sign flip. Fix the CF, don't plug the BS.

- **Circularity** (interest ↔ debt ↔ cash) — enable iterative calc *before* it errors, or use opening-balance interest.
- **Hardcoded plugs** are the single fastest way to fail; the reviewer traces one formula and finds it.
- **Formatting over substance** — don't spend 20 minutes on borders and a blank model. Function first, polish last.
- **No view** — a technically perfect model with no "cheap or dear?" conclusion still disappoints. Always land the plane.
- **Mixing real and nominal**, or effective vs marginal tax — silent value errors the reviewer will probe.

Interview drill

Q: Your model doesn't balance — walk me through debugging it. First, isolate the column where the check first goes non-zero — the error started that year. Then check the cash flow statement: does ending cash on the CF equal the cash line on the BS? Usually not, meaning a flow is missing or double-counted — most often a working-capital sign (an increase in receivables is a *use* of cash, negative in CFO) or capex not linked to both PP&E and CFI. I fix the source, never plug the balance sheet, because a plug hides the real error.

Q: You have 60 minutes for a 3-statement plus DCF — how do you spend it? Two minutes framing tabs and colour convention, ten on a driver-based income statement, twenty on the balance sheet and the cash-flow bridge, five making it balance with a visible check row, fifteen on the DCF and a sensitivity table, and the last five turning on checks and writing the one-line conclusion. I build ugly-but-linked first and only format if time remains — a balancing model with no borders beats a beautiful model that plugs.

Q: EV/EBITDA vs P/E — when each? EV multiples are capital-structure-neutral, so I use EV/EBITDA to compare companies with different leverage or to value the whole enterprise — it's above the interest line. P/E is an equity multiple, sensitive to leverage and tax, useful for comparing similar-capital-structure peers or when the market quotes the sector on earnings. For a leveraged or acquisitive name I lead with EV/EBITDA; for a stable financial or a consumer name the market prices on P/E, so I show both.

Practise free

Do the whole thing in plain Excel — no terminal required. Grab a company's five years from **Screener.in**, then rebuild the 3-statement model *from a blank sheet* against a timer (start at 90 minutes, work down to 60). Enforce the colour convention and a balance-check row every time. Replace terminal functions with their free equivalents: beta via `=SLOPE()` regression, risk-free from the RBI 10-year, ERP from Damodaran's free January dataset, comps built by hand from Screener. Rehearse the function quiz as flashcards (BDP/BDH, FA, RV, EQRV, BETA, EEO). Run a `Data Table` for the sensitivity grid. Time-box mercilessly and, at the buzzer, force yourself to write the one-line verdict — the muscle you're building is *finishing with a view under time pressure*, which is exactly what the real test grades.

SAP, Blackline & the R2R Close — Hands-On

Hands-On Drills — Do the Task, Not Just Define It (India, 2026)

Step-by-step functions, T-codes, worked models and interview drills

Contents

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7. **The GCC R2R Operating Rhythm: WD Calendar, SLAs, DTP/SOP, RPA**
8. **R2R Interview Drills + T-Code & Close Cheat-Sheet**

SAP FICO Orientation & Navigation

What you'll be able to do

After this chapter you can sit at a fresh SAP session, log into a client, drive the **SAP GUI** and **Fiori launchpad** by memory, jump to any transaction using the **command field**, and read the org structure of a company the way a GCC controller reads it: Company Code → Chart of Accounts → Controlling Area → Cost/Profit Centers. You'll know which numbers live in **FI** and which live in **CO**, and you'll be able to open a GL master (FS00), a vendor/customer master, and a cost-centre master (KS01/KS03) and explain every tab. You'll also be able to tell an interviewer the real differences between **S/4HANA** and **ECC** without bluffing.

The drill — step by step

1. Log in. SAP Logon pad → double-click the system (e.g. `PRD`, `QAS`, `DEV`) → enter **Client** (3 digits, e.g. 800), **User**, **Password**, **Language** EN. A client is a self-contained data world; a GCC often has one client for the whole group and many company codes inside it.

2. Learn the command field. Top-left white box. Type transaction codes here. - `FS00` → go to that T-code in the current window. - `/n` + code (`/nFS00`) → close current, open new in same window. - `/o` + code (`/oFB03`) → open in a **new** session (window) — keep a display session open while you post. - `/nend` → log off; `/nex` → log off without the "do you want to end" popup. - `/i` → close current session. Type `SESSION_MANAGER` or just navigate the **SAP Easy Access** menu tree (Accounting → Financial Accounting → General Ledger...) if you don't know the code.

3. Read the org structure — the spine of FICO.

Object	What it is	Config T-code to view
Company Code (4-char, e.g. `1000`)	The legal entity that produces a balance sheet & P&L. Statutory books live here.	`OX02`
Chart of Accounts (e.g. `INT` / `CAIN`)	The master list of GL account numbers, shared across company codes	`OB13`
Controlling Area (e.g. `1000`)	The CO umbrella; can span several company codes if they share a COA & fiscal year	`OKKP`
Cost Center (e.g. `IT-DEL-100`)	Where cost is *collected* — a department (IT, Finance, Facilities)	`KS03`
Profit Center (e.g. `PC-RETAIL`)	Where profit is *measured* — a business line / responsibility area	`KE53`
Segment / Business Area	Reporting slices under IFRS 8 / new GL	doc splitting config

Worked example: GCC entity **1000 – Acme India Pvt Ltd**, chart of accounts **CAIN**, controlling area **1000**. IT department cost centre **IT-DEL-100** posts salary and AMC costs; profit centre **PC-SHARED** absorbs them; at month end those costs get *assessed* to the business profit centres.

4. FI vs CO — the mental split. - **FI (Financial Accounting)** = external books. Every posting hits a **GL account**, produces the statutory Balance Sheet & P&L, feeds tax and audit. Sub-ledgers: **AP** (vendors), **AR** (customers), **Asset Accounting (AA)**, **Bank**. - **CO (Controlling)** = internal management view. The *same* expense also lands on a **cost element / cost centre / internal order** so managers see cost by department and product. In S/4HANA FI and CO are merged into one line-item table (`ACDOCA` — the Universal Journal), so a P&L GL expense account **is** the cost element (cost element is now just a GL account of type "Primary Costs/Revenue").

5. Master data — open one of each.

GL master — `FS00` : enter GL account `400000` + company code `1000` . - **Type/Description** tab: Account group (e.g. P&L Expense), P&L vs Balance-sheet flag, short/long text. - **Control Data** tab: currency, tax category, **Open Item Management** (tick for clearing accounts like GR/IR, bank clearing), **Line Item Display**, Sort key. - **Create/Bank/Interest** tab: field status group (which fields are required at posting), post automatically only.

Vendor master — `BP` in S/4HANA (the single **Business Partner** transaction; ECC used `XK01/FK01`). Roles: FI Vendor (`FLVN00/01`), Purchasing. Key fields: reconciliation account (e.g. `160000` Trade Payables), payment terms, withholding tax (TDS) type/code, bank details.

Customer master — `BP` (ECC: `XD01/FD01`). Recon account `140000` , payment terms, credit control.

Cost centre master — create `KS01` , change `KS02` , display `KS03` : enter controlling area `1000` , cost centre `IT-DEL-100` , valid-from/to dates, category (e.g. `1` Administration), person responsible, hierarchy area, profit centre link.

6. Fiori. Modern S/4HANA front-end is the **Fiori launchpad** (browser). Tiles group apps by role (GL Accountant, AP Accountant). Key apps mirror GUI T-codes: **Manage Journal Entries** (≈ FB50/FB03), **Post General Journal Entries**, **Display Line Items – General Ledger** (≈ FBL3N as the app **Display Line Item, GL**), **Trial Balance**. You can still open the classic T-code inside Fiori via the "SAP GUI for HTML" tile. Search apps in the top search bar.

The output

A one-page orientation map you can reproduce blind:

```
CLIENT 800
  └─ Company Code 1000 (Acme India) ──> statutory B/S + P&L  [FI]
        Chart of Accounts: CAIN
        Controlling Area 1000  ────────> management view       [CO]
             Cost Centers: IT-DEL-100, FIN-DEL-200 …
             Profit Centers: PC-SHARED, PC-RETAIL  …
Master data:
  GL      FS00   (e.g. 400000 Salaries, 160000 Trade Payables)
  Vendor  BP     (recon a/c 160000, TDS code, payment terms)
  Customer BP    (recon a/c 140000)
  Cost ctr KS01/03  (IT-DEL-100 → PC-SHARED)
Navigation: command field  /n /o ; Easy Access menu ; Fiori tiles
```

Checks & gotchas

- **Client vs Company Code confusion** is the classic rookie error — client is the login environment, company code is the legal entity. You post to a company code; you log into a client.
- A GL account must exist **at both** chart-of-accounts level and company-code level (`FS00` shows both) before you can post — "account not created in company code 1000" is a real error.
- **Open Item Management** must be ON for clearing accounts (GR/IR, bank clearing) or you can never clear them; it must be OFF for reconciliation accounts (you never post directly there).
- In S/4HANA you **cannot** use `XK01/FD01` to create vendors/customers — it redirects to `BP`. Saying "I'll create the vendor in XK01" in an S/4 shop flags you as ECC-only.
- Cost element vs GL account: in ECC they were separate objects (`KA01` to create a cost element); in S/4HANA cost elements are folded into GL master data (account type merged). Know this — it's the #1 "do you actually know S/4" question.

Interview drill

Q1. Difference between S/4HANA and ECC in Finance terms? "S/4HANA runs on the HANA in-memory column DB and merges FI and CO into one line-item table, the **Universal Journal** `ACDOCA` — one source of truth, no reconciliation between FI and CO, real-time reporting, no aggregate/index tables like BSEG/GLT0 for reporting. Business Partner replaces separate vendor/customer masters, cost elements become GL accounts, and the New Asset Accounting and material ledger are mandatory. ECC is the older ERP on a classic RDBMS with separate FI and CO totals tables that had to be reconciled."

Q2. Company Code vs Controlling Area vs Profit Centre? "Company code is the legal entity that files statutory accounts. Controlling area is the CO reporting umbrella and can group several company codes if they share the chart of accounts and fiscal year. Profit centre is a responsibility/business-line slice for internal profitability, cutting across cost centres."

Q3. Where does a salary expense land in both FI and CO? "FI: Dr Salaries GL (P&L expense), Cr Salary Payable / Bank. Simultaneously in CO the debit carries a cost object — the department cost centre — because that P&L account is a primary cost element. Month-end assessments then push the cost centre balance to profit centres."

Practise free

- **openSAP** (open.sap.com) — free MOOCs: "SAP S/4HANA Finance" and "Financial Accounting in SAP S/4HANA" give guided navigation and screenshots.
- **SAP Learning Hub / Learning Journeys** at learning.sap.com — free tutorials; some hands-on need a paid subscription, but the reading and videos are open.
- A live sandbox is the only true rehearsal: SAP offers **S/4HANA Fully-Activated Appliance** trials on SAP CAL (24-hour AWS/Azure instance, you pay only cloud compute ~₹100–300/session) and a **BTP free tier**. Third-party training servers (Michael Management, ERPPrep) rent GUI access hourly.
- Free-forever: install **SAP GUI** (free download with an SAP account) and practise *navigation muscle memory* — command field syntax, menu paths, session handling — even against a demo/idle connection, so on day one you move fast.

- Draw the org-structure map above from memory each morning until it's automatic; that single diagram answers 40% of FICO interview questions.

GL / AP / AR Transaction Drills (with Entries)

What you'll be able to do

You'll be able to post the everyday finance transactions of a company through SAP by T-code from memory, and — this is what separates an operator from a button-pusher — state the exact **Dr/Cr accounting entry with amounts** behind each posting *before* you hit save. You'll post a GL journal (FB50), a vendor invoice (FB60), a customer invoice (FB70), receive and make payments (F-28 / F-53), park for approval (FV50), display and drill into any document (FB03), read open items (FBL1N/FBL3N/FBL5N), and reverse a mistake (FB08). We'll carry one week of April transactions for **Acme India (company code 1000)** all the way through.

The drill — step by step

Setup for the example: COA `CAIN`, currency INR, period 01/2026 open. GLs: 400000 Salaries, 401000 Rent, 160000 Trade Payables, 140000 Trade Receivables, 200000 Sales, 100000 Bank, 154000 Input GST, 175000 Output GST.

1. Straight GL journal — `FB50` (Enter G/L Account Document). Screen: Document Date `05.04.2026`, Posting Date `05.04.2026`, Company Code 1000, Currency INR. Grid rows: G/L acct, D/C indicator (S=Debit, H=Credit), Amount, Cost Center (for P&L lines). *Txn A — book April rent accrued, no invoice yet:* - Row1: 401000 Rent, **S** 100,000, Cost Center FIN-DEL-200 - Row2: 240000 Rent Payable, **H** 100,000 Simulate (Shift+F9 / Document → Simulate), check the entry balances to 0, **Post (Ctrl+S)** → document number e.g. `100000123`. **Entry:** Dr Rent 1,00,000 / Cr Rent Payable 1,00,000.

2. Vendor invoice — `FB60` (Enter Incoming Invoice, FI only). Header: Vendor `V1000` (Sharma Supplies), Invoice date `06.04.2026`, Posting date, Amount `1,18,000` (incl GST), Calculate tax OFF (we post GST manually) or ON with tax code. Payment terms default from master. GL tab lines: - 402000 Office Supplies, **S** 1,00,000, Cost Center - 154000 Input CGST/SGST, **S** 18,000 The vendor line (Cr 160000 Trade Payables 1,18,000) is generated **automatically** from the vendor master's reconciliation account. **Entry:** Dr Office Supplies 1,00,000 + Dr Input GST 18,000 / Cr Vendor (Trade Payables) 1,18,000. *(For procurement with a PO + goods receipt you'd use MIRO instead — the 3-way match invoice; FB60 is the FI-only direct invoice with no PO.)*

3. Customer invoice — `FB70` (Enter Outgoing Invoice, FI only). Header: Customer `C2000` (Rao Retail), Amount `2,36,000`, invoice/posting date. GL tab: - 200000 Sales, **H** 2,00,000, Profit Center PC-RETAIL - 175000 Output GST, **H** 36,000 Customer line Dr 140000 Trade Receivables 2,36,000 is auto-generated from the customer recon account. **Entry:** Dr Customer (Trade Receivables) 2,36,000 / Cr Sales 2,00,000 + Cr Output GST 36,000.

4. Incoming payment from customer — `F-28`. Header: posting date `20.04.2026`, Bank acct 100000, Amount received `2,36,000`, Value date. Under "Open item selection": Account `C2000`, Account type D (customer). Click **Process Open Items** → the ₹2,36,000 invoice shows; double-click to *activate* it so "Not assigned" = 0.00. Post. **Entry:** Dr Bank 2,36,000 / Cr Customer 2,36,000 — and the invoice open item is **cleared** (fully applied).

5. Outgoing payment to vendor — `F-53` (Post Outgoing Payments). Header: Bank acct 100000, Amount `1,18,000`. Open item selection: Account `V1000`, type K (vendor). Process open items → activate the

₹1,18,000 invoice → Not assigned 0.00 → Post. **Entry:** Dr Vendor 1,18,000 / Cr Bank 1,18,000 — vendor open item cleared. (Bulk/automatic payments use the payment run **F110**.)

6. Park for approval — `FV50` (park GL) / `FV60` (park vendor). Use when a junior enters but a senior must approve. Same screen as FB50 but you **Park (document not yet posted, no GL update)**. Approver opens `FBV0` to display/post parked docs, or `FV50` change → Post. Nothing hits the ledger until posting.

7. Display line items — the reading tools. - `FBL1N` — Vendor line items (open/cleared/all). Filter V1000, "Open items at key date". - `FBL3N` — GL line items. Filter 100000 Bank to see every movement. - `FBL5N` — Customer line items. Filter C2000. (These are the ECC classics; the S/4 apps are "Display Line Items – General Ledger/Supplier/Customer". FAGLL03 is the new-GL GL line item report.)

8. Display any document — `FB03`. Enter document number 100000123 / company code / fiscal year → see header, all line items, and via Environment → Document Environment the FI/CO postings.

9. Reverse — `FB08` (Individual Reversal). Enter the wrong document number, reversal reason (e.g. `01` reversal in current period, `02` reversal in closed → posts in current), posting date. Post → SAP creates a mirror document with debits/credits flipped and links the two. Mass reversal = `F.80`.

The output — Acme April sub-ledger picture

Doc	T-code	Dr	Cr	Amount
123	FB50	Rent 401000	Rent Payable 240000	1,00,000
124	FB60	Office Supp + Input GST	Vendor V1000 (160000)	1,18,000
125	FB70	Customer C2000 (140000)	Sales + Output GST	2,36,000
126	F-28	Bank 100000	Customer C2000	2,36,000
127	F-53	Vendor V1000	Bank 100000	1,18,000

Vendor V1000 open items after: **0** (invoice cleared by pmt). Customer C2000: **0**. Bank net movement: +2,36,000 − 1,18,000 = **+1,18,000**. GST: Input 18,000, Output 36,000 → net payable 18,000.

Checks & gotchas

- **Every document must balance to zero** by company code and (with document splitting) by profit centre/segment — SAP won't post otherwise.
- Don't post directly to a **reconciliation account** (160000/140000) via FB50 — it's controlled only through the vendor/customer line. Trying gives "account is a reconciliation account, not allowed for direct posting."
- **Posting date drives the period**, not document date. Wrong posting date = wrong month = restated close.
- After F-28/F-53 confirm the item shows as **cleared** in FBL5N/FBL1N (clearing doc number populated). A payment that doesn't clear the invoice leaves *two* open items (a debit and a credit) — the classic "why is AR overstated" bug.
- FB60 vs MIRO: FB60 has no PO/GR match; using it where a PO exists breaks the 3-way match and leaves GR/IR dangling.

- FB08 with reason 01 fails if the original period is closed — use reason 02 to post the reversal in the current open period.

Interview drill

Q1. Walk me through a vendor invoice-to-pay in SAP. "PO in ME21N → goods receipt MIGO (Dr GR/IR clearing, Cr stock/expense) → invoice receipt MIRO doing the 3-way match PO-GR-Invoice (Dr GR/IR, Dr Input GST, Cr Vendor) → payment F110/F-53 (Dr Vendor, Cr Bank). For a non-PO/FI-only invoice I skip to FB60. I'd verify GR/IR nets to zero and the vendor open item clears on payment."

Q2. What's the entry behind F-28 and how do you know it worked? "Dr Bank, Cr Customer, with the specific invoice selected as an open item so it's applied, not just a cash receipt on account. It worked when FBL5N shows the invoice as cleared with a clearing document, and 'not assigned' was 0.00 at posting."

Q3. FB08 vs a manual reversing entry — why prefer FB08? "FB08 keeps a linked audit trail (original ↔ reversal), uses the correct reversal reason for period handling, and avoids fat-finger errors from re-keying. A manual opposite entry breaks that linkage and audit clarity."

Practise free

- **Excel sub-ledger simulator:** build a journal sheet with columns Doc, Date, GL, Dr, Cr, Party; a `SUMIF` trial balance; and separate vendor/customer tabs with an "open/cleared" flag. Re-post the April set above and reconcile — that *is* the FBL1N/FBL5N logic by hand.
- Post the same five transactions in **Tally** (you already know it) and note how SAP's recon-account/open-item model differs from Tally's ledger posting — great interview colour.
- **SAP CAL S/4HANA trial** or a rented training server: actually key FB50/FB60/FB70/F-28/F-53/FB08 once end-to-end; muscle memory beats theory.
- openSAP "Financial Accounting in SAP S/4HANA" has guided posting exercises with screenshots you can shadow even without a live system.

The Month-End Close in SAP

What you'll be able to do

You'll be able to run — and explain — a full FI month-end close in SAP: post and auto-reverse accruals, set up recurring entries, clear GR/IR, run foreign-currency revaluation, execute the depreciation run, push cost-centre costs through allocations/assessments, open and close posting periods safely, and produce the financial statements from a Financial Statement Version. Most importantly you'll be able to hand an interviewer a **working-day-by-working-day (WD1–WD5) close calendar mapped to exact T-codes**, which is what a GCC R2R team actually lives by.

The drill — step by step

We close **period 04/2026 (April)** for company code 1000.

1. Cut-off & open the new period — `OB52` (Posting Period Variant). `OB52` maintains per account-type (S=GL, D=customer, K=vendor, A=asset, M=material) the *from/to* period allowed. Standard practice: keep period 04 open only for the close team (authorisation group), open 05 for operations. At the very end you *close 04* by moving the lower period to 05.

2. Accruals — `FBS1` (Enter Accrual/Deferral Document). For expenses incurred but not invoiced (e.g. April electricity ₹40,000). `FBS1`: posting date `30.04.2026`, **Reversal reason `05`** and **reversal date `01.05.2026`**. - Dr 403000 Power & Fuel 40,000 (cost centre) / Cr 240100 Accrued Expenses 40,000. On WD1 of May you run `F.81` **(Reverse Accrual/Deferral Documents)** which auto-reverses every FBS1 doc dated for reversal — so the accrual self-cancels when the real invoice arrives. (The S/4 "Accrual Engine" ACEDT is the newer, schedule-based alternative.)

3. Recurring entries — `FBD1` (define) → `F.14` (post). For fixed monthly postings (rent ₹1,00,000). `FBD1`: first run 30.04.2026, last run 31.03.2027, interval 1 month, the template entry Dr Rent / Cr Rent Payable. Nothing posts yet — it's a template. Each month `F.14` (with batch input session run via `SM35`) generates the actual document. Manage/delete via `F.15`.

4. GR/IR clearing — `MR11` and `F.13`. GR/IR (goods-received/invoice-received, acct 191100) holds the timing gap between MIGO and MIRO. `F.13` **(Automatic Clearing)** nets off GR and IR lines that match (same PO, qty, value) → they clear to zero. Genuine mismatches (goods received, invoice never came, or vice-versa) are cleared/written off with `MR11` **(GR/IR Account Maintenance)**. Target: GR/IR aged report shows only in-transit, explainable items.

5. Foreign-currency revaluation — `FAGL_FCV` (New GL) / older `F.05`. Open items and balances in USD/EUR must be restated at month-end closing rate. Load rates in `OB08` first. `FAGL_FCV`: company code 1000, valuation key date 30.04.2026, valuation area, run in **test** then **update**. - Unrealised gain: Dr FC Open Item (adjustment acct) / Cr 415000 FX Gain (unrealised). - Loss: Dr 415100 FX Loss / Cr adjustment. By default the New-GL run posts a reversal on 01.05 (valuation is provisional until the item is actually settled).

6. Asset depreciation run — `AFAB` (Post Depreciation). Company code 1000, period 04/2026. First run of the month = "Planned posting run"; corrections = "Repeat/Unplanned". Test first. Posts: - Dr 410000 Depreciation Expense (cost centre) / Cr 020100 Accumulated Depreciation. Check via asset explorer `AW01N`. (S/4 uses `AFAB` still, or the Fiori "Post Depreciation" app; New Asset Accounting posts to all ledgers in real time.)

7. CO allocations — assessment `KSU5`, distribution `KSV5`. Shared-service cost centres (IT-DEL-100) must be pushed to receiving business cost/profit centres. - **Distribution (`KSV5`)** keeps the *original* primary cost element (e.g. still shows as Salaries on the receiver). - **Assessment (`KSU5`)** uses a *secondary* assessment cost element (e.g. 9AS100) and hides the primary detail — cleaner for management reporting. Define the cycle, run in test, then post. IT cost centre balance → 0 after allocation; business centres absorb the cost.

8. Reconciliations. Sub-ledger to GL: **AP** (FBL1N/S_ALR reports) = GL Trade Payables 160000; **AR** = 140000; **Assets** (`ABST2` / `AJAB` year-end) = GL. Bank GL 100000 vs bank statement (feeds Blackline — next chapter). Intercompany balances confirmed.

9. Close the period — `OB52` move GL/sub-ledger periods to 05 so April is locked. CO period lock via **`OKP1`**.

10. Financial statements — `F.01` / `S_ALR_87012284` using a Financial Statement Version (FSV). The **FSV** (config `OB58`, e.g. `INT` / `CAIN`) is the hierarchy that maps every GL account into Balance Sheet and P&L line items. Run **`F.01`** (or S/4 `FAGLB03` for GL balances, and the Fiori "Balance Sheet/P&L" app) for company code 1000, period 04, ledger 0L → the statutory-format B/S and P&L.

The output — WD-by-WD close calendar

WD	Activity	T-code
WD-1	Reverse prior-month accruals	F.81
WD1	Open/restrict periods; post recurring	OB52, F.14
WD1	Sub-ledger cut-off; last AP/AR postings	FB60/FB70
WD2	Accruals & provisions	FBS1
WD2	GR/IR review & clearing	F.13 / MR11
WD3	FX revaluation	FAGL_FCV (OB08 rates)
WD3	Depreciation run	AFAB / AW01N
WD4	CO assessments/distributions	KSU5 / KSV5
WD4	Sub-ledger↔GL recons; intercompany	FBL1N/3N/5N
WD5	Close period; lock CO	OB52 / OKP1
WD5	Trial balance, FSV, B/S & P&L	F.01 / FAGLB03

Result artefact — an April trial balance that ties, sub-ledgers agreeing to their recon GLs, GR/IR aged and clean, FX restated at closing rate, depreciation booked, shared costs allocated, and a signed-off B/S + P&L in FSV format.

Checks & gotchas

- **Accruals with no reversal date** double-count when the real invoice posts — always set the FBS1 reversal reason/date and run F.81 next month.
- Running **AFAB** in the wrong mode (planned vs repeat) either skips assets or posts nothing — always test-run and check the log first.

- **FX rates not loaded in OB08** → FAGL_FCV values everything at 1:1 or errors. Load closing rates before running.
- Closing the period in OB52 **before** all late postings are in blocks the team — coordinate the exact lock time; leave a close-team authorisation group open.
- **Assessment vs distribution** chosen wrong distorts management reports — distribution preserves the primary cost element, assessment masks it under a secondary one.
- Sub-ledger ≠ GL means a **broken reconciliation account** or a direct GL posting to a recon account — investigate before signing the TB.

Interview drill

Q1. Walk me through your SAP month-end close. "Reverse prior accruals (F.81), open/restrict periods (OB52), post recurring (F.14) and accruals (FBS1), clear GR/IR (F.13/MR11), revalue FX (FAGL_FCV), run depreciation (AFAB), do CO assessments (KSU5), reconcile every sub-ledger to its recon GL, close periods (OB52/OKP1), then produce the trial balance and the B/S/P&L via the Financial Statement Version (F.01/FAGLB03). I map it to a WD1–WD5 calendar with owners and dependencies."

Q2. Distribution vs assessment in CO? "Both move cost from sender to receiver cost objects. Distribution (KSV5) posts using the original primary cost element, so the receiver still sees 'salaries'. Assessment (KSU5) uses a secondary assessment cost element and summarises — better for management view but you lose the primary breakdown."

Q3. How does GR/IR clearing work and why does it matter? "GR/IR is the interim clearing account between goods receipt and invoice receipt. F.13 auto-clears matched GR/IR pairs; MR11 handles genuine quantity/value mismatches. A dirty GR/IR overstates payables or accruals and is an audit red flag, so it's reviewed and aged every close."

Practise free

- **Excel close pack:** build a 6-tab close — accruals (with an auto-reverse column dated next month), recurring template, GR/IR aging, FX reval (balance × closing rate − book value = gain/loss), a depreciation schedule (WDV/SLM), and a cost-allocation matrix. Running these teaches the *logic* each T-code automates.
- Replicate an **FSV** as an Excel mapping table: GL account → B/S or P&L line → subtotal — then `SUMIF` a trial balance into statements. That's exactly what OB58/F.01 do.
- **openSAP** "Financial Close with SAP S/4HANA" and SAP's Financial Closing cockpit (`CLOCO`) docs walk the sequence; the videos are free.
- On an **SAP CAL trial**, run FBS1→F.81, AFAB in test, and F.01 once — even a single dry run makes the calendar real in interviews.

Blackline Account Reconciliation Workflow (End to End)

What you'll be able to do

You'll be able to run the balance-sheet reconciliation process the way a modern GCC/shared-services R2R team runs it — in **BlackLine** — end to end: understand how SAP GL balances flow in, prepare a reconciliation with reconciling items, attach evidence, route it through **preparer → reviewer → approver**, apply **auto-certification** and **transaction matching**, raise a **journal** for a true-up, manage **tasks/SLA/aging**, and read the **variance/risk** dashboards. You'll do a worked bank-vs-GL recon with real reconciling items, and you'll be able to replicate the exact control in Excel for free.

The drill — step by step

What BlackLine is: a cloud "financial close" platform that sits *on top of* the ERP. It doesn't replace SAP — it imports SAP GL balances and transactions, then governs the *certification* of every balance-sheet account. Core modules: **Account Reconciliations**, **Transaction Matching**, **Task Management**, **Journal Entry**, **Variance Analysis**. Trintech (Cadency/Adra) and Oracle ARCS are the main alternatives; the workflow concept is identical.

1. Data in — the GL balance feed. BlackLine imports two things from SAP: the **trial balance** (GL account balances per company code/period, e.g. via a scheduled flat-file or SAP connector) and, for matching, **transaction detail** (GL line items, bank statement lines). Each SAP GL account is mapped to a BlackLine **account** with an assigned **preparer, reviewer, risk rating, and template type**.

2. Auto-certification rules. Low-risk, low-value, or zero-balance accounts don't need a human every month. Rules such as "*balance = 0*", "*balance unchanged from prior period*", or "*balance < ₹50,000 threshold*" auto-certify the account and stamp it complete — the preparer only touches exceptions. This is the single biggest efficiency lever; know it cold.

3. Choose the template & prepare. For an account needing work (say **Bank GL 100000**), open the recon. Template types: - **Account Analysis** — you list what makes up the balance. - **Balance Comparison** — GL balance vs an external source (bank statement, sub-ledger). - **Roll-forward** — opening + activity − closing (accruals, prepaids, fixed assets). For the bank recon use **Balance Comparison**: on one side the **GL balance** (from SAP), on the other the **source balance** (bank statement). BlackLine computes the **difference**, which you must fully explain with **reconciling items**.

4. Transaction Matching (auto-match). Before manual work, the Matching module pairs GL bank lines to bank-statement lines on rules (amount + date + reference). Matched items drop out; only **unmatched** items remain as candidate reconciling items — the automated bank-rec engine.

5. Enter reconciling items. For each unexplained difference add an item: type (timing / error / in-transit), amount, description, GL date, aging start, and **supporting document** (upload the bank statement PDF, the SAP FB03 screenshot). Flag items needing a correction for a **journal**.

6. Raise a journal if needed. If the recon reveals a real GL error (e.g. bank charges never booked), create a BlackLine **Journal Entry** (Dr Bank Charges / Cr Bank) that, on approval, is exported back to SAP for posting (or posted directly via connector). The reconciling item then clears next period.

7. Certify & route — the three-role flow. - **Preparer** completes the recon, ticks that reconciling items are supported and aged appropriately, and **submits**. - **Reviewer** checks quality, evidence, and item validity → **approves** or **rejects with comment** (back to preparer). - **Approver** (often a controller for high-risk accounts) gives final sign-off. Every action is timestamped with the user — a full **audit trail** (who prepared, reviewed, approved, when, with what evidence).

8. Task Management & SLA/aging. The whole close (not just recons) runs as scheduled **tasks** with owners, due dates by working day, and dependencies — a digital close calendar. **SLA/aging**: reconciling items carry an age; items older than policy (e.g. >60 or >90 days) escalate and hit the **risk dashboard**. Overdue recons flag red for management.

9. Variance / risk dashboards. Variance Analysis auto-flags accounts whose balance moved beyond a threshold (absolute or %) vs prior period and demands an explanation. Management sees a heat-map: % of accounts certified, overdue, high-risk, aged items — the close scorecard.

The output — worked Bank vs GL reconciliation (Bank GL 100000, April 2026)

	Amount (₹)
GL balance (SAP, 30.04.2026)	11,80,000
Bank statement balance	12,10,000
Difference to explain	(30,000)

Reconciling items:

#	Item	Type	Amount (₹)	Evidence	Age (days)
1	Cheque issued to vendor, not yet cleared bank	Timing (o/s cheque)	−50,000	FB03 doc 127	8
2	Customer NEFT received, not yet in GL	Timing (deposit in transit)	+18,000	Bank stmt line	3
3	Bank charges on statement, not booked	Error → journal	+2,000	Bank stmt	2
	Net reconciling items		**−30,000**		

Tie-out: GL 11,80,000 − 50,000 + 18,000 + 2,000 = **12,10,000** = bank balance. Difference fully explained → recon certifiable. Item 3 spawns Journal `Dr 404000 Bank Charges 2,000 / Cr 100000 Bank 2,000` exported to SAP; items 1 & 2 self-clear when the cheque clears and the NEFT posts.

Certification trail: *Prepared — A.Rao 02-May 10:14; Reviewed — S.Iyer 03-May 09:40; Approved — Controller 03-May 16:20.*

Checks & gotchas

- **The recon isn't done when the difference is small — it's done when every rupee of difference is explained by a supported reconciling item.** An unexplained residual, even ₹100, fails review.

- **Aged reconciling items** are the real risk: a "timing" item still open after 90 days usually isn't timing — it's an error or a missed write-off. Auditors go straight to the aging report.
- **Auto-certification thresholds** set too loose let real movements slip through uncertified — the balance/threshold rules must be defensible to audit.
- **BlackLine journals must actually post in SAP** — a journal approved in BlackLine but not exported/posted leaves BlackLine and SAP out of sync; confirm the SAP document number.
- **Segregation of duties**: preparer ≠ reviewer ≠ approver. Same person on two roles breaks the control and is an audit finding.
- Balance comparison uses the **period-end** GL and bank balances at the **same date** — mixing a 30-Apr GL with a 2-May bank balance manufactures fake reconciling items.

Interview drill

Q1. How does BlackLine tie to SAP? "BlackLine imports the SAP trial balance and transaction detail each period; each GL account maps to a BlackLine reconciliation with an owner and risk rating. We certify the SAP balance is correct and supported; corrections are raised as BlackLine journals exported back to SAP. It governs the close on top of SAP — it doesn't replace the ledger."

Q2. What is auto-certification and why is it safe? "Rule-based sign-off for low-risk accounts — zero balances, unchanged balances, or balances under a threshold certify automatically so humans focus on exceptions. It's safe when thresholds are risk-calibrated and documented, and high-risk or materially-moved accounts are always excluded from auto-cert and force manual review."

Q3. A reconciling item is 120 days old — what do you do? "Investigate root cause — an aged 'timing' item usually isn't timing. If it's an error, raise a correcting journal; if it's unrecoverable, write it off with approval; then fix the process so it doesn't recur. I'd never let it keep rolling — it's a control and audit red flag on the aging dashboard."

Practise free

BlackLine has no free tier, but the **control is fully replicable in Excel** — and doing so proves you understand it better than clicking buttons: - **Recon workbook:** one tab per account. Columns: GL balance, source balance, difference, then a reconciling-items table (type, amount, evidence link, aging-start date, `=TODAY()-start` for age). A cell check `=GL - source - SUM(items)` must equal **0** to certify. - **Auto-cert logic:** `=IF(OR(bal=0, bal=priorbal, ABS(bal)<50000), "Auto-Certified", "Manual")`. - **Workflow columns:** Preparer / Reviewer / Approver names + date stamps, with conditional formatting red when overdue vs a due-date column — mimics task management and SLA. - **Aging dashboard:** a pivot of reconciling items bucketed 0–30 / 31–60 / 61–90 / 90+ with a red flag on the last bucket. - **Transaction matching:** two lists (GL lines, bank lines), match on amount+date with `XLOOKUP`/Power Query merge; the unmatched rows are your reconciling items — exactly BlackLine's engine. Do this once against the April bank recon above and you can speak to every BlackLine screen from real experience. BlackLine and Trintech both publish free product demos and webinars (blackline.com, trintech.com) — watch two to learn the exact button names.

Intercompany & the Consolidation Submission (SAP BPC / Group Reporting)

What you'll be able to do

After this chapter you can explain — and walk an interviewer through — how a group with 40+ legal entities turns entity-level trial balances into **one consolidated financial statement**: how intercompany (IC) balances are matched and eliminated, how a foreign subsidiary's numbers get translated into the group currency, what a "BPC submission" actually is at the button level, and what a consolidation elimination journal or minority-interest calculation looks like. This is the exact domain the Dentsu "BPC Analyst" JD asks for ("intercompany matching, elimination, currency translation, monthly group submission in SAP BPC"). Be honest in interviews: the *hands-on clicking* needs employer access — but the **logic** is fully learnable, and that logic is what gets tested.

The drill — step by step

Consolidation runs on a **hierarchy**. Learn the vocabulary first because every tool (SAP BPC, SAP Group Reporting/S4HANA, Oracle HFM/FCCS) uses the same shape, only different button names.

The reporting hierarchy (dimensions): - **Entity** — each legal entity submits (e.g. IN01 India, GB01 UK, US01 US). Entities roll up to sub-groups → group total. - **Account** — the group chart of accounts (mapped from each entity's local GL). - **Time** — 2026.07 (period). - **Category / Version** — Actual, Budget, Forecast. - **Flow / Movement** — opening, additions, disposals, closing (needed for FX and fixed-asset roll-forwards). - **Intercompany (Trading Partner)** — *who* the balance is with. This is the dimension that makes elimination possible.

Step 1 — The package / submission. Each entity's controller loads its trial balance into BPC (the "**data package**" — an upload from SAP ECC/S4 via a data manager package, or a manual input schedule in EPM add-in for Excel). The controller runs **validation** (does the local TB balance to zero? do control accounts tie?) and then **submits/locks** the entity for the period. "Doing the BPC submission" = load → validate → run currency translation → confirm IC → lock. That's the monthly ritual.

Step 2 — Currency translation. A foreign sub reports in local currency; the group reports in, say, INR or USD. Apply the standard (IAS 21 / current-rate) method: - **P&L** items → **average rate** for the month. - **Assets & liabilities** (balance sheet) → **closing rate**. - **Equity / share capital** → **historical rate**. - The plug that makes the translated balance sheet balance is booked to **Currency Translation Adjustment (CTA)** / FCTR in equity (OCI).

Worked example — UK sub GB01, month 2026.07:

Item	GBP	Rate	INR (₹)
Revenue	1,000,000	105 (avg)	105,000,000
Net assets (BS)	500,000	106 (close)	53,000,000
Share capital	200,000	100 (hist)	20,000,000

The difference between the closing-rate net assets and the historical-rate equity + retained earnings flows to **CTA in equity**. BPC does this with a **rate table** + a translation logic script; you just load rates and hit "Run Currency Translation."

Step 3 — Intercompany matching. GB01 sold services to IN01. GB01 books **IC Receivable ₹8,00,000 (TP = IN01)**; IN01 books **IC Payable ₹8,00,000 (TP = GB01)**. The **IC matching report** pairs them by trading partner and flags **mismatches** (timing, FX, one side missed the invoice). Chasing and clearing those mismatches before lock is the analyst's real job — a ₹5,000 gap between the two sides holds up the whole consolidation.

Step 4 — Elimination. Once matched, the system posts **elimination entries** at the group level so intragroup items don't inflate the group: - **IC receivable** ↔ **IC payable** → eliminate. - **IC revenue** ↔ **IC cost** → eliminate. - **IC profit in closing inventory** (unrealised profit) → eliminate. - **Investment in subsidiary** ↔ **subsidiary's share capital** → eliminate at acquisition, recognise **goodwill** on the difference.

Step 5 — Minority / non-controlling interest (NCI). If the group owns 80% of a sub, 100% of the sub is consolidated line-by-line, then **20% of the sub's net assets and 20% of its profit are carved out to NCI**. Example: sub profit ₹1,00,00,000 → NCI ₹20,00,000 shown separately in equity and in the P&L split.

Step 6 — Group reports. Run the consolidated P&L, BS, and the movement/flow reports; controllers review; the submission is signed off and the group set is locked.

The output

A consolidation elimination journal (group level, 2026.07):

Line	Account	Entity	Trading Partner	Dr (₹)	Cr (₹)
1	IC Payable	IN01	GB01	8,00,000	
2	IC Receivable	GB01	IN01		8,00,000
3	IC Revenue	GB01	IN01	8,00,000	
4	IC Cost of services	IN01	GB01		8,00,000

NCI carve-out (80%-owned sub, profit ₹1,00,00,000):

Description	₹
Sub net profit (100% consolidated)	1,00,00,000
Attributable to parent (80%)	80,00,000
Non-controlling interest (20%)	**20,00,000**

Plus a translated, eliminated, consolidated set: Group Revenue, Group EBIT, Group Net Profit split parent/NCI, and CTA sitting in OCI.

Checks & gotchas

- **IC must net to zero after elimination.** If group IC receivables ≠ IC payables, a mismatch is unresolved — the classic close-blocker.

- **Translation gotcha:** using closing rate on the P&L (should be average) or average on the balance sheet is the most common error; CTA absorbs the mistake silently, so the balance still "balances" while being wrong.
- **Flow dimension:** FX movement and asset roll-forwards break if opening + movements ≠ closing.
- **Investment elimination** must use the *acquisition-date* rate for goodwill, not the current rate.
- **Rounding:** consolidated statements are often in thousands/millions — rounding differences need a defined tolerance, not manual fudging.
- **Locked periods:** once submitted and locked, changes need a **re-open + re-submit**, which is audited.

Interview drill

Q: What's the difference between elimination and translation? A: Translation converts a foreign entity's local-currency numbers into the group currency (P&L at average, BS at closing, equity at historical, difference to CTA in OCI). Elimination removes *intragroup* transactions and balances (IC receivable/payable, IC revenue/cost, investment vs share capital, unrealised profit) so the group isn't double-counting internal activity. They're sequential: translate each entity first, then eliminate at group level.

Q: How is minority/non-controlling interest handled in consolidation? A: For a partly-owned subsidiary you still consolidate 100% of assets, liabilities, income and expenses line-by-line (control, not ownership %, drives consolidation). Then you carve out the non-controlling shareholders' share of net assets and of profit — say 20% for an 80%-owned sub — presented separately within equity and as a split of profit for the year.

Q: Walk me through a BPC monthly submission. A: Load the entity trial balance via the data manager package (or EPM Excel input), run validation to confirm it balances and control accounts tie, load rates and run currency translation, review the IC matching report and clear mismatches with counterparties, submit and lock the entity. At group level, controllers run eliminations, review the consolidated set, and lock the group version.

Practise free

You can't get a free SAP BPC tenant, so **rebuild the logic in Excel** — which is exactly how many analysts prototype anyway: - Build three entity tabs (IN01, GB01, US01), each a mini trial balance with a **Trading Partner** column. - A **rates tab** (avg/closing/historical); use `SUMPRODUCT`/lookup to translate each entity to a group-currency tab; plug the difference to a CTA line. - An **IC matching tab**: `SUMIFS` each entity's IC receivable against the counterparty's IC payable, flag non-zero gaps with conditional formatting. - An **elimination tab** that posts the reversing group journals, then a **consolidation tab** = sum of entities + eliminations, with an NCI carve-out row. - Free reading to ground the vocabulary: **IAS 21** (translation), **IFRS 10** (consolidation/control), **IFRS 3** (goodwill/NCI) — the ICAI Advanced Accounting module and the IFRS Foundation summaries are free. Oracle's public **FCCS/HFM** admin guides describe the same package/journal/elimination flow if you want a second vendor's vocabulary.

Balance-Sheet Review, Commentary (BS Calls) & SOX Evidence

What you'll be able to do

After this chapter you can run a **flux (variance) analysis** on a balance sheet and P&L, write the **commentary** that gets read aloud on a monthly "BS review call", triage and clean up **aged reconciling items**, and justify **provisions/accruals** with real drivers. You'll also be able to speak **SOX / J-SOX** fluently: what a **RCM (Risk & Control Matrix)** is, what counts as **control evidence**, how **preparer/reviewer sign-off** works, and what a **control deficiency** is. This maps directly onto the Dentsu (J-SOX) and HP (SOX) R2R JDs, both of which list "balance-sheet reviews, flux commentary, SOX control evidence" as core duties.

The drill — step by step

Step 1 — Pull the two-period balances. For each BS account, get current-month closing and prior-month (and often prior-year and budget) balances. In SAP that's **FS10N** (GL balance display) or **FBL3N** (line items); export to Excel.

Step 2 — Compute the flux. Add columns: **₹ change** and **% change**. Apply a **materiality threshold** — a common rule is *explain any movement > ₹5,00,000 AND > 10%* (dual threshold so you don't chase tiny-base swings). Flag every breach.

Worked example — Accrued Expenses:

Account	Jun-26	Jul-26	Δ ₹	Δ %	Explain?
Accrued marketing spend	42,00,000	68,00,000	+26,00,000	+62%	Yes
Prepaid insurance	9,00,000	8,25,000	−75,000	−8%	No
IC receivable – GB01	8,00,000	8,00,000	0	0%	No

Step 3 — Find the driver, not the number. Don't write "accruals went up ₹26L." Open **FBL3N**, sort by posting date, and identify *what* posted: e.g. a Q3 brand-campaign accrual of ₹28L raised, partly offset by ₹2L of June accruals reversing. Now you have a *reason*.

Step 4 — Test the account against its reconciliation. Tie the GL balance to the supporting recon (from the Blackline chapter). Check the **aging** of open reconciling items:

Item	Amount ₹	Age (days)	Status	Action
Unidentified bank credit	1,20,000	95	Aged >90d	Escalate, likely customer receipt to apply
GR/IR mismatch PO 45001	3,40,000	22	Open	Chase goods receipt
Duplicate accrual	60,000	130	Aged	**Write off — reverse**

Aged items (>90 days) are a red flag on BS calls; the drill is *clean them up*, not just report them.

Step 5 — Justify provisions & accruals. For each, state (a) the driver, (b) the calculation basis, (c) evidence. Example — bad-debt provision: "ECL model, 5% on >180-day receivables = ₹12,00,000; supported by the aged debtors report dated 31-Jul-26." A provision without a documented basis fails both the BS review and SOX.

Step 6 — Write the commentary. One tight paragraph per flagged account: *what moved, by how much, why, and whether it's expected/one-off/reversing.* This is what you read on the call.

Step 7 — SOX evidence for the control. The BS review *is itself a SOX control* ("Management review of balance-sheet reconciliations, monthly"). To evidence it: - **Preparer** signs/dates the recon and flux. - **Reviewer** (independent, more senior) signs/dates, with review notes. - Attach the source reports (FBL3N export, aging, bank statement) as evidence. - Map the control to the **RCM**: risk → control → frequency → owner → evidence.

The output

A sample BS-call commentary (what actually gets circulated):

> **Account: Accrued Expenses (2100100) — Jul-26** Balance ₹68,00,000, up ₹26,00,000 (+62%) MoM. **Driver:** ₹28,00,000 accrual raised for the Q3 brand campaign (IO #4471, media plan approved 5-Jul), partly offset by ₹2,00,000 June accruals reversing on invoice receipt. Movement is **expected and one-off**; accrual will reverse on vendor invoicing in Aug–Sep. **Recon:** tied to accrual schedule v3, no unexplained items. **Aged items:** one duplicate accrual of ₹60,000 (130 days) identified — reversing this period. Prepared: A. Rao 4-Aug; Reviewed: S. Mehta 5-Aug.

A slice of the **RCM (Risk & Control Matrix):**

Risk	Control	Type	Freq	Owner	Evidence
BS balances misstated	Monthly recon prepared & independently reviewed	Detective	Monthly	R2R Lead	Signed recon + flux
Unauthorised journals	JE approval workflow before posting	Preventive	Per JE	GL Accountant	FB03 + approval log
Aged items not cleared	>90-day items escalated to controller	Detective	Monthly	Controller	Aging report + email

Checks & gotchas

- **Flux without a driver is worthless** — "up 62%" is not commentary; the *why* is.
- **Dual threshold** matters: a ₹4,000 account can swing 300% and mean nothing; a ₹2cr account can move 3% and be huge.
- **Sign-off dates must be after the close date and preparer ≠ reviewer** — same person on both is an instant SOX deficiency (segregation of duties).
- **Evidence must be dated and version-controlled**; "I reviewed it" verbally is not evidence.
- **Aged items you keep re-explaining** each month is the tell that clean-up isn't happening — auditors notice repeats.
- **Provisions must have a policy basis**; a round-number "management estimate" with no calc is a finding.

Interview drill

Q: What is flux analysis and how do you decide what to explain? A: Flux (fluctuation/variance) analysis compares each account's current balance to a baseline — prior month, prior year, or budget — and explains material movements. I apply a dual threshold, e.g. > ₹5L and > 10%, so I focus on genuinely significant swings rather than small-base noise. For each flagged item I go to the line-item detail (FBL3N) to find the actual driver — a specific accrual, reclass, or timing — and state whether it's expected, one-off, or reversing.

Q: What's a control deficiency, and give an example from R2R. A: A deficiency is a control that isn't designed or operating well enough to prevent or detect a material misstatement on a timely basis. Example: a BS reconciliation signed by the same person as preparer and reviewer — that's a segregation-of-duties failure. It escalates from a deficiency to a *significant deficiency* to a *material weakness* depending on the magnitude and likelihood of misstatement it could allow.

Q: How do you evidence a management-review control for SOX? A: The reviewer must show what they actually did, not just that they signed. So: the recon and flux with preparer name/date, an independent reviewer name/date, documented review notes or challenge, and the underlying support (GL export, aging, bank statement) attached. The key SOX expectation is "review with precision" — evidence of the questions asked and items followed up, tied to the RCM control description and frequency.

Practise free

- **Build a flux workbook** from any public company's two consecutive balance sheets (annual report PDFs are free): add Δ₹ and Δ% columns, set a threshold with conditional formatting, and write one commentary paragraph per flagged line — force yourself to hypothesise the *driver* from the notes.
- **Aging drill:** make a fake open-items list with random dates, use `=TODAY()-postingdate` to age them into 0-30/31-60/61-90/>90 buckets with `IF`, and decide an action per bucket.
- **RCM practice:** download the free **COSO 2013 framework** summary and **PCAOB AS 2201** overview; for one process (say, accruals) write 3 risks, 3 controls, and the evidence each would produce. For J-SOX specifics, the Japanese FSA's internal-control standard summaries are public.
- Rehearse the **BS call** out loud: read your commentary in 60 seconds per account as if a controller is challenging you — that's the real test.

The GCC R2R Operating Rhythm: WD Calendar, SLAs, DTP/SOP, RPA

What you'll be able to do

After this chapter you can describe — like someone who has actually sat on the desk — **how an offshore R2R team runs**: the **working-day (WD) close calendar**, the **SLAs and KPIs** you're measured on, how work is documented in a **DTP/SOP**, how requests flow through **ticket queues (ServiceNow)**, how a process gets **transitioned/knowledge-transferred** from an onshore team, and where **RPA bots (UiPath / Automation Anywhere)** fit in. This is the operating-model knowledge that Genpact, Accenture, HP and Cromwell R2R JDs implicitly test — and it's often what separates a candidate who "knows accounting" from one who "can run in a GCC from day one."

The drill — step by step

Step 1 — Learn the WD calendar. Close is measured in **working days from period-end (WD1, WD2...)**, not calendar dates, so it's consistent across months. A typical mid-market R2R close:

Day	Activity
WD-1 (pre-close)	Cut-off comms, ensure sub-ledgers ready, freeze non-essential postings
WD1	Sub-ledger close (AP/AR), accruals raised, FX revaluation (SAP **FAGL_FC_VAL**)
WD2	Intercompany postings & matching, depreciation run (**AFAB**), payroll JE
WD3	GL close, recurring/allocation journals, reclasses
WD4	Reconciliations & flux/BS review
WD5	Management reporting / BPC submission, sign-off, ledger lock

Knowing "I owned WD1–WD2 accruals and FX, WD4 recons" is exactly how you narrate experience in interviews.

Step 2 — Know your SLAs and KPIs. Offshore work is governed by a **Service Level Agreement**. The metrics you're graded on:

KPI	Typical target	What it measures
Timeliness / on-time close	≥ 98% tasks by WD deadline	Did you hit the calendar
Recon completed on time	≥ 95% by due date	Recons signed off
Accuracy / rework rate	≤ 2% error	JEs/recons redone
Aging of open items	↓ trend, nil >90d	Clean-up discipline
Ticket SLA adherence	≥ 95% within SLA	Query turnaround
Auto-cert % (Blackline)	↑ trend	Automation maturity

Step 3 — Work the DTP/SOP. Every task has a **Desktop Procedure (DTP)** / Standard Operating Procedure — a step-by-step doc: purpose, systems, T-codes, screenshots, frequency, upstream/downstream dependencies, and the control point. In a GCC you *work to the DTP and keep it current*; an out-of-date DTP is an audit and transition risk. When you learn a new task you often *write or update* the DTP — that's a concrete deliverable to mention.

Step 4 — Run the ticket queue. Business queries (a cost-centre owner questioning a charge, a vendor mis-posting, a manual JE request) arrive as **ServiceNow tickets** in a shared queue. Drill: pick up ticket → acknowledge within SLA (e.g. 4 business hours) → investigate (FBL3N/FB03) → act or route → document resolution → close. You're measured on **SLA adherence** and **first-time resolution**.

Step 5 — Understand transition / KT. New scope moves offshore through a structured **transition**: **knowledge transfer** (shadowing onshore), **reverse-shadow** (you do it, they watch), **go-live**, then **stabilisation** (hypercare). Deliverables: DTPs signed off, a **RACI**, a cutover plan. Being able to say "I was reverse-shadowed for 3 weeks before taking ownership" signals you understand the model.

Step 6 — Spot the RPA opportunities. Bots (**UiPath, Automation Anywhere, Blue Prism**) handle high-volume, rules-based, no-judgement steps in R2R: downloading bank statements, running recurring uploads, triggering FX/depreciation runs, pulling reports, doing first-pass recon matching, sending reminder emails. You feed the bot a **PDD (Process Definition Document)**. Judgement (why an accrual, is this provision reasonable) stays human. In interviews, framing "I identified the bank-recon download as a bot candidate and wrote the PDD" shows **continuous-improvement** mindset.

Step 7 — Continuous improvement (Kaizen/Lean). GCCs run CI programs — **FTE savings, cycle-time reduction, error reduction**. You're expected to log ideas, sometimes with a **green-belt** framing (DMAIC). A quantified idea ("reduced WD4 recon time 30% by templating the flux") is gold.

The output

A one-page **process ownership summary** (the artefact you'd keep, and effectively narrate in interviews):

Element	Detail
Process	R2R — GL close & BS recons, 12 entities
My WD ownership	WD1 accruals+FX, WD2 IC, WD4 recons/flux
SLA	On-time close 98%, rework ≤2%, ticket SLA 95%
Tools	SAP FICO, Blackline, BPC, ServiceNow, UiPath
DTPs owned	14, all reviewed within 6 months
CI delivered	Bank-recon bot (PDD written) → 8 hrs/month saved
KPI last quarter	Close 100% on-time, aging >90d nil, rework 1.2%

Checks & gotchas

- **WD calendar drives everything** — miss WD2 depreciation and WD3/WD4 cascade; dependencies are unforgiving.

- **SLA green ≠ quality green:** you can hit timeliness while pushing errors downstream — accuracy and aging are the honest metrics.
- **Stale DTPs** are the most common audit/transition finding; "current within 6 months" is a real control.
- **Ticket dumping:** routing a ticket without investigating just resets the clock and tanks first-time-resolution.
- **Over-automating judgement:** bots must not "decide" provisions or approve JEs — that breaks SOX and creates unmonitored risk.
- **Transition risk:** taking scope live before reverse-shadow completes is how errors spike in month one (hypercare exists for a reason).

Interview drill

Q: What does a working-day close calendar look like and why WD not dates? A: Close tasks are scheduled by working days from period-end — WD1 sub-ledger close and accruals, WD2 intercompany and depreciation, WD3 GL close and allocations, WD4 recons and flux, WD5 reporting and lock. We use WD rather than calendar dates because month-ends fall on different weekdays, so WD keeps the sequence and deadlines consistent every month and lets us measure on-time performance apples-to-apples.

Q: How are you measured in a GCC R2R role? A: Against SLAs and KPIs: on-time close (~98% of tasks by their WD deadline), recon completion rate, rework/accuracy (errors kept ≤ ~2%), aging of open reconciling items trending down with nothing over 90 days, and ticket SLA adherence with strong first-time resolution. Quality and clean-up matter as much as speed — you can be on-time but pushing errors downstream, so accuracy and aging keep you honest.

Q: Where does RPA fit in R2R, and where must it not? A: Bots suit high-volume, rules-based, judgement-free steps — statement downloads, recurring uploads, triggering standard runs, first-pass recon matching, reminders — driven by a Process Definition Document. They must not make accounting judgements or give final approval on journals or provisions; that stays human for SOX segregation-of-duties and because judgement isn't rules-based. Good CI is spotting the right bot candidates, not automating everything.

Practise free

- **Draw your own WD calendar** in Excel for a fictional 10-entity group; add a dependency column so you can see what breaks if a day slips. This alone makes you sound experienced.
- **Write one real DTP:** pick a task you know (bank recon, accrual JE), document purpose → systems → steps → control point → frequency, with screenshots. That's a portfolio piece.
- **Simulate a ticket queue:** in a free Trello/Notion board, create 10 mock queries with SLA timers and practise triage/route/close notes.
- **RPA free tier: UiPath Community Edition** and **Automation Anywhere Community** are free — automate downloading a file and dropping data into Excel to genuinely understand bot mechanics. **UiPath Academy** and Automation Anywhere University are free and give shareable certificates.
- Read free **Genpact/Accenture "R2R operating model" and SSC/GBS** whitepapers to absorb the transition/KT/CI vocabulary before you're asked about it.

R2R Interview Drills + T-Code & Close Cheat-Sheet

What you'll be able to do

After this chapter you can sit a real R2R / month-end-close interview and answer the standard rounds without hesitation: **"walk me through month-end," journal-entry and reconciliation questions, GR/IR, accruals vs provisions, a SAP T-code quiz, SOX questions, and a case ("how would you clear an aged open item")**. You'll also have a **one-page cheat-sheet** — key FICO T-codes, the WD close sequence, recon types, SOX terms — to revise the night before. These are the exact question shapes used across Genpact, Accenture, HP, Dentsu and Cromwell R2R panels.

The drill — step by step (Q&A)

Q1 — "Walk me through month-end close."

> "I work a WD calendar. Pre-close I confirm cut-off and sub-ledgers are ready. WD1: AP/AR sub-ledgers close, I raise accruals and run FX revaluation (FAGL_FC_VAL). WD2: intercompany postings and matching, depreciation run (AFAB), payroll JE. WD3: GL close — recurring, allocation and reclass journals. WD4: reconciliations and BS flux/review, clearing aged items. WD5: management reporting / BPC submission, controller sign-off, then I lock the ledger (OB52). Throughout I'm hitting SLAs — on-time close, accuracy, aging."

Q2 — "Accruals vs provisions — difference?"

> "An **accrual** is a known expense incurred but not yet invoiced — timing certain, amount fairly certain (e.g. July electricity used, bill comes August). A **provision** is a liability of uncertain timing or amount recognised because a present obligation is probable and estimable (e.g. warranty, bad-debt/ECL, restructuring — IAS 37). Both increase expense and a liability, but a provision carries genuine estimation uncertainty; an accrual is mostly a timing bridge. Both reverse — the accrual on invoice receipt, the provision on utilisation or reassessment."

Q3 — "What is GR/IR and how do you clear it?"

> "GR/IR (Goods Receipt / Invoice Receipt) is a clearing account in the P2P–GL bridge. On goods receipt: **Dr Inventory/Expense, Cr GR/IR**. On invoice receipt: **Dr GR/IR, Cr Vendor**. When both match, GR/IR nets to zero. A non-zero balance means a mismatch — goods received not invoiced, or invoiced not received, or a quantity/price gap. I analyse it with **MB5S** or **FBL3N**, chase the missing side, and clear via **F.13** (auto-clearing) or **MR11** for genuine GR/IR maintenance write-offs."

Q4 — Journal-entry quiz. "Book a ₹1,20,000 annual insurance paid 1-Jul, and the July expense."

> "On payment: **Dr Prepaid Insurance 1,20,000 / Cr Bank 1,20,000**. Monthly amortisation: **Dr Insurance Expense 10,000 / Cr Prepaid Insurance 10,000** (1,20,000 ÷ 12). By June next year prepaid is nil and full expense is recognised."

Q5 — SAP T-code quiz (rapid fire).

> FB50 — GL document entry; F-02 — general posting; FB03 — display document; FBL3N — GL line items; FS10N — GL balances; FAGL_FC_VAL — foreign currency valuation; AFAB — depreciation run; F.13 — automatic clearing; OB52 — open/close posting periods; F-04 — post with clearing; FBL1N/FBL5N — vendor/customer line items; MIRO — invoice receipt; FBRA — reset cleared items.

Q6 — SOX question. "What makes a good management-review control?"

> "It must be performed with **precision** and **evidenced**: an independent reviewer (not the preparer — segregation of duties), documented review notes/challenge, dated after the close, tied to a control in the RCM, with the underlying support attached. Signing a recon without evidence of what you checked is a deficiency. Deficiencies escalate to significant deficiency and material weakness by magnitude and likelihood."

Q7 — Case. "There's a ₹1,20,000 unidentified bank credit open for 95 days. Clear it."

> "First, investigate — pull the bank line (FBL3N/FF67) and the statement narration; check if it's a customer receipt not applied, a duplicate, a wrong-account posting, or a genuine unknown. I'd match it against open AR (FBL5N) by amount/date and contact the customer/treasury. If it's an unapplied customer receipt, apply it (F-28/F-04). If a mis-post, reclass with a corrected JE. If truly unidentifiable after reasonable effort and below policy threshold, I'd propose a write-off with controller approval and documentation. Then I'd log the root cause — a 95-day aged item usually signals a broken applied-cash process, so I'd fix upstream, not just this line."

Q8 — "Difference between reconciliation and flux?"

> "A **reconciliation** proves a GL balance is *supported* — GL vs sub-ledger/bank/schedule, with reconciling items explained. **Flux** explains the *movement* period-over-period. Recon = 'is this balance real and backed?'; flux = 'why did it change?'. Both are month-end controls; recon is completeness/accuracy, flux is analytical review."

The output — one-page cheat-sheet

Key FICO T-codes

T-code	Use
FB50 / F-02	GL journal entry
FB03	Display document
FBL1N / FBL5N / FBL3N	Vendor / Customer / GL line items
FS10N	GL account balances
FAGL_FC_VAL	Foreign currency revaluation
AFAB	Depreciation run
F.13	Automatic clearing
F-04 / F-28	Post with clearing / incoming payment
FBRA	Reset cleared items
OB52	Open/close posting periods
MIRO / MR11	Invoice receipt / GR-IR clearing
F.01 / S_ALR_87012284	Financial statement / BS report

WD close sequence

WD	Task
WD1	Sub-ledger close, accruals, FX reval
WD2	Intercompany, depreciation, payroll
WD3	GL close, allocations, reclasses
WD4	Reconciliations, BS flux/review
WD5	Reporting / BPC submission, sign-off, lock

Recon types: bank; sub-ledger-to-GL (AP/AR); balance-sheet schedule (accruals, prepaids, provisions, fixed assets); intercompany; suspense/clearing (GR/IR). Categories: *auto-certified* (system-matched) vs *manual*; risk-rated high/medium/low.

SOX terms: RCM (Risk & Control Matrix); preventive vs detective control; management-review control; segregation of duties; preparer/reviewer sign-off; control evidence; deficiency → significant deficiency → material weakness; J-SOX (Japan), SOX 404.

Accruals vs provisions: accrual = timing (incurred, not invoiced); provision = uncertain timing/amount (IAS 37, probable + estimable).

Checks & gotchas

- **Don't recite T-codes without the journal logic** — panels probe "what does F.13 actually clear?" Know the postings behind the code.
- **GR/IR direction** trips people: goods first credits GR/IR, invoice debits it — get the Dr/Cr the right way round.

- **Never say "I'd just write it off"** on the aged-item case — investigate first, write-off is last with approval.
- **Segregation of duties** is the SOX answer graders wait for — say it explicitly.
- **Reversal:** every accrual/provision answer should mention *when it reverses* — omitting it reads as shaky.

Interview drill (meta)

Q: What's the single most common R2R interview opener and how long should your answer be? A: "Walk me through month-end close." Keep it 60–90 seconds, structured by WD, and name the controls (recon, flux, sign-off, lock) and one or two T-codes — enough to show you've done it without narrating every keystroke.

Q: How do you show seniority beyond just doing tasks? A: Tie your work to metrics and improvement — "close 100% on-time, aging nil >90 days, rework 1.2%, and I automated the bank-recon download saving 8 hours a month." That shifts you from processor to owner.

Practise free

- **Flashcard the T-code table** (Anki) — cover the "Use" column and recall the code, then reverse it.
- **Mock the "walk me through close"** out loud on a 90-second timer until it's smooth; record yourself.
- **JE drills:** invent 10 scenarios (prepaid, accrual, provision, FX gain, depreciation, GR/IR clearing) and write the double entry cold; check against a free ICAI/accounting-basics reference.
- **SAP T-codes without SAP:** the public **SAP Help Portal** and **SAP Learning Hub free tier** describe every T-code's screen and fields; read the FBL3N and F.13 pages so you can talk them convincingly.
- **Pull real interview questions** from Genpact/Accenture threads on Glassdoor/AmbitionBox (free) and answer each in writing, then out loud — the aged-item case and accruals-vs-provisions come up almost every time.